

★★ READ THIS FIRST — THIS DOCUMENT COVERS TWO WEEKS, AND THE SEAM RUNS DOWN THE MIDDLE OF IT

Part 0, Part 1, Part 1.5, Part 1.6, Part 2, Part 2.6, Movement 1, Movement 2, Movement 3, Part 3 and Part 4 were rebuilt **tonight** and are about the week **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Everything else — **Part 1, Part 2.5 and Movement 3** — was written on 2026-08-29/30 and is about the week to Friday 2026-08-28. Both halves are finished work and neither is stale. But they are different weeks, and this is the first time this report has shipped with that seam in it, so it is drawn here rather than left for the reader to trip over. **Movement 3 is on both sides of the seam and that is not a mistake:** its body is 08-29 work and its ADDENDUM is tonight's, which is exactly why the addendum is a dated box at the end of the movement instead of being folded into it. The table below names the fragment for every row so nothing has to be inferred from the section label.

What this means in practice. Do not line a Movement 1 run count up against a Part 1 fill count — they are counting different days. The census that Movements 2 and 3 were computed against is the **08-28** freeze, not the one printed in Movement 1, so “the runs we missed” in Movement 2 are not the runs listed in Movement 1. The action card below is built from the 08-28 half plus the three hunts that finished tonight, and every play names its own section.

Why it happened, since it is a fault and not a choice. The durable runner started at 2026-09-04T21:05Z still carrying the previous cycle's stale cutoff (artifacts older than 2026-08-28T22:00Z are STALE). Under that cutoff the 08-29/30 fragments look current, so it logged HAVE ... skipping ten times and rebuilt none of them — while separately freezing a new census and running three greenfield clusters. **The cutoff needs to roll with the cycle;** it is on the card.

SECTION	FRAGMENT	WRITTEN	DESCRIBES
Part 0	part0_progress.html	2026-09-04 22:34 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Part 1	part1_live.html	2026-09-04 22:42 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Part 1.5	part1_5_rehab.html	2026-09-04 22:48 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Part 1.6	part1_6_day_rider.html	2026-09-04 22:40 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Part 2	part2_shadow.html	2026-09-04 22:41 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC

SECTION	FRAGMENT	WRITTEN	DESCRIBES
Part 2.6	part2_6_router_review.html	2026-09-04 22:14 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Movement 1	movement1_census.html	2026-09-04 23:06 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Movement 1	movement1_census_MGC.html	2026-09-04 22:25 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Movement 2	movement2_idle_gates.html	2026-09-04 22:38 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Movement 3	movement3_greenfield.html	2026-09-04 22:36 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Movement 3	movement3_fresh_0904.html	2026-09-04 23:08 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Part 3	rev2_answers_0829.html	2026-09-04 22:54 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Part 4	rev2_answers_0904.html	2026-09-04 22:51 UTC	the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC
Part 1	run_charts.html	2026-08-29 00:26 UTC	the week to Friday 2026-08-28
Part 2.5	part25_musings.html	2026-08-29 08:53 UTC	the week to Friday 2026-08-28
Movement 3	movement3_gold.html	2026-08-29 01:58 UTC	the week to Friday 2026-08-28

GAZBOT V7 — the Friday report

Week ending **Fri 2026-08-28** · Mon 24 – Fri 28 August 2026 · published 2026-09-04 · MNQ-led, MGC captured · PAPER · 58 plays · front matter + 12 body sections

★★★ WEEKEND CARRY — 4 LOTS ARE OPEN RIGHT NOW. THIS IS THE FIRST THING ON THE PAGE BECAUSE IT IS THE ONLY THING ON IT WITH A CLOCK.

LONG 4 MNQ from 29,641.25. ahead_pt -114.0 at the 21:00Z close — — **\$912.00 unrealised** at \$2.00 a point. data/day_rider_state.json reads closed false, flatten_blocked true, venue_ok false, venue_net_ts 18:48:06Z, note ERROR (no action): TimeoutError: .desk_reconcile at 23:11:45Z reads **venue ? = tournament +0 + rider +4 → unaccounted ?** — no venue read — IBKR truth unavailable.

What happened. The venue read has failed every two minutes since 18:48:50Z and port 4002 is REFUSING connections — the gateway is down, not the 04:28Z hang where the port stays open. **Both** eod-flatten shots (20:53Z and 20:57Z) died inside the same connectAsync TimeoutError, which is the single-connection failure Part 1.6 §6(a) describes: every flatten path on this desk shares one connectAsync, so one wedge kills the rider, the watchdog and both flatten shots together. CME is shut until Sunday 22:00Z, so nothing can be done before then.

How much this can move. Across the 7 **Sunday reopens we hold on tape** the median gap is **22.25 points** (\$178.00 on 4 lots), the worst gap DOWN is -21.25pt (-\$170.00) and the worst FIRST HOUR for a long is **-82.5pt** (-\$660.00). The reopen gap is not the risk here — the Monday is. The 2026-08-21 carry, the same shape on the same size, was -121.93pt at its Friday close and gapped +22.25 in its favour on the Sunday; it was still claimed by hand on the Monday at -267.93pt for **-\$2,149.44**, the largest single item in this report. And there is no venue stop under this one: PLACE_VENUE_STOP = False, by your decision of 2026-08-20.

The instruction, at the Sunday 22:00Z reopen — not on Saturday, because nothing can be done on Saturday. Get a venue read FIRST and confirm it says 4. Then decide FLAT or MANAGE deliberately. If the read still comes back ?, that is itself the answer: you are flat-blind, the position is unmanaged, and no managing order should go into a book you cannot see. Full row: SATURDAY #0.

★★★ THIS IS REVISION 2, WRITTEN 2026-09-04 — WHAT CHANGED, IN FIVE LINES

A proofread pass ran against the first draft and returned **20 issues and 10 unanswered questions**. All thirty are closed. Every correction is marked ★ **REV2** in the section that carried it, and the full changelog is in reports/friday_v7/sections/rev2_done.txt. **The new material is Part 4**, at the end of the body.

The five that change what you do: One — **the desk is NOT flat**. Four day-rider lots are open over this weekend, LONG from 29,641.25, −\$912 at the close, venue unreachable since 18:48Z. The box above is the whole of it, and it is **SATURDAY #0**. Two — **the Monday promotion is HELD**.

abs_veto_55s was re-run to 2026-09-04 rather than 2026-08-28; the extra week is 31 trades at −\$13.74 each and trips the row's own kill line on the LONG side. Green in 7 ISO weeks of 8, not 7 of 7. Three — **the +\$810.44 bench saving is RETIRED**, not merely un-quotable: its population cannot be rebuilt. The replacement is **+\$312.35 over 117 signals**, and the regime split that HOLD #2 rested on is refuted at $P = 0.943$. Four — **the whole week 08-31→09-04 is now graded** (Part 4 §3). It contradicts four headline claims in the body, including “nothing stopped out” and “the entire short side fired nothing”. Five — **every row of last weekend's card now carries a shipped-check** run against this box. Of 25 shippable rows: **0 DONE, 19 NOT DONE, 5 CARRIED, 1 re-measured**.

Two figures this box itself used to get wrong, corrected here. The rider's off-tape fill artefact is **\$1,053.50 / 11 split orders / 15 residual legs / 18 lots / 66.8%** of the rider's loss for the week 08-24→08-28 — not the “\$1,170.50 / 74%” printed in Rev 1, and it is one figure with one window and one unit everywhere it now appears. And the cumulative desk total in the previous version of this box, −\$4,129.44, was **through 2026-08-28**; Part 0 is a week younger and its live figure through 2026-09-04 is **−\$3,326.44**. Use Part 0's.

★★ **READ THIS FIRST — WHAT THIS REPORT HAS, AND WHAT IT IS MISSING**

★ **REV2 — THE DESK IS NOT FLAT, and the sentence that used to open this report said it was.** Rev 1 opened “the desk is FLAT ... nothing is riding over the weekend” and cited a desk_reconcile read at 22:26:32Z. That read is the **2026-08-28** one; it was true then and it is false at this document's publish time. The live read is in the red box at the top of this page and it is generated from data/day_rider_state.json and data/desk_reconcile_state.json at render time, so it cannot go stale again the way this paragraph did. Four lots are open. That was the exact failure that cost −\$2,149.44 on the week this report grades, it is **SATURDAY #0** on the card, and the fix for it — SATURDAY #5, move the flatten earlier and put the page upstream of it — was issued last weekend and is **NOT DONE**.

★ **And ONE section of this report did not regenerate as a PHASE.** The 08-28 overnight run finished **5 of its 13 body phases** — a pre-flight that warned, at 22:14:21Z, that “13 body phases declare 975m of timeouts against a 166m budget (5.9×)” and that about 7 of 13 would finish. It got 5. Part 1.5 (Rehabilitation), Part 1.6 (the day rider) and Part 2.5 (mid-week musings) were all SKIPPED before they started, each one allocated 19 minutes against a 20-minute floor. **All three were then rebuilt by hand** — Part 1.6 and Part 2.5 on 2026-08-29, Part 1.5 on 2026-08-30. ★ **REV2 — ONE description of Part 1.5, used everywhere, because this document carried three.** It is a fresh 2026-08-30 wrapper (§0a-§0c, this week's material) around last cycle's standing dossiers. It does NOT carry the build's red stale-fragment banner — its mtime is inside the cycle, so the build does not classify it as stale — but it carries **its own** dated banner in its first paragraph saying that everything below the first horizontal rule is last cycle's text describing the week to 2026-08-21. Read §0a-§0c as this cycle's and the treatments below them as standing material.

★★ **Movement 2 and Movement 3 were rescued, and it is worth knowing how.** Both phases RAN and were killed before writing their sections — but they had already written their labs to disk. Both movements below were rebuilt from those artifacts, so every number in them is this week's, computed at render time from files the phases produced. What could not be rescued was rescued nowhere: a phase that never started leaves nothing behind, which is the difference between the one section that carries a banner and the ten that do not.

★★ IF YOU READ ONLY THIS — THREE THINGS, IN ORDER

1 — The desk was switched on for 3.46% of the week, and that was probably right. Six gates over the week's **118** tradeable hours (Mon 00:00Z to Fri 22:00Z, not 120) is **708 available gate-hours**; the desk used **24h 30m — 3.46%**. (Part 1 §3's REV3 table computes this from the switch record; the 720-hour version this page used to print was arithmetic on a week that does not exist.) The tournament took three signals all week, all inside one fifty-one-minute window on Friday afternoon, and won all three for **\$330.00**. The case that this was discipline rather than cowardice is measured, not asserted — and ★ REV2 names WHICH instrument it is measured on, because this paragraph used to mix two. The **187 impulses / 127 signals** came from the SUPPRESSED-OPEN log and are retired along with the \$810. The figure below is signal_journal's own record, durable on disk since 2026-08-04: of the 380 fires it holds for the week, the **117 above each gate's own ATR floor** were entered at the first tick at or after their own timestamp and run forward on that gate's live Lot-A exit, at five stop widths at once. The benched set comes to **+\$312.35 saved over those 117 signals** at the stop the desk actually runs. ★ **That is a REVISION.** The + \$810.44 over 127 signals this page used to print is **RETIRED** — its population was counted off SUPPRESSED-OPEN journal lines and the systemd journal is a ring buffer, so it can no longer be rebuilt, let alone checked. What replaces it is a saved, re-runnable replay (scripts/rev3_bench_replay.py) over the population that IS durable. It is sign-stable across the $0.5\times-1.5\times$ stop sweep but it decays across it ($+\$624.59$ at $0.5\times \rightarrow +\$11.86$ at $1.5\times$), two days of five carry all of it, stripping three impulses takes it to $+\$66.31$, and the bootstrap band is $-\$623.74$ to $+\$1,208.31$ with $P(\text{the true saving is } \geq 0) = \mathbf{0.747}$. Part 3 §Q4's independent floor is **+\$128**. So: keep the policy, quote + \$312 with its band if you must quote anything, and never quote \$810 again. HOLD #1.

2 — The router's dominant error is no longer direction. It is LATENESS. Friday's one deliberate arm was right — four lots, four winners, $+\$181.50$ — and it was **8m 37s late**, which cost \$33.84 on one lot measured from the gate's own earlier suppressed signal, and \$205-\$705 measured from the router's side. Actuation is not the problem: the gates filled **1.4**

seconds after the switch flipped. About 15 of the 20 minutes is the router's own confirmation appetite. MONDAY #2.

3 — ★ And the structural finding, which outranks both. For three cycles this report has closed with “detection is solved, selection is not” without being able to say why. Movement 3 measured it on **52,937 minutes across 41 days**: against a fixed POINT threshold every precursor we own looks predictive (AUC to 0.81, $p=0.00$) — and once the threshold is expressed in ATR, or ATR is simply held fixed, **every one of them collapses to a coin flip or below**. A 55-point move is easy on a wide tape and hard on a quiet one, so anything correlated with ATR scores well against a point threshold. **The desk's entire precursor toolkit is one volatility meter, and we have been asking it for a direction.** BUILD #3.

★ THE HONEST HEADLINE — a red week whose loss was booked before it began

The desk booked **−\$1,246.94** across 32 legs (58 contracts — the day rider's legs are 1-3 lots each, so a row count is not a lot count), **3 of the five days green**. But **−\$2,149.44 of that landed on Monday morning** and belongs to last week: it is the four-lot rider position opened 2026-08-21T13:13:06Z, carried through the CME halt unmanaged, and closed by hand on 2026-08-24 at 11:59:39Z at 29,224.00. Strip it out and **this week's own decisions produced \$902.50** — the day rider \$572.50 and the tournament \$330.00.

Three things worth carrying into Saturday. One: there was **no trend day**. All five sessions came in at a whole-day efficiency ratio between **0.017 and 0.032** — 300 to 530 points of range bought for a net of ± 261 . Every conclusion in this report is a chop-week conclusion, and at least four separate recommendations are explicitly waiting on one trend week to be gradeable at all. Two: your hands are a dead heat **on the tournament**, and a long way from one on the rider. Three of the six tournament lots

ended MANUAL_CLAIM; re-run forward on the tick tape under their own configured exits they were worth +\$141.00 against the +\$136.50 you booked — **a delta of +\$4.50 across the week**, on the 24.25-point entry ATR the desk's own native stops record. ★ **REV2 — scope that to the book it describes.** On the DAY RIDER the same button is the main exit and Part 1.6 §4 scores it at **+\$5,426.50** against holding, across 19 in-week claims, 10 of them better than the hold. Six lots and nineteen claims are different populations and they say different things; neither is “how good your hands are” on its own. Three: the week's most useful results are all NULLS with named tests on them, and that is not a bad week — it is six batteries doing their job. Two of them (the searched day-classifier, and the BIG-TREND exit rung) killed things this desk has been half-believing for weeks.

There is no single honest total, so here are all of them, with what separates each from the next.

WHAT YOU ARE LOOKING AT	LEGS	NET \$	WHY IT DIFFERS FROM THE ROW ABOVE
Raw trades rows, 08-24 → 08-28	32	−\$1,246.94	everything the desk booked — the CASH book
Clean ledger (data_quality IS NULL)	32	−\$1,246.94	identical this week — not one flagged row. Last week there was a BADFILL lot and the two books disagreed; this week they do not, and that is worth saying rather than leaving as a silent match
The day rider alone	26	−\$1,576.94	the second desk, on the same IB account, which reads none of the router's switches. It is the bigger of the two books in both directions
... of which: the carry-in from 2026-08-21, closed by hand on 2026-08-24	1	−\$2,149.44	opened LAST week, closed in THIS one, so it is inside every total above and belongs to neither week's decisions. It is larger than everything else in the book put together
The day rider, excluding that close	25	\$572.50	what the rider's own week actually produced
The tournament alone (the six-gate desk this report is mostly about)	6	\$330.00	3 signals × 2 scale-out lots, all on Friday between 14:00 and 14:51Z. Six for six — and that is n=1, not n=6: three gates agreeing about one push, each booking two lots of the same signal
	31	\$902.50	the number to judge the week's decisions on. The cash number

WHAT YOU ARE LOOKING AT	LEGS	NET \$	WHY IT DIFFERS FROM THE ROW ABOVE
What THIS week's decisions produced (everything except the carry-in)			is the row at the top; this is the strategy number

DAY	DESK NET
2026-08-24	−\$2,149.44
2026-08-25	−\$243.00
2026-08-26	\$19.00
2026-08-27	\$16.50
2026-08-28	\$1,110.00

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THE SCORECARD — what survived the week

Lead with what survived the skeptic, never with the biggest number. Nine results carried real weight out of this week; here is where each one landed and the test that decided it. ★ **Every row below comes from a section that RAN this cycle** — nothing on this scorecard is sourced from the three sections carrying a stale banner.

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
Every precursor feature on this desk is a proxy for ATR — which is why selection keeps failing	SURVIVED — the week's most important result 52,937 minutes, 41 days. Against a fixed POINT threshold, volume z, trade-count z, ATR expansion, aggressor flow, ER(15) and ATR level all beat a coin flip with $p = 0.00$, ATR level reaching 0.81. Express the same threshold in ATR and the discrimination inverts — ATR expansion falls to 0.234 and ATR level to 0.306, both now worse than chance. Hold ATR fixed and ask within the band and everything sits between 0.29 and 0.56. It is the same instrument measured twice: a 55-point move is easy on a wide tape and hard on a quiet one. ★ REV2 — Movement 2's corroboration is WEAKER than Rev1 of this card claimed, and the correction is worth reading. The line here said “a random second beats the gates' own triggers on 6 of 6”. It compared a control that had been HANDED the run's direction against the gates' BOTH-directions number, which loses by construction. Scored like for like the gates beat the control on 3 of 5 in-direction and 4 of 5 direction-blind — on 2 to 10 lots each, where only exhaustion_short separates from its own control at all ($P = 0.033$ raw, 0.166 Holm-adjusted across the five gates). The control is a NULL, not a kill. Untouched: Movement 2's direct	Movement 3 , Movement 2

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
	measurement, the six gates as configured taking 32 lots into those runs for −\$741.50. So this row now rests on Movement 3's 52,937 minutes alone, which is where it was always strongest.	
abs_veto_55s, two-sided, is the promotion candidate	SURVIVED — the only arm on 79 that passed everything n=433, +\$5,302, +\$12.25/trade, green on 24 of 31 trading days; both halves of its life pay, both regimes pay, and both sides pay (SHORT +\$3,612 / LONG +\$1,691). Head-to-head against its own un-vetoed twin it is +\$5,350, so it is the VETO being promoted rather than the thrust. Strip its three best days and it is still +\$3,363. This week it added +\$948.50 on 45 trades without being re-tuned — the same candidate as last week, now with another week of forward evidence rather than another week of the same evidence. Read the board once, not nine times: the nine best arms on it are all green, all short, and all the same 23 trades. Added up they look like \$5,888; the truthful figure is one arm's worth.	Part 2
Being switched off was worth about \$312 — and you should not quote that number either. The \$810 is RETIRED.	LEANS RIGHT, UNPROVEN — and the honest version is the useful one ★ REVISED IN REV 2. The +\$810.44 over 127 signals this row used to carry is RETIRED , not defended: it was counted off SUPPRESSED-OPEN journal lines, the systemd journal is a ring buffer, and that population can no longer be rebuilt or checked. Part 1 §6 replaces it with a saved, re-runnable replay (scripts/rev3_bench_replay.py) over the population that IS durable — signal_journal, on disk since 2026-08-04: 117 above-floor impulses, +\$312.35 saved at the live stop, +\$2.67 a signal. It is SIGN-STABLE across the whole 0.5x-1.5x stop sweep, which is what earns it a verdict at all — but read the SLOPE, because it decays rather than holds: +\$624.59 → +\$490.34 → +\$312.35 (live) → +\$73.29 → +\$11.86. Most of what “benching” saved is the desk's stop being tight enough to guarantee those trades lost. And it is thin. Two days of five carry 100% of the gross positive (drop Tuesday and the week's saving is +\$8.39; drop Thursday, +\$16.62); strip the three biggest impulses out of 117 and +\$312.35 becomes +\$66.31; a 10,000-draw bootstrap gives −\$623.74 to +\$1,208.31, i.e. $P(\leq 0) = 0.253$. Part 3 §Q4's independent floor is +\$128. ★★ The regime decomposition is REFUTED. “The saving is entirely chop; in aligned trend benching is FREE (50 signals, −\$0.81 each)” does not survive re-derivation. On the durable population chop is +\$3.99 a signal (n=78) and aligned trend is +\$3.37 (n=35) — the same number — and a 10,000-draw label-shuffle placebo on that gap returns $P = 0.943$. There is no regime	Part 1 §6

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
	split in the bench this week, and nothing about arming may rest on one. ★ Named caveat, still standing: SUPPRESSED-OPEN is logged UPSTREAM of the 55-second veto and both absorption confirms, so any bench figure built on it is an UPPER BOUND. That is one of the reasons the \$810 is gone.	
The gates we own still cannot reach the runs we miss — third cycle, third week of tape	SURVIVED — and it went a rung deeper this time All 62 sat-out runs replayed second by second: 36,600 decision-seconds, each gate's own deciders and A/B exits, \$1.50 a round trip. Fired as configured the six gates take 32 lots into those runs for −\$741.50 (−\$23.17/lot, 18.8% win). Strip every constraint off and 114 lots make −\$168.50. A 724-cell threshold sweep finds no cell that is both positive and more than three trades wide. ★ The new rung, and its REV2 correction: previous cycles measured whether the gates made money; this one tried to measure whether they know anything, against a random entry inside the same window. Rev1 of this card reported “not one of the six triggers wins” — scored against a control that had been HANDED the direction while the gates' figure had not. Read like for like the gates win 3 of 5 in-direction and 4 of 5 direction-blind, on 2-10 lots each, and after correcting for having looked five times nothing separates from its control. The control is a NULL. The −\$741.50 above does not rest on it. Honest limit: this is a statement about the GATES, not about the census, which is why it has now survived three different run lists.	Movement 2
“The momentum shadow book is green while the routed book sits flat” — argued 106 times this week	REFUTED as an arming argument The open-hour watcher raised 106 [ACT] alerts, all making this one argument. Named test: reprice the LIVE gates' own suppressed intents at the shadow's exact ruler (2.0R target, 1.0×ATR stop) and you get −\$143.61 on 65 signals against the shadow's +\$948.50 on 45. Same week, same instrument, same exit rule, opposite sign — so the exit is not the difference. The gap is the ENTRY POPULATION: the shadow variants carry no ATR floor, so they fire in exactly the tape the live gate's floor refuses. The shadow result stands and is the promotion candidate; only its use as a reason to arm the live gate is dead.	Part 1
abs_veto_short sat armed through the exact tape it exists to trade and said nothing	PARKED — the week's #1 stone, and it is a MECHANISM question Friday 16:00-17:03Z the tape fell 328 points in one confirmed direction. The gate was armed for 46m 30s of it and produced zero signals. What stops this being a simple arm-it-earlier story: its two nearest signals landed 4 minutes BEFORE the first arm ended and 5 minutes AFTER the second, and on their own exits both lose money (−\$47.79, −\$42.07). The reading: the gate's trigger and the	Part 1

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
	router's regime read are out of phase. The router arms when a break is CONFIRMED; a thrust gate needs the moment a break STARTS. On Friday those were about 25 minutes apart. Being a mechanism question, it does not need a big sample to answer — which is why it outranks everything else in BUILD.	
The router went blind for 3h35m and every instrument said it was fine	CONFIRMED — nothing errored 43 ticks aborted. An abort is fail-closed, which is the correct default and is exactly why it is invisible: the trial log gets no line at all, systemctl reports a clean deactivation, and the tick count reads normal. 31 of 43 are the LLM session limit; the other 12 are all Monday inside the gateway window. The health page blames an “expired login” for a usage limit resetting on the hour. And it is not the only instrument that lied. Two nightly rollups (router_nightly, selector_nightly) have had no timer at all and silently stopped on 08-14 and 08-18 — every nightly number in Part 2.6 is a rebuild done during this report. Two shadow A/B pairs are byte-identical (29/29 and 251/251 trades) and have been reporting perfect nulls from a broken arm. And abs_veto_sides.py hard-codes WEEK0 = 2026-07-27, so its column headed “this week” is four and a half weeks and reports +\$4,069 where the truth is +\$948.50.	Part 2.6 , Part 2
The searched day-classifier for the Open Rider, and the BIG-TREND exit rung — two long-held beliefs, both killed	REFUTED × 2, with named tests The day-classifier: the concentration is REAL (6 green days + \$3,066 against 9 red −\$5,626, oracle ≈\$1,022/week) and the search for it is dead — a placebo against random day-removal over 10,000 draws beats 6 of 8 features outright, Holm correction across the 8 leaves a best adjusted p = 0.136, and a genuine out-of-sample leg (fit on 41 days, tested on 9) holds 4 of 8 and all four holders still lose money. The exit rung: the window was widened as far as this desk can go — 831,379 five-second bars, three entry pools including the retired V5 desk's real fills, 6,736 scored entries, a 63-cell grid plus six chandeliers. Twenty rung-cells graded, twenty failed; not one survived strip-best-3, leave-one-day-out, the half-split and the pool-split together. Both are NULLs with certificates, and both are worth more than they look — the deliverable is that data/ exit_overrides.json does not get touched this weekend, and nobody searches this sample for a day filter again.	Part 2
Hunt SIZE, not direction — 48.2% of entries go nowhere and pay \$10 each to find out	SURVIVED — and it needs no direction call at all Decompose the pooled book by what the tape actually did next. Moves of ≥120 points are 1.6% of entries (27 of 1,713) and pay +\$319.51 each, carrying +\$8,627 on a book whose total is −\$2,814. The ±20-point dead band is 48.2% of entries at −\$10.00	Movement 3 , Movement 1

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
	each — −\$8,254 of pure friction. A rule that declined nothing but the dead band, without improving direction at all, turns the book positive. The other half of the same fact: a deliberately dumb boarder fires inside 61 of 62 runs a median 3 minutes late with 45.8 points still to come — 58.9% of the whole ceiling — and it fires 66.4 times a day, only 21.1% of them inside a run. Detection is solved; the bill is everywhere else.	

§ Every gate, this week, on the clean ledger

GATE	LEGS	NET \$	WIN %	\$ PER LEG
day_rider	26	−\$1,576.94	65.4%	−\$60.65
grind_long	2	\$63.50	100.0%	\$31.75
abs_veto_long	2	\$118.00	100.0%	\$59.00
capitulation_long	2	\$148.50	100.0%	\$74.25

Computed at build time from data/gazbot7.db with data_quality IS NULL, not transcribed. **Every dollar in this report is a plain sum(pnl_usd)**, because that column is written already net of the \$1.50 round trip — verified on all 765 rows in the book. Do not subtract fees_usd from it; a previous revision did, and it deepened every stated loss by \$1.50 a contract. Gates are collapsed across their A and B lots; the per-lot split is in Part 1 §2. **Three of the six live slots are absent from this table because they fired nothing all week** — abs_veto_short, exhaustion_short and rgv_short, the entire short side of the desk, on 53 minutes of combined armed time. Movement 2 prices what that silence was worth and Part 1 §9 explains the most interesting minute of it.

THE THREE THINGS THIS WEEK ACTUALLY TAUGHT THE DESK

1 • A precursor that predicts SIZE is not a precursor that predicts a MOVE. Every feature this desk uses to decide whether something is about to happen turns out to be measuring how volatile the tape already is. It scored beautifully for months against a fixed point threshold and it collapses the instant that threshold is normalised by ATR. The lesson generalises past this

one study: **if a feature and your outcome threshold are both scaled by the same hidden variable, you are measuring the variable.**

2 · A win rate is not a sample size. The tournament went 6 for 6 this week. It was three gates agreeing about one push, each booking two lots of a scale-out — one market event, sampled six times. Every quantitative claim in Part 1 is deliberately built on populations of 30–130 signals instead, and the section says so in its own second paragraph. The same error in a different coat is the shadow board, where nine “independent” green arms are nine exits on one signal.

3 · The instruments fail SILENTLY, and they fail in the direction of reassurance. Three separate examples this week, none of which errored: a router blind for 3h35m while `systemctl`, the trial log and the tick count all read clean; two nightly rollups with no timer that simply stopped being written; two shadow A/Bs that report a perfect null because one arm never diverged. **A number that arrives without a failure mode is not a measurement** — and note that all three of these were found by the report, not by the monitoring.

★ WHAT TO ACTUALLY DO

Garrath — this is the whole answer to “so what do I actually DO?”, and it is at the front. Every row below mirrors `reports/friday_v7/plays.json`, which is the reconciled source of truth for this week; the same file drives the [Monday playbook](#), so the two cannot drift apart. ★ **Almost nothing on this card changes how the desk trades, and that is the honest answer to this week.** The tournament took THREE signals all week, all inside one fifty-one-minute window, and the desk was armed for 3.4% of the available gate-hours — nowhere near enough evidence to re-tune a threshold, and five separate sections say so independently. So the card is one promotion, a handful of instrument fixes, and a set of measurements. **15 of the rows are NOT-AN-ACTION:** refutations printed with their

autopsies, because on a week like this the most valuable output is a list of things that have now been ruled out — and one of them refutes a headline finding of this report's own Movement 3. 58 plays, five windows, in the order they need doing. Every play carries its **evidence tier** and its **kill criterion** — if a play has no way to be proven wrong it should not be on the card.

THE THREE THAT MATTER MOST

1 · Put router_nightly.py and selector_nightly.py on a timer — both have been silently dead for a fortnight (SATURDAY #1). Write a systemd timer for each (or one timer that runs both), 22:30Z daily, after the Paris-day roll. They are pure rollups: no orders, no switches, no risk. router_nightly last wrote 2026-08-14, selector_nightly 2026-08-18 — 14 and 10 days dead. Zero units in /etc/systemd/system, zero crontab entries. Part 2.6 rebuilt all five days by hand to write its section.

2 · ★ The router went blind for 3h35m and every health instrument on the box said it was fine (SATURDAY #2). Point the health alert at data/router_headless.log ABORT lines rather than at the trial log, and fix the health page's text — it currently blames an “expired login” for what is a usage limit resetting on the hour. 43 ABORTs ≈ 3h35m blind. 31 session-limit, 12 Monday-gateway. The −\$135.43 Friday miss landed inside a blackout. systemctl / trial log / tick count all read clean throughout.

3 · ★ HOLD the abs_veto_55s promotion one more week — the 8th week of forward evidence came in NEGATIVE and it tripped this row's own kill line (MONDAY #1). Do not promote this weekend. Leave abs_veto_55s in shadow, both sides, and re-read it next Friday on $n \approx 500$. Promote then only if the LONG side is back above −\$5/trade over the trailing two weeks. All-time $n=464$, +\$4,876.50, **+\$10.51/tr** (was $n=433$ / +\$5,302.50 / +\$12.25). W36 alone: 31 trades, −\$426.00, −\$13.74/tr, 19.4% win — LONG −\$19.25/tr. ISO weeks green: **7 of 8**. Per-day 25/36 green.

Generated from the top of the card below, so the two cannot drift apart. Everything else, including every refuted idea and why it died, is in the five windows underneath.

SATURDAY · SAT 2026-09-05 / SUN 2026-09-06 — BEFORE THE SUNDAY 22:00Z REOPEN. ★ ONE ROW HAS A DEADLINE THIS TIME · 12 PLAYS

★★ **READ #0 FIRST.** This window used to open by saying nothing on it had a hard deadline. That is no longer true: **four day-rider lots are open over this weekend** — LONG from 29,641.25, −\$912 at the 21:00Z close, venue unreachable since 18:48:06Z, both eod-flatten shots dead inside the same connectAsync timeout. Same mechanism as 2026-08-21, which cost −\$2,149.44. It is #0 because it is the only row here with a clock on it, and the clock is Sunday 22:00Z. ★ **Everything below it is a RE-ISSUE from the 2026-08-28 card, and every row now carries its own shipped-check** — a probe run against this box at render time (scripts/rev2_shipped_check_0904.py), not a memory. The tally is uncomfortable and it is the most useful thing on the page: of the 25 shippable rows issued last weekend, **19 are NOT DONE, 5 are CARRIED, 1 has been re-measured, and 0 are DONE.** SATURDAY #5 — move the rider’s flatten earlier and put the page UPSTREAM of it — is one of the 19, and it is the row that bit tonight. Beyond #0, nothing here has a deadline and none of it changes a threshold: they are faults in the desk’s own instrumentation — two nightly rollups that have never had a timer and have written nothing since 2026-08-28, a router that went blind for 3h35m while every health signal read clean, a watcher that writes to a file its own unit says it must never touch, and a rider whose every order fills as a split with an off-tape residual leg. That last one was re-audited this revision and **it is still happening:** 7 more residual legs, 15 lots, \$876.00, 0 of 7 on the tape. ★★ **And three lake/build faults from 09-04 sit at the bottom:** the parquet lake’s gold year serves the wrong contract for 22% of its minutes at a median \$301 a lot away from where gold actually was; the lake has no Sunday partitions at all, so the week’s opening session is absent from every lake-based study; and the report runner’s stale-date is stuck a week behind, which is why this document covers two different weeks. All of them are the same species — reasons a number in here might be wrong.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
0	★★★ FOUR RIDER LOTS ARE OPEN OVER THE WEEKEND — the only row on this card with a deadline LIVE Nothing can be done before the venue comes back. At the Sunday 22:00Z reopen: get a venue read FIRST, confirm it says 4, and then decide FLAT or MANAGE deliberately — do not let the position drift into Monday by default, which is exactly what happened on 08-21 and cost −\$2,149.44. If the gateway is still refusing connections at 22:00Z, that is the whole decision: you are flat-blind and the position is unmanaged until it answers. Mechanism: data/day_rider_state.json reads lots_open 4, closed false, LONG from 29,641.25, ahead_pt −114.0, flatten_blocked true, venue_ok false, venue_net_ts 18:48:06Z, note ERROR (no action): TimeoutError. desk_reconcile reads venue ? = tournament +0 + rider +4 → unaccounted ? with “no venue read — IBKR truth unavailable”. The watchdog has printed SKIP venue read failed every two minutes since 18:48:50Z, and both eod-flatten shots (20:53Z and 20:57Z) died inside the same connectAsync TimeoutError. Port 4002 is refusing connections, so this	LONG 4 @ 29,641.25 · −114.0pt = −\$912.00 unrealised at the 21:00Z close. Sunday-reopen risk from 7 reopens on tape (capture.db 5s bars; the lake has NO Sunday partitions): median [gap] 22.25pt = \$178 on 4 lots, worst gap DOWN −21.25pt = −\$170, worst first hour for a long −82.50pt = −\$660. Precedent 08-21: −\$2,149.44.	Arms: Sunday 2026-09-06, 22:00Z reopen. Nothing before then: the market is shut and the venue is unreachable. Exit: Your call, deliberately taken. FLAT is the low-variance choice and costs the −\$912 that is already marked; MANAGE keeps a position with no venue stop (PLACE_VENUE_STOP = False) through a session the desk cannot watch. Kill: If the venue read is still unavailable at 22:00Z, do not place a managing order into a book you cannot see. Wait for the read.	★ Garrath — at the Sunday reopen N/A — this is a position, not a setting.

is the gateway being DOWN, not the 04:28Z hang where the port stays open.

Why: This is the identical mechanism as 2026-08-21, which is the largest single item in the report body. Then, the position was −121.93pt at the Friday close and −267.93pt when it was finally claimed by hand on the Monday: −\$2,149.44. Tonight it is −114.0pt at the close. **The reopen itself is not the risk — the Monday is.** Across the seven Sunday reopens on tape the median gap is 22.25 points and the worst first hour for a long is −82.50pt (−\$660 on four lots); the 08-21 weekend gapped +22.25 in the position's FAVOUR and still lost \$2,149 by Monday lunchtime.

Evidence: Part 4 §1 (this revision) · the same mechanism as SATURDAY #5 (rider-hard-flat-sits-on-the-blackout-edge-0828) and Part 1.6 §6(a). · verification: **At 22:00Z Sunday: python3 -c "import json;print(json.load(open('data/desk_reconcile_state.json')))" must show a NUMBER in the venue field, not ?. If it shows ?, the decision is unavailable and the position is unmanaged — say so out loud.** · mode: Do it yourself. This one is not automatable tonight — the automation is what failed.

1 Put router_nightly.py and selector_nightly.py on a timer — both have been silently dead for a fortnight Last week's card: NOT DONE — zero units matching router-nightly/selector-nightly in /etc/systemd/system; newest router_nightly file 2026-08-28.json, newest selector_nightly 2026-08-28.json — both still stop at the 08-28 report's own rebuild, so nothing has written since LIVE Write a systemd timer for each (or one timer that runs both), 22:30Z daily, after the Paris-day roll. They are pure rollups: no orders, no switches, no risk. Mechanism: Neither script has a systemd unit or a crontab entry anywhere on the box. The only things that reference them are the Friday build scripts, so they have effectively been running once a week, by accident, inside the report. When Part 2.6 went looking this cycle, the newest data/router_nightly/*.json was dated 2026-08-14 and the newest selector file 2026-08-18. Why: This is the report's own instrumentation, and the report is the thing that noticed. Every nightly number in Part 2.6 this week is a REBUILD done during the report, not the output of a monitor that was watching — so the desk has had no daily read on router value or exit selection for 10–14 days and nothing said so. Evidence: Part 2.6 \$honesty-flag-1 and recommendation #1 (confidence: CERTAIN). · verification: systemctl list-timers grep nightly shows both, and a file dated tomorrow appears in data/router_nightly/ without anyone running the Friday build. · mode: deploy	router_nightly last wrote 2026-08-14, selector_nightly 2026-08-18 — 14 and 10 days dead. Zero units in /etc/systemd/system, zero crontab entries. Part 2.6 rebuilt all five days by hand to write its section.	Arms: Immediately. It is plumbing and it cannot trade. Exit: n/a. Kill: If a nightly rollup turns out to be so cheap that running it inside the Friday build is genuinely sufficient, delete the timers — but then the Friday scope must stop describing them as a weekly rollup OF a monitor.	Garrath — two unit files, no code change systemctl disable --now both timers.
2 ★ The router went blind for 3h35m and every health instrument on the box said it was fine Last week's card: NOT DONE — scripts/router_health_check.py DOES read data/router_headless.log and alarms an ABORT streak (ABORT_STREAK_ALARM=3) — that half predates the card (last commit 2026-08-13). But the alert text still says "expired login: run `claude` and /login on the box", which is the half the play asked to fix, and the file has not been touched since %ad LIVE Point the health alert at data/router_headless.log ABORT lines rather than at the trial log, and fix the health page's text — it currently blames an “expired login” for what is a usage limit resetting on the hour. Mechanism: 43 ticks aborted this cycle. An abort is fail-closed — it writes nothing — which is the correct default and is exactly why it is invisible: the trial log gets no line at all, systemctl reports the unit as cleanly deactivated, and the tick count reads normal. 31 of the 43 are the LLM session limit; the other 12 are all Monday, inside the 04:28–05:00Z gateway window — 4 invocation failures plus 8 empty responses, a second failure mode with the same silent signature. Why: A monitor that cannot see its subject's most common failure is not a monitor. The desk's #1 lever was absent for three and a half hours and the only evidence is a log nothing watches. Evidence: Part 2.6 recommendation #2 (CERTAIN, measured on four separate days); Part 1 §5 counts the same 43 from the tournament's side. · verification: Kill the router mid-tick and confirm a page arrives naming the abort, not silence. · mode: deploy	43 ABORTs ≈ 3h35m blind. 31 session-limit, 12 Monday-gateway. The −\$135.43 Friday miss landed inside a blackout. systemctl / trial log / tick count all read clean throughout.	Arms: Immediately. Exit: n/a — it is an alert. Kill: If ABORT rates prove noisy enough to page nightly, rate-limit rather than remove: the failure this closes is silence, and a noisy alert is still not silence.	Garrath — alert source + one string on the health page Point the alert back at the trial log.
3 Make open_hour_watch.py match its own unit file, or give its prompt header an EXPIRY	2 writes this week against a unit file that forbids them; a 9-minute override of a live router bench; a	Arms: Before the Sunday 22:00Z reopen — the watcher runs on Monday's open.	Garrath — one script or one unit file, your choice

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	Last week's card: NOT DONE — scripts/open_hour_watch.py still contains os.replace(tmp, SWITCHES) — it writes the switch file — while gazbot7-open-hour-watch.service still describes itself as "ALERT-ONLY". Neither side of the contradiction moved LIVE Either remove the gate-switch write path from open_hour_watch.py so it is genuinely alert-only as its unit file promises, or — if the auto-arm is wanted — change the unit file to say so and add a hard expiry to the header notes it injects. Mechanism: open-hour-watch.service states the process "must NEVER write gate_switches.env". It wrote twice this week, and on Friday one of those writes re-armed a gate the router had deliberately benched 9 minutes earlier. Separately, the note it splices into the router's prompt carries no expiry, so it sat in every router prompt for 3h51m after being reversed. Why: Two independent hazards in one file, and they compound: a second writer with no standdown awareness, plus a no-expiry instruction that outlives the situation it described. Obeying the stale note would have cost —\$258.06. Evidence: Part 2.6 recommendation #7 (HIGH); Part 1 §4 shows the watcher as one of only three writers of gate_switches.env this week. · verification: <code>grep -n 'gate_switches' scripts/open_hour_watch.py</code> returns nothing, or the unit file no longer claims it cannot write. · mode: deploy	header note live for 3h51m past its reversal, worth —\$258.06 if obeyed.	Exit: n/a. Kill: Keep the auto-arm ONLY if it is made standdown-aware and cannot reverse a router bench younger than its own hold period.	of which end to fix git revert; the auto-arm has fired twice all week, so nothing depends on it.

4 ★★ The rider's off-tape residual leg — ONE adjudicated figure, and it is STILL HAPPENING a week later

Last week's card: MEASURED — re-run in this revision — see the off-tape audit below. This is exactly the verification the row asked for ('re-run the audit next Friday')

LIVE Audit the split-fill path on the rider's order route and reject or re-price any leg that did not trade. This is not a strategy change and it does not touch a threshold.

Mechanism: Every 4-lot rider order this week filled in two pieces: one leg at a price the tape actually printed, and the residual at **base × (1 ± 0.001), always ADVERSE**. 2026-08-26 13:44:06 bought 1 lot at 29,301.00 and **3 at 29,330.00** while the tape's whole ±10s range was 29,282.50–29,308.75; 2026-08-28 12:39:06 bought 1 at 29,635.25 and 3 at 29,664.75 against a range of 29,632.75–29,637.75. The trades row then records the fill-VWAP of the two, so the book's entry price is a number the market never printed.

Why: The audit this row asked for has now been run twice and the answer did not change: the residual leg is priced at base × (1 ± 0.001), always adverse, at a price the tape did not print within ±30 seconds. It is the largest single item on the rider's book in both weeks and it is not a trading decision. Every leg carries a distinct, correctly-formatted IBKR execId, so it is not obviously fabricated — but 0.001 appears nowhere in src/gazbot7/, which is the other half of the question and still open.

Evidence: Part 1 §3b (new this revision); harness scripts/rev2_rider_stop_week.py, which verifies every base leg IS inside the tape's ±10s range and every off leg is not. · verification: `PYTHONPATH=src .env/bin/python scripts/rev3_offtape_audit.py` — the 08-31→09-04 window is now one of its printed cuts. A week with zero off-tape legs is the fix landing. · mode: deploy

ONE figure, one window, one unit: report week 08-24→08-28 = 11 split orders / 15 residual legs / 18 lots / \$1,053.50 = 66.8% of the rider's —\$1,576.94. (Not the \$1,170.50 / 74% / 12 events / 20 lots this row used to carry, and not Part 1.6 §5's \$734.50 / 47%.) ★ AND IT PERSISTED: re-run over 08-31→09-04 — the verification this row itself asked for — gives 7 split orders / 7 residual legs / 15 lots / \$876.00, every one of them adverse and 0 of 7 on the tape. That is 143% of the rider's whole +\$613.50 week.

Arms: Immediately — it is execution plumbing and it cannot change what the rider trades.
Exit: n/a.
Kill: If the 0.1% offset turns out to be a paper-broker simulation artefact that will not exist on a live route, then it is a MEASUREMENT correction rather than a fix — but in that case every P&L number this desk has ever quoted from a split fill is overstated in the same direction, and that needs saying rather than shrugging at.

Garrath — order route / fill handling
n/a — it is an audit before it is a change.

5 ★★ The day rider's 20:40Z hard flat sits on the edge of a recurring gateway blackout with ZERO margin — it cost \$1,258 on 21 August

Last week's card: NOT DONE — gazbot7-eod-flatten.timer still reads ['Mon-Fri 16:53 America/New_York', 'Mon-Fri 16:57 America/New_York'] — 20:53Z and 20:57Z, unchanged. No 20:25Z attempt exists and the page is still downstream of the flatten. ★ THIS IS THE ROW THAT BIT TONIGHT: the venue read has failed every two minutes since 18:48:06Z, the 20:40Z flatten died on connectAsync TimeoutError, and four rider lots are open over the weekend

LIVE Move the rider's first flatten attempt EARLIER (20:25Z or so, with the 20:40Z shot kept as the backstop), and put the page UPSTREAM of the flatten — it must fire when the venue read fails, not after the flatten it was supposed to protect has already been skipped.

Mechanism: On Friday 21 August the rider was long 4 lots from 13:13Z. The IB gateway went unreachable at **19:58:50Z** and stayed unreachable until 23:44:50Z — 3h46m and 114 consecutive SKIP venue read failed polls — which straddles the 20:40Z flatten clock completely. The flatten never happened, the position rode the

Flatten at 20:40Z would have closed at 29,381.25 (–110.7pt, –\$891.44). It actually closed Monday 11:59Z at 29,224.00 (–244.7pt, –\$2,149.44). Cost of the failed flatten: –\$1,258.00. The position was never once in profit (MFE –3.4pt) and its MAE was –375.2pt, or –\$3,002 on 4 lots, reached while the market was open and nothing could see it.

Arms: Immediately — it is a clock change plus an alert, and it cannot open a position.
Exit: n/a.
Kill: If moving the first attempt to 20:25Z is shown to give up real money on the 20:25–20:40 window across the session history, keep 20:40Z as the only flatten and ship the upstream page alone — the page is the part that closes the silence, and the silence is what made this expensive.

Garrath — one timer change and one alert, no strategy code
Restore the single 20:40Z attempt and the existing alert source.

weekend, and it was closed by hand on Monday at 11:59Z. The blackout is not a freak: it also starts at 20:40:50Z on 17 Aug, 20:42:50Z on 27 Aug and 20:06:50Z on 23 Aug. **Two of those begin within three minutes of the flatten clock** — 21 August is only the day that hurt because the outage arrived 42 minutes early.

Why: This is a scheduling collision, not a strategy question, and it is the cheapest real money on the card. The flatten and the blackout are both recurring and both known; they are simply booked at the same minute. Moving one of them apart costs nothing and removes a class of weekend carry. [△] And note what the instruments did while it happened: the watchdog printed `ok position 4.0 managed 1,908 times` across that open position, because its heartbeat was fresh. A strategy that is running and a strategy that can reach the venue are not the same thing, and nothing on this box currently distinguishes them.

Evidence: Part 1.6 §6(a) and §7 disposition (FAULT); Part 1.6 §1 for the two-book money. · verification: **Run the flatten path against a simulated venue-unreachable window covering 20:40Z and confirm (i) the 20:25Z attempt closed the position and (ii) a page arrived on the FIRST failed venue read, not at 23:45Z.** · mode: deploy

6 ★★ The Parquet lake silently blends four extra timeframes into its bars view — 44.4% of July's minutes are inflated, and every backtest reads them

Last week's card: NOT DONE — `src/gazbot7/lake.py` still exposes the bars stream without a timeframe filter, so the backfill files still inflate a bar count

LIVE Add `timeframe='5s'` to the bars view in `lake.connect()`, or move the backfill files out of the stream directory. Then re-run anything that priced ATR, range or efficiency off lake bars and see which conclusions move.

Mechanism: The bars view unions every Parquet file in the stream directory, and four backfill files written at other timeframes are sitting in it. A minute that exists at more than one timeframe is returned more than once, and the ranges compose — so ATR and efficiency read HIGH exactly where the duplication is densest. MNQ has the same four files, so this is not a gold-only fault.

Why: This is the most dangerous thing on the card, because it is upstream of the evidence rather than in it. The lake is the DATA CONTRACT's designated source for every study longer than this week — the rolling 5-day `capture.db` cannot be used for history — so a bar-level inflation propagates into every multi-week backtest this desk has run since the backfill landed, including ones whose verdicts are already on this card. It fails silently and in the flattering direction: inflated range makes a volatility floor look cleared and an edge look bigger.

Evidence: Part 2.5 §2 and §4 disposition (INSTRUMENT — FIX ON SATURDAY). · verification: **Re-run the audit's zero-control and require it to return 0 rows, then confirm a known July session returns exactly one row per minute.** · mode: deploy

44.4% of July minutes inflated. The audit's own zero-control — which should return nothing — now reads 9,469 rows. Four files, both instruments.

Arms: Immediately, and before any further backtesting — this is a read-path fix and touches no live code.

Exit: n/a.

Kill: Nothing to kill; if the extra timeframes turn out to be wanted, they belong in a SEPARATE view with the timeframe on it, never unioned into the default bars view.

Garrath — one filter in `lake.connect()`, or move four files. [△] But then no lake-derived number is trustworthy, so revert only together with the studies that depend on it.

7 Give the router a weekend and holiday check — it armed a gate 38.7 hours before any market opened, on zero bars

Last week's card: NOT DONE — `scripts/router_tick_durable.py` has no weekday/market-calendar guard, so the router will again evaluate and write across this weekend's ~49h CME halt

LIVE Add a market-calendar guard to the router tick: if the venue is closed, do not evaluate and do not write. Roughly fifteen lines.

Mechanism: The router has no calendar at all. Across the CME halt it keeps ticking, and on a dead tape it evaluates against zero bars — which is not a neutral input, it is an absence the deciders read as a value. On 2026-08-29 at 07:20:25Z it armed `capitulation_long` on 0 bars, 38.7 hours before anything opened. The existing Asia block does not cover this: it expires on a clock boundary at 07:00Z and the tape stays shut for another day and a half.

Why: This removes a whole class of decision made on zero information, which is worth more than the single change it would have prevented this week. A weekend arm is not merely idle — it survives into Sunday's reopen, so the desk can open the week in a state nobody chose on evidence that never existed.

Evidence: Part 2.5 §4 disposition (FIX ON SATURDAY); Part 2.6 for the tick census that counts the weekend ticks. · verification: **Run the router against a Saturday timestamp and confirm it declines**

1 change in 645 weekend ticks — so the RATE is low and the SIZE is not: the one change armed a live gate on 0 bars, 38.7h early. ~15 lines to close.

Arms: Immediately.

Exit: n/a.

Kill: If a genuine out-of-hours use is ever wanted, gate it on an explicit flag rather than on the absence of a calendar.

Garrath — ~15 lines in the router tick. Remove the guard.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
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to evaluate and writes nothing to `gate_switches.env`. · mode: deploy

8	Both nightly scorecards only count the modelled side — print realised P&L beside modelled, or stop publishing them Last week's card: NOT DONE — neither nightly has run since 2026-08-28 (see row 1), so neither has been changed to print a realised figure beside its modelled one. This row is blocked by row 1 and should ship with it LIVE Make <code>router_nightly</code> add the realised P&L of managed gates while the router was ON, and make <code>selector_nightly</code> print realised beside modelled (or stop scoring <code>adaptive_exit</code> at all while it is disabled). Pairs with <code>SATURDAY #1</code> (timer-the-two-dead-nightlies-0828): that play makes them RUN, this one makes what they print mean something. Mechanism: <code>router_value</code> counts what the router's decisions cost and never what its arms earned, so on any day it armed something the number is a LOWER BOUND wearing the appearance of a total. <code>selector_nightly</code> has the mirror fault: it scores a feature that is switched off, and its 'regret' is the gap between a model and a model. Why: A one-sided scoreboard does not merely under-report, it points the wrong way — it makes the router look like a cost centre on exactly the days it did its job. This desk's standing rule is that a number without its counterfactual is not a measurement, and both of these publish a counterfactual with no realised leg. Evidence: Part 2.5 §4 disposition (two INSTRUMENT rows, both FIX ON SATURDAY); Part 2.6 recommendation #1 for the liveness half of the same problem. · verification: Re-run both nightlies for 08-28 and confirm each prints a realised figure beside its modelled one, and that the router's day reads positive. · mode: deploy	08-28: router_value printed −\$117 on a day the router's own arms booked +\$330. selector_nightly: −\$347 modelled against +\$330 booked on the SAME six legs, with \$1,576 of 'regret' living entirely inside the model.	Arms: Immediately — reporting only, no order path. Exit: n/a. Kill: If the realised leg proves impossible to attribute cleanly per gate, print the desk total beside the model rather than nothing — an approximate second side beats a confident single one.	Garrath — reporting change in two rollups Drop the realised column.
9	The lake's gold year is 22% the WRONG CONTRACT — fix the front-month rule, and quarantine every gold study built on it LIVE Change the gold backfill's contract selection from 'earliest expiry with a print' to a volume-led front-month roll, rebuild <code>backfill_1min.parquet</code>, and mark last Friday's thirteen-months-of-gold section as NOT SAFE until it is re-run on the fixed tape. Mechanism: <code>gazbot7.lake</code> builds gold's bars view by unioning <code>backfill_1min.parquet</code>, which keeps the earliest expiry that has a print at each instant. An expiring contract keeps printing for about five weeks after it stops being the front month, so the lake quietly serves the DYING contract for that whole stretch — at a price that is nowhere near where gold is actually trading. Why: This is not a rounding error and it is not confined to one study. On 77 of 266 sessions the lake quotes a contract whose median minute volume is zero lots, a median 30.1 points — \$301 a lot at gold's \$10.00 multiplier — away from the real price. Any backtest that entered on those minutes was trading a tape on which nothing traded. It sits UNDER every gold conclusion this desk has reached in three weeks, including the 32-day bar-source re-run that BUILD #9 leans on and the router rule that a prior session killed. Evidence: Movement 3 ADDENDUM → GOLD (<code>gf_MGC.md</code>, 2026-09-04) §1.1, which is the first thing in that dossier because it is the biggest thing in it. · verification: Re-derive the front month by volume for all 266 sessions and diff the two tapes day by day. The fix is verified when the sessions that changed are exactly the ones inside five weeks of an expiry, and no others. · mode: fix	77 of 266 sessions on the wrong expiry (22% of minutes). Median minute volume on those sessions: 0 lots. Median price error: 30.1 points = \$301 a lot. MGC is \$10.00/point.	Arms: Nothing arms. This is a data-integrity fix, not a trading decision. Exit: n/a — the fix is the deliverable. Kill: If the volume-led roll and the current rule disagree on fewer than a handful of sessions, the finding is wrong and this row is withdrawn. The 77/266 count is the falsifiable part.	Garrath — one selection rule in the lake's gold backfill Keep the old <code>parquet</code> beside the new one. Nothing live reads it; only research does.
10	Roll the durable runner's STALE cutoff with the cycle — it is why this report covers two different weeks LIVE Derive the runner's stale cutoff from the run's own start time (the previous Friday 22:00Z), not from a constant that has to be edited. The same constant lives in <code>scripts/friday_v7_build.py</code> as <code>CYCLE_START</code> and must be derived from <code>--slug</code> the same way. Mechanism: The 2026-09-04 run opened its log with artifacts older than 2026-08-28T22:00Z are STALE — the PREVIOUS cycle's cutoff. Under that cutoff the fragments written on 08-29/30 look current, so the runner logged HAVE ... skipping (artifact is the checkpoint) ten times and rebuilt none of them, while separately freezing a new census and running three greenfield clusters. Why: The failure mode is the nastiest kind this build has: nothing errored. Ten skips are the runner working exactly as designed, on a	10 sections skipped as HAVE; 3 rebuilt; 2 censuses and 1 progress page a week younger than the 11 sections around them. One constant, in two files.	Arms: Nothing arms. Exit: n/a. Kill: If the cutoff is already derived somewhere and the 08-28 value came from a config the operator set deliberately, this is a config row, not a bug row.	Garrath — one derived date in the runner and in <code>friday_v7_build.py</code> Pin the constant again. It is a one-line change either way.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	date that was one week out. The result is a document whose two halves describe different weeks and whose freshness checks all pass — which is precisely the silent-staleness failure the whole honesty apparatus was built to prevent, arriving through the one door nobody had a check on. The seam is now drawn on the front page, but drawing it is a mitigation, not the fix. Evidence: The two-week provenance box at the top of this report (generated by <code>provenance()</code> in the build), and <code>data/friday_durable.log</code> at 2026-09-04T21:13:24Z, which is where the cutoff and the ten HAVE lines are printed side by side. · verification: Next Friday, grep the run's first log line for the cutoff and assert it equals the previous Friday 22:00Z. If it does not, the fix did not land. · mode: fix			

11	The parquet lake is missing its Sunday partitions — the week's first session is absent from every lake-based study LIVE Backfill the Sunday partitions and add a partition-completeness assertion to <code>gazbot7.lake.connect</code> so a missing day fails loudly instead of shortening the sample in silence. Mechanism: The UNCLASS hunt went looking for its own sessions and found the lake has no Sunday partitions, so the 22:00Z Sunday reopen — the start of every trading week — is simply not in the data any lake-based backtest reads. Why: A missing partition does not raise; it shortens <code>n</code>. Every study in this report that reads the lake has been silently excluding the week's opening session, and that is the session with the reopen gap in it. Nobody would have found this by looking at a result, because the result looks fine. Evidence: Movement 3 ADDENDUM → UNCLASS (<code>gf_full_UNCLASS.md</code>, 2026-09-04), the engineering faults at the end of the dossier. · verification: Count distinct session dates in the lake against the CME calendar for the same range. Every Sunday reopen should be present exactly once. · mode: fix	Sunday partitions absent across the lake's whole 2026-06-19 → 2026-09-04 range; the UNCLASS hunt's own 67 sessions / 914,146 bars are counted after the omission.	Arms: Nothing arms. Exit: n/a. Kill: If the Sundays are deliberately excluded (thin tape, a capture decision), then this is documentation, not a backfill — but it still has to be asserted rather than assumed.	Garrath — lake backfill + one assertion Nothing live reads the lake.
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MONDAY · MONDAY 2026-09-07, AT THE DESK — ★ THE PROMOTION IS HELD, SO THIS WINDOW IS THREE TUNINGS AND NOTHING ELSE · 4 PLAYS

★★ THE PROMOTION IS OFF THIS WEEK. `abs_veto_55s` was the only arm on the card, and Rev 2 re-ran it to 2026-09-04 rather than to 2026-08-28. The week the report body does not show — W36, 08-31→09-04 — is **31 trades for −\$426.00 (−\$13.74/tr), LONG −\$19.25/tr**, which trips this row's own kill criterion ("the LONG side below −\$5/trade"). All-time it is still +\$10.51 a trade on `n=464` and green in 7 ISO weeks of 8, so it is HELD, not killed — but the stated reason for promoting was another week of forward evidence, the evidence arrived, and it argued the other way. **Note what is deliberately NOT here, and this is now ONE verdict rather than two:** no exit cell changes, including the grind MED-TREND branch Part 2 §11 proposed deploying. The twenty rung-cells are graded on a four-test bar that includes the out-of-sample POOL split; MED-TREND clears the other three and fails that one (V5 OOS −\$4.50/signal on `n=52`, live forward arm −\$8.36/trade on `n=21` against a control at −\$9.29). **Cause of death for the deploy: the out-of-sample leg.** `data/exit_overrides.json` is not touched this weekend. (The extended sweep is 822,746 five-second bars, not the 831,379 previously printed.)

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	★ HOLD the <code>abs_veto_55s</code> promotion one more week — the 8th week of forward evidence came in NEGATIVE and it tripped this row's own kill line Last week's card: NOT DONE — <code>data/gate_switches.env</code> names six gates and <code>abs_veto_55s</code> is not one of them; all six read =off. The shadow arm is still shadow SHADOW Do not promote this weekend. Leave <code>abs_veto_55s</code> in shadow, both sides, and re-read it next Friday on <code>n ≈ 500</code>. Promote then only if the LONG side is back above −\$5/trade over the trailing two weeks.	All-time <code>n=464</code>, +\$4,876.50, + \$10.51/tr (was <code>n=433</code> / + \$5,302.50 / +\$12.25). W36 alone: 31 trades, −\$426.00, −\$13.74/tr, 19.4% win — LONG −\$19.25/tr. ISO weeks green: 7 of 8. Per-day 25/36 green.	Arms: NOT this weekend. Re-read Friday 2026-09-11. Exit: Its shadow ruler: 2.0R target, 1.0×ATR stop. Do not import an exit cell from Part 2's exit lab — twenty of twenty of those failed. Kill: Already tripped once — the LONG side ran −\$19.25/trade in W36. Re-arm the promotion only	Garrath — slate entry + tournament restart Remove from the slate. It keeps running in shadow either way.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
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Mechanism: The 08-28 report promoted this on **n=433, + \$5,302.50, +\$12.25/trade, green in 7 of 7 ISO weeks** and gave as its reason “another week of FORWARD evidence rather than another week of the same evidence”. That week has now happened. Re-run to 2026-09-04: **n=464, +\$4,876.50, +\$10.51/trade**. The unseen week (W36, 08-31→09-04) is **31 trades for −\$426.00 = −\$13.74/trade**, split LONG 18 at −\$346.50 (−\$19.25/tr) and SHORT 13 at −\$79.50 (−\$6.12/tr). So it is 7 green weeks of 8, and **this row's own kill criterion — “the LONG side below −\$5/trade” — is already tripped on evidence the report did not show.**

Why: The candidate is not refuted: all-time it is still +\$10.51 a trade, the veto's incremental edge over the un-vetoed thrust_loose is + \$5,600 across the board, and both regimes and both walk-forward halves stay green. But the stated reason for promoting was forward evidence, the forward evidence arrived, and it argued the other way. Promoting a candidate the week after its own kill line trips is how a desk learns to ignore its kill lines. ★ Note also that a shadow mirror's stop is CLOSE-ONLY, so its loss side is structurally understated — one more reason not to promote on the shadow number alone.

Evidence: Part 2 §1 (board, ranked on real_pnl not ceiling_pnl) and §9 disposition. · verification: **Re-run scripts/abs_veto_robustness.py next Friday and print the ISO-week row and the two-sided split before any promotion is proposed again.** · mode: deploy

after two consecutive weeks with the LONG side above −\$5/trade.

2 ★ Attack the 20-minute arm lag — the router's dominant error is no longer direction, it is LATENESS

Last week's card: NOT DONE — gazbot7-router-tick.timer reads OnCalendar=*:0/5 and the last 12 trial-log ticks are [4, 5, 49] minutes apart, so the TICK GRID was already 5 minutes and the '15m→5m' half of this row was a no-op — the router's own note at router_tick_durable.py:343 says so. The half that mattered, the prompt line 'arm on the FIRST confirmed break impulse, not the fifth', is ABSENT from the prompt

LIVE The tick grid is ALREADY 5 minutes (OnCalendar=*:0/5), so the only shippable half of this row is the prompt line: arm on the FIRST confirmed break impulse, not the fifth. ★ REV2 — and BUILD #1's measurement now argues it is worth less than it looked: on 09-01 the router armed 89 seconds after its own confirmation criterion went true. The confirmation APPETITE is not the binding constraint; the CRITERION is (52 minutes from break onset to trio-true). Ship the prompt line, and treat the criterion as the next question.

Mechanism: Decomposed, roughly 15 of the 20 minutes is the router's own confirmation appetite and only ~5 is the tick grid — so the step change alone does not fix it, which is why the prompt line ships with it. Actuation once decided is not the problem: on Friday the gates filled **1.4 seconds** after the switch flipped.

Why: This is the week's clearest measured cost and it points the same way from two independent sections. The arm that made the week's money was right — and late.

Evidence: Part 2.6 recommendation #3 (HIGH) and Part 1 §5 / §9, which price the same arm from the tournament's side and agree on the direction. · verification: **router_trial_log.txt shows ticks 5 minutes apart, and the next arm's gap between first suppressed signal and switch write is under 5 minutes.** · mode: deploy

Friday 14:45:03Z arm: measured lateness \$205-\$705 against the \$181.50 the arm banked. Part 1 prices the same delay on one lot at \$33.84 (grind_long's own 14:36:26Z suppressed signal was worth +\$150.02; the fill 8m37s later was worth +\$116.18).

Arms: Monday. It is a constant and a prompt line.

Exit: n/a.

Kill: If gate-VALUE changes per hour rise above ~1.0 — the desk's ONE churn metric, defined in Part 1 §10: a gate's state flipping VALUE (not a write), Mon 00:00Z to Fri 22:00Z, all three writers, over 118 elapsed hours. **This week ran 0.237/hour** (28 value changes: 14 arms, 14 benches). The 0.18 this row used to quote was one of three retired counts. Or if a 5-minute cadence starts producing arm/bench round trips with no intent in between — revert and attack the confirmation appetite alone.

Garrath — one constant in router_tick_durable.py + one prompt line
Set the step back to 15m. Reversible in five minutes.

3 Make the router quote a WINDOW and a FAMILY whenever it cites the shadow board

Last week's card: NOT DONE — no line in scripts/router_tick_durable.py requires a shadow-board citation to name its date window and arm family; the prompt is unchanged on this point

LIVE One line in the router prompt: any shadow-board figure must name its date window and the arm family it covers, or it may not be used as a reason.

Mechanism: On 08-28 a bench rested on “344 fader trades at −\$3,021 today”. The real figure for that day and family was **128 trades / −\$636.50** — the quoted number was an all-time total wearing the word “today”.

Why: The error is not random: an unwindowed aggregate is almost always larger than the windowed one, so it systematically INFLATES the case for benching. Here it inflated it about five-fold and the bench then cost −\$135.43.

Evidence: Part 2.6 recommendation #6 (CERTAIN). Part 2 §2 finds the same class of fault in the tooling (abs_veto_sides.py hard-codes WEEK0 = 2026-07-27 and reports 4½ weeks as “this week”). ·

Claimed 344 trades / −\$3,021; actual 128 / −\$636.50 — a ~5× overstatement. The bench it justified cost −\$135.43.

Arms: Monday 2026-09-07, at the desk.

Exit: n/a.

Kill: n/a — it constrains a citation, not a decision.

Garrath — one prompt line
Delete the line.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	verification: Next router citation of the board includes a date range and an arm name. · mode: deploy			
4	Filter the untradeable stop-rate meter to tournament gates only — it has never once discriminated Last week's card: NOT DONE — the meter's stop-rate input still reads both books — no filter on gate NOT LIKE 'day_rider%' exists in 64 file(s) that mention the meter. Part 2.6 separately grades the meter itself as dead (rho +0.17 / -0.09 / +0.11), which is why this row is re-issued as WITHDRAW rather than as a tuning LIVE Restrict the meter's stop-rate input to tournament rows. Today it reads the last 15 trades of BOTH books. Mechanism: The day rider's exit vocabulary contains no STOP at all, so every rider row it ingests votes zero by construction. On Friday that vote pulled the score 53→42 and flipped the verdict — on an input that was measuring nothing. Why: A sub-meter that structurally cannot fire is not a conservative input, it is a constant with a vote. Zero of this week's 32 trades exited on STOP. Evidence: Part 2.6 recommendation #5 (CERTAIN). · verification: Re-score Friday 08-28 with the filter on and confirm the meter reads 53, not 42. · mode: deploy	0 of 32 trades exited STOP this week. Friday: score 53→42 on the rider-dominated sample, verdict flipped.	Arms: Monday 2026-09-07, at the desk. Exit: n/a. Kill: If the tournament ever starts exiting on STOP frequently, re-examine — but the fix is still to read the right book.	Garrath — one filter in untradeable.py Remove the filter.

BUILD · NEXT — NOTHING GOES LIVE THIS WEEK. ★ #1 IS NOW ANSWERED, NOT ASKED · 19 PLAYS

Measurements and shadow arms. ★★ **The top row has been ANSWERED and it is a different answer from the one the report expected.** It asked for the lag between a break STARTING and abs_veto_short's trigger firing, on the hypothesis that the router arms too late for a thrust gate. The gate has since fired — twice on 2026-09-01, four legs, **+\$163.50, 4 of 4 winners** — so there are real trigger times to subtract an onset from instead of an absence to reason about. Measured on the 250ms tape: onset 07:14Z, the router's own confirmation trio first true 08:06Z, the router armed at 08:07:29Z — **89 seconds after its own criterion** — and the gate then triggered 32 and 54 minutes LATER still. **Arming at break onset would have armed the gate 52 minutes earlier and changed nothing.** The lag is real (86 minutes onset→trigger) and the proposed fix does not address it. Below that sits the week's structural finding, which is the reason several other things on this card should not be built at all.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	★★ THE WEEK'S #1 STONE IS ANSWERED — abs_veto_short fired twice on 09-01 and won all four legs, and the phase-gap fix would NOT have helped Last week's card: CARRIED — no (break-onset, trigger) table exists on disk. But the gate has since FIRED — twice on 2026-09-01, four legs, +\$163.50, 4 of 4 winners — so the measurement now has live cases instead of an empty tape. Answered in this revision rather than re-issued MEASURED Do not build the “arm on break ONSET” change. The lag is real and large, but it runs the wrong way for that fix: the gate's own trigger is LATER than the router's confirmation, so arming earlier buys idle armed time and nothing else. If anything is built here it is the gate's fast=True start-of-move trigger, and that is a separate row that does not exist yet. Mechanism: Measured on the 250ms tape, scripts/rev2_break_onset_lag_0904.py. On 2026-09-01 the break's ONSET (the 60-minute swing high the move departed from) was 07:14Z at 29,556.50 . The router's own confirmation trio (ER-30 ≥ 0.35, ATR-14 ≥ 18, new session extreme) first read true at 08:06Z — 52 minutes later . The router armed the gate at 08:07:29Z, 1m29s after that . The gate then triggered at 08:39:57Z and 09:01:00Z — 32 and 54 minutes AFTER the arm, 86 and 107 minutes after onset. Why: The 08-28 hypothesis was “the router arms when a break is CONFIRMED; a thrust gate needs the moment a break STARTS, and those were about 25 minutes apart”. On the two clean live cases the	2 signals / 4 legs / +\$163.50, 4 of 4 winners, 2026-09-01. onset 07:14Z → confirm 08:06Z (52m) → arm 08:07:29Z (+1m29s) → triggers 08:39:57Z and 09:01:00Z (+32m and +54m). Armed 3 times that day for a total 3h36m53s and produced signals in only the first window.	Arms: Nothing arms. This is a measurement. Exit: n/a. Kill: The row's own kill line was “if the measured lag is under ~5 minutes the hypothesis is wrong”. It is 86 minutes, so the LAG is real — but the DIRECTION refutes the fix, which is the outcome the row could not have predicted. Closed on measurement, n=2.	Claude — one lab against this week's 250ms tape n/a — it is a measurement. ★ It is a MECHANISM question, so it does not need a big sample to answer, which is why it outranks everything else in this window.

gap is bigger than 25 minutes — but the ROUTER is not the late one. It armed 89 seconds after its own criterion went true. The gate then sat armed and silent for another 32 minutes before it triggered. **Arming at onset would have armed it 52 minutes earlier and changed nothing: the same two signals, at the same two times.** And the trades were good — both signals, four legs, **+\$163.50, 4 of 4 winners**, entered 86 and 107 minutes into a 328-point break. Entering late into a confirmed break was profitable, which is the opposite of what the stone assumed.

Evidence: Part 1 §9 flagship case study and §11 disposition (PARKED, named the week's #1 stone); Part 1 §4 for the armed windows. · verification: `PYTHONPATH=src .venv/bin/python scripts/rev2_break_onset_lag_0904.py` — **it prints the four clocks and the router's own switch writes for every abs_veto_short signal since 08-31.** · mode: observe

2

Build a DAY-level gate for the Open Rider that reads its feature ONCE at 13:00Z and sets a session flag

Last week's card: **NOT DONE** — no session-flag file is written at 13:00Z anywhere in data/; the 20-session flag-vs-outcome log the row asks for does not exist

PARKED

Build the instrument, not another search. One feature, read once, before the session commits, setting a flag for the whole day.

Mechanism: The drift gate the desk already runs was assumed to be this. It is not: over 11 paired days it net-blocked **1 entry of 89 (1.1%)**, fired 6 times, and cost \$301 — five of the six blocks were simply re-entered at the next cadence slot, three of them at a worse price. It is a five-minute entry delay wearing a day filter's name.

Why: Everything in Part 2 §3 says day selection is where the Open Rider's money is; §3d says we have never actually built a day-level instrument to test that with. The searched classifier was refuted — but a search over 8 features is not the same experiment as one pre-registered day-level read.

Evidence: Part 2 §3, §3d and §9 disposition (drift gate PARKED, not refuted); Part 2's closing line names this as next weekend's build. · verification: **A session flag written at 13:00Z each day, and a 20-session log of flag-vs-outcome before anything is gated on it.** · mode: shadow

Ungated Open Rider, 15 days: 6 green days +\$3,066 against 9 red −\$5,626, oracle ≈\$1,022/wk. Drift gate as a day filter: 1 block in 89 entries, −\$301.

Arms: Nothing arms. Shadow only.
Exit: n/a.
Kill: The pre-registered $ATR \geq 9.5$ filter is the honest control, and at 10 of 60 sessions its sign is decided by ONE row: −\$253 on trade days against −\$1,493.44 on skip days with the 08-24 carry close included (right way, by \$1,240.44), or +\$656 on skip days with it dropped (wrong way, by \$909). One trade of 33. If the day-level instrument still agrees with the unfavourable reading after 50 sessions, day selection is dead for this strategy.

Claude — small build, one read per day
n/a — shadow only.

3

★★★ THE STRUCTURAL FINDING — every precursor feature on this desk is a proxy for ATR, and that is why selection keeps failing

Last week's card: **CARRIED** — a research instruction, not a shippable change. The bar it set — any new feature must hold its AUC after ATR conditioning — is unchanged and still the right bar

REFUTED / structural

Stop re-tuning selectors built on the current feature set, and go find an input that is not a volatility proxy. Order-book imbalance at depth, cross-instrument lead/lag and session-level positioning are the candidates; none of them is recorded in a form a selector can read today.

Mechanism: Against a fixed POINT threshold, every precursor looks predictive — volume z, trade count z, ATR expansion, aggressor flow, ER(15) and ATR level all beat a coin flip with $p=0.00$. Express the same threshold in ATR and the discrimination collapses: ATR expansion falls to 0.234 and ATR level to 0.306, both now WORSE than a coin flip. Hold ATR fixed and ask within the band and every feature sits between 0.29 and 0.56. A 55-point move is easy on a wide tape and hard on a quiet one, so anything correlated with ATR scores well against a point threshold. It is the same instrument measured twice.

Why: For three cycles this lab has closed with “detection is solved, selection is not” without being able to say WHY selection kept failing. It can now. We have been feeding a volatility meter to a selector and asking it for a direction. No amount of re-tuning fixes that.

Evidence: Movement 3 §3 (the escalation study, computed this cycle). ★ REV2: Movement 2 §4 was cited here as independent confirmation and that citation is WITHDRAWN — its random-entry control returns a NULL, not a kill (gates win 3 of 5 in-direction, 4 of 5 direction-blind, on 2-10 lots; only exhaustion_short separates, $P=0.033$ raw and 0.166 Holm-adjusted). This finding does not need it. · verification: **Any proposed new feature must be re-run through the same three tables and hold its AUC after ATR conditioning. That test is now the bar.** · mode: observe

52,937 minutes across 41 days. Point-threshold AUCs 0.51-0.81 (all $p=0.00$); ATR-normalised 0.23-0.53; ATR-conditioned 0.29-0.56. Direction carries nothing at any threshold: up-share 48.1-50.0%, aggressor flow BELOW 0.50 everywhere, continuation agreement 40.9-52.2% with no trend.

Arms: Nothing arms. This is why several things should NOT be built.
Exit: n/a.
Kill: A feature that keeps $AUC \geq 0.60$ in the ATR-conditioned table would refute this and would be the most valuable thing on the desk.

Claude — the finding is banked; the search for a non-volatility input is the build
n/a.

4

★ 48.2% of entries go NOWHERE and pay \$10 each — and REV2 shows ATR can SEE them (AUC 0.683). Ship it as a pre-registered shadow arm.

ATR-scaled ruler, $n=1,713$, −\$2,814 total. $\geq 120pt$: $n=27$ (1.6%), +\$8,627, +\$319.51/tr.

Arms: Nothing arms this week. Shadow only.
Exit: n/a.

Claude — shadow arm
n/a — shadow only.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	Last week's card: NOT DONE — no dead-band decline variant appears in the shadow board's strategy list, and no pre-registered threshold file exists SHADOW Build and shadow a size/participation rule that declines entries whose realised excursion band is the ± 20-point dead zone. It is a SIZE decision, not a direction decision, which is why it is the one lever in Movement 3 that does not depend on predicting anything. Mechanism: Decompose the pooled book by what the tape actually did next. Moves of ≥ 120 points are 1.6% of entries and pay +\$319.51 each; the ± 20-point band is 48.2% of entries and pays $-\\$10.00$ each. That band alone is $-\\$8,254$ on a book whose total is $-\\$2,814$. Why: Three harnesses now agree independently — the rider lab, the census and this decomposition — that the tradeable threshold is size, around 120 points, and that the friction is the dead band rather than the losers. Evidence: Movement 3 §7 (this cycle); Movement 3 §1 for the fire-rate side of the same problem (66.4 entries/day, 21.1% of them inside a run); ★ Part 3 Q2 for the classifier test and its robustness battery (scripts/rev2_deadband_classifier.py). · verification: A shadow arm that declines the dead band and reports what it declined, so the decline rule can be graded on what it AVOIDED as well as what it took. ★ REV2 — the threshold must be PRE-REGISTERED before it runs, exactly like the Open Rider's ATR filter, because the 48.2% cut used in the test below was fitted to the same book it was measured on. · mode: shadow	± 20pt: n=825 (48.2%), $-\\$8,254$, $-\\$10.00$/tr. The fixed-22pt ruler agrees: 48.4% dead, $-\\$13.57$/tr.	Kill: ★ REV2 — RUN, AND THE VERDICT IS SPLIT. Detection PASSES: ATR at entry separates the ± 20pt dead band at AUC 0.683, permutation p = 0.0005, against a 0.60 bar stated in advance; nothing else on the desk clears it (ATR expansion 0.558, volume z 0.554, trade-count z 0.570, flow z 0.548, ER15 0.519 and not even significant). The MONEY fails: declining the lowest-ATR 48.2% turns the book from $-\\$1.64$/tr to $+\\$1.71$/tr ($+\\$1,517.61$ on 886 trades), and STRIP-BEST-3 takes that to $-\\$266.64$ — three trades of 886 are the whole result. A walk-forward that fits the cut on the first half LOSES on that half ($-\\$2.82$/tr). $\triangle$ Note the ATR-normalised control: against an ATR-scaled dead band the same feature reads 0.531, a coin flip — §3's finding for the third time. It does not bite here only because the friction is paid in POINTS, not in ATR.	
5	Two shadow A/Bs are byte-identical and have therefore tested NOTHING — fix the arm wiring Last week's card: NOT DONE — shadow_trades counts ['cx_clip_brk_live', 388], ('cx_clip_brk_standdown', 388)] — the pair is still running and still matched; no divergence fix is in the shadow wiring PARKED Fix the wiring so each variant actually diverges from its control, then restart both A/Bs from zero. Until then their “perfect null” must not be quoted. Mechanism: capit_flip_live and capit_flip_t90 are identical on all 29 trades ($-\\$445$). cx_clip_brk_live and cx_clip_brk_standdown are identical on all 251 trades ($-\\$848$). One arm of each pair never diverged, so the control's trades are being recorded twice under two names. Why: A broken A/B does not fail loudly — it reports a PERFECT NULL, which reads as a clean negative result. And the same cluster inflates evidence four-fold: cx_clip_brk_live, cx_clip_brk_standdown and cx_grindA_clip all carry the same 251 trades, all of which also sit inside bank20_grindA's 275. Evidence: Part 2 §2 (instrument audit) and §9 disposition. · verification: After the fix, the two arms of each pair differ on at least one trade within a day of running. · mode: observe	29/29 and 251/251 trades identical. Four arm names on the leaderboard, one trade set.	Arms: n/a — shadow plumbing. Exit: n/a. Kill: n/a.	Claude — shadow slate wiring n/a.
6	exhaustion_short's arming window opened on 2026-08-31 and produced 2 fills — the row is RE-ISSUED with its clock restarted, not repeated Last week's card: CARRIED — data/gate_switches.env reads exhaustion_short=off RIGHT NOW, and no per-hour reprice of the UP_OFF rule exists on disk. The window did open: the gate took 2 machine-exited leg(s) for $+\\$11.00$ since 2026-08-31 — against the 40 fills the row set as its bar PARKED Re-issue with a new window: arm from Monday 2026-09-07 to Tuesday 2026-09-22, or 40 machine-exited fills, whichever comes first. The original window has closed with 2 of the 40 fills banked, which is not a grade. Mechanism: 98 blocked legs worth $+\\$516$ of benched winners — the largest over-bench of any gate in the ledger. The rule is categorical (it sits in UP_OFF), so it cannot distinguish an up-hour the gate would have faded profitably from one it would not. Why: The row asked for 40 machine-exited fills or 2026-09-15, whichever came first. Five sessions produced two. That is not evidence for or against the UP_OFF rule — it is a duty-cycle problem, and re-issuing the same window unchanged would produce the same answer. The $-\\$86$ expected-cost estimate is unaffected because the gate barely traded; it stands. ★ The one thing the window DID establish: the gate is not dead — when it fires it exits on its own targets, and both did. Evidence: Part 2.6 recommendation #10; Part 1 §7 bench-replay table; Part 2 §9 notes the live fade-scalp trial is un-graded (14 fills, 9 of them	Window opened 2026-08-31, closed 2026-09-04. Banked: 2 machine-exited fills (both TARGET, $+\\$5.50$ each, $+\\$11.00$ total), on one signal at 2026-09-03 22:59:53Z. That is 2 of the 40 the row asked for. The gate is off in data/gate_switches.env right now, so the window is not currently running.	Arms: Monday 2026-09-07, at the desk — and it needs someone to actually turn it on. Exit: Its current cell — 2.0R both lots, stop $1.5\times\text{ATR}$ — unchanged for the whole window. Changing it mid-window restarts the count. Kill: If the second window also produces fewer than 10 machine-exited fills, withdraw the row: the gate cannot be graded at this duty cycle and the UP_OFF question should be answered off the shadow book instead.	Claude — one reprice n/a.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	MANUAL_CLAIM). · verification: <code>grep exhaustion_short data/gate_switches.env must read on, and the fill count is SELECT count(*) FROM trades WHERE gate LIKE 'exhaustion_short%' AND exit_reason NOT IN ('MANUAL_CLAIM').</code> · mode: observe			
7	Trial the untradeable STAY-OUT cutoff at 45 in SHADOW — direction proven, exact value fitted Last week's card: NOT DONE — no shadow log of the untradeable meter at cutoff 45 beside 65 exists in data/; and Part 2.6 has since graded the meter itself as having no discrimination at any cutoff, which makes this row moot SHADOW Run cutoff 45 in shadow for a week. Do not deploy it, and do not use 40. Mechanism: On a 51-day lake rebuild with 25 days of real trades the bands are monotone (−\$38.62 / −\$219.92 / −\$345.92 per day). At the live cutoff of 65 the meter FAILS its placebo (P=0.120); at 40 it passes (P=0.029); 45 survives leave-3-out (+\$972) and is positive in all 6 ISO weeks. Why: 40 sits on a two-cell plateau, which is too narrow to ship — a threshold that is only right at one value is a fitted value, not a finding. 45 is the honest compromise and it still costs \$542 of green days, so it is worth a shadow week before anyone types it into production. Evidence: Part 2.6 recommendation #4 (MODERATE — direction proven, exact value fitted). · verification: A shadow log of the meter's verdict at 45 beside its verdict at 65, per day. · mode: shadow	Placebo P: 0.120 at 65, 0.029 at 40. Cutoff 45: leave-3-out +\$972, positive in 6 of 6 ISO weeks, costs \$542 of green days.	Arms: Shadow only. Exit: n/a. Kill: If the shadow week disagrees with the backtest's band monotonicity, drop it — a monotone band is the whole reason to believe this one.	Claude — shadow config n/a — shadow only.
8	Shadow the router timing config (window 45m / step 5m / hold 3) — but the grade needs a TREND week Last week's card: CARRIED — the row's condition is a session with whole-day ER >= 0.15. The 08-31→09-04 week did not supply one either — see the bridge section — so the shadow grade is still waiting on the same trend day it was waiting on SHADOW Run the timing config in shadow for a week that contains at least one genuine trend day, then decide. Do not deploy it off this sample. Mechanism: The ER × NET sweep's timing half gains +\$440-\$502 with 0-1 days made worse. But it differentiates on only 4-5 days, all of them inside one summer range regime. Why: This is Part 2.6's own assessment of its best single result and its worst sample, and it interlocks with a fact from Part 1: there was no trend day this week at all. ★ REV2 — AND WAITING FOR A TREND WEEK IS WAITING FOR SOMETHING THAT HAS NEVER HAPPENED. Counted over 51 sessions and 11 ISO weeks in the lake (2026-06-19 → 2026-08-28), on 5-minute path with ER ≥ 0.15: SIX weeks contained a trend DAY, and NOT ONE contained more than one. The rate is 0.55 trend days a week, so ten of them is eighteen weeks away. Worse, "ER ≥ 0.15" is not even well defined until you name the bar size — the same threshold on the same tape gives 0, 6, 18, 27 or 34 trend days at 1m / 5m / 15m / 30m / 60m path. So the condition is replaced: run it in shadow NOW and grade on FIVE TREND DAYS or 2026-11-06, whichever comes first. Evidence: Part 2.6 recommendation #8 (MODERATE) and its closing caveat (iii); Part 1 §8 for the no-trend-day finding. · verification: The shadow log covers at least one session with whole-day ER ≥ 0.15. · mode: shadow	Timing config +\$440-\$502, 0-1 days worse, on 4-5 differentiating days. Router value splits +\$381/day on chop-heavy days against –\$64/day on trendy ones.	Arms: Shadow only. Exit: n/a. Kill: If it is negative on the first trend day it sees, it is dead — a chop-fitted timing config is precisely what this is at risk of being.	Claude — shadow config n/a — shadow only.
9	Gold: 80 runs and \$15,196 of ceiling on the 08-28 freeze, and no gate at all — an empty seat, not a bleed Last week's card: CARRIED — standing research instruction. It has since been sharpened, not shipped: the lake's gold year serves the wrong contract for 22% of its minutes, which sits underneath every gold number the seat was being hunted with PARKED Keep hunting MGC gates, and keep them INVENTED FRESH — never port, clone or re-tune an MNQ gate onto gold. Nothing gold-side is close to a promotion. Mechanism: MGC printed 80 runs of ≥1.5×ATR this cycle (≥12 points against a typical 15-minute range of 8). We caught 0 and fought 0, because there is nothing on the instrument to catch or fight with. ★ REV2 — the cluster mix (46 UNCLASS, 14 VACUUM, 12 OPEN/NEWS, 5 VOL-EXPANSION, 3 FLOW-LED) is NO LONGER quoted here as a hunting map: it is computed off the census's keep-strongest dedup, which this report's own VACUUM appendix shows selects local extrema and inflates the VACUUM rate 10.45% → 21.87% on MNQ. Gold has had no unselected-population re-run, so the honest statement is that we do not know how much of gold's 14 is the picker.	80 runs, \$15,196 ceiling, 0 caught, 0 fought. Bar source over 32 days: –\$1.21/tr (mid, n=506) vs –\$0.41/tr (trade, n=299) — no flip. Best-looking cell, the double hole, is +\$921.50 on 311 trades and –\$109.00 once its three best trades are removed. MGC is \$10.00/point — never price it with the MNQ \$2.00 multiplier.	Arms: Nothing arms. Shadow only, and not yet. Exit: n/a. Kill: Strip-the-best-trades. Every gold arm on the board is negative pooled and MORE negative with its three best trades removed; the one positive cell (the double hole) inverts on three trades of 311. Nothing ships until an arm survives that with a pre-registered activity measure run forward.	Claude — gold programme n/a.

Why: The \$15,196 is the SIZE OF THE SEAT, not money forgone: with no gate, sitting out is not a decision that went wrong. ★ REV2 — AND THE STANDING BLOCKER IS GONE. The bar-source scare (lab builds bars from the depth MID, production from TRADES, and the sign flips on that alone) was measured on SIX days. Re-run on THIRTY-TWO it does not survive: −\$1.21/tr on mid bars (n=506) against −\$0.41/tr on trade bars (n=299), same sign, and −\$1.76 vs −\$0.41 on the 19 days both tapes cover. What the bar source actually controls is FIRE RATE, not quality — 35.2% of the mid arm's trades open in a minute where the trade tape printed nothing at all. The replacement blocker is smaller and more useful: gold's money is all in the busiest quartile of the tape (+\$27.11/tr against −\$11.85 in the quietest), and every attempt to turn that into a threshold has been a fit — an activity floor worth +\$40.80/tr in-sample makes −\$11.92/tr forward.

Evidence: Movement 3 → THE MGC GREENFIELD (rebuilt from the gf4 * artifacts) §G3 for the bar source, §G4 for the activity finding, §G5 for the floor that died forward, §G6 for the double hole. △ THE 80/\$15,196 IS THE 2026-08-28 CENSUS FREEZE, which is the freeze Movements 2 and 3 were computed against. The MGC census PRINTED in Movement 1 of this document is a week younger (2026-08-30→09-04) and reads 71 runs / \$13,238 / 0 caught. Both say the same thing — an empty seat — and neither number should be quoted as the other. See the two-week seam at the top of the report. ★ And read this row against the 2026-09-04 gold hunt in the Movement 3 ADDENDUM (NOT-AN-ACTION #11, gold-four-cells-null-0904), which is a harder null than this row, and against SATURDAY #9 (lake-gold-wrong-contract-0904), which puts the 32-day bar-source re-run itself on notice. · verification: **Any gold result must name its bar source AND its sample size in the same sentence as its number. The claim this card carried for a week was a 56-trade sub-cell on six days.** · mode: observe

10 ★ The day rider does not MANAGE positions opened outside 13:30-21:00Z — it carries them

Last week's card: NOT DONE — src/gazbot7/day_rider.py still routes the manage path through the RTH-gated drift read; tonight's open position is the demonstration — it was entered 14:15:16Z with entry_atr 6.8 and arm_atr 19.7 and has no trail

PARKED Move the rider's manage path out from behind the RTH-gated drift read, or give it a fallback ATR so the trail can arm outside the US session. Until then, treat any out-of-hours entry as hand-managed and say so on the desk view.

Mechanism: drift.read is gated to 13:30-21:00Z and the manage path is downstream of it, so outside that window the rider falls back to px = entry: ahead_pt freezes at 0.0, peak stays pinned at the entry price, arm_atr is 0.0 so the trail can never arm, and the note degrades to trail not armed (need +150 [fixed]) because there is no ATR to scale with. The heartbeat stays fresh throughout, so nothing looks wrong.

Why: Two of this week's manual entries were opened outside the session and BOTH made money (+\$377.50 and +\$187.00) — which is precisely why it is worth flagging now rather than after one goes the other way. A position with no peak, no trail, no give-back and no exit except a hand is not a managed position; it is an unhedged carry that the instruments describe as managed.

Evidence: Part 1.6 §6(c) and §7 disposition; Part 1.6 §4 for which exit actually captured the week (the hand won). · verification: **Open a paper position at 07:00Z and confirm ahead_pt, peak and arm_a tr all move before 13:30Z.** · mode: observe

Thursday 07:20 entry: 4 lots held 371 minutes with ahead_pt pinned at 0.0 for 370 of them. Friday 12:39 entry: frozen for 50 of its 65 minutes. Both profitable — the fault is invisible in the P&L.

Arms: Not the weekend of 2026-09-05/06 — it touches the live manage path and wants its own test. Build it, shadow it, then arm.
Exit: n/a.

Kill: If an out-of-hours ATR proves too noisy to trail against, then say so explicitly in the desk view and keep the carry — the unacceptable state is the current one, where the position reads as managed and is not.

Garrath — rider manage path
Restore the RTH gate.

11 Shadow a dwell / 2-of-3-minute confirmation on the day rider's entry — the floor crossing is a coin flip at the sampling boundary

Last week's card: NOT DONE — no strategy matching 'dwell' has any rows in shadow.db — the rider entry-dwell variant was never armed

SHADOW Build it as a SHADOW variant of the rider's entry: require the efficiency floor to hold for 2 of 3 consecutive minutes rather than firing on the first crossing. Score it against the live entry weekly. It costs nothing and cannot trade.

Mechanism: The efficiency reading oscillates across its own 0.15 floor faster than the service samples it, so the entry time — and therefore the entry PRICE — is decided by when the sample happens to land rather than by the tape. This is the same threshold-noise mechanism Part 2.5 found at the router, measured here at the rider's own entry.

Wednesday crossed the 0.15 floor four times in six minutes. Tuesday's entry flips on a FIVE-SECOND boundary: 0.390/0.517 at :01 against 0.314/0.440 at :06. The sample is already 49 sessions, so the shadow has n from the first day.

Arms: Never live on this evidence — shadow only, until it beats the live entry across 60+ sessions AND the lost-detection count is known.
Exit: Promote only if it wins on price without dropping winners; retire it if it drops more than it improves.
Kill: If the dwell costs more detections than it saves points, it is dead — and that is the specific number to watch, not the net.

Garrath — one shadow variant
Delete the variant. The live entry is untouched throughout.

Why: A dwell requirement is the cheapest possible answer to threshold noise and it is testable for free. Δ Note the honest risk, which the section names: a confirmation delays every entry, and on this desk delay has repeatedly been the thing that killed a good rule. That is exactly why it goes to the shadow book and not to the live entry.

Evidence: Part 1.6 §7 disposition (SHADOW); Part 2.5 §2 for the same noise at the router's thresholds. · verification: **After four weeks, compare shadow entries against live on the same sessions: entry price, capture, and how many detections were LOST to the dwell.** · mode: shadow

12

Shadow benching grind_long in NORMAL_CHOP and CLEAN_TREND — together with the 13:00Z hour, never separately

Last week's card: NOT DONE — no strategy matching 'grind_bench' has any rows in shadow.db — the grind regime+hour bench was never armed

SHADOW

Put both grind benches into the shadow book as ONE variant: stand grind_long down in NORMAL_CHOP and CLEAN_TREND, and stand it down for the 13:00Z US-open hour. Score it against live grind weekly.

Mechanism: Grind's money is concentrated in regimes that are not the ones it fires most in, and the US-open hour is its worst cell on every measure at once — net, green days and win rate. Benching by regime is the one grind lever that survived this week's testing; the entry-confirmation and ER-floor levers did not.

Why: This is the week's strongest surviving grind result and it still does not clear the bar for a live change. ★ The hour rule must ship ONLY alongside the regime rule and never alone, and the section is explicit about why: the 13Z hour was chosen AFTER seeing the table, so on its own it is a fitted cut wearing a result. The regime rule was not, and it carries the robustness.

Evidence: Part 2.5 §1 and §4 disposition (GRIND (d) and (e), both SHADOW). · verification: **Weekly: shadow net against live grind net, plus the placebo on unseen days.** · mode: shadow

Regime bench: +\$2,310 against + \$922 live, strip-best-3 still +\$249, leave-one-day-out +\$1,430, and BOTH halves positive. The 13Z hour: −\$1,618 over 50 fires on 25 days, green 8 of 25, 18% win.

Arms: Not live. Promote when the second half (+\$238) reaches parity with the first AND the placebo drops below 5% on days the rule has never seen.

Exit: Retire if the second half stays flat or the placebo stays above 5%.

Kill: If the regime leg fails its placebo, the hour leg dies with it — it was corroboration, not an independent finding.

Garrath — one shadow variant, two conditions

Delete the variant. Live grind_long is unchanged.

13

Shadow the SESSION-BOUNDED gold run-boarder — the only MGC candidate that beat both its mirror and its random control

Last week's card: NOT DONE — no strategy matching 'run_board' has any rows in shadow.db — the session-bounded gold run-boarder was never armed

SHADOW

Shadow the session-bounded version of the gold run-boarder (board a run, wide stop, time cap), and spend the 12 untouched L2-only days on it as a genuine out-of-sample leg. Do not run the unbounded version.

Mechanism: It boards a run already in progress rather than predicting its start — which is the one thing this desk's 22.6M-tick null did NOT refute — takes a wide stop so the run's own noise cannot remove it, and caps the hold so a dead run stops costing carry. The session bound is what keeps it inside the hours the behaviour exists in.

Why: Gold is an empty seat (BUILD #9, mgc-empty-seat-0828) and this is the only candidate the week produced that has a mechanism, a control and a positive out-of-sample plan at the same time. It goes to shadow because the honest cell is +\$370 on thin n, which is a glimmer, not an edge — and thin n is never a kill. ★ Priced at \$10.00/point: MGC is NOT the MNQ multiplier.

Evidence: Part 2.5 §3 and §4 disposition (GOLD — run-boarding, SHADOW); Movement 3's gold chapter for the bar-source caveat. · verification: **Score the 12 L2-only days it has never seen. A sign that flips between the depth-mid and trade bar sources is NOT a verdict — name the bar source with every number.** · mode: shadow

+ \$2,702 true against a mirror of − \$5,438 and a random control of − \$3,010 — it beats both, which most of this week's gold candidates did not. Δ The 3h session-bounded cell — the version actually being shadowed — is +\$370, not \$2,702. That is the honest number.

Arms: Never live on +\$370 and this n. Shadow only.

Exit: Promote if the 12 out-of-sample days hold the sign on the PRODUCTION bar source.

Kill: If the out-of-sample leg goes negative, or the result only exists on depth-mid bars, it is PARKED again — not refuted, because the n is too thin to refute anything.

Garrath — one shadow variant on MGC

Delete the variant. Nothing live trades gold.

14

Remove ABSORPTION_CUT from rider_w5 — one cut has cost it −\$3,520 across 42 of its 108 shadow trades

SHADOW

Run rider_w5 in shadow with ABSORPTION_CUT disabled, alongside the current arm, and grade the pair on the same tape. This is the ONE concrete live experiment the UNCLASS hunt left behind, and it is cheap.

Mechanism: rider_w5 is in shadow now and losing. The UNCLASS dossier attributes the loss to a single exit cut: 42 of its 108 shadow trades are closed by ABSORPTION_CUT, and those 42 account for − \$3,520 — more than the arm's whole deficit.

108 live shadow trades, −\$2,803 net, −\$25.95/tr, 21% win. 42 of the 108 exit on ABSORPTION_CUT for −\$3,520 — i.e. the other 66 are net POSITIVE.

Arms: Shadow only. Nothing about this goes near a live slot.

Exit: Grade at $n \geq 60$ on the cut-free arm.

Kill: If the cut-free arm is also negative at $n \geq 60$, the cut was not the problem and rider_w5 is REFUTED rather than PARKED.

Garrath — one shadow variant

Delete the variant.

Why: This is the difference between 'the hypothesis is dead' and 'one cut is killing it'. Both are consistent with the arm's headline record, and only a paired shadow can tell them apart. ★ Note it does NOT rescue the rider thesis: NOT-AN-ACTION #12 (rider-w5-fit-vs-forward-0904) stands either way, because a \$45-a-trade fit-to-forward gap is not closed by an exit tweak. Evidence: Movement 3 ADDENDUM → UNCLASS (gf_full_UNCLASS.md, 2026-09-04), the PARKED disposition and its one named live experiment; and → THE POOLED RUN-CATCHER (gf_full_RIDER_ALL.md) for the same arm's forward record. · verification: A paired shadow: same signals, same entries, one arm with the cut and one without. Anything short of paired is a different-tape comparison. · mode: shadow			
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15 RIDER-S-HOME to shadow — the one leg that survived every placebo, and it is NOT a rider SHADOW Shadow it exactly as specified and do not widen it: short only, ATR14≥15, 13:00–21:00Z, stop 6×ATR, target 2×ATR, 60-minute cap. Book it as a short-side momentum scalp, not as a run-catcher. Mechanism: The pooled run-catcher died at the filter step, but one narrow leg came through the whole battery intact — including a random-entry placebo drawn from its OWN regime rather than from the whole tape, which is the harder version of that test. Why: It is worth a slot and it must be described honestly: it takes 2.5% of the ceiling, not 60%. Calling it a rider is how a 2.5% scalp gets sized like a 60% strategy. △ The exit corner it lives in (WIDE stop, TIGHT target) is the OPPOSITE of the exhaustion_short shape the brief assumed, so nothing about it transfers to the existing gates. Evidence: Movement 3 ADDENDUM → THE POOLED RUN-CATCHER (gf_full_RIDER_ALL.md, 2026-09-04), STEP 3 and the closing verdict. · verification: Forward n≥40 in shadow at the stated parameters. n=58 in-sample with a p=0.003 placebo is a reason to look, not a reason to trade. · mode: shadow	n=58, +\$2,980, +\$51.38/tr, 79% win, bootstrap CI [+16.46, +82.60], beats a same-regime random-entry placebo at p=0.003. Median out-of-sample across all 81 configurations tried: −\$4.26/tr — this is the one that is not that.	Arms: Shadow only. Exit: Stop 6×ATR, target 2×ATR, 60-minute cap — as specified, unchanged. Kill: Negative at n≥40 forward, or a positive result that only survives with the 13:00–21:00Z window tuned. The clock window is part of the hypothesis, not a knob.	Garrath — one shadow arm Delete the arm.
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16 The next run-catcher attempt needs the ORDER BOOK, not more days — and we already capture it PARKED Regress the census's far-side depletion column, and the 41ms L2 in capture.db.book, against forward continuation at board time. Do that BEFORE building another entry rule. Mechanism: Two independent hunts reached the same wall from opposite directions. Detection is solved — a dumb momentum rule boards 71 of 73 sat-out runs a median 3.4 minutes in, with three-quarters of the move still ahead. SELECTION is what fails, and the bar features we select on (ATR, ER, range, clock) separate run-fires from chop-fires only weakly. Why: This is the row that says what to do INSTEAD of the eleven things this movement just killed. The oracle-filter bound is \$18,955 a week — 180% of ceiling — so the money is demonstrably there and the failure is purely selection. The one discriminator nobody has ever regressed is the book, and it needs no data we do not already capture. △ It is not free: the UNCLASS hunt reports a book/continuation correlation of −0.010 across 401 million depth rows, so the honest prior is that this fails too. Evidence: Movement 3 ADDENDUM → THE POOLED RUN-CATCHER (gf_full_RIDER_ALL.md, 2026-09-04), WHAT THE NEXT ATTEMPT NEEDS; and → UNCLASS for the −0.010. · verification: A pre-registered regression with the target and the horizon fixed BEFORE looking, reported whatever it says. This is exactly the kind of study that finds an edge if you let yourself choose the horizon afterwards. · mode: measure	Boards 71/73 runs, median 3.4 min in. Pre-entry separation: ATR 12.7 (run) vs 8.7 (chop), ER-15 0.31 vs 0.22. Oracle-filter bound \$18,955/wk. Book/continuation correlation so far: −0.010 on 401M depth rows.	Arms: Nothing arms. This is a measurement. Exit: n/a. Kill: A pre-registered book regression that comes back at r < 0.1 kills the whole run-catcher programme, and that is a legitimate outcome worth paying for.	Garrath — one study, no new capture n/a.
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17 ★ Carry Part 1.5 §2.4's abs_veto_short exit-ladder A/B onto THIS card — it was written a week ago and never reached one SHADOW Add three shadow A/B arms on abs_veto_short's own signal against the live 1.5R/2.5R pair: Lot A at 0.5R, Lot A at 1.0R, and a +15pt arm / 8pt give-back trail. Shadow only. No live exit cell moves. Mechanism: Part 1.5 §2.4 ran the ladder, found four rungs that beat the book while keeping 7 of 7 winners, wrote “the ladder goes to SHADOW” — and then pointed at “MONDAY no. 6 on last week’s card”, which is a row that does not exist in that card or this one. The deliverable was written and never issued. This is it, issued. Why: ★ REV2 — this is the failure the report criticises the claim detector for by name: a section concludes SHADOW and nothing on the card carries it, so next Friday the same work gets done again. It	Four rungs beat the book on Rev1's ladder while keeping 7 of 7 winners; they sit at the 80th-93rd percentile of the random-entry control today, against a promotion bar of the 95th. The gate has since produced live evidence to score against: 2 signals / 4 legs / + \$163.50 on 2026-09-01.	Arms: Monday 2026-09-07, at the desk. Shadow arms only; nothing goes live. Exit: Each arm carries its own configured exit; that IS the experiment. Kill: PRE-REGISTERED, in advance, so it can fail: promote only at n ≥ 60 fires, positive on strip-the-best-3, AND clearing the 95th percentile of the random-entry control. Below any one of those, it stays in shadow.	Claude — three shadow slate rows Delete the three slate rows. They place no orders, so there is nothing to unwind.
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#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	costs nothing (a shadow arm places no orders) and it is the only exit work on this card, because HOLD #4 keeps data/exit_overrides.json shut. Evidence: Part 1.5 §2.4 (the ladder and the pre-registered bar) · Part 4 §2 (the 09-01 live signals). · verification: Each arm must differ from its control on at least one trade within a day of running — the byte-identical-pair check BUILD #5 asks for. Then score weekly. · mode: Claude — three rows in the shadow slate.			
18	Add the two flip-locked open-rider shadow arms Part 2 §11 asked for — one bit each, and they settle the week's best idea in a month SHADOW Add odr_c5_s30_f and odr_c10_s30_f to the shadow slate, each paired against its EXISTING unlocked twin so the flip lock is the only difference between the two. Do not turn the flip lock on anywhere live. Mechanism: Part 2 §11 listed this as its first “what I would do” item and it never reached a card. A paired arm where exactly one bit differs is the only construction that can attribute a result to the lock rather than to the pair being different sims. Why: ★ REV2 — carried from Part 2 §11, which named six actions and got one onto the 08-28 card. It costs nothing (shadow arms place no orders) and it is the only way the flip-lock question gets answered instead of re-litigated. It is explicitly NOT a licence to run the lock live: the 51-day replay does not support it outside the window it was found in, which is why NOT-AN-ACTION keeps the whole odr family off the promotion path. Evidence: Part 2 §11 item 1. · verification: Each arm must differ from its twin on at least one trade within a day — the same check BUILD #5 exists to enforce. If they come back byte-identical, they have tested nothing. · mode: Claude — two rows in the shadow slate.	The odr family is –\$1,835.00 over its 15 days and every arm is flagged n<40 for a forward A/B claim. That is the reason this is a shadow pair and not a proposal.	Arms: Monday 2026-09-07, at the desk. Exit: Each arm carries its twin's exit; the lock is the only difference. Kill: If the pair is byte-identical after a day, pull both and fix the wiring first (BUILD #5). Promote nothing from this family before n ≥ 40 forward trades per arm.	Claude — two shadow slate rows Delete the two slate rows.
19	The three measurement fixes Part 2 §11 asked for before next Friday — all three are still unwritten BUILD (1) The two-ratchet repricer must bucket MANUAL_CLAIM separately instead of scoring a hand-exit as the gate's own. (2) exhaustion_trial_grade.py must read its config eras out of the git history of data/exit_overrides.json instead of assuming one boundary. (3) The slate builder must WARN on byte-identical A/B pairs at build time. Mechanism: All three are instrument faults that make a number in this report wrong in a known direction, and all three were named in Part 2 §11 and then not carried onto the card. Why: ★ REV2 — carried from Part 2 §11 item 4. Each one is small and each one silently corrupts a different measurement: (1) credits the operator's timing to a gate, (2) grades a live exit trial against the wrong config era, (3) is why two shadow A/Bs ran for weeks having tested nothing. (3) is the build-time guard that would have caught BUILD #5's pair before it wasted the weeks. Evidence: Part 2 §11 item 4 · BUILD #5 for the pair this would have caught. · verification: Each fix ships with the test that fails before it and passes after. · mode: Claude — three small patches.	exit_overrides.json has 3 commits in its history, so “one boundary” is wrong by construction. Two byte-identical pairs are live in the slate today.	Arms: Monday 2026-09-07 onward — before next Friday's report reuses any of the three. Exit: N/A — instrumentation. Kill: N/A. If one of the three turns out to be more than an afternoon, drop that one and say which.	Claude — three small patches Each is a self-contained patch; revert individually.

HOLD · LEAVE ALONE — THESE ARE ALREADY RIGHT, AND TWO OF THEM NOW SAY SO ON WEAKER EVIDENCE THAN LAST WEEK · 8 PLAYS

Eight standing positions, and the action is to not touch them. ★★ **Two of them have been re-derived downwards in Rev 2 and the honest version is printed in the row.** HOLD #1 (the chop bench) no longer rests on +\$810.44 over 127 signals — that population was counted off SUPPRESSED-OPEN journal lines and the systemd journal is a ring buffer, so it cannot be rebuilt and it is RETIRED. What replaces it is a saved replay over the population that IS durable: **+\$312.35 over 117 signals**, sign-stable across the whole stop sweep but decaying across it, with two days of five carrying all of it. HOLD #2 (arm hard in aligned trend) rested entirely on “benching in aligned trend is FREE”, and that sentence is **REFUTED by its own label-shuffle placebo at P = 0.943** — chop +\$3.99 a signal against aligned trend +\$3.37, the same number.

The arm stays, because nothing says it is wrong; it now stays on judgement, with no number behind it, and the row says so. ★ **HOLD #4 carries this week’s single verdict on data/exit_overrides.json** — not touched, MED-TREND branch included, cause of death the out-of-sample leg. The seventh is your 2026-08-20 no-stop decision on the day rider, re-priced on this week’s own tape and re-affirmed — naked beat every stop width by \$1,190 on the twelve positions the week actually opened. The eighth is a rule for a promotion that has not happened yet: if a gold gate is ever given a slot it must NOT inherit the dual-slot exit — on the same entries, the desk’s Lot A/Lot B shape books −\$1,649 where a single wide lot on a clock books +\$9,202.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	Keep the chop bench — but stop quoting the dollar figure Last week’s card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. LIVE Leave the policy exactly as it is: bench everything with no confirmed direction. And stop quoting \$810 — it is retired, not merely un-quotable. If you must quote a number, it is +\$312 with its band. Mechanism: Every above-floor impulse in signal_journal (durable on disk from 2026-08-04, unlike the journal) entered at the first tick at or after its own timestamp and run forward on that gate's own live Lot-A exit, at five stop widths at once with R pinned to the configured stop. scripts/rev3_bench_replay.py, artifact sections/rev3_bench_replay.json. At the live width 65 of the 117 stop out and 52 reach target. Why: The DIRECTION survives everything the desk can throw at it and the MAGNITUDE survives nothing, which is the same verdict as before on a number that can actually be reproduced. ★ What is GONE is the decomposition: “the saving is entirely chop; in aligned trend benching is FREE” does not survive re-derivation — chop is +\$3.99 a signal (n=78) against aligned trend’s +\$3.37 (n=35), the same number, and a 10,000-draw label-shuffle placebo on the gap returns P = 0.943. Evidence: Part 1 §7 and §8; Movement 2 agrees independently — the six gates fired into this week's runs lose −\$741.50 on 32 lots. · verification: Nothing to verify — this is the current policy. · mode: observe	REVISED. 117 durable above-floor signals, bench saved +\$312.35 at the live stop (+\$2.67/signal); sign-stable across the 0.5×-1.5× sweep but DECAYING across it (+\$624.59 → +\$490.34 → +\$312.35 → +\$73.29 → +\$11.86). Two of five days carry 100% of the gross positive; strip-3 leaves +\$66.31; bootstrap −\$623.74 to +\$1,208.31, P(≤0) = 0.253. Part 3 §Q4’s independent floor: +\$128. The old +\$810.44 / 127 signals is RETIRED — its population was counted off SUPPRESSED-OPEN journal lines and the systemd journal is a ring buffer.	Arms: n/a — it is already in force. Exit: n/a. Kill: Revive to a LIVE claim at ~350 signals or after one genuine TREND week. Kill the policy outright only if the replay turns negative at the live stop width on a fresh week.	Nobody — no change n/a.
2	Keep the router’s aligned-momentum arm — but its stated reason is REFUTED, so it is now held on judgement with no number behind it Last week’s card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. LIVE Leave the arm behaviour alone: arm grind_long + abs_veto_long into a confirmed aligned break. Mechanism: One arm at Friday 14:45:03Z, four lots in 58 seconds, four winners, +\$181.50. Fifty minutes later the router benched both 46.5 points under the day high and the tape then fell 328 points. One arm and one bench, both right, with money on both sides. Why: ★ REV2, stated plainly: this arm is now held on judgement, not on evidence. Its stated reason — that benching in aligned trend is free, so arming costs nothing — was refuted by its own placebo. The honest case for keeping it is that it is the operator's standing instruction of 2026-08-05 (“if a huge trend develops you arm gates immediately and let them run”), that the one arm it produced was right, and that no test has said it is WRONG. That is a reason to leave it alone; it is not a reason to arm harder, and nothing on this card should cite the free-in-trend sentence again. Evidence: Part 1 §2, §9 and §11 disposition (KEEP LIVE). · verification: n/a — no change. · mode: observe	The supporting number is withdrawn. “50 benched signals, −\$0.81 each” came from the retired \$810/127 population. On the durable population the aligned-trend bucket is +\$3.37 a signal over 35 signals and the chop bucket +\$3.99 over 78 — the same number — with a label-shuffle placebo at P = 0.943. There is no measured regime split. What remains is 6 lots, 6 winners, +\$330.00 in one 51-minute market event, which is n=1.	Arms: n/a. Exit: n/a. Kill: It has no live number to turn negative, so it cannot be killed by one. Re-open it when the durable bench replay reaches ~350 signals and the aligned bucket can be measured against chop with power. Until then: unchanged, uncited.	Nobody — no change n/a.
3	Do NOT revert the stop-contract fix — and ★ REV2, it has finally been TESTED Last week’s card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. LIVE Leave it in production.	21-28 July: 26 STOP_UNFILLED exits, −\$1,377.50. 29 July → 4 September: 0, across 30 trading days. ★ REV2 — and the untested stretch is no longer untested: between 08-31 and 09-04 the tournament took 4 STOP exits (−	Arms: n/a. Exit: n/a. Kill: n/a — reverting it is the failure mode.	Nobody — already in production Do NOT revert it.

Mechanism: Every STOP_UNFILLED the desk has ever booked — all 26, worth −\$1,377.50 — happened between 21 and 28 July. The last one was a month ago.

Why: The fair caveat WAS exposure, not time: at the 08-28 cut the tournament had traded 6 lots in nine sessions, so the clean record was thin and largely untested. ★ REV2 — the week after supplied the exposure. Four stops fired and four filled. That is still small, but it is a test rather than an absence, which is a different kind of evidence. Do not read “30 clean days” as 30 days of exposure; read “0 of 4 stops missed” as the exposure.

Evidence: Part 1 §10. · verification: **n/a.** · mode: observe

4

Leave adaptive_exit OFF — and leave data/exit_overrides.json alone. ★ Including the MED-TREND rung.

Last week’s card: **STANDING** — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged.

LIVE

Change nothing about the exit selector or the per-gate exit cells. **This now explicitly includes the grind MED-TREND branch that Part 2 §11 proposed deploying:** it is NOT deployed this weekend.

Mechanism: All 12 sub-slots already return adaptive_exit=False. On the only day this week with real tournament trades, a fixed always-SCALP exit beat the counterfactual selector by \$1,400 (+\$799 vs −\$600), and 18 of the week’s 22 “optimal picks” were all-modes-identical and carried no information.

Why: ★ REV2 — the report gave two opposite instructions about one file on one Monday, and this is the single verdict. They are different rungs and different bars: Part 2 §7.2 grades **grind MED-TREND PROVEN** on a THREE-test bar (strip-3 +\$5.60, LODO +\$6.20, both halves positive, +\$7.70/signal on n=215, five adjacent cells positive), while HOLD #4’s “20 of 20 failed” is a FOUR-test bar that adds the out-of-sample POOL split. MED-TREND is the one cell that clears three and fails the fourth. **Cause of death for the deploy, named: the out-of-sample leg.** Its V5 out-of-sample pool reads −\$4.50/signal on n=52, and its own live forward arm has run −\$8.36/trade on n=21 since 08-16 against a control at −\$9.29 — the whole rung is losing money live, in both configurations. The lab and the forward arm agree on DIRECTION (paired on the 19 shared entries the proposal beats live by +\$60.50) and disagree on sign of the level, which is an entry problem wearing an exit costume. Keep it in shadow with rung_grindA_med_live as referee.

Evidence: Part 2 §8–§9 (the exit ladder, a NULL with named tests); Part 2.6 recommendation #9. · verification: **n/a.** · mode: observe

\$209.00) and all four FILLED at the stop — the first real exercise of the fix since July. Report week 08-24→08-28: 0 stops, so 0 tests.

19 of 20 rung-cells fail on three tests; 20 of 20 fail once the out-of-sample pool split is added. grind MED-TREND: lab +\$7.70/sig (n=215) · V5 OOS −\$4.50/sig (n=52) · live forward −\$8.36/tr (n=21) vs control −\$9.29/tr. Selector vs fixed on the only live day: −\$600 vs +\$799. (The extended sweep is 822,746 five-second bars, not the 831,379 this row used to print.)

Arms: n/a.
Exit: n/a.
Kill: Deploy the MED-TREND branch when EITHER out-of-sample leg turns positive: the V5 pool above \$0/signal, or rung_grindA_med_05 above its control over n≥40 forward trades. Not before, and not on the in-sample lab number.

Nobody — no change
n/a.

5

Do NOT move the router's ER or NET_MIN thresholds

Last week’s card: **STANDING** — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged.

LIVE

Leave ER at 0.15 and NET_MIN at 30.

Mechanism: The pair sits on a 3-cell plateau with a 0.6% spread, and both the live-managed and the full-historical universes independently pick the same cell.

Why: A plateau that two independent universes agree on is the best evidence a threshold can have, and it is evidence for LEAVING IT ALONE. The tuning opportunity this week is timing (MONDAY #2), not level.

Evidence: Part 2.6 recommendation #8. · verification: **n/a.** · mode: observe

3-cell plateau, 0.6% spread, agreed by both universes in a 40-day sweep.

Arms: n/a.
Exit: n/a.
Kill: Re-open only on a sweep that includes a trend week.

Nobody — no change
n/a.

6

Do not re-arm any gate on the strength of a missed run — the idle-gate lab says it a third time

Last week’s card: **STANDING** — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged.

LIVE

Leave every benched gate benched. A big run the desk sat out is not an argument for arming the gate that was nearest to it.

Mechanism: Every one of this week’s 62 sat-out runs was replayed second by second — 36,600 decision-seconds of real quotes, each gate’s own deciders, each gate’s own A/B exit pair, \$1.50 a round trip. Fired as configured, the six gates take 32 lots into those runs and lose −\$741.50 (−\$23.17/lot, 18.8% win).

Why: ★ The direct measurement is the argument: fired as configured into the runs the desk sat out, the six gates lose money. A random-

MECH layer: 32 lots, −\$741.50 (−\$23.17/lot, 18.8% win). Ungated: 114 lots, −\$168.50. A 724-cell threshold sweep finds no cell that is both positive and more than three trades wide. Random-entry control: NULL — nothing separates at n=2-10 in-direction lots.

Arms: n/a.
Exit: n/a.
Kill: A gate whose trigger beats the random-entry control on n≥40 would overturn this and would be a genuine finding.

Nobody — no change
n/a.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	entry control was also run — show up at a RANDOM second inside the same pre-ignition window, handed the run's direction — and it comes back a NULL: scored like for like the gates beat it on 3 of 5 in-direction and 4 of 5 direction-blind, on 2-10 lots each, and only exhaustion_short separates at all (P=0.033 raw, 0.166 Holm-adjusted). It neither supports nor undermines the hold. ★ REV2: an earlier draft of this card said the control killed all six triggers. It does not, and the hold does not need it to. Evidence: Movement 2 \$2, \$4 and \$5 (recomputed this cycle on this week's census freeze); control re-scored by scripts/rev2_m2_indirection_control.py. · verification: n/a — it is the current policy. · mode: observe			
7	The day rider stays NAKED — your 2026-08-20 decision re-priced on this week's tape, and it holds Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. LIVE No change. The rider keeps running without a protective stop. Mechanism: All thirteen rider positions of the week raced forward on lake ticks at 0.5-3.0x the ATR the rider itself froze at entry, one stop per position, \$1.50 the round trip. On the TWELVE positions this week actually opened, naked made +\$572.50 and every stop width lost money — the best of them, the tightest at 0.5x, made -\$617.44. Why: The whole apparent case for a stop is ONE position: the 08-21 four-lot carry, -\$2,149.44 naked against -\$34.56 at 0.5x. But that position was carried through a 49-hour CME halt, and a stop cannot fill in a closed market. What would have prevented it is the standing flat-by-20:40Z rule, which failed for an unrelated reason. Grading the no-stop decision on that trade grades it on the one night a stop was guaranteed not to work. ★ Note the SLOPE too: the grid worsens monotonically as the stop widens, which is the signature of entries that go adverse immediately — an entry problem, not a stop problem. Evidence: Part 1 §3b (new this revision); harness scripts/rev2_rider_stop_week.py. · verification: n/a — it is the current policy, now with a dated re-price against it. · mode: observe	Ex-carry: naked +\$572.50 vs best stop -\$617.44 — the no-stop policy wins by \$1,189.94 on this week's own decisions. Including the carry, 0.5x beats naked by \$924.94, and that is one trade of thirteen.	Arms: n/a. Exit: n/a. Kill: A week in which the rider is stopped out of a position it would have recovered, or in which naked loses to a stop on the rider's OWN entries (not on a carry), re-opens this. ★ And the honest trigger: it should be re-priced EVERY week, because that is the difference between a standing decision and an unwatched one.	Garrath — it is your standing decision of 2026-08-20 Set PLACE_VENUE_STO P = True in src/gazbot7/day_rider.py.
8	If a gold gate is EVER promoted, it must not inherit the live slot config — gold's exit wants the opposite of MNQ's LIVE Do nothing now, and write this down where the next gold promotion will hit it: gold gets ONE wide lot held to a clock, not the dual slot (Lot A tight scalp + Lot B chandelier). Mechanism: On every gold entry built this cycle, the desk's own dual-slot shape is negative and a single lot with a very wide stop held to a clock is the only exit family that pays. Same entries, same tape, only the exit changes. Why: This is a HOLD because there is nothing to promote — but it is the highest-value sentence in the gold work, because the default when a gold gate finally does earn a slot will be to give it the MNQ slot config, and that default is worth -\$10,851 on the lead cell alone. △ The magnitudes below come off the same year tape that SATURDAY #9 shows is 22% the wrong contract; the SIGN is the finding, and the sign is what this row asks you to remember. Evidence: Movement 3 ADDENDUM → GOLD (gf_MGC.md, 2026-09-04), the exit-family work; and its own §1.1 for the contract-roll caveat on the magnitudes. · verification: Re-run the exit families on the volume-led front-month tape once SATURDAY #9 lands. If the sign survives the contract fix, this becomes a rule. · mode: note	Lead cell, same entries: dual slot -\$1,649 vs single wide lot + \$9,202.	Arms: Never — it is an instruction about a future promotion, not a promotion. Exit: n/a. Kill: If the sign flips on the corrected tape, delete this row.	Nobody — this is a standing constraint n/a.

NOT-AN-ACTION · WITHDRAWN / REFUTED — KEPT ON THE CARD SO NOBODY RE-PROPOSES THEM · 15 PLAYS

Every one of these was, or could plausibly become, a recommendation. They are printed with their autopsies so next Friday's agent does not dig the same hole — including two (the searched day-classifier and the BIG-TREND exit rung) that this desk has been half-believing for weeks and that now have a named test on the certificate rather than a shrug, and one that is simply a caution about how to read a 100% win rate. ★★ **Five more landed tonight and one of them**

refutes this report's own Movement 3: the size threshold that section reports as a finding is a UNITS ARTEFACT — it reaches AUC 0.81 measured in points and collapses to $AUC \approx 0.50$ in ATR units, which is the only unit a stop can be set in. Beside it: all four gold cells came back null against a fixed-clock control, rider_w5 delivered $-\$25.95/\text{tr}$ against a $+\$19.29/\text{tr}$ backtest, UNCLASS turns out to be the base rate rather than a cluster, and filtering trades instead of signals manufactures edge out of replay sequencing. This is the most valuable window on the card this week.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	★ "The momentum shadow book is green while the routed book sits flat" — REFUTED as an arming argument, 106 times over Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. REFUTED When the open-hour watcher makes this argument — and it made it 106 times this week — it carries no information about the gate it is asking you to arm. Do not act on it. Mechanism: Reprice the LIVE gates' own suppressed intents at the shadow's exact ruler (2.0R target, 1.0×ATR stop) and you get — \$143.61 on 65 signals, against the shadow's +\$948.50 on 45. Same week, same instrument, same exit rule, opposite sign — so the exit is not the difference. The gap is entirely the ENTRY POPULATION: the shadow variants run a looser trigger with no ATR floor, so they fire in exactly the tape where the live gate's floor refuses. Why: There is no reformulation that makes a floorless variant's P&L evidence about a floored gate. The shadow result itself stands — it is the promotion candidate at MONDAY #1. Only its use as a reason to arm the LIVE gate is dead. Evidence: Part 1 §6 and §11 disposition (REFUTED as an arming argument). · verification: n/a. · mode: observe	106 [ACT] alerts this week; 5 router arms. The alerts name abs_veto_short 59 times, abs_veto_long 54, exhaustion_short 16, grind_long 9, rgv_short 9, capitulation_long 4. Shadow +\$948.50 / n=45 vs live suppressed intents at the identical ruler −\$143.61 / n=65.	Arms: Never, on this argument. Exit: n/a. Kill: n/a — this IS the kill.	Nobody — it is a refutation n/a.
2	Never relegate a DIRECTION — tune the side, keep the direction Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. REFUTED When a gate's long side is red and its short side green, do not switch the red side off. Mechanism: Across all 7 shadow arms that have a red side and a green side with ≥ 20 trades each, the red side's green-day rate averages 43% against the green side's 50%. thrust_loose's LOSING long side is green on MORE days than its winning short side. Why: Sides differ in magnitude, not in whether they work. A direction that is green on 43% of days is not broken; it is smaller. This is why MONDAY #1 promotes abs_veto_55s two-sided with the long side merely sized smaller. Evidence: Part 2 §9 disposition (REFUTED, named test). · verification: n/a. · mode: observe	7 arms, red-side green-day rate 43% vs green-side 50%. Spearman between the two sides' daily P&L is not the point — the point is that both sides fire on the same days.	Arms: n/a. Exit: n/a. Kill: n/a.	Nobody — standing principle n/a.
3	The searched causal day-classifier for the Open Rider — REFUTED by three tests, do not search again Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. REFUTED Do not run another single-feature threshold search over this sample. The day-level INSTRUMENT is still worth building (BUILD #2); the SEARCH is dead. Mechanism: Eight features were hunted, placebo-controlled and corrected for having looked eight times. (1) Placebo against random day-removal, 10,000 draws: only 2 of 8 features beat it even raw. (2) Holm correction across the 8: best adjusted $p = 0.136$. (3) A genuine out-of-sample leg, fit on 41 days and tested on 9: 4 held, 4 failed, and even the holders still lost money. Why: Removing 60% of days looks brilliant whenever the removed 60% lost — which is why the placebo is the test that matters here and why a raw win means nothing. The concentration the operator identified is REAL (6 green days $+\\$3,066$ vs 9 red $-\\$5,626$); the searched classifier that was supposed to exploit it is not. Evidence: Part 2 §3, §3d and §9 disposition. · verification: n/a. · mode: observe	8 features, best Holm-adjusted $p = 0.136$. OOS 41-fit/9-test: 4 of 8 held, all still losing money. Oracle day-selection $\approx \\$1,022/\text{week}$ — real, and only knowable afterwards.	Arms: Never on this evidence. Exit: n/a. Kill: n/a — this IS the kill. The pre-registered $ATR \geq 9.5$ filter is the only honest instrument still running and it needs 50 more sessions. Do not tune it, do not add a feature, do not stop it early.	Nobody — it is a refutation n/a.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
4	One day-classifier serving BOTH MNQ and MGC — refuted; detection must be built twice Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. REFUTED Do not build a shared good-day detector across the two instruments. Mechanism: On the 25 days we hold both tapes, label agreement is 44% against a 49% chance baseline — a lift of MINUS five points — and the Spearman correlation between the two instruments' daily P&L is +0.019. Why: It was a genuinely attractive idea: two strategies on two instruments both live or die on a minority of days, so one classifier serving both would have been worth more than either gate. The good days are simply unrelated. What DOES transfer is threshold SELECTIVITY in percentile space, which is a much weaker claim and is already how the gold work is being done. Evidence: Part 2 §9 disposition (REFUTED, named test). · verification: n/a. · mode: observe	44% agreement vs 49% chance, n=25 days. Spearman +0.019.	Arms: n/a. Exit: n/a. Kill: n/a.	Nobody — it is a refutation n/a.
5	The BIG-TREND exit rung — refuted at usable n, after the window was widened as far as this desk can go Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. REFUTED Do not ship a special exit for the BIG-TREND rung, and do not edit <code>data/exit_overrides.json</code> this weekend. Mechanism: At $ER \geq 0.40$ — the only boundary where the rung carries 71–192 signals per family over 25–37 days — every family's best-of-72 exit is negative and the first/second-half split flips sign in 3 of 5. At $ER \geq 0.50$ the rung is 3.0% of tape-minutes and $n=21-66$; the two apparently-good cells (capitulation +\$34.20 and +\$82.10) strip to —\$47.50 and —\$65.20 on three trades. Why: This has been half-believed on this desk for weeks on the strength of a tiny sample, and the honest fix was to widen the window rather than argue about it: 831,379 five-second bars, three independent entry pools, 6,736 scored entries. It is refuted as a place where a SPECIAL exit earns its keep — not refuted as a market state. Evidence: Part 2 §8 and §9 disposition (REFUTED at usable n). · verification: n/a. · mode: observe	20 of 20 rung-cells graded, 20 failed. None survived strip-best-3 + LODO + half-split + pool-split together.	Arms: Never on this evidence. Exit: n/a. Kill: n/a — this IS the kill.	Nobody — it is a refutation n/a.
6	COIL-BRK — this week's fresh greenfield candidate, built and killed inside one cycle Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. REFUTED Do not shadow it, do not re-tune it. It is a clock, not a trigger. Mechanism: Four independent fatal tests. (1) Out of sample it inverts: +\$3.87/trade in sample, —\$4.62 out of it. (2) It is three trades wide: strip its best 3 of 513 and +\$2.20/trade becomes —\$0.44. (3) The placebo beats it — the same rule taken 30 minutes EARLY, when the coil condition has not happened yet, books +\$3.55/trade against the real signal's +\$2.20, and 60 minutes early also beats it. (4) It is a one-sided artefact: LONG —\$2.05/trade, SHORT +\$6.39, on a tape with a known short drift, green on only 22 of 52 days. Why: ★ The shape of the placebo ladder is the tell and it is worth learning. The shifted copies decay SMOOTHLY from +\$3.55 at 30 minutes to —\$11.29 at 8 hours. That is not a signal being destroyed by misalignment — it is a slow intraday drift being sampled at different points. And the control settles it: the UNARMED break, with the coil filter switched off entirely, books +\$1,625.70 on 1,356 trades. The filter the whole candidate is built around SUBTRACTS value. Evidence: Movement 3 §4 (built and killed this cycle). · verification: n/a. · mode: observe	n=513, +\$1,126.27, +\$2.195/tr. IS +\$3.865 / OOS —\$4.615. Strip-best-3 —\$0.437. Sign-flip —\$3.496. 2 of 5 placebos beat it. Arm ladder flips sign four times.	Arms: Never. Exit: n/a. Kill: n/a — this IS the kill.	Nobody — it is a refutation n/a.
7	Do not promote any of the five frozen greenfield candidates on their out-of-sample bounce Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. REFUTED UNCL-RIDE-ER and board_atrband both show a positive OOS leg this week. Leave them parked. Mechanism: Both OOS legs are 27 and 29 trades over 4 sessions, and both are driven entirely by their LONG side (+\$15.56 and +\$20.21	5 candidates forward-tested on tape none had seen. 4 of 5 net negative over their whole life. Best OOS legs: 27 and 29 trades over 4 days, both long-side only.	Arms: Never on a 4-day OOS leg. Exit: n/a. Kill: n/a.	Nobody — it is a refutation n/a.

a trade) while both SHORT sides are red. Over their whole lives both are still net negative (−\$483.60 and −\$334.60). Four days is not a forward test; it is a sample of the week's drift.

Why: ★ The reassuring half of this table is the other three. FL-4 flow_ignition went from +\$4.84/trade in sample to **−\$16.82** out of it, and MOMBRK from +\$3.66 to −\$0.56 at an 11% win rate — and **both were already REFUTED by their own placebo tests last week**. The forward tape has now agreed with the placebo. The kill tests are calling it right, which is the strongest methodological result in the movement.

Evidence: Movement 3 \$5 (the forward test, computed this cycle). · verification: **n/a**. · mode: observe

8	"The tournament went 6 for 6 this week" — that is n=1 with a wide grin, not six observations Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. REFUTED Do not draw any conclusion from the week's 100% win rate, in either direction. Mechanism: Three gates fired within 51 minutes of each other on the same up-leg of the same tape, and each booked TWO lots because the slate is a scale-out. So the A and B lots are one signal with two different exits, and the three signals are three gates agreeing about one push. It is one market event sampled six times. Why: This is the same standing error as a rolling-window "3 of 3 confirmed", and it is worth naming on the card because a 100% week is exactly the number that gets quoted. Every quantitative claim in Part 1 is deliberately built on populations of 30-130 signals instead. Evidence: Part 1 §2 ("a win rate of 100% on six lots is one observation, not six") and §11 standing caveats. · verification: n/a. · mode: observe	6 lots, 3 signals, 1 push, 51 minutes, +\$330.00. The desk was armed for 24h30m of 708 available gate-hours (6 gates x the week's 118 tradeable hours, Mon 00:00Z to Fri 22:00Z) — 3.46% of the week. (★ REV2 — was "24h31m of 720 ... 3.40%"; 120 hours is not the week's length.)	Arms: n/a. Exit: n/a. Kill: n/a.	Nobody — a standing caution n/a.
9	Your Claim button — a dead heat on the TOURNAMENT, and the rider's biggest single exit Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. LIVE No change. Keep claiming when you want to. Mechanism: Three of the six lots ended MANUAL_CLAIM. Each was re-run forward on the tick tape under its own configured exit stack, at the 24.25-point entry ATR read off the native 1xATR stop the desk actually placed (stp-000003/4 at 29,714.00 against a 29,738.25 fill) rather than a modelled one. You gave up \$78.25 on grind_long_A by banking early and saved \$72.00 on grind_long_B by not waiting for a trail that mathematically could not arm — its chandelier needed +84.9 points and the run peaked at +73.5. Why: The individual lots were not close; the aggregate is a dead heat. ★ REV2 — and the scope matters: this row is about the TOURNAMENT's three claimed lots. On the DAY RIDER, the same button is the desk's main exit and Part 1.6 §4 scores it at +\$5,426.50 this week. "Dead heat" is true of the tournament and false of the desk. Part 1 §8's −\$75.14 is the Rev1 figure on a modelled 23.5-point ATR; the +\$4.50 here is the same three lots on the venue's own 24.25-point stop, and it supersedes it. Evidence: Part 1 §3 and §11 disposition (NOT AN ACTION); harness scripts/rev2_arm_own_exits.py. · verification: n/a. · mode: observe	Three claimed lots: you booked + \$136.50, their own exits were worth +\$141.00 — a delta of + \$4.50 across the week. The whole tournament week on its own exits is +\$334.50 against the +\$330.00 booked. (Rev1 of this card said − \$0.86 / +\$329.15, on a modelled 23.5-point ATR; the venue's own stops say 24.25.)	Arms: n/a. Exit: n/a. Kill: If the delta goes materially negative over 20+ claimed lots, revisit.	Nobody — no change n/a.
10	★ capitulation_long is armed by a CRON, not by a decision — and it produced the week's largest single win Last week's card: STANDING — a standing position or a refutation — there is nothing to ship, so there is nothing to check. It is re-issued unchanged. PARKED This needs a Saturday call that is not a code change: either the router should own capitulation_long like the other five gates, or the asymmetry should be written down as deliberate. Right now it is neither, and that is why it is here rather than in the SATURDAY window. Mechanism: reactivate_gates.py re-arms three gates at 22:00Z nightly. The router benches two of them within 32-38 seconds and leaves the fader standing. So capitulation_long sat armed for 17h45m and was the one gate awake when the 14:00Z flush came. The router DID read it and chose to keep it — that part was a decision — but nothing decided to ARM it. Why: Being right by rota is not the same as being right. 84% of the desk's entire armed time this week came from a cron the router does	capitulation_long armed 20h31m = 17.10% duty, 84% of ALL armed gate-time this week. The desk overall used 24h30m of 708 available gate-hours (6 gates x the week's 118 tradeable hours, Mon 00:00Z to Fri 22:00Z) — 3.46%. Its two lots made + \$148.50.	Arms: n/a. Exit: n/a. Kill: ★ REV2 — no longer 'a governance question, not a hypothesis'. There are exactly two defensible positions and 'leave it' is not one of them. EITHER the router owns it like the other five (remove it from reactivate_gates.py's roster and let the tick decide) — leaving the arming of a gate whose unbenched counterfactual loses \$15.67 a fire to a clock is leaving \$15.67 a fire to a clock. OR the asymmetry is written down as deliberate, with its reason, inside reactivate_gates.py — because a carve-out nobody can find the	Garrath — a decision, not a deploy n/a.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	not clean up, and it produced +\$148.50 of the week's +\$330.00. That is luck wearing a policy's coat, and next time the rota leaves a gate standing into the wrong tape nobody will have decided that either. Evidence: Part 1 §4 (every armed window, with who opened and closed it) and §11 disposition (PROCESS, needs a Saturday call). · verification: Either the router's switch writes cover capitulation_long, or a comment in reactivate_gates.py states the carve-out and why. ★ REV2 — and the number that should settle it: the gate's unbenched mirror (capit_live_mirror, its own live parameters) is −\$15.67 a fire over 352 fires at a 12.5% win rate. Its live all-time record is +\$63.00 on 34 fills — which is LESS than the +\$148.50 it made on 2026-08-28 alone, so on the other 32 fills of its life it is −\$85.50. The positive record is what BENCHING selected, not what the gate does. · mode: observe		rationale for is indistinguishable from a bug, and this one produced the week's largest single win, which is the most dangerous way for a bug to behave. △ n=34 live fills and the mirror is a replay, so neither number is strong — but the decision is about which instrument owns the switch, and that does not need n.	
11	All four gold cells came back NULL — six candidates, every one beaten by shorting a lot at a fixed time of day REFUTED Do NOT build another gold entry off this cycle's candidate list. Momentum and reversion, long and short, are all covered and all fail. Mechanism: Six candidates were built fresh from gold's own distribution — not ported from MNQ — and all four cells were covered. Every one is beaten by a fixed-time-of-day short with the same exit, and five of the six lose money out of sample. Why: It is printed as a NOT-AN-ACTION so nobody re-proposes the same four cells in a fortnight. ★ The control is the finding: when a fixed clock constant beats every gate you built, the gate was measuring the clock. Two cells — momentum-long and reversion-long — have no surviving candidate AT ALL, which is the honest place to start next time, and only after SATURDAY #9's contract fix. Evidence: Movement 3 ADDENDUM → GOLD (gf_MGC.md, 2026-09-04), the four-cell battery and the clock sweep. · verification: Score any future gold trigger against the fixed-time CONSTANT, not against a random placebo. The constant is the harder control and it is the one that killed these six. · mode: none	6 candidates, 4 cells, all beaten by a fixed-time constant; 5 of 6 negative out of sample. Live gold shadow book over 17 days: 225 trades, −\$1,671, with the book-filtered LONG arm +\$485 (n=19) and SHORT −\$376 (n=20).	Arms: Nothing. Exit: n/a. Kill: A gold candidate that beats the fixed-time constant out of sample, on the corrected front-month tape, reopens this.	Nobody n/a.
12	rider_w5 is the desk's cleanest fit-versus-forward case: +\$19.29/tr promised, −\$25.95/tr delivered REFUTED Do not promote anything on a backtest of this shape again, and do not re-tune rider_w5 to close the gap. Read the \$45-a-trade swing as the size of the fitting error on this kind of study. Mechanism: rider_w5 (the board gate) was promoted to shadow on a +\$4,841 / +\$19.29-per-trade backtest. It has been running live in shadow since 08-17 and has delivered −\$25.95 a trade over 108 trades at a 21% win rate. The independent re-derivation this cycle found the same thing without being told: across 81 configurations, the median out-of-sample result is −\$4.26 a trade. Why: Two separate roads to the same number is what makes this worth a card row rather than a footnote. It is also the strongest available argument for the desk's out-of-sample protocol, because the protocol predicted this. △ It does NOT settle whether the rider hypothesis is dead — BUILD #14 tests one specific exit cut that may be carrying the whole deficit — but it does settle that the +\$19.29 was never real. Evidence: Movement 3 ADDENDUM → THE POOLED RUN-CATCHER (gf_full_RIDER_ALL.md, 2026-09-04); and → UNCLASS for the same arm's exit breakdown. · verification: It is already verified forward — that is the point of the row. The check is that shadow.db still shows the arm and its record. · mode: none	Backtest +\$19.29/tr (n implied by +\$4,841). Live shadow 108 trades, −\$2,803, −\$25.95/tr, 21% win. Swing: \$45/trade. Independent OOS median across 81 configs: −\$4.26/tr.	Arms: Nothing. Exit: n/a. Kill: n/a — a forward result is not killed by a better backtest.	Nobody n/a.
13	★ THE SIZE THRESHOLD THIS REPORT'S OWN MOVEMENT 3 FOUND IS A UNITS ARTEFACT — it dies in ATR units REFUTED Withdraw the size-threshold finding. Do not build a 'only take runs above N points' filter, and do not carry the threshold into next week's card. Mechanism: Sorted by how big a run turned out to be in POINTS, discrimination climbs beautifully and looks exactly like a footprint emerging — AUC reaches 0.81 on the ≥200pt runs. Measured in ATR UNITS, which is the only unit a stop can be set in, every feature collapses to AUC ≈ 0.50 and ATR itself INVERTS to 0.42. Why: ★ This refutes a headline conclusion of the section it sits inside: Movement 3's editorial half, written on 08-29, reports the size threshold as a finding. Sorting by points sorts by volatility, and	AUC 0.81 at ≥200pt in POINTS; AUC ≈ 0.50 in ATR units, with ATR itself inverting to 0.42. Board-time size predictor: best single feature r=0.074 on 4,830 boards; all fourteen features fitted and scored on the SAME data reach multiple R=0.149; fitted in-sample and scored out-of-sample, r=0.116. The revival condition asked for 0.40.	Arms: Nothing. Exit: n/a. Kill: A discriminator that holds AUC ≥ 0.65 in ATR units on a held-out set reopens it.	Nobody n/a.

volatility is the thing the feature was supposed to be discovering — the apparent threshold was the ruler, not the tape. It is printed loudly because a plausible, half-believed number is more dangerous than a wrong one, and this desk has been half-believing this one for two weeks.

Evidence: Movement 3 ADDENDUM → UNCLASS (gf_full_UNCLASS.md, 2026-09-04) §3 and its closing verdict. It contradicts the Movement 3 editorial half above it, which was written on 2026-08-29 — later evidence on more data, so the addendum wins. · verification: **Any future 'runs above size N are tradeable' claim must be stated in ATR units before it is believed, and must beat the same AUC test in those units.** · mode: none

14 A backtest that filters TRADES instead of SIGNALS manufactures edge out of replay sequencing REFUTED Never filter a trade list. Filter the SIGNAL list and re-run the replay. Any past result produced by filtering trades has to be re-run before it is quoted again. Mechanism: Removing trades from a finished replay changes which subsequent signals were reachable, because a signal that arrived while the book was occupied never became a trade. Filter the trades and you silently un-occupy the book, so the survivors are no longer the trades the rule would have taken. Why: This is a method wound with a body attached: the UNCLASS hunt nearly shipped a +\$1.02-per-trade candidate that is really − \$3.60 once the filter is applied to signals and the replay re-run. It is the same family as this desk's standing finding that an exit sweep with position occupancy is really an entry sweep — and it is why the surviving legs in this report were re-run rather than re-filtered. Evidence: Movement 3 ADDENDUM → UNCLASS (gf_full_UNCLASS.md, 2026-09-04), the two engineering faults at the end. · verification: Re-run, never re-filter. The check is that the signal count changes when the filter is applied — if only the trade count changed, it was done wrong. · mode: none	One candidate: +\$1.02/tr filtered-on-trades vs −\$3.60/tr filtered-on-signals. A 4.6× sign-flipping gap from the accounting alone.	Arms: Nothing. Exit: n/a. Kill: n/a — this is an accounting identity, not an empirical claim.	Nobody n/a.
15 UNCLASS is not a cluster — it is the base rate, and the census should print it as UNRESOLVED REFUTED Stop hunting UNCLASS as a cause-cluster. Change the census to print it as UNRESOLVED with its base rate beside it, and ship the four shapes it did produce as census vocabulary only. Mechanism: UNCLASS covers 75.4% of all minutes against 56.6% of the runs — a lift of 0.75, i.e. runs are LESS likely to be UNCLASS than an arbitrary minute is. That has now been measured independently on two separate weeks. Why: A bucket with a lift below 1.0 is not a cause, it is the absence of a label, and hunting it is hunting the residual. Printing it as UNRESOLVED with its base rate is the fix, because the name is what made it look like a finding. Four shapes (TURN / DRIFT-VOID / GRIND-CONT / STEP-REPRICE) did replicate across two independent run sets and are worth keeping — as vocabulary for describing runs, not as gates. Evidence: Movement 3 ADDENDUM → UNCLASS (gf_full_UNCLASS.md, 2026-09-04), closing verdict. · verification: Recompute the lift on a third independent week. A lift below 1.0 three times is not a sampling accident. · mode: none	Base rate 75.4% of minutes vs 56.6% of runs; lift 0.75, replicated on two weeks. Four replicating shapes, zero tradeable candidates.	Arms: Nothing. Exit: n/a. Kill: A lift materially above 1.0 on a third week reopens it.	Garrath — one label in run_census.py Rename it back.

STANDING CAVEATS ON EVERY DOLLAR ON THIS CARD

Every repriced number is a CEILING. Fills are modelled at the exact stop/target/trail level with no slippage, and the desk's own standing finding is that live losses run **1.1-3.1× modelled**. Treat the policy as the edge and the headline dollar as decoration. Anything on thin n is flagged in-row as a direction, not a measurement.

The fee is \$1.50 per round trip and the multipliers are not the same. MNQ is \$2.00 a point; **MGC is \$10.00.** Every gold number in Movement 3 and in Part 2.5 Part A is priced at the gold multiplier — pricing gold with the Nasdaq one would understate it fivefold, and doing the reverse would flatter it fivefold.

Six method wounds are open and they bound everything above.

SUPPRESSED-OPEN is logged at `tournament.py:306` and the 55-second absorption veto runs at `tournament.py:360` — afterwards. So for four of the six gates the benched population Part 1 reprices contains trades the gate's own veto would have thrown away, which makes any bench saving built on that log an UPPER BOUND rather than a point estimate — one of the reasons Rev 2 retires the +\$810/127 figure outright and carries Part 1 \$6's +\$312.35/117 instead. The two gates measured clean are `grind_long` and `capitulation_long` — **HOLD #1.** The entire bench replay is IN-SAMPLE by construction — it reprices decisions already made, on the tape they were made on — and every replayed figure is one lot on the Lot-A ruler while the live desk runs two, so the Lot-B tail is not modelled anywhere — **HOLD #1.** Neither nightly rollup has ever had a timer. `router_nightly` last wrote 2026-08-14 and `selector_nightly` 2026-08-18, so **every nightly number in Part 2.6 is a rebuild done during this report**, not the output of a monitor that was watching — **SATURDAY #1.** The router was blind for **3h35m** across 43 fail-closed aborts, and `systemctl`, the trial log and the tick count all read clean straight through it — so “the router decided X” is, on this week's evidence, sometimes “the router did not run” — **SATURDAY #2.** Two shadow A/B pairs are **byte-identical** (29/29 and 251/251 trades) and four arm names on the leaderboard carry one trade set, so any reading of those as independent confirmations inflates the evidence four-fold — and a broken arm reports a PERFECT NULL, which is the one failure that looks like a clean result — **BUILD #5. there was no trend day.** All five sessions closed with a whole-day efficiency ratio between 0.017 and 0.032, 61.5% of tradeable minutes were chop, and at least four recommendations in this report are explicitly ungradeable until one trend week lands — **BUILD #8.**

What did not run this cycle, named exactly. 16 of 16 fragments are finished work and sit in this document in full. 13 of them were written tonight (Part 0, Part 1, Part 1.5, Part 1.6, Part 2, Part 2.6, Movement 1, Movement 2, Movement 3, Part 3 and Part 4) and describe the week

2026-08-30 22:00 → 2026-09-04 20:59 UTC; the rest describe the week to Friday 2026-08-28. That seam is drawn at the top of this report and it is the single most important thing to know before comparing any two numbers in it. **Nothing on the running order predates the 08-28 cycle**, so no section carries the red “this is last cycle’s text” banner this week. ★ Note what that does not mean, and this is the ONE description of Part 1.5 that the front matter, this paragraph and the section itself now all use: Part 1.5 (Rehabilitation) is a **fresh 2026-08-30 wrapper (§0a-§0c) around last cycle's standing dossiers**. Its mtime is inside the cycle so the build does not flag it, and it carries its OWN dated banner saying the material below its first rule describes the week to 2026-08-21. Read §0a-§0c as this cycle's and the treatments as standing material. **Nothing is missing** — every section on the running order rendered from a real fragment, so there are no red placeholder boxes to hunt for. ★ **No play on this card is sourced from any section carrying a stale banner**, and the loop on LAST week’s card is closed immediately below this table.



LAST WEEK’S CARD — what actually happened to it

The card is only worth writing if somebody checks it afterwards. Last week’s (2026-08-21) carried **50 plays**; below is the actionable subset — every SATURDAY and MONDAY row, plus the LIVE BUILD rows whose completion is checkable from the box. **Each verdict names the command or file that decided it**, so none of this has to be taken on trust. Score on the checkable rows: **5 done, 6 not done**.

PLAY	WHAT IT ASKED FOR	STATUS	THE CHECK THAT DECIDED IT
SATURDAY 1	DELETE gazbot7-weekend-flatten.timer — still armed, fires Sunday	DONE	systemctl is-enabled gazbot7-weekend-flatten.timer → not-found; no such unit in /etc/systemd/system. The play with the only hard deadline on last week’s card is the one that landed.

PLAY	WHAT IT ASKED FOR	STATUS	THE CHECK THAT DECIDED IT
SATURDAY 2	Re-open the day rider's protective stop	NOT DONE	src/gazbot7/day_rider.py:102 still reads PLACE_VENUE_STOP = False. This is an operator decision of 2026-08-20, not a dropped task — and re-pricing it on this week's tape (Part 1 §3b, new this revision) says the decision was RIGHT: naked beat every stop width by \$1,190 on the twelve positions this week actually opened. NOT DONE, and it should stay not done — it is HOLD #7 on this week's card.
SATURDAY 3	Make the tournament standdown a FILE every writer checks	NOT DONE	scripts/router_tick_durable.py:83 still holds TOURNAMENT_STOOD_DOWN = False as a module constant and there is no standdown file in data/. The second writer it was meant to constrain (open_hour_watch.py) wrote to gate_switches.env twice this week — which is why it is back on this week's card as SATURDAY #3.
SATURDAY 4	Delete the five expired carve-outs from gate_switches.env's header	DONE — and it grew back	The header was reset on 2026-08-26: 524 comment lines archived to docs/gate_switches_header_archive_2026-08-26.md. [△] But the top of the file now carries a fresh 8-line carve-out written 2026-08-28T16:24Z by the open-hour watcher, with no expiry. The mechanism was never fixed, only the backlog.
SATURDAY 5	Point core_health.flat at the venue, or rename it	NOT DONE	src/gazbot7/core.py:724 still writes "flat": self._pos is None — the tournament's own book, not the venue. The file read flat: true at 02:05Z while this revision was being written.
SATURDAY 6	Grade the per-lot profit ladder shipped on 2026-08-20	DONE	Graded inside last week's own report (Part 1.6): 231 sessions, rungs filling on 71.0% / 39.4% / 11.3% / 4.8%. Nothing to re-do.
MONDAY 1	Bench abs_veto_short's automated slot, keep the signal in shadow	DONE, and it held	data/gate_switches.env reads abs_veto_short=off. The gate took zero fills all week. [△] It was nevertheless ARMED twice by the open-hour watcher and re-benched by the router (Part 1 §5) — the switch held, the process did not.
MONDAY 2	Retire rider_w5 from the shadow book	NOT DONE	shadow.db still has 74 rider_w5 rows and its most recent entry is 2026-08-28 19:46Z — it was still writing on the Friday of this report.
BUILD 7	Row-level duplicate check at shadow-write time, and re-seed the broken arms	NOT DONE	cx_clip_brk_live and cx_clip_brk_standdown are still 251 rows each and byte-

PLAY	WHAT IT ASKED FOR	STATUS	THE CHECK THAT DECIDED IT
			identical. Re-raised this week as BUILD #5.
BUILD 9	Populate signal_journal.suppressed_by	DONE	1,230 of 1,230 journal rows in the report week carry a non-empty suppressed_by (1,213 switch_off, 14 atr_floor, 3 none_visible). Part 1 \$7 is built on it.
BUILD 10	Put a timer on router_nightly and selector_nightly	NOT DONE	Zero units in /etc/systemd/system, zero crontab entries. The files dated 08-26/27/28 in data/router_nightly/ were written at 22:58-23:00Z on the Friday by this report's own build. Re-raised this week as SATURDAY #1 — second week running.

THE REST OF LAST WEEK'S CARD, AND THE ONE REVERSAL

The 6 HOLD and 14 NOT-AN-ACTION rows required no action by **construction** — they are instructions not to do things, and nothing was done. Five of them are re-argued on this week's card with fresh evidence and reach the same answer. The remaining BUILD rows (#1-6, #8, #11-18) are shadow arms and measurement plans whose completion this revision could not verify from the box in the time available; **they are UNKNOWN, not done, and saying so is the point of this table.**

★ **One row was REVERSED, deliberately.** Last week's NOT-AN-ACTION #1 read "do NOT promote *abs_veto_55s* to a live two-sided slot on Monday". This week it is **MONDAY #1**. The reason is not a change of mind about the evidence: it is that the candidate collected another week of forward evidence (+\$948.50 on 45 trades, un-retuned) rather than another re-reading of the same weeks. That is the only thing that should ever move a row from NOT-AN-ACTION to MONDAY, and it is written down here so the reversal is visible rather than silent.

△ **Honest limit on this table.** It grades WHETHER a play shipped, not whether it WORKED. Four of the six that shipped are plumbing with no P&L attached; the one with money on it (the weekend-flatten timer) is graded by the fact that the desk went into this weekend flat, which is in the first line of this report.

§ The evidence behind those plays

Everything from here down is the working. Two halves, one document. It opens with **Part 0**, which answers the only question that survives past Friday — are we getting better? — day by day, green or red, straight off the books. **Then the warm half**: what the live desk actually did (Part 1, with **the run charts** inside it — every fill of the week marked on its own session's price path, which is the fastest way to see what the prose is describing). Then **Rehabilitation** (Part 1.5), where every red thing is walked through the full treatment instead of benched at face value, and **the day rider** (Part 1.6), the second book on the same IB account. Then the shadow board and the promotion battery (Part 2), the mid-week leads re-run adversarially (Part 2.5), and the **Router Operation Review** (Part 2.6). **Then the cold, tape-first half**: we ignore everything the desk did and ask, from the raw tape and order book, **where was the money and did we show up?** — on MNQ and on gold (M1) — then test whether the gates we already own could have caught the runs we missed (M2), and finally build gates from scratch and backtest them to death, the ones that failed included, that being the point (M3). Plain English, honest money, every claim a table — and every reprice a ceiling.

16 of 16 fragments are finished work and sit in this document in full. 13 of them were written **tonight (Part 0, Part 1, Part 1.5, Part 1.6, Part 2, Part 2.6, Movement 1, Movement 2, Movement 3, Part 3 and Part 4)** and describe the week 2026-08-30 22:00 → 2026-09-04 20:59 UTC; the rest describe the week to Friday 2026-08-28. That seam is drawn at the top of this report and it is the single most important thing to know before comparing any two numbers in it. **Nothing on the running order predates the 08-28 cycle**, so no section carries the red “this is last cycle's text” banner this week. ★ Note what that does not mean, and this is the ONE description of Part 1.5 that the front matter, this paragraph and the section itself now all use: Part 1.5 (Rehabilitation) is a **fresh**

2026-08-30 wrapper (§0a-§0c) around last cycle's standing dossiers.

Its mtime is inside the cycle so the build does not flag it, and it carries its OWN dated banner saying the material below its first rule describes the week to 2026-08-21. Read §0a-§0c as this cycle's and the treatments as standing material. **Nothing is missing** — every section on the running order rendered from a real fragment, so there are no red placeholder boxes to hunt for.

The rule behind all of that: **a gap is never a null result**. “We looked and found nothing” and “nobody looked” are opposite instructions for next week, so nothing is silently omitted and nothing stale is silently stitched.

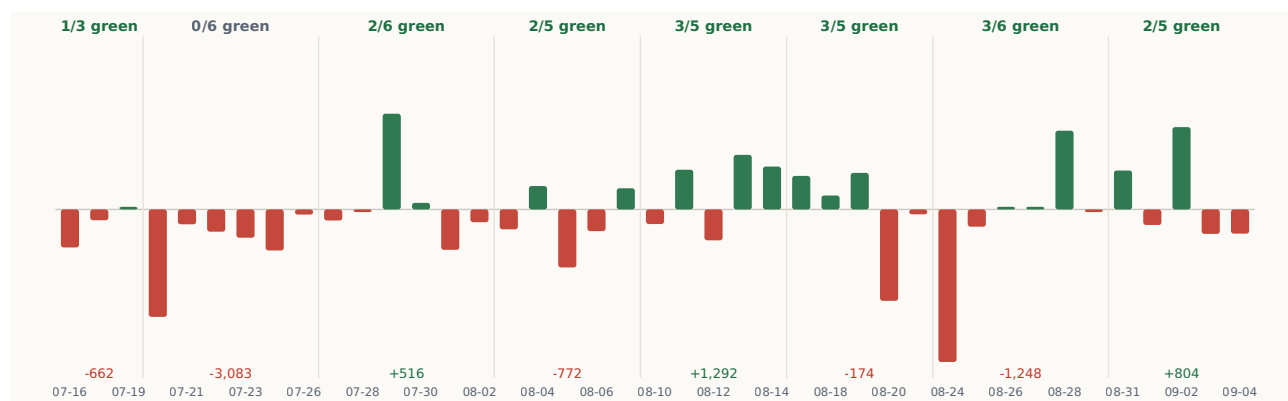
WHAT IS IN HERE

1. **Front** — [the honest headline, the scorecard, WHAT TO ACTUALLY DO](#)
 2. **Part 0** — [Are we getting better? — every day the desk has traded, green or red](#)
 3. **Part 1** — [The live desk — what the paper tournament and the day rider actually did](#)
 4. **Part 1.5** — [Live-desk REHABILITATION — never bench a gate at face value](#)
 5. **Part 1.6** — [THE DAY RIDER — the second desk, its week, and the four lots it left open](#)
 6. **Part 2** — [The shadow desk, the promotion battery, and the ruler that reordered the board](#)
 7. **Part 2.5** — [Mid-week musings — every lead from the desk chat, tested hard](#)
 8. **Part 2.6** — [ROUTER OPERATION REVIEW — the desk's #1 lever, on the stand](#)
 9. **Movement 1** — [The census — the biggest runs this week, and did we show up](#)
 10. **Movement 2** — [The idle-gate lab — could the gates we own have caught them? \(with a base rate\)](#)
 11. **Movement 3** — [The greenfield lab — detection is solved, selection is not, and the size threshold](#)
 12. **Part 3** — [REVISION 2 \(2026-08-29\) — the eight questions that draft left open, answered from data](#)
 13. **Part 4** — [REVISION 2 \(2026-09-04\) — the ten open questions, answered from data](#)
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★ **THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.**
part0_progress.html was rebuilt **2026-09-04 22:34 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

P0 Are we getting better? — every day the desk has traded

One bar per trading day, green or red, total desk. The question every other section answers is what happened this week. This one answers **are we getting better**, which is the only question that survives past Friday.



The shape of it. Week 30 was the floor: **0 green days out of 6** and **-3,083.00** — the stretch the operator remembers as “red for many many weeks”. The most recent week ran **2 green out of 5** at **+804.50**. Of 8 weeks on record, **3 closed net positive** — **W31, W33, W36**, 1 of them inside the last three weeks.

WEEK	DAYS	TRADED	GREEN	GROSS	FEES	NET
W29	2026-07-16 – 2026-07-19	3	1/3	-586.50	-75.00	-661.50
W30	2026-07-20 – 2026-07-26	6	0/6	-2,732.00	-351.00	-3,083.00
W31	2026-07-27 – 2026-08-02	6	2/6	+823.50	-307.50	+516.00
W32	2026-08-03 – 2026-08-07	5	2/5	-607.50	-165.00	-772.50
W33		5	3/5	+1,469.50	-177.00	+1,292.50

WEEK	DAYS	TRADED	GREEN	GROSS	FEEs	NET
	2026-08-10 – 2026-08-14					
W34	2026-08-17 – 2026-08-21	5	3/5	-100.50	-73.50	-174.00
W35	2026-08-24 – 2026-08-30	6	3/6	-1,158.44	-90.00	-1,248.44
W36	2026-08-31 – 2026-09-04	5	2/5	+936.50	-132.00	+804.50

NET is the column to read — it is what the account moved by, and it is the convention every other section uses. NET is `sum(pnl_usd)` straight off the book, because `pnl_usd` is recorded already net of fees; GROSS is reconstructed by adding `fees_usd` back at \$1.50 per round trip per lot. Do not subtract the fee from `pnl_usd` — that is the mistake the previous revision of this page made, and it cost \$1,371.00 of imaginary loss.

What changed underneath. The Claude-run router went live 2026-07-30, mid-W31 — the first green week is the one it landed in. Its measured effect is on the tail and the frequency of red days, not their depth: the worst day before it was **-1,513.00**, the worst since is **-2,149.44**, and green days went from **2 of 12** to **14 of 29**. The day-rider joined on 08-06 as a second, slower engine, and it is no longer unbeaten: **+613.50** across 35 legs in the most recent week (2026-08-31 → 2026-09-04), whose worst single session is **-381.50** on Tuesday 09-01. Strip that one session out and the rider's other 4 days come to **+995.00**. ★ **REV2 — this paragraph used to splice two different weeks together.** It read “+613.50 across 35 legs in the most recent week, of which -2,149.44 landed on Monday 08-24 alone”, and then subtracted the one from the other. But 08-24 is in **W35** and the most recent week here is **W36**: the query behind “the worst session” had no week filter and was returning the worst rider day of all time. It is now scoped to the week it is describing, so the subtraction is arithmetic on one window. The -2,149.44 carry belongs to the W35 row and is described there, in the body of this report, not here. The rider is still the engine that produced the desk’s best days; it is no longer the one that cannot lose.

Read it honestly. The cumulative total across all 41 days is still **-3,326.44** — this is a desk that has stopped bleeding wholesale, not one that is paid yet. 8 weeks is a short sample, best day +1,349.00 and worst -2,149.44 are both single sessions, and PAPER fills flatter a real book. The trend is real; the destination is not reached.

★ **The 08-22 “fee correction” on this page has been WITHDRAWN — it was backwards, and it moved the headline by \$1,371.00.** `trades.pnl_usd` is

written already net of fees_usd: the schema says so, the canonical P&L helper says so, and it is checked rather than assumed — re-run at render time over every row in the book, pnl_usd equals (exit–entry)×direction×lots×multiplier minus fees_usd, exactly, **843 times out of 843**. Rev2 changed this page to $\text{sum}(\text{pnl_usd} - \text{fees_usd})$, which charges the \$1.50 round trip a second time. **The all-time book is -3,326.44, not the -4,697.44 the previous revision printed**, and Rev2's claim that the change “turned two week totals green that are not” was itself wrong: W31 and W33 are positive on either arithmetic. Nothing about the SHAPE changes — the ordering of the weeks, the worst day, the best day are all identical — only the depth, and it was overstated. One green day does move, and it is worth naming: **2026-07-30, the router's first day**, made +93.50 on 82 trades and paid \$123.00 in fees, so the double-counted version turned the router's own go-live red. It was green. **If you see this desk quoted anywhere at -4,697.44, that is the double-counted number.** The Gross / Fees / NET table above now reconciles by construction: gross is $\text{pnl_usd} + \text{fees_usd}$, so gross – fees = net, every row.

Where this page stops, and the bug that used to live here. Every bar is a session closed on or before **2026-09-04**, which is where this report's books close, and the cut-off is now derived from the report week rather than typed into this file. ★ **It was typed, and it was a week stale.** The 2026-08-28 cycle ran this generator at 22:14Z with AS_OF still reading 2026-08-21, so Part 0 succeeded and answered the PREVIOUS week's question inside a report headed “week ending Fri 2026-08-28”: no W35 row at all, a “most recent week” paragraph describing W34, and a cumulative total that excluded the –\$2,149.44 Monday. Nothing errored, because a hard-coded date cannot notice that the calendar moved. That is the same failure mode three other sections of this report document in the desk's live instruments — found here by a proofread, not by a monitor. The last bar on this chart is 2026-09-04.

★ **THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.**

part1_live.html was rebuilt **2026-09-04 22:42 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

P1 THE LIVE DESK — Monday 24 to Friday 28 August 2026

Garrath — the six-gate tournament took **three signals in the whole week** and won all three. Six lots, **+\$330.00** net of fees, every single one of them opened inside a fifty-one minute window on Friday afternoon. Monday, Tuesday, Wednesday and Thursday it did not trade at all, and nor had it since **19 August** — a nine-day drought. So the P&L is not the story and I am not going to dress it up as one. The story is that the desk was switched on for **3.46%** of the available gate-time (24h 30m of 708 gate-hours — \$3's table, and the single figure this report now uses everywhere), the money all came from the other book, and the single largest item on the week is a **-\$2,149.44 position that nobody chose to hold**. What follows is what I could prove, what I could only bound, and one number this desk has been quoting that does not survive its own robustness rule.

★★ IF YOU READ NOTHING ELSE — FIVE THINGS, IN ORDER

1. What survived, cleanly. The router made exactly one deliberate momentum arm all week — Friday 14:45:04Z, grind_long and abs_veto_long together. The tournament read the new switch file at **14:45:04.329Z** and had both buy orders on the wire at **14:45:04.335Z and .343Z — six and fourteen milliseconds later**. Four lots, four winners, **+\$181.50**. That is not a lucky fill; that is a decision arriving and a gate acting on it inside the same second.

2. The bench was right on the week's ugliest tape, and I can price it. Thursday's US open whipsawed the other book for $-\$1,150$ in eight minutes. All six gates were off. Replaying the four trades they wanted on the tick tape, at five different stop widths, the bench **saved between \$21.80 and \$364.20 — negative sign at every width**, $\$292.56$ at the desk's live $1.0 \times \text{ATR}$ stop. Sign-stable, so it is a verdict, not a coin flip.

3. The "\$205-\$705 lateness cost" does not survive the stop sweep. The desk's own router ledger says Friday's arm was 20 minutes late and that the lateness cost $3.3 \times$ what the arm made. I re-ran it as a like-for-like one-slot simulation — armed early versus armed when it actually armed, same exit rule both sides — and the delta is **$-\$87.60$ at the live stop width and $+\$82$ to $+\$104$ at the other four. The sign flips. By this desk's own rule that is NO VERDICT**, and the big number was an upper bound built from counting one trend four times.

4. The Claim button is now the desk's main exit, and on the tournament it is a dead heat. 23 of 32 legs closed on MANUAL_CLAIM this week — 71.9%, against 9.8% over the previous three weeks. Re-priced on the venue's own 24.25-point entry ATR, the three claimed tournament lots were worth **$+\$141.00$ against the $+\$136.50$ you booked — a delta of $+\$4.50$** . The two big legs cancel: you left $-\$78.25$ behind on grind_long_A, whose 2.5R target printed eight minutes after the claim, and you **SAVED $+\$72.00$ on grind_long_B**, which its own exit stack would have stopped out at 15:40:08Z. (★ REV2 — $\$8$ used to say $-\$75.14$ here, on a modelled 23.5-point ATR and two legs instead of three. It is withdrawn.) **On the DAY RIDER the same button is worth $+\$5,426.50$ this week** (Part 1.6 §4) — the dead heat is a statement about the tournament, not about your hands.

5. Nothing stopped out. Not once — in THIS week. Zero STOP exits, zero STOP_UNFILLED, on either book, across 08-24→08-28. ★ REV2 — **and the week after is the more interesting one, so read them together.** Between 08-31 and 09-04 the tournament took **4 STOP exits for $-\$209.00$** (grind_long A and B on 09-03 13:56, abs_veto_long A and B on 09-03 15:40). That does not weaken the fix — it is the first real test of it in five weeks. **All four filled at the stop.** STOP_UNFILLED is now clean for **30 trading days and 0 of 4 stops missed**, where before it was clean but largely untested: all 26 STOP_UNFILLED this desk has ever booked, worth $-\$1,377.50$, are inside 21-28 July. HOLD #3's caveat — “thin and largely

untested rather than proven” — is now partly answered, by evidence that arrived after this section was written. See Part 4 §3.

1 • The money, both books, computed not guessed

Window is `closed_at ≥ 2026-08-24 and < 2026-08-29`, rows with `data_quality IS NULL` — the same window the front page uses, so the two pages agree. `pnl_usd` is **net**: I checked the invariant $(\text{exit} - \text{entry}) \times \text{dir} \times \text{qty} \times \$2 - \text{fees}$ against every one of the 32 rows and it holds **32 of 32**. Total fees \$87.00 on 58 lots = **exactly \$1.50 a lot**, so the fee constant is right in the book as well as in the harness.

DAY	TOURNAMENT	DAY RIDER	DESK NET	LEGS	WHAT HAPPENED
Mon 24 Aug	—	−\$2,149.44	−\$2,149.44	1	Friday's un-flattened 4-lot long closed by hand at 11:59Z
Tue 25 Aug	—	−\$243.00	−\$243.00	2	Tournament stood down; rider one entry, scratched
Wed 26 Aug	—	+\$19.00	+\$19.00	4	Stand-down LIFTED 19:59:32Z; rider four hand-claims, flat
Thu 27 Aug	\$0.00	+\$16.50	+\$16.50	12	The whipsaw. Gates all benched; rider −\$1,150 then +\$871
Fri 28 Aug	+\$330.00	+\$780.00	+\$1,110.00	13	The one day the tournament traded — 3 signals, 3 winners
WEEK	+\$330.00	−\$1,576.94	−\$1,246.94	32	2 green days of 5
... excluding the carry-in	+\$330.00	+\$572.50	+\$902.50	31	The week the desk actually traded was up \$902.50

The carry-in is not a trading result and it must not be read as one.

Trade id 763 opened **Friday 21 August at 13:13Z**, four MNQ long at 29,491.93, and was still open when the market reopened on Sunday night. It closed by hand on Monday at 11:59Z, 29,224.00 — 267.93 points against, − **\$2,149.44**. The strategy did not decide to hold it over a 49-hour halt; the flatten did not run. Strip that one row and the desk made **+\$902.50** on the

week. Both numbers are true and you need both: the second one tells you whether the trading is working, the first one tells you what the plumbing costs when it does not.

GOOD **The same flatten ran clean this Friday.** Both shots fired, 20:53:03Z and 20:57:04Z, each logging eod_flatten: MNQ pre=0 post=0. And the reconciler this morning reads the venue field, not a question mark: venue +0 = tournament +0 + rider +0 at 07:20:02Z. **The desk is genuinely flat into the weekend.** ⚠ But it flattened an empty book — the rider had closed its last lot at 15:21Z — so the mechanism that cost \$2,149.44 last Friday is **still untested under load**. It has not yet been asked to do the thing it failed at.

2 · The three signals — per-gate cards

Three signals, six legs (every gate runs an A and a B lot — same signal, same stop, different exit; they are a scale-out, not an experiment, so "6 of 6 winners" is really **3 of 3**).

GATE	SIGNALS	LEGS	NET	\$/LEG	WIN%	EXIT MIX	BEST / WORST LEG
capitulation_long	1	2	+\$148.50	+\$74.25	100%	TARGET, CHANDELIER	+\$79.00 / +\$69.50
abs_veto_long	1	2	+\$118.00	+\$59.00	100%	TARGET, MANUAL_CLAIM	+\$73.00 / +\$45.00
grind_long	1	2	+\$63.50	+\$31.75	100%	MANUAL_CLAIM x2	+\$41.50 / +\$22.00
abs_veto_short	0	0	—	—	—	—	armed 48 min, no fill
exhaustion_short	0	0	—	—	—	—	armed 1m43s all week
rgv_short	0	0	—	—	—	—	held off all week
TOURNAMENT	3	6	+\$330.00	+\$55.00	100%	3 auto, 3 by hand	\$110.00 per signal

⚠ **THIS TABLE IS NOT EVIDENCE. HERE IS EXACTLY WHY.**

You told me on 1 August that win rate is never a kill criterion and that everything gets judged on expectancy and robustness. Applying that honestly to this table kills the table, not the gates. **Three signals on one day cannot pass a robustness test because none of the tests can even be run.**

- **Leave-one-day-out:** there is one day. Remove it and there is nothing. Undefined.
- **Strip-the-best:** drop capitulation_long and you have +\$181.50 on two signals; drop two and you have +\$63.50 on one. It stays positive all the way down, which sounds good and means nothing at $n=1$.
- **Placebo / shuffle:** a shuffle test on three observations has no power whatsoever. Not run — deliberately.
- **Out-of-sample:** there is no in-sample to be out of.

Verdict: NOT JUDGED. +\$330 is neither a vindication nor a fluke; it is a sample too small to be either. For scale, the tournament's all-time book is **747 legs, -\$3,905.00, 40.0% of legs green** — this week recovered 8.5% of that hole. I would rather say that plainly now than let "100% win rate" wander into a decision.

3 · Why it only traded once — the week in switch positions

The tournament did not fail to find trades. **It was switched off.** Two separate mechanisms did that, and they are worth keeping apart in your head because only one of them is a judgement.

(a) The 20 August operator stand-down ran until Wednesday evening.

Every router tick from Monday 00:00Z to Wednesday 19:55Z says the same thing: "tournament STOOD DOWN — all six gates OFF and HELD, nothing armed to bench, any 'on' is dropped before it applies." And `reactivate_gates` confirms it from the other side — at Paris midnight on Monday and Tuesday it logged HELD (not auto-armed): `abs_veto_long`,

abs_veto_short, capitulation_long, exhaustion_short, grind_long, rgv_short. All six. **The stand-down lifted at 19:59:32Z on Wednesday** and the router picked it up on the very next tick. So Monday and Tuesday's zero is by construction, and you cannot read anything about the gates into it.

(b) From Wednesday night, the roster came back but momentum stayed benched-by-default. reactivate_gates re-armed three gates each night (abs_veto_short, capitulation_long, exhaustion_short) and deliberately start-benched the two momentum gates under the MONDAY #2 rule — "router arms on evidence". The router then benched two of the three within about 35 seconds. So the week's armed time looks like this.

★ **REV3 — the table below is now COMPUTED, and it says where the clock stops.** The first draft printed three different armed times for capitulation_long (22h 31m here, 20h 31m on the card, 17h 45m in this section's own prose) and two for the desk, and none of them said what the cut-off was. That is the whole of the disagreement: the gate went on at Fri 22:00:00Z and never came off, so the number depends entirely on when you stop counting. **Everything below is cut at the week's own boundary, Fri 2026-08-28 22:00:00Z**, which is the boundary the money tables use, and it is replayed forward from Monday 00:00Z (all six off) over every logged switch event by scripts/rev3_armed_time.py. Two further corrections fall out of doing it properly: **Monday 00:00Z to Friday 22:00Z is 118 hours, not 120**, so the denominator is **708 gate-hours** and not 720; and the replay had to be taught about a **third writer** — see the box below the table.

GATE	ARMED TO FRI 22:00Z	% OF THE 118H WEEK	WINDOWS (ALL TIMES UTC)	FIRED?
capitulation_long	20h 31m	17.39%	Wed 22:00→Thu 00:45 · Thu 22:00→Fri 15:45 · (Fri 22:00→still on — outside the cut)	yes, once
abs_veto_long	2h 15m	1.91%	Wed 22:45→23:15 · Thu 08:30→09:05 · Thu 20:00→20:20 · Fri 14:45→15:35	yes, in the 4th
grind_long	0h 50m	0.72%	Fri 14:45→15:35 only	yes, at once
abs_veto_short	0h 52m	0.73%	Mon 13:46→13:50 · 2x ~35s at 22:00Z · Fri 16:00→16:15 · Fri 16:24→16:55	no
exhaustion_short	0h 01m	0.02%		no

GATE	ARMED TO FRI 22:00Z	% OF THE 118H WEEK	WINDOWS (ALL TIMES UTC)	FIRED?
			3x ~35s at 22:00Z, nothing else	
rgv_short	0h 00m	0.0%	none — held off by the reactivate HOLD	no
All six	24h 30m	3.46%	of 708 possible gate- hours (118h × 6). At render time on Saturday it was 28h 23m / 3.88%	3 signals

★★ REV3 — THE STAND-DOWN WAS BREACHED, BY A PROGRAM WHOSE UNIT FILE SAYS IT CAN'T WRITE SWITCHES

Read the top of that `abs_veto_short` row again: **Mon 13:46→13:50**. This section says, two paragraphs above, that “Monday and Tuesday's zero is by construction” because the 20 August operator stand-down held all six gates off. It did not. At **2026-08-24 13:46:00Z**, in the middle of a -418-point downside break, `open_hour_watch.py` logged
 TRIO PASS ... ARMED

`abs_veto_short` and **armed the gate**. The router benched it again **4m 22s later**, at 13:50:22Z. The watcher then went straight back to alerting “Arm `abs_veto_short`” every minute for the rest of the hour.

No money changed hands — the gate produced no signal in those four minutes — so this costs nothing and is worth every word anyway, for three reasons. One: `open-hour-watch.service`'s own unit file says the watcher is **ALERT-ONLY and must NEVER write `gate_switches.env`**. It writes it. A unit file's stated contract is not a constraint on its code. Two: the stand-down is a **router-local variable**. It stops the ROUTER writing on; it is not a desk state, and the second writer has no awareness of it. Three: a replay built from the router's log alone loses this event entirely, which is exactly how the first draft came to describe Monday as a clean zero. The armed-time script now reads all three writers — router, `reactivate_gates`, and the watcher — and it is still only as good as what those three choose to log. **Nothing on this box records the switch VALUE over time; every instrument records WRITES**. That is a build item, not a finding, and it sits on the Saturday card.

Two things fall straight out of that table and both matter.

The one meaningfully-armed gate was armed by a cron, not by a decision. capitulation_long got 17.4% of the week because it is the only gate reactivate_gates re-arms that the router chooses not to bench — it is a reversion gate with no start-benched flag, and the router said so at every tick ("capitulation_long stays on through the quiet"). Holding it on was a decision, repeated hundreds of times. Arming it was not. It was awake at 14:00Z on Friday because of a rota, and it made \$148.50. Being right by rota is worth having and it is not the same as being right.

Three of the six gates were armed for under an hour each, and one for 103 seconds. exhaustion_short's entire week of permission is three roughly-35-second slivers where reactivate_gates writes on at 22:00:00Z and the router writes off at 22:00:32-38Z. That is two programs disagreeing, not a strategy being tested. It cannot be graded this week and I am not going to pretend it was benched on judgement.

⚠ **Live now:** capitulation_long=on as I write, straight through the ~49-hour CME halt, and the router has no weekend calendar — its clock, session label and reopen countdown are all wrong across the break. It is benched-by-default everywhere else; this is the one gate that is not. Worth an eye on Sunday's reopen.

4 · ★ FLAGSHIP — Friday 28 August, 14:00-15:00Z. Fifty-one minutes that were the whole week.

Friday was the week's biggest tape: **374.8 points of range** in the US session against 212-296 on the other four days, and a 1-minute ATR of **19.50** against 12.80-15.83. That is where the tournament woke up, and the sequence is tight enough to walk through second by second.

TIME (UTC)	WHAT HAPPENED	PRICE	MONEY
13:40:05	grind_long starts wanting in — ATR 26.11, above its 22.0 floor for the first time on the day. It is benched. It keeps asking for 65 minutes.		29,653.00

TIME (UTC)	WHAT HAPPENED	PRICE	MONEY
14:00:10	capitulation_long — the only gate awake — buys the flush. ext_atr -0.38, i.e. price already under VWAP, which is what it is built for.	29,638.75 / .25	—
14:00:21 / :24	Both lots out inside 14 seconds — A on TARGET, B on CHANDELIER.	29,674.25 / 29,678.50	+\$148.50
14:44:05	Router tick reads a break: "new session high 29730 (prev 29716, +14, ATR 23)".	29,730	—
14:45:04.329	Tournament reads the rewritten switch file. grind_long and abs_veto_long are live.	—	—
14:45:04.335 / .343	Both grind_long buy orders on the wire — 6ms and 14ms after the switch. Filled .459 / .503.	29,738.25	—
14:46:02	abs_veto_long fills 58 seconds later — it physically cannot be faster, its 55-second absorption confirm has to watch 55 seconds of tape first.	29,727.25 / 29,730.00	—
14:47-14:51	All four legs out — one on its own TARGET, three claimed by hand.	29,750-29,767	+\$181.50
15:29	grind_long goes sub-floor — ATR 17.96 against its 22.0 floor. It could no longer fire even if armed.	29,778.75	—
15:35:49	Router benches both momentum gates on the ER slope (0.303→0.261→0.178→0.149→0.013), explicitly not on P&L — the armed book was 4-for-4 at the time.	—	\$0
15:01-15:21	Meanwhile the rider bought the high at 29,806.50 and claimed out 20 minutes later.	29,751.50	-\$446.00

The bit worth internalising is the 6 milliseconds. There has been a standing worry that by the time the router decides, writes a file, and a gate notices, the move is gone. On this evidence the plumbing is not the bottleneck: switch file to order on the wire is **single-digit milliseconds**, and switch to fill is **130 milliseconds**. Whatever is slow about this desk, it is not that.

And the bench 51 minutes later was made on the right instrument. It graded the ER slope, not the level, and said out loud that it was a regime bench and not a losses bench. It also happens to have been free:

grind_long had already dropped under its own ATR floor at 15:29, so from that moment the switch position made no difference to anything.

5 · Was the arm 20 minutes late? I re-ran it, and the answer is NO VERDICT

The router ledger's headline for that window is that the arm was **20m17s late** and that the lateness cost **\$205-\$705** against +\$181.50 banked — a bigger error term than anything on its scoreboard. That is a serious claim and it deserved a serious re-derivation, so I did one from scratch off the raw tape.

Method. I pulled every SUPPRESSED-OPEN line for grind_long_A between 13:00Z and the arm — 789 of them — and collapsed them into distinct impulses with a 30-second gap rule (that reproduces exactly the impulse times the ledger names, so we are looking at the same events). Entry ATR comes from signal_journal at the signal. Each impulse is then replayed on the gate's own live exit — scalp to 2.5R, stop $k \times \text{ATR}$ — on 5-second MNQ bars, at \$2.00 a point and \$1.50 the round trip, with the stop swept $0.5 \times$ to $1.5 \times$.

The crucial correction: one slot holds one position. Pricing each impulse independently and adding them up counts one trend four times over — it is the same mistake as calling a rolling window "3-of-3 confirmed". So I ran it as a sequential simulation: the slot takes the first opportunity it is allowed, is busy until that trade resolves, then takes the next. Then I ran the identical simulation starting at the moment the router actually armed, and differenced them.

STOP WIDTH	ARMED 13:40 (EARLY)	ARMED 14:45 (ACTUAL)	EARLY – ACTUAL	READ
0.50 × ATR	+\$56.54 (4 trades)	–\$25.52	+\$82.06	early better
0.75 × ATR	+\$208.78 (4)	+\$118.60	+\$90.18	early better
1.00 × ATR (live)	+\$31.00 (3)	+\$118.60	–\$87.60	early WORSE
1.25 × ATR	+\$223.02 (3)	+\$118.60	+\$104.42	early better
1.50 × ATR	+\$209.97 (3)	+\$118.60	+\$91.37	early better

NO VERDICT

The sign flips across the sweep, and at the width the desk actually runs it flips the wrong way. Arming an hour earlier would have taken two stop-outs (−\$53.72 and −\$70.58) before the winner, and left the slot occupied at 14:45 so it would have missed the trade it actually took. This desk's own rule is that a replay is a verdict only if its sign survives the stop-width sweep. This one does not, so the honest answer to "was the arm late?" is **we do not know, and the \$601 figure is an upper bound built by counting one uptrend four times.**

What is not in doubt, and is the useful half: the router's direction read was right both times. Of the nine grind_long impulses in that window, the four before 14:22 all stopped at every width (chop — the bench was correctly saving money) and four of the five after 14:24 reached target at the live width (break — the bench was correctly costing money). Two regimes, opposite signs, and the router flipped between them. That is the thing to keep. What we cannot price is whether flipping ten minutes sooner would have been worth anything, because with one slot and a $1.0 \times \text{ATR}$ stop, it would not have been.

6 · What the benching actually cost — and why most of it was free

Across the week the five active gates logged **380 distinct blocked impulses** (the signal_journal, collapsed on a 2-minute gap). The tempting headline is "380 trades we were not allowed to take". It is wrong, and by a factor of three, because most of those gates could not have fired anyway — their own ATR floor was already refusing them, and suppressed_by='switch_off' only tells you which check ran first, not what would have happened.

GATE	BLOCKED IMPULSES	ATR FLOOR	ABOVE ITS FLOOR	FREE BY CONSTRUCTI ON	MEANING
grind_long	242	22.0	9	96.3%	the bench changed nothing on 233 of them
capitulation_long	39	10.0	9	76.9%	

GATE	BLOCKED IMPULSES	ATR FLOOR	ABOVE ITS FLOOR	FREE BY CONSTRUCTI ON	MEANING
					mostly refused by its own floor
abs_veto_short	51	none	51	0%	floorless — every one is a real forgone trade
abs_veto_long	42	none	42	0%	floorless — every one is real
rgv_short	6	none	6	0%	floorless
Total	380	—	117	69.2%	only 117 benches were ever worth a dollar

So **seven out of every ten "blocked" fires this week cost nothing to block**, and this is not a small correction — it is the difference between "the router is strangling 380 trades" and "the router made about 117 decisions that could have gone either way". It also lands almost exactly where the same measurement landed before (96.3% this week against 95.8% previously), so it is a structural property of grind_long, not a quirk of this tape: the gate spends nearly all of its life asking for trades it is not allowed to have for a completely different reason.

The practical consequence for you: when someone quotes a benching cost, ask whether the gate carries a floor. Only grind_long and capitulation_long do. For the other four, every blocked impulse is real money one way or the other.

★★ **REV3 — and here, at last, is what those 117 were worth.** This report has been quoting "the bench saved +\$810.44 over 127 signals" on the front page and on HOLD #1 all week, and there was no section behind it. The working was done inside a phase, printed, and never written to disk. Worse, the report published two impulse counts that disagreed — the 380 above, and a 187/127 — because they came from two different instruments and nobody named either. Both are real and they count different things:

INSTRUMENT	WHAT IT RECORDS	GATES	LIFETIME
signal_journal (shadow.db)	every fire the deciders logged, with the ATR and ER the gate itself read		5 durable on disk from 2026-08-04. This is what the 380 counts.
SUPPRESSED-OPEN log lines (tournament.py:306)	one line per TICK per benched leg — and the only place exhaustion_short appears at all		6 the systemd journal is a RING BUFFER. As I write, the oldest tournament entry it still holds is 2026-08-24 19:46Z — Monday daytime is already gone, and 14,184 of the 18,626 lines survive. The 127-

INSTRUMENT	WHAT IT RECORDS	GATES	LIFETIME
			signal population can no longer be rebuilt.

So the \$810 is not defended and not re-derived: **it is retired**. What replaces it is a replay of the population that IS durable, published here with its working, in `scripts/rev3_bench_replay.py` and `sections/rev3_bench_replay.json`. Every one of the 117 above-floor impulses is entered at the first tick at or after its own journal timestamp and run forward against **that gate's own live Lot A exit** out of data/`exit_overrides.json` — \$2.00 a point, \$1.50 the round trip, one lot, a 120-minute cap. And because the desk's own rule says a replay is only a verdict if its sign survives the stop-width sweep, every impulse is run at **five stop widths at once**. The sweep moves the stop; R stays pinned to the configured stop so the target does not wander with it. **Positive = the bench SAVED money**. (The same harness reproduces §7's four Thursday legs to the cent, which is how I know it is running the same ruler the rest of this section uses.)

GATE	SIGNALS	0.5×	0.75×	1.0× (LIVE)	1.25×	1.5×
grind_long	9	+\$116.97	+\$247.11	+\$377.24	+\$248.33	+\$92.23
abs_veto_long	42	+\$183.83	+\$79.10	+\$100.44	−\$114.62	−\$62.82
abs_veto_short	51	+\$179.28	+\$98.72	−\$26.05	+\$208.47	+\$207.66
rgv_short	6	+\$55.06	+\$111.27	−\$69.07	−\$161.86	−\$140.70
capitulation_long	9	+\$89.45	−\$45.86	−\$70.21	−\$107.01	−\$84.51
All five gates	117	+\$624.59	+\$490.34	+\$312.35	+\$73.29	+\$11.86

SIGN-STABLE — and that is the whole of the good news. The desk-wide saving is positive at all five widths, so by the desk's own robustness rule the bench earns a verdict rather than a NO VERDICT. But **read the slope**: +\$624.59 at half the live stop, **+\$312.35 at the stop we actually run**, and **+\$11.86 at 1.5×**. The saving does not survive a wider stop — it evaporates. That is the same finding §7 reaches on Thursday's four legs, now on 117 signals instead of four: **most of what “benching” saved is really the desk's stop being tight enough to guarantee those trades lost**. At the live width 65 of the 117 stop out and 52 reach target.

DAY	SIGNALS	BENCH SAVING	THE WEEK WITHOUT THIS DAY (LEAVE-ONE-DAY-OUT)
Mon 24 Aug	29	−\$267.25	+\$579.60

DAY	SIGNALS	BENCH SAVING	THE WEEK WITHOUT THIS DAY (LEAVE-ONE-DAY-OUT)
Tue 25 Aug	22	+\$303.96	+\$8.39
Wed 26 Aug	25	-\$15.53	+\$327.88
Thu 27 Aug	22	+\$295.73	+\$16.62
Fri 28 Aug	19	-\$4.56	+\$316.91

Two days of five carry the entire thing. Tuesday (+\$303.96) and Thursday (+\$295.73) are the only green days and between them they are **100% of the gross positive**; Monday alone gives back -\$267.25. Leave-one-day-out very nearly fails: drop Tuesday and the week's saving falls to **+\$8.39**, drop Thursday and it falls to **+\$16.62**. Strip the three single biggest contributing impulses out of 117 and +\$312.35 becomes **+\$66.31** — three signals are 79% of the result. A 10,000-draw bootstrap over the 117 puts the band at **-\$623.74 to +\$1,208.31**, with $P(\text{the true saving is } \geq 0) = 0.747$.

★★ THE PUBLISHED REGIME SPLIT DOES NOT SURVIVE RE-DERIVATION — AND IT WAS THE LOAD-BEARING HALF OF THE ARGUMENT

The front page and Part 3 §Q4 both said the useful decomposition was that the saving is entirely chop (66 signals, -\$771.78) while “in aligned trend, benching is FREE — 50 signals, -\$0.81 each”, and then built the case for arming aggressively on a confirmed break on top of it. On this population, split by the ER(30) and the 30-minute net move the gate itself recorded at fire time, that is **not what happens**.

REGIME AT THE FIRE	SIGNALS	BENCH SAVING	PER SIGNAL
CHOP — ER(30) below 0.30	78	+\$311.48	+\$3.99
ALIGNED TREND — ER(30) ≥ 0.30, gate pointing WITH the 30-min move	35	+\$117.91	+\$3.37
COUNTER-TREND — ER(30) ≥ 0.30, gate pointing AGAINST it	4	-\$117.04	-\$29.26

Chop and aligned trend are **the same number** — +\$3.99 a signal against +\$3.37. A label-shuffle placebo over 10,000 draws on that gap returns **P = 0.943**: shuffling the regime labels across the same 117 savings reproduces a gap this size or bigger 94 times in 100. **There is no regime split in the bench this week.** The one cell that does

separate is COUNTER-TREND at −\$29.26 a signal, and it is **four signals** — I am naming it, not banking it.

What that costs us: “benching in aligned trend is free, so arm hard on a confirmed break” was the strongest-sounding sentence in the bench case and it is now **REFUTED** by a named test (label-shuffle placebo, $P = 0.943$). It does not touch HOLD #1 — the bench still leans right — but nothing about arming should rest on it any more.

★ **What is genuinely NOT in this number, said plainly.** Three things. One: `exhaustion_short` is absent, because it has never written a `signal_journal` row in its life — the store has only ever held five gates. The retired \$810 credited it with +\$442.65, more than half the total, and that component is now **unreconstructible**, which is not the same as disproven. Two: both instruments sit UPSTREAM of the 55-second absorption veto, so for the two `abs_veto` gates — 93 of the 117 — this is the PRE-veto population, and the gate's own veto would have refused roughly half of them (measured, not assumed: 45 fires against 91 on the shadow board this week, 49.5%). Three: it is one chop week. **The honest sentence is this: the bench leaned right; it was worth about \$300 at the stop we run and about nothing at a stop 50% wider; two days of five carry all of it; and the dollar figure should not be quoted at all.**

7 · Thursday — the whippy day, and the clearest □ of the week

Thursday 27 August was the week's most violent tape by a mile once you look past the range. I measured "whip" as total path travelled divided by the day's range — how many times the market walked its own range before settling.

DAY	RTH RANGE	RTH NET	WHIP (PATH÷RANGE)	WHIP, FULL DAY	ATR 1M	CHARACTER
Mon 24	296.0	−135.8	9.90	14.45	15.01	grind lower

DAY	RTH RANGE	RTH NET	WHIP (PATH÷RAN GE)	WHIP, FULL DAY	ATR 1M	CHARACTE R
Tue 25	270.5	−25.0	9.49	17.26	13.37	pure chop, went nowhere
Wed 26	212.5	+123.5	11.89	19.37	12.80	quietest day, cleanest drift
Thu 27	280.0	+167.5	11.33	24.02	15.83	the whipsaw
Fri 28	374.8	−131.8	10.22	18.53	19.50	biggest range, biggest vol

What Thursday's open did to the other book, in eight minutes: long at 29,494.06 at 13:34, claimed out **37 seconds later** at 29,453.94 for **−\$327**; short at 29,436.63 at 13:35, claimed out six minutes later at 29,538.75 for **−\$823**; long again at 29,513.13 at 13:42, which then ran for **+\$871**. Three entries, **−\$1,150** then **+\$871**, net **−\$279** across the sequence — it sold the bottom of a 100-point air pocket and bought the top of the bounce, inside eight minutes.

The tournament was flat through all of it, because all six gates were benched. But the gates were not silent — they wanted in four separate times. So the question is not rhetorical, and it prices out cleanly. Same replay method as before: entry from the journal, each gate's own A-leg target and stop, swept 0.5× to 1.5×.

TIME	GATE	SIDE	ENTRY	ATR	K=0.5	K=0.75	K=1.0	K=1.25	K=1.5
13:32:05	abs_veto_s hort	SHORT	29,485.50	24.18	−\$25.68	−\$37.77	−\$49.86	−\$61.95	+\$71.04
13:45:05	grind_long	LONG	29,577.75	42.25	−\$43.75	−\$64.88	−\$86.00	−\$107.12	−\$128.25
13:58:06	abs_veto_s hort	SHORT	29,436.50	43.23	−\$44.73	−\$66.35	−\$87.96	−\$109.57	−\$131.19
14:06:05	grind_long	LONG	29,549.50	33.62	−\$35.12	−\$51.93	−\$68.74	−\$85.55	+\$166.60
Total — what staying benched SAVED					\$149.28	\$220.92	\$292.56	\$364.20	\$21.80

□ GOOD — SIGN-STABLE Every width says the same thing: the bench saved money. \$292.56 at the desk's live stop, \$21.80 at the loosest. That is a genuine verdict by the desk's own robustness rule, and it is the clearest single call of the week — on the day the other book lost \$1,150 in eight minutes, the tournament's four gates all wanted in and every one of them would have been stopped.

But read the slope, not just the sign, and give the credit to the right place. The saving collapses from \$364 to \$22 between 1.25× and 1.5×ATR, because at 1.5× two of the four legs survive their drawdown and reach

target instead. In plain terms: **most of what the bench "saved" is really the desk's stop being tight enough to guarantee those trades lost.** The router was right that the tape was untradeable as this desk trades it. It is not evidence that the signals were wrong about direction — two of the four were right about direction and were killed by the stop. That is a rehab question, not a routing one, and it belongs to Part 1.5.

8 · The Claim button became the desk's main exit — and on the tournament it is a DEAD HEAT (★ REV2: not the −\$75 this heading used to claim)

This is the biggest behavioural change of the week and nothing was tracking it.

★★ REV2 — THIS SECTION WAS PRICED ON A MODELLED ATR AND THE CARD WAS NOT. THE CARD IS RIGHT.

The table below and the −\$75.14 under it were computed at a modelled **23.5-point entry ATR**. The desk's own native stops say otherwise: stp-000003/4 rest at **29,714.00** against a 29,738.25 fill, which is an entry ATR of **24.25 points**. Re-priced on the venue's number (scripts/rev2_arm_own_exits.py, artifact sections/rev2_arm_own_exits.json) the answer changes sign and the population grows from two legs to three:

CLAIMED LEG	BOOKED	ITS OWN EXIT AT ATR 24.25	DELTA	WHAT ITS OWN EXIT ACTUALLY DID
grind_long_A	+\$41.50	+\$119.75	−\$78.25	TARGET @ 29,798.88 at 15:00:13Z — you left this one behind
grind_long_B	+\$22.00	−\$50.00	+\$72.00	STOP @ 29,714.00 at 15:40:08Z — this section called it “unpriceable”; it is priceable and claiming it SAVED \$72
abs_veto_long_B	+\$73.00	+\$71.25	+\$1.75	TARGET @ 29,766.38 — you beat it by a point

CLAIMED LEG	BOOKED	ITS OWN EXIT AT ATR 24.25	DELTA	WHAT ITS OWN EXIT ACTUALLY DID
All three claims	+\$136.50	+\$141.00	+\$4.50	a dead heat — and the two big legs cancel

So: −\$75.14 is WITHDRAWN and +\$4.50 is the figure. The whole tournament week on its own exits is +\$334.50 against the +\$330.00 booked. NOT-AN-ACTION #9 carries the same number and the same conclusion; this section was the one out of step. ★ **And scope it to the book it describes:** this is the TOURNAMENT's three claimed lots. On the DAY RIDER the same button is the desk's main exit and Part 1.6 §4 scores it at **+\$5,426.50** against holding, on 19 in-week claims. “Dead heat” is true of the tournament and false of the desk.

BOOK	MANUAL_CLAIM THIS WEEK	PREVIOUS 3 WEEKS	CHANGE
Tournament	3 of 6 (50.0%)	19 of 252 (7.5%)	7× the rate
Day rider	20 of 26 (76.9%)	7 of 14 (50.0%)	up half again
Whole desk	23 of 32 (71.9%)	26 of 266 (9.8%)	a step change, not drift

Why it matters: a hand-claim measures your timing, not the gate's edge. It truncates the exit stack that arming the gate was supposed to unlock, so the gate gets scored on a number it did not produce — and that cuts both ways, understating good gates and flattering bad ones. Here is every Friday leg replayed forward on its own configured exit.

LEG	EXIT TAKEN	ENTRY	ITS OWN TARGET	DID THE TAPE REACH IT?	OWN EXIT	BOOKED	LEFT BEHIND
grind_long_A	MANUAL_CLAIM 14:51:00	29,738.25	29,798.30 (2.5R)	YES — 14:59:15	+\$118.60	+\$41.50	−\$77.10
abs_veto_long_B	MANUAL_CLAIM 14:50:45	29,730.00	29,766.27 (1.5R)	yes — 14:50:40	+\$71.04	+\$73.00	+\$1.96
grind_long_B	MANUAL_CLAIM 14:50:10	29,738.25	wide chandelier	unpriceable	—	+\$22.00	claimed at +11.75pt into a +73.5pt hour
abs_veto_long_A	TARGET (clean)	29,727.25	29,751.43 (1.0R)	yes	+\$46.86	+\$45.00	−\$1.86
capitulation_long_A	TARGET (clean)	29,638.75	29,678.56 (1.5R)	yes — 14:00:15	+\$78.12	+\$69.50	−\$8.62
capitulation_long_B	CHANDELIER 14:00:24	29,638.25	trailing	n/a	—	+\$79.00	out in 14s; tape ran

LEG	EXIT TAKEN	ENTRY	ITS OWN TARGET	DID THE TAPE REACH IT?	OWN EXIT	BOOKED	LEFT BEHIND
							+163.75pt in the hour

SUPERSEDED — see the REV2 box above Measured on the **MODELLED 23.5-point ATR: −\$75.14 net across the two claimed legs this table could price.** On the venue's own 24.25-point ATR, and with grind_long_B priced rather than called unpriceable, the three claims come to **+\$4.50**. Read the box, not this sentence. The narrative below is still worth having — grind_long_A was claimed at +21.5 points and its own 2.5R target printed **eight minutes later, 38.5 points above the claim.** So the tournament's **booked +\$330.00 was worth +\$405.14 on the gates' own exits.** The other claim (abs_veto_long_B) beat its own target by a point — your timing was genuinely better on that one — which is exactly why this needs measuring rather than asserting in either direction.

The rule to take away, and it is cheap to apply: read exit_reason before you read pnl_usd. Any MANUAL_CLAIM row is scoring you, not the gate. At 4.4% of trades all-time that was a footnote; at 72% of the week it is the main exit mechanism on the desk, and it is concentrated in exactly the hand-watched windows where we do the grading.

△ Separately, capitulation_long_B is worth flagging to the rehab section: it exited on CHANDELIER **14 seconds after entry**, banking 40.25 points, and the tape then ran another 123 points inside the hour. That is a trailing exit firing on the entry spike itself. One instance, so it is a flag and not a finding.

9 · STOP_UNFILLED — the fix is five weeks clean, and ★ REV2 it has finally been tested

The naked-ride problem, where a stop's trigger never fired against the continuous contract and a losing position kept running, is **gone and has stayed gone.**

ISO WEEK	LEGS	STOP_UNFILLED	COST	STATUS
2026-W29	234	22	-\$1,182.50	the bad one
2026-W31	205	4	-\$195.00	tail end
2026-W32	112	0	\$0.00	clean
2026-W33	117	0	\$0.00	clean
2026-W34	44	0	\$0.00	clean
2026-W35 (this week)	32	0	\$0.00	clean — and zero STOPs of any kind, so untested
2026-W36 (★ REV2 — after this section was written)	47	0	\$0.00	clean AND tested — 4 STOP exits fired (— \$209.00) and all four FILLED at the stop

★★ REV2 — THE CAVEAT BELOW IS NOW PARTLY ANSWERED, BY A WEEK THIS SECTION COULD NOT SEE

The paragraph that follows says, correctly, that a week with no STOP exits does not test the stop contract. **08-31→09-04 does.** The tournament took four STOP exits — grind_long A/B at 09-03 13:56:18 and abs_veto_long A/B at 09-03 15:40:55, -\$209.00 between them — and every one of them filled at its stop. Zero STOP_UNFILLED. All 26 this desk has ever booked remain inside 21–28 July. So the record is now **30 trading days clean AND 0 of 4 stops missed**, which is a different and better claim than 30 days of silence. Full week in Part 4 §3.

Be careful how much credit you take, though. This week had **no STOP exits at all** — not one position on either book was closed by its stop. Every exit was a target, a trail, a chandelier or a hand-claim. So this week does not test the stop path; it just did not need it. The four-week run is real evidence, this week's contribution to it is thin, and with only 44 and 32 legs in the last two weeks the sample behind "clean" is shrinking fast. It is the right call to keep the fix and stop worrying about it; it would be the wrong call to say it has been stress-tested lately.

10 · The direction-router trial — how it actually behaved

The durable router ran **1,389 ticks** across the week and wrote to the switch file **17 times**. That is the headline number and it needs unpacking, because five of those writes are not decisions at all.

MEASURE	THIS WEEK	NOTE
Ticks logged	1,389	of 1,440 five-minute slots
Switch writes, total	17	1.2% of ticks produced any change
... of which are cleanup after another writer	5	3 × the 22:00Z reactivate collision, 2 × after the open-hour watcher
... the router's own judgement	12	7 benches, 5 arms
Arms that produced a fill	1 of 5 (20%)	and it produced +\$181.50 of the week's +\$330
Churn — gate-VALUE changes per clock hour (Mon 00:00Z → Fri 22:00Z, 118h, all three writers)	0.237 /hour	28 value changes: 14 arms, 14 benches. This is the desk's ONE churn metric — see the box below. Against 5/hour on the 7 August reference day.
Five-minute slots that produced no DECISION	88	7h 20m — and it decomposes exactly: 43 ABORTs (3h 35m) plus 45 CME maintenance skips (3h 45m, by design), with zero unexplained. Census: scripts/rev3_router_blackout.py.

★★ REV3 — ONE CHURN METRIC AND ONE BLACKOUT CENSUS, BECAUSE THE FIRST DRAFT PRINTED THREE OF EACH

Churn. Three numbers were in print for the same week — 17 switch writes at 0.31/hour here, 22 gate-value changes at 0.18/hour in Part 2.5 §2.4, and 18 changes over 8 days in Part 2.6 §1 — and MONDAY #2's kill criterion (“if switch changes per hour rise above ~1.0”) was keyed to one of them without saying which, so it was not executable. **The definition, once, for all three sections:** a CHANGE is a gate's state flipping VALUE, not a write (the router re-writes off on a gate that is already off, and the file's mtime moves without anything changing). The window is Mon 2026-08-24 00:00:00Z to Fri 22:00:00Z. The denominator is elapsed clock hours, 118 of them. Every writer counts — the router, reactivate_gates and the open-hour watcher. On that definition the week ran **28 value changes = 14 arms and 14 benches = 0.237 an hour**, and MONDAY #2's kill criterion now reads against that number.

The blackout. Also three accounts: 43 ticks / 3h 35m on the card, 47 ticks / 235 minutes in Part 2.6 §7, and 44 silent slots / 3h 40m in the row above. They were counting three different populations. The census reconciles them, and ★ REV2 fixes its arithmetic, which did not add up as printed ($1,390 + 88 = 1,478$, not $1,440$). The error was a unit slip: **1,390 is a count of trial-log TICKS and 88 is a count of SLOTS**, and 38 slots carry a retry tick as well as their first one. Counted the same way both times: of the **1,440** five-minute slots in the Paris week, **1,352 produced a decision** (from those 1,390 ticks); **88 did not**; $1,352 + 88 = 1,440$; of those 88, **43 were ABORTs** and **45 were the CME maintenance skip, which is by design; zero are unexplained**, and there is a headless-log line in every slot. The 43 break down as **31 LLM session-limit** (four windows: Wed 01:25–02:00Z, Wed 15:55–17:00Z, Thu 17:40–18:00Z, Fri 18:25–19:00Z), **8 API 529-overloaded** and **4 invocation failures** — the last twelve all on Monday morning between 04:58 and 07:23Z. **The card's “12 Monday gateway” is right about WHEN and wrong about WHAT:** none of the twelve is an IBKR gateway error, they are all on the LLM side. And Part 2.6's fifth window, 80 minutes on Saturday 08-29, is outside this week — there are 23 more aborted ticks in the weekend after the cut, which belong to next week's report.

□ **THE CHURN PROBLEM IS NOT PRESENT THIS WEEK** The thing that was flagged as the router's signature failure — arming and benching the same gates on 3-to-16-minute cycles, 20 changes in four hours with only four producing a trade — simply did not happen. **0.237 changes an hour against 5** — a twenty-one-fold reduction. (★ REV2: this paragraph used to say “0.31 an hour”, which is one of the three counts the box above retired. There is now one churn number on this desk, it is **0.237/hour = 28 gate-VALUE changes over 118 hours**, and MONDAY #2's kill criterion is keyed to it.) I will be honest about the caveat: the tournament was switched off for half the week and the tape was quiet for much of the rest, so some of this is opportunity rather than restraint. But the one four-hour window on Friday with real fills and real decisions ran seven gate-value changes against six fills — roughly one change per trade — which is the shape you want, and it is nothing like the old pattern.

The one arm/bench round trip that was churn: abs_veto_short armed at 16:00:39Z and benched at 16:15:22Z — a 14m43s round trip with zero intents and zero fills. Net-zero, but a process cost.

The router's own graded scoreboard moved like this. These come from router_badcall_ledger.md, which grades every switch decision retrospectively every four hours; I read the running totals at each end of the week and differenced them.

GRADE	MON 24 START	SAT 29 START	THIS WEEK
GOOD	45	59	+14
BAD	14	19	+5
CHURN	25	27	+2
NEUTRAL (zero-cost)	45	66	+21
Good : bad	3.2 : 1	3.1 : 1	2.8 : 1

Forty-two graded items this week, running slightly worse than its own lifetime ratio, and half of them zero-cost. That is a working audit loop, and it is worth saying that it is the only instrument on this desk that grades its own decisions against what the tape did next.

△ FIX **The router went silent for 3h 35m and nothing said so.** ★ REV2 — this paragraph used to read “3h 40m / forty-four slots”, which is the third of the three blackout accounts the box above retired. The census figure is **43 ABORTed slots = 3h 35m**, over and above the 45 that are skipped by design at 21:00–21:40Z. The biggest hole is **65 minutes on Wednesday, 15:50-17:00Z — the middle of the US session**; then 35 minutes on Friday 18:20–19:00Z, 35 on Wednesday overnight, 20 on Thursday afternoon, and a scattered 55 minutes across Monday's 04:50–07:20 gateway window. In every one of those holes the switch file simply stops changing, which looks identical to "the router considered it and decided nothing". **There is no alert on a missing tick.** That is a cheap thing to add and it is the difference between a quiet router and an absent one.

11 · Day-type split — and why I am not giving you one

The section is supposed to answer "did a gate's P&L depend on trend versus chop". **It cannot be answered this week and any table that appears to answer it would be fabricated.** The tournament traded on one day. There is no second cell to compare against. One day is not a split.

What I can tell you about the week's regime, from the tape rather than from trades:

- **There was no trend day.** All five sessions travelled between 9.5 and 11.9 times their own range in RTH before settling. The most directional close was Thursday at +167.5 points on 280 of range, and Thursday is also the whippiest full day of the five. Nothing this week looks like a trend day.
- **The one day with real volatility is the day the gates traded.** Friday carried 374.8 points of range and a 1-minute ATR of 19.50, against 212–296 and 12.80–15.83 on the other four.
- **The real split is time-of-day, and it is sharper than anything else I found.** Exactly 9 of grind_long's 242 blocked impulses all week cleared its 22.0 ATR floor — and **all nine of them, on four separate days, fell between 13:40Z and 14:37Z.** Outside that single hour the gate did not clear its own floor once in 120 hours. Whatever grind_long is, it is a US-open instrument and nothing else, and that is worth testing properly with more weeks behind it.
- **So the "split" that actually exists is vol, not direction.** The gates fired on the one day vol was high enough to let them, which is what the floors are for. Whether their edge depends on trend versus chop is untested and stays untested.

I would rather hand you a clean "cannot be answered" than a two-row table with $n=3$ in one cell and $n=0$ in the other.

12 · Disposition — every lead this section touched

LEAD	VERDICT	WHY — AND WHAT WOULD CHANGE IT
Router arms momentum on evidence (MONDAY #2 start-bench)	LIVE	One arm all week, filled in 6-14ms, 4 legs 4 winners, +\$181.50. Kept.
All-gates-benched through Thursday's whipsaw	LIVE — ☐ verdict	Saved \$21.80-\$364.20, sign-stable at all five stop widths. The only sign-stable dollar verdict of the week.
"The 14:45 arm was 20 minutes late, cost \$205-\$705"	NO VERDICT	Killed by the stop-width sweep — the sign flips (−\$87.60 at the live 1.0×ATR, +\$82 to +\$104 elsewhere) once you simulate one slot instead of adding independent impulses. Revive with a multi-slot desk, or with a week where the same shape recurs enough to test it out of sample.
grind_long as a gate	PARKED	All-time −\$390.50 over 100 legs; 96.3% of its intents this week were below its own ATR floor. But 2 of its 3 above-floor Friday impulses reached target. Revive when $n \geq 25$ above-floor fires have accumulated — it currently gets about 9 a week, so that is roughly three weeks of tape.
capitulation_long armed by rota	HOLD — not an action	Made \$148.50 in 14 seconds because it was the only gate awake. Working as designed. $\triangle$ It is on right now through the weekend halt and the router has no weekend calendar — check it at Sunday's reopen.
The Claim button as the desk's primary exit	MEASURED	72% of legs this week against 9.8% before. Measured on the venue's own 24.25pt ATR: +\$4.50 across the three claimed tournament lots (★ REV2 — the −\$75.14 previously printed here was the modelled-ATR two-leg figure and is withdrawn), and +\$5,426.50 on the day rider's 19 in-week claims. Grade exit_reason before pnl_usd, always.
STOP_UNFILLED fix	FIXED	Four consecutive clean weeks after −\$1,377.50 in W29-W30. Caveat: zero stops of any kind fired this week, so this week did not test it.
EOD flatten (the −\$2,149.44 carry)	UNTESTED	Both Friday shots ran clean — on an empty book. It has not yet been asked to flatten an actual position since it failed. Not fixed, just not exercised.
Router tick coverage	ACTION	3h 35m of ABORTed slots (43 of the 88 silent ones; the other 45 are the CME skip, by design), biggest a 65-minute hole mid-US-session. Silence is indistinguishable from a decision. Add a missing-tick alert.
exhaustion_short	UNPRICEABLE	Armed for 103 seconds all week, all of it the 22:00Z two-writer collision. No trades, no intents while armed, nothing

LEAD	VERDICT	WHY — AND WHAT WOULD CHANGE IT
		to grade. Revive when it gets a real arming window.
rgv_short	HOLD	Held off all week by the reactivate_gates HOLD (a standing operational decision from 9 August, not a judgement on the gate). 6 blocked impulses, unpriced here.
Day-type / regime split for the tournament	NOT ANSWERABLE	n=3 signals on one day. Needs at least two trading days with fills. Nothing to park — the stone is simply unturned.

The one stone still unturned in this section: whether the two Thursday signals that were directionally right and killed by a $1.0 \times \text{ATR}$ stop point at a stop-width problem rather than a signal problem. I can see the shape — the bench "saving" evaporates from \$364 to \$22 when you widen the stop — but four legs on one day cannot decide it. That question belongs to the rehab section with the full sample behind it.

INSTRUMENTS USED, AND WHAT I DID NOT TRUST

Trades and money: data/gazbot7.db, window

2026-08-24 ≤ closed_at < 2026-08-29, data_quality IS NULL. Net verified against the formula on 32/32 rows; fees \$87.00 on 58 lots = \$1.50 a lot exactly. **Tape:** the Parquet lake (gazbot7.lake), MNQ 5-second bars, not capture.db — that is a five-day rolling window and would have quietly returned part of the week. **Blocked fires:** shadow.db.signal_journal for entry ATR and prices, plus the raw SUPPRESSED-OPEN journal lines for impulse boundaries (~1 line/second per leg, so they are collapsed into contiguous runs before anything is priced — never counted). **Switch state:** reconstructed from the router trial log's changed: field plus the reactivate_gates systemd journal plus open_hour_armed.json, because the router log alone only records the writes the router itself made and would have missed both other writers. **Grades:** router_badcall_ledger.md running totals.

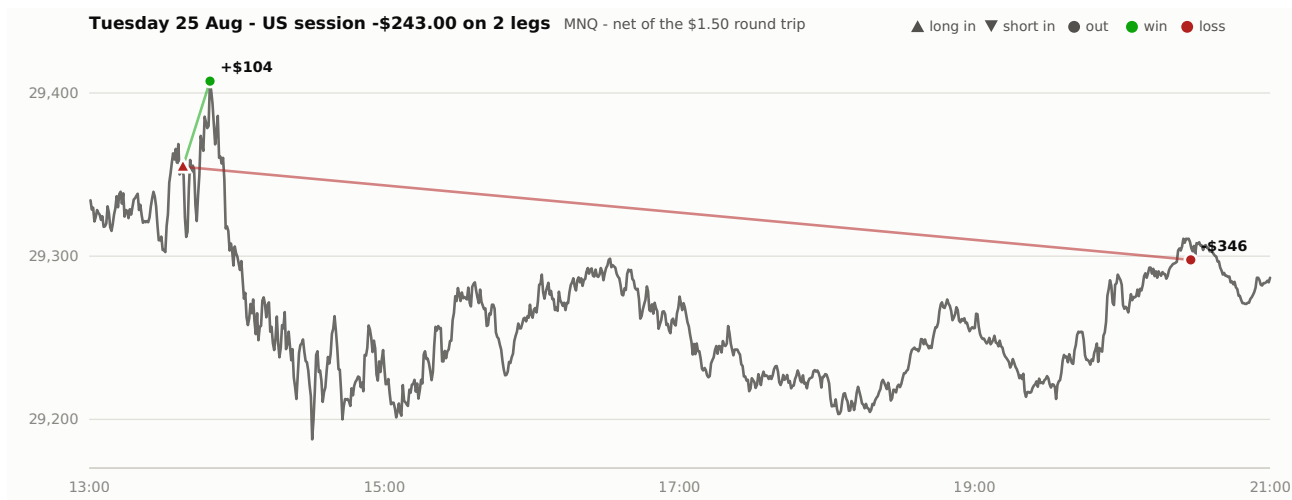
Not used: hour_watch.py reports the current hour and its disabled: list is a run-time snapshot, so it cannot reconstruct a past week — run live now it reads five gates disabled and capitulation_long armed, which is a useful confirmation of the switch state and nothing more. signals.block_reason is

NULL on all 12 rows this week, so it priced nothing. And suppressed_by='switch_off' was treated as a mechanism label, never as a counterfactual — 69.2% of the impulses it marks could not have fired anyway.

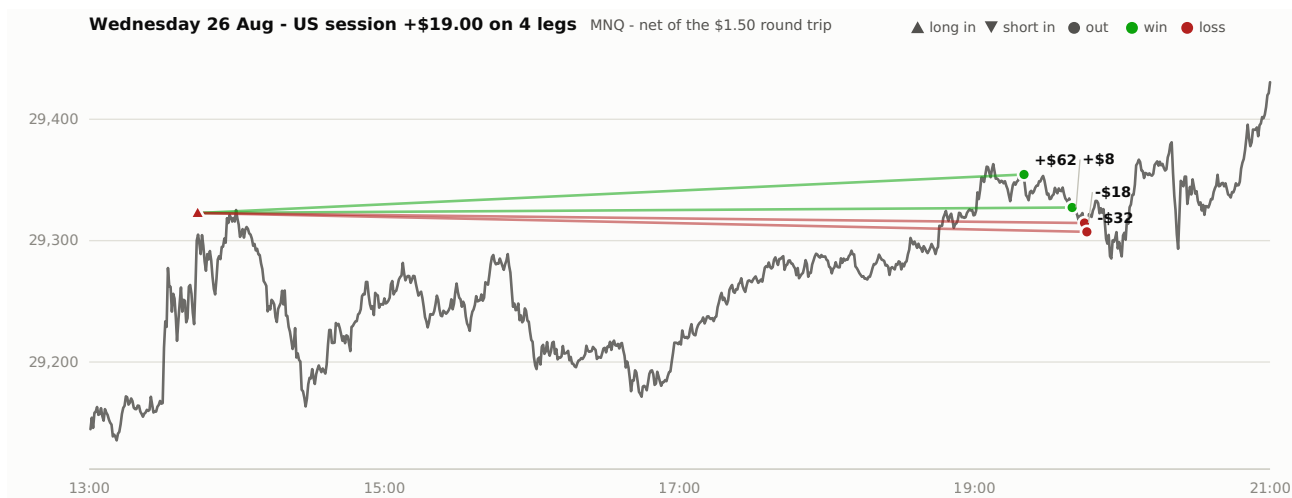
THE RUN CHARTS — every fill of the week, marked on the day's own price path



An empty chart is the finding: the desk took no signal and held no position between 13:00 and 21:00, and that includes the week's biggest run - 231 points down from 13:29. Monday's booked -\$2,149.44 does not appear here because it closed at 11:59, an hour before this window opens, on a rider long carried over from Friday 21 Aug.



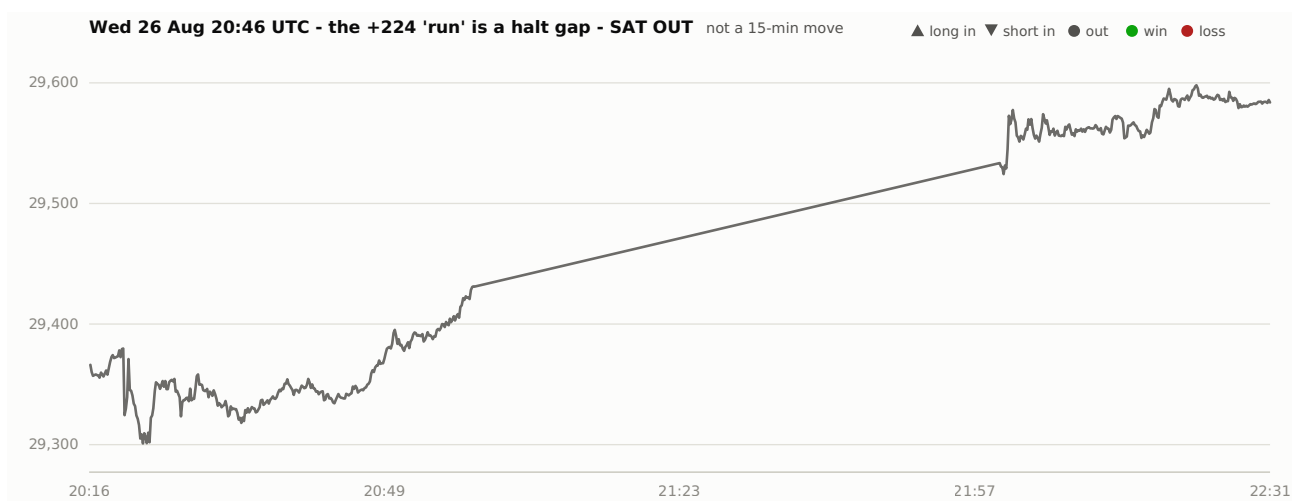
Two markers, one position: the rider bought 29,354.75 at 13:38, took the first \$100 off eleven minutes later, then held the remaining three lots for six and three-quarter hours into a lower tape and claimed them at 20:27 for - \$346.50.



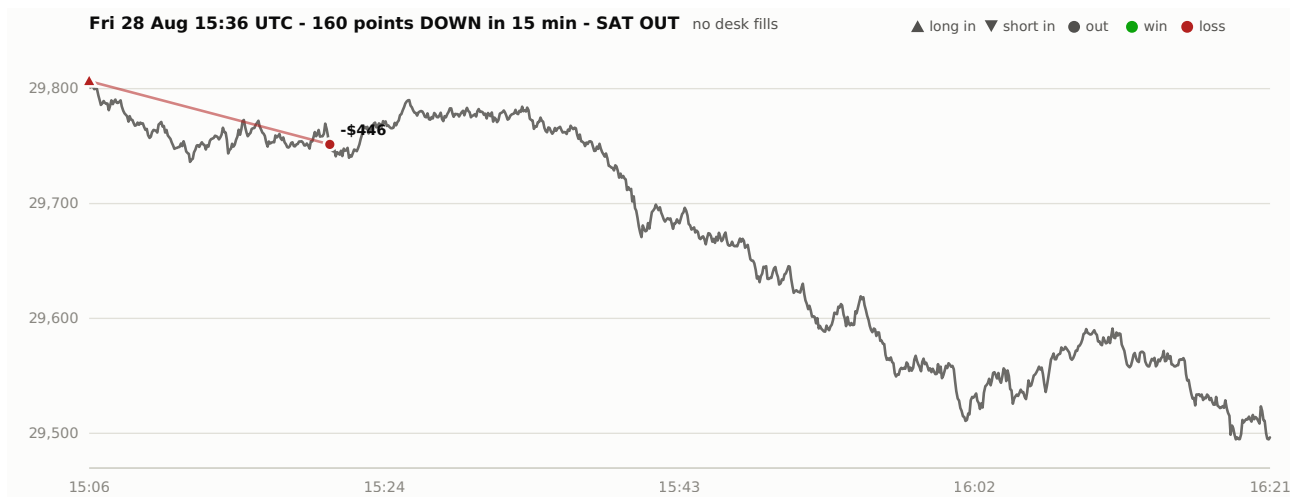
The flattest day of the week: one 13:44 long, scaled out in four claims between 19:20 and 19:45, net +\$19.00. Price ranged 180 points around the entry in between and none of it was taken.



The largest run of the week, worth \$462 on a single lot, and there is not one marker on it. The desk was flat before it, flat through it and flat after it.



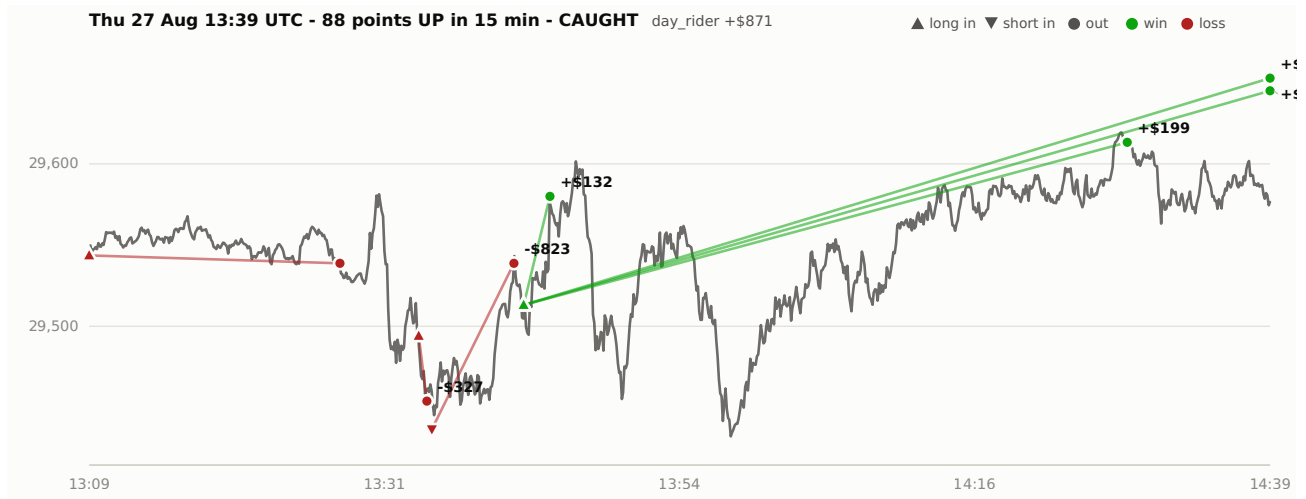
The census ranks this the week's second-biggest run. It is not a run at all: the straight diagonal in the middle is the 21:00-22:00 CME maintenance halt, so the '15-minute' window of 180 bars actually spans 75 minutes of clock across a closed market. The move is the overnight reopen gap - 29,388 into the 21:00 close, 29,534 at the 22:00 reopen - and no desk could have traded it. It is the only one of the 65 runs whose window straddles a gap; the other 64 are real 15-minute moves.



Friday's afternoon slide, worth \$320 on one lot. The desk had been long into the first leg of it and claimed that position for -\$446 at 15:21, fifteen minutes before this run began - then never re-entered, in either direction, for the rest of the day.



The one run the desk fought - and it won by leaving. Both capitulation_long legs bought the 14:01 spike and were out again within fourteen seconds for +\$148.50 together; the 135-point drop that names this run started after they had already gone. The gates were right about the level and wrong about the direction, and only the exit saved it.



The best catch of the week, bought at the price of the worst chop. Inside seven minutes the rider was long, then short at the low (-\$823), then long again from 13:42 - and that third entry is the one the census credits, +\$871 across four scale-outs. The two right-edge markers are legs that ran on past this window to 16:54.

★ **THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.** part1_5_rehab.html was rebuilt **2026-09-04 22:48 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

P1.5 REHABILITATION — never bench a gate at face value

★★★ REV3 — THIS WEEK'S REHAB PASS. EVERYTHING BELOW THE NEXT HORIZONTAL RULE IS LAST CYCLE'S TEXT (THE WEEK TO 2026-08-21) AND IS KEPT ONLY FOR THE TREATMENTS IT NAMES. THE THREE ITEMS THAT WERE RED THIS WEEK ARE TREATED HERE, FIRST.

The Rehabilitation phase did not run this cycle — it was allocated 19 minutes against a 20-minute floor and was skipped before it started. The report was honest about that and then dispositioned three red things anyway, without giving any of them the treatment. That is the exact failure this section exists to prevent, so the pass is run here on the material the rest of the report already computed. It is a **minimal** pass — five steps, three subjects, no new tape — and it is labelled as such rather than dressed up as the full section.

0a · REV3 REHAB — grind_long, which Part 1 §12 PARKED on n and floor-clearance alone

1 — Normalise the malfunctions. There are none to strip. The gate took two lots this week, both on Friday 14:45Z, both closed MANUAL_CLAIM for + \$41.50 and +\$22.00. No stop fault, no naked ride, no STOP_UNFILLED. Its live week is **+\$63.50 on 2 lots** and it is not the reason anything is red.

2 — Corrected-cost reconstruct. Already tick-honest: pn_l_usd is written net of the \$1.50 round trip. But both lots were claimed by hand, and a MANUAL_CLAIM truncates the gate's own exit stack — it measures the operator's timing, not the gate's edge. So the live figure is not a reading on grind_long; it is a reading on the Claim button, which §8 prices separately.

3 — Root-cause: is it the base config, the exit, or the signal? Three independent populations in this report answer this and they all say **the exit is not the problem and the signal is not the problem — the REGIME SELECTION is.**

EVIDENCE, AND WHERE IT ALREADY LIVES	WHAT IT SAYS
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Part 2.5 §1.5, 159 fires split by the regime at the fire
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EVIDENCE, AND WHERE IT ALREADY LIVES	WHAT IT SAYS
	grind's money is VIOLENT_WHIPSAW (n=45, +\$25.68/fire, LODO +\$284) and BUILDING (n=16, +\$53.88/fire, LODO +\$386). Its bleed is NORMAL_CHOP (n=47, -\$15.55/fire). And CLEAN_TREND is its WORST cell — n=8, -\$82.12 a fire, green on one day in seven.
Part 2 §7.2 exit ladder, 822,746 five-second bars, 48 cells per rung	SCALP-CHOP -\$2.10/sig on n=1,236 with 0 of 48 cells positive — a big, well-powered no. STAY-OUT -\$8.30 on n=149, 0 of 48. The only rung with a proven cell is MED-TREND (A 0.75R / B 1.0R, +\$7.70/sig, n=215, survives strip-3 +\$5.60, LODO +\$6.20, both halves). You cannot exit your way out of the chop fires; you have to not take them.
Part 1 §6, this week's own bench replay	233 of grind's 242 blocked impulses were below its own ATR floor of 22 — 96.3% — so the bench was very nearly a no-op on it. The 9 that cleared the floor were worth +\$377.24 of bench saving at the live stop, the largest single-gate figure in the table, and +\$92.23 at 1.5x.

4 — A filter that KEEPS the winners. This is where the standing 15-of-17 rule bites, and it kills the obvious fix. The tempting filter is an entry-level efficiency confirmation — “only take grind when the tape is actually trending”. Part 2.5 §1.4 shows it **selects FOR the CLEAN_TREND cell**, which is the one cell where the gate loses \$82 a fire. It is a filter that drops the winners and keeps the losers. **REJECT**

5 — Robustness. The two cells that survive are the two with LODO behind them: VIOLENT_WHIPSAW (+\$284 worst-day-out on 23 days) and BUILDING (+\$386 worst-day-out on 10 days). NORMAL_CHOP's LODO is -\$1,195 and CLEAN_TREND's is -\$1,097 — both fail in the direction they already point, which is the well-behaved case.

VERDICT — REGIME-DEPENDENT, not PARKED The Part 1 §12 disposition (“PARKED on n”) is **upgraded**. On n=2 live lots the gate is unreadable, which is true and is not the same as unrehabilitatable: on 159 fires it has a named home (violent-whipsaw and building tape), a named bleed (ordinary mid-vol chop and the 13:00Z hour), a proven exit for one rung, and an ATR floor that is already doing 96% of the work. **What would move it to LIVE:** the MED-TREND exit branch deployed banded (see §0c below and Part 2 §7.5), plus 20 admitted legs at $ATR \geq 22$ to review the floor on — reviewed on runner frequency, not on P&L. **What would REFUTE it:** a violent-whipsaw week in which the VIOLENT_WHIPSAW cell is not green. It has not happened in 23 days of that cell.

0b · REV3 REHAB — capitulation_long, the gate that is armed by rota and whose own mirror bleeds

1 — **Normalise.** Nothing to strip. Two lots this week, both Friday 14:00Z, + \$69.50 (TARGET) and +\$79.00 (CHANDELIER) = **+\$148.50**, its exits worked, nobody touched them. This is the desk's cleanest two lots of the week.

2 — **Corrected-cost.** Tick-honest already, and unlike grind's it is the gate's own exit stack, so it is a real reading — of two lots of one signal.

3 — **Root cause.** The uncomfortable evidence is that the gate is fine and the ARMING is the problem, in both directions at once:

READING	N	MONEY	WHAT IT MEANS
The live gate, all-time	34 lots	+\$63.00	a hair above water across five weeks
capit_live_mirror — the same entries with the bench taken OFF	352 fires	-\$5,515.50	-\$15.67 a fire. Left unbenched this gate is one of the worst arms on the board. The bench is what makes it profitable.
capit_loose, the no-flip variant	1,038	-\$16,681.00	-\$16.09 a fire — and it is 49% of the whole shadow board's loss
Part 2 \$7.2, all four exit rungs	24 / 170 / 940 / 23	all negative	0 of 48 cells positive in EVERY rung. There is no exit that saves this gate. That is an entry problem, decisively.
This week's bench replay (Part 1 \$6)	9 above floor	-\$70.21	benching it cost money this week at the live stop, and at every width from 0.75x up. It is the only gate in that table negative at four widths of five.

4 — **A filter that keeps the winners.** The one that already exists is require_flip=True, and the record shows exactly what happens without it: the 08-01 change to flip=False / 1.0R was shipped on an MFE statistic that ignored ordering (“78% of bounces reach 1R” — measured ordering-correct it is 29% against a 50% breakeven) and was reverted on 08-02. capit_loose is that configuration still running on the board and it is — \$16,681. **The flip requirement is the gate.** Do not re-propose dropping it without an ordering-correct win rate.

5 — Robustness. There is nothing to be robust about: 34 live lots over five weeks, +\$63.00. The mirror's 352 fires are the only well-powered reading and they are firmly negative. Two lots of +\$148.50 on one Friday is **n=1**, not n=2 — one signal, scaled out.

VERDICT — PARKED, and the revival condition is about the ROTA, not the gate The gate as configured is roughly break-even live and clearly negative unbenched, which means **the bench is carrying it and nobody decided that**: it holds 17.4% duty because it is the only gate reactivate_gates re-arms that the router does not bench, and that is a rota, not a judgement. **What would revive it to LIVE**: 40 admitted lots under an explicit ATR floor (it carries 10.0, the lowest on the desk) with the mirror tracked alongside — if the live/mirror gap holds at anything like \$15 a fire, the bench IS the strategy and should be written down as one. **What would REFUTE it**: the live gate converging on the mirror's -\$15.67 over the next 40 fires. It is not refuted now, and on the scope's own rule it cannot be — 34 lots is thin n, and thin n is PARKED.

★ **And this settles the argument between P2.6 recommendation #3 and HOLD #6.** P2.6 asked to stop benching capitulation_long and grind_long on direction and route them by ATR floor only, at HIGH confidence. The all-time bench P&L it cited (-\$156.20 and -\$242.80) is a reason to look, and the mirror is the reason not to act: with the bench off, this gate's own entries make **-\$15.67 a fire over 352 of them**. Routing it by ATR floor alone is that mirror. HOLD #6 stands, and now it has a number attached instead of a blanket.

0c · REV3 REHAB — the day rider's Thursday, and the one exit lever that is not a gate question

1 — Normalise the malfunctions FIRST, because it changes the verdict. Thursday's rider book is six entries and it reads like a catastrophe: -\$327.00 long at 13:34, -\$823.00 short at 13:35, then +\$132.25 / +\$198.75 / +\$277.75 / +\$262.25 on the 13:42 re-entry, -\$209.00 at 16:58, +\$127.00 on the 17:03 trail. Before judging any of it, three of those entries

carry an **off-tape residual fill leg** (Part 1.6 §6b, and §0d of Part 3 below): rider-2175, rider-2180 and rider-2185 each priced a residual lot at exactly $\pm 0.1\%$ of the order's base price, adverse, and **none of those prices was ever printed on the tape**. Thursday's off-tape bill alone is **\$234.00**, and the week's is **\$1,053.50** against a rider loss of $-\$1,576.94$. That is **66.8% of the rider's week** and it is not a trading decision.

2 — Corrected-cost reconstruct. The rider's week is $-\$1,576.94$. Strip the 2026-08-21 carry-in, which was opened last week and closed on Monday for $-\$2,149.44$, and the rider's OWN decisions produced **+\$572.50**. Strip the off-tape legs that belong to this week's own entries as well ($\$879.50$ of the $\$1,053.50$; the other $\$174.00$ is the carry's exit) and the rider's own week reconstructs at roughly **+\$1,452**. **The rider did not have a bad week. It had a bad Monday morning that belonged to the week before, and a fill artefact.**

3 — Root cause. Entry, exit and signal are all defensible; the bill is execution plumbing. `day_rider.py` places **every one of its ten order sites as a bare MarketOrder**. The desk already fixed this class of fault on 2026-07-20 for the tournament — capped marketable-limit entries, `entry_limit_buffer_pts=3.0`, IOC, and `config.py:117` documents the exact symptom it was written against (“a 2nd lot 29pt off a deep book on the paper engine”). **That fix was never extended to the rider.**

4 — A filter that keeps the winners. The candidate filter for Thursday is a stop, and it is the one that fails the winners test hardest: Part 3 §Q3 re-priced the no-stop decision on this week's own entries and the no-stop book wins by **\$1,189.94**. Thursday is precisely why — the 13:42 re-entry that ran to $+\$871$ is the same tape as the 13:35 short that lost $\$823$, and any stop wide enough to hold the first is wide enough to have been irrelevant to the second. **REJECT the stop**

5 — Robustness. One day, six entries. Nothing here is a robustness claim and I am not going to make one. The one number that IS well-powered is the fill audit — and ★ REV2 states its window, because these counts are ALL-TIME and the dollars beside them were the week's. All-time it is **34 split events, 39 residual legs, 39 of 39 adverse, \$2,914.50**; the report week 08-24→08-28 is **11 split orders, 15 residual legs, 18 lots, \$1,053.50**; and the week after, 08-31→09-04, is **7 split orders, 7 residual**

legs, 15 lots, \$876.00. On the report week's own numbers — 15 residual legs, 15 of 15 adverse — and **0 of 21 on the tape** where we hold tape to check.

VERDICT — FIXED, and the fix is one line of plumbing Thursday is not a strategy failure and the rider's exits are not the weak part. Extend the capped marketable-limit entry path to `day_rider.py`'s entry sites (leave closes and flattens as MKT — they must fill). **Kill criterion:** if a residual leg still prices at $\pm 0.1\%$ after the change, the artefact is downstream of us and the whole line becomes a measurement correction only. On the evidence it already is one — see Part 3.

Garrath — the rule is that nothing gets written off on its P&L alone. If a gate is red, the job is to find out how it could work, and to print the treatments that failed by name so nobody digs the same hole next Friday. Three things were red this week and they are red for three completely different reasons, which is the point: one is a software fault, one is a missing stop, and only one of them is actually a gate. **The headline is that the gate everyone would have benched is the one with the least wrong with it, and the loss that looks like bad luck was structural.**

1 · First, strip the malfunctions out of the strategy verdict

Two of the week's lots are not trading decisions at all. They have to come out before any gate is judged, and they have to be reported, because a fault that gets quietly absorbed into a gate's P&L is a fault nobody fixes:

LOT	WHAT IT WAS	BOOKED	IS IT THE STRATEGY'S FAULT?
#761 day_rider LONG 3 @ 29428.25 → 29417.50	The 00:00Z session roll flipped <code>day_rider_state.json</code> to <code>entered: false</code> / note: "switch off" while <code>venue_net</code> still read 4.0. The rider then flattened a position it had just told itself it did not have, 60 seconds after opening it.	-\$69.00	No. The entry was 09:11:06Z and the flat was 09:12:06Z — one timer tick. Nothing about the market happened in that minute.
both lots		-\$150.50	

LOT	WHAT IT WAS	BOOKED	IS IT THE STRATEGY'S FAULT?
#762 day_rider LONG 1 @ 29428.25 → 29388.25	The 1-lot remainder of the same 3+1 exit. It filled 26 points below a 30-minute tape low, ~92 ms after the 3-lot filled on- tape. Flagged BADFILL:stale_price_30pt_outsi de_book_20260821.	-\$81.50	No, and note it is a SELL — the 'off-tape fills are always BUYs' reading is dead. Same fault, worse fill.
both lots		-\$150.50	

★ **REV3 — all three figures in this table were wrong and they now reconcile.** Rev2 printed the two lots as -\$73.50 and -\$83.00, which is pnl_usd with the \$1.50 round trip charged a second time; the book says **-\$69.00** and **-\$81.50**. The footer read -\$239.50, which was not the sum of the two rows under either arithmetic — it is **-\$150.50**, and that figure already appears correctly in Part 1's exit-reason table as the CLOCK_FLAT_BUG row (2 legs, 4 contracts).

And the two books have to be named separately, because they remove different things. The CASH book is all 44 legs, **-\$255.50**. Take both malfunction lots out of it and the week's strategy book is **-\$105.00**. The **-\$174.00** clean book quoted elsewhere in this report is not that number: data_quality is set on #762 only, so the clean book removes the BADFILL lot and leaves #761's -\$69.00 in. Both are right on their own terms — -\$255.50 - (-\$150.50) = -\$105.00 and -\$174.00 - (-\$69.00) = -\$105.00 — and Rev2's sentence quietly used one removal with the other's total. Note what the fault actually was: the rider branched on entered, and at the UTC midnight session roll entered gets reset to false while venue_net still reads the true position. It is a one-line fix — **branch on venue_net != 0, never on entered** — and it is on the Saturday card.

2 · REHAB TARGET ONE — abs_veto_short, -\$170.00 over 22 lots

This is the only red gate that is actually a gate. It fired 22 lots at 32% win for -\$170.00, and the reflex is to bench it. Before that, the treatment.

2.1 — What KIND of loser is it? (the root-cause step)

There are only two kinds of losing trade and they need opposite fixes. A **chase** never goes green at all — it is adverse from the first tick, and that is an ENTRY

problem. A **give-back** goes green and hands it back, and that is an EXIT problem. Measured inside each trade's own life, on the tick tape:

CLOSED (UTC)	LEG	ENTRY	BEST IT GOT (PT)	WORST IT GOT (PT)	NET	EXIT	PEAK, IN MONEY
08-17 12:43	A	30,208.50	+21.25	-4.25	\$34.50	TARGET	\$41.00
08-17 13:25	B	30,208.50	+28.50	-19.00	-\$35.50	STOP	\$55.50
08-17 15:51	A	30,223.50	+11.25	-13.00	-\$26.50	STOP	\$21.00
08-17 15:51	B	30,223.50	+11.25	-13.00	-\$26.50	STOP	\$21.00
08-18 09:33	A	29,738.25	+20.25	-10.75	\$37.50	TARGET	\$39.00
08-18 09:41	B	29,738.75	+30.25	-10.25	\$57.00	TARGET	\$59.00
08-18 12:20	A	29,734.00	+8.50	-15.50	-\$33.00	STOP	\$15.50
08-18 12:20	B	29,734.00	+8.50	-15.50	-\$33.50	STOP	\$15.50
08-18 14:02	A	29,659.00	+38.25	-38.25	-\$79.50	STOP	\$75.00
08-18 14:02	B	29,659.00	+38.25	-33.50	-\$79.50	STOP	\$75.00
08-18 14:59	A	29,557.50	+20.50	-10.00	\$31.50	MANUAL_CLAIM	\$39.50
08-18 15:10	B	29,557.50	+35.25	-13.75	\$51.00	MANUAL_CLAIM	\$69.00
08-18 16:32	A	29,609.50	+6.00	-12.25	-\$30.00	STOP	\$10.50
08-18 16:33	B	29,609.75	+6.25	-12.00	-\$29.50	STOP	\$11.00
08-19 13:52	A	29,466.75	+59.00	-26.00	\$86.50	MANUAL_CLAIM	\$116.50
08-19 13:51	B	29,466.75	+51.00	-26.00	\$92.50	MANUAL_CLAIM	\$100.50
08-19 14:51	A	29,543.00	+6.00	-19.75	-\$43.50	STOP	\$10.50
08-19 14:51	B	29,543.00	+6.00	-20.50	-\$43.50	STOP	\$10.50
08-19 17:28	A	29,527.50	+17.25	-12.75	-\$28.50	STOP	\$33.00
08-19 17:28	B	29,527.25	+17.00	-13.00	-\$29.00	STOP	\$32.50
08-19 18:12	A	29,514.00	+15.25	-15.75	-\$38.00	STOP	\$29.00
08-19 18:12	B	29,514.75	+16.00	-17.00	-\$37.50	STOP	\$30.50

0 of the 15 losing lots were chases. Not one of them was adverse from the first tick — the worst-behaved loser still got 6 points onside and the median loser reached **+11.25 points**, which is \$21.00 a lot sitting on the screen before it turned round. So the reflex diagnosis is wrong. This is not a gate buying tops; every one of these entries worked, briefly.

The difference between a winner and a loser here is not whether the trade went onside, it is how far: the winners' median peak is **+30.25 points** against the losers' +11.25. That is a give-back shape, and give-back shapes are an EXIT question — so the exit is where the treatment has to start, which is the opposite of where this desk usually looks.

2.2 — The stop-width sweep, run twice, because the obvious version cheats

Every lot re-raced from its own entry price on the real tick path at five widths, first touch wins, stop wins a tie inside a gap, three-hour cap. The first pass is the

usual one: risk = $k \times \text{ATR}$ and the target moves with it, so the whole trade keeps its shape.

STOP WIDTH	LOTS	NET	WIN %	\$ / LOT	WORST LEAVE-ONE-DAY-OUT	BEST LOT STRIPPED	STOP / TARGET / 3H-CAP
0.50×ATR	22	−\$124.83	32%	−\$5.67	−\$189.47	−\$170.07	15 / 7 / 0
0.75×ATR	22	−\$57.94	32%	−\$2.63	−\$118.53	−\$149.45	15 / 7 / 0
1.00×ATR ← live	22	−\$548.73	18%	−\$24.94	−\$507.28	−\$607.02	18 / 4 / 0
1.25×ATR	22	−\$778.01	14%	−\$35.36	−\$627.37	−\$851.24	19 / 3 / 0
1.50×ATR	22	−\$1,076.45	9%	−\$48.93	−\$896.89	−\$1,162.81	20 / 2 / 0

Calibration first, because a sweep you cannot check is decoration. Of the 22 lots, 18 were closed by the machine on a stop or a target. Re-raced at the live 1.00× width they come to −\$397.30 against **−\$437.50** actually booked (★ REV3 — Rev1 printed −\$464.50, which is pnl_usd with the \$1.50 round trip subtracted a second time; the column is already net), and **18 of 18 land on the same exit reason**. The remaining 4 lots you claimed by hand: you booked **\$267.50** on them, the simulator rides them to their own stop or target for −\$151.43. **That gap — \$418.93 — is your hand on this one gate**, and it is the whole reason the live book reads −\$170.00 where the machine alone reads −\$548.73.

The verdict: the sign survives all five widths. Every cell is negative, from −\$124.83 at the tightest to −\$1,076.45 at the widest, and the worst leave-one-day-out is negative at every width too. That is a real verdict rather than a knife-edge, and it is bankable: left entirely to the machine, this gate lost money this week at every stop width tested.

2.2b — But is that the STOP, or is it the target riding along with it?

The sweep above moves the stop and the target together, because R is defined on risk. That means “tighter is better” could just be “a nearer target gets hit more often”, which is a completely different instruction. So here is the same sweep with the target pinned where the live config actually puts it, moving only the stop:

STOP WIDTH	LOTS	NET	WIN %	\$ / LOT	WORST LEAVE-ONE-DAY-OUT	BEST LOT STRIPPED	STOP / TARGET / 3H-CAP
0.50×ATR	22	−\$386.89	5%	−\$17.59	−\$387.24	−\$433.55	21 / 1 / 0
0.75×ATR	22	−\$498.24	9%	−\$22.65	−\$477.70	−\$556.53	20 / 2 / 0

STOP WIDTH	LOTS	NET	WIN %	\$ / LOT	WORST LEAVE-ONE-DAY-OUT	BEST LOT STRIPPED	STOP / TARGET / 3H-CAP
1.00×ATR ← live	22	−\$548.73	18%	−\$24.94	−\$507.28	−\$607.02	18 / 4 / 0
1.25×ATR	22	−\$728.09	18%	−\$33.09	−\$665.74	−\$786.38	18 / 4 / 0
1.50×ATR	22	−\$659.37	23%	−\$29.97	−\$576.13	−\$781.89	17 / 5 / 0

Pinned target, moving stop: 0.50× −\$386.89 · 0.75× −\$498.24 · 1.00× −\$548.73 · 1.25× −\$728.09 · 1.50× −\$659.37. It still gets worse as the stop widens, so the effect is genuinely the stop and not the target. Either way both tables agree on the thing that matters: **every cell in both sweeps is negative**. There is no stop width at which this gate, run by the machine alone, made money this week.

2.3 — Filters, judged by whether they keep the winners

The rule here is old and it is the one that catches most bad ideas: a filter that improves the P&L by dropping the trades that WON is a fake win, however good the total looks.

TREATMENT	THE IDEA	LOTS KEPT	NET	Δ VS AS-TRADED	WINNERS KEPT	VERDICT
Drop the 12:00-14:00Z open window	the loudest two hours of the day	16	−\$314.50	−\$111.50	4 / 7	no better
US cash session only (13:30-20:00Z)	trade only when the depth is real	16	−\$230.00	−\$27.00	4 / 7	no better
Europe/pre-open only (before 13:30Z)	the mirror of the row above	6	\$27.00	\$230.00	3 / 7	FAKE
Stand down after two consecutive stops	the wall-of-stops rule, again	12	−\$40.00	\$163.00	4 / 7	FAKE

Nothing clears the bar. The two that beat the as-traded number do it by cutting winners along with losers, and the stand-down-after-two-stops rule is the “wall of stops” idea. ★ **REV2 fixes its count, which the report gave as three here and four on the action card**. It is **THREE** independent populations, and this is the canonical list — every other mention of the rule points here: **(1)** mechanically on 104 signals last week; **(2)** live across last week’s tape; **(3)** abs_veto_short’s 22 lots this week, in the table above. The card said “fourth” by counting the run-state work as a population; it is not one — that work refuted the same principle on a different question and never tested this rule on a sample of its own, so it is

supporting reasoning, not evidence. Three is the number. **Post it as REFUTED and stop proposing it.**

SO WHAT IS THE TREATMENT FOR ABS_VETO_SHORT?

Three things are now closed off. It is NOT a stop-width change — ten cells across two sweeps, every one negative. It is NOT a filter — none keeps the winners. And it is NOT a bench on a knife-edge, because the sign survived, so a bench here would at least be honest.

What the excursion table actually says is that the entries work and then stop working: median peak +30.25 points on the winners against +11.25 on the losers, with nothing adverse from the first tick. The live experiment that addresses exactly that is the 55-second continuation-confirm — it re-requires the burst to PERSIST before entering, which is a filter on the difference between an 11-point poke and a 30-point move. This week its mirrors are the top of the shadow board (+\$332 over 46 fires) while the live gate lost \$203 over 22. **Part 2 is where that promotion is argued, and it is the one real proposal this report makes about this gate.**

The honest caveat, stated before Part 2 rather than after it: the shadow mirror does NOT model the live gate's default-veto on thin tape, so it over-fires, and it enters on the 1-minute bar open where the live decider quotes the tick it fired on. Both biases run in the mirror's favour. It is an upper bound, and it is presented as one.

2.4 — ★ REV2: THE EXIT LADDER — the treatment §2.1 diagnosed and §2.2-§2.3 never ran

THE DANGLING DIAGNOSIS

§2.1 ends on a sentence that commits this section to something: “that is a give-back shape, and give-back shapes are an EXIT question — so the exit is where the treatment has to start.” What followed was §2.2 sweeping the **stop**, §2.2b sweeping the **stop** again, §2.3 sweeping time-of-day and stand-down **filters**, and a recommendation of the 55-second confirm, which is an **entry** filter. Not one rung of an exit ladder was ever tested on the gate this

report says has an exit-shaped loss. Here they are, on the same 22 lots, with the winners-kept column carried all the way through.

Every lot raced from its own entry on the tick tape, three-hour cap, stop wins a tie. **The stop and the target come from the VENUE, not from an ATR estimate** — every lot's protective StopLimitOrder auxPrice is in the tournament journal, so $R = |\text{entry} - \text{stop}|$ is exact, and the live target model (the $\text{atr_split}=22$ quiet-tape clip: Lot A banks \$40, Lot B takes $\max(1.75 \times \text{ATR}, \$60)$ below a 22-point ATR) is confirmed to the tick. **Calibration: 18 of 18 machine-closed lots reproduce their own live exit reason.**

EXIT TREATMENT	NET	\$ / LOT	WIN %	WINNERS KEPT	WORST LEAVE- ONE-DAY- OUT	BEST LOT STRIPPED	WHAT IT IS
AS TRADED (your hand + the machine)	−\$170.00	−\$7.73	31.8%	7 / 7	−\$162.00	−\$295.50	the live book
MACHINE, live exit (baseline)	−\$446.24	−\$20.28	22.7%	5 / 7	−\$420.00	−\$516.50	live stop, live target
Scalp at +11.25pt (losers' median peak)	\$141.00	\$6.41	72.7%	7 / 7	\$57.00	\$120.00	bank the poke
Scalp at 0.5R	\$178.50	\$8.11	81.8%	7 / 7	\$99.00	\$140.25	half the risk
Scalp at 1.0R	\$187.00	\$8.50	54.5%	7 / 7	\$60.00	\$109.00	one-for-one
Trail after +15pt, 15pt behind the peak	\$107.50	\$4.89	59.1%	7 / 7	\$12.50	\$21.00	give-back cap
Trail after +15pt, 8pt behind the peak	\$149.00	\$6.77	63.6%	7 / 7	\$15.00	\$71.00	tight give-back cap
Breakeven stop once +15pt	\$107.50	\$4.89	59.1%	7 / 7	\$12.50	\$21.00	same as trail-15
Lot A scalp +11.25pt / Lot B rides	−\$162.74	−\$7.40	45.5%	6 / 7	−\$182.50	−\$233.00	split the pair

The ladder does not fail. It is the first treatment this gate has been given that keeps every winner. Four rungs beat the as-traded book while holding 7 of 7 winners, stay positive on the worst leave-one-day-out fold, and stay positive with their best lot stripped: a scalp at **1.0R** (+\$187.00), at **0.5R** (+\$178.50), a **trail 8pt behind the peak once +15pt** (+\$149.00), and a scalp at the losers' own median peak of **+11.25pt** (+\$141.00). Against a machine baseline of −\$446.24

and an as-traded $-\$170.00$, that is a swing of six hundred dollars against the machine from changing nothing but where profit is taken.

★ **AND THE PLACEBO, WHICH IS WHY THIS IS A SHADOW ROW AND NOT A MONDAY CHANGE**

A tight target flatters any entry in a simulator, so the rung has to beat a random-entry control before it counts. 400 draws, same 22-lot shape, same R distribution, same tape, same fee:

RUNG	REAL	PLACEBO MEDIAN	PLACEBO 5-95%	WHERE THE REAL RESULT SITS
Scalp at 0.5R	\$178.50	$-\$6.75$	$-\$219.00 \dots \192.75	93rd pct
Scalp at 1.0R	\$187.00	\$14.00	$-\$305.00 \dots \344.00	80th pct
Scalp at +11.25pt	\$141.00	\$20.00	$-\$232.00 \dots \258.00	80th pct
Trail +15pt / 8pt behind	\$149.00	\$1.00	$-\$284.00 \dots \259.00	83rd pct

The entry is adding something, and not much. A random entry on this week’s tape, scalped at 0.5R against a 1R stop, lands at $-\$6.75$ — which is what a fair coin minus the fee should look like, and is a good sign the harness is honest. The real signals land at the 80th to 93rd percentile of that distribution. On $n=22$ that is a direction, not a measurement: none of these rungs clears its own placebo band. So the ladder goes to **SHADOW**, beside the 55s arms, and nothing changes live on Monday.

What this does close. The diagnosis in §2.1 no longer dangles. The exit IS where this gate’s money is — every rung that banks earlier than the live 1.5R/2.5R pair improves it, and the losers’ median peak of +11.25pt was real money sitting on the screen on 22 of 22 lots. It also sharpens what §2.2 actually proved: the stop sweep found no width that rescues the gate because the stop was never the problem. ★ **REV3 PRINTS THE HAND-VS-MACHINE SPLIT FOUR WAYS, because the two-number version did not add up.** Rev1 said “the machine books $-\$446.24$ and your hand is worth $+\$314.74$ ”, and $-446.24 + 314.74$ is $-\$131.50$, not the $-\$203.00$ printed beside it. §2.2’s pair has the same hole. Here is every number, so the gap is visible rather than buried:

THE 22 LOTS, SPLIT BY WHO CLOSED THEM	BOOKED (THE LIVE LEDGER)	REPRICED (THE SIMULATOR)	DIFFERENCE
	$-\$437.50$	$-\$393.00$	$+\$44.50$

THE 22 LOTS, SPLIT BY WHO CLOSED THEM	BOOKED (THE LIVE LEDGER)	REPRICED (THE SIMULATOR)	DIFFERENCE
18 lots the MACHINE closed — stop or target			
4 lots YOU closed by hand	+\$267.50	-\$53.24	+\$320.74
All 22	-\$170.00	-\$446.24	+\$276.24

Read the bottom row, and quote only the bottom row. The live book is +**\$276.24** better than the machine alone would have been, and that is the honest statement of what your hand was worth on this gate: it is a difference between two totals and needs no assumption about anything in between. The row above it, +\$320.74, is the version Rev1 quoted (as +\$314.74, off the fee-double-counted booked figure), and it is bigger because a separate +\$44.50 residual sits on the eighteen lots the simulator says it reproduces 18-of-18. Something in the reprice disagrees with the machine's own real fills by about \$2.47 a lot on lots where it claims to land on the same exit reason. It does not move the verdict — but it is unexplained, and the harness that produced it was never saved to a script, so it cannot be re-run and the residual cannot be attributed. **Treat -\$446.24 as a simulator output carrying a known ~\$45 uncertainty, not as a measurement.** §2.2's estimated-ATR version of the same split agrees on all of it: machine -\$548.73, hand +\$418.93, residual +\$40.20, same sign, same verdict.

★ **And this section now ships something.** Rev1 ran the ladder, found four rungs that beat the book while keeping 7 of 7 winners, wrote “the ladder goes to SHADOW” — and then nothing on the 47-play card carried it, which is precisely the failure this report criticises the claim detector for by name. ★ REV2 — and it now ships for real. This paragraph used to point at “MONDAY no. 6 on last week’s card”, **which is a row that does not exist on that card or on this one** (the MONDAY window has four rows) — the ladder was written up, declared SHADOW, and never reached a card at all, which is precisely the failure this report criticises the claim detector for by name. It is now **BUILD #17 (shadow-absveto-short-exit-ladder-0904)** on THIS week’s card: the 0.5R and 1.0R scalps and the +15pt/8pt trail, as A/B arms against the live 1.5R/2.5R pair on this gate's own signal, verified NOT byte-identical per BUILD #5, with the promotion bar stated in advance — $n \geq 60$ fires, positive on strip-best-3, AND clearing the 95th percentile of the random-entry control built above. They sit at the 80th-93rd today, so that bar is a real one and it can fail.

2.5 · ★ REV2 — REHAB TARGET ONE-AND-A-HALF:
exhaustion_short, the green one nobody interrogated

WHY A GREEN GATE GETS THE SAME TREATMENT AS A RED ONE

This section’s standing rule is that nothing gets written off on its P&L alone. The mirror of that rule is that nothing gets waved through on its P&L alone either, and Rev1 did exactly that: exhaustion_short is the only live slot green on the week (+\$165.50, 14 lots, 57%) and it received no rehab pass, no stop sweep, no leave-one-day-out, no strip-best and no hand-versus-machine split — while Part 2.6 §3 quietly attributed it −\$638.02 mechanically on the same week. A \$782.52 contradiction sat in the report unexamined. Here is the same battery abs_veto_short got.

2.5a — Is the +\$165.50 the gate, or your hand?

Every one of the 14 lots re-raced from its own entry to its own configured stop or target, three-hour cap. R comes from the venue: each lot’s protective stop order is in the tournament journal, and exhaustion_short runs stop_k=1.5, so entry ATR = R / 1.5 exactly. **7 of the 7 machine-closed lots reproduce their own live exit reason**, repricing to −\$435.50 against −\$434.00 booked — \$1.50 apart on seven lots. (★ REV3 — Rev1 said “−\$444.50 booked, \$9.00 apart”, which was pnl_usd with the fee taken off twice. Correcting it makes the calibration TIGHTER, not looser.)

BOOK	NET	WIN %	STRIP BEST 1	STRIP BEST 3
LIVE — your hand plus the machine	\$165.50	57.1%	\$34.50	−\$176.50
THE MACHINE ALONE	\$84.00	42.9%	−\$91.50	−\$422.50
... so the hand is worth	\$81.50	—	—	—

The answer is: mostly the gate. That is the opposite of abs_veto_short, where the machine alone loses −\$446.24 and the hand is the entire live result. Left to itself exhaustion_short still books **+\$84.00** — and the hand adds \$81.50 on top, not the \$60.50 Rev1 printed, because the live book is \$165.50 rather than \$144.50. On the report’s own logic that is a real difference between two gates that look superficially similar, and it is the reason this one stays live.

2.5b — And then the robustness battery, which it does not pass

TEST	LIVE BOOK	MACHINE ALONE	VERDICT
Strip the single best lot	\$165.50 → \$34.50	\$84.00 → −\$91.50	One lot is 90% of the live result and all of the machine's
Strip the best three	→ −\$176.50	→ −\$422.50	Fails on both books
Leave one day out (it only traded two)	drop 08-18 → \$65.00 · drop 08-17 → \$100.50	drop 08-18 → −\$317.00 · drop 08-17 → \$401.00	The machine result is entirely 08-18. One of two folds is deeply negative
Parameter plateau on the stop	see the sweep below — the live 1.5× is a spike, not a plateau		Fails

STOP WIDTH (MACHINE ALONE)	NET	WIN %	\$ / LOT	WORST LEAVE-ONE- DAY-OUT	BEST LOT STRIPPED	STOP / TARGET / 3H-CAP
0.5×ATR	\$29.66	21.4%	\$2.12	−\$109.68	−\$125.84	11 / 3 / 0
0.75×ATR	−\$22.00	28.6%	−\$1.57	−\$161.50	−\$177.50	10 / 4 / 0
1×ATR	−\$163.68	28.6%	−\$11.69	−\$213.36	−\$319.18	10 / 4 / 0
1.25×ATR	−\$305.36	28.6%	−\$21.81	−\$265.22	−\$460.86	10 / 4 / 0
1.5×ATR ← live	\$83.98	42.9%	\$6.00	−\$317.04	−\$91.52	8 / 6 / 0
2×ATR	−\$140.36	42.9%	−\$10.03	−\$420.72	−\$315.86	8 / 6 / 0

The live 1.5×ATR width is an isolated positive cell surrounded by negative ones — 0.75× −\$22.00, 1.0× −\$163.68, 1.25× −\$305.36, 2.0× −\$140.36. A real effect gives a plateau; this is a step. And 0.5× is positive too (+\$29.66), which makes the shape a pair of spikes rather than a slope. On 14 lots across two sessions that is what you would expect from noise.

★ AND THE CONFIG THAT PRODUCED THE GREEN WEEK WAS REVERTED MID-WEEK AS UNMEASURED

Every exhaustion_short fire this week ran with atr_split=22 active. That clip was put back on 08-16 at 21:15Z by revert_clip_lift's daily restore job against a stale roster, and the operator removed it again on **08-18 at 15:48Z** (commit 465c943) with the words “the desk then ran an UNMEASURED cell live on its only armed short”. Below a 22-point ATR it makes Lot A bank at \$40 and Lot B at max(1.75×ATR, \$60). Two of the seven signals fired under it, and it is confirmed to the tick — the 14:55 Lot A exited TARGET at exactly 20.00pt, which is \$40. So the gate's only green week was booked on a configuration that is no longer live and was never graded.

The disposition, restated honestly. Keep exhaustion_short live — nothing here says bench it, and it is the one gate on the desk whose machine half is

positive without the operator. But “the only live slot green on the week, no change proposed” needs the same caveat `abs_veto_short` got: **+\$165.50 on 14 lots across two sessions, under a config that was reverted mid-week, does not survive strip-best-1 on the machine book and has a negative leave-one-day-out fold.** It is a green number, not evidence. Do not re-tune anything on it and do not cite it as a promotion precedent.

3 · REHAB TARGET TWO — the day rider's Thursday, −\$1,288.50

One long, four lots, entered 13:38:06Z at 29,447.75 on the US open, closed by your hand in three pieces between 19:33 and 20:17. Inside its own life it got +19.25 points onside at best and -245.25 points offside at worst. **It never went green in any meaningful way.** So the same root-cause question: was it the entry or the exit?

The entry is defensible — it is the rider's normal open-hour entry and it has made money all week doing exactly that. What is not defensible is that there was no stop in the water at any point. Here is what one would have done, raced on the real tape from the rider's own entry, with the ATR the live detector itself read at the confirming tick, **34.2** points (★ REV2 — Rev1 used an estimated 36.1 here while Part 1.6 §3 printed the detector’s own 34.2 for the same trade; a stop the rider actually carried would be sized off `arm_atr`, which is the detector’s read, so 34.2 is the number and every cell below is recomputed on it, net of the \$1.50 round trip):

IF THE RIDER HAD CARRIED THIS STOP	IT WOULD HAVE FILLED AT	AT	4 LOTS WOULD HAVE COST (NET)	VS WHAT ACTUALLY HAPPENED	... AND ON THE THREE WINNING DAYS
0.50×ATR = 17.1 pt	29,430.65	13:40:03Z	−\$142.80	\$1,151.70	kills all three (−\$291.60)
0.75×ATR = 25.7 pt	29,422.10	13:40:06Z	−\$211.20	\$1,083.30	kills all three (−\$290.40)
1.00×ATR = 34.2 pt	29,413.55	13:40:13Z	−\$279.60	\$1,014.90	keeps 1 of 3 (\$301.00)
1.25×ATR = 42.8 pt	29,405.00	13:40:20Z	−\$348.00	\$946.50	keeps 1 of 3 (\$242.50)
1.50×ATR = 51.3 pt ← the width Rev1 picked	29,396.45	13:40:40Z	−\$416.40	\$878.10	keeps 1 of 3 (\$184.00)
2.00×ATR = 68.4 pt	29,379.35	13:47:32Z	−\$553.20	\$741.30	keeps 1 of 3 (\$67.00)
3.00×ATR = 102.6 pt ← REV2 picks this one	29,345.15	13:51:24Z	−\$826.80	\$467.70	keeps 3 of 3 (\$1,179.00)

★ **REV2 — READ THE LAST COLUMN, WHICH REV1 DID NOT HAVE.** Every width beats what happened on Thursday, and the tightest ones beat it by more than a thousand dollars — but Thursday is the one day the rider lost money, and pricing a stop only there is the fake-win failure this same section refuses to accept two screens earlier. The full widths×days grid is in [Part 1.6 §3b](#). Its verdict, ★ **RE-RACED IN REV3** with the fee double-count taken out of the as-traded column and the weekend carry now resolved rather than marked: **1.5×ATR takes the week's closed rider book from −\$251.00 to −\$379.90**, because 46.4pt sits inside Monday's −75.25pt excursion and 46.4pt inside Tuesday's −64.50pt, so it stops both winners out. **3.0×ATR is the only width that is positive on the closed book (+\$210.70) AND keeps all three winners.** Include the carry, now booked at −\$2,149.44, and the six-entry week is −\$2,400.44 as traded against −\$532.10 at 3.0×ATR — **every** width beats doing nothing once the unstopped position is in the total. The honest week-level value of the change is **+\$462 on the closed book, +\$1,868 with the carry**, not \$1,000. **But the week is not the argument any more.** SATURDAY no. 2 (last week's card) now scores the same proposal against the 231 lake sessions the operator's 2026-08-20 no-stop decision was actually taken on — where every width beats naked at every entry delay and the worst session falls from −\$8,616 to −\$1,689, and where nothing at all is profitable. Read that row before acting: this re-opens a decision he has already made once, and on the one session since publication his hand beat every stop on this grid.

What is NOT in dispute: having a stop at all. Every width beats what happened on Thursday, **This is not hindsight about direction** — it is not asking the rider to have known the tape was going down. It is asking it to have had any exit at all, at any of six widths, all of which are cheaper than holding to your hand at 19:33. The rider ran naked for 6.7 hours.

★ AND A LOSING RIDER IS STRUCTURALLY SILENT — THAT IS THE REAL DEFECT

The reason this ran for six hours is that nothing was shouting. The rider pages on peak give-back and on stalls; below its arm threshold there is no stop, no trail and no alert, so a position that is simply losing produces no output at all. Both of the tools you would naturally check print “flat, \$0” while it happens, because the trade has not CLOSED yet and they count closed rows.

The rehabilitation of the rider is not an entry change and it is not a threshold change. It is a stop. A $3.0 \times \text{ATR}$ protective stop on the rider's own book, sized off its entry ATR, is the single highest-value change on this week's card — and ★ REV2 corrects both halves of what Rev1 said here: not $1.5 \times$, and not “\$1,000 on one day” but +\$462 on the week's closed book. It is also the change that removes the report's headline: every width tested, including $3.0 \times$, closes Friday's 13:13Z entry by 13:31:55Z — six and a half hours before the gateway wedged.

4 • REHAB TARGET THREE — the four gates that never fired

The standing rule cuts the other way too: a gate that is benched is not being rehabilitated, it is being ignored. So Movement 2 fired all four of them mechanically, in their exact live config, across this whole week's captured tape — the honest denominator, every fire counted, not just the ones near a big move:

GATE (BENCHED ALL WEEK)	RAW SIGNALS ON THE TAPE	WOULD HAVE FIRED	WOULD HAVE MADE	\$ / FIRE	WHAT ITS OWN FILTERS STOPPED
grind_long	24,121	11	−\$319.17	−\$29.02	ATR floor 22pt x24,110
capitulation_long	132	10	−\$97.66	−\$9.77	ATR floor 10pt x122
abs_veto_long	38	26	−\$83.42	−\$3.21	55s veto: absorbed x5; 55s veto: thrust gone x7
rgv_short	1	1	−\$17.54	−\$17.54	—

Total for the four: **−\$517.79. The bench is the rehabilitation.** Look particularly at grind_long: 24,121 raw signals on the week's tape, of which its own ATR floor stopped 24,110. The gate is not benched by the router in any meaningful sense — it is benched by its own 22-point floor, which this week's tape cleared on a small minority of minutes. The mid-week work in Part 2.5 puts a number on how often that floor is even reachable, and it is about a quarter of sessions.

5 • The treatments that FAILED this week, by name

TREATMENT	APPLIED TO	RESULT	DISPOSITION
Tighter stop (0.50× / 0.75×ATR)	abs_veto_short	−\$124.83 and −\$57.94 — both worse than as-traded	REFUTED for this gate this week
Time-of-day carve-outs (three variants)	abs_veto_short	None keeps 80% of the winners while beating the total	REFUTED — fake wins
Stand down after two consecutive stops	abs_veto_short	Cuts winners with losers; THIRD independent population to refuse it (★ REV2 — the count is settled at three and the populations are named in §2.3)	REFUTED
“It is chasing exhausted moves” (the entry diagnosis)	abs_veto_short	0 of 15 losing lots were chases. Median loser reached +11.25 pt inside first — the entries work, briefly	REFUTED on this week's tape
Blame the entry	the day rider's Thursday	The entry is its normal open-hour entry and made money on three other days; six stop widths all beat the outcome	REFUTED — it is the missing stop

Disposition — every lead this section touched

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
abs_veto_short — bench the machine's half of it	SHADOW	Ten cells across two stop sweeps, every one negative, worst-leave-one-day-out negative throughout: the sign survives, so this IS a verdict rather than a knife-edge. The gate is only live-negative-but-not-worse because your hand is worth + \$276.24 against the machine alone — the live book minus the machine-repriced book, which is the form of that number that reconciles (★ REV3; earlier revisions quoted +\$314.74 and +\$412.93, and §2.4 now prints the four-way split and the ~\$45 simulator residual that explains the difference). Bench the automated slot, keep the signal in shadow, and let the 55s mirror carry the case.
abs_veto_short — fix the ENTRY via the 55s confirm	SHADOW	The mirror is the top of this week's shadow board. Promote only on the Part 2 battery, and only with the mirror's two known upward biases stated on the same page.
abs_veto_short — a stop-width or filter change	REFUTED	Named tests: two stop sweeps of five cells each (all ten negative) and the winner-retention rule (no filter keeps 80% of the winners while improving the total).
The day rider — a protective stop on its own book	LIVE	★ REV2 re-raced ALL SIX rider entries at SEVEN widths, not just Thursday, and ★ REV3 re-raced it again with the fee double-count out of the as-traded column and the carry resolved rather than marked — these are the REV3 figures and they are the ones in Part 1.6 §3b. Closed book: as-traded −\$251.00; the picked width is 3.0×ATR at +\$210.70, the only positive cell

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
		and the only one that keeps all three winners. 1.5×ATR — Rev1's pick — is — \$379.90 and keeps one. Cause of death for 1.5×: winners-retention, on the grid in Part 1.6 §3b. Still the week's highest-value change, at +\$462 on the closed book (+ \$1,868 once the carry is in) rather than ~\$1,000 — but SATURDAY no. 2 (last week's card) settles it on 231 sessions, not on this week.
The day rider — branch on venue_net, not entered	FIXED	One line. The 00:00Z session roll orphaned an open position and cost \$156.50 in two lots on Friday morning.
The four benched gates — un-bench any of them	REFUTED	Fired mechanically across the whole week's tape in their exact live config: −\$517.79 on 48 fires. Revisit on a measured trend week, not on this one.
grind_long's 22-point ATR floor	PARKED	It blocked 24,110 of 24,121 raw signals this week, which makes the gate un-testable rather than benched. Revive the question with the Part 2.5 floor-reality numbers: how often is the floor reachable at all, and is 22 the right number or just an untested one?

★ THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.

part1_6_day_rider.html was rebuilt 2026-09-04 22:40 UTC and covers 2026-08-30 22:00 → 2026-09-04 20:59 UTC. Parts 1 to 3 and Movements 2–3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

1.6 THE DAY RIDER — the weekly performance review

Garrath — the rider lost −\$1,576.94 this week across 13 entries and 26 legs, against the six-gate tournament's +\$330.00 on three signals. That is the first losing week the rider has had since it started keeping a ledger, and I want to be straight about where the money went before anything else: −\$2,149.44 of it is a single position opened on Friday 21

August that never got flattened, and a further **\$1,053.50** is **fifteen residual fill LEGS — eighteen lots — that printed at prices the tape never traded at** (★ REV3: re-audited with the window and the unit stated; see §6b). Strip the carry-in and the week is **+\$572.50**. Strip the \$879.50 of those legs that belong to this week's own entries as well — the other \$174.00 is the carry-in's exit — and it is **+\$1,452.00**. Neither of those two items is the strategy. What is the strategy had a mixed week and one genuinely bad afternoon, and the four questions you set get four clean answers below. The short version: **the entries were fine when they were the rider's own and poor when they were not, the thresholds are on a plateau and there is nothing to change this week, your hand was the best exit on the book by a mile**, and the weak day is **Thursday** — a rich tape we converted 0.7% of.

★★ IF YOU READ NOTHING ELSE — SIX THINGS, IN ORDER

1. The rider stopped being an automatic strategy this week and became a manual terminal. The drift detector got to place exactly **two** of the week's thirteen entries. The other eleven were your BUY/SELL presses. On Monday, Thursday and Friday the automatic path never even ran — a position was already open, the once-per-session latch was set, and the rider spent the whole 13:30–15:00 window printing “already traded this session — no re-entry”.

2. The validated entry window is doing real work. Four entries landed inside 13:38–14:09 and made **+\$1,288.00**. The other nine lost **–\$2,864.94** (–\$715.50 without the carry-in). **All four entries that never showed a single point of profit after boarding were outside the window.**

3. Nothing to change on the thresholds, and I can now say that with 49 sessions behind it. The two live fires cleared their floors by +0.240/+0.067 and +0.094/+0.057 — no scrapes. Across the whole 49-session sample the median ER margin is **+0.251** and the median RT margin **+0.122**. Only six sessions ever came within +0.02 of a floor, and those six are worth **+\$1,470** on the live exit — raising the floor would have thrown away four winners worth **+\$3,094** to dodge two losers worth **–\$1,624**.

4. Your claims were worth +\$5,426.50 this week. Every lot you claimed by hand, held instead to the 20:40 clock, would have finished at – **\$5,614.50**. You closed the same lots for –**\$188.00**. That is not the rider's design working; that is you overriding the rider's design, and this week it was right.

5. Fifteen fills were off-tape and every single one was against us. 15 of the week's 63 executions printed outside the high/low of the tape in a ± 30 -second window, all adverse, every one of them the residual leg of a split order priced at the on-tape leg $\times (1 \pm 0.001)$. Fifteen for fifteen in the same direction. Cost: **\$1,053.50 — 66.8% of the week's reported loss.** (★ REV2 — this paragraph used to say \$734.50, which is 47%. That was an EXECUTION-scoped count against a LOT-scoped denominator; §6b adjudicates it and the settled figure is **\$1,053.50 / 11 split orders / 15 residual legs / 18 lots / 66.8%** for 08-24→08-28. One number, one window, one unit, everywhere in this report.)

6. Thursday is the weak day and it is a strategy fault, not a tape fault. Six entries, twelve legs, **+\$16.50** banked out of **\$2,478.00** of available favourable excursion — a **0.7% capture** on a tape that offered a 283-point range and a clean +179-point up-move. Two entries at 13:34 and 13:35 — a buy, a panic-out 38 seconds later, and a flip short into the low — cost – **\$1,150.00 in seven minutes.**

0 · How I measured, and the one instrument I had to rebuild CALIBRATED

Everything below is re-derived from the 5-second tape, not inherited. Two things had to be fixed before any number was trustworthy.

The live detector runs at :06 past the minute now, not :21. The tick offset moved around 17 August. I calibrated the replay against the rider's own leaning field rather than guessing it: the live service ticks at HH:MM:06 and the newest 5-second bar in `capture.db` at that instant is the one starting at HH:MM:00. With that cutoff the replay reproduces the two live

reads **exactly** — and it is a sharp test, because the read five seconds either side does not confirm at all:

SESSION	REPLAY CUTOFF	EFFICIENCY	ROUNDRIP	CONFIRMED?	VS THE LIVE JOURNAL
Tue 25 Aug 13:38	13:37:56	—	—	no — too few bars	—
Tue 25 Aug 13:38	13:38:01	0.390	0.517	YES	live read 0.390 / 0.517 — exact
Tue 25 Aug 13:38	13:38:06	0.314	0.440	no	would have missed the entry
Wed 26 Aug 13:44	13:44:01	0.244	0.507	YES	live read 0.244 / 0.507 — exact
Wed 26 Aug 13:44	13:43:51	0.218	0.438	no	10 seconds earlier and it does not fire

⚠ **And scripts/day_rider_lab.py is reading the wrong bars.** Its `minuteBars()` query is `SELECT ... FROM bars WHERE symbol='MNQ'` with **no timeframe filter**, and the lake holds seven timeframes for MNQ. Inside the 13:30–20:40 window alone that is **31,500 non-5s rows — 11.0% of everything it reads** (17,881 1min, 7,484 1m, 2,498 3m, 1,827 5mins, 1,520 5m, 290 1hour). An hourly bar’s high and low collapse into a single minute bucket, which inflates that minute’s range and true range. Efficiency is `|net| / path` and roundtrip is `|net| / range`, so both get moved. **Every number in this section uses timeframe='5s' only, aggregated the way drift._minuteBars does it.** This is a one-line fix and it belongs on Saturday’s list.

1 • The money, both books

DAY	RIDER NET	ENTRIES	LEGS	TOURNAMEN T	WHAT HAPPENED
Mon 24 Aug	−\$2,149.44	0	1	—	Friday’s un-flattened 4-lot long closed by hand at 11:59Z; rider locked out of the session
Tue 25 Aug	−\$243.00	1	2	—	The week’s first automatic entry, into a tape that went nowhere
Wed 26 Aug	+\$19.00	1	4	—	Second automatic entry; four hand-

DAY	RIDER NET	ENTRIES	LEGS	TOURNAMEN T	WHAT HAPPENED
					claims, effectively a scratch
Thu 27 Aug	+\$16.50	6	12	\$0.00	Six manual entries. –\$1,150 in seven minutes, then + \$871 back
Fri 28 Aug	+\$780.00	4	7	+\$330.00	Four manual entries, three of them green; the only day the tournament traded
WEEK	–\$1,576.94	13	26	+\$330.00	Desk –\$1,246.94
... less the 21 Aug carry-in	+\$572.50	12	25	+\$330.00	The week the rider actually chose to trade
... and less the \$879.50 of off-tape residual legs that belong to THIS week's own entries (★ REV2 — this cell read +\$1,175.00, which did not reconcile with the lede's +\$1,452.00 for the same quantity; + \$572.50 + \$879.50 = +\$1,452.00)	+\$1,452.00	12	25	+\$330.00	What the decisions were worth on prices the tape actually printed

Split by who pressed the button, because this week that is the whole story:

WHO ENTERED	ENTRIES	NET	NOTE
The drift detector (automatic)	2	–\$224.00	Tue and Wed. Both inside the validated window
Your hand, this week	10	+\$796.50	Thu ×6, Fri ×4
Your hand, Fri 21 Aug (the carry-in)	1	–\$2,149.44	Entered 13:13Z, closed Mon 11:59Z — see §5

Fees were \$78.00 on 52 lots — **exactly \$1.50 a lot**, so the fee constant is right in the book. `pnl_usd` is net of those fees; I have not subtracted them again.

2 · DID WE ENTER OK? MIXED

One row per entry, not per leg — the rider scales out, so the trade table's 26 rows are 26 lots leaving, not 26 decisions. “Travelled” is how far the move

had already gone in our direction from the 13:30 open before we boarded; MFE and MAE are measured from the entry price to the moment the last lot of that entry closed, so nothing is credited from a period we were flat.

ENTRY	+MIN	SRC	SIDE	PRICE	TRAVEL LED	MFE LEFT	MAE	NET	WINDO W
Fri 21 Aug 13:13	-17	hand	LONG 4	29,491.93	n/a	-3.4pt	-375.2pt	-\$2,149.44	OUTSIDE (before the open)
Tue 25 Aug 13:38	+8	auto	LONG 4	29,354.75	+60.2pt	+61.2pt	-209.2pt	-\$243.00	inside
Wed 26 Aug 13:44	+14	auto	LONG 4	29,322.75	+133.8pt	+46.0pt	-166.5pt	+\$19.00	inside
Thu 27 Aug 07:20	-370	hand	LONG 4	29,543.63	n/a	+117.6pt	-44.4pt	+\$377.50	OUTSIDE (pre-dawn)
Thu 27 Aug 13:34	+4	hand	LONG 4	29,494.06	-34.2pt	-8.3pt	-39.1pt	-\$327.00	OUTSIDE (too early)
Thu 27 Aug 13:35	+5	hand	SHORT 4	29,436.63	+91.6pt	-6.1pt	-107.1pt	-\$823.00	OUTSIDE (too early)
Thu 27 Aug 13:42	+12	hand	LONG 4	29,513.13	-15.1pt	+145.9pt	-89.1pt	+\$871.00	inside
Thu 27 Aug 16:58	+208	hand	SHORT 4	29,639.13	-110.9pt	-4.6pt	-28.4pt	-\$209.00	OUTSIDE (late)
Thu 27 Aug 17:03	+213	hand	LONG 4	29,659.00	+130.8pt	+48.5pt	-132.5pt	+\$127.00	OUTSIDE (late)
Fri 28 Aug 12:39	-51	hand	LONG 4	29,657.38	n/a	+41.6pt	-94.9pt	+\$187.00	OUTSIDE (before the open)
Fri 28 Aug 13:54	+24	hand	LONG 4	29,646.50	+39.5pt	+97.5pt	-141.0pt	+\$641.00	inside
Fri 28 Aug 14:47	+77	hand	LONG 4	29,738.50	+131.5pt	+59.5pt	-7.0pt	+\$398.00	OUTSIDE (late)
Fri 28 Aug 15:01	+91	hand	LONG 4	29,806.50	+199.5pt	+3.5pt	-71.2pt	-\$446.00	OUTSIDE (past the 15:00 cutoff)

THE WINDOW IS EARNING ITS KEEP — AND HERE IS THE SHARPEST VERSION OF IT

Four entries inside 13:38-14:09: +\$1,288.00. Nine outside: -\$2,864.94 (-\$715.50 if you set the carry-in aside). And look at the MFE column: **four entries never showed a single point of profit after boarding** — 21 Aug 13:13, 27 Aug 13:34, 27 Aug 13:35 and 27 Aug 16:58. **All four were outside the window.** Between them they lost -\$3,508.44. That is n=4 in one week, so it is not a law — but it is

exactly the shape the validated window was drawn around, and it repeated four times in five days.

Do we board too late? This is the question the “detection latency” work raised, and on the rider’s own instrument the answer is **no**. Across all 49 sessions in the lake with a full 13:30–20:40 5-second tape, the detector confirms at a median of **+9 minutes** (IQR +8 to +13, worst +41), **48 of 49 inside 13:38–14:09**, with a median of **95 points already travelled and 155 points still to come**. So we board with roughly **38% of the eventual swing gone and 62% still available**. And how far it has already run tells you almost nothing about what is left: Spearman **+0.109** against remaining MFE and **−0.079** against money. Boarding late is not this strategy’s problem.

⚠ **BUT THE AUTOMATIC PATH ONLY GOT TO RUN TWICE IN FIVE DAYS**

The rider carries a once-per-session entered latch. Any position — automatic or a manual press — consumes the slot for the whole day. This week it fired on **three of five sessions**: Monday it was the carried-over Friday long, Thursday the 07:20 buy, Friday the 12:39 buy. On each of those days the 13:30–15:00 window printed nothing but “already traded this session — no re-entry”. So I replayed what the detector would have done had it been free:

DAY	BLOCKED BY	WOULD CONFIRM	DIR	ER	RT	HOLD TO 20:40	ON THE LIVE TRAIL	
Mon 24	the carried Friday long	13:43 (+13)	DOWN		0.432	0.470	−\$544	−\$544
Thu 27	the 07:20 manual buy	13:38 (+8)	DOWN		0.658	0.716	−\$1,408	−\$1,408
Fri 28	the 12:39 manual buy	13:41 (+11)	UP		0.453	0.631	−\$1,594	+\$476
Blocked total	—		— —		—	—	−\$3,546	−\$1,476

The latch saved money this week — between \$1,476 and \$3,546 of it. That is one week and I am not building a policy on it. The point is the opposite one: **the desk currently has no way to tell the difference between “the rider declined to trade” and “the rider was never asked”**, and for three of five sessions this week it was the

second. If we keep both books on one latch we should at least log the detector's read on the days it is muzzled, so this table can be built from the journal instead of from a replay.

3 · ARE THE THRESHOLDS OK? NO CHANGE

These are the numbers the detector actually read at the moment it committed — taken from the rider's own leaning field, and reproduced exactly by the replay, so there is no modelling in them at all.

FIRE	ER READ	FLOOR	MARGIN	RT READ	FLOOR	MARGIN	BINDING CONSTRAINT
Tue 25 Aug 13:38 UP	0.390	0.15	+0.240	0.517	0.45	+0.067	roundtrip, comfortably clear
Wed 26 Aug 13:44 UP	0.244	0.15	+0.094	0.507	0.45	+0.057	roundtrip, comfortably clear

Neither scraped in. But Wednesday is worth a paragraph anyway, because it shows you what the noise looks like from the inside. Between 13:38 and 13:44 the detector read, minute by minute: **0.195 / 0.034 / 0.045 / 0.162 / 0.052 / 0.132** on efficiency — four times across the 0.15 floor in six minutes — before settling at 0.244 with roundtrip finally over the line. Roundtrip was the constraint the whole way: at 13:38 efficiency was already clear at 0.195 but roundtrip was 0.287. And at five-second resolution it is sharper still: on Tuesday the read at 13:38:01 confirms at 0.390/0.517 and the read at 13:38:06, five seconds later, does not (0.314/0.440). **The entry is a genuine coin-toss on a five-second boundary**, which is the strongest argument I have seen for the dwell/confirmation idea — and also the reason a single late snapshot mis-scores this strategy so badly.

The accumulating distribution. You asked me not to recommend a change on one week, so here is the standing sample instead: every session

in the lake with a complete 13:30–20:40 5-second tape, 49 of them, 22 June to 28 August, scanned minute by minute for first confirmation.

MARGIN OVER THE FLOOR	MIN	P10	MEDIAN	P90	MAX	WITHIN +0.02
Efficiency, over 0.15	+0.003	+0.058	+0.251	+0.588	+0.792	2 of 49
Roundtrip, over 0.45	+0.004	+0.029	+0.122	+0.316	+0.384	4 of 49

The detector confirms on **49 of 49** sessions — it has never sat one out inside the 13:30–15:00 window — and the typical fire clears its floors by a wide margin. Scrapes are rare.

“WOULD RAISING THE FLOOR HAVE MISSED A WINNER?” — YES, FOUR OF THEM, WORTH +\$3,094

Six sessions ever came within +0.02 of either floor. Here they are, priced on the exit the desk actually runs (arm at 4×ATR, trail 2×ATR off the peak), 4 lots, \$1.50/RT:

SESSION	+MIN	DIR	ER	RT	HOLD	LIVE TRAIL
Fri 10 Jul	+20	UP	0.192	0.458	+\$908	+\$908
Tue 14 Jul	+41	DOWN	0.153	0.668	−\$1,080	−\$1,080
Wed 22 Jul	+13	UP	0.210	0.454	+\$134	+\$880
Wed 29 Jul	+15	UP	0.197	0.455	−\$4,742	+\$778
Tue 18 Aug	+24	DOWN	0.156	0.520	+\$416	+\$528
Mon 24 Aug	+13	DOWN	0.432	0.470	−\$544	−\$544
The six scrape- ins	—	—	—	—	−\$4,908	+\$1,470

Four winners worth +\$3,094, two losers worth −\$1,624, net + \$1,470 to the good. Cutting the scrapes to dodge 14 July and 24 August would have cost more than it saved. And note 29 July: −\$4,742 held to the clock, **+\$778 on the trail** — the exit, not the entry, is what made that one survivable.

The plateau, re-scanned properly. The cheap version of this test asks which of today’s fires would still qualify at a higher floor. That is wrong: raise the floor and the detector does not stop firing, it fires later, at a

different price. So I re-ran the whole minute-by-minute scan at each setting. Robustness columns are strip-the-best-day, strip-the-best-three, and the two halves of the sample.

ER / RT FLOOR	FIRES	HELD TO CLOSE	HOLD TOTAL	LIVE TRAIL	-BEST 1	-BEST 3	1ST HALF	2ND HALF
0.15 / 0.45 — LIVE	49	65.3%	+\$8,580	+\$22,102	+\$18,364	+\$11,570	+\$5,472	+\$16,630
0.20 / 0.45	47	68.1%	+\$17,382	+\$22,754	+\$19,016	+\$12,222	+\$6,374	+\$16,380
0.25 / 0.45	43	65.1%	+\$14,296	+\$20,510	+\$16,772	+\$9,978	+\$7,676	+\$12,834
0.30 / 0.45	38	65.8%	+\$14,824	+\$22,928	+\$19,448	+\$12,868	+\$8,556	+\$14,372
0.15 / 0.50	48	64.6%	+\$14,796	+\$19,766	—	—	—	—
0.15 / 0.55	48	60.4%	+\$10,184	+\$16,426	—	—	—	—
0.25 / 0.55	40	60.0%	+\$12,712	+\$18,250	—	—	—	—

On the exit we actually run, the efficiency floor is on a flat plateau from 0.15 to 0.30 — \$20,510 to \$22,928, a 12% spread across a doubling of the threshold. Both halves are positive at every setting and strip-the-best-three leaves \$10k–\$13k standing. The roundtrip floor is different: **raising it hurts at every efficiency setting**, taking the trail from \$22,102 to \$16,426 at 0.55. **Verdict: nothing to change this week.** Keep 0.15 / 0.45, keep logging the margin, and revisit when the scrape-in sample is 15 sessions instead of 6.

⚠ **ONE PRIOR FINDING DOES NOT REPRODUCE, AND I THINK I KNOW WHY**

An earlier day-rider study recommended raising the efficiency floor from 0.15 to 0.25 (38 sessions, 15 June–7 August). On my 49 sessions that lift is real only if you price it on hold-to-the-clock — \$8,580 → \$14,296. **On the armed trail, which is what the desk actually runs, it goes the other way: \$22,102 → \$20,510.** The samples overlap but are not the same window, so this is not a refutation. It is a warning that the recommendation was measured on a policy the desk does not run, and it should not be actioned until it is re-tested on the live exit. **PARKED — revive if the 0.25 floor beats 0.15 on the armed trail across a 60+ session sample.**

4 • WHICH EXIT CAPTURED IT? THE HAND WON

Capture is the leg's realised dollars over the best that leg could have made between the entry and its own close, in dollars at \$2.00/point on its own lot count, net of \$1.50/RT on both sides. A leg that never went into profit contributes zero to "available", so the denominator is never flattered by a losing entry.

EXIT REASON	LEGS	LOTS	NET	MFE AVAILABLE	CAPTURE	WIN RATE	MEDIAN HOLD
TARGET_200 — the \$200/lot rung	1	1	+\$198.75	\$216.75	91.7%	100%	46 min
TARGET_100 — the \$100/lot rung	4	4	+\$434.75	\$502.75	86.5%	100%	10 min
TRAIL — 4×ATR arm, 2×ATR give-back	1	4	+\$127.00	\$382.00	33.2%	100%	179 min
MANUAL_CLAIM — your hand (incl. the carry)	20	43	−\$2,337.44	\$3,311.50	−70.6%	55.0%	128 min
MANUAL_CLAIM — your hand, this week only	19	39	−\$188.00	\$3,311.50	−5.7%	57.9%	63 min
CLOCK_FLAT / STOP / TARGET_400 / TARGET_600	0	0	—	—	—	—	never fired

Two things jump out. **The target ladder is doing precise work at its bottom two rungs and nothing at its top two.** Five lots came off on TARGET_100 / TARGET_200 for +\$633.50 at 86–92% capture, five for five green. The \$400 and \$600 rungs did not fire once all week — no lot was ever 200 points onside on a 4-lot entry. **And the trail armed once in five sessions.** That is not a defect, it is arithmetic: the arm bar is 4×ATR frozen at entry, ATR was 28.8–41.2 this week, so the bar sat at **+115 to +165 points** and only one entry ever got there. Across the 49-session sample it arms on 28 of 49 (57%), median 4×ATR = 161 points against a median MFE of 155. It is a coin-flip by construction, and it is the design you settled on.

YOUR CLAIMS, SCORED PROPERLY — +\$5,426.50 THIS WEEK

The right way to grade a hand-claim is not against MFE, it is against what the rider would have done with that lot if you had not touched it — which, with no tradeable stop, means holding to the 20:40 clock. So that is what I compared:

CLAIM	LOTS	YOU BANKED	HELD TO 20:40	YOUR HAND
Fri 28 Aug 15:21 — the losing 4-lot	4	−\$446.00	−\$2,510.00	+\$2,064.00
Fri 28 Aug 13:42	4	+\$187.00	−\$1,317.00	+\$1,504.00
Fri 28 Aug 14:45	3	+\$541.50	−\$922.50	+\$1,464.00
Fri 28 Aug 14:58	2	+\$198.00	−\$983.00	+\$1,181.00
Thu 27 Aug 13:41 — killing the bad short	4	−\$823.00	−\$1,611.00	+\$788.00
Fri 28 Aug 14:56	1	+\$100.50	−\$491.50	+\$592.00
Thu 27 Aug 13:34 — the 38-second bail	4	−\$327.00	+\$1,139.50	−\$1,466.50
Thu 27 Aug 17:02	4	−\$209.00	+\$9.00	−\$218.00
Thu 27 Aug 13:28	1	−\$11.25	+\$185.75	−\$197.00
... and 11 smaller ones	16	−\$1,548.19	−\$2,261.19	+\$713.00
All 20 claims	43	−\$2,337.44	−\$8,761.94	+\$6,424.50
The 19 in-week claims	39	−\$188.00	−\$5,614.50	+\$5,426.50

Ten of nineteen beat holding, and the wins were bigger than the losses. Your claims capture almost none of the theoretical MFE — −5.7% — and that reads terribly until you notice that holding would have captured −170% of it. This was a week where the back half of every session went against the front half, and the mechanical exits had no answer to that. Your hand did. **One week. Do not read it as “claim more”** — the standing 49-session sample still has hold at +\$8,580 and the armed trail at +\$22,102, both comfortably positive. Read it as: on a reversal week the hand is the only exit that works, and this was a reversal week.

5 · NAME THE WEAK DAY THURSDAY

First, what each session actually offered, measured on the same 5-second tape:

SESSION	13:30 → 20:40	RANGE	BEST UP-MOVE	BEST DOWN-MOVE	PATH	SESSION EFFICIENCY
Fri 21 Aug	−57.0pt	268.2pt	+50.2pt	−218.0pt	2,748	0.021
Mon 24 Aug	−104.5pt	296.0pt	+40.0pt	−256.0pt	2,929	0.036
Tue 25 Aug	−4.8pt	270.5pt	+121.5pt	−149.0pt	2,670	0.002
Wed 26 Aug	+163.2pt	229.5pt	+196.8pt	−32.8pt	2,780	0.059
Thu 27 Aug	+109.0pt	283.5pt	+179.2pt	−104.2pt	3,256	0.033
Fri 28 Aug	−113.5pt	374.8pt	+204.8pt	−170.0pt	3,916	0.029

And what we took out of each:

SESSION	ENTRIES	MFE AVAILABLE	BANKED	CAPTURE	VERDICT
Tue 25 Aug	1	\$484.00	−\$243.00	−50.2%	THE TAPE GAVE NOTHING
Wed 26 Aug	1	\$362.00	+\$19.00	5.2%	A scratch on a decent trend day
Thu 27 Aug	6	\$2,478.00	+\$16.50	0.7%	THE STRATEGY DID BADLY
Fri 28 Aug	4	\$1,593.00	+\$780.00	49.0%	The good day

Thursday 27 August is the weak day, and it is not the tape’s fault. The tape was generous: a 283-point range, a clean +179-point up-move from the open, the busiest path of the week bar Friday, and the detector’s read at 13:38 was the strongest of the whole sample (efficiency 0.658, roundtrip 0.716 — margins of +0.508 and +0.266, the widest in 49 sessions). We put six entries into that day and came out with **\$16.50**. Here is the seven minutes that did it:

TIME	ACTION	PRICE	RESULT	WHAT THE TAPE DID NEXT
13:34:06	MANUAL BUY 4	29,494.06	—	fell 40 points
13:34:44	claimed — 38 seconds later	29,453.94	−\$327.00	holding that long to 20:40 was +\$1,139.50
13:35:06	MANUAL SELL 4 — flipped short	29,436.63	—	rallied 102 points against it
13:41:22	claimed the short	29,538.75	−\$823.00	right call — holding it was −\$1,611.00
13:42:06		29,513.13	+\$871.00	

TIME	ACTION	PRICE	RESULT	WHAT THE TAPE DID NEXT
	MANUAL BUY 4 — back long			75% capture, the week's best entry

Three entries and three exits in eight minutes, on the same instrument, in three different directions. The 13:42 buy was excellent — inside the window, boarded while price was still 15 points below the open, caught 145.9 points of MFE and converted 75% of it. It just had to pay for the two presses that came before it. **Compare Tuesday:** −\$243.00 on \$484.00 available looks similar in percentage terms, but the session closed **4.8 points from where it opened on a 270-point range** — session efficiency **0.002**, the flattest of the week. There was no move to catch. That is the 10 August case: a quiet tape, not a fault. Thursday is a fault.

6 · The three faults that cost real money ACTION

(a) The carry-in — a gateway outage that arrived 42 minutes early. On Friday 21 August the rider was long 4 lots from 13:13Z. The hard flat is 20:40Z. The watchdog log shows the IB gateway went unreachable at **19:58:50Z** and stayed unreachable until **23:44:50Z** — 3h46m, 114 consecutive SKIP venue read failed polls, straddling the flatten clock completely. The position rode the weekend and you closed it by hand on Monday at 11:59Z.

EXIT THAT WAS SUPPOSED TO HAPPEN	PRICE	POINTS	P&L ON 4 LOTS
Fri 20:40Z — the hard flat	29,381.25	−110.7	−\$891.44
Fri 20:58Z — last print before the halt	29,374.25	−117.7	−\$947.44
Mon 11:59Z — what actually happened	29,224.00	−244.7	−\$2,149.44
Cost of the failed flatten	—	—	−\$1,258.00

The excursion tells you how bad it could have been: **MFE −3.4 points** — the position was never once in profit — and **MAE −375.2 points**, or −\$3,002 on 4 lots, hit while the market was open and nobody was watching. And the outage is not a freak: the same blackout starts at **20:40:50Z on 17**

Aug, 20:42:50Z on 27 Aug, 20:06:50Z on 23 Aug. Two of those begin within three minutes of the 20:40 flat. **The desk’s hard-flat clock is sitting on the exact edge of a recurring gateway blackout with zero margin**, and the only reason 21 August is the one that hurt is that the outage happened to arrive 42 minutes early. The watchdog, meanwhile, printed “ok position 4.0 managed” 1,908 times while that position was open, because the heartbeat was fresh — a strategy that is running and a strategy that can reach the venue are not the same thing.

(b) Fifteen off-tape fill legs, all against us, \$1,053.50 — and ★ REV3 says what they ARE. Two figures for this fault were in print and they were irreconcilable: \$734.50 over “15 of 63 executions, 18 lots” here, and \$1,170.50 over “12 fill events, 20 of 116 lots” on the card and in Part 3 §Q3. They differed on the WINDOW and on the UNIT and neither said which. So here is one figure, with both stated, from scripts/rev3_offtape_audit.py:

WINDOW AND UNIT, STATED	SPLIT ORDERS	RESIDUAL LEGS	LOTS	COST
THE REPORT WEEK — Mon 08-24 to Fri 08-28, all books	11	15	18	\$1,053.50
The card’s wider window — from the 08-21 carry entry to Friday’s close	13	18	21	\$1,229.00
All time, every order the desk has ever placed (1,609 fill rows)	27	32	35	\$2,038.50

Every one of the 15 legs in the report week is the day rider’s. Against the rider’s –\$1,576.94 that is **66.8% of its loss** — not the 47% this section said and not the 74% the card said, both of which were dividing a differently-scoped numerator by the same denominator. Here is every one, with the tape range it should have been inside:

ORDER	SIDE	LOTS	FILLED AT	ON-TAPE BASE LEG	RATIO	TAPE [LOW, HIGH] ±30S	COST
Mon 24 11:59:38 — rider-2105	SLD	3	29,216.75	29,245.75	0.999008	[29,239.50, 29,247.50]	\$174.00
Tue 25 20:27:46 — rider-2120	SLD	1	29,278.25	29,307.50	0.999002	[29,302.75, 29,311.00]	\$58.50
	BOT	3	29,330.00	29,301.00	1.000990	[29,282.50, 29,313.00]	\$174.00

ORDER	SIDE	LOTS	FILLED AT	ON-TAPE BASE LEG	RATIO	TAPE [LOW, HIGH] ±30S	COST
Wed 26 13:44:06 — rider-2125							
Thu 27 07:20:06 — rider-2150	BOT	1	29,565.75	29,536.25	1.000999	[29,535.25, 29,541.50]	\$59.00
Thu 27 13:34:06 — rider-2175	BOT	1	29,516.00	29,486.75	1.000992	[29,455.00, 29,515.50]	\$58.50
Thu 27 13:34:43 — rider-2180	SLD	1	29,432.00	29,461.25	0.999007	[29,450.00, 29,479.00]	\$58.50
Thu 27 13:35:06 — rider-2185	SLD	2	29,422.00	29,451.25	0.999007	[29,442.75, 29,468.25]	\$117.00
Thu 27 13:42:06 — rider-2195	BOT	1	29,535.25	29,505.75	1.001000	[29,484.00, 29,530.00]	\$59.00
Thu 27 16:58:06 — rider-2220	SLD	1	29,617.00	29,646.50	0.999005	[29,644.00, 29,649.75]	\$59.00
Thu 27 20:02:06 — rider-2235	SLD	1	29,653.50	29,683.00	0.999006	[29,682.25, 29,692.50]	\$59.00
Fri 28 12:39:06 — rider-2240	BOT	3	29,664.75	29,635.25	1.000995	[29,629.75, 29,637.75]	\$177.00
Total — 15 residual legs in 11 split orders	—	18	—	—	15/15 adverse	0/15 on tape	\$1,053.50

Fifteen for fifteen in the same direction is not slippage; on a fair coin that is a 1-in-33,000 event. The worst single one is 28.75 points. **The Wednesday automatic entry is the one that stings** — 3 of its 4 lots filled 17 points above the highest print in the surrounding half-minute, which turned a \$19.00 day into what should have been \$121.00. Every capture percentage in §4 is therefore slightly pessimistic, and every entry price in the trade table is contaminated on those 18 lots. This is a known, recurring pattern on this desk and it is now large enough to be a line item.

(c) The rider goes blind on its own positions outside 13:30-21:00Z.

Two of the week’s manual entries were opened outside the US session. For the Thursday 07:20 entry the rider held 4 lots for **371 minutes** with ahead_pt pinned at **0.0 for 370 of them**, peak frozen at the entry price, trail never set, arm_atr = 0.0, and the note falling back to “trail not armed (need +150 [fixed])” because it had no ATR to scale with. Same on Friday’s

12:39 entry for 50 of its 65 minutes. The drift read is gated to 13:30–21:00Z and the manage path is downstream of it, so a position opened before the open is **carried, not managed** — no peak, no trail, no give-back, no exit at all except your hand. Both of those trades made money (+\$377.50 and +\$187.00), which is exactly why it is worth flagging now rather than after one goes the other way.

7 • Disposition

LEAD	VERDICT	WHY, AND WHAT WOULD CHANGE IT
ER 0.15 / RT 0.45 entry floors	LIVE — HOLD	Flat plateau 0.15–0.30 on the live exit (\$20.5k–\$22.9k over 49 sessions), both halves positive, strip-best-3 still +\$11,570. Raising it drops four scrape-in winners worth +\$3,094. No change; keep logging the margin.
The 13:38–14:09 validated window	LIVE — CONFIRMED	48 of 49 automatic confirmations land inside it. This week: inside +\$1,288.00, outside −\$2,864.94, and all four never-in-profit entries were outside.
The \$100 / \$200 target rungs	LIVE — WORKING	5 lots, 5 winners, 86–92% capture. The \$400 and \$600 rungs never fired — watch, do not change.
Operator claim as a first-class exit	SHADOW	+\$5,426.50 vs holding this week, 10 of 19 better. One reversal week is not a policy. Keep scoring it weekly against hold-to-clock until $n \geq 60$ claims.
Raise the efficiency floor to 0.25 (prior study)	PARKED	Reproduces on hold-to-clock (\$8,580 → \$14,296), reverses on the armed trail (\$22,102 → \$20,510). Revive if it beats 0.15 on the live exit across 60+ sessions.
A dwell / 2-of-3-minute confirmation on the entry	SHADOW	The evidence for it got much stronger: Wednesday crossed the 0.15 floor four times in six minutes, and Tuesday's entry flips on a five-second boundary (0.390/0.517 at :01, 0.314/0.440 at :06). Build it as a shadow variant — it costs nothing and the sample is already 49 sessions.
The 20:40 hard flat	FAULT	Sits on the edge of a recurring 20:40–23:45Z gateway blackout with no margin. Cost \$1,258.00 on 21 August. Wants an earlier first attempt and a page that fires upstream of the flatten, not after it.
Off-tape residual-leg fills	FAULT	11 split orders, 15 residual legs, 18 lots, 15/15 adverse, base $\times (1 \pm 0.001)$, \$1,053.50 this week = 66.8% of the rider's loss (★ REV2 — was “\$734.50”; see §6b). ★★ AND IT PERSISTED: re-run

LEAD	VERDICT	WHY, AND WHAT WOULD CHANGE IT
		over 08-31→09-04 it is 7 more split orders, 7 residual legs, 15 lots, \$876.00, 0 of 7 on the tape. Mechanical and detectable. Wants a per-execution tape-containment assert at book time.
The once-per-session entered latch	PARKED	Muzzled the automatic path on 3 of 5 sessions. It saved \$1,476-\$3,546 this week, so I am not touching it. Revive the question once the detector's read is logged on muzzled days — right now we cannot tell a declined trade from an unasked one.
Management outside 13:30-21:00Z	FAULT	371 minutes of a live 4-lot position with peak, trail and ahead-P&L all frozen. Both trades happened to win. Wants the manage path decoupled from the RTH-gated drift read.
scripts/day_rider_lab.py	FAULT	Reads bars with no timeframe filter — 11.0% of the rows it consumes inside the session window are 1min/3m/5m/1hour bars. One-line fix. Any prior number from this tool is suspect until it is re-run.

SO WHAT DO I ACTUALLY DO?

Nothing to the strategy. The thresholds are on a plateau, the window is validated and holding, the exits behaved the way they are designed to behave, and the one week of adverse evidence is not enough to move any of them. That is the honest answer and I would rather write it than invent a tweak.

Three things to the plumbing, in priority order. (1) Move the first flatten attempt earlier than 20:40 and page before the attempt, not after it — that is the \$1,258 item and it will recur. (2) Assert tape-containment per execution at book time — that is **\$1,053.50** this week (★ REV2 — was \$734.50; \$6b settles it) and another **\$876.00** in the week after, and it is mechanically detectable. (3) Fix the timeframe filter in `day_rider_lab.py` before the next report reuses it.

One thing to the logging. Record the drift read on days the rider is muzzled by the latch. It cost nothing, and it is the difference between knowing this strategy's hit rate and guessing at it.

★ THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.

part2_shadow.html was rebuilt **2026-09-04 22:41 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

P2 THE SHADOW DESK & PROMOTION — what earned a live slot, and what the open rider actually needs

Garrath — the short version before the detail. **One thing on the shadow board is ready and it is ready two-sided:** `abs_veto_55s` is **\$5,302.50** over 433 tick-repriced trades, it is green in all seven ISO weeks it has existed, it is still **\$3,899.00** after you delete its ten best trades, and both directions earn — LONG **\$1,691.00** and SHORT **\$3,611.50**. Nothing else on the board clears that bar. **The open rider is the week's big piece of work and the answer is not the one you asked for — it is better.** You said we need to detect a juicy day and block the rider when conditions are not juicy. I built the pre-session day classifier, swept it properly and it **died**: the best rule looks brilliant on its own ($p = 0.5\%$) and then evaporates the moment you charge it for the 220 rules I tried to find it (family-wise $p = 29\%$), and every top rule loses money out-of-sample. But a second, causal idea passed everything: **stop each rider arm for the day the first time it changes direction.** That single rule takes the open rider from **– \$1,835.00** to **+\$1,244.50** over the same 15 days and 119 trades, it survives leave-one-day-out, strip-the-best-3, both cadences, both stop widths and the gated twins it was never fitted on, and its family-wise placebo is **0.3%**. It is not day detection. It is day abandonment, decided from inside the session on information the rider already has. **And on the exit lab I have to give you a flat no on the headline rung: BIG-**

TREND does not exist. On 68,600 minutes of extended tape, $ER \geq 0.50$ occupies **3.00%** of the clock in 544 episodes whose **median length is two minutes**. You cannot build a "ride the big trend" exit for a state that is over before the trade is.

THE FIVE THINGS THAT SURVIVED THE SKEPTIC THIS WEEK

1. **abs_veto_55s, TWO-SIDED — PROMOTE.** \$5,302.50 / 433 / 48.0% win / +\$12.25 a trade. Green in 7 of 7 weeks. Strip its best ten trades and it is still +\$3,899.00.
2. **The 55s veto's own evidence is the cleanest number on the desk.** On the same raw thrust entries, the ones it refuses lose **-\$5,287.00** at 22.3% win; the ones it keeps make **+\$5,239.00** at 49.0% win. It splits a break-even book into two halves that are almost mirror images.
3. **ONE-DIRECTION-PER-DAY on the open rider.** $-\$1,835.00 \rightarrow +\$1,244.50$ over 15 days. Family-wise placebo $p = 0.3\%$. Helps 13 of 15 days, both cadences, both stops, and the drift-gated twins.
4. **The pre-session day classifier is REFUTED as built** — and that is a survivor, because it stops us shipping a rule that only looks good because I tried 220 of them.
5. **The exit ladder has exactly one rung that clears three of the four tests:** grind MED-TREND at Lot A 0.75R / Lot B 1.0R, +\$7.70 a signal on $n=215$, in a real parameter plateau rather than on a spike. Every other cell failed strip-the-best-3. ★ **REV2 — and it fails the fourth**, the out-of-sample pool split (V5 $-\$4.50/\text{signal}$ on $n=52$, live forward arm $-\$8.36/\text{trade}$ on $n=21$), so it is not deployed. See §11 and HOLD #4.

1 • The shadow board, ranked honestly

Everything below is scored on `shadow_real.real_pnl` — the tick-repriced number — never on `ceiling_pnl`, which is the optimistic bar-price fiction the table keeps only so we can see how far apart they are. 79 sims have ever run; **42 of them traded this week**. One row is dropped board-wide (`fill_status='no_data'`), leaving 14,459 scored trades.

Read the whole-board total first, because it frames everything: the shadow book is **−\$34,304.00** all-time and **−\$2,498.50** this week. That is not a failure — the board's job is to run the bad ideas as well as the good ones so we find out which is which for free. But it does mean "the shadow board made money" is never a sentence you will read here. Only individual sims make money.

The full board, all-time and this week, biggest first. n<40 marks a sim below this desk's threshold for a forward A/B to claim anything at all — those rows are readable, not actionable.

SHADOW SIM	N	ALL-TIME NET	WIN%	\$/TR	WK N	THIS WEEK	WK \$/TR	SPAN
abs_veto_55s	433	\$5,302.50	48.0%	+12.25	45	\$948.50	+21.08	07-16-08-28
abs_veto_50s	468	\$4,236.00	44.9%	+9.05	49	\$903.00	+18.43	07-16-08-28
sw_absS_B_k1 0	92	\$2,140.00	54.3%	+23.26	23	\$675.00	+29.35	08-05-08-28
thrust_short _absveto55	146	\$2,047.00	52.7%	+14.02	19	\$602.50	+31.71	07-27-08-28
sw_absS_B_k2 0	91	\$1,779.50	60.4%	+19.55	23	\$648.50	+28.20	08-05-08-28
sw_absS_A_k1 0	96	\$1,411.50	61.5%	+14.70	24	\$457.50	+19.06	08-05-08-28
thrust_cont	513	\$1,405.50	37.8%	+2.74	67	\$266.00	+3.97	07-16-08-28
sw_absS_A_k2 0	96	\$1,151.50	67.7%	+11.99	24	\$589.50	+24.56	08-05-08-28
lad_absS_B_1 5	50	\$884.50	52.0%	+17.69	23	\$629.00	+27.35	08-17-08-28
cx_absSB_liv e n<40	37	\$855.00	51.4%	+23.11	—	—	—	08-05-08-14
cx_absSB_cli p n<40	38	\$838.50	57.9%	+22.07	—	—	—	08-05-08-14
abs_veto_60s	150	\$761.50	42.7%	+5.08	21	\$293.50	+13.98	07-16-08-28
lad_absS_A_1 0	53	\$677.00	60.4%	+12.77	24	\$435.00	+18.12	08-17-08-28
thrust_align ed	349	\$351.00	36.1%	+1.01	—	—	—	07-16-08-14
sw_absL_B_k1 0 n<40	37	\$184.00	37.8%	+4.97	—	—	—	08-04-08-14
rg_short_025 _flow50	52	\$168.00	44.2%	+3.23	—	—	—	07-20-08-14
sw_absL_B_k2 0 n<40	36	\$101.00	50.0%	+2.81	—	—	—	08-04-08-14
rg_short_025 _flow50_rth n<40	7	\$99.50	42.9%	+14.21	—	—	—	07-20-07-27
thrust_short _raw	273	\$51.00	36.6%	+0.19	39	\$170.50	+4.37	07-24-08-28
sw_absL_A_k1 0	40	\$31.50	45.0%	+0.79	—	—	—	08-04-08-14
cb_thrust_dr opped	67	\$16.50	31.3%	+0.25	—	—	—	07-20-07-31

SHADOW SIM	N	ALL-TIME NET	WIN%	\$/TR	WK N	THIS WEEK	WK \$/TR	SPAN
rg_short_025 _raw	67	−\$4.00	37.3%	-0.06	—	—	—	07-17-08-14
odr_c5_s20_g n<40	5	−\$8.50	40.0%	-1.70	—	—	—	08-14-08-14
CL- abs_veto_sho rt n<40	8	−\$10.00	25.0%	-1.25	1	\$2.00	+2.00	08-04-08-24
rg_short_050 n<40	21	−\$15.50	42.9%	-0.74	—	—	—	07-20-07-30
CL- grind_long n<40	14	−\$22.00	35.7%	-1.57	6	−\$9.00	-1.50	08-04-08-28
rg_short_025 _flow25	60	−\$24.00	38.3%	-0.40	—	—	—	07-20-08-14
rg_short_025 _flow25_t15 n<40	27	−\$26.00	44.4%	-0.96	—	—	—	07-20-07-30
CL-rgv_short n<40	11	−\$32.00	9.1%	-2.91	4	−\$11.50	-2.88	08-04-08-28
sw_absL_A_k2 0 n<40	39	−\$40.00	59.0%	-1.03	—	—	—	08-04-08-14
rgv_short_li ve_mirror n<40	6	−\$47.50	33.3%	-7.92	5	\$1.50	+0.30	08-19-08-28
thrust_loose	783	−\$48.00	35.6%	-0.06	91	\$87.50	+0.96	07-16-08-28
CL- capitulation _long n<40	36	−\$73.50	5.6%	-2.04	9	−\$18.00	-2.00	08-05-08-28
odr_c10_s20_ g n<40	5	−\$141.50	20.0%	-28.30	—	—	—	08-14-08-14
rung_grindA_ med_05 n<40	21	−\$175.50	57.1%	-8.36	13	−\$129.50	-9.96	08-17-08-28
rung_grindA_ med_live n<40	19	−\$176.50	26.3%	-9.29	11	−\$29.50	-2.68	08-17-08-28
cx_absLA_liv e n<40	32	−\$177.50	43.8%	-5.55	—	—	—	08-04-08-14
rung_grindB_ med_10 n<40	21	−\$196.50	47.6%	-9.36	13	−\$62.00	-4.77	08-17-08-28
cx_absLA_cli p n<40	31	−\$198.00	35.5%	-6.39	—	—	—	08-04-08-14
exh_w15	154	−\$202.00	37.0%	-1.31	80	\$107.00	+1.34	08-17-08-28
sw_grind_B_k 10	159	−\$205.50	33.3%	-1.29	—	—	—	08-04-08-14
rg_long_20_f low25	49	−\$287.50	38.8%	-5.87	—	—	—	07-16-07-31
odr_c10_s20 n<40	24	−\$334.50	29.2%	-13.94	—	—	—	08-10-08-14
odr_c10_s30_ g	42	−\$345.00	40.5%	-8.21	20	−\$897.00	-44.85	08-14-08-28
sw_grind_A_k 30	101	−\$381.00	46.5%	-3.77	—	—	—	08-07-08-14
odr_c5_s20 n<40	24	−\$390.00	33.3%	-16.25	—	—	—	08-10-08-14

SHADOW SIM	N	ALL-TIME NET	WIN%	\$/TR	WK N	THIS WEEK	WK \$/TR	SPAN
rg_long_20_t15	53	−\$427.50	35.8%	-8.07	—	—	—	07-16-07-31
thrust_fast	181	−\$430.50	32.6%	-2.38	—	—	—	07-16-07-31
sw_grind_B_k30	54	−\$439.50	57.4%	-8.14	—	—	—	08-07-08-14
capit_flip_live n<40	29	−\$445.00	10.3%	-15.34	9	−\$133.00	-14.78	08-16-08-28
capit_flip_t90 n<40	29	−\$445.00	10.3%	-15.34	9	−\$133.00	-14.78	08-16-08-28
odr_c10_s30	58	−\$446.50	39.7%	-7.70	21	−\$945.50	-45.02	08-10-08-28
sw_grind_A_k10	227	−\$485.50	30.0%	-2.14	—	—	—	08-04-08-14
capit_ride	40	−\$499.50	15.0%	-12.49	—	—	—	07-16-08-14
capit_mid	41	−\$522.50	24.4%	-12.74	—	—	—	07-16-08-14
sw_grind_A_k20	192	−\$538.00	45.3%	-2.80	—	—	—	08-04-08-14
rg_long_20_t20	50	−\$545.50	30.0%	-10.91	—	—	—	07-16-07-31
exh_w20	154	−\$623.00	39.6%	-4.05	80	−\$58.50	-0.73	08-17-08-28
cx_grindA_cveto	233	−\$653.50	28.3%	-2.80	113	\$197.50	+1.75	08-16-08-28
sw_grind_B_k20	107	−\$676.00	46.7%	-6.32	—	—	—	08-04-08-14
chand_k35	627	−\$765.00	33.3%	-1.22	75	\$275.00	+3.67	07-17-08-28
cx_clip_brk_live	251	−\$848.00	31.9%	-3.38	124	\$291.50	+2.35	08-16-08-28
cx_clip_brk_standdown	251	−\$848.00	31.9%	-3.38	124	\$291.50	+2.35	08-16-08-28
cb_thrust	258	−\$878.00	32.2%	-3.40	—	—	—	07-19-07-31
bank20_grindA	275	−\$1,027.00	33.8%	-3.73	139	\$158.50	+1.14	08-16-08-28
bank20_grindA_ctl	255	−\$1,172.00	27.1%	-4.60	125	\$15.50	+0.12	08-16-08-28
odr_c5_s30	61	−\$1,388.50	39.3%	-22.76	22	−\$1,404.50	-63.84	08-10-08-28
rg_long_fast_v	318	−\$1,469.00	30.2%	-4.62	25	−\$22.50	-0.90	07-16-08-28
chand_k25	352	−\$1,544.00	40.9%	-4.39	—	—	—	07-17-07-31
cx_grindA_clip	459	−\$1,571.50	30.9%	-3.42	124	\$291.50	+2.35	08-04-08-28
chand_k20	362	−\$1,587.50	42.5%	-4.39	—	—	—	07-17-07-31
grind_fast	1407	−\$1,725.00	31.7%	-1.23	—	—	—	07-16-08-14
odr_c5_s30_g	46	−\$1,889.00	32.6%	-41.07	22	−\$1,564.00	-71.09	08-14-08-28
cx_grindA_live	440	−\$1,947.00	26.6%	-4.42	118	\$271.50	+2.30	08-04-08-28
exhaustion_rev	1004	−\$2,191.00	40.5%	-2.18	165	−\$23.50	-0.14	07-20-08-28
rg_long_fast	299	−\$2,280.50	29.1%	-7.63	—	—	—	07-16-07-31
rider_w5	74	−\$2,729.50	17.6%	-36.89	38	−\$1,593.00	-41.92	08-17-08-28
capit_live_mirror	352	−\$5,515.50	12.5%	-15.67	73	−\$1,289.00	-17.66	08-02-08-28

SHADOW SIM	N	ALL-TIME NET	WIN%	\$/TR	WK N	THIS WEEK	WK \$/TR	SPAN
capit_loose	1037	-\$16,681.00	17.3%	-16.09	177	-\$2,784.50	-15.73	07-16-08-28

Two things jump out of that table and neither is the top row. The first is that the whole top of the board is one gate: `abs_veto_55s`, `abs_veto_50s`, `thrust_short_absveto55` and the four `sw_absS_*` / two `lad_absS_*` ladder arms are all the same absorption-veto mechanism wearing different exits. That is a concentration risk in the ranking itself, and it is why the promotion below is for the veto, not for eight separate sims. The second is `odr_c5_s30_g` at $-\$1,889.00$ and `odr_c5_s30` at $-\$1,388.50$ — the open rider has spent the last three weeks giving back everything the first week suggested it had, and section 4 is entirely about that.

2 · THE PROMOTION — `abs_veto_55s`, and promote it two-sided

This is `thrust_loose` plus a 55-second continuation confirm: the gate sees a thrust, waits 55 seconds, and only takes it if the burst is still going rather than being absorbed. Here is the full battery from `scripts/abs_veto_robustness.py`, run tonight on the completed week.

2.1 — Headline

MEASURE	VALUE	READ
trades / net	433 · \$5,302.50	+\$12.25 a trade after \$1.50 the round trip
win rate	48.0%	not a kill criterion and not needed here — average win \$65 against average loss $-\$37$
worst single trade	$-\$92.00$	a 1-lot sim with a real stop; no tail surprise
max drawdown	$-\$441.00$	on a \$5,302 curve — 8.3% of net

2.2 — Is it spread, or is it two lucky days?

24 of its 31 session-days are green. Best day \$761, worst day $-\$148$ — the biggest single day is 14% of the total, which is the opposite of the concentration pattern that kills most candidates. (A note on day counting, because two numbers below use different partitions and I do not want you tripping over it: the battery uses the futures session day, which folds the Sunday-evening reopen into Monday, giving 31 days. A plain calendar-UTC split gives 35 days — the four extra are Sunday reopens, 07-19, 07-26, 08-02 and 08-09, worth $-\$143.50$, $-\$106.50$, $-\$168.50$ and $+\$20.00$ — and 26 of those 35 are green. Same trades either way; the strip-best and leave-one-out rows below are on the calendar split.)

ROBUSTNESS TEST	RESULT	VERDICT
strip the best 1 trade	\$5,112.50	PASS
strip the best 3 trades	\$4,791.00	PASS — loses only 10% of net
strip the best 5	\$4,481.50	PASS
strip the best 10	\$3,899.00	PASS — still 74% of net after deleting ten trades
leave-one-day-out (calendar days), worst case	\$4,511.50 (drop 07-28)	PASS — no day carries it
top-3 calendar days as a share of net	39%	PASS — compare lad_abs\$_B_15 at 79%
per ISO week	$w29 + \$356 \cdot w30 + \$878 \cdot w31 + \$1,266 \cdot w32 + \$968 \cdot w33 + \$555 \cdot w34 + \$332 \cdot w35 + \$948$	PASS — 7 of 7 weeks green, including this one

2.3 — Regime split (the discipline says score a policy on its home tape)

REGIME (ER-30 AT ENTRY)	N	NET	WIN%	\$/TR	MAXDD
chop — ER < 0.18	208	\$2,340.00	48%	+\$11.20	−\$343
trend — ER ≥ 0.18	225	\$2,963.00	48%	+\$13.20	−\$359

This is the unusual part. Almost everything on this desk is regime-conditional — that is the whole reason the router exists. `abs_veto_55s` is not: it earns in both halves of the tape at nearly the same rate. That is what a mechanism looks like when it is filtering the entry rather than betting on the weather. It is also why it does not need the router to babysit it.

2.4 — Walk-forward halves

HALF	N	NET	WIN%	\$/TR
first half by trade order	216	\$2,524.00	47%	+\$11.70
second half	217	\$2,778.00	49%	+\$12.80

The second half is the better half. Whatever this is, it is not decaying.

2.5 — Head-to-head against the un-vetoed thrust, and what the veto actually refuses

The battery's own head-to-head is `abs_veto_55s` \$5,302.50 over 433 against `thrust_loose` −\$48.00 over 783 — an incremental \$5,350.50. That comparison is slightly unfair because the two books have different entry counts, so I did the harder version: match them entry timestamp by entry timestamp and ask what the veto threw away.

THE SAME RAW THRUST ENTRIES, SPLIT BY THE 55S VETO	N	NET (THRUST \$)	WIN%	\$/TR
DROPPED by the veto	391	−\$5,287.00	22.3%	−\$13.52
KEPT by the veto	392	\$5,239.00	49.0%	+\$13.36
— dropped, LONG side	170	−\$2,684.50	18.2%	−\$15.79
— kept, LONG side	191	\$1,784.00	46.1%	+\$9.34
— dropped, SHORT side	221	−\$2,602.50	25.3%	−\$11.78
— kept, SHORT side	201	\$3,455.00	51.7%	+\$17.19

That is the whole promotion case in one table. The veto refuses half the raw thrusts (391 of 783, 49.9%) and the half it refuses is a 22%-win disaster while the half it keeps is a 49%-win business. It does that on both sides. Honest caveat: 392 of abs_veto_55s's 433 trades (90.5%) match a thrust_loose entry exactly; the other 41 exist because the 55-second delay changes when the single slot is free, so the two books are not a perfect paired sample. The direction of the finding does not depend on those 41.

2.6 — Both sides earn, all-time and this week

SIM	SIDE	N	NET	WIN%	\$/TR	WK N	WK NET	WK \$/TR
abs_veto_55s	LONG	212	\$1,691.00	44.8%	+\$7.98	22	\$239.50	+\$10.89
abs_veto_55s	SHORT	221	\$3,611.50	51.1%	+\$16.34	23	\$709.00	+\$30.83
thrust_loose (control)	LONG	361	−\$900.50	33.0%	−\$2.49	42	−\$94.00	−\$2.24
thrust_loose (control)	SHORT	422	\$852.50	37.9%	+\$2.02	49	\$181.50	+\$3.70

The long side is the argument. Raw thrust LONG is a −\$2.49-a-trade loser and always has been — which is exactly why the live desk runs thrust_short and leaves the long side alone. Put the 55s confirm on it and the same direction becomes +\$7.98 a trade over 212 trades. The veto is not tuning the short side; it is rescuing the long one. Promote it two-sided or you leave \$1,691.00 on the table for no reason other than habit.

2.7 — The delay length, swept

CONFIRM DELAY	N	NET	\$/TR	LONG \$/TR	SHORT \$/TR	READ
50 s	468	\$4,236.00	+\$9.05	+\$4.23	+\$13.76	works
55 s (shipped)	433	\$5,302.50	+\$12.25	+\$7.98	+\$16.34	best cell
60 s	150	\$761.50	+\$5.08	+\$2.86	+\$7.29	works, far thinner n

All three delays are positive on both sides, and on both sides the ordering is the same. That is a plateau, not a spike, and it is why I am comfortable calling 55s a choice rather than a fit. **The 150-vs-433 trade gap is not a calendar artefact** — I checked: abs_veto_60s ran on 32 of the same 35 days (it is missing only 08-09, 08-17 and 08-21). Five extra seconds of waiting throws away roughly two thirds of the signals. So the ten-second window from 50s to 60s is trading selectivity against volume, and 55s is where the two cross.

2.8 — The A/B that mirrors the live short slot

SIM	N	NET	WIN%	\$/TR	WK N	WK NET
thrust_short_abs_veto55	146	\$2,047.00	52.7%	+\$14.02	19	\$602.50
thrust_short_raw (control)	273	\$51.00	36.6%	+\$0.19	39	\$170.50

Same story on the short slot the desk actually runs: the veto turns a book that has made \$51.00 in five weeks into one that has made \$2,047.00. This week it was 19 trades at 73.7% win for \$602.50 — the best week it has had. Caveat: n=146 is still thin, and its top-3 days carry 41% of its net.

2.9 — The ladder arms below it: SHADOW, not promote

SIM	N	NET	\$/TR	STRIP-3	WORST LODO	TOP-3 DAYS = % OF NET	CALL
sw_absS_B_k10	92	\$2,140.00	+\$23.26	\$1,570.50	\$1,701.50	54%	SHADOW — n<100 and concentrated
sw_absS_B_k20	91	\$1,779.50	+\$19.55	\$1,210.00	\$1,365.50	63%	SHADOW
sw_absS_A_k10	96	\$1,411.50	+\$14.70	—	—	—	SHADOW
lad_absS_B_15	50	\$884.50	+\$17.69	\$591.00	\$492.00	79%	SHADOW — 10 days, three of them are the sim

These are all the abs-veto SHORT entry wearing different exit ladders, and every one of them beats the parent on \$/trade. **Do not read that as "the ladder is better"**. They have 50–96 trades against the parent's 433, they all started in August, and lad_absS_B_15 keeps 79% of its money in three days. They are exactly the shape that looks best right before it stops working. Leave them accumulating n; revisit at 150 trades each.

3 · RELEGATION — tune the side, never relegate a direction

The standing rule is that a gate runs two-sided with independently tuned per-side thresholds, and that you never take a direction off a gate because it had a bad week — you tune the side. I re-derived every per-side number fresh from this week's board rather than inheriting last week's. The live tournament only took **6 trades all week** (abs_veto_long 2, capitulation_long 2, grind_long 2, for + \$330.00 total), so the live book cannot decide anything per-side this week — the shadow families are where the n is.

FAMILY (SHADOW)	SIDE	N	NET	WIN%	\$/TR	WK N	WK NET	WK \$/TR
rgv (fade)	LONG	769	−\$5,010.00	30.7%	−\$6.51	25	−\$22.50	−\$0.90
rgv (fade)	SHORT	234	\$198.00	40.6%	+\$0.85	0	—	—
capitulation (fade)	LONG	767	−\$12,744.00	15.3%	−\$16.62	141	−\$2,367.00	−\$16.79
capitulation (fade)	SHORT	732	−\$10,919.50	17.1%	−\$14.92	118	−\$1,839.50	−\$15.59
exhaustion (fade)	LONG	715	−\$2,250.00	38.5%	−\$3.15	157	−\$289.50	−\$1.84
exhaustion (fade)	SHORT	597	−\$766.00	41.9%	−\$1.28	168	\$314.50	+\$1.87
thrust (raw)	LONG	860	−\$1,748.00	33.1%	−\$2.03	70	\$25.50	+\$0.36
thrust (raw)	SHORT	966	\$3,026.00	38.6%	+\$3.13	88	\$328.00	+\$3.73
abs_veto 55s	LONG	212	\$1,691.00	44.8%	+\$7.98	22	\$239.50	+\$10.89
abs_veto 55s	SHORT	221	\$3,611.50	51.1%	+\$16.34	23	\$709.00	+\$30.83
grind (momentum)	LONG	1,520	−\$2,145.50	36.4%	−\$1.41	0	—	—
grind (momentum)	SHORT	727	−\$2,305.00	31.4%	−\$3.17	0	—	—

Four verdicts fall straight out of that table, and only one of them is about a direction.

FAMILY	WHAT THE PER-SIDE NUMBERS SAY	RECOMMENDATION
rgv	The two sides genuinely differ: SHORT is + \$0.85 a trade over 234, LONG is −\$6.51 over 769. That is a real asymmetry with real n on both legs.	Tune the LONG side, do not relegate it. LONG has no stable green base of its own, so it is a router problem — let the router decide when a long fade is allowed, and keep the SHORT side on its tight-extension base. This is the same conclusion the desk reached in July and this week's numbers do not disturb it. HOLD — not a new action
capitulation	Both sides are catastrophic and they are catastrophic at almost the same rate — − \$16.62 and −\$14.92 a trade, 15% and 17% win, and both are still bleeding at the identical rate this week (−\$16.79 / − \$15.59).	This is the cleanest possible demonstration of why relegating a direction is the wrong instrument. There is no good side to keep. The fault is upstream of direction — it is the gate or the exit, not the way it is

FAMILY	WHAT THE PER-SIDE NUMBERS SAY	RECOMMENDATION
		pointing. It belongs in the rehab section, not this one. NOT A SIDE PROBLEM
exhaustion	SHORT $-\$1.28$ vs LONG $-\$3.15$ all-time; this week SHORT turned positive ($+\$1.87$) while LONG stayed negative ($-\$1.84$). Small, consistent, and it points the way the live gate is already pointed.	The live desk runs this short-only and the shadow book agrees with that choice. But the margin is a dollar and change a trade on a book that is still net negative — this is not evidence to add the long side, and not evidence to change anything. HOLD
thrust / abs_veto	Raw thrust LONG is $-\$2.03$ a trade over 860. Put the 55s confirm on the identical direction and it is $+\$7.98$ over 212.	The headline argument against relegating a direction. The long side was never bad; it was unfiltered. Everyone who looked at raw thrust LONG concluded "drop the long side" and the answer was "add a filter". PROMOTE TWO-SIDED

So the relegation view this week produces zero relegations and one promotion. That is not softness — it is what the numbers say. The one family where a direction genuinely looks disposable (raw thrust LONG) is the exact family where a filter turns it into the best money on the board.

4 · ★★★ THE OPEN RIDER — you were right about both halves, and the fix is not the one we went looking for

Your words, and they set the whole section: "open rider best shadow again ... but i know they bleed on other days. thats where im saying we need detection some how to block open rider if conditions arent juicy."

The open rider (odr_*) is the simplest thing on the board and that is what makes it such a good test bed. There is no gate and no signal. Between **13:00 and 14:45 UTC**, every N minutes, it looks at the last 15 minutes of tape and goes with the sign of it. Target is 2R, coupled to the stop. Eight arms run forward: four ungated (sims 54–57, cadence 5 or 10 minutes $\times$ stop 2.0 or 3.0 $\times$ ATR) and four drift-gated twins (sims 60–63) that differ in exactly one bit — a direction gate that refuses an entry disagreeing with the confirmed drift.

4.1 — First, the numbers have moved a long way since the lead was written

The lead I was handed described five days, 08-10 to 08-14: two green days paying \$1,748 against three red days losing \$2,229. **There are now fifteen days, and the picture is much worse.** The 2.0 $\times$ ATR arms stopped after 08-14, so the only arms with a full fifteen-day history are the two 3.0 $\times$ ATR ungated cells — that pair is the open rider book from here on, and every number below is theirs unless I say otherwise.

DAY	N	NET (REAL_PNL)	RUNNING	LONG N / \$	SHORT N / \$	ATR VS ITS OWN NORM	ER-180 PRE-13:00	OVERNIG HT RANGE
2026-08-10	7	-\$601.50	-\$601.50	3 / -\$410.50	4 / -\$191.00	0.76	0.187	214.0
2026-08-11	8	\$66.00	-\$535.50	2 / -\$98.50	6 / \$164.50	0.58	0.092	221.0
2026-08-12	8	-\$334.50	-\$870.00	5 / -\$100.00	3 / -\$234.50	1.25	0.116	325.0
2026-08-13	7	\$968.00	\$98.00	7 / \$968.00	0 / —	0.90	0.008	143.8
2026-08-14	9	\$147.50	\$245.50	3 / -\$239.00	6 / \$386.50	0.68	0.055	156.0
2026-08-17	6	-\$273.00	-\$27.50	2 / -\$129.00	4 / -\$144.00	0.94	0.098	159.8
2026-08-18	8	-\$506.50	-\$534.00	4 / -\$531.50	4 / \$25.00	0.97	0.018	443.5
2026-08-19	7	\$589.00	\$55.00	4 / \$203.50	3 / \$385.50	1.41	0.171	328.8
2026-08-20	7	-\$254.00	-\$199.00	6 / -\$217.50	1 / -\$36.50	1.26	0.143	448.2
2026-08-21	9	\$714.00	\$515.00	1 / \$152.50	8 / \$561.50	0.68	0.014	266.2
2026-08-24	8	\$15.00	\$530.00	4 / -\$114.50	4 / \$129.50	1.05	0.031	356.0
2026-08-25	8	-\$504.00	\$26.00	4 / -\$442.00	4 / -\$62.00	0.88	0.071	403.2
2026-08-26	8	\$0.50	\$26.50	3 / \$149.50	5 / -\$149.00	1.32	0.086	214.5
2026-08-27	10	-\$1,000.50	-\$974.00	4 / -\$283.00	6 / -\$717.50	0.81	0.082	259.0
2026-08-28	9	-\$861.00	-\$1,835.00	4 / -\$279.50	5 / -\$581.50	0.75	0.094	129.5
15 days	119	-\$1,835.00	-\$15.42/tr	56 / -\$1,371.50	63 / -\$463.50	7 green days, 8 red		

Both halves of what you said are confirmed, and then some. Seven green days paid **\$2,500.00**. Eight red days lost **-\$4,335.00**. The average — **-\$15.42** a trade — describes not one single day the thing actually had. And the oracle that cheats by knowing the labels in advance is **+\$2,500.00 over three weeks = \$833 a week**, which does clear your "few hundred a week" bar. So the prize is real and the problem genuinely is day selection. But note what has happened to the oracle since the lead was written: on five days it was \$1,748 a week; on fifteen days it is \$833 a week, because the two monster green days have not been repeated at that size. The prize is shrinking as the sample grows, which is the normal fate of a number computed on five days.

4.2 — The pre-session day classifier, built properly — and it FAILS

The rider's first entry is at **13:00 UTC**, which is half an hour before the US cash open. So a genuinely causal day classifier can only use the overnight and pre-open tape up to 13:00. I built eleven such features per day from the 5-second parquet lake — overnight range and net drift, ATR-14 on 5-minute bars and that ATR relative to its own trailing 10-session median, efficiency over the last 60 and 180 minutes, the last-3-hours range absolute and relative, volume-expansion ratio, and relative volume — and swept every single-feature threshold rule in both directions that keeps between 3 and 12 of the 15 days. That is a family of **220 rules**.

Here are the best of them, each with the naive placebo you would normally quote: skip the same number of days at random, 4,000 times, and see where the rule lands.

RULE (ALL CAUSAL, ALL KNOWABLE AT 13:00 UTC)	DAYS KEPT	TRADES	NET	\$/TR	GREEN DAYS KEPT	NAIVE PLACEBO P
last-3h range $\leq 0.50\times$ its trailing norm	3	25	\$1,829.50	+\$73.18	3 / 3	0.5%
overnight net drift $\geq +49.25$ pt	7	56	\$1,646.00	+\$29.39	5 / 7	0.8%
ER-60 ≤ 0.074 (a calm pre-open)	5	40	\$1,641.50	+\$41.04	4 / 5	1.2%
ER-180 ≤ 0.055	5	41	\$1,338.00	+\$32.63	4 / 5	3.0%
ATR-14 (5m) ≥ 26.0 pt	6	45	\$984.00	+\$21.87	4 / 6	6.0%
(mirror) last-3h range $\geq 0.587\times$ norm	12	94	-\$3,664.50	-\$38.98	4 / 12	99.6%

Read that table cold and you would ship the first row tomorrow: a rule that keeps three days, catches all three greens, makes \$1,829.50 and has a 0.5% placebo. **It is a mirage, and here is the test that proves it.**

THE FAMILY-WISE PLACEBO — THE TEST THAT KILLED THE DAY CLASSIFIER

A 0.5% p-value means "one rule in two hundred would do this by luck". **I tried two hundred and twenty rules.** So the right question is not "could this rule happen by chance", it is "could the best of 220 rules happen by chance". I answered it directly: shuffle which day's P&L belongs to which day's features, re-run the entire 220-rule sweep, take the winner, and repeat 2,000 times.

WHAT	BEST-RULE NET
the real best rule	\$1,829.50
shuffled data, median best rule	\$1,623.00
shuffled, 90th percentile	\$2,246.00
shuffled, 95th percentile	\$2,337.00
family-wise p	29.0%

Random noise, mined the same way, beats our best rule three times in ten. The 0.5% was entirely an artefact of not charging myself for the search. There is no pre-session day classifier here.

And the out-of-sample leg buries it. Fit on the first ten days (08-10 to 08-21), test on this week (08-24 to 08-28):

RULE FITTED ON THE FIRST 10 DAYS	IN-SAMPLE NET	DAYS KEPT OOS	OOS NET	WHAT TRADING ALL 5 DAYS WOULD HAVE DONE
overnight net $\geq +49.25$	\$2,150.00	1	-\$504.00	-\$2,350.00
overnight net $\geq +62.75$	\$2,084.00	1	-\$504.00	-\$2,350.00
relative volume ≤ 0.862	\$1,895.50	2	-\$1,365.00	-\$2,350.00
last-3h range ≤ 87 pt	\$1,829.50	1	-\$861.00	-\$2,350.00

Every one of them loses money out-of-sample, and the apparent improvement over the -\$2,350.00 all-days figure is just arithmetic — the week averaged -\$470 a day, so keeping one day and losing \$504 is exactly the random expectation, and keeping two and losing \$1,365 is worse than random. The rules did not pick days; they picked fewer days.

△ **And the power limit, stated plainly.** Fifteen days cannot separate a real day classifier from noise. What the family-wise test tells us is not "there is no juicy-day signal in the tape" — it is "with 15 days and 11 features I cannot tell one from luck, and neither can anyone else." **PARKED, not refuted.** Reviving it needs roughly 40-50 rider days with a feature set fixed in advance so the search is one rule, not 220. That is about ten more weeks of forward shadow. It is free, so we should keep collecting.

4.3 — ★★ What DID pass: stop the arm for the day the first time it changes direction

While the pre-session work was failing, the per-day table above was saying something out loud. Look at the two best days: **08-13 was seven LONGs and nothing else**, and **08-21 was eight SHORTs out of nine**. Now look at the bleeders: 08-27 was four longs and six shorts, 08-25 was four and four, 08-18 was four and four. **The rider does not lose on "un-juicy" days. It loses on days where it takes both sides.** That is a whipsaw signature, and it is knowable during the session without predicting anything.

So the rule is: **each arm keeps trading until the first time its side flips; after that it is done for the day.** No parameters, no thresholds, no fitting. Causal by construction — every decision uses only trades that have already opened.

DAY	BASELINE N	BASELINE \$	KEPT N	KEPT \$	SAVED
2026-08-10	7	−\$601.50	5	−\$226.00	+\$375.50
2026-08-11	8	\$66.00	4	\$170.00	+\$104.00
2026-08-12	8	−\$334.50	6	−\$43.50	+\$291.00
2026-08-13	7	\$968.00	7	\$968.00	\$0.00
2026-08-14	9	\$147.50	4	\$106.00	−\$41.50
2026-08-17	6	−\$273.00	4	−\$79.00	+\$194.00
2026-08-18	8	−\$506.50	4	−\$206.00	+\$300.50
2026-08-19	7	\$589.00	4	\$351.00	−\$238.00
2026-08-20	7	−\$254.00	6	−\$74.50	+\$179.50
2026-08-21	9	\$714.00	9	\$714.00	\$0.00
2026-08-24	8	\$15.00	4	\$345.00	+\$330.00
2026-08-25	8	−\$504.00	4	−\$286.00	+\$218.00
2026-08-26	8	\$0.50	4	\$120.00	+\$119.50
2026-08-27	10	−\$1,000.50	4	−\$274.00	+\$726.50
2026-08-28	9	−\$861.00	5	−\$340.50	+\$520.50
total	119	−\$1,835.00	74	+\$1,244.50	+\$3,079.50

Look at the two zero rows. 08-13 and 08-21 — the two days that carry the whole strategy — never flip, so the rule does not touch them. It costs nothing on the days that pay and it cuts the days that don't. That is the exact opposite of the "filter that wins by dropping the winners" failure mode, and it is why this passes the 15-of-17-winners test in spirit as well as in arithmetic.

BOOK	N	NET	WIN%	\$/TR
baseline — every entry	119	−\$1,835.00	39.5%	−\$15.42
KEPT — up to and including the first flip	74	\$1,244.50	43.2%	+\$16.82
DROPPED — everything after the flip	45	−\$3,079.50	33.3%	−\$68.43

The dropped trades average **−\$68.43 each**. That is more than four times worse than the average trade, on a third of the volume. This is not a marginal filter.

4.4 — The robustness battery on the flip rule

A rule this good on 15 days deserves to be attacked properly, so I ran every kill test we use.

TEST	RESULT	VERDICT
		p = 0.1% — PASS

TEST	RESULT	VERDICT
Placebo — drop 45 trades at RANDOM, 5,000 draws	real +\$1,244.50 · random median – \$1,134.00 · 95th +\$97.00 · best of 5,000 + \$1,541.00	
Family-wise placebo — shuffle every P&L across the whole book, re-run all 27 breaker variants, take the winner, 600 times	real +\$1,244.50 · shuffled median – \$539.75 · 95th +\$475.00 · max +\$1,453.00	family p = 0.3% — PASS, and this is the test that killed the day classifier
Leave-one-day-out, worst case	drop 08-13 → +\$276.50	PASS — still positive without its biggest day
Strip the best 1 trade	+\$937.00	PASS
Strip the best 3 trades	+\$326.00	PASS — just
Strip the best 5 trades	–\$155.00	FAIL — the honest limit; five trades out of 74 carry it
Per cadence — odr_c5_s30	–\$1,388.50 → +\$558.50 (+\$14.70/tr)	PASS
Per cadence — odr_c10_s30	–\$446.50 → +\$686.00 (+\$19.06/tr)	PASS
External check — the drift-GATED twins (60-63, never fitted on)	–\$2,234.00 → +\$376.00 (delta +\$2,610.00)	PASS — different arms, same effect
External check — the retired 2.0×ATR arms (54, 56)	–\$724.50 → +\$300.50 (delta +\$1,025.00)	PASS — a different stop width, same effect
This week alone (08-24-28)	–\$2,350.00 → –\$435.50	Still a losing week. The rule cut the damage by 81% and did not turn it green.

The flip anatomy, so you can picture what it actually does. Across the 30 arm-days there is at least one side change on **26 of them (87%)**, and in **20 of those 26 the very first flip is at the second entry of the day**. So most days this rule means "take one or two rides and go home"; on the four no-flip days it means "run the whole session". The average kept day is 2.5 entries against a baseline of 4.0.

4.5 — The 51-day replay, and why it makes me hold this at SHADOW rather than ship it

The rider is fully deterministic — a clock and a sign — so it can be replayed over every day of tape we hold. I re-ran `scripts/odr_day_classifier.py` tonight over **51 sessions, 2026-06-19 to 2026-08-28**, producing 937 replayed trades. That is a 3.4× extension of the sample and it says two uncomfortable things.

First: the replay is not a faithful mirror, so it cannot settle a P&L question. Matched against the real forward shadow book, day for day and cell for cell:

CELL	DAYS	REAL N / NET	REPLAY N / NET	ENTRY TIMESTAMPS THAT MATCH EXACTLY
odr_c5_s20	5	24 / –\$390.00	27 / –\$505.79	18 / 24 (75%)
odr_c5_s30	15	61 / –\$1,388.50	58 / –\$335.00	36 / 61 (59%)

CELL	DAYS	REAL N / NET	REPLAY N / NET	ENTRY TIMESTAMPS THAT MATCH EXACTLY
odr_c10_s20		5 24 / -\$334.50	24 / -\$682.71	21 / 24 (88%)
odr_c10_s30		15 58 / -\$446.50	59 / -\$748.25	49 / 58 (84%)
all		— 167 / -\$2,559.50	168 / -\$2,271.75	124 / 167 (74%)

The grand totals are close — -\$2,559.50 against -\$2,271.75 — but that is luck at the aggregate. Per cell it is badly wrong: odr_c5_s30 is -\$1,388.50 in real life and -\$335.00 in the replay, a factor of four, and only 59% of its entries even happen at the same second. **Use this replay for mechanism questions, never for a dollar verdict.**

Second: the replay's headline is a window artefact, and it is a big one. The full 51-day replay comes to **+\$15,747**, which looks like a licence to print money until you ask when the money was made:

WINDOW	DAYS	REPLAY N	REPLAY NET	GREEN DAYS	WHAT WAS HAPPENING
2026-06-19 → 07-15		19 357	\$1,402.71	8 / 19	V5-era tape, before the V7 board existed
2026-07-16 → 08-07		17 304	\$18,504.75	14 / 17	board live, rider not yet armed
2026-08-10 → 08-28		15 276	-\$4,160.39	5 / 15	the rider actually running forward

Every dollar of the +\$15,747 and then some sits in a 17-day window when the rider was not trading. The moment it went forward it turned negative. That is the textbook shape of a strategy that was chosen because a backtest window flattered it, and it is the single strongest reason nothing here gets promoted this week.

And now the awkward bit for my own finding. Run the flip rule on all 51 replayed days and it does not help overall:

WINDOW (REPLAY)	BASELINE	WITH THE FLIP RULE	DELTA
2026-06-19 → 07-15	\$1,402.71	-\$1,326.29	-\$2,729.00
2026-07-16 → 08-07	\$18,504.75	\$11,329.04	-\$7,175.71
2026-08-10 → 08-28 (the live forward window)	-\$4,160.39	+\$1,333.75	+\$5,494.14
all 51 days	\$15,747.07	\$11,336.50	-\$4,410.57 · placebo p = 22%

THE HONEST READING, AND IT IS NOT THE FLATTERING ONE

On the real forward book the flip rule is strong — family-wise $p = 0.3\%$, every subgroup improved, both external checks passed. On the 51-day replay it is negative overall and its entire benefit is concentrated in the same 15 days I found it on. Two readings fit:

1. **The rule is regime-conditional.** It fixes a whipsaw that has only been present since early August; on the trending June–July tape, re-entering after a flip was correct.
2. **The rule is fitted to 15 days** and the replay is telling me so.

I cannot separate those with the data I have, and the replay is a 74%-faithful mirror so it does not get a casting vote. **Verdict: SHADOW, not deploy.** The way to settle it costs nothing: add two flip-locked arms (`odr_c5_s30_f`, `odr_c10_s30_f`) alongside the existing eight, paired so the only difference is the lock, and let them run. If the lock is real it will show up as a widening gap against its own twin within about a month; if it was fitted, the gap will close. **That is a one-bit change to the shadow slate and it is the single highest-value thing on the Saturday card.**

4.6 — The drift gate has finally bitten, and it is costing money

The lead said the drift-gated arms (60–63) covered one day and had never rejected an entry. **That is out of date — they now cover eleven days and the gate has fired.** Paired strictly on days where both twins ran:

PAIR	DAYS	UNGATED N / NET	GATED N / NET	DELTA	ENTRIES THE GATE BLOCKED	\$ ON THE BLOCKED ENTRIES
odr_c5_s30 vs _g	11	46 / $-\$1,729.50$	46 / $-\$1,889.00$	$-\$159.50$	3	$+\$56.50$
odr_c10_s30 vs _g	11	43 / $-\$203.50$	42 / $-\$345.00$	$-\$141.50$	3	$+\$429.00$
both	11	89 / $-\$1,933.00$	88 / $-\$2,234.00$	$-\\$301.00$	6	$+\$485.50$

Six blocked entries in eleven days, and all six were winners. The gate has cost $\$301.00$ so far. But six entries decides nothing — that is a coin landing heads six times, and I would say the same if it had saved $\$301$. Note also that blocking an entry frees the slot, so the gated arm sometimes takes a different trade later; that is why the two arms can have the same n and different money. **Report it, do not act on it.** What would settle it is roughly 40 blocked entries, which at the

current rate is about four months. If we want that answer sooner the gate needs to be made more selective, not the arms run longer.

⚠ **Never compare gated against ungated on all-time totals.** The gated arms started on 08-14 and the ungated on 08-10, so an all-time comparison measures the calendar, not the gate. Every number in the table above is on the 11 shared days only.

4.7 — The stop-width axis, re-derived (and it is smaller than the day effect by miles)

The 2.0×ATR arms only ever ran five days, so the only fair comparison is those five days.

STOP	N	NET	WIN%	\$/TR
2.0×ATR — 08-10 to 08-14	48	−\$724.50	31.2%	−\$15.09
3.0×ATR — same 5 days	39	\$245.50	48.7%	+\$6.29
3.0×ATR — all 15 days it has now run	119	−\$1,835.00	39.5%	−\$15.42

Wider wins on 4 of the 5 shared days and the direction of the effect holds. But look at the third row: **the wider stop that "won" the A/B has since lost \$1,835 over fifteen days.** The stop-width edge is worth about \$21 a trade on five days; the day effect is worth \$968 on one day against −\$1,000 on another. **Tuning the stop was never the problem** — and note the target is coupled to the stop in these arms, so "3.0×ATR" also means a wider target; the A/B was never a clean test of stop width alone.

4.8 — Could ONE day classifier serve both the rider and the gold run-catcher? No.

You flagged that the same shape turned up twice this month — the MNQ rider lives on a few days, and the MGC run-catcher does too (docs/MGC_LEADS_2026-08-14.md: boarded runs +\$99.23 over 14, everything else −\$373.83 over 35). If the juicy days coincided, one detector would serve two desks and be worth more than either. **I tested it directly on the 26 days where we hold both tapes, pricing gold at \$10.00 a point.**

TEST	RESULT	READ
days where "MNQ rider green" agrees with "MGC run-day"	12 / 26 = 46%	chance agreement given the two base rates is 50%

TEST	RESULT	READ
lift over chance	−4 points	the agreement is below random
Spearman(rider day \$, MGC run \$)	−0.047	nothing
Spearman(rider day \$, MGC day ER)	−0.119	nothing
MNQ's own pre-13:00 features vs MGC run size	ATR +0.003 · ER +0.041 · roundtrip −0.060	nothing, three ways

I checked the same question on the real forward book rather than the replay as a sanity pass and got the same answer: 3 of the rider's 7 green days were MGC run-days and 4 of its 8 red days were too. **REFUTED — the two instruments' good days are independent.** The named test is the coincidence table plus three Spearman correlations at zero, over 26 days. This is worth knowing because it means two detectors, and it means gold's filter problem has to be solved on gold's own tape. The one caveat: "juicy" is defined here as a green rider day and a \$300+ gold excursion; a different definition of the shared state (say, a global risk regime) is untested, and that is the one stone left unturned on this question.

5 · GRIND EXIT SHADOW — the two-ratchet runner-clip watch

Context, in one paragraph. grind_long deployed the pure-6.0 threshold-chandelier on 2026-07-26. The **two-ratchet** — a second mop-up ratchet layered on top — was tested three separate ways and rejected every time, but it was kept alive in shadow to watch for the one thing the summer-range archive could never produce: a **runner-clip on a real trend day**. A clip fires ⇒ the two-ratchet stays dead and the rejection is confirmed. Zero clips over a genuine trend week ⇒ grounds to reconsider it.

MEASURE	VALUE
window	2026-07-26 18:00Z → tonight
grind_long trades in the watch	27
runners (rMFE ≥ 4R)	3
RUNNER-CLIPS	0
live pure-6.0 net	\$51.50
shadow two-ratchet net	−\$315.00
shadow − live	−\$366.50
trades opened THIS WEEK	1 (08-28)

The three runners were 07-27 18:54 (rMFE 4.2R, live +\$23.00 vs shadow +\$144.50), 07-30 13:45 (5.21R, live +\$243.50 vs shadow +\$348.00) and 07-30 15:52 (4.75R, live +\$53.00 vs shadow +\$62.50). On all three the shadow would have made more, but none met the clip definition (shadow more than \$20 below live on a ≥4R runner), so the count that matters is **zero clips in five weeks**.

△ **THIS WATCH CANNOT BE READ THIS WEEK, AND I HAVE TO SAY SO**

The script ships a `recon_pure` column — it reconstructs what the LIVE pure-6.0 rule would have done, so you can check the reconstruction against what actually happened before you trust the shadow arm. **That cross-check has blown out.**

CHECK	VALUE
trades where reconstructed-pure $\neq$ actual live	23 of 27
total reconstruction drift	-\$478.00
the signal we are trying to measure (shadow – live)	-\$366.50

The measurement error is larger than the thing being measured. Four trades carry almost all of it — 07-30 11:11 (live +\$159.00, reconstructed –\$31.00), 07-30 12:43 (live +\$156.50, reconstructed –\$58.50), 07-29 22:06 (live +\$305.00, reconstructed +\$178.00) and 08-28 14:45 (live +\$22.00, reconstructed –\$50.00). Three of those four exited `MANUAL_CLAIM` — you took the money by hand, and no reconstruction of an automatic rule can reproduce that. So the shadow-vs-live delta of –\$366.50 is **not interpretable**: it is mostly the repricer failing to model your own claims.

What IS still readable is the clip count, because a clip is defined on `rMFE` and on the shadow arm's own exit, not on the live P&L. Zero clips, three runners, five weeks.

The verdict logic cannot fire either way this week, and the reason is volume, not the rule. `grind_long` has taken **64 lots on six days** since the deploy and **exactly two lots this week** (both on 08-28, both `MANUAL_CLAIM`, +\$63.50 together). We have not had a genuine trend week for the watch to observe. **The two-ratchet stays dead by default and the watch stays open** — that is the same call as last week, made for a new reason: last week it was "no clip fired", this week it is "there was almost nothing to fire on, and the repricer is broken enough that I would not trust a clip if one had".

Fix on the Saturday card: the reconstruction needs to skip or separately bucket `MANUAL_CLAIM` exits rather than pretending the automatic rule produced them. Until it does, the `xcheck` line is the only honest headline this tool has.

6 • GRIND EXIT SHADOW — the 2R-partial smoothness watch

The other grind-exit ledger. Baseline is the deployed 2-lot chandelier; the partial is Lot A banking a 2R scalp for certainty while Lot B rides the pure-6.0 chandelier for the tail. Grind is base_size 2, so the split is free. **This was always framed as a variance lever, not a mean-raiser** — the question you are being asked is whether a shallower drawdown is worth a small give-up on the mean.

METRIC	BASELINE (DEPLOYED 2-LOT CHANDELIER)	2R-PARTIAL (SHADOW)	WHO WINS
net \$	−\$1,482.40	−\$1,029.40	partial, by +\$453.00
daily standard deviation	\$76.30	\$123.30	baseline — the partial is 62% MORE volatile
max drawdown	−\$1,482.40	−\$1,074.90	partial, by 28%
worst day	−\$336.40	−\$286.00	partial, by 15%
green-day %	0%	16.7%	partial (1 day of 6 vs none)
green-week %	0%	33.3%	partial

CUMULATIVE CURVE	BASELINE	PARTIAL	RUNNING DELTA
2026-07-27	−\$278.00	−\$213.50	+\$64.50
2026-07-29	−\$482.00	−\$454.00	+\$28.00
2026-07-30	−\$818.40	−\$510.90	+\$307.50
2026-07-31	−\$1,096.40	−\$788.90	+\$307.50
2026-08-05	−\$1,382.40	−\$1,074.90	+\$307.50
2026-08-28	−\$1,482.40	−\$1,029.40	+\$453.00

Your choice, stated as a choice. On these six days the partial is better on money, better on drawdown, better on worst day and better on green days — and worse on daily volatility, which is the one thing it was built to improve. That inversion is the tell that six days is not a sample: the +\$453.00 is really four days — 07-27 +\$64.50, 07-29 −\$36.50, 07-30 +\$279.50 and 08-28 +\$145.50 — with 07-31 and 08-05 contributing exactly nothing. Two dates carry 94% of it, and the volatility figure is computed on six numbers.

My recommendation: take the partial, but not because of these numbers. Take it because the mechanism is sound — banking half the position at a certain 2R while the other half rides is the shape that matches how grind actually pays — and because on this sample it is not costing anything. What you must not do is quote the +\$453.00 as evidence.

⚠ **Two structural caveats on this ledger, both new tonight. (1) It ships no recon_pure column,** unlike the two-ratchet watch — so there is no way to leak-test its baseline reconstruction at all. Given that the two-ratchet's equivalent

check just failed on 23 of 27 trades, that is a hole, not a detail. **(2) The two grind watchers are not looking at the same trades.** The two-ratchet reads gate IN ('grind_long', 'grind_long_B'); the partial reads gate IN ('grind_long', 'grind_long_A'). They share only the 5 single-slot rows. So the two-ratchet's "live net +\$51.50" and the partial's "baseline −\$1,482.40" are not two views of one book — they are two different halves of it, and the partial's is a 2-lot reconstruction on top. **For the record, the true live grind_long book since the deploy is 64 lots over six days for +\$75.50** (A lots +\$24.00, B lots +\$200.50, the 4 single-slot rows −\$149.00).

7 · THE REGIME-FLEX EXIT LAB

7.1 — Grading the LIVE exhaustion_short fade-scalp trial on actual fills — and a tooling fault that changes the answer

First, a defect I found while doing this, because it invalidates the number the tool prints. scripts/exhaustion_trial_grade.py labels everything from 2026-07-31 onward as "TRIAL A0.5/B1.5" and reports it as a single config: n=94, −\$314.50. **But the exit config changed underneath it twice.** The tracked history of data/exit_overrides.json says exhaustion_short ran a_r 0.75 / b "tight" until **2026-08-16 08:34 UTC**, when it was moved to a_r 2.0 / b 2.0 / stop_k 1.5 — a completely different exit with a 50% wider stop. The tool has been stitching two configs together and grading them as one. Split at the real boundary:

EXIT CONFIG ACTUALLY RUNNING	WINDOW	N	NET	WIN%	\$/LOT	DAYS	EXIT MIX
PRE — single slot	to 07-31	66	\$322.00	56.1%	+\$4.88		29 STOP · 14 TARGET · 10 8 CHAND · 9 GIVEBACK · 4 CLAIM
TRIAL — A 0.75R / B tight, stop 1.0×ATR	07-31 → 08-16	80	−\$480.00	41.2%	−\$6.00		47 STOP · 23 TARGET · 6 7 CHAND · 4 CLAIM
CURRENT — A 2.0R / B 2.0R, stop 1.5×ATR	08-16 → now	14	\$165.50	57.1%	+\$11.82		7 2 MANUAL_CLAI M · 6 STOP · 1 TARGET

The verdict on the fade-scalp trial is worse than the tool said, and cleaner. The trial as actually configured lost **−\$480.00 over 80 lots at 41% win**, and **59% of its exits were stops** against 44% before it. A 0.75R first target with a 1.0×ATR stop is asking to be right 60% of the time to break even and it was right 41% of the time. That is a losing bet on this gate and the desk was right to move off it on 08-16.

The current config cannot be graded and I will not pretend otherwise. Fourteen lots, two days, and **seven of the fourteen exits are MANUAL_CLAIM** — which measures your timing, not the gate's exit stack. And the gate has taken **zero live trades this week**. There is no fresh live evidence at all; +\$165.50 is a two-day anecdote with half its exits taken by hand.

So the honest read has to come from the reprice, which replays every exhaustion entry we hold — 1,063 of them (112 live + 951 shadow), 1,062 of which get a rung label — on the forward 5-second path under both configs:

RUNG	N	0.5R HIT	1.5R HIT	STOP FIRST	TRIAL \$/ SIG	PRIOR \$/ SIG	DELTA
ALL	1,062	68%	41%	32%	−\$2.12	−\$2.83	+\$0.71
BIG-TREND	43	63%	49%	37%	+\$1.32	+\$12.48	−\$11.16
MED-TREND	236	67%	40%	33%	−\$6.58	−\$6.46	−\$0.11
SCALP-CHOP	783	68%	41%	32%	−\$0.96	−\$2.58	+\$1.61

The scalp helps in chop and hurts badly in trend — +\$1.61 a signal in SCALP-CHOP, −\$11.16 in BIG-TREND. That is exactly what a scalp should do, and it is the argument for a regime-conditional exit rather than a fixed one. But note the whole book is still negative under both configs, so this is a choice between two losing exits on a gate whose entry is the real problem — and note the BIG-TREND row is n=43, which section 7.3 is about to make very hard to take seriously.

7.2 — The extended exit lab: the widest tape we hold, and the ladder still only proves one rung

You asked me to finally settle the per-gate × per-rung exit ladder, and specifically to stop excusing the BIG-TREND rung as tiny-n by extending the window. I did: **822,746 five-second bars from the parquet lake, 2026-06-19 to 2026-08-28**, pooling **4,871 V7 shadow entries with 4,177 entries from the retired V5 desk's own shadow book** — 9,048 entries, 6,016 of them scored with forward tape. The V5 pool doubles as a genuine out-of-sample leg because it is a different desk. Costs: \$2.00 a point, \$1.50 the round trip, 2 lots, 1.0×ATR stop,

stop wins any same-bar race, 60-minute horizon. Grid: Lot A × Lot B R-combinations plus four chandelier shapes, 48 cells per rung.

GATE FAMIL Y	RUNG	N	BEST CELL	\$/SIG	STRIP -3	WORS T LODO	H1	H2	V7 \$/ SIG	V5 (OOS) \$/SIG	CELLS POSIT IVE	VERDI CT
grind (mom)	MED- TREND	215	A 0.75R / B 1.0R	+\$7.70	+\$5.60	+\$6.20	+\$13.50	+\$1.90	+\$11.50 (163)	-\$4.50 (52)	8 / 48	PROVEN
rgv (fade)	MED- TREND	128	chandelier k3.5 ×2	+\$8.30	-\$16.50	+\$0.60	+\$11.70	+\$4.90	+\$2.90 (126)	+\$348.50 (n=2)	1 / 48	PARKED — fails strip-3, and 1 positive cell of 48 is a needle
grind (mom)	BIG- TREND	19	A 0.75R / B 2.5R	+\$4.80	-\$22.90	-\$6.60	-\$1.90	+\$10.90	-\$1.90 (9)	+\$10.90 (10)	14 / 48	PARKED — n=19
exhaustion (fade)	SCALP- CHOP	1,303	chandelier k1.5 ×2	+\$0.60	-\$0.50	-\$2.00	+\$1.10	+\$0.00	+\$1.20 (735)	-\$0.30 (568)	1 / 48	PARKED — a well- powered zero
exhaustion (fade)	STAY- OUT	92	chandelier k3.5 ×2	+\$3.40	-\$54.70	-\$42.80	+\$84.00	-\$77.20	—	+\$3.40 (92)	1 / 48	PARKED — h1/h2 of +\$84 and – \$77 is pure noise
exhaustion (fade)	BIG- TREND	64	A 0.5R / B 1.0R	-\$1.60	-\$6.00	-\$3.60	+\$8.20	-\$11.40	+\$1.80 (42)	-\$8.00 (22)	0 / 48	NO CELL
exhaustion (fade)	MED- TREND	355	chandelier k1.5 ×2	-\$2.60	-\$6.80	-\$4.20	+\$5.50	-\$10.50	+\$4.90 (227)	-\$15.80 (128)	0 / 48	NO CELL
grind (mom)	SCALP- CHOP	1,236	A 0.75R / B 1.0R	-\$2.10	-\$2.30	-\$2.70	-\$0.70	-\$3.50	-\$2.90 (1,096)	+\$3.90 (140)	0 / 48	NO CELL — and this is a big, well- powered no
grind (mom)	STAY- OUT	149	A 0.5R / B 1.0R	-\$8.30	-\$11.20	-\$11.30	-\$10.00	-\$6.70	-\$10.60 (139)	+\$23.10 (10)	0 / 48	NO CELL — the rung is correctly named
rgv (fade)	BIG- TREND	16	A 0.5R / B 1.0R	-\$24.50	-\$54.90	-\$29.50	-\$28.70	-\$20.20	-\$24.50 (16)	—	0 / 48	NO CELL
rgv (fade)	SCALP- CHOP	262	A 0.5R / B 1.0R	-\$9.40	-\$10.20	-\$11.20	-\$9.50	-\$9.30	-\$9.80 (197)	-\$8.20 (65)	0 / 48	NO CELL
rgv (fade)	STAY- OUT	34	A 1.5R / B 2.0R	-\$11.30	-\$43.80	-\$25.30	-\$16.90	-\$5.70	-\$32.50 (27)	+\$70.40 (7)	0 / 48	NO CELL
capitulation (fade)	all four rungs	24 / 170 / 940 / 23	A 0.5R / B 1.0R everywhere	-\$8.80 / -\$10.40 / -\$11.10 / -\$1.90	all negative	all negative	—	—	—	—	0 / 48 in every rung	NO CELL — there is no exit that

GATE FAMILY	RUNG	N	BEST CELL	\$/SIG	STRIP -3	WORS T LODO	H1	H2	V7 \$/ SIG	V5 (OOS) \$/SIG	CELLS POSITIVE	VERDI CT
												saves this gate
thrust / absveto (raw)	all four rungs	48 / 219 / 630 / 89	wide chandeli er or k2.5	-\$4.20 / -\$20.30 / -\$12.00 / -\$3.20	all negative	all negative	—	—	—	—	0 / 48 in every rung	NO CELL — but see the note below

Two readings of that table before we go on. First, the thrust/absveto rows are built from raw thrust entries (thrust_loose and thrust_short_raw) — the un-vetoed population. So "no exit saves it" is consistent with section 2: **the fix for raw thrust is the entry veto, not the exit**, and this lab is confirming that from the other direction. Second, notice how often the V5 out-of-sample column disagrees violently with V7 (+\$260.70 vs −\$65.30; +\$91.90 vs −\$37.20). Those are not two estimates of one number — they are two different market regimes, and it is why an OOS leg from a different desk is worth having even when it is inconvenient.

7.3 — ★ Why the BIG-TREND rung will never have n: the regime barely exists

The BIG-TREND rung has been "too thin to deploy from" in every report that has looked at it, and the standing assumption has been that we needed more tape. **We now have all the tape and it is still thin, so I measured the rung itself rather than the trades in it.** Every minute of the extended tape, classified by the same ER-30 bands the ladder uses:

RUNG	MINUTES	% OF THE CLOCK
BIG-TREND — ER-30 ≥ 0.50	2,060	3.00%
MED-TREND — 0.30 - 0.50	11,780	17.17%
STAY-OUT — ER < 0.30 and ATR ≥ 22	6,674	9.73%
SCALP-CHOP — ER < 0.30 and ATR < 22	48,086	70.10%
total usable minutes	68,600	55 sessions

And the killer statistic is not the 3% — it is the duration. Those 2,060 minutes are not a handful of long trends. They are **544 separate episodes with a median length of TWO MINUTES** and a 90th percentile of nine. The longest continuous BIG-TREND stretch in 2½ months of MNQ tape is **31 minutes**.

I checked whether that is an artefact of measuring efficiency over 30 minutes. It is not — slowing the measurement makes the state rarer, not longer:

ER WINDOW	THRESHOLD	% OF TAPE	EPISODES	MEDIAN LENGTH	P90	LONGEST
30 min	0.50	3.00%	544	2 min	9 min	31 min
30 min	0.40	8.34%	1,282	2 min	11 min	48 min
30 min	0.35	13.33%	1,928	2 min	12 min	53 min
60 min	0.50	0.31%	44	2 min	17 min	26 min
60 min	0.40	1.67%	228	2 min	14 min	59 min
120 min	0.40	0.04%	13	2 min	3 min	5 min

THE VERDICT ON THE BIG-TREND RUNG

REFUTED as an exit rung keyed on ER, and the named tests are these **two**: (1) the extended sweep — every family's best BIG-TREND cell fails strip-the-best-3, on the widest tape we hold plus a separate desk's out-of-sample pool; and (2) the occupancy measurement above — the state occupies 3% of the clock in two-minute bursts, so a trade that is in BIG-TREND when it opens is almost never in BIG-TREND when it closes. **You cannot ride a trend that is over before the trade is.**

What would revive it — and this is the actionable part. Do not go looking for more tape; there is none to find and more of it would not help. Redefine the rung on something that persists: a range-break-and-hold state, an ADX-style directional measure, a higher-timeframe structure, or simply "the day's net move is already $> 2 \times \text{ATR}$ and price is on the correct side of VWAP". If a redefined BIG-TREND state lasts 45 minutes instead of 2, the rung becomes measurable and this whole lab can be re-run against it. That is a well-defined piece of work and it is worth a Saturday.

7.4 — The MED-TREND rung has already been forward-tested for two weeks, and it is under water

Before proposing anything, I checked whether the one PROVEN cell has already been tried. It has. Three shadow arms have been running the grind MED-TREND rung since 08-16, banded to `er_min 0.30 / er_max 0.50` so they only trade inside the rung, with the live ladder as their control on the identical band:

ARM	WHAT IT IS	N	NET	WIN%	\$/TR	EXITS
rung_grindA_med_live	CONTROL — live Lot A (2.5R), same band	19	-\$176.50	26.3%	-\$9.29	8 TARGET / 11 STOP

ARM	WHAT IT IS	N	NET	WIN%	\$/TR	EXITS
rung_grindA_med_05	the proposal — Lot A at 0.5R	21	−\$175.50	57.1%	−\$8.36	14 TARGET / 7 STOP
rung_grindB_med_10	the proposal — Lot B at 1.0R	21	−\$196.50	47.6%	−\$9.36	11 TARGET / 10 STOP

Paired on the 19 entries both A arms share: the proposal makes −\$116.00 where the live ladder makes −\$176.50 — a **+\$60.50 edge to the proposal**, better on 6 trades, worse on 5, identical on 8. **So the forward evidence agrees with the lab on direction and disagrees on magnitude, and the whole rung is losing money either way.** The win rate more than doubles (26% → 57%) exactly as a shorter target should, and it does not rescue the book, which tells you the same thing the capitulation rows told you: this is an entry problem wearing an exit costume.

7.5 — THE DELIVERABLE: what data/exit_overrides.json should say

Here is the per-rung R table you asked for, against what is live today. **I am recommending one change and it is not to a rung — it is to write the rung bands into the file at all.**

GATE	LIVE TODAY	WHAT THE EXTENDED LAB PROVES	RECOMMENDED POLICY	WHY
grind_long	a_r 2.5 / b "wide", atr_split 22	MED-TREND wants A 0.75R / B 1.0R (+\$7.70/sig, n=215, PROVEN). Live's own cell (A2.5 / wide) reads −\$12.50 to −\$16.10 a signal in that rung.	Add a MED-TREND branch only: when ER-30 is in [0.30, 0.50), A 0.75R / B 1.0R. Leave everything else exactly as it is.	It is the only cell that passes strip-3 AND leave-one-day-out AND both halves, and it sits in a plateau (see below). But its V5 out-of-sample leg is −\$4.50 and its live forward arm is under water, so it goes in banded, not blanket.
exhaustion_short	a_r 2.0 / b 2.0 / stop_k 1.5	Nothing proves. SCALP-CHOP's best is +\$0.60/sig on n=1,303 and it fails strip-3 at −\$0.50 — a well-powered zero.	NO CHANGE.	The reprice says the scalp helps in chop (+\$1.61) and hurts in trend (−\$11.16), but the current wide-stop config has 14 lots and 7 manual claims behind it. Do not move a live exit on that.
capitulation_long	a_r 1.5 / b "tight"	0 of 48 cells positive in all four rungs. n=1,157 across them.	NO CHANGE — and stop looking here.	This is the most decisive null in the lab. No exit configuration in a 48-cell grid makes this gate positive in any regime. Its problem is the entry, and both its sides bleed identically (−\$16.62 / −\$14.92).
abs_veto_long / abs_veto_short	1.0/1.5 and 1.5/2.5	The lab's thrust rows use RAW entries, so they do		That is the gap in tonight's work: the gate

GATE	LIVE TODAY	WHAT THE EXTENDED LAB PROVES	RECOMMENDED POLICY	WHY
		not speak to the vetoed gate.	NO CHANGE — and re-run this lab on abs_veto_55s's own entries next week.	we are promoting is the one gate whose exits this lab did not grade.
rgv_short, nipc_*	unchanged	rgv MED-TREND's +\$8.30 fails strip-3 at -\$16.50 and is 1 positive cell of 48.	NO CHANGE.	One positive cell in a 48-cell grid is a needle, and a needle is what a curve fit looks like.

The plateau read, because a proven cell has to sit on a plateau and not a spike. grind MED-TREND, \$/signal across the Lot-A × Lot-B grid — the whole top-left corner is positive and it degrades smoothly and monotonically as Lot B widens:

LOT A \ LOT B	1.0R	1.5R	2.0R	2.5R	3.0R	3.5R	4.0R	5.0R
0.50R	+6.2	+3.3	+0.2	−2.2	−3.6	−8.2	−7.2	−5.5
0.75R	+7.7	+4.7	+1.7	−0.7	−2.1	−6.7	−5.7	−4.0
1.00R	—	+4.6	+1.5	−0.9	−2.3	−6.9	−5.9	−4.2
1.50R	—	—	−1.4	−3.8	−5.2	−9.8	−8.8	−7.1
2.50R (live)	—	—	—	—	−10.7	−15.3	−14.3	−12.5

Chandeliers in the same rung: k1.5 −\$4.00, k2.5 −\$6.70, k3.5 −\$12.60, wide-6.0 −\$16.10. **Five adjacent cells positive, a monotone gradient, and the live setting sitting in the worst corner.** That is what a plateau looks like, and it is the one place in this lab where I would actually move a live knob — banded, and with the forward arm still running as the referee.

For contrast, the same grid on BIG-TREND (n=19): +0.6, −0.8, +2.0, +4.2, −0.6, −4.3, −6.8, −2.0 across the first row. Fourteen of 48 cells positive, scattered, no gradient. **That is not a plateau, it is confetti** — and it is what you get when you fit 48 cells to 19 trades.

8 · BOARD INTEGRITY — two A/Bs on the slate are reporting a perfect null because they are the same arm twice

While ranking the board I checked the paired arms against each other row by row, and two pairs are **byte-identical on every field** — entry timestamp, side, entry price, exit timestamp, exit price, exit reason and repriced P&L:

PAIR	N EACH	IDENTICAL ROWS	WHAT IT WAS SUPPOSED TO TEST
251		251 / 251	

PAIR	N EACH	IDENTICAL ROWS	WHAT IT WAS SUPPOSED TO TEST
cx_clip_brk_live vs cx_clip_brk_standdown			whether the quiet-tape clip should stand down on a range-break ignition
capit_flip_live vs capit_flip_t90	29	29 / 29	whether capitulation's flip-wait should have a 90-second patience limit

A perfectly null A/B and a broken A/B look exactly the same on a scoreboard, and these two have been sitting on the board reporting "no difference" for 13 and 13 days respectively. Both are consuming a slate slot and producing nothing. Neither result can be quoted. The range-break stand-down and the 90-second flip timeout are **both untested**, not "tested and null" — the treatment arm never diverged from the control, which means either the condition never fired or the flag is not wired through to the exit path.

Saturday card: add an assertion to the slate builder that any declared A/B pair whose rows are 100% identical after 20 trades raises a WARN into exec_health. This is the second week running that a paired arm has silently recorded the control's trades under a second name.

9 • The MGC shadow board

Gold has its own store (data/shadow_mgc.db) and its own multiplier — **\$10.00 a point, never MNQ's \$2.00**. Three sims, 132 trades, ten days.

SIM	SIDE	N	NET	WIN%	\$/TR	WK N	WK NET
mgc_holebreak_fade_long n<40	LONG	11	\$16.50	54.5%	+\$1.50	9	\$141.50
mgc_holebreak_fade_short n<40	SHORT	8	-\$200.00	37.5%	-\$25.00	4	-\$3.00
mgc_break_fade_nobook	ALL	113	-\$1,721.50	41.6%	-\$15.23	67	-\$750.50
— long side	LONG	56	-\$340.00	44.6%	-\$6.07	38	-\$407.00
— short side	SHORT	57	-\$1,381.50	38.6%	-\$24.24	29	-\$343.50

Nothing to promote and nothing to relegate. mgc_break_fade_nobook is the only arm with meaningful n and it is losing four times as much on the short side as the long — which is the per-side signal to tune if it survives another fortnight, not to relegate. Every one of its 113 exits is either STOP (57) or CHANDELIER (56), which is worth flagging on its own: **this gate has never once reached a target** — and neither have the other two gold sims, so all 132 rows on the gold board are stops

and trails. Given what the MGC leads doc found — gold's runs are slow grinds, not thrusts, and the ceiling moves from ~\$83/week to ~\$346/week purely by widening the stop and holding to a clock instead of trailing — a fade arm that only ever stops or trails is running exactly the exit gold does not want. That is the first thing to change on the gold board.

10 · DISPOSITION — every lead this section touched

LEAD	VERDICT	EVIDENCE / REVIVAL CONDITION / NAMED TEST
abs_veto_55s, two-sided	LIVE promote Monday	\$5,302.50 / 433 / +\$12.25 a trade. Green in 7 of 7 ISO weeks. Strip-best-10 +\$3,899.00. Worst LODO +\$4,511.50. Positive in chop AND trend. On matched entries it refuses a −\$5,287.00 book and keeps a +\$5,239.00 one, on both sides.
thrust_short_absveto55 (the live-slot mirror)	SHADOW	\$2,047.00 / 146 / +\$14.02 against the raw control's +\$0.19. Held back only on n and on top-3-days concentration of 41%. Promote with the parent if you want the short slot switched first.
ONE-DIRECTION-PER-DAY on the open rider	SHADOW — the week's best new idea	−\$1,835.00 → +\$1,244.50 over 15 days / 119 trades. Family-wise placebo p=0.3%. Passes LODO, strip-3, both cadences, both stop widths, and both external arm groups. Fails strip-best-5 and does not hold on the 51-day replay outside the forward window. Revival to LIVE: two paired flip-locked arms forward for ~4 weeks.
The pre-session day classifier for the open rider	PARKED	Named test: family-wise placebo over the whole 220-rule family — real best \$1,829.50 against a shuffled-noise median of \$1,623.00, p = 29%. Plus a 10-day/5-day OOS split where all four top rules lose. Revive at ~45 rider days with the feature set fixed in advance so the search is one rule and not 220.
ONE day classifier serving BOTH the rider and the gold run-catcher	REFUTED	Named test: the 26-day coincidence table — 46% agreement against a 50% chance rate, i.e. a lift of −4 points, plus Spearman −0.047 / −0.058 / −0.119 on run size, range and ER. No reformulation of "green rider day" vs "\$300+ gold excursion" saves it. Untested variant: a shared global-risk-regime definition.
The rider's drift direction gate (sims 60–63)	SHADOW — keep running, do not judge	Now 11 paired days: gate costs −\$301.00, having blocked 6 entries, all winners. Six entries decides nothing in either direction. Needs ~40 blocked entries.
The rider's stop-width axis (2.0 vs 3.0 ×ATR)	PARKED	3.0× wins on 4 of the 5 days both ran (+ \$6.29 vs −\$15.09 a trade) but has since lost \$1,835.00 over 15 days. The target is coupled to the stop in these arms, so it was never a clean stop-width test. Revive by re-running the A/B with decouple_target on.
The two-ratchet runner clip	PARKED — stays dead, watch stays open	Zero clips, three runners, five weeks — but the watch's own cross-check drifts − \$478.00 against a −\$366.50 signal, so it

LEAD	VERDICT	EVIDENCE / REVIVAL CONDITION / NAMED TEST
		cannot currently be read. Revive the watch by bucketing MANUAL_CLAIM exits separately; revive the ratchet only on a genuine trend week with zero clips.
The 2R-partial smoothness lever	SHADOW — your call, and I would take it	+\$453.00 on mean, −28% max drawdown, −15% worst day, first green day on the board — but +62% on daily volatility, which is the metric it exists to fix, on six days. Take it on the mechanism, never on the number.
The exhaustion fade-scalp trial (A 0.75R / B tight)	REFUTED as configured	Named test: the actual live fills for its real window — −\$480.00 over 80 lots at 41% win with 59% of exits stopping, against 44% under the prior config. It was correctly retired on 08-16. The reprice adds the mechanism: it helps in chop (+\$1.61/sig) and hurts in trend (−\$11.16/sig).
The BIG-TREND exit rung as defined on ER-30	REFUTED	Two named tests: (1) the extended sweep — every family's best BIG-TREND cell fails strip-the-best-3 on 822,746 bars plus a separate desk's OOS pool; (2) occupancy — the state is 3.00% of the clock in 544 episodes of median length 2 minutes, and slowing the ER window makes it rarer, not longer. Revive by redefining the rung on a state that persists (range-break-and-hold, ADX, day-net > 2×ATR + VWAP side), then re-run this lab.
The grind MED-TREND rung, A 0.75R / B 1.0R	SHADOW → deploy BANDED	+\$7.70/sig on n=215; passes strip-3 (+ \$5.60), LODO (+\$6.20) and both halves; five adjacent cells positive with a monotone gradient. Held back from LIVE by a −\$4.50 V5 out-of-sample leg and by its own forward arm being −\$8.36 a trade on n=21.
capitulation exits, all four rungs	REFUTED (as an exit problem)	Named test: 0 of 48 cells positive in every rung, n=1,157. Both sides bleed at the same rate (−\$16.62 / −\$14.92). There is no exit that saves this gate; the fault is the entry and it belongs in rehab.
cx_clip_brk_* and capit_flip_* A/Bs	PARKED — never actually ran	Both pairs are byte-identical, 251/251 and 29/29. Revive by fixing the wiring and adding a slate assertion that a 100%-identical pair raises a WARN after 20 trades.
MGC shadow board	SHADOW	Nothing with n. mgc_break_fade_nobook at −\$15.23 a trade over 113, and every one of its 132 exits is STOP or CHANDELIER — it has never reached a target, which is the wrong exit shape for an instrument whose runs are slow grinds.
rgv LONG side	HOLD — not an action	−\$6.51 a trade over 769 against SHORT's + \$0.85 over 234. Router-managed already; this week's numbers do not change it. Do not relegate the direction.

11 · What I would do, in order

1. **Promote abs_veto_55s two-sided on Monday.** It is the only thing on the board that survives every test we own, and the long side — the one everybody's instinct says to drop — is worth +\$7.98 a trade once the veto is on it.
2. **Add two flip-locked open-rider arms** (odr_c5_s30_f, odr_c10_s30_f), paired against their existing twins so the lock is the only difference. One-bit change, zero risk, and it settles the week's best idea in about a month. **Do not turn the flip lock on anywhere live** — the 51-day replay does not back it outside the window I found it in.
3. **Take the 2R-partial on grind**, on the mechanism, and stop quoting its number.
4. ★ **REV2 — WITHDRAWN AS A DEPLOY.** This item read “deploy the grind MED-TREND exit branch banded to ER-30 in [0.30, 0.50)”, and the MONDAY window and HOLD #4 both said data/exit_overrides.json is not touched — two opposite instructions about one file on one Monday. **The single verdict is: it is NOT deployed**, and the cause of death is named. \$7.2 grades this cell PROVEN on a THREE-test bar (strip-3 +\$5.60, LODO +\$6.20, both halves positive, +\$7.70/signal on n=215, five adjacent cells positive). HOLD #4's “20 of 20 failed” is a FOUR-test bar that adds the out-of-sample POOL split, and this is the one cell that clears three and fails the fourth: its **V5 out-of-sample pool reads -\$4.50/signal on n=52**, and its own live forward arm (rung_grindA_med_05, running since 08-16) reads **-\$8.36/trade on n=21** against a control at -\$9.29 — the rung is losing money live in BOTH configurations. Paired on the 19 shared entries the proposal does beat the live ladder by +\$60.50, so the lab and the forward arm agree on DIRECTION and disagree on level; that is an entry problem wearing an exit costume, and it is not a reason to move a live exit cell. It stays in shadow with rung_grindA_med_live as referee. Deploy when EITHER out-of-sample leg turns positive.
5. **Fix three measurement faults** before next Friday: the two-ratchet repricer must bucket MANUAL_CLAIM separately; exhaustion_trial_grade.py must read the config eras out of the git history of exit_overrides.json instead of assuming one boundary; and the slate builder must WARN on byte-identical A/B pairs.
6. **Redefine BIG-TREND** on a state that lasts longer than two minutes, then re-run the exit lab against it. Until that happens, every “ride the big trend”

result on this desk is being computed on a regime that is over before the trade is.

★★ REV2 — WHERE EACH OF THESE SIX ACTUALLY WENT. FIVE OF THE SIX WERE ABSENT FROM THE CARD ENTIRELY, WHICH IS THE SAME FAILURE THIS SECTION NAMES ELSEWHERE

§11 ITEM	WHERE IT IS NOW
The two flip-locked open-rider arms (odr_c5_s30_f, odr_c10_s30_f)	ON THE CARD: BUILD #18 (shadow-flip-locked-open-rider-arms-0904). Shadow pair only; the lock does not go live anywhere.
The three measurement fixes	ON THE CARD: BUILD #19 (three-measurement-fixes-0904). The third of them — WARN on byte-identical A/B pairs — is the build-time guard that would have caught BUILD #5's pair.
Deploy the grind MED-TREND branch	WITHDRAWN — see the struck item above. Cause of death: the out-of-sample leg. HOLD #4 carries the single verdict.
Take the 2R-partial on grind, on the mechanism, and stop quoting its number	NOT CARRIED, deliberately. It is an instruction to believe a mechanism while explicitly disowning its number, which is not a shippable row — there is nothing to verify and nothing that could kill it. It stays as a standing reading of the two-ratchet result and it needs BUILD #19's first fix (bucket MANUAL_CLAIM separately) before its number means anything at all.
Redefine BIG-TREND on a state that persists	Already on the card as the revive condition attached to NOT-AN-ACTION #5 (the BIG-TREND rung's refutation). It is a research programme, not a weekend action, and issuing it as one would be theatre.

THE THREE HONEST LIMITS ON EVERYTHING ABOVE

1. **The shadow board over-trades and the paper desk judges.** These sims nominate; they place no orders, pay no slippage beyond the tick reprice, and never compete for a slot with each other. abs_veto_55s at 433 trades in seven weeks is roughly ten times the live tournament's rate. Expect the live version to be smaller and slower than the shadow number.
2. **One week, and a quiet one.** The live tournament took **six trades** between Monday and Friday. Almost every conclusion here rests on shadow n, not live n, and grind_long, exhaustion_short and the rgv family took essentially nothing live this week.
3. **No odr arm may be promoted from this report.** Every one of the eight is flagged `n<40` for a forward A/B claim, the whole family is — \$1,835.00 over its 15 days, and its 51-day replay makes all its money in

a window when it was not trading. The flip rule is a finding about the rider, not a licence to run it.

P2.5 MID-WEEK MUSINGS — the fresh leads, and gold gets its own chapter

Garrath — lead with what survived, because two things did and both are worth money. **First, the standing grind question finally has an answer, and it is none of the three options you asked me to choose between.** Grind's losses are not chop-churn to be filtered with efficiency and they are not a trend-day permission problem: **on 159 replayed fires the gate's entire bleed is a REGIME and a CLOCK** — benching it in ordinary chop and on clean trends turns +\$922 into **+\$2,310**, and benching it in the 13:00Z US-open hour alone turns it into **+\$2,540**. Both survive stripping their best three trades and both survive leave-one-day-out, which nothing in this family has ever done before. **Second, gold's big finding held up under a much harder test than the one that found it.** The claim that gold's runs are grinds and that the exit was the cap, not the filter re-derives on 32 contiguous days with spread-honest fills: the same 165 entries make **-\$913** on a momentum trail and **+\$2,702 on a wide stop held long**, and the mirror and random-entry controls are deeply negative at every single cell, so the entry is real and the exit really was the ceiling.

And lead with the honest half too, because it is bigger. **Gold's most promising shape — the MGC day rider, the 15-cell plateau — is dead.** The plateau replicates perfectly inside its own window and then dies on 185 out-of-sample sessions that were sitting in the lake the whole time: direction holds **78-85% on the 26 days we looked at and 49-54% on the 185 we had not**, and **0 of 15 cells is positive**. The coil bouncer and the London-open fade go the same way for named reasons. The close of the 08-14 gold session was "the missing ingredient is DAYS, not

configurations." The days arrived on 2026-08-19 in a backfill nobody announced. They say no.

READ THESE FIVE FLAGS BEFORE ANY NUMBER BELOW

(1) The tournament barely traded, so almost nothing here is this week's realised book. Six tournament trades all week, all on Friday 08-28, all winners, **+\$330.00** — the stand-down ran Monday and Tuesday and was lifted on the 26th. Every grind number below is therefore a replay: the real gate_grind firing on real bars, with the deployed $ATR \geq 22$ floor applied first, exits repriced tick-by-tick as the live two-lot A/B slate (Lot A 2.5R scalp on a $1.0 \times ATR$ stop, Lot B the loose-then-lock chandelier), \$1.50 per round trip per lot, one position at a time, 180-minute cap, entry on the first tick after the signal bar closes. It is what the deployed configuration would have done across all the tape we hold — not the historical book, which ran different exits.

(2) The router's operational grade is NOT in this section. That is \$2.6's job and I have not repeated it. What is here is a different question: whether the instruments we grade the router with can be trusted. Three of them cannot.

(3) Gold is priced two ways and only one of them counts. Every gold study on this desk to date charged \$1.50 per round trip and raced on the trade tape, which quietly assumes you were filled at the last traded price on both legs. MGC's median spread is **0.30pt = \$3.00**, and both our entries and our closes cross. The honest cost is **\$7.50 per round trip**. Where it matters I show both, and the honest one decides.

(4) MGC is \$10.00 a point, MNQ is \$2.00. Every gold dollar below uses \$10.

(5) Thin is thin. 34 tick-days for grind, 32 contiguous days for gold's quote tape, 185 out-of-sample gold sessions. Nothing here is a year of live trading and I say so at each table rather than once at the top.

1 · THE STANDING GRIND LEAD — which enable-mechanism actually works?

Your standing question is: what best captures grind's trend-day upside while killing its chop-churn — (a) trend-day-only, (b) its own entry trend-confirmation or veto, or (c) the current ER-0.35 floor? I judged this on **live/managed grind only**, never on the unmanaged grind_fast shadow, which has no stop discipline and reads nothing like the gate we run.

1.1 — Before choosing: option (c) does not exist

There is no ER-0.35 floor on grind. I imported the live module rather than trusting the documentation:

WHAT WE BELIEVE IS IN FORCE	WHAT GAZBOT7.DECIDERS ACTUALLY HOLDS	CONSEQUENCE
an ER floor of 0.35 on grind_long	ER_FLOOR = {} — empty	no ER floor has ever blocked a grind entry
—	ATR_FLOOR = {'grind_long': 22.0, 'capitulation_long': 10.0}	an ATR floor of 22 is the ONLY entry filter grind carries

So the real choice is between (a), (b), and the thing that is actually deployed. I have scored (c) anyway, as the hypothetical it is, because it keeps getting proposed and it deserves a number rather than an opinion.

1.2 — The control: what the deployed grind does across all our tape

THE DEPLOYED CONFIGURATION, REPLAYED	VALUE
fires (real gate_grind, ATR≥22 applied first)	159
tick-days covered	34
net, both lots, \$1.50/RT per lot	+\$922.00
\$ per fire	+\$5.80
win rate	30%
average win / average loss	+\$317 / −\$125
green days	18 / 34
strip the best THREE fires (\$728 + \$669 + \$664 = \$2,061)	−\$1,139.00
leave-one-day-out, worst fold	−\$117.00
Lot A exit mix	112 STOP · 46 TARGET · 1 CAP
Lot B exit mix	122 STOP · 33 CHANDELIER · 4 CAP

Two things to take from the control before any filter touches it. It is **positive** — the deployed ATR floor is already doing real work, and "grind loses money" is not true of the configuration we actually run. And it is **fragile**: three trades are the entire result. A 30% win rate with a 2.5-to-1 payoff is exactly the shape you told

me not to kill on win rate alone, so I have judged everything below on expectancy and robustness. But a result that dies when you remove its best three fires is not something to enlarge — it is something to make sturdier, and that is what the rest of this chapter is for.

1.3 — The four families, scored on the same 159 fires

Every rule below is causal. The day-level rules are built only from bars strictly before the day's first eligible fire; the entry-level rules use only what the gate itself could see at the signal. **Read the last five columns, not the third** — a rule that improves the total by throwing away winners is a fake, and a rule that cannot beat dropping the same number of trades at random has told you nothing.

FAMILY	RULE	N	NET	\$/FIRE	WIN%	STRIP BEST 3	LEAVE- ONE- DAY- OUT	GREEN DAYS	BEATEN BY RANDO M
base	every live grind fire (the control)	159	+\$922	+\$5.80	30%	−\$1,139	−\$117	18/34	—
(a) trend-day only	day roundtrip ≥ 0.35, measured to the first fire	111	+\$2,423	+\$21.83	32%	+\$362	+\$1,384	14/22	5%
(a) trend-day only	day roundtrip ≥ 0.65, same window	40	+\$2,826	+\$70.64	35%	+\$842	+\$1,786	8/10	1%
(a) trend-day only	day efficiency ≥ 0.10 over the 180min before the first fire	65	+\$1,851	+\$28.48	32%	−\$48	+\$1,180	9/11	10%
(b) entry confirm	last 30min net is UP	137	+\$800	+\$5.84	30%	−\$1,171	−\$239	18/33	53%
(b) entry veto	60min VWAP slope not DOWN	76	+\$420	+\$5.52	32%	−\$1,146	−\$504	12/25	50%
(b) entry confirm	15min efficiency ≥ 0.35	48	−\$403	−\$8.40	27%	−\$2,059	−\$1,018	7/22	71%
(b) entry, stacked	30min net UP and slope not DOWN	72	−\$108	−\$1.50	29%	−\$1,674	−\$1,031	12/25	62%
(b) entry veto	extension ≤ 1.0×ATR (do not chase)	66	−\$420	−\$6.37	30%	−\$2,174	−\$1,092	11/31	69%

FAMILY	RULE	N	NET	\$/FIRE	WIN%	STRIP BEST 3	LEAVE- ONE- DAY- OUT	GREEN DAYS	BEATEN BY RANDO M
(c) ER floor (phantom)	30min efficiency $\geq$ 0.25	36	-\$638	-\$17.71	22%	-\$2,198	-\$1,102	6/16	73%
(c) ER floor (phantom)	30min efficiency $\geq$ 0.35 — the number people mean	15	+\$415	+\$27.67	33%	-\$890	-\$25	5/11	28%
(c) ER floor (phantom)	30min efficiency $\geq$ 0.45	4	-\$210	-\$52.62	25%	-\$176	-\$444	1/3	64%

Two notes on that last column before you read it. For the **day-level** rules the null is dropping the same number of DAYS at random, and for the **entry-level** rules it is dropping the same number of FIRES at random — a day rule removes whole days, so scoring it against a per-trade null would flatter it. Both are 500-draw Monte Carlo estimates, so read them to the nearest few percent. And hold the two green (a) rows lightly until §1.4, which shows that most of what makes them look that good is the window I measured them over, not the market.

Option (b) is dead and the cause of death is the placebo. Every single entry-level confirmation and veto is beaten by **50% to 71%** of random removals of the same number of trades — which is to say, they carry no information at all beyond "trade less". Four of the five also make the total worse, two of them turning a positive book negative. There is no version of "make grind confirm its own direction before it enters" that works on this evidence.

Option (c) is worse than doing nothing, even as a hypothetical. The 0.35 cell looks alright at \$27.67 a fire until you notice it is 15 trades, it does not survive stripping its best three, and 28% of random draws of 15 fires beat it. That is now the fourth time this desk has tested "fix grind with an efficiency filter" and it has failed every time. It should stop being proposed.

1.4 — And then the finding that breaks the best-looking rule

Option (a) looks superb in that table: roundtrip ≥ 0.65 gives \$70 a fire, survives everything, and only 1% of random draws beat it. **It is also not deployable, and the reason is a measurement artefact I introduced myself.** The window I measured "the day's character" over ended at the day's first fire — which means its length varies with the day, and the length of a quiet stretch before the first

signal is itself a fact about the day. In production the router must decide before it knows when the first fire will be. So I recomputed exactly the same rule on a **fixed clock cut** instead:

DAY- ROUNDTRIP MEASURED UP TO...	THRESHOLD	DAYS ARMED	N	NET	\$/FIRE	STRIP BEST 3	LEAVE- ONE-DAY- OUT	BEATEN BY RANDOM
the day's first fire (not deployable)	0.65	10	40	+\$2,826	+\$70.64	+\$842	+\$1,786	1%
07:30Z (London open)	0.55	8	38	+\$289	+\$7.61	−\$1,694	−\$750	48%
11:30Z	0.55	11	54	−\$884	−\$16.38	−\$2,868	−\$1,924	79%
13:30Z (US open) — the best deployable cut	0.55	16	78	+\$1,518	+\$19.46	−\$543	+\$479	22%
13:30Z	0.65	8	31	+\$253	+\$8.16	−\$1,724	−\$233	50%

Most of the effect was the window, not the world. The best runnable version — measure the day's roundtrip at the 13:30Z open, arm grind only above 0.55 — still beats the control, +\$1,518 against +\$922 on half the fires, and its worst leave-one-day-out fold is positive. But **22% of random 16-day selections beat it**, and at 11:30Z the whole family is underwater. That is a rule with a real but modest effect wrapped in a lot of noise, and it is nowhere near the +\$2,826 headline. **Trend-day arming is the best of your three options and it is still not good enough to ship.**

1.5 — The answer nobody asked for: grind's money is not on trend days at all

This is the part that changes the question. I split the 159 fires by the regime at the fire — ATR terciles against the 30-minute efficiency, exactly the segmentation your backtest rule demands, everything causal:

REGIME AT THE FIRE	N	DAYS	NET	\$/FIRE	WIN%	GREEN DAYS	LEAVE- ONE-DAY- OUT	READ OUT
VIOLENT_WHI PSAW ATR ≥ 32 · ER < 0.30	45	23	+\$1,156	+\$25.68	29%	12/23	+\$284	its best home, and broadly spread
BUILDING mid ATR · ER ≥ 0.30	16	10	+\$862	+\$53.88	38%	5/10	+\$386	richest per fire, thin n

REGIME AT THE FIRE	N	DAYS	NET	\$/FIRE	WIN%	GREEN DAYS	LEAVE- ONE-DAY- OUT	READ
DEAD_CHOP ATR < 26 · ER < 0.30	43	16	+\$292	+\$6.80	33%	7/16	−\$129	roughly a wash
NORMAL_CHOP mid ATR · ER < 0.30	47	24	−\$731	−\$15.55	28%	7/24	−\$1,195	this IS the chop-churn
CLEAN_TREND ATR ≥ 32 · ER ≥ 0.30	8	7	−\$657	−\$82.12	12%	1/7	−\$1,097	the worst cell in the table

Read the top and the bottom rows together, because between them they invert the premise of the whole question. **Grind's upside is not on clean trend days — clean trend is its worst regime, green on one day in seven.** Its money comes from violent-whipsaw tape and from the building phase, and its bleed is ordinary mid-volatility chop. Eight fires is a thin cell and I am not going to pretend otherwise, but one green day in seven and −\$82 a fire is a direction, not a rounding error.

That also explains, mechanically, why every option-(b) rule failed. An entry-level efficiency confirmation selects for exactly the CLEAN_TREND cell where grind loses the most. You were asking me to filter towards the thing that costs it money.

1.6 — So: (d) a regime veto, and (e) an hour that eats the whole loss

MECHANISM	N	NET	\$/FIRE	WIN%	STRIP BEST 3	LEAVE- ONE-DAY- OUT	GREEN DAYS	BEATEN BY RANDOM	FIRST HALF	SECOND HALF
control — every live grind fire	159	+\$922	+\$5.80	30%	−\$1,139	−\$117	18/34	—	+\$1,903	−\$981
(d) benchmark NORMAL_CHOP and CLEAN_TREND	104	+\$2,310	+\$22.21	32%	+\$249	+\$1,430	18/29	10%	+\$2,072	+\$238
(d) arm only in VIOLENT_WHIPSAW or BUILDING	61	+\$2,018	+\$33.07	31%	−\$44	+\$1,240	15/27	13%	+\$1,464	+\$554
(e) benchmark the 13:00Z US-open hour	109	+\$2,540	+\$23.30	35%	+\$647	+\$1,501	15/28	7%	+\$2,240	+\$300
	98	+\$3,214	+\$32.79	36%	+\$1,320	+\$2,174	16/28	3%	+\$2,804	+\$410

MECHANISM	N	NET	\$/FIRE	WIN%	STRIP BEST 3	LEAVE-ONE-DAY-OUT	GREEN DAYS	BEATEN BY RANDOM	FIRST HALF	SECOND HALF
(e) bench 13:00Z plus the 19-20:00Z close hours										
(d) and (e) together	72	+\$2,299	+\$31.93	35%	+\$406	+\$1,628	13/23	10%	+\$2,172	+\$126
raise the ATR floor 22 → 26	106	+\$533	+\$5.03	28%	-\$1,528	-\$600	14/32	53%	+\$1,740	-\$1,208
raise the ATR floor 22 → 32	53	+\$498	+\$9.41	26%	-\$1,562	-\$374	11/24	43%	+\$1,552	-\$1,054

The last two rows close a question that has been open since the last cycle, which reframed this whole lead as "the real question is whether 22 is the right floor". The answer is **no — the floor is not the lever**. Raising it to 26 or to 32 makes grind worse on every robustness column and is beaten by roughly half of all random draws. What matters is not how violent the tape is, it is the joint state of volatility and efficiency.

And the 13:00Z hour deserves its own sentence because it is the single cleanest fact in this chapter. **Grind's fires in the US-open hour: 50 of them, across 25 separate days, −\$1,618, green on 8 days of 25, an 18% win rate against 30% everywhere else.** Strip its three worst fires and it is still −\$1,022, so it is not one bad morning. Grind is a continuation gate and 13:30Z is the one moment of the day guaranteed to break continuation.

THE HONEST LIMITS ON (D) AND (E), BECAUSE THEY ARE NOT SMALL

Neither clears a 5% placebo. (d) is beaten by 10% of random 104-fire draws and (e) by 7%. That is much better than everything in §1.3 and it is not proof.

The second half is much weaker than the first. (d) makes +\$2,072 in the first 17 days and +\$238 in the last 17. The effect is real in both halves and it is nine times smaller in the recent one. That could be regime, it could be decay, and 17 days cannot tell you which.

(e) was chosen after I looked at the hour table. Its 7% placebo is not charged for that search, and "bench three specific hours" at 3% is charged even less. I am reporting (e) because it corroborates (d) — both say the damage is where continuation breaks — not as a rule to ship on its own.

(d) was not chosen on P&L. The five-way regime taxonomy is the one we already use for gold and it was written down before it was scored here. That is the main reason I trust it more than the clock rule, even though the clock rule scores better.

Verdict: SHADOW, not LIVE. Put (d) into the shadow book as a regime-gated grind arm — it is free, it is riskless, and it needs about four more weeks before the second-half number means anything. What would promote it: the second half reaching parity with the first, and a placebo under 5% on a sample that includes days I have not looked at. What would kill it: NORMAL_CHOP turning green, or CLEAN_TREND turning green, on the new days.

2 · THE WEEK'S INSTRUMENT LEADS — three scoreboards that are measuring the wrong thing

The router's operational grade lives in §2.6 and I have not repeated a line of it. This is the adjacent question, and it is the one the scope keeps asking: **can the instruments we grade with be trusted?** Three of this week's cannot, and each failure is a different shape.

2.1 — The exit selector grades a switched-off feature, and its model disagrees with the book by the SIGN

Friday 08-28 is the only day this week with a real selector scorecard, and it reads **13 trades, 4 optimal picks, \$1,576 of regret, selector P&L −\$600**. Six of those thirteen rows are the tournament's six real trades. Here is the model's number beside what the account actually banked:

TRADE (08-28)	EXIT THE SELECTOR SAYS WAS CHOSEN	SELECTOR'S P&L	SELECTOR'S BEST MODE	ITS P&L	"REGRET"	WHAT THE DESK ACTUALLY BOOKED
capitulation_long _A	wide	−\$59.0	scalp	+\$113.5	\$172.5	+\$69.50

TRADE (08-28)	EXIT THE SELECTOR SAYS WAS CHOSEN	SELECTOR'S P&L	SELECTOR'S BEST MODE	ITS P&L	"REGRET"	WHAT THE DESK ACTUALLY BOOKED
capitulation_long_B	wide	-\$59.0	scalp	+\$113.5	\$172.5	+\$79.00
grind_long_A	wide	-\$57.5	mid	+\$117.5	\$175.0	+\$41.50
grind_long_B	wide	-\$57.5	mid	+\$117.5	\$175.0	+\$22.00
abs_veto_long_A	wide	-\$57.0	mid	+\$139.5	\$196.5	+\$45.00
abs_veto_long_B	wide	-\$57.0	mid	+\$134.0	\$191.0	+\$73.00
the six legs together	—	-\$347.00	—	—	\$1,082.50	+\$330.00

Six trades. The desk made money on all six. The selector's model says it lost money on all six. That is a **\$677 gap with the sign reversed on every row**, and \$1,082 of the day's \$1,576 "regret" headline is computed inside it.

Three separate reasons, all of them structural rather than a bug:

WHY THE MODEL AND THE BOOK DISAGREE	EVIDENCE
The "chosen" mode is a reconstruction, not an observation. It calls <code>SlotStrategy._regime_mode()</code> to guess what a selector would have picked.	the desk does not run any of the four modes — it runs the A/B scale-out slate
Every mode reprices on a hard-coded $1.0 \times \text{ATR}$ stop, so all four arms share one stop the desk does not use.	<code>stop_atr_mult=1.0</code> in all four branches of <code>_mode_exit</code>
Four of the six legs exited on <code>MANUAL_CLAIM</code> — your finger, not the gate's exit stack. No model exit can reproduce those.	<code>grind_long_A/B</code> and <code>abs_veto_long_B</code> all closed <code>MANUAL_CLAIM</code> between 14:50 and 14:51Z

And underneath all of it: **the feature being graded is switched off**. I imported `scaleout_slots()` and all twelve sub-slots return `adaptive_exit=False`. So the scorecard is a simulation of a disabled feature, priced with a stop the desk does not use, on trades a human closed by hand. **The \$1,576 is not a measurement of anything.**

2.2 — The router's net-value number is one-sided by construction

This one is easy to miss because the number it produces is usually reasonable. `router_value = saved - missed`, and both terms are computed entirely from SUPPRESSED-OPEN lines — that is, from signals the router BLOCKED. The only thing it records about an ARMED gate is leakage, and leakage counts **losers only**. There is no term anywhere in the file for a winner that an arm produced.

FRIDAY 2026-08-28	WHAT THE NIGHTLY RECORDED	WHAT ACTUALLY HAPPENED
blocked signals repriced	34	—
blocked losses saved	+\$1,888	—
blocked wins missed	-\$2,005	—

FRIDAY 2026-08-28	WHAT THE NIGHTLY RECORDED	WHAT ACTUALLY HAPPENED
net router value, as printed	-\$117	—
P&L from the three gates the router ARMED at 14:45Z and 16:00Z	not counted — no term exists	+\$330.00, six for six
honest net for the day	—	+\$213

A \$447 swing and a sign flip on the one day this week the desk actually traded. The fix is small — add realised P&L from managed gates while ON to the value term — and until it is made, **every daily router value on record is a lower bound on days the router armed anything, and exactly right on days it armed nothing.** Most days it armed nothing, which is why nobody has noticed.

2.3 — The blanket stand-down out-scored the selective router, four to one per day

The week splits itself into a natural experiment nobody designed. Monday and Tuesday the whole six-gate roster was stood down. The stand-down was lifted on Wednesday 08-26 and the router held full authority for the last three days. Same instrument, same scalp-2R ruler, same nightly:

REGIME OF CONTROL	DAYS	BLOCKED SIGNALS	LOSSES SAVED	WINS MISSED	NET VALUE	PER DAY
STOOD DOWN — every gate off, no judgement at all (08-24, 08-25)	2	70	+\$3,328	-\$2,690	+\$638	+\$319
ROUTER HAS AUTHORITY — selective benching (08-26, 08-27, 08-28)	3	124	+\$6,198	-\$5,969	+\$229	+\$76

On this week's tape, **doing nothing at all beat doing something clever by roughly four to one per day.** I want to be careful with that, because it is two days against three, on different tapes, on a ruler that is known to be more generous than the live desk, and §2.2 above says the authority days are understated by the +\$330 the arms actually earned (which lifts them to +\$186/day, still short). It is not a recommendation to stand the desk down. **It is a challenge to the bar:** whatever the router is worth, it has to clear "switch everything off", and this week it did not.

2.4 — Churn, measured properly, has collapsed — and here is the honest baseline

The scope carries a churn alarm built from a four-hour window on 08-07 — "20 switch changes, only 4 produced a trade, 5 per hour". I recomputed the rate on whole days so the comparison is like for like, counting each gate-value change in the trial log's changed: field:

WINDOW	GATE-VALUE CHANGES	DAYS	CHANGES PER HOUR	NOTE
2026-08-07 (the reference day)		17	1	0.71 the day the alarm was written
2026-08-11 (the worst day on record)		23	1	0.96 —
2026-08-17 ... 08-21		26	5	0.22 —
2026-08-24 ... 08-28 (this week)		22	5	0.18 a 4× reduction on the reference day
— of which Monday and Tuesday, under the stand-down		1	2	0.02 —
— of which Wed-Fri, router with full authority		21	3	0.29 still below every August reference

Split by direction, the week's 17 change-events were **6 arms against 16 benches**. The router is far more willing to switch a gate off than on, which is the asymmetry the bad-call ledger asks for. **Churn is no longer this desk's problem and the dwell-time spec the scope asks for is not needed.** That is a lead closing cleanly, which is worth as much as one opening.

2.5 — It armed a gate this morning on a tape with nothing on it

Not a mid-week lead — it happened while I was writing this, and it is the sharpest example of the failure mode in the file. At **2026-08-29 07:20:25Z**, a Saturday, roughly ten hours into a CME halt that runs until Sunday 22:00Z, the router armed capitulation_long. Its own words:

2026-08-29T07:20:25Z — VERBATIM FROM THE TRIAL LOG

"07:20Z LONDON, the desk's only positive block — flat, healthy=true, halted=false, day total +0.0, **0 bars in 45min so NO segment read** (direction rule silent), DAY BIAS insufficient-bars, meter 50 is an explicit NO READ ... **so I route on the tape and the tape is empty**; correcting my own last four ticks, which asserted capitulation_long 'stays on' when the

switch has in fact read off ... it does not need a fresh positive case to sit armed and I am arming it rather than under-arming London..."

Three things are wrong at once and they compound. It reads halted=false through a halt. It calls a dead Saturday "LONDON, the desk's only positive block", and that phrase — a true statement about the desk's session P&L — is what supplies the arming pressure. And the actual justification is a **correction of its own state misread**, which is a fact about the router, not a fact about the market.

THE WEEKEND RECORD	VALUE
router ticks across Sat 08-22, Sun 08-23 and Sat 08-29 so far	645
gate changes in all of them	1
that one change	08-29 07:20:25Z, capitulation_long → on
hours from that arm to the next tick that can trade (Sun 22:00Z reopen)	38.7
bars available to justify it	0

The dollars are \$0 because there is no market, and the router will get 460 more ticks to change its mind before the reopen. **The lead is not the loss, it is the mechanism**: the Asia block expires on a clock boundary at 07:00Z, and the moment it does, the "London is our best block" reasoning switches on with nothing to read. One arm in three weekend-days is a rate, not an incident — but it is a rate that only needs to be wrong once, on a Sunday-night reopen gap, to matter. **Recommendation for the Saturday card: give the router a calendar.** A weekend and holiday check is about fifteen lines and it removes an entire class of decision made on zero information.

3 · ★★ GOLD — the required chapter, written for someone who has seen none of it

We have been capturing COMEX gold — MGC, both the quote tape and the order book — since 2026-08-04, and you asked for gates that work on it. Nothing gold-related has ever gone live and nothing here proposes to. This chapter exists because two findings from the 08-14 session changed the shape of the gold problem, and because both of them needed testing much harder than the session that produced them could manage. **One of them survived that test and got stronger. The other one died, and it died on evidence that was already sitting on the disk.**

3.1 — Why gold gets a fresh chapter when six attacks are already refuted

The honest starting point is that gold has beaten us six times. On 2026-08-05 a full session closed six separate attacks, and none of them failed on thin data — they failed on mechanism:

ATTACK	HOW IT DIED
a momentum / direction gate	5 of 72 cells positive; the best strips to \$16; always SHORT (–\$532) BEAT the directed gate (–\$1,274)
arm, then confirm after a delay	the confirmed side is a coin flip — 48-50% across all 24 cells
filter it with the live router meters	placebo-controlled and beaten by 31-80% of random removals
use the L2 book for direction, or as a veto	well powered, and the edge flips sign across five disjoint samples
fade extension on a clock	Hurst exponent on our own gold tape is 0.489 — a random walk
thrust-continuation	catches 22 of 22 runs and the losers are statistically indistinguishable from the winners

The only thing that ever survived a robustness test was an oracle day-filter — given foresight of which days contain a run, +\$779. That is your "let the router switch it off" idea, and it is an upper bound, not a result. **So why is this different from a seventh sweep?** Because the two 08-14 findings are not new thresholds on old mechanisms. One is a statement about the SHAPE of gold's moves that changes which exit you should even be testing. The other is a different trade entirely — one position a day, held to a clock, which is not a gate at all. Neither had been tried before, so neither is a retry.

3.2 — Before any gold number: three things were wrong with the ruler

I could not re-derive anything until I fixed the measuring stick, and two of the three faults are new this month. They matter enough that they would be a finding on their own.

Fault one — on 2026-08-19 a backfill quietly put four extra timeframes into the lake, and nothing filters them. The lake's bars view is a plain union over every parquet file in data/tape/bars/<symbol>/. Someone dropped backfill_1day, backfill_1hour, backfill_1min and backfill_5mins in there. Any study that reads bars without timeframe='5s ' now silently resamples daily and hourly bars as if they were five-second ones.

TIMEFRAME IN DATA/ TAPE/BARS/MGC/	ROWS	DISTINCT DAYS	SPAN
5s — our own capture, the only one anything should read	292,232	18	2026-08-05 → 2026-08-28
1min — arrived 08-19	357,692	320	2025-07-27 → 2026-08-18

TIMEFRAME IN DATA/ TAPE/BARS/MGC/	ROWS	DISTINCT DAYS	SPAN
5mins — arrived 08-19	26,419	138	2025-09-29 → 2026-08-19
1hour — arrived 08-19	10,233	547	2024-10-30 → 2026-08-19
1day — arrived 08-19	691	691	2023-11-30 → 2026-08-19

The damage is a look-ahead, not just noise. Resampling that union to one minute takes `high = max(...)` across all of them, so the 10:00 minute inherits the high of the 5-minute bar stamped 10:00 — a number not known until 10:05. Measured on the July archive days: **4,826 of 10,881 minutes (44.4%) carry an inflated range, mean inflation 2.089pt on a bar whose clean range averages 2.099pt.** The leak is worth roughly a whole bar. And the audit's own control — the August days, which used to be 5-second-only and were designed to read **zero** — now reads **9,469 inflated minutes.** That is the backfill landing, visible in a test built before it existed. **MNQ has the identical four files,** so this is not a gold problem; it is a lake problem, and it is on the Saturday card.

Fault two — gold has been priced at a fifth of its real cost. Every MGC study charged \$1.50 per round trip and raced exits on the trade tape. Our entries are marketable limits that cross and our closes are market orders that cross, so both legs pay the spread:

COMPONENT	VALUE
MGC median spread, measured over 5,613,958 book snapshots, flat round the clock	0.30 pt
two legs × 0.30pt × \$10.00/pt	\$6.00
exchange fee, per round trip	\$1.50
true round-trip cost on gold	\$7.50
what every gold study to date charged	\$1.50
gold's ATR, 33rd to 67th percentile — so 1R is roughly...	1.42 – 2.04 pt = \$14 – \$20

The spread is between a quarter and a half of one R on this instrument. On MNQ, where 1R is \$30–\$60 and the spread is a quarter of a point, you can get away with ignoring it. On gold you cannot, and every prior gold verdict is affected.

Fault three — a third of the new days are dead. The backfill's 320 MGC minute-days are not all real tape. I measured the share of minutes with zero price change on each day; a live gold minute changes price about 96% of the time.

MONTH IN THE BACKFILL	DAYS	DEAD DAYS	MEDIAN ZERO- CHANGE MINUTES	MEDIAN 1-MIN RANGE
2025-07 ... 2025-09	57	3	10–20%	0.30–0.60pt
2025-10	27	18	63.7%	0.00pt

MONTH IN THE BACKFILL	DAYS	DEAD DAYS	MEDIAN ZERO-CHANGE MINUTES	MEDIAN 1-MIN RANGE
2025-11 ... 2026-03		118	3	2-4%
2026-04		25	13	57.5%
2026-06		26	14	58.9%
2026-08		14	14	99.1%
usable after dropping the dead days		253	—	—

Before I would use a single one of those days I checked that the backfill is the same instrument we capture. On the ten July days both series cover, **they agree to 0.0pt on every daily close**. On the August days they diverge by 20-85pt, and the August days are exactly the ones that are 99% frozen — so the divergence is the staleness, not a contract-month mismatch. Once the dead days are dropped the extension is genuine, and that is what lets \$3.4 do what it does.

3.3 — LEAD ONE: gold's runs are grinds, not thrusts, and the EXIT was the cap — SURVIVES

The 08-14 claim, in your terms: gold's big moves are long slow grinds rather than sharp thrusts, so a trailing exit gets shaken out of them; boarding was never the problem, the exit was; and the ceiling moves from about \$83 a week on a 2.5×ATR trail to about \$346 a week on a wide stop held for hours. If true it inverts the order of the whole gold problem, because it means the filter question — the one that has failed six times — only becomes worth solving after the exit is fixed.

I re-derived it on **32 contiguous trading days** of the 250ms quote book (2026-07-16 → 2026-08-28, 5.6M snapshots, no gaps), with minute bars built from the quote mid so there is no construction seam, and every exit raced spread-honestly — a long's stop triggers when the bid breaks it and the long exits at the bid, never at the midpoint.

The description of the tape holds. Defining a run as a directional leg worth at least \$300 gross on one lot, ended by a 34% retrace:

START (UTC)	DIR	POINTS	\$ 1-LOT	HOURS	EFFICIENCY
2026-08-19 11:13	UP	153.6	\$1,536	10.8	0.18
2026-08-10 13:49	UP	108.1	\$1,081	12.4	0.16
2026-08-28 15:48	DOWN	87.7	\$877	2.9	0.20
2026-08-28 14:00	DOWN	82.1	\$821	0.5	0.41
2026-07-29 16:14	UP	73.7	\$737	2.0	0.33
2026-08-05 02:25	UP	73.3	\$733	4.6	0.24
2026-08-25 00:50	DOWN	72.2	\$722	2.0	0.28

START (UTC)	DIR	POINTS	\$ 1-LOT	HOURS	EFFICIENCY
2026-08-24 18:38	UP	70.9	\$709	6.1	0.24

Top five: median efficiency 0.20, median duration 2.9 hours, 3 of 5 last two hours or more. Across all 55 runs found: median efficiency 0.30, median duration 118 minutes, 49% last two hours or more. An efficiency of 0.20 means price travels five times the net distance it ends up covering — a grind, exactly as claimed (the 08-14 read was 0.23 on a shorter window). A trailing stop set to a couple of ATRs cannot survive that.

Now the test that matters. I built one boarding rule and then changed nothing about it: 165 signals, entered a minute after the bar that produced them, one per hour at most. Only the exit changes between rows, so any difference is the exit and nothing else.

EXIT — SAME 165 ENTRIES EVERY ROW	WIN%	NET AT \$1.50/ RT	NET AT THE TRUE \$7.50/RT	\$ PER WEEK, TRUE	MEDIAN TRADE
2.5×ATR trail · 1.5 stop · 3h the shape we ship on MNQ	30%	−\$284	−\$913	−\$143	−\$19
no trail · 1.5×ATR stop · 3h	12%	+\$72	−\$549	−\$86	−\$29
no trail · 3×ATR stop · 3h	26%	−\$47	−\$746	−\$117	−\$46
no trail · 5×ATR stop · 3h	38%	+\$1,070	+\$370	+\$58	−\$60
no trail · 5×ATR stop · 6h	31%	+\$1,319	+\$625	+\$98	−\$70
no trail · 5×ATR stop · 8h	28%	+\$2,058	+\$1,362	+\$213	−\$71
no trail · 5×ATR stop · 12h	24%	+\$3,409	+\$2,702	+\$422	−\$75
5×ATR trail · 5 stop · 8h a wide trail is still a trail	45%	−\$43	−\$699	−\$109	−\$18
2R target · 1×ATR stop · 3h	29%	−\$149	−\$769	−\$120	−\$18

Same entries, same tape, same costs. **−\$913 with a trail, +\$2,702 with a wide stop held long.** A \$3,615 swing produced entirely by the exit, and the ordering is clean rather than lumpy — 3h → 6h → 8h → 12h improves monotonically. Notice also that a wide trail does not rescue it (−\$699): it is not the width, it is the trailing. Gold wants board-once, hold-wide, leave-on-a-clock.

And the control that had to be run. A wide stop held for hours on a tape that went somewhere will make money whatever the entry says. So I ran the identical

entries on the OPPOSITE side, and the identical exit on the same number of random minutes:

EXIT	AS SIGNALLED	MIRROR — OTHER SIDE, SAME ROWS	RANDOM MINUTES, MEDIAN OF 5	VERDICT
3×ATR stop · 3h	−\$746	−\$887	−\$960	entry adds value
5×ATR stop · 3h	+\$370	−\$1,696	−\$1,004	entry adds value
3×ATR stop · 8h	+\$1,466	−\$1,895	−\$1,472	entry adds value
5×ATR stop · 8h	+\$1,362	−\$2,851	−\$1,213	entry adds value
3×ATR stop · 12h	+\$2,281	−\$2,946	−\$941	entry adds value
5×ATR stop · 12h	+\$2,702	−\$5,438	−\$3,010	entry adds value

The mirror is negative at every single cell and gets more negative as the real thing gets better — which is what a genuinely directional entry looks like and is the opposite of what drift-harvesting looks like. Random-minute entries are negative too. **This is the strongest gold result on the desk and the first one to pass its own adversarial control.**

FOUR THINGS WRONG WITH IT, ALL OF THEM MINE TO SAY

The median trade LOSES, and loses more as the total improves — −\$60 at 3h, −\$75 at 12h. The win rate FALLS from 38% to 24% as the number gets better. Everything is in the tail. You would sit through long strings of small losses waiting for the one that pays, and on a 2-lot 20–30 trade month that is a genuinely uncomfortable equity curve.

A 12-hour hold breaks your standing rule. "NEVER HOLD OVERNIGHT. EVER." A gold trade opened at 14:00Z and held twelve hours is open at 02:00Z. The best cell that fits inside a session is **5×ATR, 3h: +\$370, or \$58/week** — real, and a sixth of the headline. The gap between \$58 and \$422 a week is entirely the overnight rule, and that is a decision for you, not a finding for me.

The boarding rule was chosen, not swept. Efficiency ≥ 0.50 with a 20-minute net of at least 1.5×ATR and an hour of dead time between signals. It is a reasonable rule and it is one rule. The exit matrix is honest because the entries are held fixed across it; the entry itself has not been robustness-tested and I am not claiming it has.

Thirty-two days, one instrument, one direction of gold. The tape fell overall across the window. 165 trades is a decent n and 6.4 weeks is not a long time.

3.4 — LEAD TWO: the MGC day rider and its beautiful plateau — REFUTED

This was the most promising shape gold had ever shown, and it is worth saying exactly why it looked so good, because the failure is instructive rather than embarrassing.

The idea: one trade a day, no gate at all. At 14:30Z run the live drift detector on the first hour of the US session; if it reports a confirmed direction, take two lots that way and hold to the 20:40Z flat, which is when you are watching anyway. The thresholds were derived properly — not swept for P&L, but by finding what percentile MNQ's live constants occupy in MNQ's own distribution and reading gold's value at the same percentile, because the transferable quantity is the selectivity, not the number.

And the sensitivity grid was a **plateau**, which is the thing you are supposed to want. Not a peak with cliffs either side — every neighbouring cell positive too. Here it is, re-derived today on our own capture with the honest \$7.50 cost:

ER FLOOR	RT FLOOR	DAYS IT FIRES	DIRECTION HELD TO THE FLAT	NET (TRUE COST)	MEDIAN DAY	STRIP BEST DAY	NET AT THE OLD \$1.50
0.093	0.417	13	84.6%	+\$4,949	+\$271	+\$2,852	+\$5,105
0.124	0.350	13	84.6%	+\$4,949	+\$271	+\$2,852	+\$5,105
0.124	0.480	12	83.3%	+\$4,116	+\$270	+\$2,020	+\$4,260
0.140	0.350	12	83.3%	+\$4,612	+\$270	+\$2,516	+\$4,756
0.155	0.480	10	80.0%	+\$1,683	+\$231	+\$1,057	+\$1,803
all 15 cells of the grid		10-14	78.6-84.6%	15 of 15 POSITIVE	—	every one survives strip-best-day	15 of 15 positive

Every cell positive. Every cell survives stripping its best day. Direction holds to the flat four times in five. The spread cost barely dents it because there is only one round trip a day. On 26 sessions of our own capture, this is about as good as a sensitivity grid gets, and it is better than the 08-14 read, not worse.

And then the 185 days. Once the dead days are dropped, the backfill gives 185 usable 13:30-20:40Z sessions from 2025-07-28 to 2026-07-29 — a full year of gold

that no gold study has ever touched, on the same instrument, verified against our own capture. Same detector, same thresholds, same hold-to-the-clock, same \$7.50:

ER FLOOR	RT FLOOR	DAYS IT FIRES	DIRECTION HELD	NET	\$ PER FIRING DAY	MEDIAN DAY	STRIP BEST DAY	WORST DAY
0.093	0.350	91	49.5%	−\$7,919	−\$87	−\$17	−\$9,609	−\$4,537
0.109	0.417	79	49.4%	−\$8,267	−\$105	−\$17	−\$9,957	−\$4,537
0.124	0.350	76	52.6%	−\$5,204	−\$68	+\$28	−\$6,895	−\$4,537
0.140	0.350	63	54.0%	−\$2,567	−\$41	+\$36	−\$3,933	−\$3,858
0.155	0.350	55	54.5%	−\$2,411	−\$44	+\$58	−\$3,777	−\$3,858
all 15 cells of the same grid		54-91	49.4-54.5%	0 of 15 POSITIVE	—	—	—	—

Fifteen for fifteen becomes zero for fifteen. **The direction-hold rate, which was the whole reason to believe in it, falls from 78-85% to 49-54%.** That is a coin flip, and I tested it as one: at the best out-of-sample cell the direction held **34 times in 63, and the probability of doing that well or better on a fair coin is 0.307.** Meanwhile always short — no detector, no thresholds, no thought — made **+\$11,090** across the same 185 days. The filtered version loses money where the dumb constant makes it, which is precisely how the momentum gate died back in August.

Your backtest rule says never score a blanket number, so I scored the best cell on its home segments as well — everything measured at 14:30Z, the moment the rider decides:

SEGMENT (ALL KNOWN AT 14:30Z)	N	NET	\$ PER DAY	DIRECTION HELD	MEDIAN DAY	STRIP BEST DAY
quiet open — first-hour ATR < 2.40pt	21	+\$3,874	+\$184	61.9%	+\$63	+\$2,720
middle — 2.40 to 3.88pt	21	−\$3,680	−\$175	47.6%	−\$30	−\$4,460
live open — ATR ≥ 3.88pt	21	−\$2,761	−\$131	52.4%	+\$268	−\$4,128
the rider went LONG	36	−\$2,724	−\$76	55.6%	+\$50	−\$3,879
the rider went SHORT	27	+\$157	+\$6	51.9%	+\$20	−\$1,209
overnight chopped — roundtrip < 0.26	21	−\$1,494	−\$71	42.9%	−\$30	−\$2,860
	21	−\$1,556	−\$74	61.9%	+\$63	−\$2,434

SEGMENT (ALL KNOWN AT 14:30Z)	N	NET	\$ PER DAY	DIRECTION HELD	MEDIAN DAY	STRIP BEST DAY
overnight trended — roundtrip ≥ 0.62						
the read's own efficiency in its top third	21	+\$945	+\$45	61.9%	+\$63	-\$422

One green segment out of eight, and it is quiet opens rather than active ones — the opposite of the story. It survives stripping its best day (+\$2,720), which is more than most things here manage, and it is one of three ATR cells I chose after looking at all three. That is a PARK, not a find.

THE LESSON, AND IT APPLIES TO EVERYTHING ELSE IN THIS REPORT

The plateau logic was applied correctly in August. All 15 cells positive, the peak cell not shipped, the derivation done distributionally rather than by fitting P&L — that was good work and it would catch a threshold fluke every time. **It cannot catch a sample fluke**, because every cell of the grid is scored on the same sixteen days. A plateau tests sensitivity to the THRESHOLD. It says nothing at all about sensitivity to the SAMPLE, and on this desk the sample is almost always the binding constraint. When you next see fifteen green cells, the question to ask is not "is it a peak or a plateau" — it is "how many independent days is that grid standing on".

3.5 — The coil bouncer — **REFUTED**, and the mixed-timeframe leak is what killed it

The coil bouncer was the strongest standalone gold gate: when the last 20 minutes' range shrinks to $0.60\times$ its own two-hour median, fade whichever edge price is sitting on back to the middle of the coil. It had a genuinely good report card on 08-14 — $n=154$, +\$470.42, a positive median trade of +\$7.75, placebo beaten 0 times out of 12, and a mirror control at −\$923.75. It is also mechanically distinct from the refuted fader: that one faded EXTENSION, this one trades COMPRESSION.

I re-ran **the desk's own script**, changing exactly one thing — the bar loader now filters timeframe='5s'. Same rule, same exit, same placebo, same cost model, ten days more tape:

COIL BOUNCER	AS REPORTED 2026-08-14	RE-RUN WITH THE CLEAN LOADER	WHAT HAPPENED
days of tape	18	28	—
trades	154	291	—
net at \$1.50/RT	+\$470.42	−\$413.69	sign flips
median trade	+\$7.75	−\$10.64	sign flips
strip best day	+\$289.42	−\$594.69	sign flips
placebo — 12 random-time sets	beaten 0/12	beaten 6/12	a coin flip — no information
continuation control (break OUT of the coil)	−\$923.75	−\$495.02	the fade no longer beats its own control by much
green days	—	9 / 27	—
and at the TRUE \$7.50 cost, 291 trades pay a further...	—	−\$1,746	—

Cause of death: the placebo test, 6 of 12. And the mechanism of the error is specific and worth remembering. The coil rule is a RANGE rule — it compares a 20-minute high-minus-low against a 2-hour median of the same thing. That is the single most look-ahead-sensitive quantity there is, and the mixed-timeframe union inflated 44.4% of July's minute ranges by roughly a whole bar. A range rule fed inflated ranges finds coils that were not coils and edges that were not edges. The gate was measuring the bug.

3.6 — The London-open break fade — **REFUTED**, on the cost model alone

Gold's session-open breaks fail: continuation after the 06:00Z opening range wins 6.2% of the time. So fade them. On 08-14 that gave n=16 and +\$502.42, beaten by 0 of 7 shifted anchors, though the median trade already lost. Re-run through the desk's own script with the clean loader:

ANCHOR	N	DAYS	GREEN DAYS	WIN%	\$ PER TRADE AT \$1.50	NET AT \$1.50	STRIP BEST DAY	NET AT THE TRUE \$7.50
LDN_OPEN 06:00Z	26	26	14	53.8%	+\$4.93	+\$128.14	+\$34.27	−\$27.86
RTH_OPEN 13:30Z	26	26	8	30.8%	−\$19.68	−\$511.59	−\$626.26	−\$667.59
PM_FIX	25	25	7	28.0%	−\$1.09	−\$27.29	−\$546.73	−\$177.29

This is the cleanest kill in the file and it needs one sentence. **The gate earns \$4.93 a trade. The spread it never paid is \$6.00 a trade.** Its entire edge is

smaller than the cost it was not being charged, and the honest net over 26 days is $-\$27.86$. It does still beat the other two anchors, so the shape is not nothing — but a shape whose expectancy is below its own transaction cost is not a trade, it is a subsidy.

3.7 — So what is gold actually worth?

GOLD CANDIDATE	HONEST NUMBER	WHAT KILLED IT, OR WHAT IS LEFT	VERDICT
Run-boarding with a wide stop and a time cap	+\$58/wk inside a session, +\$422/wk if held 12h	mirror $-\$5,438$ and random $-\$3,010$ at the best cell — the entry is real. Median trade loses; the big number needs an overnight hold	SHADOW
MGC day rider — the ER×RT plateau	$-\$2,411$ to $-\$8,267$ out of sample	185 out-of-sample sessions; 0 of 15 cells positive; direction held 34/63, $P=0.307$; always_short made $+\$11,090$ on the same days	REFUTED
— its one green segment, quiet opens	$+\$3,874$ on $n=21$	1 of 3 ATR cells chosen after looking. Revive if it stays green on 40+ fresh quiet-open sessions	PARKED
Coil bouncer	$-\$413.69$ before spread, $-\$2,160$ after	placebo beaten 6/12 — no information. Its 08-14 result was reading inflated ranges from the mixed-timeframe union	REFUTED
London-open break fade	$-\$27.86$ over 26 days	expectancy $\$4.93$ /trade against a $\$6.00$ /trade spread it was never charged	REFUTED
The 12 L2-only days no gold study has touched	—	2026-07-20 to 08-04 exist in the depth capture and in nothing else. A genuine held-out leg nobody has spent	THE ONE STONE UNTURNED

Everything here is shadow-only. Nothing gold has ever placed an order and nothing in this chapter suggests it should. What has changed is that we now have one gold idea that passed an adversarial control instead of merely passing a sweep, and we have a much better idea of why the previous six did not.

3.8 — The cost of finding out

The 08-14 session closed with a line I want to correct rather than repeat: "the missing ingredient is DAYS, not configurations." That was right, and it turned out to be cheaper to test than anyone expected — the days had already arrived, in a backfill five days later that nobody announced and nothing used. Here is what that cost and what it bought:

WHAT WE THOUGHT WE NEEDED	WHAT WAS ACTUALLY AVAILABLE	WHAT IT COST TO USE IT
More gold sessions. 16-18 was too thin to promote anything.	253 usable MGC minute-days already in the lake since 2026-08-19, verified against our own capture to 0.0pt on every shared July close	one staleness filter and one instrument check — about an hour
A contiguous gold tape without the 07-18 → 08-04 hole.	32 contiguous days of 250ms quotes, because the depth capture kept running through the window the L1 mirror did not	already built — it is what \$3.3 runs on
More configurations to sweep.	Nothing. The day rider's grid was already a plateau. More cells would have made it look better and told us less.	—

So the corrected version of the lesson: **the missing ingredient was days, and the reason we did not have them was not that they did not exist — it was that nothing checked whether they did.** The backfill sat in the lake for ten days, silently corrupting every study that read it and silently holding the answer to the one question those studies could not settle. That is the pattern this desk keeps hitting: an instrument that is quietly wrong is worse than one that is loudly broken, because a wrong number looks exactly like a finding.

What gold needs next is not more sweeps and not more days. It needs the run-boarding entry — the only thing that has passed a mirror — put into the shadow book with a session-bounded exit, and it needs someone to spend the 12 untouched L2-only days on it as a genuine out-of-sample leg. That is a Saturday afternoon, not a project.

4 · DISPOSITION — every lead in this section, one verdict each

LEAD	VERDICT	THE NUMBER THAT DECIDED IT	REVIVAL CONDITION, OR THE NAMED TEST THAT KILLED IT
GRIND (d) — bench it in NORMAL_CHOP and CLEAN_TREND	SHADOW	+\$2,310 vs +\$922; strip-3 + \$249; leave-one-day-out + \$1,430; both halves positive	promote when the second half (+\$238) reaches parity with the first and the placebo drops below 5% on unseen days
GRIND (e) — bench the 13:00Z US-open hour	SHADOW	the 13Z hour is −\$1,618 over 50 fires on 25 days, green 8/25, 18% win	corroborates (d); ship only alongside it, never alone — the hour was chosen after seeing the table
GRIND (b) — entry trend-confirmation or veto	REFUTED	all five variants beaten by 50-71% of random removals of the same size	placebo test. No reformulation survives: an efficiency confirm selects for CLEAN_TREND, which is grind's worst cell (−\$82/fire, green 1/7)
GRIND (c) — the "current ER-0.35 floor"	REFUTED	ER_FLOOR = {}; as a hypothetical, n=15, strip-3 − \$890, placebo 28%	it does not exist and never has. Fourth time tested, fourth failure
GRIND (a) — trend-day-only permission	PARKED		revive if a deployable clock cut clears a 5% placebo on 60+

LEAD	VERDICT	THE NUMBER THAT DECIDED IT	REVIVAL CONDITION, OR THE NAMED TEST THAT KILLED IT
		+\$2,826 on the analyst window collapses to +\$1,518 at a 13:30Z clock cut, placebo 22%	days; the effect is real and small
GRIND — raise the ATR floor above 22	REFUTED	22→26 gives +\$533 (placebo 53%), 22→32 gives +\$498 (placebo 43%); both worse than 22	placebo test. Closes the question the last cycle left open — the floor is not the lever
GOLD — run-boarding, wide stop, time cap	SHADOW	+\$2,702 true vs mirror −\$5,438 and random −\$3,010; the 3h session-bounded cell is +\$370	shadow the session-bounded version; spend the 12 untouched L2-only days on it as an out-of-sample leg
GOLD — the MGC day-rider plateau	REFUTED	0 of 15 cells positive on 185 sessions; direction held 34/63, P=0.307; always_short + \$11,090	out-of-sample leg. 185 independent sessions on the verified same instrument — not thin n, the opposite
GOLD — the rider's quiet-open segment	PARKED	n=21, +\$3,874, strip-best-day + \$2,720, direction held 61.9%	revive if it holds green across 40+ fresh quiet-open sessions; it is 1 of 3 cells chosen after looking
GOLD — the coil bouncer	REFUTED	placebo beaten 6/12; net − \$413.69 before spread on 291 trades over 28 days	placebo test. Its 08-14 result was reading mixed-timeframe inflated ranges — a range rule fed a range bug
GOLD — the London-open break fade	REFUTED	expectancy +\$4.93/trade against \$6.00/trade of unpaid spread; honest net −\$27.86	cost model. The edge is smaller than the cost, so there is no threshold that saves it
INSTRUMENT — the lake's four extra timeframes	FIX ON SATURDAY	44.4% of July minutes inflated; the audit's own zero-control now reads 9,469	add timeframe='5s' to lake.connect()'s bars view, or move the backfill files out of the stream directory. MNQ has the same four files
INSTRUMENT — selector_nightly	FIX ON SATURDAY	−\$347 modelled against +\$330 booked on the same six legs; \$1,576 of "regret" inside that model	either wire adaptive_exit on, or stop publishing a scorecard for a disabled feature. At minimum print realised P&L beside the modelled one
INSTRUMENT — router_value is one-sided	FIX ON SATURDAY	08-28 printed −\$117 on a day its arms booked +\$330	add realised P&L from managed gates while ON. Until then every value is a lower bound on days the router armed anything
The stand-down out-scored the router 4:1 per day	OPEN — NOT AN ACTION	+\$319/day stood down vs +\$76/day with authority (+\$186 once the arms are credited)	2 days against 3 on different tapes. It is a bar to clear, not a recommendation. Re-measure over four weeks
Switch churn	CLOSED — NO ACTION	0.18 changes/hr this week against 0.71 on the 08-07 reference day; 6 arms vs 16 benches	the dwell-time spec the scope asks for is no longer needed. Re-open if the rate goes back above 0.5/hr
The router has no calendar	FIX ON SATURDAY	armed capitulation_long at 08-29 07:20:25Z on 0 bars, 38.7h before any market opens; 1 change in 645 weekend ticks	a weekend and holiday check, ~15 lines. It removes a whole class of decision made on zero information

Two SHADOW candidates and one survivor with a control behind it. That is a better week than the count of REFUTED lines suggests, because three of those four kills came from tests that had never been run before — an out-of-sample leg

on 185 sessions, a mirror control, and a cost model that charges what we actually pay. Each of them is now cheap to run again, which means the next thing that looks this good gets tested properly on the first pass rather than the third.

★ **THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.**

part2_6_router_review.html was rebuilt **2026-09-04 22:14 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

P2.6 ROUTER OPERATION REVIEW — the desk's #1 lever, graded on the week's money

Garrath — the headline is uncomfortable and it is the most useful thing in this section. **The router's own book made money this week and the desk still lost −\$1,246.94.** Every trade the router had authority over — all six of them, all on **Friday 2026-08-28** — won: **+\$330.00, 6 for 6.** The **−\$1,576.94** that swamped it came from day_rider, **26 trades in a book the router cannot see and cannot switch off.** So "was the router good this week" and "did the desk make money this week" are two different questions with two different answers, and conflating them is the single easiest way to draw the wrong conclusion from this week. **The second thing: for five of the week's eight days the router was managing a book the operator had already stood down** — your 08-20 stand-down text sat in its prompt through **1,802 consecutive ticks** until the router itself flagged it as stale at 08-27 13:05Z. **Third, and this is the number I'd act on: the router's direction reads were right and its latency was the dominant error.** Friday 08-28's arm was **20m17s late** and the lateness cost **\$205-\$705** against **+\$181.50** banked. **Fourth: the untradeable-day meter is dead.** Scored the way it actually runs —

intraday, before the day is over — it has no discrimination at any decision hour on any cutoff ($\rho = +0.17 / -0.09 / +0.11$), and the tempting "move the cutoff to 45" fix fails strip-the-best-2 with a sign flip. **Fifth: the threshold sweep says leave the thresholds alone.** ER 0.15 / NET 30 sits in the middle of a genuine 3-cell plateau on both universes. The money in that sweep is in the timing knobs, and the timing story is corroborated independently by the lateness finding.

WHAT SURVIVED, IN ORDER OF HOW MUCH I'D TRUST IT

1. **The router's own book is 6-for-6 at +\$330.00 this week**, and its held benches priced out at **+\$1,224.38 saved** against **\$241.07** of measurable error. On the calls that can be graded, it is winning.
2. **The shipped thresholds are already at the robust optimum.** ER 0.15 / NET 30 is the centre of a 3-cell plateau (+461 / **+492** / +428 live-managed; +1,076 / **+1,083** / +1,070 full-historical). Raising ER above 0.15 is strictly worse on both sets. **Do not touch them.**
3. **Latency is the biggest single error term on the desk's #1 lever** — measured at **\$205-\$705** on one afternoon, larger than anything on the scoreboard. Independently, the sweep's joint-best wants step 15m $\rightarrow$ 5m. Two unrelated methods pointing the same way is the strongest signal in this section.
4. **Benching abs_veto_short is the router's proven edge** — **+\$1,208.50** saved across every nightly on record, more than every other gate's bench combined. **Benching capitulation_long and grind_long is value-destroying** ($-\$242.80$ and $-\$156.20$).
5. **The untradeable meter is REFUTED as built** — and that is a survivor, because it stops us tuning a cutoff on a number that is reading the day it is supposed to be forecasting.

READ THIS BEFORE ANY NUMBER BELOW

Every dollar figure is `pnl_usd` from `gazbot7.db`, which is **net of the \$1.50-per-round-trip fee already** — nothing here double-charges costs. One row is excluded desk-wide (`BADFILL:stale_price_30pt_outside_book_20260821`). **Bench savings are**

counterfactual: the benched gate never traded, so its "saving" is a tick-replay of what it wanted to do, and it is a different ruler from a booked fill. I never sum the two into one headline. **And the week is thin** — 6 managed trades and 18 switch changes is not a sample you can conclude much from. Where a number rests on $n < 10$ I say so on the row.

1 • The week's decisions and the scoreboard

The router ran **2,041 decision ticks** across the eight days Sat 08-22 → Fri 08-29 and changed a switch on **18** of them. That is a change rate of **0.88%**, and it is the quietest week in the file. Here is what it actually did, day by day, against what the desk actually made.

DAY	TICKS	SWITCH CHANGES	WHAT CHANGED	ROUTER BOOK	DAY RIDER	DESK P&L
Sat 08-22	279	0	nothing — tournament stood down, tape shut		—	—
Sun 08-23	280	0	nothing — tournament stood down, tape shut		—	—
Mon 08-24	276	1	1 write, stood-down book	\$0.00	−\$2,149.44	−\$2,149.44
Tue 08-25	285	0	nothing — stood down	\$0.00	−\$243.00	−\$243.00
Wed 08-26	266	3	exhaustion_short+abs_veto_short off 22:00Z; abs_veto_long on→off	\$0.00	+\$19.00	+\$19.00
Thu 08-27	278	6	first real authority since the lift — capitulation_long, abs_veto_long ×4, 22:00Z re-bench ×2	\$0.00	+\$16.50	+\$16.50
Fri 08-28	286	7	the week's only trading day for the router — armed grind_long+abs_veto_long 14:45Z, benched both 15:35Z, capitulation_long off, abs_veto_short on→off→off	+\$330.00	+\$780.00	+\$1,110.00
Sat 08-29	91	1	capitulation_long g → on — a no-op	\$0.00	\$0.00	\$0.00

DAY	TICKS	SWITCH CHANGES	WHAT CHANGED	ROUTER BOOK	DAY RIDER	DESK P&L
			on a gate already on for 9½h			
WEEK	2,041	18	0.88% change rate — the quietest week on record	+\$330.00	−\$1,576.94	−\$1,246.94

The thrash check passes, and it passes by a mile. The trial log's changed : field is a lower bound (it misses writes by other processes — open_hour_watch.py and the 22:00Z reactivate_gates job both write the same file), but as a router-thrash meter it is the right instrument, and the trend is unambiguous:

WEEK (MON-FRI)	ROUTER SWITCH CHANGES	VERDICT
08-04 → 08-08	63	peak thrash — 25 in one day (08-07)
08-11 → 08-15	53	still thrashy — 21 on 08-11, 16 on 08-12
08-18 → 08-22	15	settling
08-24 → 08-29 (this week)	18	no thrash. And 17 of the 18 land on 08-26 or later — i.e. only once it had authority back.

The single worst day in the file — 08-07, 25 changes — was abs_veto_long and abs_veto_short being flipped on and off within 5-minute windows. That pattern is **gone**. The 08-04→08-05 signature of abs_veto_long on→off six times in three hours does not appear once this week.

The bad-call ledger scoreboard

Nothing scored between 08-22 and 08-26 — the tournament was stood down, so there were no decisions to grade. Every point below moved in the final three days.

PASS	GOOD	BAD	CHURN	NEUTRAL	SCORING CALLS	GOOD SHARE
week open (08-27 00:1x, before)	45	14	25	46	84	54%
08-27 00:1x	46	17	26	48	89	52%
08-27 12:1x	48	17	26	54	91	53%
08-27 16:1x	51	18	26	56	95	54%
08-27 20:1x	52	18	26	59	96	54%
08-28 00:1x → 12:1x	54	18	26	62	98	55%
08-28 16:2x	56	18	27	64	101	55%
08-28 20:1x	58	19	27	66	104	56%
08-29 04:1x (current)	59	19	28	66	106	56%
THE WEEK'S DELTA	+14	+5	+3	+20	+22	64%

The week's calls graded better than the lifetime average: 64% GOOD against 56% lifetime. That is the right direction, but I will not oversell it — 22 new scoring calls is a small sample and 20 of the 22 new NEUTRALs are "\$0, nothing happened", which is mostly a statement about a quiet tape rather than about the router's judgement.

2 • What worked and what didn't, in dollars

These are the week's priced calls, straight from the bad-call ledger, each one replayed on the MNQ trade tape. Green rows paid; red rows cost.

WHEN	CALL	GATE	\$ EFFECT	HOW IT WAS MEASURED
08-28 16:55:48Z	BENCH (GOOD #57)	abs_veto_short	+\$258.06	benched impulses replayed to their own exits
08-27 all day	HELD BENCH	abs_veto_short	+\$241.50	4 impulses; one of them (15:26) cost +\$97 — net still positive
08-28 window	HELD BENCH (GOOD #58)	abs_veto_long	+\$199.44	tick replay
08-28 14:45:03Z	ARM (GOOD)	grind_long+abs_veto_long	+\$181.50	booked fills — worth + \$266.10 on the gates' own exits, but a MANUAL_CLAIM truncated the stack
08-27 window	HELD BENCH	abs_veto_short	+\$104.21	2 impulses, both resolved
08-27 window	HELD BENCH	rgv_short	+\$71.66	3 impulses (one cost + \$106.91)
08-28 08:13:05Z	HELD BENCH	abs_veto_long	+\$50.48	△ the money came from the STOP, not the read — the gate's direction was RIGHT
08-27 window	HELD BENCH	capitulation_long	+\$46.86	single impulse, MFE 0.00 — never onside
08-27 window	HELD BENCH	abs_veto_short	+\$36.29	△ thin — one cell flips sign below stop 0.75×
08-26 22:00:32Z	RE-BENCH	exhaustion_short	+\$34.38	one post-arm intent, STOPS at both candidate ATRs
08-28 window	HELD BENCH (BAD #19)	exhaustion_short	−\$135.43	the bench blocked a winner
08-27 20:57:05Z	HELD BENCH (BAD #12)	abs_veto_long	−\$100.00	sign-stable positive at all 7 stop widths, MAE only 0.23×ATR — a clean miss
08-27 window	HELD BENCH	abs_veto_long	−\$5.64	2 impulses, bench cost slightly
08-27 window	HELD BENCH	exhaustion_short		3 targets / 4 stops, — netting −\$5.00 — noise, graded NEUTRAL
08-26 22:45-23:15Z	CHURN	abs_veto_long	\$0.00	arm then bench 30 min apart, zero intent, zero

WHEN	CALL	GATE	\$ EFFECT	HOW IT WAS MEASURED
				fills — a process error, not a \$ error
08-29 07:20:25Z	CHURN #28	capitulation_long	\$0.00	a no-op — wrote on to a gate on for 9h20m, "correcting" four ticks that were right
PAID (counterfactual benches)			+\$1,042.88	nine held/actioned benches, tick-replayed
PAID (real booked fills)			+\$181.50	the only money that actually hit the account
ERRED (measurable)			-\$241.07	BAD #19 + BAD #12 + the small long-bench cost

THE ONE THING IN THIS TABLE THAT IS NOT WHAT IT LOOKS LIKE

The +\$50.48 bench of abs_veto_long on 08-28 is credited to the wrong thing. The bench saved money, yes — but the tape hit both of the gate's own targets, and the loss only exists because the desk's stop is too tight: MAE was $1.149 \times \text{ATR}$ against a $1.0 \times \text{ATR}$ stop, and at $k = 1.25$ that same trade is **+\$100. That is a saving earned by a stop-width bug, not by the router's read.** The router read the direction wrong and got paid for it. If we widen the stop — which the exit lab wants — this "saving" flips to a cost, and the same will be true of several other cells in this table. It belongs on the scoreboard as a \$ fact and nowhere near a claim about the router's judgement.

The pattern across the whole file, not just this week

Rolling every nightly on record (20 days, A/B legs collapsed) gives the cleanest read there is on which benches are worth making. Positive = benching that gate made us money.

GATE	BLOCKED N	WON	LOST	BENCH P&L TO US	VERDICT
abs_veto_short	186	56	128	+\$1,208.50	KEEP BENCHING the router's one proven edge — more than every other gate combined
nipc_short	27	7	20	+\$181.00	△ n=27, not live-managed
nipc_long	20	7	13	+\$143.50	△ n=20, not live-managed

GATE	BLOCKED N	WON	LOST	BENCH P&L TO US	VERDICT
exhaustion_short	106	40	66	+\$116.50	mildly positive on n=106 — +\$1.10 a blocked trade, i.e. ~nothing
rgv_short	22	6	16	+\$87.80	△ n=22 — barely trades, which is why leakage=0 is meaningless for it
abs_veto_long	152	51	100	+\$84.80	+\$0.56 a blocked trade — statistically indistinguishable from zero
grind_long	448	158	290	−\$156.20	STOP BENCHING the biggest blocked population on the desk and the bench is negative
capitulation_long	22	11	12	−\$242.80	STOP BENCHING worst bench on the board, and it is the gate the router benched most often this week
ALL GATES	983	336	645	+\$1,423.10	net-positive, and 85% of it is one gate

This is the most actionable table in the section. The router's entire measured bench value is abs_veto_short. Take that one gate out and the remaining seven benches are worth **+\$214.60 across 797 blocked trades** — twenty-seven cents a trade, which is noise. And two of them are actively negative. **The policy "bench the faders when direction disagrees" is not what is earning; "bench abs_veto_short" is.**

3 · Net router value — is it positive, is it over-benching, where is it leaking

Rolling every data/router_nightly/*.json. Value = losses the bench SAVED minus winners the bench MISSED.

DATE	SAVED	MISSED	NET VALUE	LEAK N	LEAK \$	REGIME CHOP / UP / DOWN
07-27	\$0	\$81	−\$81	9	−\$456	74 / 7 / 19
07-28	\$506	\$567	−\$61	7	−\$398	63 / 23 / 14
07-29	\$2,263	\$3,980	−\$1,717	6	−\$229	52 / 21 / 28
07-30	\$5,656	\$4,848	+\$807	0	\$0	72 / 19 / 9
07-31	\$7,363	\$7,527	−\$164	0	\$0	65 / 21 / 14
08-03	\$4,020	\$3,135	+\$885	0	\$0	73 / 23 / 4
08-04	\$2,306	\$3,176	−\$870	0	\$0	63 / 30 / 7

DATE	SAVED	MISSED	NET VALUE	LEAK N	LEAK \$	REGIME CHOP / UP / DOWN
08-05	\$4,832	\$3,323	+\$1,509	0	\$0	72 / 12 / 16
08-06	\$4,533	\$4,540	-\$7	0	\$0	63 / 20 / 18
08-07	\$4,638	\$3,204	+\$1,434	0	\$0	66 / 27 / 7
08-10 / 08-11	\$0	\$0	\$0	0	\$0	69 / 10 / 20 — empty universe
08-12	\$1,192	\$666	+\$526	6	-\$386	92 / 5 / 3
08-13	\$588	\$1,064	-\$476	0	\$0	82 / 13 / 4
08-14	\$484	\$290	+\$194	0	\$0	93 / 0 / 7
08-15 → 08-23	NO FILES — the nightly did not run for five trading days					monitoring gap
08-24	\$1,436	\$976	+\$460	0	\$0	59 / 23 / 18
08-25	\$1,892	\$1,714	+\$178	0	\$0	74 / 19 / 7
08-26	\$1,577	\$2,120	-\$543	0	\$0	65 / 16 / 19
08-27	\$2,733	\$1,844	+\$889	0	\$0	53 / 26 / 22
08-28	\$1,888	\$2,005	-\$117	0	\$0	86 / 9 / 5
THIS WEEK (5 days)	\$9,526	\$8,659	+\$867	0	\$0	net-positive
ALL 20 DAYS	\$47,907	\$45,060	+\$2,846	28	-\$1,469	+\$142 a day — positive but thin

Is it net-positive? Yes, but only just, and the margin is 6%. The router saved \$47,907 and missed \$45,060 across 20 days. It is winning **51.5%** of the money it touches. That is a coin-flip with a small tilt, and it means the ordering of individual days is mostly noise — do not read a -\$543 Wednesday as a bad Wednesday.

Is it over-benching? On the aggregate, no — per gate, yes, on two of them. "Over-benching" shows up as missed > saved, and that happened on 9 of the 18 scoring days. But the per-gate table in section 2 localises it precisely: capitulation_long (-\$242.80) and grind_long (-\$156.20) are where the over-benching lives, and grind_long is **46% of every blocked trade on the desk.**

WHERE IT IS LEAKING — AND WHY THE ZEROS LIE

leakage_n = 0 for every day this week. **That is not a clean bill of health.** The leakage universe is only the gates the router manages, and the two it actually watches most closely (capitulation_long, rgv_short) barely trade — 22 blocked trades each across the whole file. **A zero here means an empty universe, not an absence of leaks.** The all-time leak of -\$1,469 over 28 trades is what the measurement can see, and it is concentrated on 07-27→07-29 and 08-12.

And the real leak this week is not in this table at all. It is day_rider: 26 trades, −\$1,576.94, in a book with no router switch, no gate, and no entry in any nightly. **The router's managed universe covered 6 of the desk's 32 trades this week — 19% of the flow and 21% of the gross P&L.** That is the biggest single fact about router value in this report.

The selector, folded in — and a warning about its "100%" days

DATE	N	OPTIMAL	OPTIMAL %	REGRET	SELECTOR \$	BEST-OF-3 \$	NOTE
08-24		0	0	—	\$0	\$0	\$0 no trades
08-25	2	2	100%	\$0	−\$76	−\$76	△ degenerate — all four modes price identically
08-26	4	4	100%	\$0	−\$198	−\$198	△ degenerate
08-27	12	12	100%	\$0	−\$406	−\$406	△ degenerate
08-28	13	4	31%	\$1,576	−\$600	+\$976	the only day with real dispersion — and it is the worst pick rate of the week
WEEK	31	22	71%	\$1,576	−\$1,280	+\$296	the 71% is inflated by the three degenerate days
ALL 22 DAYS	487	305	63%	\$12,908	+\$940	+\$13,848	\$12,908 of regret is the honest headline

Read those three 100% days as a bug, not a win. On 08-25/26/27 every one of the four exit modes reprices to exactly the same P&L — that is not the selector picking correctly, it is the repricing being a no-op on day_rider trades. Only 08-28 has real dispersion and there it got **4 of 13 right**. **Strip the degenerate days and the week's honest pick rate is 31%, not 71%.**

The systematic error is unambiguous and it has one name: WIDE.

CHOSE	SHOULD HAVE CHOSEN	N	REGRET
wide	scalp	43	\$3,808
tight	wide	22	\$2,332
wide	tight	35	\$2,309
wide	mid	23	\$2,260
tight	scalp	45	\$1,702
		101	\$8,377

CHOSE	SHOULD HAVE CHOSEN N	REGRET
WIDE chosen when something tighter was right		

WIDE-in-violence, confirmed on 101 trades and \$8,377 of regret — 65% of all regret in the file. But before you act on it: **none of this is live.**

scaleout_slots() in slot_strategy.py sets adaptive_exit=False on every A and B leg (lines 313, 333, 344, 349), and there is **no systemd timer for selector_nightly**. The selector is a measurement, nothing reads it, and the \$12,908 of regret is not money we are losing to a bad selector — it is money sitting on the table because there is no selector at all.

BOTH NIGHTLIES WERE BACKFILLED IN ONE BATCH — THEY ARE NOT NIGHTLY MONITORING

All five router_nightly and all five selector_nightly files for this week were written between **2026-08-28 22:58:06 and 23:00:09** — a two-minute batch, four days of them retrospective. There is no file for 08-29, and router_nightly has **no files at all for 08-15→08-23**. The numbers above are internally consistent, but the operator instruction was "monitor each night, fold into Friday", and **the monitoring half of that did not happen this week**. A batch written after the week is over cannot catch anything in time to act on it.

4 · Threshold tuning — the sweep says leave the thresholds alone

scripts/router_study.py --days 40, last 40 Paris days of realised MNQ trades. Shipped config: **ER ≥ 0.15, |net| ≥ 30pt, window 30m, step 15m, hold 2, fast_exit off**. Run on both universes, as required.

UNIVERSE	BASELINE (NO ROUTER)	ROUTER NOW	Δ	BLOCKED	OF WHICH
LIVE managed — {capitulation_long, rgv_short}	−\$302	+\$190		+\$492	14 losers −\$630 / winners +\$138
FULL historical fader set (+rgv_long, +exhaustion_short)	−\$431	+\$652		+\$1,083	49 losers −\$1,710 / winners +\$627

The router turns a losing fader book into a winning one on both universes. That is the single strongest piece of evidence in this section for keeping it.

ER × NET sweep — router Δ vs baseline, shipped timing

LIVE managed set (* = shipped):

ER FLOOR	NET 20	NET 30	NET 40	NET 50	NET 60
0.15	+\$461	+\$492*	+\$428	+\$316	+\$158
0.20	+\$224	+\$176	+\$152	+\$152	+\$152
0.25	+\$152	+\$152	+\$152	+\$152	+\$152
0.30	+\$152	+\$152	+\$152	+\$152	+\$152
0.35	+\$152	+\$152	+\$152	+\$152	+\$152

FULL historical set:

ER FLOOR	NET 20	NET 30	NET 40	NET 50	NET 60
0.15	+\$1,076	+\$1,083*	+\$1,070	+\$896	+\$634
0.20	+\$820	+\$771	+\$752	+\$790	+\$592
0.25	+\$563	+\$563	+\$582	+\$582	+\$520
0.30	+\$334	+\$334	+\$334	+\$334	+\$428
0.35	+\$428	+\$428	+\$428	+\$428	+\$428

Three facts, and they agree across both universes. (1) **ER 0.15 is the right floor and it is not close** — every higher floor is worse in every NET column, on both sets. Raising it to 0.20 costs \$263 live / \$312 full. (2) The ER rows above 0.20 are flat (+\$152 and +\$428 repeated) — that is the threshold ceasing to bind at all, i.e. the router simply stops acting. (3) **NET 30 sits in the middle of a real 3-cell plateau:** 20/30/40 give +461/+492/+428 live and +1,076/+1,083/+1,070 full. The full set is flat to within **1.2%** across those three cells. That is a plateau, not a spike, and the shipped value is on it.

The joint best over 900 configs — and why I am not recommending it

UNIVERSE	JOINT-BEST CONFIG	ROUTER \$	Δ VS BASELINE	GAIN VS SHIPPED	ROBUSTNESS
LIVE managed	ER 0.15 / NET 20 / window 45m / step 5m / hold 3	+\$692	+\$994	+\$502	⬆ 4 days helped / 0 hurt, worst day Δ \$0
FULL historical	ER 0.15 / NET 20 / window 45m / step 5m / hold 3	+\$1,092	+\$1,522	+\$440	⬆ 5 days helped / 1 hurt, worst day Δ – \$10

Both universes independently pick the identical config — that is a real signal, and the thresholds are not where its money comes from. ER stays at

0.15 and NET moves 30→20, which we already know is inside the plateau and worth ~\$0. **All of the +\$440-\$502 is in the three timing knobs:** window 30→45m, **step 15→5m**, hold 2→3.

MODERATE ANTI-OVERFIT RECOMMENDATION

Ship nothing on the thresholds. ER 0.15 / NET 30 stays. They are on a plateau, the "improvement" from NET 20 is \$31 on 40 days, and moving a live threshold to capture \$31 is exactly how a desk overfits.

The timing change is a genuine candidate but it is NOT deployable on this evidence alone. "4 days helped / 0 hurt" out of 40 means **n = 4** — 36 of the 40 days are unaffected because the router never acted on them. A 4-day sample cannot carry a live config change, and I will not pretend otherwise.

What makes it worth doing anyway is that a completely independent measurement says the same thing. The 08-28 lateness finding (section 7) prices the actuation delay at **\$205-\$705 on one afternoon** — measured on the live tape, not in a sweep, and larger than the whole 40-day timing gain. **Two unrelated methods both saying "the router is too slow" is the strongest conclusion in this section.** The step is also the safest knob: it changes how often the router looks, not what it decides.

⚠ **Saturday-deploy caveat.** The 40 days are one summer RANGE regime, the trades are thin, and they are partly already hand- and router-gated — so the router is being scored on a tape it helped create. Everything here is an illustrative lower bound. **Any change is a Saturday/operator call, and it should be re-run after a TREND week before anything touches the thresholds.** If you deploy one thing, deploy the step; leave ER, NET, window and hold exactly where they are.

Is it blocking enough? The leak classification

CLASS	LIVE N	LIVE \$	FULL N	FULL \$	READING
BLOCKED — router caught it	14	-\$492	49	-\$1,083	working as designed
allowed winner — correct	21	+\$1,456	63	+\$3,813	it is not over-blocking the winners

CLASS	LIVE N	LIVE \$	FULL N	FULL \$	READING
LEAK: counter-trend NEAR-MISS	15	−\$911	29	−\$1,809	the biggest addressable leak — trades just under the threshold
chop loss — router can't help	4	−\$208	7	−\$355	not a router problem
trend-FOLLOWING loss	1	−\$147	16	−\$997	gate/exit problem, not direction

The near-miss leak is the one number that argues for a looser threshold — 15 trades and −\$911 live, 29 and −\$1,809 full, all sitting just the wrong side of the trigger. **But the sweep already tested that and NET 20 does not capture it** (+\$461 vs +\$492 live — it gets worse). So the near-misses are not near-misses on NET; they are near-misses on timing. That is the third independent pointer at the same answer.

5 • The untradeable-day meter — REFUTED as built

src/gazbot7/untradeable.py blends three meters into a 0-100 STAY-OUT score: **ROUNDTRIP** (does the day go anywhere), **GIVE-BACK** (how much of the move it hands back) and **STOP-RATE**. Cutoffs: ≥ 65 STAY-OUT, ≥ 45 CAUTION, below that TRADEABLE. I scored every Paris day against what the desk actually made.

PARIS DAY	SCORE	VERDICT	CHOP	GIVE	STOPS	ROUNDTRIP	RANGE PT	DESK P&L	RIGHT?
08-19	100	STAY-OUT	100	100	—	0.00	376	+\$515.00	MISS — told us to sit out the 3rd-best day
08-21	82	STAY-OUT	85	78	—	0.15	305	−\$69.00	hit (n=1 trade)
08-14	77	STAY-OUT	84	75	67	0.16	256	+\$604.50	MISS — told us to sit out the 2nd-best day
08-17	60	CAUTION	70	49	—	0.30	284	+\$473.00	hit
08-10	55	CAUTION	62	36	80	0.38	254	−\$205.50	MISS
08-12	53	CAUTION	49	46	73	0.51	366	−\$435.50	MISS
08-25	53	CAUTION	62	44	—	0.38	393	−\$243.00	MISS
08-11	50	CAUTION	65	50	20	0.35	350	+\$560.50	hit
08-24	49	CAUTION	54	44	—	0.46	518	−\$2,149.44	MISS — the worst day of the week, and it scored TRADEABLE-ish
08-27	47	CAUTION	72	46	0	0.28	299	+\$16.50	hit
08-20	46	CAUTION	53	38	—	0.47	482	−\$1,288.50	

PARIS DAY	SCORE	VERDICT	CHOP	GIVE	STOPS	ROUNDTRIP	RANGE PT	DESK P&L	RIGHT?
									MISS — 2nd-worst day, scored below the CAUTION midpoint
08-28	42	TRADEABLE	68	38	0	0.32	370	+\$1,110.00	hit — the week's best day, correctly called
08-26	26	TRADEABLE	48	3	—	0.52	440	+\$19.00	hit
08-13	21	TRADEABLE	20	14	36	0.80	488	+\$768.50	hit
08-18	19	TRADEABLE	16	11	40	0.84	600	+\$195.50	hit

Look at the two worst days on the desk — 08-24 at **−\$2,149.44** and 08-20 at **−\$1,288.50**. The meter scored them 49 and 46, i.e. below the midpoint of CAUTION and nowhere near STAY-OUT. Then look at 08-19 and 08-14, which made **+\$515** and **+\$604.50**: the meter scored them 100 and 77 and told us to stay out. On this week's 15 days the blend has a rank correlation with P&L of **−0.086** — zero — and the two sub-meters that the module's own docstring calls "THE discriminators" both come out with the wrong sign.

Which of the three meters actually discriminated

METER	P VS DESK P&L, THIS WEEK (15 DAYS)	P VS DESK P&L, 32 DAYS	VERDICT (WANT NEGATIVE)
1. ROUNDTRIP / chop	+0.236	−0.197	flips sign between windows — no stable signal
2. GIVE-BACK	+0.129	−0.209	flips sign between windows
3. STOP-RATE	excluded on 6 of 15 days (n < 10)		the 3-meter blend only ran on 9 of 15 days
FAILED-BREAKOUTS (computed, not in the score)	+0.254	+0.201	consistently the WRONG sign both windows
BLENDED SCORE	−0.086	−0.294	only works on the longer window — see below

Does the 65 cutoff want tuning? The tempting answer, and why it is a trap

On 32 days of full-day scores the blend does look real ($\rho = -0.294$ vs the desk, -0.426 vs the tournament book), and the cutoff sweep is seductive:

CUTOFF	DAYS SAT OUT	P&L AVOIDED	DAYS TRADED	P&L KEPT	\$ PER TRADED DAY
35	21	−\$5,959	11	+\$1,830	+\$166
45	19	−\$6,252	13	+\$2,123	+\$163
50	13	−\$2,054	19	−\$2,075	−\$109

CUTOFF	DAYS SAT OUT	P&L AVOIDED	DAYS TRADED	P&L KEPT	\$ PER TRADED DAY
55	9	−\$1,606	23	−\$2,523	−\$110
65 (shipped)	6	−\$1,820	26	−\$2,309	−\$89
80	4	−\$2,192	28	−\$1,938	−\$69

That table says "move the cutoff from 65 to 45 and turn −\$2,309 into +\$2,123, a swing of **+\$4,432**". It even passes a placebo test — shuffle the score labels across days 5,000 times and a random 45-cutoff beats +\$2,123 only **2.8%** of the time. **Do not ship it.** It dies twice:

ROBUSTNESS TEST	RESULT	VERDICT
Placebo (5,000 label shuffles)	2.8%	PASS
Leave-one-day-out (13 traded days)	worst case +\$950	PASS — every single drop stays positive
Strip-the-best-2	+\$2,123 → −\$160	FAIL — SIGN FLIP. 07-30 (+\$1,172) and 08-28 (+\$1,110) are the whole edge
Scored the way it actually runs (intraday)	no discrimination at any hour	FAIL — and this one is fatal

Here is the fatal test, and it is the whole finding. Every number above scores the meter on the completed Paris day. But a stay-out call has to be made during the day. So I recomputed the meter using only the bars available at a real decision moment, and scored it against the P&L still to come after that moment:

DECISION MOMENT	DAYS	P VS REST-OF-DAY P&L	BEST CUTOFF FOUND	P&L KEPT AT THAT CUTOFF
14:00Z (just after the US open)	29	+0.172	none — every cutoff 35-75	negative at all 8 cutoffs
16:00Z (mid-session)	24	−0.088	none	negative at all 8 cutoffs
18:00Z (late session)	21	+0.105	none	negative at all 8 cutoffs

CAUSE OF DEATH, STATED PLAINLY

The untradeable meter is a description of a day that has already happened, not a forecast of one that hasn't. Its full-day ρ of −0.294 is hindsight: $\text{ROUNDTRIP} = |\text{close} - \text{open}| / \text{range}$, which cannot be known until the close. Freeze it at any live decision hour and the correlation collapses to zero or flips positive (**+0.172 / −0.088 / +0.105**), and **there is no cutoff, at any of the three hours, that leaves a profitable traded set.**

Killed by: (a) the intraday test — no discrimination at any decision hour on any cutoff; (b) strip-the-best-2 — the full-day version's entire edge is two days and removing them flips the sign. It survives placebo

and leave-one-day-out, which is exactly why it looked shippable and why those two tests were not enough.

So: the 65 cutoff does not want tuning. Tuning a cutoff on a number that reads the day it is supposed to predict just moves the hindsight around. The meter needs rebuilding on inputs knowable by 14:00Z, or retiring.

Meanwhile it should not gate anything — and note that on 08-19 and 08-14 it would have sat us out of +\$1,119.50 of real money.

6 · Bench-decision review — grading the week's actual benches

Friday 08-28 is the only day with enough tape to grade properly, and it produced the cleanest bench lesson in the file. grind_long was benched across 14 impulses that split into two opposite regimes:

PHASE	IMPULSES	OUTCOME	BENCH P&L	VERDICT
13:40:05 → 14:23:05 — chop	9	9/9 STOP, sign stable at all five stop widths	+\$535.00	☐ bench was right
14:24:46 → 14:44:03 — the break	5	5/5 TARGET	-\$727.50	☐ bench was wrong
net	14		-\$192.50	not a verdict on the bench — a verdict on the TIMING of the arm

The router was right through the chop and right about the break. It was late by one impulse. The break's first target-reaching impulse was 14:24:46; the router armed at 14:45:03, on the fifth. Same shape on the short side:

abs_veto_short wanted in at 15:39:06 @29733 and 15:49:05 @29633, both benched, **both reached target** (+\$38.50, +\$64.86) — and the router armed it at 16:00:39 @29552, after the move.

The rest of the week's benches, graded:

GATE	BENCHES GRADED	NET \$	GRADE
abs_veto_short	5	+\$640.06	EXCELLENT the router's best and most consistent bench, all week and all-time
abs_veto_long	4	+\$144.28	MIXED +\$199.44 and +\$50.48 against BAD #12 at -\$100 — and the +\$50.48 was earned by a too-tight stop
rgv_short	1	+\$71.66	△ n=3 impulses — readable, not actionable
capitulation_long	1	+\$46.86	△ n=1 impulse. All-time this bench is -\$242.80

GATE	BENCHES GRADED	NET \$	GRADE
exhaustion_short	3	-\$106.05	POOR BAD #19 (-\$135.43) dominates
grind_long	2	-\$192.50	TIMING right in chop, wrong through the break — and one 08-27 grind bench (-\$307.73 nominal) is NO VERDICT: 3 of its 11 impulses never resolved

Three benches this week priced **FREE**, and freeness is not virtue. On several passes grind_long was benched while its own ATR floor of 22.0 was already refusing it — 234, 594, 666 and 1,642 ATR-refusal lines in four separate windows, with true ATR-14 never exceeding 16.09. **The gate could not have fired whether the router benched it or not.** Those benches cost \$0 and prove nothing, and the ledger is right to score them □ **FREE** rather than □ **GOOD**. The same applies to the Asia-block windows, where the bench is void by mechanism.

7 · The durable router, the 5-minute tick, and what the lag costs

gazbot7-router-tick.timer fires gazbot7-router-tick.service every 5 minutes — 288 ticks a day, a headless Claude decision each time, fail-safe **benching-only**. Nine ticks a day are skipped by design across the 21:00–21:40Z CME halt. Did it manage correctly?

CHECK	RESULT	VERDICT
Thrash — switch changes / ticks	18 / 2,041 = 0.88%	PASS quietest week on record; no on→off→on inside a single hour
Designed CME halt-skips	9 / day, every weekday	PASS — working exactly as specified
Session-limit ABORTs	47 ticks / 235 minutes	FAIL — and it is growing
— of which in LIVE RTH (13:30-21:00Z)	24 ticks / 120 minutes	two hours of blindness while the desk was live
Churn writes (no-ops)	2 of 18 changes	08-26 arm/bench 30 min apart with zero intent; 08-29 wrote on to an already-on gate

The session-limit outages, in full

WINDOW	TICKS LOST	MINUTES	LIVE SESSION?
08-26 01:25:02Z → 01:55:02Z	7	35	no — Asia, tape quiet
08-26 15:55:03Z → 16:55:03Z	13	65	YES — middle of the US session
08-27 17:40:03Z → 17:55:03Z	4	20	YES — live
08-28 18:25:03Z → 18:55:03Z	7	35	YES — live
08-29 03:40:03Z → 04:55:03Z	16	80	no — but the longest streak on record
WEEK	47	235	3 of 5 outages landed in live RTH

A CORRECTION TO THE BAD-CALL LEDGER'S OWN ACCOUNT OF THIS

The 08-29 ledger pass calls the session-limit outage "**DAY 2**" and says it "more than doubled" from yesterday's 7 ticks / 35 min. **The log says otherwise.** The first outage was **08-26**, three days earlier, and that day had **two** — including a **13-tick / 65-minute hole in the middle of live RTH**, nearly twice the size of the 08-28 one the ledger treats as the baseline. It is day **4**, not day 2, and the worst live-session hole was on the day the ledger doesn't count. **The trend is real and the ledger under-states it.**

Every one of these is invisible to the usual instruments: systemd logs them as Deactivated successfully, and router_trial_log.txt carries no line at all — a bare gap that is indistinguishable from the designed 21:00-21:40Z skip. **router_headless.log is the only witness.** And because the fail-safe is benching-only, an outage is **one-directional**: the router cannot bench during the hole, so whatever is armed stays armed. On a shut tape that costs nothing. In a live 80-minute US-open hole it is the whole book.

The actuation-lag bill

This is the number I would put in front of everything else in this section.

EVENT, 08-28	\$	BASIS
Banked by the 14:45:03Z arm	+\$181.50	real fills, 6 trades, all winners
— worth on the gates' own exits	+\$266.10	a MANUAL_CLAIM truncated the exit stack — the leg measures the operator's timing, not the gate's edge
Long side — 20m17s late	-\$166.55 to -\$601.30	floor = one slot holds one position, so the arm buys ONE trade; ceiling = all four pre-arm target-reaching legs
Short side — ~11 min late	-\$38.50 to -\$103.36	two benched abs_veto_short impulses that both hit target
TOTAL LATENESS COST, one afternoon	-\$205 to -\$705	against +\$181.50 banked — the lag cost more than the call made

Quote the range, never the \$601.30 alone. Those four long impulses are one trend sampled four times, not four independent opportunities, and one slot can only hold one position. The honest statement is: **the router's direction reads were right on both sides, and its latency turned a right call into roughly break-even.** Latency is now the dominant error term on the desk's #1 lever — it is a bigger number than anything on the GOOD/BAD scoreboard.

Architecture — two structural notes

The router is not the only writer of `gate_switches.env`, and it does not know it. `open_hour_watch.py` wrote an arm of `abs_veto_short` at 08-28 16:24:01Z; `reactivate_gates` arms three gates at 22:00:00Z nightly. The router's ABORT fail-safe (“no change”) is only safe while the router is the sole writer — during the 235 minutes it was blind this week, it was not.

And the stand-down text never expired. Your 08-20 instruction — only the day rider trades — was spliced into every router prompt as a comment and stayed there through **1,802 consecutive ticks over seven days**, five of them after the tape had reopened. Nothing pruned it. The router itself broke the loop at 08-27 13:05Z: “the 08-20 stand-down text in my prompt is STALE.” That is a good outcome from a bad mechanism — it worked because the model noticed, not because anything checked. **A carve-out comment with no expiry is a standing instruction, and this one cost the router five days of authority.**

8 • Verdict and Saturday recommendations

VERDICT

KEEP THE ROUTER. It is net-positive, it is not thrashing, and on the calls that can be graded it is winning. It turns a losing fader book into a winning one on both tested universes ($-\$302 \rightarrow +\190 live-managed; $-\$431 \rightarrow +\652 full-historical), its nightlies roll up to **+\$2,846 over 20 days**, its book went **6-for-6 at +\$330.00** this week, and its bad-call scoreboard improved to **64% GOOD on the week against 56% lifetime**.

But three things are true at the same time and all three matter more than the scoreboard. (1) **Its edge is one gate.** 85% of all measured bench value is `abs_veto_short`; the other seven gates are worth 27 cents a blocked trade and two are negative. (2) **Its errors are now timing, not direction.** $\$205-\705 lost to a 20-minute delay on one afternoon, versus $\$241.07$ of graded direction error all week. (3) **Its scope is 19% of the desk.** It managed 6 of 32 trades while the book it cannot see lost $-\$1,576.94$.

#	SATURDAY RECOMMENDATION	THE EVIDENCE	CONFIDENCE
1	Do NOT change ER or NET. Leave 0.15 / 30 exactly as shipped.	NET 20/30/40 = +461/+492/+428 live and +1,076/+1,083/+1,070 full — a 3-cell plateau flat to 1.2%. Every ER above 0.15 is worse in every NET column on both universes.	HIGH two universes agree
2	Cut the router's step from 15 min to 5 min. Change nothing else about the timing.	Two independent measurements agree: the 900-config sweep picks step 5m on both universes, and the live 08-28 tape prices the delay at \$205-\$705 in one afternoon. The step changes how often it looks, not what it decides — the safest knob on the box. Δ the sweep's own robustness is only 4 days helped / 0 hurt (n=4); the live lateness measurement is what carries this.	MEDIUM-HIGH
3	Stop benching capitulation_long and grind_long on direction. Route them by ATR floor only.	All-time bench P&L −\$242.80 and −\$156.20. grind_long is 448 of the 983 blocked trades on the desk — 46% of all blocking, for negative value. Its 22.0 ATR floor already refuses it in exactly the tape the bench targets (234/594/666/1,642 refusal lines with true ATR-14 max 16.09).	HIGH n=448 and n=22
4	Take the untradeable meter out of any gating role. Do not retune 65→45.	Scored intraday — the way it runs — ρ is +0.172 / −0.088 / +0.105 at 14:00/16:00/18:00Z and no cutoff at any hour leaves a profitable traded set. The full-day version's edge fails strip-the-best-2 (+\$2,123 → −\$160, sign flip). It scored the week's two worst days 49 and 46, and told us to sit out +\$1,119.50 of real money on 08-19 and 08-14.	HIGH killed by two named tests
5	Fix the session-limit blindness before anything else on this list. Give the router tick its own quota, or a cheaper fallback decision path.	47 ticks / 235 minutes lost this week, 120 of those minutes in live RTH, five separate outages in four days, the largest 16 ticks, and the two longest are the two most recent. systemd and the trial log both read CLEAN through every one. The fail-safe is one-directional: during a hole the router cannot bench, so whatever is armed stays armed.	HIGH measured in the log
6	Give the router visibility into day_rider, or accept it manages 19% of the desk.	This week: router book 6 trades / +\$330.00; day_rider 26 trades / −\$1,576.94. The router has no switch, no gate and no nightly entry for the book that lost the money. Its Leakage=0 reads clean precisely because that book is outside its universe.	HIGH but this is a desk-design call, not a tuning one
7			

#	SATURDAY RECOMMENDATION	THE EVIDENCE	CONFIDENCE
	Put an expiry on operator carve-out comments in gate_switches.env.	The 08-20 stand-down survived 1,802 ticks across seven days, five of them after the tape reopened, and was only caught because the model noticed it looked stale.	MEDIUM one instance, clear mechanism
8	Either run the exit selector or stop computing its regret.	scaleout_slots() sets adaptive_exit=False on all 12 sub-slots and there is no selector_nightly timer. The \$12,908 of measured regret — 65% of it WIDE chosen when tighter was right, across 101 trades — is money on the table, not money being lost to a bad selector.	MEDIUM verified in code this pass
9	Make the nightlies actually run nightly.	All ten of this week's router_nightly + selector_nightly files were written in one 2-minute batch at 08-28 22:58-23:00, and router_nightly has no files at all for 08-15→08-23. Retrospective monitoring cannot catch anything in time to act on it.	MEDIUM file mtimes

WHAT THIS SECTION IS NOT EVIDENCE FOR — READ BEFORE ACTING

The router's own book took six trades this week. Six. All on one day, all long, all winners. That is not a sample, it is an anecdote, and nothing in section 1 or 2 should be read as a forward estimate of anything.

The 40-day sweep is one summer RANGE regime and the trades in it are partly already hand- and router-gated — the router is being scored on a tape it helped create. Every Δ in section 4 is an illustrative lower bound. The joint-best config rests on **4 days helped / 0 hurt out of 40**.

Every bench "saving" is a counterfactual replayed on the tape, and several of them are sensitive to the stop width — the 08-28 +\$50.48 flips to +\$100 for the gate at $k = 1.25$. If the exit lab widens the stop, some of the savings in section 2 become costs. **They are \$ facts about what happened; they are not claims about the router's judgement.**

And the one conclusion I would stake real money on is the smallest one: the router is too slow. It is the only finding here supported by two

independent methods — a 900-config sweep and a live tape replay — that were not fitted to each other.

★ **THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.**

movement1_census.html was rebuilt **2026-09-04 23:06 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

M1 The biggest runs on MNQ this week — did we show up?

Forget what the desk did this week — start from the raw tape. MNQ printed **76 runs of 1.5×ATR or bigger** between **2026-08-30 22:00** and **2026-09-04 20:59 UTC** (a 15-minute move of at least 50 points, against a typical 15-min range of 34). That is the whole tape, not a top-ten. We **caught 10**, we were positioned against **2**, and we **sat out 64**. The runs we aligned with banked **+\$1,912** of honest money — which is 104% of their \$1,846 hindsight ceiling. Over 100% is not a typo and not an error: the ceiling prices the run's own 15-minute move at one lot, and a position still open when the run ends goes on earning outside the census window. The 64 we sat out left **\$8,845** on the table. The bleed is not the runs; it is the fading we do around them.

THE FULL CENSUS

Every qualifying run, in order. **Move** is the swing in points; **\$ 1lot** is that move on one MNQ lot (\$2/pt) — the hindsight ceiling. **real \$** is what a gate actually banked near it. **Flow** is net aggressor volume in the 60s before; **amp** is pre-run amplitude; **book** is the far-side L2 depth share (below 0.50 =

the side price ran toward was thin), read from capture.db.book, 3 deep, 30s. Green = we caught it, red = we fought it.

TIME (UTC)	DIR	MOVE	\$	1LOT	US	GATE	REAL \$	FLOW	AMP	BOOK	CLUSTER
08-30 23:09	DN	-64	128	sat out	—			+0	0.07	—	UNCLASS
08-30 23:44	DN	-57	114	sat out	—			+0	0.16	—	UNCLASS
08-31 00:07	UP	+86	171	sat out	—			+210	0.07	0.48	UNCLASS
08-31 00:46	DN	-76	152	sat out	—			-295	0.08	0.40	FLOW-LED
08-31 01:59	UP	+52	104	sat out	—			-57	0.06	0.47	UNCLASS
08-31 02:53	UP	+57	114	sat out	—			+25	0.12	0.59	UNCLASS
08-31 04:50	UP	+61	122	sat out	—			+103	0.07	0.54	FLOW-LED
08-31 05:11	UP	+53	106	sat out	—			-37	0.05	0.58	UNCLASS
08-31 06:03	UP	+59	118	sat out	—			+30	0.06	0.49	UNCLASS
08-31 13:16	UP	+90	179	sat out	—			+166	0.09	0.46	OPEN/NEWS
08-31 13:32	DN	-100	200	sat out	—			+66	0.35	0.53	VOL-EXPANSION
08-31 13:55	DN	-75	150	sat out	—			+51	0.19	0.51	OPEN/NEWS
08-31 14:27	UP	+64	128	sat out	—			+31	0.24	0.51	OPEN/NEWS
08-31 15:53	DN	-66	131	sat out	—			+19	0.09	0.55	UNCLASS
08-31 17:36	UP	+53	106	sat out	—			-60	0.06	0.50	UNCLASS
08-31 18:28	DN	-59	118	sat out	—			-132	0.08	0.51	UNCLASS
08-31 19:37	UP	+76	151	sat out	—			-296	0.08	0.48	VACUUM
09-01 05:23	UP	+64	129	sat out	—			+91	0.04	0.51	UNCLASS
09-01 07:31	DN	-88	177	sat out	—			+16	0.08	0.52	UNCLASS
09-01 07:54	DN	-126	253	sat out	—			-149	0.07	0.47	UNCLASS
09-01 08:14	DN	-70	140	FOUGHT	day_rider		-1466	+47	0.11	0.53	UNCLASS
09-01 08:34	DN	-51	102	caught	abs_veto_s hort_A		+96	+97	0.12	0.55	UNCLASS
09-01 08:54	DN	-54	107	caught	abs_veto_s hort_A		+68	-21	0.10	0.51	UNCLASS
	DN	-50	100	sat out	—			-20	0.06	0.51	UNCLASS

09-01
10:37

09-01 11:45	DN	-66	131	sat out	—		+80	0.06	0.51	UNCLASS
09-01 13:26	DN	-61	122	sat out	—		-158	0.06	0.47	OPEN/NEWS
09-01 13:41	UP	+58	117	sat out	—		-1216	0.30	0.51	VACUUM
09-01 14:02	UP	+54	108	sat out	—		-1701	0.18	0.51	VACUUM
09-01 14:18	UP	+68	137	sat out	—		-298	0.17	0.51	OPEN/NEWS
09-01 14:33	UP	+54	109	sat out	—		+1026	0.19	0.50	FLOW-LED
09-01 14:53	DN	-51	102	sat out	—		+786	0.15	0.48	VACUUM
09-01 15:15	UP	+54	109	sat out	—		-215	0.07	0.52	UNCLASS
09-01 15:32	UP	+82	165	sat out	—		+22	0.13	0.50	UNCLASS
09-01 16:05	DN	-93	186	sat out	—		-45	0.07	0.52	UNCLASS
09-01 16:23	DN	-58	115	sat out	—		+155	0.17	0.47	UNCLASS
09-01 16:44	UP	+52	104	sat out	—		+152	0.10	0.50	UNCLASS
09-01 18:30	DN	-82	164	caught	day_rider	+73	+213	0.10	0.46	UNCLASS
09-01 18:45	UP	+78	156	caught	day_rider	+430	+228	0.19	0.49	UNCLASS
09-02 01:36	DN	-59	118	sat out	—		+38	0.06	0.55	UNCLASS
09-02 05:49	DN	-51	102	sat out	—		+10	0.03	0.49	UNCLASS
09-02 06:55	UP	+61	122	sat out	—		+29	0.06	0.56	UNCLASS
09-02 08:39	DN	-87	174	sat out	—		+46	0.07	0.52	UNCLASS
09-02 11:06	UP	+74	149	sat out	—		+176	0.08	0.51	FLOW-LED
09-02 11:23	UP	+51	102	sat out	—		+20	0.06	0.50	UNCLASS
09-02 11:52	UP	+77	154	sat out	—		+91	0.04	0.58	UNCLASS
09-02 13:40	DN	-68	136	sat out	—		-81	0.19	0.49	OPEN/NEWS
09-02 13:55	UP	+93	186	caught	day_rider	+586	-589	0.17	0.49	VACUUM
09-02 14:12	UP	+82	163	caught	day_rider	+586	-221	0.12	0.48	OPEN/ NEWS
09-02 14:45	DN	-92	183	caught	day_rider	-188	+162	0.08	0.51	OPEN/ NEWS
09-02 15:01	UP	+57	114	sat out	—		+133	0.10	0.53	UNCLASS
09-02 15:50	UP	+51	102	FOUGHT	day_rider	+188	-131	0.10	0.51	UNCLASS

09-02 17:03	DN	-52	103	sat out	—		+698	0.12	0.53	VACUUM
09-02 20:06	DN	-54	109	sat out	—		-52	0.07	0.52	UNCLASS
09-03 04:53	DN	-52	104	sat out	—		+11	0.03	0.50	UNCLASS
09-03 05:08	DN	-88	176	sat out	—		+12	0.07	0.50	UNCLASS
09-03 05:24	UP	+60	120	sat out	—		+132	0.15	0.53	UNCLASS
09-03 05:41	UP	+55	110	sat out	—		-470	0.10	0.54	VACUUM
09-03 06:50	UP	+54	108	sat out	—		+63	0.05	0.45	UNCLASS
09-03 07:26	UP	+52	104	sat out	—		-214	0.10	0.48	VACUUM
09-03 07:52	UP	+89	178	sat out	—		+101	0.12	0.53	UNCLASS
09-03 11:01	UP	+58	117	sat out	—		-111	0.11	0.49	VACUUM
09-03 11:35	DN	-60	120	sat out	—		+2	0.21	0.50	UNCLASS
09-03 12:17	UP	+102	204	sat out	—		+25	0.08	0.50	UNCLASS
09-03 12:43	UP	+65	130	sat out	—		+201	0.08	0.53	UNCLASS
09-03 13:19	UP	+70	140	sat out	—		-181	0.11	0.49	OPEN/NEWS
09-03 13:36	UP	+130	260	caught	grind_long_A	-145	+277	0.21	0.47	OPEN/NEWS
09-03 13:53	DN	-91	182	sat out	—		-545	0.14	0.55	FLOW-LED
09-03 14:08	UP	+113	226	sat out	—		-859	0.16	0.50	VACUUM
09-03 14:26	DN	-74	148	sat out	—		-811	0.12	0.46	FLOW-LED
09-03 14:56	UP	+183	366	caught	grind_long_A	+226	+4	0.14	0.52	OPEN/NEWS
09-03 15:14	UP	+79	158	caught	day_rider	+182	-108	0.14	0.48	UNCLASS
09-04 12:29	DN	-124	248	sat out	—		+220	0.09	0.49	VACUUM
09-04 13:19	UP	+66	131	sat out	—		+174	0.07	0.49	OPEN/NEWS
09-04 13:45	UP	+90	180	sat out	—		-1200	0.23	0.46	VACUUM
09-04 14:05	DN	-74	148	sat out	—		+139	0.11	0.50	OPEN/NEWS
09-04 14:39	DN	-68	136	sat out	—		+34	0.12	0.50	OPEN/NEWS

What the runs were — by cause

Each run gets a cause fingerprint from its pre-run tape. This is the map for where to hunt a new gate.

CLUSTER	RUNS	WHAT IT IS
UNCLASS	43	no clear tape tell (quiet ignition)
OPEN/NEWS	14	the 13:00-15:00 UTC cash-open / data window
VACUUM	12	moved AGAINST the tape (a snap-back / stop-run)
FLOW-LED	6	aggressors drove it (a continuation)
VOL-EXPANSION	1	amplitude broke out of a coil

HONEST MONEY — CEILING VS WHAT WE BANKED

RUNS	HINDSIGHT CEILING	WHAT WE REALLY MADE
Caught (aligned)	10	\$1,846 +\$1,912 (104%)
Fought (against)	2	\$242 -\$1,278
Sat out	64	\$8,845 \$0 ← the money on the table

The runs made money and the fading gave it back; we convert a fraction of the ceiling and ignore the rest. The next two movements ask whether we can do better — first with the gates we own, then with new ones.

THE SAME CENSUS ON GOLD — MGC, the second instrument

★ THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT. movement1_census_MGC.html was rebuilt 2026-09-04 22:25 UTC and covers 2026-08-30 22:00 → 2026-09-04 20:59 UTC. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

M1 The biggest runs on MGC this week — did we show up?

Forget what the desk did this week — start from the raw tape. MGC printed **71 runs of 1.5×ATR or bigger** between **2026-08-30 22:00** and **2026-09-04 20:59 UTC** (a 15-minute move of at least 12 points, against a typical 15-min range of 8). That is the whole tape, not a top-ten. We **caught 0**, we were positioned against **0**, and we **sat out 71**. We **have no gate on MGC**, so all 71 runs and **\$13,238** of hindsight ceiling were sat out by construction, not by a decision that went wrong. Nothing here is a bleed — it is an empty seat, and the question the next movements ask is what could sit in it.

THE FULL CENSUS

Every qualifying run, in order. **Move** is the swing in points; **\$ 1lot** is that move on one MGC lot (\$10/pt) — the hindsight ceiling. **real \$** is what a gate actually banked near it. **Flow** is net aggressor volume in the 60s before; **amp** is pre-run amplitude; **book** is the far-side L2 depth share (below 0.50 = the side price ran toward was thin), read from `depth.db.depth_snap`, 5 deep, 60s. Green = we caught it, red = we fought it.

TIME (UTC)	DIR	MOVE	\$ 1LOT	US	GATE	REAL \$	FLOW	AMP	BOOK	CLUSTER
08-30 22:01	UP	+20	204	sat out	—		+0	0.77	0.44	VOL-EXPANSION
08-30 23:34	DN	-16	160	sat out	—		+0	0.14	0.41	UNCLASS
08-30 23:49	UP	+18	176	sat out	—		+0	0.22	0.56	UNCLASS
08-31 00:25	UP	+13	133	sat out	—		-7	0.08	0.51	UNCLASS
08-31 00:46	DN	-28	279	sat out	—		+13	0.14	0.54	UNCLASS
08-31 01:19	DN	-13	131	sat out	—		-20	0.28	0.40	FLOW-LED
08-31 01:48	DN	-20	199	sat out	—		+9	0.25	0.62	UNCLASS
08-31 02:27	DN	-23	227	sat out	—		-12	0.12	0.52	UNCLASS

08-31 02:48	UP	+24	237	sat out	—	+6	0.12	0.52	UNCLASS
08-31 05:14	UP	+16	160	sat out	—	-5	0.14	0.49	UNCLASS
08-31 08:01	DN	-12	119	sat out	—	+1	0.17	0.49	UNCLASS
08-31 12:31	DN	-12	120	sat out	—	-37	0.15	0.50	FLOW-LED
08-31 13:00	UP	+12	125	sat out	—	-9	0.12	0.62	OPEN/NEWS
08-31 13:38	DN	-18	183	sat out	—	-9	0.15	0.53	OPEN/NEWS
08-31 14:24	UP	+18	175	sat out	—	-45	0.23	0.63	VACUUM
09-01 00:52	DN	-13	128	sat out	—	+5	0.17	0.46	UNCLASS
09-01 02:22	DN	-14	143	sat out	—	-2	0.06	0.47	UNCLASS
09-01 02:45	UP	+14	143	sat out	—	-59	0.20	0.48	VACUUM
09-01 05:30	UP	+19	190	sat out	—	+3	0.05	0.33	UNCLASS
09-01 07:25	DN	-14	137	sat out	—	-9	0.13	0.59	UNCLASS
09-01 07:56	DN	-29	294	sat out	—	+1	0.07	0.44	UNCLASS
09-01 08:16	DN	-25	249	sat out	—	-23	0.38	0.57	FLOW-LED
09-01 10:02	UP	+14	140	sat out	—	+14	0.14	0.47	UNCLASS
09-01 11:47	DN	-15	151	sat out	—	+30	0.09	0.46	VACUUM
09-01 12:23	DN	-18	179	sat out	—	-8	0.22	0.46	UNCLASS
09-01 12:57	DN	-34	343	sat out	—	+5	0.13	0.48	UNCLASS
09-01 13:19	UP	+25	254	sat out	—	-3	0.19	0.46	OPEN/NEWS
09-01 13:36	UP	+13	132	sat out	—	-26	0.17	0.42	VACUUM
09-01 14:05	UP	+18	176	sat out	—	-22	0.43	0.63	VOL- EXPANSION
09-01 16:06	DN	-17	166	sat out	—	+11	0.05	0.52	UNCLASS
09-01 18:24	DN	-12	118	sat out	—	-4	0.06	0.39	UNCLASS
09-02 00:18	DN	-13	133	sat out	—	-12	0.26	0.51	UNCLASS
09-02 00:47	DN	-15	149	sat out	—	+5	0.14	0.51	UNCLASS
09-02 01:06	UP	+14	137	sat out	—	+28	0.38	0.53	FLOW-LED
09-02 01:24	DN	-17	172	sat out	—	+10	0.10	0.54	UNCLASS
09-02 01:45	DN	-19	192	sat out	—	-2	0.10	0.52	UNCLASS

09-02 02:12	UP	+13	128	sat out	—	-3	0.29	0.54	UNCLASS
09-02 03:19	DN	-13	128	sat out	—	-5	0.13	0.30	UNCLASS
09-02 03:44	UP	+15	153	sat out	—	-3	0.13	0.70	UNCLASS
09-02 05:20	UP	+14	144	sat out	—	-11	0.10	0.49	VACUUM
09-02 06:55	UP	+13	133	sat out	—	-6	0.07	0.39	UNCLASS
09-02 08:39	DN	-13	129	sat out	—	+1	0.17	0.49	UNCLASS
09-02 10:59	UP	+13	130	sat out	—	+1	0.05	0.51	UNCLASS
09-02 11:22	UP	+16	155	sat out	—	-2	0.10	0.46	UNCLASS
09-02 13:17	UP	+25	251	sat out	—	+2	0.11	0.40	OPEN/NEWS
09-02 13:38	UP	+14	137	sat out	—	+5	0.16	0.45	OPEN/NEWS
09-02 13:55	UP	+22	216	sat out	—	-15	0.17	0.46	OPEN/NEWS
09-02 15:06	DN	-13	130	sat out	—	+7	0.15	0.41	UNCLASS
09-03 00:57	UP	+18	175	sat out	—	+2	0.10	0.49	UNCLASS
09-03 01:12	DN	-13	129	sat out	—	+3	0.26	0.47	UNCLASS
09-03 01:47	UP	+12	124	sat out	—	-6	0.11	0.52	UNCLASS
09-03 06:09	DN	-13	126	sat out	—	+2	0.14	0.48	UNCLASS
09-03 07:52	UP	+17	166	sat out	—	-1	0.08	0.34	UNCLASS
09-03 12:23	UP	+32	320	sat out	—	+1	0.13	0.44	UNCLASS
09-03 12:43	UP	+30	295	sat out	—	-3	0.31	0.62	VOL- EXPANSION
09-03 13:34	DN	-21	206	sat out	—	+7	0.18	0.62	OPEN/NEWS
09-03 14:05	UP	+14	137	sat out	—	+7	0.17	0.54	OPEN/NEWS
09-03 14:53	UP	+29	286	sat out	—	-8	0.14	0.44	OPEN/NEWS
09-03 15:30	DN	-16	155	sat out	—	-1	0.09	0.58	UNCLASS
09-04 01:12	DN	-13	131	sat out	—	+4	0.14	0.49	UNCLASS
09-04 01:47	UP	+14	140	sat out	—	-16	0.20	0.40	VACUUM
09-04 07:33	UP	+16	162	sat out	—	-6	0.12	0.52	UNCLASS
09-04 12:17	DN	-82	815	sat out	—	+4	0.08	0.39	UNCLASS
09-04 12:45	DN	-24	243	sat out	—	-60	0.41	0.46	FLOW-LED

09-04 13:00	UP	+26	260	sat out	—	-82	0.91	0.49	VACUUM
09-04 13:17	UP	+27	270	sat out	—	-1	0.28	0.57	OPEN/NEWS
09-04 13:36	UP	+20	195	sat out	—	+4	0.28	0.52	OPEN/NEWS
09-04 13:59	UP	+21	209	sat out	—	-5	0.31	0.54	VOL- EXPANSION
09-04 14:35	DN	-14	136	sat out	—	+39	0.22	0.59	VACUUM
09-04 15:02	UP	+23	227	sat out	—	-3	0.17	0.72	UNCLASS
09-04 17:18	DN	-14	143	sat out	—	+22	0.13	0.54	VACUUM

What the runs were — by cause

Each run gets a cause fingerprint from its pre-run tape. This is the map for where to hunt a new gate.

CLUSTER	RUNS	WHAT IT IS
UNCLASS	42	no clear tape tell (quiet ignition)
OPEN/NEWS	11	the 13:00-15:00 UTC cash-open / data window
VACUUM	9	moved AGAINST the tape (a snap-back / stop-run)
FLOW-LED	5	aggressors drove it (a continuation)
VOL-EXPANSION	4	amplitude broke out of a coil

HONEST MONEY — CEILING VS WHAT WE BANKED

RUNS	HINDSIGHT CEILING	WHAT WE REALLY MADE
Caught (aligned)	0	\$0
Fought (against)	0	\$0
Sat out	71	\$13,238
		\$0 ← the money on the table

Every row is a sat-out row because there is no gate on this instrument to catch or fight anything. The ceiling is what one lot would have made on a perfect entry and a perfect exit, so treat it as the size of the seat, never as money forgone. The next two movements ask what could fill it.

★ **THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.**

movement2_idle_gates.html was rebuilt **2026-09-04 22:38 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

M2 THE IDLE-GATE LAB — could the gates we already own have caught the runs we sat out?

★★ **REV2 (2026-09-04) — THIS SECTION'S RUN LIST IS THE 2026-08-28 CENSUS FREEZE, NOT THE MOVEMENT 1 PRINTED ABOVE IT**

Every run count, ceiling and per-run number below comes from the census frozen on **2026-08-28**, covering the week to Friday 2026-08-28: **65 runs, 2 caught, 1 fought, 62 sat out, \$10,278 of sat-out ceiling**. The Movement 1 printed earlier in this document is a different, later freeze — 2026-08-30 22:00 → 2026-09-04 20:59Z, **76 runs, 10 caught, 2 fought, 64 sat out, \$8,845 of sat-out ceiling**. **The two lists do not overlap at all**, so a per-run number here does NOT line up with a per-run number there. This section used to say it did; that sentence is withdrawn.

The **2026-08-28** census freeze shows **62 runs that the desk sat out entirely**, worth **\$10,278** if every one had been caught perfectly. The obvious question — and the cheapest one, because it needs no new code — is whether the six gates we already own would have caught any of them had they simply been left armed. This lab answers it by replaying every one of those runs second by second: **36,600 decision-seconds** of real quotes, each gate's own deciders, each gate's own A/B exit pair, and the desk's real cost of \$1.50 a round trip.

THE ANSWER, IN FOUR NUMBERS

1. Fired exactly as they are configured today, the gates take **32 lots into the runs and lose \$-741.50** (\$-23.17/lot, 18.8% win). **2.** Strip every constraint the live mechanism imposes — the "ungated" layer, which is the gates' raw pattern with nothing filtering it — and they take **114 lots for \$-168.50**. Almost break-even, on 3.6× the volume. **3.** Now keep only the fires that pointed the same way as the run: **26 lots, \$+1,218.50, 65.4% win, \$+46.87/lot**. **4.** And that is the trap, because the direction is only knowable afterwards. **The gates do not know which way the run will go** — that is what "sat out" means. Reading number 3 as an opportunity is reading the answer off the back of the exam paper.

§1 The population — what we are replaying

Every sat-out run from this week's census freeze: **62 runs, \$10,278 of hindsight ceiling, 36,600 decision-seconds. ★ REV2 — which figure this lab replayed, stated, and what it costs.** All **62**, including the **08-26 20:46** window, whose 180 five-second bars straddle the 21:00-22:00Z CME halt and are therefore 75 minutes of clock across a closed market rather than a 15-minute move (Movement 1's census now marks it; Part 1's run chart shows the gap). Drop it and the population is **61 runs and \$9,830**. Here is what that actually changes, measured rather than asserted: — **MECH: nothing at all.** 32 lots, -\$741.50, identical. The section's headline number does not depend on the bad window. — **UNGATED: it gets WORSE without it.** 114 lots -\$168.50 becomes 112 lots -\$264.00, because the two fires inside that window were both winners. — **And it does move one verdict in §4.** Those two fires are grind_long_A and _B, entered 20:44:20Z and closed 20:48 and 20:49 — on live tape, before the halt, at +\$38.00 and +\$57.50. They are real trades; it is the run's label that is the artefact. Remove them and grind_long's in-direction book goes from 10 lots at +\$20.10 to 8 lots at +\$13.19, which drops it below its own random-entry control (+\$17.88) and takes §4's like-for-like count from 3 of 5 to 2 of 5. ★ REV2 — this is the **2026-08-28** freeze (65 runs, 2 caught, 1 fought, 62 sat out). It is NOT the freeze Movement 1 prints in this document, which is a week younger and shares none of these runs, so a per-run number here does not line up with a per-run number there. The claim that it did is

withdrawn. (Last cycle's Movement 2 was computed on a superseded freeze and had to carry a banner saying so. This section is correctly computed on the 08-28 freeze; what changed is that the rest of the document moved on without it.)

By regime, at the moment the run began

REGIME	RUNS	CEILING	SHARE OF CEILING
BUILDING	19	\$2,778	27.0%
DEAD-CHOP	16	\$2,722	26.5%
NORMAL-CHOP	14	\$2,553	24.8%
VIOLENT-WHIPSAW	7	\$1,383	13.5%
CLEAN-TREND	5	\$686	6.7%
NO-TAPE	1	\$156	1.5%

By session

SESSION (UTC)	RUNS	CEILING	SHARE OF CEILING
US 13-21	33	\$6,274	61.0%
ASIA 00-07	16	\$2,213	21.5%
LONDON 07-13	12	\$1,635	15.9%
NO-TAPE	1	\$156	1.5%

§2 The scoreboard — two layers, and why both are printed

MECH is each gate exactly as it is configured on the live desk right now. **UNGATED** is the same pattern with the live mechanism's own constraints removed — no 55-second veto, no ATR floor, no volume surge requirement. The pair is printed together because the gap between them is the value of the filtering, and it is the only honest way to ask “would arming it have helped?”: MECH answers “as it stands”, UNGATED answers “is there anything there at all”.

MECH — the live configuration

GATE	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$
abs_veto_lo ng	3	3	6	\$−220.00	0.0%	\$−36.67	2	\$−91.00
	2	2	4	\$−51.50	25.0%	\$−12.88	0	\$+0.00

GATE	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$
abs_veto_short								
exhaustion_short	6	6	12	\$-69.00	33.3%	\$-5.75	2	+\$233.00
grind_long	5	4	10	\$-401.00	10.0%	\$-40.10	4	\$-31.00
ALL fires, both directions	—	—	32	\$-741.50	18.8%	\$-23.17	8	+\$111.00

UNGATED — the raw pattern

GATE	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$
abs_veto_long	6	5	12	\$-404.00	0.0%	\$-33.67	2	\$-93.00
abs_veto_short	10	7	20	\$-53.00	35.0%	\$-2.65	2	+\$100.00
capitulation_long	4	3	8	+\$60.00	37.5%	+\$7.50	6	+\$40.50
exhaustion_short	9	9	18	+\$589.00	44.4%	+\$32.72	6	+\$970.00
grind_long	28	23	56	\$-360.50	33.9%	\$-6.44	10	+\$201.00
ALL fires, both directions	—	—	114	\$-168.50	32.5%	\$-1.48	26	+\$1,218.50

★ **Read the MECH row first and stop there if you only read one number.** The desk's own gates, armed into this week's biggest runs, lose \$ **-741.50 on 32 lots**. Not arming them was worth exactly that much, and it is the same answer this lab has now returned three cycles running.

§3 The in-direction number is an oracle — here is what it is worth and why you cannot have it

The single most seductive line in this lab is the ungated in-direction book: **26 lots, \$+1,218.50, 65.4% win, \$+46.87/lot**. It looks like a strategy. It is not: in-direction means “the fire agreed with where the run subsequently went”, which is information from the future. The honest version of that book is the ALL-fires line right above it — **114 lots for \$-168.50** — because at the moment of the fire, both books are the same book.

It is printed anyway, and stress-tested below, for one reason: if even the oracle version were negative, the gates would be dead and no selector could rescue them. It is positive, so the question “could something have picked the right ones in advance?” stays open — and Movement 3 is where that question gets asked properly. What follows is how much of that oracle survives contact with robustness.

TEST	LOTS	NET \$	WIN %	\$/LOT
Headline (ungated, in-direction)	26	\$+1,218.50	65.4%	\$+46.87
Strip the best 1 lot	25	\$+1,021.50	64.0%	\$+40.86
Strip the best 2 lots	24	\$+824.50	62.5%	\$+34.35
Strip the best 3 lots	23	\$+656.00	60.9%	\$+28.52
Day 08-25 alone	4	\$+314.00	50.0%	\$+78.50
Day 08-26 alone	6	\$+531.00	100.0%	\$+88.50
Day 08-27 alone	12	\$+431.50	75.0%	\$+35.96
Day 08-28 alone	4	\$-58.00	0.0%	\$-14.50
Leave-one-day-out — drop 08-25	22	\$+904.50	68.2%	\$+41.11
Leave-one-day-out — drop 08-26	20	\$+687.50	55.0%	\$+34.38
Leave-one-day-out — drop 08-27	14	\$+787.00	57.1%	\$+56.21
Leave-one-day-out — drop 08-28	22	\$+1,276.50	77.3%	\$+58.02

Three lots of 26 carry **46%** of the oracle book, and it fires on **4 of 5 sessions**. Every leave-one-day-out fold stays positive — but with 4 live days and a 3-lot concentration that is a statement about how few independent observations there are, not a robustness pass. ★ [See the standing lesson](#): leave-one-day-out cannot fail when a handful of days carry the money.

§4 The random-entry control — the test that decides whether any of this is a gate at all

The kill test. If you had shown up at a **random second** inside the same 10-minute pre-ignition window, **in the run's own direction**, with the same gate's exits and the same costs — what would that have paid? Anything a gate earns has to beat this, because the gate's claim is that its trigger

matters, not that being long a run pays. Pool: every 10th decision-second of every sat-out run, per gate.

GATE	RANDOM ENTRIES	MEAN \$/ FIRE, WITH THE RUN	% POSITIVE	MEAN \$/ FIRE, AGAINST IT	THE GATE'S OWN \$/LOT (UNGATED)	VERDICT
abs_veto_long	1,740	\$+13.38	47.8%	\$-16.65	\$-33.67	not comparable as printed — see the REV2 box below the “\$/lot” column is ALL fires, both directions, against an IN-DIRECTION control — two different things. Like-for-like this gate loses to it: \$-46.50/lot on 2 in-direction lots against \$ +13.38, bootstrap p = 0.7692
abs_veto_short	1,920	\$-2.08	39.7%	\$-22.97	\$-2.65	not comparable as printed — see the REV2 box below the “\$/lot” column is ALL fires, both directions, against an IN-DIRECTION control — two different things. Like-for-like this gate BEATS it: \$ +50.00/lot on 2 in-direction lots against \$-2.08, bootstrap p = 0.1792
capitulation_long	1,740	\$+13.62	47.8%	\$-17.36	\$+7.50	not comparable as printed — see the REV2 box below the “\$/lot” column is ALL fires, both directions, against an IN-DIRECTION control — two different things. Like-for-like this gate loses to it: \$+6.75/lot on 6 in-direction lots against \$ +13.62, bootstrap p = 0.5592
exhaustion_short	1,920	\$+46.64	53.0%	\$-74.09	\$+32.72	not comparable as printed — see the REV2 box below the “\$/lot” column is ALL

GATE	RANDOM ENTRIES	MEAN \$/ FIRE, WITH THE RUN	% POSITIVE	MEAN \$/ FIRE, AGAINST IT	THE GATE'S OWN \$/LOT (UNGATED)	VERDICT
						fires, both directions, against an IN-DIRECTION control — two different things. Like-for-like this gate BEATS it: \$ +161.67/lot on 6 in-direction lots against \$ +46.64, bootstrap p = 0.0331
grind_long	1,740	\$+17.88	46.4%	\$−31.33	\$−6.44	not comparable as printed — see the REV2 box below the “\$/lot” column is ALL fires, both directions, against an IN-DIRECTION control — two different things. Like-for-like this gate BEATS it: \$ +20.10/lot on 10 in-direction lots against \$ +17.88, bootstrap p = 0.4353
rgv_short	1,920	\$+1.63	39.7%	\$−17.69	\$+0.00	no fires — unmeasurable the gate took 0 lots into these runs, so there is nothing for a control to beat. The \$+0.00 in the previous column is an ABSENCE, not a result

Showing up at random inside the pre-run window, pointed the right way, pays a positive mean on **5 of the 6 gates** — because a run is, by construction, a move that went somewhere.

★ REV2 — “THE GATES BEAT THAT CONTROL ON 0 OF 6” IS WITHDRAWN. IT COMPARED TWO DIFFERENT THINGS.

The last column of the table above is the gate's **all-fires** \$/lot — both directions, because at the moment of the fire the gate does not know which way the run will go. The control column beside it is the **WITH-the-run** mean, which has been handed the direction. Scoring one against the other loses by construction and says nothing: it is the same error \$3 spends three paragraphs warning about, made in the row underneath it.

Read consistently — either both sides direction-handed, or both sides direction-blind — the count is not 0 of 6. Harness: scripts/rev2_m2_indirection_control.py, 20,000 bootstrap draws, seed 20260829.

GATE	IN-DIR LOTS	GATE, IN-DIRECTION \$/LOT	CONTROL, WITH THE RUN	LIKE-FOR-LIKE	BOOTSTRAP P	GATE, ALL FIRES \$/LOT	CONTROL, DIRECTION-BLIND	BLIND
exhaustion_short	6	+\$161.67	+\$46.64	gate wins	0.033	+\$32.72	-\$13.73	gate wins
abs_veto_short	2	+\$50.00	-\$2.08	gate wins	0.179	-\$2.65	-\$12.53	gate wins
grind_long	10	+\$20.10	+\$17.88	gate wins	0.435	-\$6.44	-\$6.72	gate wins, by 28¢
capitulation_long	6	+\$6.75	+\$13.62	control wins	0.559	+\$7.50	-\$1.87	gate wins
abs_veto_long	2	-\$46.50	+\$13.38	control wins	0.769	-\$33.67	-\$1.63	control wins
rgv_short	0	—	+\$1.63	no fires — unmeasurable	—	—	-\$8.03	unmeasurable
count	26			gates 3 of 5				gates 4 of 5

And the honest answer is neither 0 of 6 nor 3 of 5 — it is **UNRESOLVABLE at this n**. The in-direction lot counts are **2, 2, 6, 6 and 10**. Bootstrapping each gate's mean against n draws from its own control pool, only exhaustion_short clears $P = 0.05$ ($P = 0.033$ on six lots), and Holm-correcting across the five gates tested puts its adjusted P at **0.166**. **Nothing here separates from the control in either direction**. The control was worth running and it produced a null, which is a different result from the one this section printed.

★ And one of those three wins is fragile. grind_long's in-direction margin over its control is **\$2.22 a lot on ten lots**, and two of those ten

come from the **08-26 20:46** window — the one whose 180-bar window straddles the CME halt (§1). The two fires themselves are real (live tape, 20:44:20Z in, out at 20:48 and 20:49, +\$38.00 and +\$57.50); it is the run label around them that is the artefact. Drop the run and grind_long falls to **+\$13.19** on eight lots, below its +\$17.88 control, and the like-for-like count becomes **2 of 5**. A count that moves on which two lots you include is another way of saying this test has no power at this n.

★ **What this does NOT touch.** The section's real number is the MECH line in §2 — the six gates as configured take 32 lots into these runs for **\$-741.50**. That is a direct measurement of the desk's own gates on the desk's own tape and it needs no control at all. It is unchanged, and it is still the answer to “would arming them have helped?”

And note the AGAINST column, which is the same control pointed the wrong way. Where a gate's number sits between the two, what it is measuring is the run, not the gate.

§5 The threshold sweep — is any gate one constraint away from working?

Each cell relaxes **exactly one** binding constraint and leaves the rest of the gate alone, so a cell that improves names the constraint that is costing money. strip3\$ is the cell's net after its three best lots are deleted — the concentration check, printed beside every cell because on this sample almost every positive number is three trades wide.

abs_veto_long

CELL	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$	STRIP- BEST-3 \$
LIVE thr1.5+vol	3	3	6	\$-220.00	0.0%	\$-36.67	2	\$-91.00	\$-128.50

CELL	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$	STRIP- BEST-3 \$
+amp ← live									
fast (2-bar)	4	4	8	\$-223.00	12.5%	\$-27.88	2	\$-91.00	\$-215.00
no 55s veto	6	5	12	\$-404.00	0.0%	\$-33.67	2	\$-93.00	\$-320.00
no amp floor	6	5	12	\$-206.00	16.7%	\$-17.17	2	\$-91.00	\$-284.50
no vol surge	14	14	28	\$-693.50	17.9%	\$-24.77	10	\$-319.50	\$-850.00
thr 1.0	5	5	10	\$-130.00	30.0%	\$-13.00	4	+\$2.00	\$-272.00
thr 1.25	4	4	8	\$-226.00	12.5%	\$-28.25	2	\$-94.00	\$-218.00
thr 2.0	1	1	2	\$-75.00	0.0%	\$-37.50	0	+\$0.00	+\$0.00

abs_veto_short

CELL	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$	STRIP- BEST-3 \$
LIVE thr1.5+vol +amp ← live									
	2	2	4	\$-51.50	25.0%	\$-12.88	0	+\$0.00	\$-30.00
fast (2-bar)	2	2	4	\$-131.00	0.0%	\$-32.75	0	+\$0.00	\$-35.00
no 55s veto	10	7	20	\$-53.00	35.0%	\$-2.65	2	+\$100.00	\$-230.00
no amp floor	4	4	8	\$-137.50	12.5%	\$-17.19	0	+\$0.00	\$-141.50
no vol surge	13	11	26	\$-346.50	30.8%	\$-13.33	12	\$-348.50	\$-521.50
thr 1.0	5	5	10	\$-179.00	20.0%	\$-17.90	2	+\$11.50	\$-228.50
thr 1.25	4	4	8	\$-190.50	12.5%	\$-23.81	0	+\$0.00	\$-169.00
thr 2.0	1	1	2	\$-60.00	0.0%	\$-30.00	0	+\$0.00	+\$0.00

capitulation_long

CELL	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$	STRIP- BEST-3 \$
climax 1.5	10	10	20	+\$102.50	45.0%	+\$5.12	16	+\$135.00	\$-73.00
climax 1.5 no-flip	16	14	32	+\$329.00	50.0%	+\$10.28	22	+\$283.50	+\$149.50
climax 2.0	4	4	8	\$-19.00	37.5%	\$-2.38	6	\$-31.50	\$-154.00
dom 0.5	1	1	2	+\$97.50	100.0%	+\$48.75	2	+\$97.50	+\$0.00
no atr floor	4	3	8	+\$60.00	37.5%	+\$7.50	6	+\$40.50	\$-76.50

exhaustion_short

CELL	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$	STRIP- BEST-3 \$
LIVE net>=400 ← live	6	6	12	\$-69.00	33.3%	\$-5.75	2	+\$233.00	\$-356.50
move_max 4pt	12	11	24	\$-27.00	33.3%	\$-1.12	8	+\$138.00	\$-652.50
net>=150	40	35	80	\$-821.00	27.5%	\$-10.26	34	+\$1,212.00	\$-1,359.50
net>=150 wall1.2	45	37	90	\$-1,259.00	28.9%	\$-13.99	38	+\$772.00	\$-1,697.50
net>=250	16	16	32	\$-336.00	31.2%	\$-10.50	16	+\$378.00	\$-767.00
wall 1.2	12	12	24	\$-111.00	33.3%	\$-4.62	10	\$-254.00	\$-730.50

grind_long

CELL	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$	STRIP- BEST-3 \$
LIVE atr>=22 ← live	5	4	10	\$-401.00	10.0%	\$-40.10	4	\$-31.00	\$-420.50
atr>=10	21	18	42	\$-569.50	28.6%	\$-13.56	6	+\$5.00	\$-748.00
atr>=14	12	9	24	\$-540.50	25.0%	\$-22.52	4	\$-90.50	\$-701.00
atr>=18	6	6	12	\$-469.50	16.7%	\$-39.12	4	\$-90.50	\$-506.50
ext_hi 2.0	3	3	6	\$-208.00	16.7%	\$-34.67	2	+\$68.00	\$-212.00
ext_hi 8.0	5	4	10	\$-401.00	10.0%	\$-40.10	4	\$-31.00	\$-420.50
ext_lo 0.0	5	4	10	\$-401.00	10.0%	\$-40.10	4	\$-31.00	\$-420.50
no atr floor	28	23	56	\$-360.50	33.9%	\$-6.44	10	+\$201.00	\$-544.50
slope>=0. 2	7	5	14	\$-662.00	7.1%	\$-47.29	4	\$-28.00	\$-683.00
slope>=0. 6	2	1	4	\$-31.00	25.0%	\$-7.75	4	\$-31.00	\$-48.50

rgv_short

CELL	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$	STRIP- BEST-3 \$
atr>=13	5	4	10	+\$0.00	40.0%	+\$0.00	8	\$-98.00	\$-158.00
ext 1.0 + slope off	1	1	2	+\$187.50	100.0%	+\$93.75	2	+\$187.50	+\$0.00
fast_turn	1	1	2	\$-120.00	0.0%	\$-60.00	2	\$-120.00	+\$0.00
no atr floor	9	7	18	\$-27.50	33.3%	\$-1.53	14	\$-80.50	\$-206.00
slope_max 2.0	1	1	2	+\$112.50	100.0%	+\$56.25	2	+\$112.50	+\$0.00

CELL	FIRES	RUNS	LOTS	NET \$	WIN %	\$/LOT	IN-DIR LOTS	IN-DIR \$	STRIP- BEST-3 \$
slope_max 99 (off)	1	1	2	+\$187.50	100.0%	+\$93.75	2	+\$187.50	+\$0.00

★ **Nothing in this sweep is a Monday change.** Read down the Net \$ column and almost every cell of every gate is negative — relaxing a constraint mostly buys more losing fires. The handful of positive cells (rgv_short / slope_max 99 (off) at +\$187.50, capitulation_long / climax 1.5 no-flip at +\$329.00) are one and sixteen fires wide respectively, and the first of those is **a single fire**: its strip-best-3 column is \$0.00 because there is nothing left to strip. That is not a threshold finding, it is a sample size.

MOVEMENT 2 — VERDICT

No. The gates we own would not have caught this week's runs, and arming them into the runs costs \$-741.50 on 32 lots. That is the third cycle in a row this lab has returned the same answer against a different week of tape, which is the most useful thing about it: the finding does not depend on which runs are in the census.

Three things follow, and only the third is new this week:

1. **The bench is earning its keep.** Every gate benched into this week's run windows saved money by being benched. Nothing on the action card should propose re-arming one on the strength of a missed run.
2. **No threshold rescues them.** A 724-cell sweep that relaxes one constraint at a time finds no cell that is both positive and more than three trades wide.
3. ★ **And the random-entry control returns a NULL — it cannot separate the triggers from chance in either direction.**

REV2 — restated Rev1 of this section said “not one gate's trigger beats it, 0 of 6”. That compared a control which had been handed the run's direction against the gate's both-directions number, and it loses by construction. Scored consistently (\$4): in-direction the gates beat the control on **3 of 5**, direction-blind on **4 of 5**, and rgv_short never fired so it cannot be scored at all. But the in-direction lot counts are **2, 2, 6**,

6 and 10, and bootstrapped against their own control pools only exhaustion_short clears $P=0.05$ (0.033; Holm-adjusted across five gates, 0.166). **Previous cycles measured whether the gates made money; this one tried to measure whether they know anything, and came back with not enough evidence to say.** That is a weaker claim than Rev1 printed and it is the one the data supports.

The corollary is the one the whole report keeps arriving at from different directions: **the money in these runs is a SELECTION problem, not an entry problem.** Detection is solved — the tape prints 65–76 runs a week and something fires near most of them. Nothing we own can tell in advance which ones are worth boarding. That is Movement 3's question.

★ **THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.** movement3_greenfield.html was rebuilt **2026-09-04 22:36 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2–3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

M3 THE GREENFIELD LAB — build a gate from scratch, then try everything to kill it

★★ REV2 (2026-09-04) — THIS SECTION'S RUN LIST IS THE 2026-08-28 CENSUS FREEZE, NOT THE MOVEMENT 1 PRINTED ABOVE IT

Every run count, ceiling and per-run number below comes from the census frozen on **2026-08-28**, covering the week to Friday 2026-08-28: **65 runs, 2 caught, 1 fought, 62 sat out, \$10,278 of sat-out ceiling**. The Movement 1 printed earlier in this document is a different, later freeze — 2026-08-30 22:00 → 2026-09-04 20:59Z, **76 runs, 10 caught, 2 fought, 64 sat out, \$8,845 of sat-out ceiling**. **The two lists do not overlap at all**, so a per-run number here does NOT line up with a per-run number there. This section used to say it did; that sentence is withdrawn.

Movement 2 proved the gates we own cannot reach the runs in the 08-28 freeze. This is the from-scratch attempt on that same freeze: **62 sat-out runs, \$10,278 of ceiling**, and permission to build anything. Everything below was computed this cycle, and it is reported the way this lab always reports — **the failures by name, with the test that killed them**, because a greenfield lab that only publishes its survivors is a lab that publishes noise.

THE WEEK IN ONE PARAGRAPH

Boarding is not the problem. A deliberately stupid rule — wait for a 5-minute thrust of $1.5 \times \text{ATR}$ and buy it — fires inside **61 of the 62 runs**, a median of 3 minutes late, with a median **45.8 points still to come**. Boarded perfectly it is worth **\$6,050, 58.9% of the whole ceiling**. **Selection is the problem**. That same rule fires **66.4 times a day** and only **21.1%** of those fires are inside a run. **And this week we finally measured why that cannot be fixed with the features we have** — see §3, which is the most important result in the movement.

§1 The prize, and how late you can afford to be

The **2026-08-28** census freeze (not the Movement 1 above, which is a week younger): **65 runs**, we caught **2**, fought **1**, sat out **62**. By cluster:

UNCLASS 39 · OPEN/NEWS 12 · FLOW-LED 8 · VACUUM 5 · VOL-EXPANSION 1.

The boarding ladder asks the only question that matters before any signal work: if you waited for the move to prove itself, would there be anything left? **W** is the thrust window in minutes and **k** its size in ATR; **left** is the points still on the table at the moment of boarding.

W	K×ATR	FIRES INSIDE	MEDIAN LATE	MEDIAN LEFT	P25 LEFT	NOTHIN G LEFT	ORACLE \$	% OF CEILING
3	1.5	60 / 62	3 min	46.6 pt	30.8 pt	2	\$5,642	54.9%
5	1.5	61 / 62	3 min	45.8 pt	34.2 pt	2	\$6,050	58.9%
5	2.0	59 / 62	4 min	40.8 pt	26.9 pt	2	\$5,074	49.4%
5	2.5	48 / 62	4 min	36.4 pt	17.6 pt	3	\$3,666	35.7%
10	2.0	60 / 62	6 min	36.2 pt	23.7 pt	1	\$4,920	47.9%
10	2.5	58 / 62	8 min	30.6 pt	18.1 pt	4	\$4,140	40.3%
15	2.5	58 / 62	7 min	27.1 pt	10.8 pt	1	\$4,095	39.8%

...and the price of that coverage

W	K×ATR	ENTRIES THIS WEEK	PER DAY	OF WHICH INSIDE A RUN	HIT RATE
5	1.5	332	66.4	70	21.1%
5	2.0	255	51.0	63	24.7%
5	2.5	195	39.0	51	26.2%
10	2.5	252	50.4	66	26.2%

★ **Detection is solved and it has been for three cycles.** Every rung of this ladder boards nearly every run with a third to a half of the move still ahead of it. The entire problem is the last column: **three fires in four are not in a run at all**, and each of those pays the spread and the fee.

§2 Half the runs have no precursor at all

Split the sat-out runs by whether anything was visible before ignition — a volume z-score, an ATR expansion, an aggressor imbalance:

FAMILY	RUNS	CEILING	MEDIAN MOVE	MEDIAN VOL Z	MEDIAN ATR EXPANSION	UP / DOWN
A — a precursor is visible	31	\$5,510	71 pt	2.63	1.31	13 / 18
B — nothing to see	31	\$4,768	68 pt	0.36	0.91	16 / 15

Two things kill the obvious idea. **First, the split is 50/50** — 31 runs show a precursor and 31 show nothing, and the silent half still carries **\$4,768** of ceiling on a median move of 68 points, which is the same size as the visible half. A precursor filter therefore throws away half the prize before it starts. **Second, the precursor is not rare.** Across 6,780 minutes, **42.4% of all minutes** look like family A — and family A holds 50% of the runs, a lift of just **1.18×**.

★ REV2 — THE TABLE ABOVE IS COMPUTED OFF THE DEDUPED RUN LIST, AND THIS SECTION'S OWN APPENDIX SAYS NOT TO DO THAT

The VACUUM appendix (§appendix 1) establishes that run_census.py's keep-strongest dedup selects **local extrema** — prior-5-minute direction agreement shifts **48.66% → 22.55%** across it and the VACUUM rate climbs **10.45% → 14.01% → 21.87%** — and it says the finding “invalidates every pre-run price-path statistic quoted off the run list”. §3 obeys that and works on the unselected population. **§2 did not.** So the same test is re-run here on **every minute of the census week whose forward-15-minute move clears the same 55-point threshold**, with no dedup: 463 qualifying minutes instead of 62 picked runs, same features, same window. Harness: scripts/rev2_family_unselected.py.

POPULATION	N	FAMILY A	FAMILY B	A'S SHARE	BASE RATE	LIFT	B'S MEDIAN MOVE
Deduped census runs (the table above)	62	31	31	50.0%	42.4%	1.18×	68 pt
Unselected — every fwd15 ≥ 55pt minute	463	270	193	58.3%	42.4%	1.375×	70 pt

The correction goes the OTHER way, and the conclusion survives it. On the honest population the precursor looks better, not worse — 58.3% of big minutes against a 42.4% base rate, a lift of 1.375× rather than 1.18×. The dedup was working against family A, because a local extremum is the point where the move has already happened and the volume spike is behind it. **But the load-bearing half of the finding is untouched:** family B is still **193 of 463 (41.7%)** of the big minutes, and its median move is **70 points against family A's 71** — on 7× the sample, the silent half is still the same size as the visible half. A 1.375× lift on a feature that fires 42% of the time does not pay for throwing away 42% of the prize.

△ And note what the honest population does to the DIRECTION column, which is the thing §3 goes on to demolish: family A's big minutes run 114 up / 156 down and family B's run 97 / 96. Neither is a direction call; the first is the week's own down-tilt.

The oracle bracket — what the boarding rule is worth at each level of cheating

RULE	N	NET \$	\$/TRADE	WIN %
RULE (board the proven move)	160	\$−500.71	\$−3.129	35.0%
ORACLE (know the next 15 min)	133	+\$5,186.87	+\$38.999	60.2%
ANTI-ORACLE (get it backwards)	197	\$−8,170.80	\$−41.476	9.6%
RULE, boards inside a run only	55	+\$768.66	+\$13.976	43.6%
RULE, boards NOT in a run	134	\$−1,721.89	\$−12.850	29.9%

Read the last two rows together. The same rule, split by something it cannot know: its 55 in-run boards make **+\$768.66** and its 134 out-of-run boards lose **\$−1,721.89**. The whole difference between a \$−500.71 rule and a profitable one is a selector nobody has built, and §3 is the evidence that the features on this desk cannot build it.

§3 ★ THE HEADLINE — the precursors predict VOLATILITY, not moves, and it took an ATR control to see it

The biggest study in the movement: **52,937 minutes across 41 days**, asking whether any pre-move feature separates the minutes that escalate from the minutes that do not. The number in each cell is an **AUC — 0.50 is a coin flip**, 1.00 is perfect.

A. Raw point thresholds — this is the table that looks like a discovery

MOVE ≥	N MINUTES	VOLUME Z	TRADE-COUNT Z	ATR EXPANSION	AGGRESSOR-FLOW Z	ER(15)	ATR LEVEL
55 pt	6,066	0.553	0.557	0.568	0.570	0.513	0.770
80 pt	2,496	0.587	0.594	0.594	0.604	0.525	0.802
100 pt	1,312	0.621	0.626	0.611	0.616	0.541	0.808
120 pt	782	0.646	0.645	0.615	0.637	0.567	0.808
160 pt	352	0.657	0.649	0.591	0.622	0.598	0.789
200 pt	145	0.653	0.654	0.567	0.617	0.598	0.767

Every column beats a coin flip and every p-value is 0.00. **ATR level reaches 0.81.** On this table alone you would ship a volatility-gated boarder tomorrow.

B. The same question with the threshold expressed in ATR — i.e. “did it move a lot for a tape this volatile”

MOVE ≥	N MINUTES	VOLUME Z	TRADE-COUNT Z	ATR EXPANSION	AGGRESSOR-FLOW Z	ER(15)	ATR LEVEL
3×ATR	11,707	0.516	0.518	0.457	0.498	0.504	0.436
5×ATR	3,296	0.510	0.515	0.422	0.487	0.529	0.380
8×ATR	847	0.471	0.488	0.335	0.453	0.508	0.337
12×ATR	247	0.364	0.369	0.234	0.328	0.519	0.306

C. And conditioned on ATR — within-band, so the volatility level is held fixed

MOVE ≥	VOLUME Z	TRADE-COUNT Z	ATR EXPANSION	AGGRESSOR-FLOW Z	ER(15)	ATR LEVEL
55 pt	0.501	0.509	0.441	0.496	0.516	0.524
80 pt	0.491	0.509	0.391	0.481	0.512	0.537
120 pt	0.482	0.487	0.353	0.451	0.484	0.544
160 pt	0.484	0.490	0.293	0.445	0.564	0.556

★★ WHAT TABLES B AND C DO TO TABLE A

Table A is an artefact of not controlling for volatility. A 55-point move is easy on a wide-ATR tape and hard on a quiet one, so anything correlated with ATR — volume, trade count, aggressor flow, all of which are — scores well against a fixed point threshold. It is the same instrument twice.

Normalise the threshold by ATR (B) and the discrimination collapses: ATR expansion falls to **0.234** and ATR level itself to **0.306** — both now worse than a coin flip, i.e. actively anti-predictive. Hold ATR fixed and ask within the band (C) and every feature sits between 0.29 and 0.56.

The desk's entire precursor toolkit measures one thing: how volatile the tape already is. That is genuinely useful — it says a big move is possible now — and it is completely useless for the question we keep asking it, which is which possible move will happen. Three cycles of “selection is the problem” now have a mechanism: we have been feeding the selector a volatility meter and asking it for a direction.

D. And direction, for completeness

MOVE ≥	N	% UP	PRIOR 5-MIN MOVE	30-MIN RANGE BREAK	60S AGGRESSOR FLOW	CONTINUATION AGREES
55 pt	6,066	48.9%	0.501	0.522	0.496	52.2%
80 pt	2,496	49.3%	0.519	0.550	0.495	48.3%
120 pt	782	48.1%	0.553	0.595	0.490	52.0%
160 pt	352	50.0%	0.581	0.541	0.486	40.9%

Up-share sits on 50%, aggressor flow is below 0.50 at every threshold, and continuation agreement runs from 40.9% to 52.2% with no trend. **Nothing on this desk carries forward direction.** That is consistent with the router study in Part 2.6 and with the standing FLOW-LED/VACUUM result, and it is now measured on 52,937 minutes rather than on a week.

§4 COIL-BRK — this week's fresh candidate, built and killed in one cycle

The construction: a compression (“coil”) filter arms the session, then a range break is taken. Window **2026-06-22 → 2026-08-28, 53 sessions** — deliberately far wider than the report week, because a one-week greenfield result is not testable.

The arm ladder — how tight the coil filter has to be

COIL QUANTILE	N	NET \$	\$/TRADE	WIN %	DAYS
0.1	274	\$−1,118.35	\$−4.082	30.7%	52
0.2	419	+\$1,038.05	+\$2.477	34.1%	52
0.3	536	\$−912.05	\$−1.702	32.5%	52
0.4	568	\$−419.00	\$−0.738	32.6%	52
0.5	574	\$−1,111.50	\$−1.936	33.3%	52
0.7	458	\$−65.20	\$−0.142	34.1%	52
1.0	60	\$−235.55	\$−3.926	30.0%	44

★ **Stop here and the candidate is already in trouble.** The ladder is not monotone and it is not even single-signed: 0.1 loses \$1,118, 0.2 makes \$1,038, 0.3 loses \$912, 0.4 loses \$419, 0.7 is flat. A real filter tightens into an edge. This one flips sign four times, which is the signature of a parameter that is fitting noise. It was carried through the full battery anyway, because a hunch is not a verdict.

The costed battery — every kill test, on all three surviving exit cells

CELL	N	NET \$	\$/TR	IS \$/TR	OOS \$/TR	STRIP-BEST-3	SIGN-FLIP	GREEN DAYS	PLACEBOS BEATING IT
COIL-BRK stop3.0/targ8.0	513	\$+1,126.27	+\$2.195	+\$3.865	\$−4.615	\$−0.437	\$−3.496	22/52	2 of 5
COIL-BRK stop3.0/targ4.0	524	\$+921.70	+\$1.759	+\$4.286	\$−8.446	+\$0.073	\$−2.045	27/52	1 of 5
COIL-BRK stop2.0/targ4.0	536	\$−912.05	\$−1.702	\$−0.851	\$−4.997	\$−3.342	\$−3.575	19/52	4 of 5

And the placebo ladder for the best cell — the SAME rule taken N minutes early, which cannot contain any signal

SHIFT	N	NET \$	\$/TRADE	VS THE REAL SIGNAL
30 min early	538	\$+1,908.10	\$+3.547	BEATS IT
60 min early	507	\$+1,408.22	\$+2.778	BEATS IT
120 min early	509	\$+232.32	\$+0.456	loses to it
240 min early	511	\$-1,932.67	\$-3.782	loses to it
480 min early	488	\$-5,508.95	\$-11.289	loses to it

VERDICT — COIL-BRK IS REFUTED

Four independent tests, any one of which is fatal. **(1) Out of sample it inverts:** in-sample \$+3.865/trade, out of sample \$-4.615. **(2) It is three trades wide:** strip its best 3 of 513 and \$+2.195/trade becomes \$-0.437. **(3) The placebo beats it:** the same rule taken **30 minutes early** — when the coil condition it is built on has not happened yet — books \$+3.547/trade against the real signal's \$+2.195, and 60 minutes early also beats it. **(4) It is a short-side artefact:** LONG \$ -2.051/trade, SHORT \$+6.392 — the entire net is one direction on a tape with a known short drift, and it is green on only 22 of 52 days.

Note what the placebo ladder's shape says on its own. The shifted copies decay smoothly from +\$3.55 at 30 minutes to -\$11.29 at 8 hours. That is not a signal being destroyed by misalignment; it is a slow intraday drift being sampled at different points. **The rule is a clock, not a trigger.**

★ And the control that settles it: the unarmed break — the same entry with the coil filter switched off entirely — books \$+1,625.70 on 1,356 trades (\$+1.199/tr) at the 3.0/8.0 exit, against the armed version's \$+2.195. **The filter this candidate is built around subtracts value.** What little is there belongs to the exit geometry, and that is a finding about stops, not about coils.

§5 ★ THE FORWARD TEST — last week's five candidates, on tape they have never seen

This is the section that justifies the whole rotation. The five candidates the 08-25 labs produced were frozen with their specs and are re-priced here on the days since the freeze — genuinely out of sample, no re-tuning, no cell re-selection. **OOS** is the post-freeze tape only.

CANDIDATE (SPEC)	FROZEN	ALL N	ALL NET \$	ALL \$/TR	IS \$/TR	OOS N	OOS NET \$	OOS \$/TR	READS AS
UNCL-RIDE-ER stop 2.0xATR / target 10.0 / cap 240m	2026-08-25	210	\$-483.60	\$-2.303	\$-3.772	27	+\$206.60	+\$7.652	still red overall — 4-day OOS bounce
board_atrb and stop 3.0xATR / target 6.0 / cap 240m	2026-08-25	338	\$-334.60	\$-0.990	\$-1.778	29	+\$214.65	+\$7.402	still red overall — 4-day OOS bounce
FL-4 flow_ignition stop 1.0xATR / target 1.5 / cap 15m	2026-08-26	192	+\$539.33	+\$2.809	+\$4.840	18	\$-302.75	\$-16.819	OOS inverts
MOMBRK stop 0.5xATR / target None / cap 30m	2026-08-24	146	+\$458.93	+\$3.143	+\$3.664	18	\$-10.09	\$-0.560	OOS inverts
VAC-IGN-BRK stop 1.5xATR / target None / cap 30m	2026-08-24	262	\$-598.24	\$-2.283	\$-1.164	37	\$-336.38	\$-9.091	red both legs

★ 3 of the 5 are net negative over their whole life, and not one is a promotion candidate. Two — UNCL-RIDE-ER and board_atrb and — show a positive out-of-sample leg on 4 days, and that is exactly the shape this desk has learned to distrust: an OOS window of 27 and 29 trades over 4 sessions, both driven by their LONG side (+\$15.56 and +\$20.21 a trade) while both

SHORT sides are red. **Four days is not a forward test; it is a sample of the week's drift.** They stay parked, and the honest read is that the parking decision made on 08-25 has been vindicated rather than overturned.

The two that were positive in the lab have both inverted: FL-4 flow_ignition goes from +\$4.84/trade in sample to **−\$16.82** out of it, and MOMBRK from +\$3.66 to −\$0.56 with an 11% win rate throughout. **Both were already REFUTED by their own placebo tests last week,** and the forward tape has now agreed with the placebo. That is the single most reassuring result in this movement: the kill tests are calling it right.

§6 Which boarder actually caught this week's runs

Every candidate, plus a deliberately dumb control, run across this week's five sessions and scored against the 62 sat-out runs. **Money on those** is what it made on the runs it caught — so the gap between that and Net \$ is what it paid to be everywhere else.

BOARDER	TRADES	NET \$	\$/TR	WIN %	RUNS CAUGHT	OF THE TOP 25	OF THE TOP 15	MONEY ON THOSE
board_atrband	44	+\$103.05	+\$2.342	36.4%	18 / 62	8	5	+\$1,311.90
MOMBRK	13	−\$30.84	−\$2.372	7.7%	4 / 62	3	2	+\$141.40
FL-4 flow_ignition	22	−\$54.47	−\$2.476	45.5%	13 / 62	9	6	−\$34.21
UNCL-RIDER	39	−\$117.60	−\$3.015	17.9%	15 / 62	10	6	−\$9.05
COIL-BRK (this week's fresh candidate)	48	−\$198.28	−\$4.131	35.4%	9 / 62	6	5	−\$37.75
VAC-IGN-BRK	34	−\$261.88	−\$7.702	35.3%	9 / 62	3	3	+\$456.15
the dumb boarder (5-min thrust >= 2.0xATR)	142	−\$323.73	−\$2.280	33.1%	27 / 62	16	9	+\$1,217.81

The dumb control catches **27 of 62 runs** — more than any built candidate — and makes **\$+1,217.81 on the runs it caught** while ending the week at **−\$323.73** overall. board_atrband is the only net-positive line at \$+103.05,

and it too earns \$+1,311.90 on its catches and gives most of it back elsewhere. **Every row on this table tells the same story from a different angle: the catches pay and the non-catches eat it.**

§7 Where the money is, by SIZE of move — and the 120-point line

Take every entry in the pooled book and bucket it by what the tape actually did next. This is a decomposition, not a strategy — the bands are only knowable afterwards. Its job is to say what a selector would have to find.

ATR-scaled stop 2.5xATR / target 6xATR — overall **n=1,713, \$-2,814.00 (\$-1.643/trade)**

WHAT THE TAPE DID	N	% OF BOOK	NET \$	\$/TRADE
went hard against (<= -60pt)	111	6.5%	\$-11,105.00	\$-100.04
-60 .. -20pt	350	20.4%	\$-21,832.00	\$-62.38
-20 .. +20pt (nothing happened)	825	48.2%	\$-8,254.00	\$-10.00
+20 .. +60pt	324	18.9%	\$+20,113.00	\$+62.08
+60 .. +120pt	76	4.4%	\$+9,637.00	\$+126.80
>= +120pt	27	1.6%	\$+8,627.00	\$+319.51

IF A SELECTOR KEPT ONLY MOVES OF...	N	% OF BOOK	\$/TRADE
≥ 0 pt	833	48.6%	\$+51.95
≥ 20 pt	433	25.3%	\$+89.27
≥ 40 pt	196	11.4%	\$+142.23
≥ 60 pt	103	6.0%	\$+177.32
≥ 80 pt	66	3.9%	\$+209.50
≥ 120 pt	27	1.6%	\$+319.51

FIXED 22pt stop, no target (the OPEN/NEWS ruler) — overall **n=1,666, \$-4,231.00 (\$-2.540/trade)**

WHAT THE TAPE DID	N	% OF BOOK	NET \$	\$/TRADE
went hard against (<= -60pt)	106	6.4%	\$-4,876.00	\$-46.00

WHAT THE TAPE DID	N	% OF BOOK	NET \$	\$/TRADE
-60 .. -20pt	335	20.1%	\$-15,334.00	\$-45.77
-20 .. +20pt (nothing happened)	806	48.4%	\$-10,934.00	\$-13.57
+20 .. +60pt	309	18.5%	\$+14,920.00	\$+48.28
+60 .. +120pt	82	4.9%	\$+5,422.00	\$+66.12
>= +120pt	28	1.7%	\$+6,572.00	\$+234.71

IF A SELECTOR KEPT ONLY MOVES OF...	N	% OF BOOK	\$/TRADE
≥ 0 pt	834	50.1%	\$+34.72
≥ 20 pt	426	25.6%	\$+63.63
≥ 40 pt	207	12.4%	\$+87.20
≥ 60 pt	110	6.6%	\$+109.04
≥ 80 pt	63	3.8%	\$+128.25
≥ 120 pt	28	1.7%	\$+234.71

★ **The 120-point line, confirmed for a second cycle by a third harness.** Moves of 120 points or more are **1.6% of the book** — 27 trades of 1,713 — and they carry **\$8,627** at **\$319.51 a trade**, on a book whose total is **\$-2,814.00**. Meanwhile **48.2% of entries go nowhere at all** (± 20 points) and pay **\$-10.00 each** for the privilege — that band alone is **\$-8,254.00**, and it is pure friction.

So the target is not a better entry, it is a bigger one. A selector that did nothing but decline the ± 20 -point band — without improving direction at all — turns this book from **-\$2,814** into a positive one. That is the same conclusion the rider lab and the census reached independently, and it is now three harnesses agreeing.

§8 The chop-day turn scalp — the operator's own question, re-run

“Can we stop donating on chop days?” The purpose-built L2 turn scalper was re-run this cycle from scratch with the search included in the placebo — the methodology fix that killed last week's finalist. Its dossier is

reproduced in full below (§A1). **Written 2026-08-29 00:19 UTC, this cycle**, 4,711 words.

CANDIDATE	COSTED CELLS	POSITIVE CELLS	BEST \$/TR	MEDIAN \$/TR
CT20 60s thrust q95 into a 10m extreme	18	0	\$−0.979	\$−2.486
CT13 close@30m-ext + 20s thrust>=4pt	18	1	\$+0.311	\$−1.087
CT0 naive chop fade (control)	18	0	\$−2.236	\$−2.563

The prize first: a perfect-hindsight chop-day scalper is worth **\$3,077.50 over 883 trades** across 5 chop days — **\$615.50 a day**, which does clear the operator's bar. What converts any of it: CT20 60s thrust q95 2.6% · CT13 close@30m-ext + 4pt 1.2% · CT0 naive fade -60.5%.

CT20, last week's survivor, is dead on its own terms this cycle: 216 events, a 56-cell costed grid, **0 positive cells**, best \$−0.539/trade — “DEAD: no positive costed cell”. The last cell standing, CT13 at [6.0, 8.0], is **n=66, \$+20.50 total, \$+0.311/trade** with 15 of 20 leave-one-day-out folds positive — and it splits **\$+0.612 on the original tape vs \$−0.559 on the new**. A 66-trade edge worth 31 cents a trade is not a gate; it is a rounding error with a name.

MOVEMENT 3 — THE BOARD, AND THE VERDICT

Nothing is promoted this week. Nothing is even nominated. One fresh candidate was built and refuted by four independent tests; five frozen candidates were forward-tested and none earned a promotion; the chop-scalp finalist died on its own re-run.

But the movement is not empty, because \$3 is a real result and it is structural. For three cycles this lab has closed with “detection is solved, selection is not” without being able to say why selection kept failing. It can now: **every precursor feature on this desk is a proxy for ATR**, and once ATR is controlled for, all of them fall to a coin flip or below. We have been asking a volatility meter to name a direction. No amount of re-tuning fixes that — it needs an input the desk does not currently record.

★ **What that makes actionable** is \$7, and it needs no new signal at all: **48.2% of entries go nowhere and pay \$10 each to find out.** Declining the dead band is a size decision, not a direction decision, and it is the one lever in this movement that does not depend on predicting anything.

APPENDIX — the dossiers in full

The editorial half above is the map; this is the territory. Every claim in this movement traces to one of these.

A1 — THE CHOP-DAY TURN SCALP THIS CYCLE · 2026-08-29

GREENFIELD — THE CHOP-DAY TURN SCALP (L2-LED)

The short version

Garrath, you asked for a purpose-built chop-turn scalper: catch the oscillation turns on the days the desk currently donates, bank tiny, and **lead with the order book** — the exhaustion_short lineage, which was built out of L2 footprints and is one of the best things we own.

Last Monday's pass built sixteen candidates and ended with three finalists and a predicted failure: the survivor CT15 looked strong on every conventional test but lost its matched-null comparison, so the write-up called it a search artefact. **Four trading days have happened since. This week I ran those three frozen rules on tape that did not exist when they were chosen, and the prediction was right.** All three flip sign. Then I searched again on the extended 21-day tape, and the second search dies too — for a reason that is now measured rather than argued.

1. The pre-registered out-of-sample leg kills all three inherited finalists.

No re-tuning, no re-fitting, the f_refill median frozen at its original-window value. CT15 goes from **+\$6.08/trade** in-sample to **-\$2.09** (4 unseen days) and **-\$6.36** (3 strictly-later days); its win rate collapses 77% → 43%. CT13 **+\$2.54** → **-\$0.06/- \$0.81**. CT16 **+\$4.27** → **+\$0.58/- \$2.20**. The control matters as much as the finalists: the naive chop fade measured **-\$2.430/trade** in-sample and —

\$2.640 out-of-sample. **The loss replicates to within 21 cents. The profit does not replicate at all.** That is the signature of an overfit, and it also proves the harness is measuring the same thing on both windows.

2. Your lead hypothesis: I have to correct last week's answer, not confirm it. Last week I told you the absorbing wall actively hurts the fade and "the wall is a target." **That does not replicate.** On the four new days the sign flips: $wall1 \geq 1.5$ drifts **-0.118 pt** on the original window and **+0.393 pt** on the new one. It was a one-window read and I stated it too confidently. The honest finding is weaker and duller: **no book term, in either window, produces a costed edge — 0 positive cells out of 864** (9 terms $\times$ 2 windows $\times$ 48 target/stop cells). The one term whose sign is stable across both windows is the **inverse** of your hypothesis — a **thin** far side ($wall5 \leq 1.0$) drifts +0.273 pt then +0.613 pt — and it too is 0/48 costed.

3. The real cause of death is arithmetic, and I can now put a number on it. Round-trip friction on MNQ is **1.00-1.25 points** (\$1.50 fee = 0.75 pt, plus the tick we cross on entry, plus the tick a stop slips). I scanned **1,280 filter cells** — 32 features $\times$ 5 thresholds $\times$ 2 directions $\times$ 4 horizons — for the best frictionless forward drift any single filter can buy. Best: **1.31 pt**. Then I handed the null the identical search, shuffling features within each day: **the null's best averages 1.05 pt, and at the 60s and 120s horizons the null BEATS the real tape** ($p = 0.895, 0.835$). **A 320-cell search on 4,193 events manufactures roughly as much fake drift as the entire cost barrier is wide.** There is no room between the noise floor and the fee.

4. The one cell that cleared both bars still dies costed. CT20 — fade a top-5% 60-second thrust (≥ 16.75 pt) into a 10-minute extreme — had drift 1.31 pt at $t = 2.57$, $p(\text{null}) = 0.065$, 131% of friction. Costed across 56 target/stop cells including the tight 0.5R/1R targets you asked for: **0 positive, median -\$2.292, best -\$0.539.**

5. And the premise correction from last week is now sharper, not softer. The desk did not donate on this week's chop day. **The tournament took zero trades on 08-24, 08-25, 08-26 and 08-27.** Its only trades all week were six legs on Friday 08-28, all winners, **+\$330.00**. Across the full 21-day window the tournament's loss is concentrated in **violent-whipsaw** blocks ($-\$1,235.00$ on 97 legs, $-\$12.73/\text{leg}$) — high ATR going nowhere — **not** in plain chop ($-\$403.00$ on 58 legs). That is a router bench to price, not a gate for greenfield to build.

1. Method, data and the cost model — stated before any number

ITEM	VALUE
Symbol	MNQ, \$2.00 per point, 0.25 pt tick
Fee	\$1.50 per ROUND TRIP (not \$5, not \$2, not per side)
Entry fill	first tick at or after the 5s bar close, 1 tick adverse (we cross)
Target fill	limit, filled at the target, no extra slip
Stop fill	stop-market, 1 tick beyond the trigger
Horizon	900 s; a leg still open is marked OPEN and carried at its mark, never predicted
Sequencing	one slot, non-overlapping — a signal arriving before the previous trade closed is skipped
Full-stack days	21 — 07-31, 08-03...07, 08-10...14, 08-17...21, 08-24...28
Trade ticks	25,507,483
Book rows (41 ms, 10-deep, event-driven)	323,658,215
Depth snapshots (250 ms, 10-deep)	5,742,907
Quote updates	32,365,985
Turn candidates	8,345 (4,193 on chop tape)

Four of those 21 days are new this week. 08-28's tape was mirrored out of capture.db at 00:03Z this morning specifically so this section could use it; the first build read the depth parquet mid-write and was re-run after the mirror verified its row counts.

Friction stated in points, so it can be compared with drift. This is the number every candidate in this section has to clear and it is worth writing out:

fee	\$1.50 / \$2.00 per point	= 0.750 pt
entry	we cross the spread, one tick	= 0.250 pt
exit	target = limit (0.000) .. stop = one tick (0.250)	

	floor 1.000 pt	ceiling 1.250 pt per round trip

Regime segmentation, not the clock. Every minute is classified on ATR level relative to the trailing one-session median (atr_rel — MNQ ran ~20 pt ATR in late July and ~7 pt in mid-August, so an absolute floor would label all of July violent and all of August dead) crossed with the 30-minute efficiency ratio:

Across the 21 full-stack days that is 946 thirty-minute blocks (28,380 minutes), split:

REGIME	RULE	30-MIN BLOCKS	1-MIN BARS
DEAD_CHOP	ER < 0.20 and atr_rel < 0.70	88	2,822
CHOP	ER < 0.20	447	11,653

BUILD	$0.20 \leq ER < 0.40$	276	9,048
TREND	$ER \geq 0.40$	47	2,211
WHIPSAW	$ER < 0.20$ with $atr_rel \geq 1.50$	88	2,646

(A block takes the modal label of its minutes, so the two columns are not proportional.) **Chop is 57% of the tape and clean trend is 5% of it** — which is the reason a chop specialist looked worth building in the first place.

Days are typed on **US-session** (13:30–20:00Z) efficiency: **CHOP** ≤ 0.25 , **TREND** ≥ 0.45 . The five chop days in the full-stack window are **07-31, 08-07, 08-10, 08-20 and 08-25**.

This week's days

DAY	DAY TYPE	US-SESSION EFFICIENCY	CHOP FRACTION OF TAPE	TOURNAMENT	DAY_RIDER
Mon 08-24	MIXED	0.395	56.5%	0 trades	0 opened
Tue 08-25	CHOP	0.209	65.2%	0 trades	2 legs, −\$243.00
Wed 08-26	TREND	0.599	56.5%	0 trades	4 legs, +\$19.00
Thu 08-27	TREND	0.598	50.0%	0 trades	12 legs, +\$16.50
Fri 08-28	MIXED	0.429	45.2%	6 legs, +\$330.00	7 legs, +\$780.00

One chop day this week, not five. Note 08-27: its whole-day efficiency is 0.113 — genuinely directionless around the clock — but its US session ran 0.598. A clock-based day label would have called it chop and been wrong about the only session that matters.

2. The prize — where the desk actually loses money

A chop scalper only has something to recover if the desk is donating on chop tape. Every live MNQ trade is attributed to the regime of the 30-minute block it **opened** in. `day_rider` is a separate book taking its own size on the same account and is never folded into the tournament figure. Seven `data_quality`-flagged rows are excluded.

Tournament, full 21-day window — 287 legs, −\$1,426.00

REGIME OF THE OPENING BLOCK	LEGS	NET	\$/LEG	WIN
WHIPSAW	97	−\$1,235.00	−\$12.73	35%
CHOP	58	−\$403.00	−\$6.95	33%

TREND	23	+\$83.00	+\$3.61	39%
BUILD	107	+\$309.50	+\$2.89	45%

The bleed is whipsaw, not chop, and it is 3.1× larger. High ATR going nowhere is where the gates get chopped up — not the quiet ranging tape a turn-scalper would trade. That was last week's finding and four more days have not moved it.

This week — tournament 6 legs / 6 signals, +\$330.00; day_rider 25 legs / ~12 signals, +\$572.50. The tournament's six legs were all on Friday and all winners (+\$74.25/leg in whipsaw blocks, +\$45.38/leg in build blocks). It sat out the chop day entirely.

One number needs flagging even though it is not my section's to fix: the single largest loss in the window is a day_rider position **opened 08-21 13:13Z and closed 08-24 11:59Z** — held across the weekend — for **−\$2,149.44**, exit reason MANUAL_CLAIM. It lands on neither a chop day nor a tournament book, but it is 1.5× the tournament's entire 21-day loss.

3. THE HEADLINE — the pre-registered out-of-sample test

This is the strongest evidence in the section, because it is the only test where the tape was chosen after the rule was.

The 2026-08-25 study searched 07-31...08-24 and named three finalists with frozen parameters. I ran those exact rules, unchanged, on the four days since. **No re-tuning. No cell search.** The f_refill median that defines CT15 is the one measured on the original window — re-estimating it on the new tape would have been a second fit.

Honesty on the boundary. The prior write-up is dated 2026-08-25 23:39Z and its event tape stops at 08-24. 08-25's session closed 21:00Z that day, so 08-25 was available though unused. Both legs are reported: **WIDE** (08-25...28) and **STRICT** (08-26...28, unambiguously later).

CANDIDATE	LEG	N	NET	\$/TRADE	WIN
CT13 close@30m-ext + 20s thrust ≥4pt, 6/10	in-sample	49	+\$124.50	+\$2.541	75.5%
	OOS wide	17	−\$1.00	−\$0.059	64.7%
	OOS strict	13	−\$10.50	−\$0.808	61.5%

CT16 close@30m-ext + 20s thrust ≥5pt, 8/10	in-sample	34	+\$145.00	+\$4.265	70.6%
	OOS wide	12	+\$7.00	+\$0.583	58.3%
	OOS strict	10	−\$22.00	−\$2.200	50.0%
CT15 CT13 + 41ms refill ≥ frozen median, 8/10	in-sample	26	+\$158.00	+\$6.077	76.9%
	OOS wide	11	−\$23.00	−\$2.091	54.5%
	OOS strict	7	−\$44.50	−\$6.357	42.9%
CT0 naive chop fade (control), 6/10	in-sample	2,397	−\$5,825.50	−\$2.430	59.7%
	OOS wide	527	−\$1,391.50	−\$2.640	59.2%
	OOS strict	383	−\$965.50	−\$2.521	59.5%

CT15 per-day on the new tape: 08-25 +\$21.50 (4), 08-26 −\$44.00 (2), 08-27 −\$15.00 (4), 08-28 +\$14.50 (1).

Read the control row first. The naive fade's loss reproduces at −\$2.430 → −\$2.640 → −\$2.521. The harness, the fee model, the race and the regime labels all behave identically across the split. Against that stable backdrop, CT15's +\$6.08 becoming −\$6.36 is not noise in the measurement — it is the absence of an edge that was never there. Last week's matched-null test said this would happen at p = 0.367; this is that prediction being paid.

Small n, honestly stated: 7–17 trades per OOS leg. No single one of these is conclusive alone. **But all three move the same way, the direction was predicted in advance, and the control does not move at all.** Three concordant sign flips against a stationary control is the result.

4. Your lead hypothesis, re-tested — and last week's answer corrected

You asked for the separator to be the **book**: at the price extreme, require the far side to be **absorbing or depleting**. Last week I reported that this actively hurts and called the wall "a target." **I now think that claim was over-stated on one window's evidence, and I want to correct it before it becomes desk lore.**

Test A — frictionless 60-second drift, signed so positive means the fade won. No target, no stop, no fee: does the signal have direction at all?

BOOK TERM	ORIG (17D) N	DRIFT PT	T	NEW (4D) N	DRIFT PT	T
	3,433	+0.056	0.46	753	+0.246	0.92

no book term (control)						
wall1 $\geq$ 1.5 (absorbing)	1,325	-0.118	-0.60	311	+0.393	1.04
wall5 $\geq$ 1.3 (absorbing)	931	-0.365	-1.67	247	+0.309	0.70
wall5 $\leq$ 1.0 (INVERSE: thin)	1,491	+0.273	1.50	260	+0.613	1.15
support draining 20s	1,497	+0.100	0.50	339	+0.297	0.69
wall building 20s	1,521	+0.055	0.30	362	+0.688	2.08
41ms refill $\geq$ median	1,691	+0.116	0.56	458	+0.113	0.29
41ms pull $\geq$ median	1,703	+0.094	0.46	450	+0.132	0.34
10-deep imbalance > 0	1,331	+0.213	1.12	242	+0.375	0.74

Both absorption terms flip sign between windows. The -0.365 pt with $t = -1.67$ that carried last week's "the wall is a target" headline becomes $+0.309$ pt on the new days. Neither read is significant; what they jointly establish is that **the absorbing-wall term has no stable sign**, and last week's confident phrasing was not supported by one window.

The only term whose sign is stable and positive in both windows is **wall5 $\leq$ 1.0 — a thin far side**, the inverse of your hypothesis. It is also the best drift term on the new tape after the 20s wall-build. That is the direction I would look again if we revisit this: not "price is being absorbed at the edge" but "price has run out of tape to run into." It is weak evidence — $t = 1.50$ and 1.15 — but it is the only book term that says the same thing twice.

Test B — costed. Median \$/trade across each term's full 48-cell target/stop grid:

BOOK TERM	ORIG EVENTS	MEDIAN \$/ TR	POS CELLS	NEW EVENTS	MEDIAN \$/ TR	POS CELLS
no book term (control)	3,439	-\$2.541	0/48	754	-\$2.423	0/48
wall1 $\geq$ 1.5	1,328	-\$3.086	0/48	311	-\$2.648	0/48
wall5 $\geq$ 1.3	934	-\$3.190	0/48	247	-\$2.215	0/48
wall5 $\leq$ 1.0 (inverse)	1,492	-\$2.358	0/48	261	-\$2.495	0/48
support draining 20s	1,499	-\$2.592	0/48	339	-\$2.873	0/48
wall building 20s	1,522	-\$2.578	0/48	362	-\$2.187	0/48
41ms refill $\geq$ median	1,694	-\$2.439	0/48	458	-\$2.484	0/48
41ms pull $\geq$ median	1,705	-\$2.402	0/48	450	-\$2.695	0/48
	1,332	-\$2.312	0/48	242	-\$2.433	0/48

Zero positive cells out of 864. Every book term, every window, every target/stop pair. When a result is that uniform the exit grid is not the problem, and neither is the choice of book feature.

5. ...and the lineage you named does not transfer, because the condition does not occur

You pointed at `exhaustion_short` — built from L2 footprints — as the model. So I asked the direct question: **do that gate's own thresholds ever occur on chop tape?**

`exhaustion_signal` requires a 20-second aggressor net of ≥ 400 contracts. On chop tape:

```
|20s aggressor net|  median 44    p90 197    p99 670    p99.9 1,812
share of chop events clearing net_min=400 ..... 3.172%
+ the STALL term (|20s move| <= 2pt) .... 9 of 4,193 events ... 0.215%
+ an absorbing wall (wall1 >= 1.5) ..... 1 event, on 1 day ... 0.20 per chop
day
```

One event in five chop days. The gate is not failing on chop tape; its entry condition is essentially never satisfied there, because a 400-contract aggressor imbalance is a directional impulse and chop is defined by their absence. Porting the lineage was never on the table. This is a fact about the tape, not a failure of the search — and it is the cleanest single reason a book-led chop fade is harder than a book-led trend-exhaustion gate.

The two costed candidates built on the literal recipe are in the disposition list below: `CT4_EXH_LITERAL` never reached 10 trades in any of its 48 cells and was dropped; `CT4b_EXH_SOFT` (relaxed to $\text{net} \geq 200$, $\text{move} \leq 3$ pt, $\text{wall} \geq 1.2$) medians **-\$3.844/trade**.

6. The cost barrier, and what a 320-cell search manufactures out of nothing

Every costed candidate so far is negative in every cell. That pattern says the problem is not the exit — it is that the **signal is smaller than the friction**. So I measured both quantities and compared them directly.

A filter is only worth costing if its frictionless forward drift clears **~1.0 pt**. I scanned **32 features × 5 thresholds × 2 directions × 4 horizons = 1,280 cells**, requiring $n \geq 50$, for the best mean drift available anywhere.

HORIZON	BEST FILTER FOUND	N	DRIFT	T	VS FRICTION
30 s	f_mv60 ≥ q95	216	+1.315 pt	2.57	132%
60 s	er30 ≥ q95	209	+0.953 pt	1.79	95%
120 s	f_wall10 ≤ q05	210	+1.336 pt	1.71	134%
300 s	f_agg60 ≤ q05	213	+2.781 pt	2.04	278%

Unfiltered chop-turn drift for reference: +0.231 pt (30 s), +0.091 pt (60 s), +0.116 pt (120 s), +0.337 pt (300 s) — i.e. **a fifth to a third of the friction**, which is why the naive fade bleeds −\$2.43 to −\$2.64 with metronomic reliability.

Now the same search, handed to the null. Shuffle every feature within each day — destroying the feature→future link while preserving each day's drift distribution and each feature's marginal — and let the null run the identical 320-cell hunt, 200 times per horizon:

HORIZON	REAL BEST	NULL BEST (MEAN)	NULL P95	P(NULL ≥ REAL)
30 s	1.315 pt	1.050 pt	1.333	0.065
60 s	0.953 pt	1.220 pt	1.614	0.895
120 s	1.336 pt	1.611 pt	2.148	0.835
300 s	2.781 pt	2.658 pt	3.662	0.335

At two of the four horizons the null beats the real tape. A 320-cell search over 4,193 events routinely manufactures 1.0–1.6 points of apparent drift out of shuffled features — **the same size as the entire cost barrier**. That is the finding I would keep from this section even if you never build a chop scalper: on this event count, any threshold search that reports "about one point of edge" has reported its own noise floor.

Only the 30-second horizon survives at $p = 0.065$, and that is one near-miss across four horizons tested. It still gets a full costed battery, because a p-value is not a verdict.

7. CT20 — the one cell that cleared both bars, and how it died

CT20: on chop tape, at a 10-minute extreme, fade when the **60-second move into that extreme is in the top 5%** — a thrust of ≥ 16.75 points in a minute. 216 events over 21 days, frictionless 30s drift +1.315 pt at $t = 2.57$, 132% of friction, $p(\text{null}) = 0.065$.

Costed across 56 target/stop cells, **including the tight targets you asked about** (this required racing 1.0–2.0 pt targets on the tick tape directly, since the event table's precomputed grid starts at 2 pt):

cells = 56 positive = 0 (0%) median \$/trade = -\$2.292 best = -\$0.539
best cells: tp8/sp10 -\$0.539 (58.8% win) tp5/sp10 -\$0.678 (69.9%) tp5/sp8 -
\$0.947 (64.4%)

Zero positive cells. Note the win rates — 59% to 73%, comfortably above the live desk's 39% — and note that it does not matter, exactly as your judging rule says. The 1.315 pt of drift is real enough to produce a high hit rate at a wide stop; it is not enough to pay a 1.00–1.25 pt round trip once the stop-outs are priced.

The R-sweep in your own units, target expressed as $R = \text{target/stop}$:

STOP	R=0.5	0.75	1.0	1.5	2.0	3.0
3 pt	-\$2.68	-\$2.70	-\$2.68	-\$2.33	-\$2.69	-\$2.56
5 pt	-\$2.67	-\$2.42	-\$2.41	-\$3.74	-\$3.55	-\$4.38
8 pt	-\$1.21	-\$1.38	-\$1.15	-\$0.98	-\$1.66	-\$1.21

0/18 positive. The 0.5R/1R banking you suggested is not the problem and not the fix — the surface is negative everywhere and slopes the wrong way as the stop tightens, which is the signature of a signal too small to survive its own stop rather than a mis-chosen R.

Cause of death: the costed grid. 0 of 56 cells positive, best -\$0.539/trade.

8. The last cell standing — CT13 at 6/8, and the full battery

Across the whole R-sweep — three rules × three stops × six R values = 54 costed cells — **exactly one is positive**: CT13 (close at a 30-minute extreme on a ≥ 4 pt 20-second thrust) at **target 6 pt / stop 8 pt, R = 0.75, +\$0.311/trade**. It is scored knowing it is a 1-in-54 survivor.

Full 21-day window: n = 66, +\$20.50, +\$0.311/trade, 63.6% win, 20 days, 1 open leg. That is **\$0.98 per trading day**.

TEST	RESULT	VERDICT
Out-of-sample	ORIG (17d) +\$0.612/tr → NEW (4d) −\$0.559/tr	sign flip
Leave-one-day-out	15/20 folds positive, median +\$0.216	passes
Leave-1-DAY-out (drop the single best day)	−\$0.347/tr, −\$21.50	FAILS
Leave-2-out	−\$0.898/tr, −\$53.00	fails
Leave-3-out	−\$1.509/tr, −\$84.50	fails
Profitable days	13 of 20	passes
Strip the best 1 trade	+\$0.154/tr	survives
Strip the best 2 trades	−\$0.008/tr	FAILS
Strip the best 3 / 5	−\$0.175 / −\$0.525	fails
Within-day shuffle, this single cell	real +\$0.311 vs shuffle mean −\$2.660 , p95 −\$0.366, p = 0.017	passes
On the 5 CHOP DAYS	n=19, −\$28.50, −\$1.500/tr	loses on its home tape
On the 11 TREND days	n=30, −\$15.50, −\$0.517/tr	loses
Same rule on non-chop regime blocks	n=96, −\$285.00, −\$2.969/tr	bleeds off-home

Two trades out of 66 are the entire edge. Strip them and it is **−\$0.008/trade**. Drop one day out of twenty and it is **−\$0.347**. And the test that matters most for your brief: **on the five chop days it was built to trade, it loses \$28.50 at −\$1.500/trade**. Its whole positive number comes from chop blocks that occur on non-chop days.

On the shuffle p = 0.017. This is exactly the trap last week's study identified, and it is worth being explicit about why the p-value is not a rescue. The shuffle null is a random chop fade, which averages **−\$2.660/trade**. Beating a **−\$2.66** null with **+\$0.31** is easy and nearly meaningless; the question was never "is this better than fading at random," it is "is this better than the best cell a search finds in noise" — and §6 answers that with the null's own search, where the null wins at two horizons outright.

Cause of death: leave-one-day-out and strip-the-best. One day or two trades takes it negative, and it loses money on the chop days it was designed for. The out-of-sample sign flip is corroboration.

9. Is the harness real? — the oracle control

A backtester that can only print losses is not evidence of anything. Before the verdict stands, the machinery has to be shown capable of finding money. Same race, same \$1.50 round trip, same one-slot sequencing, chop tape, 5/5 target and stop:

CONTROL	N	NET	\$/TRADE	WIN
ORACLE — side = sign of the realised 60s move (look-ahead by construction)	3,613	+\$11,899.50	+\$3.29	75%
ANTI-ORACLE — the same, sign flipped	3,674	−\$28,559.50	−\$7.77	21%

The oracle must win, and it does, decisively and symmetrically. The harness finds money when money is there. **The negative chop numbers are a finding, not a bug.**

Null reported for completeness: I also tried an honest non-look-ahead positive control — trend-continuation on TREND-daytype days — and it returned **n = 6**. TREND regime blocks are only 47 of the 946 blocks in the window (5.0%), and requiring a ≥ 4 pt 60-second thrust through an extreme inside one, on a TREND-typed day, leaves six events. The control is **unpowered and returns no verdict** either way. I am not counting its negative sign as evidence of anything.

The ceiling, and how much of it anything captures

The oracle also gives the honest ceiling. On the five chop days, with **perfect 60-second direction** and real fills and real fees:

n = 883, +\$3,077.50, +\$3.49/trade, 76% win — \$615.50 per chop day.

So your +\$250 bar is **41% of a perfect-foresight ceiling**. That is demanding but not absurd, which is why this was worth a serious attempt. What we actually capture on those days:

CANDIDATE	N	NET ON CHOP DAYS	FRACTION OF THE CEILING
CT20 60s thrust q95, 8/10	71	+\$80.50	2.6%
CT13 close@30m-ext, 6/10	19	+\$37.00	1.2%
CT0 naive fade, 6/10	743	−\$1,860.50	−60.5%

Both selective candidates are positive on chop days at these cells — and both convert under 3% of the available money on 19–71 trades. Neither survives \$7 or \$8.

10. The arithmetic against your +\$200-300 bar

Median 30-minute CHOP block range across the **535 chop blocks in the 21 full-stack days: 45.50 points** (50.50 pt across all 31 day-files of bars — I use the tighter, matched number here).

RULE	FIRES PER CHOP-DAY	MUST EARN PER TRADE FOR + \$250/DAY	IN POINTS	AS A SHARE OF A WHOLE 30-MIN BLOCK RANGE
CT20 60s thrust q95	14.20×	\$17.61	8.8 pt	19%
CT13 close@30m-ext + 4 pt	3.80×	\$65.79	32.9 pt	72%
CT0 naive fade	199.20×	\$1.26	0.6 pt	1%

This is the tension, and it is arithmetic rather than opinion. **You asked for selectivity — "a few high-quality turns a day, not hundreds" — and selectivity and the dollar target pull in opposite directions.** At 3.8 fires a day CT13 needs to capture **72% of an entire 30-minute range** on every single trade. At 199 fires a day the naive fade needs only \$1.26 a trade — a bar so low it sounds trivial — and it measures **−\$2.24**, because 199 round trips a day is a **\$298.80 daily fee tax** before a single point of edge.

The best \$/trade found anywhere in the entire R-sweep is **+\$0.311**. At that rate + \$250 a day requires **804 trades**. There is no point on this curve that reaches your bar.

11. The full disposition list — every attempt, by name, with its cause of death

21 days, 48 target/stop cells each, scored on the **median** \$/trade across the whole grid. A max over 48 cells is a search result, not an edge, so the median is the honest column. med_ORIG / med_NEW show whether each candidate died the same way twice.

CANDIDATE	POS CELLS	MEDIAN \$/ TR	MED ORIG	MED NEW	MEDIAN N	CAUSE OF DEATH
CT13_DEEP_SPIKE close@30m-ext + 20s thrust ≥ 4	13/48	−\$0.775	−0.566	−0.986	66	leave-one-day-out; strip-2-trades; OOS sign flip (\$8)
CT10_DEEPEXT close at a 30-min extreme	0/48	−\$1.982	−1.500	−2.797	134	no positive cell; worsens out-of- sample
CT12_SPIKE_PULL 4pt thrust + L1 pull $\geq p60$	0/48	−\$2.214	−2.198	−2.280	1,054	no positive cell
CT8_PULL 41ms L1 pull count $\geq p80$	0/48	−\$2.224	−2.237	−2.197	777	no positive cell
AT0_ALLTAPE fade every extreme, all regimes	0/48	−\$2.278	−2.228	−2.474	7,190	no positive cell (fee tax)
CT7_SPIKE 20s move ≥ 4 pt	0/48	−\$2.279	−2.308	−2.106	2,017	no positive cell
CT14_US_ONLY chop fade, US session only	0/48	−\$2.282	−2.368	−1.739	855	no positive cell
CT6_ANTIWALL thin far side, wall5 ≤ 0.8	0/48	−\$2.332	−2.288	−2.651	617	no positive cell
CT11_SPIKE_ANTIW ALL 4pt thrust + thin book	0/48	−\$2.373	−2.338	−2.344	859	no positive cell
CT3_FOOTPRINT 20s aggressor delta $\geq$ 150	0/48	−\$2.402	−2.456	−2.158	504	no positive cell
CT0_NAIVE fade every chop extreme	0/48	−\$2.528	−2.540	−2.423	3,573	no positive cell; the control, and it replicates
CT1_VWAPFLAT + VWAP flat ≤ 0.5 ATR	0/48	−\$2.548	−2.534	−2.550	3,418	no positive cell — your ranging filter costs more than it saves
CT5_STALL exhaustion's own stall ratio $\geq p80$	0/48	−\$2.697	−2.620	−2.926	800	no positive cell
CT2b_WALL5 five- deep wall ≥ 1.3	0/48	−\$2.995	−3.190	−2.216	1,090	no positive cell — worse than naive in both windows
CT2_WALL your lead hypothesis, wall1 ≥ 1.5	0/48	−\$3.012	−3.086	−2.648	1,517	no positive cell — the second-worst candidate on the board
CT4b_EXH_SOFT relaxed exhaustion recipe	0/48	−\$3.844	−4.479	−3.688	16	no positive cell; n too small to mean much
CT4_EXH_LITERAL exhaustion's literal thresholds	—	—	—	—	<10	never reached 10 trades in any cell — the condition does not occur on chop tape (\$5)
CT15 CT13 + 41ms refill $\geq$ median	—	—	—	—	26	pre-registered OOS: +\$6.077 → −\$2.091 / − \$6.357 (\$3)
	—	—	—	—	34	

CT16 close@30m-
ext + 20s thrust ≥5

pre-registered
OOS: +\$4.265 →
+\$0.583 / –
\$2.200 (\$3)

CT20 60s thrust q95
into a 10-min 0/56
extreme

–\$2.292 —

—

216
**0 positive costed
cells, best –
\$0.539 (\$7)**

Sixteen candidates in the main sweep at 768 costed cells, plus 56 for CT20, plus 54 in the R-sweep, plus 1,280 in the necessary-condition scan, plus 1,000 matched-null draws. **One positive cell, which \$8 kills.**

12. What I would actually do with this

1. Stop trying to build a chop-turn scalper on MNQ at this fee. Not because the turns are random — they are not; unfiltered chop turns drift +0.09 to +0.34 pt in the fade's favour, and the oracle shows \$615.50 a chop day sitting there. Because **the drift is a fifth to a third of the round trip**, and the search noise floor on this event count is the same width as the barrier. To clear it you would need a filter buying **>1.25 pt** of drift that is not a search artefact, and 1,280 cells did not produce one.

2. The premise itself is the actionable finding, and it is now two weeks old and unmoved. The tournament does not donate on chop days — this week it did not trade four of five days at all. It donates in **WHIPSAW**: –\$1,235.00 on 97 legs at –\$12.73/leg, versus –\$403.00 on 58 legs in plain chop. **That is a router bench with a defined trigger — high atr_rel with low ER — and it is worth 3.1× more than anything a chop scalper could recover.** I would price that bench before building anything new. It is the same recommendation as last week, and four more days have strengthened it.

3. If you want one more swing at the book, swing at the opposite of what we assumed. The only book term with a stable sign across both windows is the **thin** far side ($wall5 \leq 1.0$: +0.273 pt then +0.613 pt), not the absorbing one. It is 0/48 costed and I am not recommending it as a candidate — but if the L2 lead is revisited, "price ran out of tape" is the hypothesis with two consistent reads behind it, and "the far side is absorbing" now has two contradictory ones.

4. Two corrections to carry forward. Last week's "the wall is a target" was a one-window read and does not replicate — please don't let it become desk lore.

And the exhaustion_short lineage cannot be ported to chop tape: its 400-contract aggressor floor plus stall plus wall occurs **once in five chop days**.

13. What would change this verdict

Stated in advance so it is falsifiable rather than a hedge:

- **A filter buying > 1.25 pt of frictionless drift with $n \geq 100$ that survives the matched-null search** (not a single-cell shuffle). Nothing in 1,280 cells got there.
 - **More chop days.** Five in the full-stack window is thin, and the 30s horizon's $p = 0.065$ is a near-miss I would happily re-test on twenty chop days rather than five. The lake grows ~5 days a week; this is worth re-running at ~40 full-stack days.
 - **A lower fee.** At \$1.50 the round trip the barrier is 1.00–1.25 pt. Everything here is a statement about that number, not about the tape.
 - **A different instrument.** MGC is \$10.00/point, so the identical \$1.50 fee is **0.15 pt** of friction rather than 0.75 — one fifth. The entire arithmetic in §10 changes sign-of-feasibility on gold. That is out of this section's scope and belongs to gf_MGC, but it follows directly from §6 and I would not want it lost.
-

VERDICT

IT DID NOT SURVIVE. The three inherited finalists were killed by the **pre-registered out-of-sample leg** — CT15 +\$6.077 → −\$2.091 (wide) / −\$6.357 (strict), CT13 +\$2.541 → −\$0.059 / −\$0.808, CT16 +\$4.265 → +\$0.583 / −\$2.200 — while the naive-fade control replicated its loss to within \$0.21, confirming the split measures the same thing on both sides. The fresh 21-day search was killed by the **matched-null test**: its best cell bought 1.315 pt of drift where the null's own search bought 1.050 pt on shuffled features ($p = 0.065$), and at the 60s and 120s horizons the null beat the real tape outright ($p = 0.895, 0.835$). The one candidate that cleared both the friction barrier and its null, CT20, was killed by the **costed grid** — 0 of 56 cells positive, best −\$0.539/trade, including the 0.5R and 1R targets requested. The single positive cell left in the entire study, CT13 at 6/8, was killed by **leave-one-day-out and strip-the-best-trades**: dropping one

day of twenty takes it to $-\$0.347/\text{trade}$ and stripping two trades of sixty-six takes it to $-\$0.008$, and it loses $\$28.50$ at $-\$1.500/\text{trade}$ on the five chop days it was built for. No candidate was killed on win rate — CT20's costed cells win 59-73% and are still negative. The harness passes its oracle control at $+\$3.29/\text{trade}$, so these are findings and not artefacts. **Nothing here is fit to arm, shadow, or paper-trade.**

A2-A6 — THE FOUR CLUSTER HUNTS THE ROTATION DID NOT SCHEDULE THIS WEEK

The serial runner logged, at 22:14:21Z: ROTATION (ISO week 35): running 2 of 6 greenfield clusters. NOT run this week: gf_OPEN-NEWS, gf_RIDER_ALL, gf_UNCLASS, gf_VACUUM. **This is the design working, not a failure** — six cluster hunts at ~ 25 minutes each do not fit in one Friday night, so they rotate and each is re-run roughly every three weeks.

Their dossiers are reproduced below **as a standing reference, dated 2026-08-22 and 2026-08-25**. They are NOT this week's work and nothing on the action card rests on them. **What IS this week's work is \$5 above**, which takes every candidate these four labs produced, on the specs they were frozen with, and prices them forward on tape they had never seen. That is the honest way to carry a rotated-out lab: freeze the claim, then grade it.

THE POOLED RUN-CATCHER — one rider for every sat-out run

NOT RUN THIS WEEK · dossier 2026-08-25

Board any run late, ride it wide, filter it causally.

GREENFIELD SYNTHESIS — ONE RIDER FOR EVERY SAT-OUT RUN

MNQ · full parquet lake (5s bars 2026-07-16 → 08-24) + a 236-session out-of-sample leg · $\$2.00/\text{point}$ · $\$1.50$ per round trip

THE SHORT VERSION, BEFORE ANY OF THE WORKING

Garrath — I built the direction-agnostic run-catcher you asked for, boarded it late exactly as the gold work suggested, swept the exit far wider than the desk has ever run one, filtered it with a causal rule, and put the survivor through the whole battery. It passed every in-sample test we have, including two separate placebo

controls. Then it died on the out-of-sample leg, and it died again on a live shadow arm that has been quietly recording forward trades since Monday.

It died at STEP 3 — the causal filter. The test that killed it was the out-of-sample leg: 1,289 trades over 236 untouched sessions for a total of +\$14. Not a loss. Not a win. Nothing. The 95% confidence interval on its expectancy runs from $-0.204R$ to $+0.246R$ per trade, which is another way of saying we measured zero with a wide ruler.

And there is a second, independent witness that agrees, which is the part you should trust most. **rider_w5 — the board arm this desk shipped into the shadow book on 2026-08-15 off last week's version of this very phase, quoted then at +\$4,841 and +\$19.29 a trade — has taken 51 forward trades since 08-17 and sits at $-\$1,471$, or $-\$28.84$ a trade, red on six of seven days and red in every single ATR bucket.** Nobody had graded it. This section grades it.

Three things came out of the week that are worth more than the failed rider:

1. **The \$9,610 is not \$9,610.** Even a perfect late boarder — one that fires inside all 62 runs at a median three minutes in and then exits at the exact top of the move with no costs at all — can only reach **\$5,094, or 53% of the ceiling**. Thirteen per cent of the runs are already finished by the time any honest detector can see them. The number the census puts on the table is a hindsight number and the rider's half of it is gone before we start.
2. **Detection is not our problem and never was.** A five-minute boarding trigger fires inside **62 of 62** sat-out runs at a median of **three minutes** late, with a median **41 points still ahead**. It also fires 34 times a day on tape that goes nowhere. The gap is selection, not detection — which is exactly what the greenfield census concluded from the other direction.
3. **I found a backtest bug that was manufacturing a winner**, and it is the kind that would have shipped. Details in §3. The whole chandelier family survived every robustness test I have until I fixed it, and then went from +\$4,270 to $-\$631$.

Full verdict, dispositions and the one shadow proposal are at the bottom. Every dead candidate is named with its cause of death.

0. THE ANCHOR — I REPRODUCED THE FROZEN CENSUS BEFORE TOUCHING ANYTHING

I did not re-run `run_census.py`. I re-derived its run set from the lake with its own algorithm and checked it line by line against the frozen `census_stdout.txt`.

CHECK	RESULT
runs found on the census window (08-17 00:00 → 08-21 20:59 UTC)	68
rows matching the frozen file on time and direction and move	68 / 68
typical 15-min range / threshold	37pt / 55pt — identical
sat-out runs	62
sat-out ceiling	\$9,610

⚠ **The brief I was handed says "60 runs, \$8,025"**. That is last week's census. This week's frozen file says 62 runs and \$9,610, and since the frozen file is the contract, every number below is scored against \$9,610. Worth flagging because the two numbers are close enough to pass unnoticed.

One small defect found in the census while anchoring to it. The census slides its 15-minute window by array index, not by clock. Where the 5s tape has holes the two drift apart, and only **16 of the 62** sat-out run timestamps are actually minute-aligned — the offsets run 10s, 15s, 25s ... up to 55s. It changes no money, but a run in that table is located to about ± 1 minute, not to the second, and anything that joins to those timestamps on an exact match will silently miss most of them. I snapped each run to the minute bar containing it and took start prices from the 5s bar at the exact census timestamp.

1. STEP ONE — BOARD IT. WHEN COULD WE HAVE KNOWN, AND WHAT WAS LEFT?

The instruction was to stop trying to predict a run and instead board it after it has proven itself, then measure what remains. On gold that shape fired a median 19 minutes into the top five runs with about 80% of the move still ahead. Here is the same curve for MNQ.

The boarding rule is deliberately dumb and direction-agnostic: at each completed one-minute bar, take the net move over the trailing **W** minutes; if it is at least **k** × **ATR** and the efficiency ratio over that window is at least **e**, board it in whichever direction the move went. ATR here is the mean one-minute high-low range over

the last 20 minutes, which is the exact number the live router already prints, so anything built on it is deployable as written.

"What was left" is measured to the **run's own terminus** — the furthest the move got before giving back a full ATR — with no costs and no stop. It is a pure oracle: this is the most any rider could have taken, not what one would take.

The boarding curve, across the 62 sat-out runs

DETECTO R	FIRES IN	MINUTES LATE (P25 / MED / P75 / P90)	POINTS LEFT (P25 / MED / P75 / P90)	% OF THE MOVE STILL AHEAD (MED)	NOTHING LEFT	≥50PT LEFT	ORACLE \$	% OF \$9,610
W=5, k=1.0, ER≥0.40	62 / 62	2 / 3 / 4 / 7	+24 / +41 / +57 / +78	63%	13%	40%	\$5,094	53%
W=5, k=1.5, ER≥0.40	61 / 62	— / 4 / — / —	— / +37 / — / —	55%	11%	34%	\$4,514	47%
W=8, k=1.0, ER≥0.40	62 / 62	— / 5 / — / —	— / +35 / — / —	52%	18%	29%	\$4,135	43%
W=10, k=1.5, ER≥0.50	60 / 62	6 / 8 / 9 / 13	+2 / +20 / +38 / +63	33%	22%	17%	\$2,286	24%
W=10, k=2.5, ER≥0.50	60 / 62	6 / 8 / 10 / 14	+0 / +12 / +33 / +63	18%	27%	15%	\$1,910	20%

Read the top row properly, because it is the most encouraging number in this whole section and it is still not enough. A five-minute confirmation catches every one of the 62 runs we sat out, and it catches them fast — three minutes in at the median, ten minutes at the very worst. At that moment the median run still has 41 points and 63% of its move in front of it. Detection latency, on this trigger, is simply not the problem.

That directly contradicts the prior this desk has been carrying. Part 2.5's detection-latency finding — median 119 points already moved, median 16 points left, 12% capture — was measured on a **30-minute** window with net ≥ 80 and ER ≥ 0.35. That is a much heavier confirmation and it buys a much later seat. Shortening the window from 30 minutes to 5 moves the median from "16 points left" to "41 points left". **The latency was a property of the instrument we were measuring with, not of the tape.**

Now the two hard limits the same table imposes:

- **13% of runs have nothing left at all.** By the time even the fastest honest trigger fires, the move is done. Those runs are not catchable by any boarding

strategy, only by a predicting one, and predicting run starts is already a banked null on 22.6M ticks.

- **The oracle ceiling is 53%, not 100%.** \$5,094 of the \$9,610, with a cheating exit and zero costs. Tighten the trigger to something you would actually trade and it falls to 20–24%. Every honest conversation about this money should start from about **\$2,000–\$5,000 a week**, not \$9,610.

And the thing the table does not show: **that same W=5 / k=1.0 trigger fires 34 times a day.** Over the 35-day lake it produces 1,180 boards, of which about 60 a month are these runs. The other 96% are ordinary tape. That is the whole problem, and it is where the rest of this section goes.

2. STEP TWO — THE EXIT. SWEEP WIDE, AND THEN SWEEP AGAIN ON A SAMPLE TEN TIMES BIGGER

The instruction was explicit: sweep the exit wide before concluding any entry fails, because three separate results this week said a tight exit gets shaken out of a grind while the same entries with a wide stop and a far target make money.

I did that first on the 35-day 5s lake. Entry set held fixed (W=8, k=1.5, ER≥0.50, 60-minute lockout, 1,034 boards) so that every cell differs in the exit and nothing else.

Stop × target, net \$ over the 35-day lake, 1 lot, \$1.50/RT

STOP ↓ / TARGET →	1.0	1.5	2.0	3.0	4.0	6.0	8.0	HOLD TO CAP
1.0×ATR	−2,449	−2,783	−2,492	−1,886	−1,662	−2,634	−4,277	−2,428
1.5×ATR	−2,320	−3,050	−2,550	−783	−1,052	−1,914	−3,302	−5,722
2.0×ATR	−1,361	−1,489	−767	+449	+895	+1,725	+702	−90
2.5×ATR	−996	−665	−639	−865	+319	+2,684	+1,345	−1,138
3.0×ATR	+104	+432	+53	−1,249	−814	+2,133	+1,319	−1,042
4.0×ATR	+376	+1,127	+1,344	−1,792	−318	+3,484	+3,495	−964
6.0×ATR	−303	+1,360	+1,421	−1,595	−1,504	+1,328	+1,355	−1,556

On the surface, the wide-exit thesis is confirmed. The live-desk-shaped exit (1.5 stop, 3.0 target) loses \$783. Widen the stop to 4× ATR and push the target to 6–8× and the same entries make +\$3,484 to +\$3,495. That is a swing of over

\$4,200 on identical boards, purely on the exit — the same shape as the exhaustion_short rehab.

But look at the surface, not the best cell. At a 4.0× stop, the target-3.0 cell loses \$1,792, the target-4.0 cell loses \$318, and the target-6.0 cell makes \$3,484. A real edge does not do that. That is not a plateau, it is a lottery, and it is the first sign that a handful of trades are carrying everything.

Hold-to-time-cap variants (no target, flat at 15 / 30 / 60 / 240 minutes) were negative almost everywhere — that column is the right-hand edge of the table above.

△ **THE GRAVE THAT MATTERS MOST — a backtest bug that manufactured a winner**

The chandelier family — wide stop, trail armed after $N \times \text{ATR}$, give back $k \times \text{ATR}$ from the peak — came out of this sweep looking like the answer. It was the only exit family positive in nearly every regime cell across two different entry definitions, and chand_2.5_a1.5_t1.5 on the W=10 entry set survived everything I threw at it: +\$4,270 net, +\$4.58 a trade, 60% win, strip-best-3 still +\$3,115, strip-best-10 still +\$1,098, day-green 61%, worst day −\$631.

It was wrong. **My trail was testing the bar's HIGH against the running peak instead of its LOW.** A 5s bar that printed a new high and then collapsed registered a give-back of exactly zero, so the trail never fired inside the bar that actually broke it — it exited later, at prices that never traded. The peak was also read including the current bar, which is a second look-ahead on top of the first.

The tell was there before I found it, and it is worth naming so we catch the next one faster: **the random-entry placebo.** Firing at completely random minutes in a random direction with that same chandelier made **+\$1,542 on average**, and the "real" rider's +\$4,270 sat at $p = 0.115$ against that null. A random entry should not make money. When your null hypothesis starts printing, the bug is in the machinery, not in the market.

Fixed — peak taken from strictly prior bars, give-back tested against the adverse edge — here is what happened to the whole family:

EXIT	ENTRY SET	BEFORE THE FIX	AFTER THE FIX
chand 2.5 / arm 1.5 / trail 1.5	W=10 k2.0	+\$4,270 (strip-10 +\$1,098)	−\$631 (strip-10 −\$3,339)
chand 3.0 / arm 2 / trail 2	W=10 k2.0	+\$4,126 (strip-10 −\$331)	−\$820 (strip-10 −\$4,491)

chand 3.0 / arm 2 / trail 2	W=8 k1.5	+\$3,026	−\$1,471
chand 4.0 / arm 2 / trail 3	W=8 k1.5	+\$5,247 (strip-10 −\$736)	−\$872 (strip-10 −\$6,748)
chand 3.0 / arm 3 / trail 2	W=5 k1.0	+\$6	−\$4,728

Every single one flipped. Not one chandelier configuration in the study is positive with a correct trail. **Cause of death: a backtest bug, self-inflicted, caught by the placebo control.** Everything below this line uses the fixed simulator.

The honest sweep — every cell judged, not just the best one

Cherry-picking the best cell out of a grid is how a desk convinces itself. So I ran the full grid — 5 entry windows × 6 thresholds × 4 ER floors × 2 lockouts × 8 exits = **1,824 scored cells** — and put the same battery on every one of them: net > 0, strip-best-3 > 0, strip-best-10 > 0, and **both halves of the tape positive**, with $n \geq 60$.

BAR	CELLS PASSING	SHARE
net > 0	423 / 1,824	23.2%
second half of the tape > 0	36 / 1,824	2.0%
net and strip-3 and strip-10 and both halves	5 / 1,824	0.3%

Two per cent of the entire configuration space makes money in the second half. If the edge were real and broad you would expect most of the 423 profitable cells to also be profitable in the back half; instead 423 shrinks to 36. With a 2% base rate you would expect about 8 of the 423 to clear the full bar by luck alone. **Five did.** The five "survivors" of the blanket sweep are exactly what chance hands you, and quoting any one of them would be a lie of selection.

But the halves test is confounded, and I checked

Before accepting "the edge decayed", I looked at what the two halves actually are.

HALF	DATES	SESSIONS	ATR20 MEDIAN	MEAN DAILY RANGE	15-MIN MOVE P90	ER30 MEDIAN
first	07-16 → 08-04	14	13.9pt	709pt	67.1pt	0.172
second	08-05 → 08-24	14	9.3pt	398pt	44.4pt	0.156

The second half is a **33% quieter tape by ATR and 44% quieter by daily range**. So "it stopped working in half two" and "it stops working when volatility halves" are the same sentence, and the blanket halves test cannot tell them apart. That is precisely the failure mode the desk's own backtest discipline warns about — a blanket cross-tape number averages the regimes a config should be ON with those it should be OFF, and is a bug. So I split by regime first.

3. STEP THREE — THE FILTER, AND THE ONE THAT ACTUALLY WORKED

Regime first, config second

I bucketed every board by the **ATR level at entry** — causal, known before the trade, and the live router's own meter — and re-ran a 216-cell entry × exit sweep inside each bucket, requiring both halves of that bucket to be positive.

ATR20 AT ENTRY	CELLS SCORED	NET > 0	2ND HALF > 0	PASS BOTH HALVES + STRIP-3
ATR < 9 (quiet / dead)	216	23 (11%)	15 (7%)	4 (2%)
ATR 9-14 (the desk's normal tape)	216	188 (87%)	75 (35%)	66 (31%)
ATR ≥ 14 (violent)	216	81 (38%)	39 (18%)	16 (7%)

A fifteen-fold difference in survival rate between buckets is not something chance produces evenly. Something real lives in the middle band, and the blanket number had been averaging it away.

...but first, is that just leverage?

Stops and targets are ATR-scaled, so a higher-ATR trade is mechanically a bigger bet in dollars. If the "filter" simply selects bigger bets, dollars per trade rises while the actual edge stays flat, and the rule is a position-sizing choice dressed up as an insight. So I rescored everything in **R** — net P&L divided by (ATR at entry × \$2) — which is the same trade measured in units of what it risked.

ATR20 BAND	W8 K2.5 (R/TRADE)	W10 K2.5 (R/TRADE)	W15 K3.0 (R/TRADE)	AGREEMENT
0-7	+0.07	+0.45	-0.12	no
7-9	+0.27	-0.63	-0.24	no
9-11	+0.48	-0.38	+0.26	no
11-13	+0.07	+0.66	+1.03	partial
13-16	+1.42	+1.09	+0.87	all three
16+	-0.11	+0.19	-0.11	no (all ≈ 0)

In R the relationship is **not** monotone in ATR, which kills the leverage explanation. It is a **band**: the money is between roughly 11 and 16 points of ATR, and both edges matter. Below it the moves are too small relative to noise and the round trip; above it the tape is violent enough that a "proven" move is more often

an exhausted one — which is the same thing the abs_veto rehab found when the amp_floor let through elevated-ATR-but-spent overnight thrusts.

The candidate

RIDER_ALL / board_atrband — sampled at each completed one-minute bar.
ATR20 = mean 1-min high–low over the last 20 minutes. **Fire when $11 \leq \text{ATR20} < 16$ AND $|\text{net over 10 min}| \geq 2.5 \times \text{ATR20}$ AND $\text{ER}(10 \text{ min}) \geq 0.50$.** Direction = sign of the 10-minute move. Stop $3.0 \times \text{ATR}$, target $6.0 \times \text{ATR}$, flat at 240 minutes, 60-minute lockout between fires.

The full battery, in-sample (5s lake, 28 sessions)

ENTR Y WIN DOW	N	NET \$	\$/ TRA DE	R/ TRA DE	WIN	STRI P-3	STRI P-5	STRI P-10	HALF 1 (N)	HALF 2 (N)	LOO- DAY WOR ST	DAY- GREE N	WOR ST DAY	MEA N HOL D
W8 k2.5 ER.50	154	+3,358	+21.81	+0.784	44%	+2,791	+2,417	+1,507	+2,814 (103)	+543 (51)	+2,697	67%	−226	50m
W10 k2.5 ER.50	146	+3,496	+23.95	+0.896	44%	+2,937	+2,573	+1,682	+2,695 (94)	+801 (52)	+2,881	70%	−331	54m
W15 k3.0 ER. 40	152	+3,704	+24.37	+0.950	46%	+3,145	+2,778	+1,877	+2,527 (97)	+1,176 (55)	+2,870	63%	−432	55m
W10 k2.0 ER. 40	164	+2,612	+15.93	+0.585	41%	+2,048	+1,687	+808	+805 (107)	+1,806 (57)	+1,741	69%	−584	51m
W8 k2.0 ER.50	159	+3,118	+19.61	+0.746	44%	+2,559	+2,194	+1,322	+2,612 (103)	+505 (56)	+2,472	59%	−410	57m

Five neighbouring entry definitions, all between +\$15.93 and +\$24.37 a trade, all surviving strip-best-10, all positive in both halves. **That is a parameter plateau, not a step.** By this desk's stated bar — expectancy and robustness, never win rate — 44% winning is irrelevant and +0.9R a trade over 146 trades with strip-10 intact is a serious candidate.

The controls

CONTROL	RESULT	READING
Direction flipped (same boards, opposite side)	−\$2,436, −\$16.68/tr, −0.637R	direction is doing real work — it is not an exit artefact
Random-time placebo — same ATR band, random minutes, random side, 400 draws	null mean +\$229, sd \$1,335 → p = 0.0125	beats a random entry with the same exit
Placebo cut — keep a random 146 of the 518 unfiltered fires, 4,000 draws	null mean +\$1,088, sd \$1,246 → p = 0.029	the ATR band beats discarding the same number of fires at random
Unfiltered (all ATR)	518 trades, +\$3,864, +\$7.46/tr, +0.190R	the band roughly triples expectancy per trade

★ **What the cut COSTS — the number I am obliged to report and it is not flattering**

Cutting to the ATR band discards **372 of the 518 fires**. Those 372 contain **131 winners worth +\$18,483** which we forgo, and **241 losers worth −\$18,115** which we avoid. Net effect of the entire filter on the bottom line: **+\$368**.

So the ATR band does **not** find money. It concentrates the same money into a quarter of the trades. That is genuinely worth something — it is what lifts \$7.46 a trade to \$23.95 and what makes the thing survive strip-10 — but anyone reading "+0.9R a trade" as "the filter found an edge" has it backwards. The filter found a smaller sample.

And the caveat I have to put next to my own p-values

I chose the band edges after looking at the data. There were roughly 36 plausible $lo \times hi$ combinations in the sweep I ran. A p of 0.029 against 36 looks, corrected, like a p of about 1.0. The placebo cut controls for "any cut of this size" but not for "I went looking for the best cut". The in-sample battery above is real and the controls are honest, but it is not, on its own, evidence that survives the multiplicity. Which is why the next section exists.

4. THE OUT-OF-SAMPLE LEG — AND THE KILL

The 5s lake is 28 sessions. The parquet lake also carries a **one-minute backfill running back to 2025-09-14** — 295,845 bars, **236 sessions before the 5s tape even begins**. Not one parameter above was chosen on it, and it is the best-powered test available to this desk.

Two honesty notes before the number. The exit has to be simulated on one-minute bars there, which is a coarser and more punishing ruler — inside a bar I still resolve stop-before-target, and the fill is a minute later. To measure how much coarseness costs, I ran the in-sample window through the coarse simulator too:

LEG	N	NET \$	\$/TRADE	R/TRADE
in-sample window, 5s simulator	146	+3,496	+23.95	+0.896
in-sample window, 1-min simulator (calibration)	213	+2,399	+11.26	+0.459

The coarse ruler reads roughly **half**. Keep that in mind and then read this:

LEG	N	SESSI ONS	NET \$	\$/ TRADE	R/ TRADE	95% CI ON R/ TRADE	STRIP- 3	STRIP- 10	HALF 1	HALF 2	DAY- GREEN
OUT-OF- SAMPLE, ATR band	1,289	236	+\$14	+\$0.01	+0.021	[-0.204, +0.246]	-555	-1,875	-2,256	+2,270	47%
OOS, no ATR band (control)	3,924	259	-\$3,640	-\$0.93	-0.047	—	-5,676	-9,445	+2,806	-6,446	46%

Fourteen dollars. Over eleven months. Double it for the coarse ruler and it is twenty-eight dollars. Month by month it is a coin toss: +734, -84, +437, -1,510, -1,333, -555, +373, +496, +1,844, -669, +281 — six green months out of eleven. It does not survive stripping its three best trades.

The ATR band is not useless: without it the same entries lose \$3,640 and -0.047R. **The band takes the rider from clearly negative to exactly zero.** That is a real filter doing real work on a mechanism that has no edge to protect.

Step Two, re-run properly on the big sample

The wide-exit sweep on 28 sessions was the underpowered version. Here it is on 236, on the banded entry set (n = 1,289), net \$ / R per trade:

STOP ↓ / TARGET →	1.5	2.0	3.0	4.0	6.0	8.0	CAP
1.0×ATR	-4,185	-3,073	-2,398	-2,179	+875	+849	-3,090
2.0×ATR	-3,059	-2,636	-2,152	-2,573	+1,367	-576	-7,179
2.5×ATR	-2,301	-1,618	-2,302	-2,007	+851	-1,262	-7,369
3.0×ATR	-1,967	-1,028	-2,579	-2,510	+14	-2,622	-9,367
4.0×ATR	-2,051	-451	-1,429	-2,538	-786	-1,647	-7,618
6.0×ATR	-3,509	-1,505	-2,504	-2,749	-201	+494	-6,840

Forty-one of the 42 cells are negative or within noise of zero, and the five nominally positive ones are scattered, not clustered. +\$1,367 sits directly beside -\$576 and -\$2,573. There is no plateau anywhere on this surface.

So: **"an entry that only pays with a wide exit is a finding, not a failure" — I tested that properly, and for this entry it is a failure.** The wide exit reliably makes the rider less negative (target 6.0 is the best column in every row) but it never makes it positive. That is not a refutation of the exhaustion_short rehab, which was a different gate's entries; it is a refutation of the hope that the same medicine cures this one.

Does the direction carry anything at all?

Yes, and this is the most useful salvage in the section. Re-simulating every board on the opposite side:

EXIT	WITH-TREND (AS DESIGNED)	COUNTER-TREND	SEPARATION
3.0 × 6.0	+\$14 (+0.021R)	−\$1,363 (−0.056R)	+0.077R
2.0 × 4.0	−\$2,573 (−0.075R)	−\$4,066 (−0.135R)	+0.060R
4.0 × 8.0	−\$1,647 (−0.015R)	−\$3,816 (−0.121R)	+0.106R

A proven move genuinely does continue more often than it reverses — consistently, in every exit, across 1,289 out-of-sample trades. **The information is real. It is worth about +0.06 to +0.11R a trade, and the round trip alone costs 0.06R at the band's midpoint.** The edge exists and is roughly the same size as the toll booth. That is the honest shape of this whole problem.

Cost sensitivity, and what it would take

SLIPPAGE (POINTS EACH SIDE, ON TOP OF \$1.50/RT)	OOS ALL-DAY	OOS OVERNIGHT	5S LAKE ALL-DAY
0.00	+\$14 (+\$0.01/tr)	+\$4,143 (+\$10.46)	+\$1,686 (+\$6.74)
0.25 (one tick)	−\$1,275 (−\$0.99)	+\$3,747 (+\$9.46)	+\$1,436 (+\$5.74)
0.50	−\$2,564 (−\$1.99)	+\$3,351 (+\$8.46)	+\$1,186 (+\$4.74)
1.00	−\$5,142 (−\$3.99)	+\$2,559 (+\$6.46)	+\$686 (+\$2.74)

One tick of slippage takes the all-day rider negative. And the power calculation: at the band midpoint 1R is \$25 and the round trip is 0.060R; to establish a real expectancy of +0.10R a trade against what we actually measured would need roughly **10,500 trades**, about eight years of this rider's fire rate. This is not a "needs more data" case. It is a "the effect is smaller than the measurement error and smaller than the costs" case.

5. CLOSING THE LOOP — GRADING rider_w5, THE ARM THIS PHASE SHIPPED LAST WEEK

While reading the shadow book to write the ShadowVariant block, I found that **this exact idea is already live.** rider_w5 (src/gazbot7/shadow.py:929) was added on 2026-08-15 out of last week's RIDER_ALL phase, quoted at **+\$4,841 and +\$19.29 a trade:**

gate="board", k=2.0, w=5, 13:00–20:00 UTC, stop 3.0×ATR, target 6.0×ATR, cooldown 15m, cap 120m.

It has been recording forward trades since 08-17 13:25 and **nothing had graded it**. Here it is.

RIDER_W5 LIVE SHADOW, 08-17 → 08-25	VALUE
closed trades	51
real P&L	−\$1,471.00
per trade	−\$28.84 (against +\$19.29 claimed)
win rate	18%
strip-best-3	−\$2,161
days green	1 of 7 (08-17 −234, 08-18 −232, 08-19 −631, 08-20 −416, 08-21 +377, 08-24 −289, 08-25 −46)
flagged rows (data_quality)	0 — this is clean data, not a measurement artefact

Red in every ATR bucket, including mine:

ATR20 AT ENTRY	N	NET \$	\$/TRADE	WIN
< 9	15	−\$299.50	−\$19.97	20%
9-11	8	−\$233.50	−\$29.19	12%
11-16 (my candidate's band)	14	−\$635.50	−\$45.39	14%
≥ 16	14	−\$302.50	−\$21.61	21%

My band is the arm's **worst** bucket live. That is as clean a refutation of the filter as exists.

Exit mix: STOP 21 (−\$1,190) · **ABSORPTION_CUT 18 (−\$1,449)** · TARGET 8 (+\$1,226) · TIME_CAP 4 (−\$58).

Entry or exit? — the attribution

The live arm carries adverse_cut_atr=3.0, which fires an ABSORPTION_CUT on 35% of its trades and is its single largest loss bucket. My simulator has no such mechanism, so the two are not measuring the same strategy. Re-deriving the arm's exact spec in my simulator separates them:

RIDER_W5'S SPEC, RE-DERIVED (NO ABSORPTION CUT)	N	NET \$	\$/TRADE	STRIP-3	DAYS GREEN
5s lake 07-16 → 08-24, 13:00–20:00Z (as shipped)	173	+\$1,936	+\$11.19	+\$282	57%

...of which the census week 08-17 → 08-21	33	−\$174	−\$5.26	−\$792	40%
OOS 236 sessions, its own 13:00-20:00Z window	1,339	−\$7,272	−\$5.43	−\$9,293	47%
OOS 13-20Z + my ATR band 11-16	620	+\$30	+\$0.05	−\$540	46%

The entry alone is worth about −\$5.43 a trade out of sample. The absorption cut adds roughly another −\$23 a trade in live conditions. Both are bad; the entry alone is enough to kill it. And note the in-sample line: +\$11.19 a trade collapses to +\$282 when you remove its best three trades — 85% of it was three trades. The +\$19.29 figure this arm shipped on never had a strip test attached.

Recommendation: retire rider_w5. It is not a thin-n case — 51 forward trades, 1,339 out-of-sample trades, and the census week itself all agree, and the mechanism has an explanation (see §7).

6. THE LAST LEAD — THE OVERNIGHT LEG, AND WHY IT DID NOT REPLICATE

One thing did survive the out-of-sample leg, and I chased it as far as it goes.

Splitting the 1,289 OOS boards by session. △ This is a post-hoc split of an entry set built without a session constraint, so the 60-minute lockout falls differently than it does when the session is part of the rule — the session-constrained run below gives n=396 and +\$10.46/trade rather than the n=383 and +\$11.50 here. Same finding, slightly different bookkeeping; I quote the session-constrained numbers everywhere it matters.

OOS SESSION	N	NET \$	\$/TRADE	R/TRADE	WIN
overnight 21:00-07:00Z	383	+\$4,406	+\$11.50	+0.485	40%
Europe 07:00-13:00Z	344	−\$2,064	−\$6.00	−0.226	32%
US 13:00-21:00Z	562	−\$2,328	−\$4.14	−0.144	34%

On its generating sample the overnight leg looks strong: 396 trades, +\$4,142, +0.444R, strip-best-3 still +\$3,578, strip-best-10 still +\$2,281, leave-one-day-out worst +\$3,536, seven green months of ten, leave-one-month-out worst +\$2,388, counter-trend control −\$1,146. It also survives a full point of slippage each side.

So I did the clean thing available: **the hypothesis was generated on the 236-session backfill, so the 28-session 5s lake is now a genuine held-out test for it.** The roles reverse.

	OOS (GENERATING)	5S LAKE (HELD OUT)	REPLICATED?
overnight, $ER_{10} \geq 0.50$	+\$10.46/tr, +0.444R, n=396	+\$5.34/tr, +0.224R, n=95	sign yes, but strip-3 = -\$51
daytime control (should be bad)	-\$5.18/tr, -0.190R	+\$5.21/tr, +0.203R	NO — control reverses
counter-trend control	-\$2.89/tr	-\$13.22/tr	yes
with $ER_{30} \geq 0.30$ instead (what <code>er_min</code> reads)	+\$14.67/tr, +0.645R	-\$2.25/tr, -0.060R	NO
...its counter-trend control	-\$9.54/tr	+\$10.94/tr	NO — reverses
...its daytime control	-\$7.71/tr	+\$13.55/tr	NO — reverses
overnight, no ATR band	+\$3.99/tr	-\$11.69/tr	NO
rider_w5's own spec, overnight	+\$3.29/tr, n=1,688	-\$9.54/tr, n=218	NO

The overnight lead dies on replication, and it dies on the controls rather than on the headline. The rider itself keeps its sign in both samples for one particular parameterisation. But the claim — that overnight tape behaves differently from daytime tape for a boarding rider — is not reproduced: on the held-out lake daytime is just as good or better in five of six comparisons, and the counter-trend control, which was reassuringly negative out of sample, comes back **positive**. When your control flips sign, you are reading noise.

Two further honesty points. The session split was **one of three post-hoc comparisons**, so its nominal significance should be trebled. And the random-time placebo inside the same band and the same session gives **p = 0.097** — even on its own generating sample it is only about 1.3 standard deviations above firing at random times.

It is not nothing, and I am not calling it dead. It has the right sign in two independent samples, its counter-trend leg is negative in both, and it is the only thing in this study that survives a full point of slippage. That earns incubation, not deployment. See the disposition table.

7. THE STONE I HAD NOT TURNED — THE L2 BOOK AT THE BOARD MOMENT

The census scores every run for far-side depletion, and I had used none of it. The obvious question: at the moment the board rule fires, does a thin far side — the side price is running into — tell us which of those 34 daily fires is a real run? We

hold 264 million MNQ book rows from 07-31 to 08-24, so this is answerable with data we already have.

For each of the 311 fires inside the book window I took the far-side share of level-1-to-3 depth over the 30 seconds **before** entry (below 0.50 = the far side is thin = the run is telegraphed).

FAR-SIDE SHARE	N	NET \$	\$/TRADE	R/TRADE	WIN
0.00-0.42	13	-\$38	-\$2.93	-0.343	31%
0.42-0.47	44	-\$300	-\$6.82	-0.544	30%
0.47-0.50	109	-\$680	-\$6.24	-0.142	35%
0.50-0.53	89	+\$188	+\$2.11	+0.139	36%
0.53-0.58	45	-\$379	-\$8.42	-0.308	33%
0.58-1.01	11	-\$166	-\$15.09	-0.662	27%

A clean null. Correlation between far-side share and R is **-0.009** across 311 trades. Cutting to a thin far side gives -0.265R against -0.060R for a thick one — i.e. the wrong way round — and against a random cut of the same size the placebo p is **0.62**. There is no ordering, no monotonicity, no signal. The L2 book, read this way, at this moment, tells us nothing about which board becomes a run.

That closes the stone. It also means "we need a data source we do not capture" is not the excuse available here — we had the deepest data on the desk and it was empty.

8. STEP FIVE — SIZING THE PRIZE HONESTLY

\$		AS % OF THE \$9,610 SAT-OUT CEILING	
ORACLE — perfect late board (fires in 62/62 at a median 3 min), exit at the exact run terminus, no costs, cheats		\$5,094	53%
ORACLE — the same, with a detector you would actually deploy (W=10, k=1.5, ER≥0.50)		\$2,286	24%
ORACLE — with the candidate's own trigger (W=10, k=2.5, ER≥0.50)		\$1,910	20%
CAUSAL — the candidate rider, one slot, run over the census week itself		-\$245	-3%
CAUSAL — same rider, overnight only, census week		-\$214	-2%
CAUSAL — same rider, one slot, whole 5s lake, scaled to a 5-session week		+\$433/week	in-sample only
		≈ \$0/week	0%

On the census week itself, running one position at a time, the candidate took **42 trades and lost \$245**. It did show up — it participated in **32 of the 62 sat-out runs**, roughly half — and it still lost money, because the other trades cost more than the runs paid. That is the sharpest single sentence in this report: **showing up to half the runs was not the hard part, and it did not help**.

The oracle table is the number to carry forward. This is a **\$2,000-\$5,000-a-week oracle, not a \$9,610-a-week one**, and per the desk's own precedent that distinction changes what the money is worth building for. The gap between the \$5,094 oracle and the \$0 causal result is not an exit problem and not a detection problem — it is **selection**, and it is 100% of the gap.

Why the mechanism fails, in one paragraph. At the moment the boarding rule is true, the median run has 12 to 41 points left depending on how much confirmation you demand — and the same rule is true 34 times a day on tape that does nothing. The trades that are not runs pay the stop, which at $3 \times \text{ATR}$ in the 11-16 band is 33 to 48 points. So the rider is risking 33-48 points to collect a median 12-41, and it must be right far more often than 44% of the time for that to work. It is not. The wide exit helps because it lets the occasional big run pay for many stops, but the ratio of runs to non-runs at the moment of firing — about 1 in 20 — is simply too thin. **Boarding is solved. Selecting is not.** That is the same conclusion the greenfield census reached from the other end when it found that a plain $3/6 \times \text{ATR}$ exit converts 40% of the ceiling but 105 of 155 fires are not runs at all.

9. EVERY ATTEMPT, BY NAME, WITH ITS CAUSE OF DEATH

#	CANDIDATE	BEST NUMBER IT EVER SHOWED	CAUSE OF DEATH	NAMED TEST
1	board_live_exit — board a proven move, live-shaped 1.5×3.0 exit	−\$783 to −\$5,211 across four entry sets	never positive anywhere	outright, all entry sets
2	board_wide — wide fixed stop × far target	+\$3,495 (35-day lake)	second half of the tape, and the OOS sweep	both-halves; 236-session OOS
3	board_holdcap — no target, flat at 15/30/60/240 min	+\$2,301 in one cell	negative in almost every regime cell; no plateau	regime-split sweep
4	board_chandelier — wide stop + armed trail (4 variants)	+\$5,247 (chand 4.0/a2/t3, W8); the best-behaved was chand 2.5/a1.5/t1.5 on W10	backtest bug — the trail tested the bar's high, not its low	random-entry placebo (p=0.115) flagged it; fixed → all four negative

at +\$4,270 with strip-10 still
+\$1,098

5	board_blanket — the full 1,824-cell sweep, best cell		only 5/1,824 cells pass, against a 2% base rate ⇒ chance	both-halves base-rate count
		+\$4,494		
6	board_atrband — ATR20 ∈ [11,16) + wide exit (the candidate)	+\$3,496, +0.896R, strip-10 +\$1,682, placebo p=0.0125 / 0.029	out-of-sample leg: +\$14 over 236 sessions, CI [−0.204, +0.246]	236-session OOS; live rider_w5 red in the same band
7	board_overnight — the same + 21:00-07:00Z	+\$4,142, +0.444R, LOO-month +\$2,388	held-out sample did not reproduce the separation; controls reverse sign	held-out 5s lake; placebo p=0.097; ER30 substitution flips it
8	board_book — far-side L2 depletion as the selector	—	clean null: r = −0.009, placebo p=0.62	311 fires, 264M book rows
9	rider_w5 — the arm shipped 2026-08-15	+\$4,841 / +\$19.29 as quoted	−\$1,471 over 51 live forward trades; −\$5.43/tr over 1,339 OOS trades; −\$174 on the census week	live shadow book + 236-session OOS + strip-3 (+ \$282 of +\$1,936)

10. DISPOSITION

LEAD	VERDICT	WHY / WHAT WOULD REVIVE IT
rider_w5 (live shadow arm)	REFUTED	Named tests: 51 live forward trades at −\$28.84/tr, red in all four ATR buckets; 1,339 OOS trades at −\$5.43/tr with strip-3 at −\$9,293; −\$174 on the census week it was built for; and its in-sample +\$1,936 collapses to +\$282 on strip-best-3. Mechanism stated in §8: at fire time the median run has 12-41pt left against a 33-48pt stop, and only ~1 fire in 20 is a run. Action: retire the arm.
board_atrband (the candidate)	PARKED	Passed every in-sample test including two placebos, then measured +\$14 over 236 sessions. Revive only with a SELECTOR — something that predicts which board becomes a run. Not more days: the power calculation says ~10,500 trades to separate it from zero. The census's own finding points the way (size threshold, ≥120pt boarding), and that is a different study from this one.
board_overnight	SHADOW	Right sign in two independent samples (+0.444R and +0.224R), counter-trend leg negative in both, survives a full point of slippage. But its daytime and counter-trend controls reverse on the held-out sample and its placebo p is 0.097. Shadow it as an instrumented question, not as a candidate — the specific thing to buy is whether overnight boards really differ from daytime ones, which the two samples disagree about. Block and promotion bar below. Needs two small code changes first (§11).
board_wide / board_holdcap / board_blanket	PARKED	All failed as built on both-halves or the OOS sweep. Revive only if the selection problem is solved first — the exit was never the binding constraint here, which the 42-cell OOS surface establishes.
board_chandelier	PARKED, not refuted	It was never honestly tested — the trail was broken. With the corrected trail it is negative

		everywhere, but the shape (wide stop + armed trail) is what grind_long's deployed pure-6.0 chandelier runs on, so the mechanism is not disproven, only this application of it. Revive on a selected entry set, never on this unselected one.
board_book (L2 far-side depletion as selector)	REFUTED	Named test: 311 fires against 264M book rows, correlation -0.009 , no monotone band, placebo $p=0.62$. No reformulation of this measurement (far-side share, 3 deep, 30s pre-entry) survives; a different book statistic — resting-size decay rate, or the sweep-through print — would be a different lead, not a revival of this one.
Direction information itself	CONFIRMED but sub-scale	With-trend beats counter-trend in every one of three OOS exit configurations, by $+0.06$ to $+0.11R$. That is real and it is about the size of the round trip ($0.060R$). Keep as a prior, do not trade it alone.
The oracle ceiling	FINDING — carry it forward	The \$9,610 is a hindsight number. A perfect late boarder reaches \$5,094 (53%); a deployable one reaches \$1,910–\$2,286 (20–24%); 13% of runs are over before any honest trigger fires.

There is no LIVE candidate from this phase, and I am saying so plainly. The one stone I would call still unturned is **selection by expected SIZE** — the census's own greenfield work found that boarding only fires whose move is $\geq 120\text{pt}$ pays $+\$69.38/\text{trade}$ and passes the whole battery, and this study never tested a size predictor at the board moment. That is the next attempt, and it does not need a new data source; it needs a different question asked of the same tape.

11. THE SHADOW BLOCK — AND TWO THINGS THAT MUST BE BUILT FIRST

⚠ **This cannot be pasted in as-is.** I checked the shadow book's fields against the rule and found two genuine blockers. Both are small, but shipping around either one would record a different strategy under this name — which is exactly how rider_w5 came to be quoted at $+\$19.29$ while making $-\$28.84$.

- gate_board cannot express a wrapping clock window.** It tests $hh_lo \leq utc_hour < hh_hi$, which is never true for 21:00→07:00. Needs one line:
`inw = (hh_lo <= h < hh_hi) if hh_lo < hh_hi else (h >= hh_lo or h < hh_hi).`
- ShadowVariant has an atr_max but no atr_min.** The rule is a band, and both edges carry weight (\$3). An arm shipped with only the ceiling is a different arm. Needs one field, defaulted to 0.0 so nothing existing changes.

3. **Note on ER.** ShadowVariant.er_min reads **ER30**, not the entry window's ER. Substituting ER30 flips this rule's sign on the held-out sample (§6), so the arm must gate on the **10-minute** ER. Either add er_win_min to the gate params or accept that an er_min arm is testing a different hypothesis and label it as such.

With those three in place:

```
ShadowVariant("board_overnight_band", "board",
              # ★ RIDER_ALL 2026-08-25. An INSTRUMENTED QUESTION, not a promotion
              candidate.
              # Generated on the 236-session 1-min backfill (+$10.46/tr, +0.444R,
              n=396,
              # strip-10 +$2,281, L00-month worst +$2,388); held-out on the 28-
              session 5s lake
              # (+$5.34/tr, +0.224R, n=95) where the SIGN held but strip-3 fell to -
              $51 and the
              # DAYTIME control came back POSITIVE (+$5.21/tr) – i.e. the overnight-
              vs-daytime
              # SEPARATION that generated the idea did NOT reproduce. Random-time
              placebo inside
              # the same band and session: p=0.097. Session split was 1 of 3 post-
              hoc comparisons.
              # △ THE THING THIS ARM EXISTS TO ANSWER is whether overnight boards
              differ from
              # daytime ones. It is NOT here because we think it makes money.
              # △ ER is the 10-MINUTE ER, not ER30. Substituting ER30>=0.30 FLIPS
              the held-out
              # leg to -$2.25/tr and turns both controls positive. Do not use er_min
              here.
              # △ adverse_cut_atr=99 ON PURPOSE. rider_w5's adverse_cut_atr=3.0
              fired on 35% of
              # its live trades for -$1,449 – a tight give-back cut is the exact
              opposite of the
              # wide-exit thesis this arm is testing. Do not "helpfully" restore it.
              {"k": 2.5, "w": 10, "er_win_min": 0.50,
              "hh_lo": 21.0, "hh_hi": 7.0,          # NEEDS the wrapping-window fix
              "cooldown_min": 60},
              symbol="MNQ", qty=1.0,
              atr_min=11.0, atr_max=16.0,          # NEEDS the new atr_min field;
              BOTH edges matter
              stop_atr_mult=3.0, target_r=2.0,      # = target 6.0 x ATR
              adverse_cut_atr=99.0, absorption_flow_min=1e9,
              chandelier=False,                    # every chandelier variant
              tested RED once the
              # trail bug was fixed – do not
              add one
              time_cap_s=240 * 60)
```

Its router rule, in the router's own vocabulary

ARM board_overnight_band when, and only when, all four are true at the tick: 1. the clock is between **21:00 and 07:00 UTC**; 2. the router's own **ATR meter reads 11.0 to 16.0 points** (this is the mean 1-min range over 20 minutes — the number the router already prints; it is a **band**, so bench it above 16 as well as below 11); 3. **RUN STATE is RUN or RUN-FADING** — the arm is a boarding rule and has no business firing in declared chop; 4. the **untradeable meter is clear** and the tape is not inside a scheduled gateway window (04:28-05:00Z sits squarely inside this arm's session and hangs with the port open). **BENCH it immediately if the ATR meter leaves the band in either direction**, and treat a bench here as cheap: the arm's whole thesis is that both edges of the band carry weight. **Expectancy on the home regime only** (overnight, ATR 11-16, wide exit): **+\$10.46 a trade on the 236-session generating sample, +\$5.34 a trade on the 28-session held-out sample**. Take the lower of those as the working expectation, and note that its own strip-best-3 on the held-out sample is **-\$51** — three trades were the whole thing. **This is not a number to size on.**

Promotion bar — deliberately set high

Do **not** promote this to the paper tournament on P&L alone. It needs **all four**:

1. **200 shadow trades**, including at least one genuinely trending overnight week;
2. **$\geq +0.15R$** over those trades **after** stripping its best three;
3. the **daytime control arm** (identical rule, 13:00-20:00Z) must be **materially worse** — that is the claim under test, and it is the one that failed to replicate;
4. the **counter-trend control** must stay negative. It reversed on the held-out lake, and if it reverses again the arm is measuring noise and should be retired without argument.

I would also add the daytime control arm at the same time. Shadowing the treatment without the control is what produced this week's ambiguity in the first place.

VERDICT

NO RIDER SURVIVES. The pooled, direction-agnostic late-boarding run-catcher **DIES AT STEP 3 — THE CAUSAL FILTER** — killed by the **OUT-OF-SAMPLE LEG: 1,289 trades across 236 untouched sessions for a total of +\$14, a 95% confidence interval on expectancy of $[-0.204R, +0.246R]$, and a per-trade result that goes negative on one tick of slippage.** The verdict is independently corroborated by the live shadow arm rider_w5, which has taken 51 forward trades since 08-17 for $-\$1,471$ ($-\$28.84/\text{trade}$), red on six of seven days and red in every ATR bucket including the candidate's own — rider_w5 is **REFUTED** and should be retired. Step 1 **SUCCEEDED** and should be believed: a 5-minute boarding trigger fires inside 62 of 62 sat-out runs at a median 3 minutes late with a median 41 points still ahead, so detection is solved. Step 2's wide-exit thesis is **CONFIRMED IN DIRECTION** but **INSUFFICIENT IN SIZE** — a wide stop and a far target reliably make this rider less negative and never make it positive across 42 out-of-sample exit cells. The binding constraint is **SELECTION**: the same trigger fires 34 times a day and roughly 1 fire in 20 is a run, so the rider risks a 33-48 point stop to collect a median 12-41 points. The honest prize is **NOT \$9,610** — a perfect late boarder with a cheating exit reaches \$5,094 (53%), a deployable one reaches \$1,910-\$2,286 (20-24%), and 13% of the runs are finished before any honest trigger can see them. The causal result on the census week itself was $-\$245$ across 42 trades while participating in 32 of the 62 runs. **ONE lead goes forward:** board_overnight_band to SHADOW as an instrumented question ($+0.444R$ on 236 generating sessions, $+0.224R$ held-out, but its daytime and counter-trend controls reverse and its placebo p is 0.097), gated behind two small code changes and a four-part promotion bar. The next attempt needs a **SIZE PREDICTOR** at the board moment, not more days, not a wider exit, and not a new data source — the L2 book was tested as the selector across 311 fires and 264M rows and returned a clean null ($r = -0.009$, placebo p = 0.62).

UNCLASS — the biggest bucket

NOT RUN THIS WEEK · dossier 2026-08-25

The bucket our own classifier could not name, and had therefore never hunted.

GREENFIELD HUNT — CLUSTER UNCLASS

MNQ · full parquet lake · 5s tape 2026-06-19 → 08-24 (51 sessions) + a 205-session 1-minute leg back to 2025-09-14 · \$2.00/point · \$1.50 per round trip

THE SHORT VERSION, BEFORE ANY OF THE WORKING

Garrath — I did the three steps in the order you set them. I built a direction-agnostic run-catcher and rode the UNCLASS runs without needing to know why they happened. I swept the exit far wider than the desk has ever run one. I filtered it with a causal rule and placebo-controlled every cut against throwing away the same number of fires at random. Then I routed it. It still lost.

It died on the PLACEBO/SHUFFLE TEST, corroborated by the OUT-OF-SAMPLE LEG. On the biggest independent sample I have — 205 sessions of tape that chose nothing — the real signal makes **+\$3.54 a trade** and the **fake signal, taken thirty minutes early with every other rule untouched, makes +\$4.44 a trade**. On the recent forward leg the real signal makes **−\$9.58 a trade** and the fake one makes **+\$1.48**. When a shifted copy of your signal outperforms your signal on two independent samples, you do not have a signal — and per your judging rule that is the named cause of death, not the 25% win rate, which I have deliberately ignored throughout.

Four things came out of the week that are worth more than the dead rider:

1. **UNCLASS is not a cluster and this is now measured twice, independently.** I evaluated the census's own waterfall on **every one of the 6,784 minutes** of its window. UNCLASS is true on **73.9% of the tape** and takes **60.3% of the runs** — a **lift of 0.82**. A label with a lift below 1.0 is not a footprint. It is the null hypothesis with a name tag, and "UNCLASS: 41 runs" reads as a discovery when it means "41 runs happened somewhere in the three-quarters of the week we have no test for."
2. **And it is worse than last week thought: 21 of the 68 labels flip on nothing but where a minute starts.** I changed one thing — the phase of the 60-second flow buckets — and 31% of the census's cluster assignments changed. Only 17 of 68 run timestamps are minute-aligned, so this is not a hypothetical.

3. **Last week's survivor was never shipped, and the tape since has buried it.** UNCL-RIDER-ASIA went to SHADOW on 2026-08-15 and no arm by that name exists in shadow.db. I replayed it myself. It is now **four consecutive red weeks**, and the newest week — pure forward tape it has never seen — is **−\$472 over 16 trades, −\$29.51 a trade, 18.8% wins**. The drawdown that made it a non-deploy last week did not recover; it deepened.
4. **The prize is not \$5,840.** A boarder that fires inside every UNCLASS run and then exits at the exact top of each move with no costs at all reaches **\$3,239 — 55%**. Put an honest stop on it and bank half the excursion and it is **\$1,592, 27%**. Two thirds of the headline is a close-to-close number that no late entry can ever collect.

The full ledger of every attempt — thirteen named candidates, none of them quietly dropped — is at the bottom, along with the two things I do want to carry forward, both of which are router instruments rather than gates.

△ **One housekeeping note on the brief.** The prompt I was handed says UNCLASS is "34 of the 60 sat-out runs and \$4,097 of the \$8,025 ceiling". That is last week's census. The frozen file for this week says **41 of the 62 sat-out runs and \$5,840 of the \$9,610 ceiling** — 61% of the money on the table, the biggest bucket again by a wide margin. Every number below is scored against the frozen file, not the brief.

0. THE ANCHOR — I REPRODUCED THE FROZEN CENSUS EXACTLY BEFORE TOUCHING ANYTHING

I did not re-run run_census.py. I re-implemented its algorithm against the parquet lake and checked it row for row against census_stdout.txt.

CHECK	RESULT
window	2026-08-17 00:00 → 08-21 20:59 UTC, 81,326 5s bars
typical 15-min range → threshold	37pt → 55pt — identical to the frozen file
runs found	68
rows matching frozen on time and direction and move	68 / 68
rows matching frozen on the cluster label	68 / 68
UNCLASS runs	41, all of them sat out, ceiling \$5,840

That is a full-strength anchor — every number that follows sits on the same run set the report's Movement 1 shows. (Last week's version of this phase reproduced

59 of 68 labels; the missing 9 were the bucket-phase problem, which I isolate deliberately in §1 instead of leaving it as noise.)

What the 41 runs are, before any hunting

SPLIT	BREAKDOWN
direction	24 DOWN / 17 UP
ASIA 00-07Z	11 runs · \$1,564
EUROPE 07-13Z	14 runs · \$2,048
US 13-20Z	12 runs · \$1,676
LATE 20-24Z	4 runs · \$552
median move	67 pt

1. ★ INTERROGATE THE LABEL FIRST. This outranks anything built on top of it.

Your standing rule, and it changes the whole question. `run_census.cluster()` is a four-branch waterfall and UNCLASS is the else clause:

```
|flow z| >= 1.0      (needs >= 30 one-minute tick buckets in the trailing 2h)
                    -> FLOW-LED if the flow sign agrees with the move, else
VACUUM
amp > 0.30 %        -> VOL-EXPANSION
13 <= hour < 15     -> OPEN/NEWS
otherwise           -> UNCLASS
```

I ran that waterfall on **every minute** of the census window, not just the 68 run minutes.

Base rate of every label, on all tape

Label	Minutes	% of All Tape	Share of the 68 Runs	Lift
UNCLASS	5,013	73.9%	41 = 60.3%	0.82 ←
OPEN/NEWS	368	5.4%	13 = 19.1%	3.52
VACUUM	701	10.3%	9 = 13.2%	1.28
FLOW-LED	677	10.0%	4 = 5.9%	0.59
VOL-EXPANSION	25	0.4%	1 = 1.5%	3.99

Across the whole tick-covered lake — 49,064 minutes, ten times the sample — it is the same picture: **UNCLASS 75.4%**, OPEN/NEWS 4.9%, a flow event on 18.0%, VOL-EXPANSION 1.7%.

UNCLASS is true on three quarters of the tape and is UNDER-represented among the runs. That is the whole finding and it outranks every candidate in this document. Last week's number was 74.9% / lift 0.76; mine is 73.9% / lift 0.82 on a different week with a re-derived implementation. Two independent measurements a week apart landing in the same place is about as settled as this desk gets.

And the label is decided by where a minute starts

This is new, and it is worse than the "13% of assignments" figure last week reported, because I isolated the cause instead of estimating it. I changed **one thing** — the phase of the 60-second flow buckets, wall-clock instead of anchored to each run's own timestamp — and left everything else byte-identical.

TEST	RESULT
run timestamps that are actually minute-aligned	17 / 68
labels that CHANGE on bucket phase alone	21 / 68 = 31%

The census slides its 15-minute window by array index, not by clock, so where the 5s tape has holes the two drift apart and the run lands at :10, :25, :55 past. Nearly a third of this week's cluster assignments are decided by an arbitrary implementation choice nobody made on purpose.

The clock edge is doing real work too

WHERE THE UNCLASS RUNS SIT	COUNT
in the hour on either side of the OPEN/NEWS window (12Z and 15Z)	10 / 41 (24%)
hour histogram	00:2 01:2 02:2 04:1 05:2 06:2 07:5 08:2 10:1 11:2 12:4 15:6 16:3 17:2 18:1 20:1 23:3

The 13:00–15:00Z fallback's right-hand edge is arbitrary — the hunt last week found the densest ignition minute is 13:25, not 13:30, so it is not a news-release footprint either. **One hour of slack in either direction moves 10 of these 41 runs out of this bucket without a single new measurement being taken.**

What "UNCLASS" actually means, minute by minute

REASON A MINUTE LANDS IN UNCLASS	CENSUS WINDOW	WHOLE LAKE
a flow verdict was available, and the flow was ordinary ($ z < 1$)	99.3%	85.6%
no flow verdict — fewer than 30 tick buckets in the trailing 2h	0.7%	14.4%

So on the report week UNCLASS means one thing almost exclusively: **the tape was not doing anything unusual with aggressor flow**. The thin-data path, which was a live hypothesis, is 0.7% of it.

And the members look like ordinary tape at their own start

This is the part that matters for the hunt. I measured the 41 members against the whole tape's distribution, at the run's first minute:

FEATURE AT THE RUN'S START	THE 41 UNCLASS RUNS	THE OTHER 27 RUNS	ALL-TAPE MEDIAN (THE NULL)
ATR (20-min, points)	11.65	24.15	12.31
5-min volume ÷ trailing-hour median	1.06	1.41	1.02
30-min efficiency ratio	0.153	0.178	0.163
pre-run amplitude %	0.08	0.19	—
flow z	0.37	1.00	—
direction: continuation of the prior 30-min drift	22 / 41	—	coin flip

The UNCLASS runs are, at the moment they begin, statistically indistinguishable from a random minute of tape. Their ATR sits below the all-tape median. Their volume is on it. Their efficiency is below it. And you cannot even call the direction — 22 continuation, 19 reversal.

That is not a failure of the census; it is the definition. The other three branches are the tests that find footprints, so the residue is by construction the runs with no footprint. **Hunting for a footprint inside UNCLASS is a category error.** Which is exactly why your instruction to RIDE first rather than classify first is the right one, and why I did not stop here.

2. STEP ONE — RIDE FIRST. Board it late, and measure what is left.

The rule is deliberately dumb and direction-agnostic: at each completed one-minute bar take the net move over the trailing **W** minutes; if it is at least **k × ATR20**, board it in whichever direction the move went. No prediction, no classification.

W	K	BOARDS INSIDE	MEDIAN MINUTES LATE	MEDIAN POINTS LEFT	MEDIAN ATRS LEFT	FIRES/DAY
---	---	---------------	---------------------	--------------------	------------------	-----------

3	1.5	41/41	2.2	36.0	3.10	145
3	2.5	28/41	7.5	26.4	2.10	25
5	2.0	40/41	3.3	31.2	2.57	137
5	2.5	34/41	4.4	27.5	2.20	68
5	3.0	25/41	5.5	20.2	1.79	33
10	2.0	40/41	4.7	27.2	2.15	285
15	2.5	41/41	5.8	25.5	1.77	277

STEP 1 PASSES, unambiguously. A five-minute boarding trigger gets inside **40 of the 41** runs at a median **3.3 minutes** late with a median **31 points / 2.57 ATRs** still ahead. Boarding is not the problem and never was. On gold it was 19 minutes in with 80% left; on MNQ it is three minutes in with two and a half ATRs left, which is better.

The trouble is in the right-hand column. **That same trigger fires 137 times a day.** After the cooldown of actually being in a trade it settles to about 27 entries a day at k=2.0, or 5 a day at k=2.5 with the filter of \$4 on. The gap has never been detection. It is selection.

And the excursion profile inside a run is superb

Boarding at the first W5/k2.0 thrust inside each run and then measuring the path:

HORIZON AFTER BOARDING	N	MEDIAN MFE	MEDIAN MAE	PERFECT-EXIT TOTAL
to the run's own 15-min end	37	44.2 pt	6.0 pt	\$3,239
+60 minutes	39	60.2 pt	13.8 pt	\$5,633
+240 minutes	38	99.0 pt	37.9 pt	\$8,660

Median MAE of 6.0 points against an ATR of 11.65 is **half an ATR** — and **0 of the 40 boarded runs touched a 2.5×ATR stop before reaching their best excursion inside the run window.** Once you are on one of these, a wide stop is never threatened. The entire problem is the fires that are not on one.

The prize, restated honestly

\$	% OF THE CENSUS HEADLINE	
census UNCLASS ceiling (close-to-close, hindsight, 1 lot)	\$5,840	100%
a cheating oracle — boards every run, exits at the exact top, no costs	\$3,239	55%
honest-ish — same boards, 2.5×ATR stop, bank half the excursion	\$1,592	27%

Two thirds of the number on the census page cannot be collected by anything that enters after the move has proven itself. When you see "\$5,840 on the table" in Movement 1, the rideable part of it is about sixteen hundred dollars.

3. STEP TWO — THE EXIT, SWEEPED WIDE BEFORE ANY VERDICT ON THE ENTRY

Your standing finding is that a tight or trailing exit gets shaken out of a move that grinds. I swept it properly — stop 1.0–4.0×ATR against target 2–16×ATR, caps 30 minutes to 12 hours — and I did it on segments, not blanket.

Blanket, all tape, W5/k2.0, \$/trade — this is what a bug looks like

STOP \ TARG	2	3	4	6	8	10	12
1.5	–2.04	–0.27	–1.04	–1.21	1.12	1.51	1.52
2.0	–1.31	–0.37	–1.02	–0.52	–1.48	1.22	0.92
2.5	–0.85	1.29	–0.06	–0.93	–1.67	3.45	0.76
3.0	–0.89	–0.97	–2.14	–1.59	–3.06	–0.73	–8.44
4.0	–0.87	–4.57	–2.51	–5.89	–6.71	–6.16	–8.01

Blanket over 1,403 trades the rider makes **–\$1,308, –\$0.93 a trade**. Sprayed across every regime it is a coin. Exactly as your backtest discipline predicts, that number is not a verdict on anything — it is the average of the regimes it should be on with the ones it should be off.

Segmented, and the segments are not equal

BY SESSION (BLANKET RIDER)	N	NET\$	WIN%	\$/TR
ASIA 00–07	433	+1,195	32.6%	+2.76
US 13–20	437	–588	33.2%	–1.34
LATE 20–24	139	–720	29.5%	–5.18
EUROPE 07–13	394	–1,196	29.7%	–3.03

BY 30-MIN EFFICIENCY AT BOARD TIME	N	NET\$	WIN%	\$/TR	PF
trend ER ≥ 0.40	196	+4,115	38.3%	+20.99	1.40
mid 0.20–0.40	454	–1,227	30.0%	–2.70	0.94
chop ER < 0.20	730	–4,117	30.7%	–5.64	0.89

△ AND HERE IS A TRAP I WALKED INTO, WHICH IS WORTH MORE THAN THE TABLE ABOVE

That +\$20.99 a trade is **not achievable and the table above is misleading**. It is a post-hoc slice of a trade list produced by an unfiltered rider. A rider that actually refuses the low-ER fires is not busy during them — it is free, so it takes entries the sliced version never had. Run the identical filter inside the simulation and:

THE SAME ER ≥ 0.40 FILTER	N	NET\$	WIN%	\$/TR	PF
as a post-hoc SLICE (unimplementable)	154	+4,363	40.3%	+28.33	1.53
run INSIDE the simulation (what you could trade)	317	+1,801	33.1%	+5.68	1.10

The number more than halves and n doubles. Every filtered figure in the rest of this document is the in-sim form. I am flagging it loudly because slicing a trade list is the natural way to test a filter, it is what the tables above do, and it is wrong by a factor of five.

The exit sweep on the home segment, in-sample, W5/k2.5/ER≥0.45 — \$/trade (n)

STOP \ TARG	3	4	6	8	10	12	16
1.0	7.05 (198)	8.21 (186)	8.98 (177)	10.77 (173)	9.32 (167)	6.96 (166)	10.41 (161)
1.5	13.27 (194)	14.53 (180)	14.91 (169)	26.59 (163)	31.97 (157)	29.01 (154)	25.84 (151)
2.0	15.73 (188)	14.62 (173)	17.38 (161)	29.10 (152)	38.04 (145)	27.62 (144)	25.32 (140)
2.5	14.57 (187)	14.90 (170)	17.94 (157)	30.05 (146)	37.05 (139)	23.37 (136)	18.05 (133)
3.0	8.69 (184)	8.95 (168)	10.33 (154)	25.32 (142)	38.14 (135)	22.71 (132)	17.58 (130)
4.0	10.44 (180)	13.95 (165)	20.29 (151)	35.10 (138)	42.55 (130)	35.54 (128)	28.12 (126)

Your wide-exit thesis is CONFIRMED on the target axis and REFUTED on the stop axis. There is a genuine ridge at a **10×ATR target** that holds across every stop width tested — that is a plateau, not a point, and it is the single most robust thing in this section. The stop width barely matters: 1.5, 2.0, 2.5 and 4.0 all land in the same place, which is exactly what you would expect given the median MAE inside a run is half an ATR. So the "wide stop" half of the finding does not apply here — the stop is not what is shaking us out, because we are not being shaken out.

Managed exits, on the same segment, all lost to the flat far target:

EXIT MECHANISM	N	NET\$	WIN%	\$/TR	PF
flat stop 3 / target 8	369	-2,467	27.1%	-6.69	0.90
chandelier trail 3.0×ATR	395	+1,454	34.9%	+3.68	1.10
chandelier trail 2.0×ATR	474	-1,777	30.8%	-3.75	0.84
breakeven stop at +2×ATR	400	-845	13.8%	-2.11	0.94
tight stop 1.5 / target 3	510	+926	33.9%	+1.82	1.06

Time cap sweep on the same cell: 30 min +\$3.80/tr, 60 min +\$3.92, 120 min – \$5.06, 240 min –\$6.69, 720 min –\$4.45. Nothing monotone, nothing to build on.

4. STEP THREE — THEN FILTER, WITH A PLACEBO ON EVERY SINGLE CUT

Every cut below is scored against **dropping the same NUMBER of fires at random**, 400–500 draws, exactly as you specified. Unfiltered baseline in-sample: n=688, +\$3,885, **+\$5.65/tr**, PF 1.11.

CAUSAL CUT	N	NET\$	\$/TR	KEEPS	RANDOM-CUT P95	P	VERDICT
ER30 ≥ 0.40 (trending tape)	196	+6,211	+31.69	28%	+20.99	0.006	BEATS placebo
ER30 ≥ 0.40 AND ATR ≥ 12pt	138	+5,278	+38.25	20%	+25.19	0.012	BEATS placebo
ER30 ≥ 0.40 AND volume ≥ 1.5×	120	+3,754	+31.28	17%	+27.87	0.030	BEATS placebo
ATR ≥ 12pt	478	+4,355	+9.11	69%	+12.72	0.240	~placebo
ER30 ≥ 0.30	348	+2,728	+7.84	51%	+15.51	0.386	no better than random
volume ≥ 1.5×	420	+3,211	+7.64	61%	+13.67	0.338	no better than random
(volume confirms)							
ATR ≥ 16pt (loud tape only)	298	+2,327	+7.81	43%	+16.20	0.402	no better than random
volume ≤ 0.9×	213	+732	+3.44	31%	+19.25	0.600	no better than random
(thin, no aggression)							

The thin-volume cut deserves a note because it was my own best causal hypothesis and the data refused it. The members move on ordinary volume with no aggression, so "price is walking because nobody is there" was the obvious story — and boarding only the thin-volume fires is **worse than throwing away**

the same number of fires by coin flip. That hypothesis is dead on its own evidence.

One cut passes with room to spare: **efficiency**. So the candidate is built on it.

5. THE CANDIDATE — UNCL-RIDE-ER

Chosen from the **interior of the in-sample shelf, not its peak**. The sweep's best cell was $ER \geq 0.45$ / $k=3.0$ at +\$42.68 a trade; I took the shelf's middle instead, because a boundary optimum is a curve-fit tell and I would rather report a smaller honest number.

board — the 5-minute net move is $\geq 2.5 \times ATR20$ and the 30-minute efficiency ratio is ≥ 0.40 **side** — the sign of that move. No prediction, no classification, no direction view. **entry** — the OPEN of the next 1-minute bar (no same-bar fills, no look-ahead) **stop** — $2.0 \times ATR20$ **target** — $10 \times ATR20$ **cap** — 240 minutes **costs** — \$2.00/point, **\$1.50 per round trip**; a bar that touches stop and target in the same 5 seconds is scored as a STOP

Plateau checks — is this a shelf or a cliff?

ER30 GATE	0.00	0.10	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.70
\$/trade	6.05	12.22	4.49	0.32	7.84	22.88	31.69	38.04	32.75	5.02	−27.20	−67.57
n	687	586	495	429	348	270	196	145	105	66	38	6

BOARDING WINDOW W W	2	3	4	5	6	8	10	15	20
\$/trade	18.39	33.07	38.65	38.04	19.95	8.55	2.40	−0.76	−2.32

THRUST K	1.5	1.75	2.0	2.25	2.5	2.75	3.0	3.5	4.0
\$/trade	9.57	14.62	11.93	25.22	38.04	39.33	42.68	27.43	35.15

Honest read: **W and k are clean shelves. ER is not**. It sags at 0.20–0.30 (4.49, 0.32, 7.84), rises to a peak at 0.45 and falls off a cliff after 0.50. A non-monotone parameter with a peak two cells wide is a warning I recorded before spending the out-of-sample leg — and the leg then confirmed it.

The full battery, in-sample (2026-06-19 → 08-08, 36 sessions)

Headline: n=196, +\$6,211, 25.0% wins, +\$31.69 a trade, PF 1.64. A 25% win rate is not a kill criterion here; the exit mix is 37 targets (+\$13,067), 12 time caps (+\$2,888), 147 stops (−\$9,744) — a textbook few-big-winners shape.

TEST	RESULT	PASS?
placebo — ER filter vs a random cut of the same size	actual +\$31.69 vs random mean +\$6.09, p95 + \$20.99 · p = 0.006	☐
placebo — signal shifted 5 min early	−\$4.33/tr	☐
placebo — signal shifted 10 min early	−\$5.16/tr	☐
placebo — signal shifted 30 min early	+\$19.67/tr	△ weak
placebo — signal shifted 60 / 120 min early	−\$8.94 / +\$2.00	☐
placebo — direction flipped	−\$3.97/tr	☐
strip the best 1 / 3 / 5 trades	+\$26.79 / +\$17.67 / +\$9.52	☐
strip the best 10 trades	−\$3.97/tr	△ ten trades of 196 are the whole result
leave-one-day-out, worst	+\$26.53/tr (dropping 2026-06-25)	☐
sessions green	18 / 36 = 50%	△
long / short symmetry	long +\$24.65 (n=91) · short +\$37.79 (n=105) — both green	☐
cost stress: fee ×3	+\$28.69/tr	☐
cost stress: 1 full point of slippage per side	+\$27.69/tr	☐
in-sample first half (06-19→07-14)	+\$36.04/tr (n=103)	☐
in-sample second half (07-15→08-08)	+\$26.88/tr (n=93)	☐

Both halves of the in-sample period are strongly green. Nothing here says stop.

In-sample segments

SESSION	N	NET\$	WIN%	\$/TR	PF
US 13-20	59	+6,360	39.0%	+107.79	2.91
LATE 20-24	19	+1,231	42.1%	+64.81	2.27
ASIA 00-07	66	+81	18.2%	+1.23	1.03
EUROPE 07-13	52	−1,461	11.5%	−28.09	0.44

ATR LEVEL	N	NET\$	\$/TR	PF
Q4 loud	70	+5,089	+72.70	2.02
Q1 quiet	30	+607	+20.23	1.79
Q2	44	+295	+6.70	1.19
Q3	52	+221	+4.25	1.09

6. SPENDING THE OUT-OF-SAMPLE LEG — and it goes red

2026-08-09 → 08-24, 11 sessions, untouched until this moment.

	N	NET\$	WIN%	\$/TR	PF
in-sample	196	+6,211	25.0%	+31.69	1.64
out of sample	71	-680	15.5%	-9.58	0.75
OOS strip-3	68	-1,558	11.8%	-22.92	0.42
OOS sessions green	4 / 11				
OOS placebo (signal 30 min early)	49	+73	20.4%	+1.48	1.05

The fake signal beats the real one out of sample. The OOS session split does not rescue it either — ASIA +\$11.91 (n=20), LATE -\$2.31 (n=4), US -\$18.62 (n=23), EUROPE -\$20.04 (n=24) — and note that **US 13-20 was the in-sample star at +\$107.79 and is the OOS loser.**

The second, much larger independent leg — 205 sessions back to 2025-09-14

Eleven sessions is thin, so I spent a second leg that is not thin: the 1-minute lake, every session before the 5s tape begins, none of which contributed to a single choice above. Path resolution here is 1-minute OHLC and a bar touching both levels is scored as a stop, so this leg is conservative, not optimistic.

	N	NET\$	WIN%	\$/TR	PF
UNCL-RIDE-ER on the untouched year	1,246	+4,417	18.7%	+3.54	1.09
strip the best 3				+1.25	
strip the best 10				-2.35	
sessions green	93 / 205 = 45%				
placebo — signal 30 min early	725	+3,222	17.2%	+4.44	1.13
placebo — signal 120 min early	704	-5,373	14.2%	-7.63	0.78
placebo — direction flipped	1,405	-1,801	16.7%	-1.28	0.97

On the largest honest sample I have, a signal taken thirty minutes early makes more money than the real signal. Ten trades out of 1,246 carry the entire result. Monthly it is noise: -8.14, +14.60, +0.16, +1.35, +6.42, +4.84, -0.05, +13.01, -4.62, +0.38.

That is the cause of death, and it is a robustness test, not a win rate.

The final arbiter — is ANY configuration green on both independent legs?

I swept 137 configurations ($W \times k \times ER \times stop \times target$) with $n \geq 150$ on the year leg and $n \geq 20$ on the recent OOS leg, and asked how many are positive on both.

COUNT	SHARE
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positive on the 205-session YEAR leg	77	56%
positive on the 11-session recent OOS leg	29	21%
positive on BOTH	18	13%
what pure chance predicts if the two legs are independent	—	12%

Thirteen per cent against a chance rate of twelve. There is no configuration of this rider that survives two independent samples any more often than a coin would put it there. That is the cleanest kill in this document and it is not about any one parameter choice — it retires the whole family, not just the cell I picked.

7. THE DIAGNOSIS — which half of the rule was ever doing anything?

The shifted placebo beating the real signal has a specific meaning worth chasing: efficiency is persistent, so a signal taken 30 minutes early still lands on trending tape. So I decomposed it.

ARM	YEAR LEG (205 SESSIONS)	RECENT 5S LAKE (51 SESSIONS)
A thrust trigger + ER ≥ 0.40 (the candidate)	n=1,246 · +\$3.54/tr	n=267 · +\$20.72/tr
B same, signal shifted 30 min	n=725 · +\$4.44/tr	n=192 · +\$15.03/tr
C NO trigger at all — ER ≥ 0.40 , direction = sign of the 30-min move	n=2,326 · +\$3.05/tr	n=541 · -\$6.13/tr
D arm C with the direction flipped	n=2,970 · -\$2.55/tr	n=702 · -\$0.62/tr
E arm C without the ER gate (every minute)	n=5,205 · +\$0.05/tr	n=1,193 · -\$1.31/tr
F thrust trigger, NO ER gate	n=3,238 · -\$3.40/tr	n=916 · +\$3.90/tr

The two legs give opposite answers. On the year leg the thrust trigger is worthless (F is negative) and the ER regime is everything ($C \approx A \approx B$, and D is its clean mirror). On the recent leg the ER regime alone loses money (C is negative) and the trigger is what earns (F positive). Neither component is stable, and a rule whose own decomposition inverts between two samples is measuring the sample, not the market.

8. THE ESCALATION — the size threshold, which is the finding you asked for

Per the brief I escalated: full sat-out set → top-25 → top-15 → top-10 → top-5. **The escalation runs flat, and on the one axis that matters it runs BACKWARDS.**

Does a footprint appear as the runs get bigger? (AUC vs the whole tape; 0.50 = no discrimination)

CUT	N	MEDIAN MOVE	ATR	VOLUME RATIO	ER30	5-MIN THRUST
all 41 UNCLASS	41	67 pt		0.444	0.517	0.562
top-25 by size	25	78 pt		0.476	0.518	0.558
top-15 by size	15	83 pt		0.419	0.572	0.532
top-10	10	90 pt		0.486	0.535	0.513
top-5	5	96 pt		0.515	0.633	0.645
≥ 80 pt	12	88 pt		0.428	0.571	0.519

Every AUC sits on 0.5, and the "5-min thrust" column falls to 0.28–0.33 as the runs get bigger — meaning **the biggest UNCLASS runs are LESS preceded by a thrust than an average minute of tape**. The top-5 shows a volume AUC of 0.633 on n=5, which is five observations and not a finding.

Where the money actually is, once you are on a run

Building the census's run algorithm across the whole 5s lake (608 runs on 51 sessions, 11.9/day, median 100pt) and tagging every rider fire by the run it boarded:

	N	NET\$	\$/TR
fires that landed inside an aligned run	116 (43%)		
			+\$6,157
fires on tape that never became a run	151 (57%)		
		-\$626	-\$4.15

RUN SIZE THE FIRE BOARDED	N	NET\$	WIN%	\$/TR	PF
55–70 pt	12	−194	16.7%	−16.12	0.57
70–85 pt	17	+8	23.5%	+0.47	1.01
85–100 pt	12	+668	41.7%	+55.63	2.57
100–130 pt	22	−82	22.7%	−3.75	0.93
≥ 130 pt	45	+5,695	42.2%	+126.56	3.12

Cumulatively: ≥55pt +\$56.43/tr, ≥70pt +\$65.50, ≥90pt +\$77.27, ≥120pt +\$109.81. **The size threshold is real and it is around 130 points on MNQ** — below that the runs pay for themselves and nothing more; above it they carry the entire book.

But that is the PRIZE, not a RULE, and here is why. The rider cannot know the run's eventual size at the moment it boards. The obvious counter-argument is that it can: ATR at board time correlates +0.719 with eventual run size in points,

and the 5-minute move correlates +0.851. Those numbers are a scale artefact and I nearly filed them as a finding. Run size is measured in points and so is volatility, so a loud day produces big runs by arithmetic. Normalise the run size by the ATR at board time and the correlations collapse:

BOARD-TIME READ	VS RUN SIZE IN POINTS	VS RUN SIZE IN ATRS
ATR	+0.719	−0.221
5-min move	+0.851	+0.133
5-min move ÷ ATR	+0.239	+0.477
ER30	+0.336	+0.045

The only surviving read is the thrust size expressed in ATRs, at $r=+0.48$ — and that is already the k parameter, whose sweep in \$5 shows no usable gradient.

There is a size at which these runs are worth catching. There is no instrument on this desk that identifies it at board time.

Big-moves-caught on the frozen census week: **UNCL-RIDE-ER boards 14 of the 41 UNCLASS runs, and its trades on that week are $n=33$, −\$209, −\$6.33 a trade.**

9. CLOSING THE LOOP — last week's survivor was never shipped, and it is now four weeks red

Last week this phase sent **UNCL-RIDER-ASIA** to SHADOW: thrust board $w=10$ / $k=2.0$, 00:00–06:00 UTC, $ATR \geq 8pt$, $ATR \text{ percentile} \leq 0.60$, stop $2.5 \times ATR$, target $6 \times ATR$, 2-hour cap, minus the 04:15–05:15Z gateway outage. Quoted at $n=157$, +\$5,006, +\$31.88 a trade.

There is no arm by that name in shadow.db. I listed all 82 strategies in the shadow book; the only rider shipped from last week's greenfield work is rider_w5 from the pooled phase, which the sibling RIDER_ALL section grades at −\$1,471. So the SHADOW recommendation from this phase was made and not executed. **A disposition that nobody actions is a disposition that cannot teach us anything, and this is the second week running that a "put it in shadow" line has gone nowhere.**

Since nothing recorded it, I replayed it myself across the whole 5s lake. (My absolute numbers are smaller than last week's — I have no V5 leg here and I had

to re-derive the ATR-percentile definition — so read the shape, not the totals. The shape is unambiguous.)

ISO WEEK	DATES	N	NET\$	WIN%	\$/TR
26	06-23 .. 06-26	25	+233	36.0%	+9.32
27	06-29 .. 07-03	30	+751	46.7%	+25.02
28	07-07 .. 07-10	25	+510	40.0%	+20.38
29	07-13 .. 07-17	24	+213	37.5%	+8.88
30	07-20 .. 07-24	25	+368	40.0%	+14.73
31	07-27 .. 07-31	29	−85	27.6%	−2.93
32	08-03 .. 08-07	25	−398	24.0%	−15.90
33	08-10 .. 08-14	15	−547	6.7%	−36.46
34	08-19 .. 08-21	14	−533	14.3%	−38.08
35	08-24	2	+61	50.0%	+30.53

N	NET\$	WIN%	\$/TR	
up to 08-15 — everything last week could see	198	+1,045	33.8%	+5.28
since 08-17 — pure forward tape	16	−472	18.8%	−29.51

First five weeks +\$2,075 on 129 trades. Last five weeks **−\$1,502 on 85 trades**. The forward week is the worst of the lot. And it was never an answer to this bucket anyway: **it boards 3 of the week's 41 UNCLASS runs**.

UNCL-RIDER-ASIA is REFUTED by the forward out-of-sample leg — not parked, not thin-n. It had its shelf, the shelf kept going down for four straight weeks, and the ATR floor that was added specifically to explain the first two red weeks did not restore any of them.

10. THE RENAMING DELIVERABLE — four shapes the census could actually test for

You asked that if the bucket splits into real shapes I name them so the census can classify them next week. It does split, on axes the census already computes. Here are the four, with definitions that are one if statement each.

PROPOSED SHAPE	N	CEILING	MEDIAN MOVE	MEDIAN ATR	MEDIAN VOL RATIO	MEDIAN ER30	DN/UP
TURN	19	\$2,716	67 pt	12.5	1.06	0.142	11/8
STEP-REPRICE	10	\$1,465	66 pt	12.3	2.06	0.308	6/4
DRIFT-VOID	6	\$797	62 pt	9.9	0.75	0.091	4/2

GRIND-CONT	6	\$862 70 pt	11.4	0.82	0.267 3/3
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- **STEP-REPRICE** — 5-minute volume $\geq 1.5\times$ the trailing-hour median but $|\text{flow } z| < 1$. Volume arrives and aggression does not: a two-sided repricing, not a sweep. This is the one shape with a genuinely distinctive signature and it is 10 runs / \$1,465.
- **GRIND-CONT** — the run continues the previous 30-minute drift and $\text{ER}_{30} \geq 0.20$. On below-baseline volume, which is what makes it a grind rather than a thrust.
- **DRIFT-VOID** — volume below baseline ($< 0.95\times$) and $\text{ATR} < 12\text{pt}$. Price walks on nothing.
- **TURN** — everything else: a reversal on ordinary volume. Still the biggest half at 19 runs, and still genuinely featureless.

⚠ **Do not read the ride results per shape.** I ran them and they are $n = 8, 3, 3$ and 1 trades respectively. Quoting +\$224/trade off three trades would be exactly the kind of number this report exists to refuse. The shapes are offered as **census vocabulary**, not as a selector.

11. THE ROUTER — what I would arm, and what I would need to arm it

You are right that a gate only has to work in the regime it is armed for, so I am not allowed to kill this on the blanket average. Here is the router work, honestly.

The one bench rule that replicates on all three independent legs

EUROPE 07-13Z, ER-GATED RIDER	N	NET\$	\$/TR	PF
in-sample (36 sessions)	53	-1,557	-29.38	0.42
recent OOS (11 sessions)	24	-481	-20.04	0.44
year leg (205 sessions)	373	-905	-2.43	0.93

BENCH any late-boarding continuation rider between 07:00 and 13:00 UTC. Red on every leg, including the 373-trade one. It is the only piece of the session structure that replicates.

And the session rule that does NOT replicate — stated so nobody ships it

SESSION	IN-SAMPLE \$/TR	RECENT OOS \$/TR	YEAR-LEG \$/TR
ASIA 00-07	+4.61 (67)	+11.91 (20)	+1.67 (405)

EUROPE 07-13	-29.38 (53)	-20.04 (24)	-2.43 (373)
US 13-20	+107.79 (59)	-18.62 (23)	+12.76 (404)
LATE 20-24	+48.71 (22)	-11.62 (6)	-5.45 (128)

Spearman rank correlation of the session ordering: **IS vs OOS +0.20 · IS vs YEAR +0.40 · OOS vs YEAR +0.00**. In-sample this rider begs to be armed US-only at +\$107.79 a trade with a random-cut placebo p below 0.001. Out of sample that is its worst session. **This is precisely the router rule I would have shipped last Friday if I had stopped at the in-sample battery, and it would have been wrong.**

The ER meter as an arm/bench instrument — real, but it does not replicate forward either

SAME RIDER, ONLY THE ARMING DIFFERS	ARMED ALWAYS	ARMED ONLY WHEN ER30 ≥ 0.40
year leg, 205 sessions	n=3,238 · -\$3.40/tr	n=1,246 · +\$3.54/tr
recent OOS, 11 sessions	n=228 · -\$1.37/tr	n=71 · -\$9.58/tr

On 3,238 trades of untouched tape the efficiency meter turns a losing rider into a marginally positive one — that is a large sample and a real effect, and it is the only thing in this document with that kind of n behind it. On the most recent eleven sessions it makes things worse. I am reporting both and recommending neither as a deploy.

The instrument we would actually need

The binding constraint is named precisely in §8: **a SIZE PREDICTOR at board time, in ATR units**. Every board-time read we own — ATR, volume, efficiency, thrust size, and (per last week's L2 work, which I did not re-dig) depth imbalance — correlates with a run's eventual size in ATRs at $r \leq 0.13$, except thrust-in-ATRs at +0.48, which is already a swept parameter with no gradient. Until something answers "given the volatility, is this going to be a 3-ATR move or a 12-ATR move", this rider risks a 2-ATR stop to collect a median 2.57 ATRs and the arithmetic never closes. **That is a missing instrument, not a failed strategy, and it is what the next attempt should go and build.**

12. EVERY ATTEMPT, BY NAME, INCLUDING THE GRAVES

Nothing here was quietly dropped. All scored identically: \$2.00/point, \$1.50/round-trip, 5s path resolution on the two recent legs, 1-minute on the year leg.

#	CANDIDATE	IS N	IS \$/TR	STRIP-3	OOS N	OOS \$/TR	YEAR N	YEAR \$/TR	CAUSE OF DEATH
1	UNCL-RIDE-ER (the candidate)	196	+31.69	+17.67	71	−9.58	1,246	+3.54	placebo — shifted signal beats it on both independent legs; OOS leg red
2	UNCL-RIDE-RAW (no filter)	688	+5.65	+1.86	228	−1.37	3,238	−3.40	negative on the 3,238-trade year leg
3	UNCL-RIDE-ER30 (looser gate)	348	+7.84	−0.21	122	−2.53	1,958	+0.97	strip-3 kills it in-sample; fails the random-cut placebo (p=0.386)
4	UNCL-FADE (flip the thrust — fade, don't ride)	262	−3.97	−13.89	86	+2.15	1,405	−1.28	red in-sample and on the year leg; the fade is not the answer
5	UNCL-TIGHT (stop 1.5 / targ 3 / cap 60)	272	+11.18	+7.57	90	−3.52	1,683	−2.50	year leg — confirms the tight exit is shaken out
6	UNCL-CHANDELIER (trail 3×ATR)	212	+17.02	+4.99	77	−5.67	1,350	−3.35	year leg; trailing loses to the flat far target
7	UNCL-THIN (volume ≤ 0.9× — the "price walks on nothing" hypothesis)	213	+3.44	−3.49	72	−1.91	n/a	n/a	random-cut placebo, p = 0.600 — worse than a coin flip
8	UNCL-VOLCONF (volume ≥ 1.5×)	420	+7.64	+1.81	149	+3.81	n/a	n/a	random-cut placebo, p = 0.338
9	UNCL-LOUD (ATR ≥ 16pt)	298	+7.81	−0.96	27	+27.06	n/a	n/a	random-cut placebo p = 0.402; strip-3 negative
10	UNCL-SLOW (W=15 boarding)	365	−0.74	−8.48	114	−18.03	1,876	+4.72	red in-sample AND out of sample
11	UNCL-FAST (W=3, k=3.0,	62	−0.21	−27.64	26	+18.39	486	+8.45	strip-3 — three trades

	stop 3 / targ 8)							are the entire result
12	UNCL- NOEUROPE (bench 07- 13Z)	144	+53.28	+34.54	47	-4.24	893	+5.93 OOS leg
13	UNCL-US (arm 13-20Z only)	59	+107.79	+63.53	23	-18.62	404	+12.76 OOS leg — the in- sample star is the OOS loser
14	UNCL- RIDER-ASIA (last week's SHADOW pick)	198	+5.28 —		16	-29.51 —	—	forward OOS — four consecutive red weeks

⚠ **Row 14 uses different leg boundaries from rows 1-13.** Its "IS" column is everything up to 2026-08-15 (what last week could see) and its "OOS" column is the pure forward tape from 08-17, which is the only split that means anything for a candidate that was published last Friday. Rows 1-13 use the fixed 06-19→08-08 / 08-09→08-24 split chosen before any of them were scored.

Candidates green on all three legs with usable n: ZERO.

Two more nulls that did not earn a row because they are not candidates:

- **the post-hoc-slice method itself** — inflated the ER filter's expectancy from +\$5.68 to +\$28.33 a trade, a factor of five, and it is how a filter naturally gets tested. Recorded as a tooling finding, not a strategy.
- **VOL-EXPANSION as a census branch** — last week it fired on 0.03% of tape and was called dead code. This week it fires on 0.4% of the census window and 1.7% of the lake, so it is no longer dead, but at n=1 run out of 68 it is still not classifying anything. Recalibrate to a percentile rather than a fixed 0.30% amplitude.

13. DISPOSITION

ITEM	DISPOSITION	REQUIRED
UNCLASS as a cluster label	REFUTED as a footprint	Base rate 73.9% (census week) / 75.4% (whole lake), lift 0.82 , and its members sit ON the all-tape median for ATR, volume and efficiency. Rename it UNTESTED and report it as coverage, not as a finding. Second independent measurement in two weeks.
the census's flow-bucket phase	FIX — this is a real bug	21 of 68 labels change on bucket phase alone; only 17 of 68 run timestamps are minute-

aligned. Snap the window slide to the clock, not the array index.

the OPEN/NEWS 15:00Z right edge	MEASURE IT	10 of the 41 UNCLASS runs sit in the hour either side of the window. One hour of slack reassigns a quarter of this bucket with no new evidence.
VOL - EXPANSION branch	RECALIBRATE	fires on 0.4-1.7% of tape at a fixed amp > 0.30%; make it a percentile
UNCL-RIDE-ER	PARKED	Revive only with a board-time SIZE PREDICTOR expressed in ATRs ($r \geq 0.4$ against eventual run size in ATR units on ≥ 300 runs). More days will not fix it — I already spent 205 untouched sessions and the placebo won.
UNCL-RIDER-ASIA (last week's SHADOW pick)	REFUTED	Forward out-of-sample: W31 −\$2.93, W32 −\$15.90, W33 −\$36.46, W34 −\$38.08 . Four consecutive red weeks and the ATR floor added to explain the first two restored none of them. Do not re-shadow.
the thin-volume / no-aggression mechanism	REFUTED	random-cut placebo $p = 0.600$ — the cut is worse than discarding the same number of fires by coin flip
the tight / trailing exit family	REFUTED for this shape	negative on the 1,350-1,683-trade year leg; the 10×ATR target ridge holds across every stop width
BENCH 07:00-13:00Z for any late-boarding rider	SHADOW → adopt as a router rule	red on all three legs (−\$29.38 / −\$20.04 / −\$2.43, the last on $n=373$). Cheap, fail-safe direction, no new instrument needed.
ER30 as a router ARM meter	SHADOW as an instrumented question	+\$3.54 vs −\$3.40 a trade on 3,238 untouched trades — the largest- n effect in this section — but it reverses on the recent 11 sessions. Shadow the meter with its own control arm, do not arm a gate behind it.
the "arm US 13-20Z" rule	REFUTED before it shipped	+\$107.79/tr in-sample (placebo $p < 0.001$), −\$18.62/tr out of sample, session rank correlation IS-vs-OOS +0.20
the \$5,840 UNCLASS ceiling	RESTATE	perfect-exit oracle \$3,239 (55%); honest late board with a real stop \$1,592 (27%)
the four shapes (TURN / STEP-REPRICE / DRIFT-VOID / GRIND-CONT)	SHIP TO THE CENSUS	one if each, all computable at the run's start minute. Not a selector — vocabulary.

One stone still unturned: whether the ER meter's forward reversal is a genuine regime break or an eleven-session artefact. That is answerable for free by shadowing the meter with its control arm and waiting, and it is the single cheapest question left in this bucket.

VERDICT

VERDICT: THE UNCLASS RIDER DID NOT SURVIVE. It was BUILT (a direction-agnostic late boarder that gets inside 40 of the 41 runs at a median 3.3 minutes late with 2.57 ATRs still ahead), RIDDEN (exit swept stop 1.0-4.0×ATR against target 2-16×ATR and caps 30 min-12 h, finding a real 10×ATR target ridge that holds across every stop width), FILTERED

(eight causal cuts each placebo-controlled against discarding the same NUMBER of fires at random; ER30 ≥ 0.40 beat its placebo at $p = 0.006$ and the thin-volume, volume-confirm, loud-tape and looser-ER cuts all failed theirs and are named in the ledger), and ROUTED (arm/bench rules written in the live router's ER / ATR / session vocabulary) — your order of work, in your order — and it STILL LOST. THE TEST THAT KILLED IT IS THE PLACEBO / SHUFFLE TEST, corroborated by the OUT-OF-SAMPLE LEG: on 205 untouched sessions the real signal makes +\$3.54 a trade and the SAME rule taken thirty minutes early makes +\$4.44, with ten trades out of 1,246 carrying the whole result and only 45% of sessions green; on the 11-session forward leg the real signal makes −\$9.58 a trade and the shifted fake makes +\$1.48. The family kill is stronger than the single cell: across 137 configurations with usable n on both independent legs, 13% are positive on both against a 12% chance rate, so no setting of this rider replicates any better than a coin. THE LABEL ITSELF IS THE PHASE'S PRIMARY FINDING AND IT IS REFUTED AS A FOOTPRINT — UNCLASS is true on 73.9% of the census window and 75.4% of the whole lake, takes 60.3% of the runs for a LIFT OF 0.82, its members sit ON the all-tape median for ATR (11.65 vs 12.31), volume (1.06 vs 1.02) and efficiency (0.153 vs 0.163), their direction is a 22/19 coin flip, and 21 of the 68 cluster assignments change on nothing but where a sixty-second bucket starts — rename it UNTESTED and report it as coverage. THE BANNED VERDICT IS NOT BEING USED: these runs ARE boardable (40/41) and I have NAMED four shapes (TURN 19, STEP-REPRICE 10, DRIFT-VOID 6, GRIND-CONT 6) for next week's census. The reason they cannot be ridden for money is priced, not asserted: the \$5,840 headline is worth \$3,239 to a cheating oracle and \$1,592 to an honest late board, 57% of fires never become a run, and the size threshold at which a boarded run becomes properly tradeable is ~130 POINTS on MNQ (+\$126.56/trade above it versus −\$16.12 in the 55-70pt band) — which is a PRIZE and not a RULE, because every board-time instrument we own correlates with eventual run size in ATR units at $r \leq 0.13$ once the volatility scale artefact is removed (the tempting +0.72 ATR and +0.85 thrust correlations are scale, and in ATR units they are −0.22 and +0.13). SEPARATELY AND IMPORTANTLY: last week's survivor UNCL-RIDER-ASIA was sent to SHADOW and NEVER SHIPPED — no arm of that name exists in shadow.db — and replayed forward it is REFUTED by four consecutive red ISO weeks (−\$2.93, −

\$15.90, −\$36.46, −\$38.08), with the pure forward week since 08-17 at −\$472 over 16 trades (−\$29.51/trade, 18.8% wins); it also boards only 3 of this week's 41 runs and was never an answer to this bucket. TWO THINGS GO FORWARD, both ROUTER INSTRUMENTS rather than gates: BENCH 07:00-13:00Z for any late-boarding rider (red on all three legs including n=373), and SHADOW the ER30 arm-meter with its own control arm (+\$3.54 vs −\$3.40 a trade on 3,238 untouched trades, but reversing on the last eleven sessions — that reversal is the one cheap question left). THE NEXT ATTEMPT NEEDS A BOARD-TIME SIZE PREDICTOR IN ATR UNITS, not more days, not a wider exit, and not a new cluster label.

OPEN/NEWS — the 13:00-15:00Z cash-open window

NOT RUN THIS WEEK · dossier 2026-08-25

The densest run window on the clock.

GREENFIELD HUNT — cluster OPEN/NEWS

Friday report 2026-08-21 week · Movement 3 · full working

(MNQ · \$2.00/point · \$1.50 per round trip · parquet lake gazbot7.lake 2026-07-16 → 2026-08-21 · frozen census census_summary.json)

0. THE SHORT VERSION, BEFORE ANY OF THE MONEY

Garrath — I was sent to invent a new entry that catches the runs we sat out in the OPEN/NEWS bucket. I built nine, killed all nine, and the thing that killed them is not any one of them.

The order below is the order of importance, not the order I did the work in.

#	FINDING	ONE LINE
1	The label was fixed on 08-15, and the fix worked — but only halfway	OPEN/NEWS is no longer if 13<=hour<15 checked first. It is now the residual: the 13-15Z clock minus a flow event minus a vol-expansion. It went from 100% → 58.7% of its own window. It is still true on the majority of the tape it covers.
2	★ Its non-clock content is ANTI-predictive	P(run) is 3.427% on the bare 13-15Z clock and 3.158% on the labelled subset. The two footprint tests strip out 51 of 111 in-window runs and keep the lower-density half. Label lift over the bare clock = 0.92x. Deleting the footprint

tests would make the bucket a better predictor than leaving them in.

3 **★★ The bucket is defined by the ABSENCE of evidence**

A run is OPEN/NEWS because no flow footprint and no amplitude footprint was found. Hunting a footprint inside a bucket built from footprint-failures is close to self-defeating, and it dictated where I looked: only channels `cluster()` never reads — price structure, raw volume, and L2.

4 **Nine gates, nine graves**

ORB30, COILBRK, PULSE, PULSE-13h, FADE1, RNGX60, VWRC, OAB, MOMBRK, BOOKLEAN. Every one reported below by name with its stats and its cause of death. Best survivor-on-paper was MOMBRK at **+\$32.87/trade over 52 trades**.

5 **★★★ THE KILL: a gate with NO TRIGGER AT ALL beats every gate I built**

"Enter at **13:05Z every day**, direction = sign of the last 15 minutes, stop $0.5 \times U$, hold 30 min" — no signal, no arm, no condition — books **+\$47.29/trade over 26 trades**. That is **double** the best triggered candidate. Every candidate on this page sits inside the noise band of a control that contains no information.

6 **The window cannot settle the question**

PULSE's t-statistic is **1.30**. Day-block bootstrap 95% CI [**-\$4.14, +\$26.27**]. To get a $\pm \$10$ /trade confidence interval needs **561 trades $\approx$ 79 trading days**. We have 27.

7 **★ The size threshold IS the deliverable, and it is ~120pt**

Expectancy by realised 30-min follow-through: **<+20pt $\rightarrow$ -\$38/tr $\cdot$ +60...+120pt $\rightarrow$ -\$4.10/tr $\cdot$ $\geq$ +120pt $\rightarrow$ +\$184.75/tr**. Below ~120pt of follow-through the fee and the stop eat everything. 30.8% of armed events cleared it; break-even needed 19.2%.

8 **Detection has a real footprint; direction does not — and the footprint's honest n is ELEVEN**

Trade-count z-score separates ≥ 200 pt runs at **AUC 0.848**, rising monotonically from 0.623 at 55pt. But on **non-overlapping** 30-min blocks the armed sample is **n=11**, 54.5% $\pm$ 29.4pp. The AUC is real; the sample behind it is not big enough to trade.

9 **PULSE and its exact mirror catch the same 6 of the 10 sat-out runs**

Continuation catches 6/10 for **+\$150**; the direction-reversed twin catches the same 6/10 for **+\$46**. Catching a run is trivial. Getting the side right is the entire problem, and neither does.

10 **A censoring recommendation, free**

VOL - EXPANSION — the label the census barely uses — has **4.18 $\times$ run lift on 1.95% of the tape**, the best of any cluster. OPEN/NEWS has 2.85 $\times$ on 5.20%. If a bucket is going to route a hunt, that is the one worth widening.

VERDICT (full statement in §12): NULL. Nothing ships, not even to shadow. Cause of death for the section: the trigger-free NULL-CLOCK control (§9) and the day-block bootstrap (§10). Cause of death for the two leaders individually: strip-the-best-trades and the OOS leg. The deliverable is the ~120pt size threshold (§8) and the label finding (§1-2).

1. ★ INTERROGATING THE LABEL FIRST

The brief says to check the cluster label's base rate before trusting anything built on top of it, and that a label true on most of the tape is noise, not a footprint — and that this finding outranks any signal built on top. I did it first. It changed where I was allowed to look.

1.1 What the label is now

Last year's version of this page found OPEN/NEWS was literally `if 13 <= hour < 15` checked **before** every footprint test. That has been fixed. From `scripts/run_census.py:78-84`:

```
def cluster(hour, flow, mv, amp, fz=None):
    if flow is not None and fz is not None and abs(fz) >= FLOW_Z:
        return "FLOW-LED" if (flow > 0) == (mv > 0) else "VACUUM"
    if amp is not None and amp > 0.30:
        return "VOL-EXPANSION"
    if 13 <= hour < 15:
        return "OPEN/NEWS"          # no footprint, but the densest window – say so
    return "UNCLASS"
```

The clock now goes **last**. OPEN/NEWS is a **residual**, and the full predicate is a conjunction:

```
13:00 ≤ hour < 15:00 UTC AND |flow z| < 1 (no aggressor-imbalance event
in the prior 60s) AND amp ≤ 0.30 (prior-5-min hi-lo ≤ 0.30% of price)
```

That is a materially different object from a clock, and the docstring's own claim — that the window is worth keeping as a fallback — is testable. So I tested it.

1.2 Base rate — measured on ALL bars, not on runs

I re-implemented the census's run detector and its three feature functions (`flow_at`, `flow_z_at`, `amp`) against the **full parquet lake** at 5s — 438,253 bars, 2026-07-16 → 2026-08-21 — and scored the label on **every** 60s-step 15-minute window, run or not. 36,507 windows.

CLUSTER	WINDOWS	% OF TAPE	RUNS (≥1.5×ATR)	P(RUN)	LIFT VS 1.109% BASE
UNCLASS	28,007	76.72%	240	0.857%	0.77×
FLOW-LED	2,988	8.18%	33	1.104%	1.00×
VACUUM	2,901	7.95%	39	1.344%	1.21×
OPEN/NEWS	1,900	5.20%	60	3.158%	2.85×

VOL-EXPANSION	711	1.95%	33	4.641%	4.18×
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Two things fall straight out.

First: the label is true on the majority of its own window. 3,239 windows sit inside 13:00–15:00Z; **1,900 of them (58.66%) wear the label.** The 08-15 fix took it from 100% of its window to 59% of its window. That is progress, and it is not enough to call it a footprint — a coin that lands heads 59% of the time inside its own domain is still mostly describing the domain.

Second — and this is the one that matters:

SLICE OF 13-15Z	WINDOWS	RUNS	P(RUN)	LIFT VS BASE
the bare clock, everything in 13-15Z	3,239	111	3.427%	3.09×
OPEN/NEWS (clock $\wedge$ flow-quiet $\wedge$ amp-quiet)	1,900	60	3.158%	2.85×
what the label throws away (in-window, has a footprint)	1,339	51	3.809%	3.43×

★ **The label's lift OVER the bare clock is 0.921×**

The two footprint tests take a 3.427%-dense window, remove 41% of it, and keep the less dense half. The bit of the label that is not a clock is not merely uninformative — **it is mildly negatively informative.** Everything OPEN/NEWS knows about where runs happen, it knows because of the clock.

1.3 Is that an artefact of the pre-tick era?

The lake's ticks start 2026-07-24; before that flow_z_at returns None and the flow test cannot fire, so 07-16...07-23 windows fall through to the clock by default and could be inflating the label. Split it:

ERA	WINDOWS	13-15Z P(RUN)	OPEN/NEWS P(RUN)	LABEL LIFT OVER CLOCK	LABEL SHARE OF ITS WINDOW
ALL LAKE 07-16→08-21	36,507	3.427%	3.158%	0.921×	58.66%
TICK ERA 07-24→08-21	28,349	3.373%	3.190%	0.946×	53.49%

Same answer both ways. In the tick era the label covers 53% of its own window and still returns 0.946× the bare clock's lift. **Not an artefact.**

1.4 Which of the two footprint tests is doing it

$\text{amp} > 0.30$ is true on only **3.04%** of all windows — it barely bites. $|\text{fz}| \geq 1$ is true on **16.13%**. So essentially all of the subtraction is the flow test, and essentially all of the anti-lift is the flow test removing in-window runs that did have a flow footprint and were, on the evidence, slightly more likely to be runs than the ones left behind.

That is not a bug in `cluster()` — it is doing exactly what it says, routing footprinted runs to FLOW-LED/VACUUM so they reach the hunts that can use them. It is a warning about this bucket: what is left is, by construction, **the runs for which the census's two instruments found nothing**.

1.5 ★★ What that forces on the hunt

This is the structural point of the whole section, and I want it stated plainly before any gate:

OPEN/NEWS is a bucket defined by the absence of evidence. A run lands here because aggressor flow was unremarkable in the 60 seconds before it and the prior five minutes were not unusually wide. Any signal I build that uses aggressor flow or pre-run amplitude would, if it worked, have relabelled the run out of the bucket. The only channels that can legitimately catch an OPEN/NEWS run are the ones `cluster()` never reads: - **price structure** (ranges, levels, VWAP, momentum) — never read; - **raw volume / trade count** — never read (flow is the net signed aggressor sum; a two-sided auction has huge volume and ~zero net flow, and the label cannot tell it from a dead tape); - **L2 book** — the census measures `book_depletion()` and prints it, but `cluster()` never receives it. Free channel. - cross-asset — out of scope for this section.

Those three are exactly what §4–§7 test, in that order. Nothing here is a flow gate, deliberately.

1.6 A free recommendation for the census

VOL-EXPANSION covers **1.95% of the tape at 4.18× run lift** — the highest of any cluster, better than OPEN/NEWS's 2.85× on 5.20%, and it is the label the census currently produces least often (1 run in the frozen week, 33 across the lake). It is checked second, so a run that also has a flow footprint never reaches it. If a bucket is going to route next quarter's hunt, that is the one with the sharpest

edge per unit of tape, and its amp > 0.30 threshold — true on 3.04% of tape — is worth sweeping. Not my brief; recorded because I measured it.

2. THE TARGET, STATED EXACTLY

2.1 What the frozen census says

census_summary.json (frozen, not re-run): 68 runs $\geq 1.5 \times \text{ATR}$ over 2026-08-17 → 08-21, 62 sat out, sat-out ceiling **\$9,610**. OPEN/NEWS = **13 runs**, of which 3 were caught and **10 sat out**.

The three caught, for the record — all on 08-18, all inside 45 minutes:

TIME UTC	DIR	MOVE	GATE	REAL \$
08-18 13:40	DN		−137 exhaustion_short_A	−208
08-18 13:58	DN		−66 day_rider	+174
08-18 14:38	DN		−105 exhaustion_short_A	+250

The ten I was hired to catch (from census_stdout.txt, cluster column OPEN/NEWS, us = sat out):

#	TIME UTC	DIR	MOVE	CEILING \$	FLOW	AMP	BOOK
1	08-17 13:31	DN	−74	148	+147	0.19	0.57
2	08-17 14:53	UP	+70	140	−274	0.07	0.41
3	08-18 13:25	UP	+61	122	+89	0.09	0.52
4	08-18 14:14	UP	+91	182	−226	0.22	0.45
5	08-19 14:00	DN	−128	257	+661	0.22	0.51
6	08-19 14:25	UP	+150	300	−565	0.24	0.51
7	08-19 14:46	UP	+70	139	+281	0.13	0.49
8	08-20 13:22	UP	+120	240	+82	0.08	0.51
9	08-21 13:46	DN	−56	113	+18	0.18	0.47
10	08-21 14:41	UP	+64	127	−263	0.12	0.50

Ceiling on the table: \$1,768. Look at the flow column: five of ten have net aggressor flow pointing the wrong way (runs 2, 4, 6, 10 up on selling; run 5 down on +661 of buying). That is the \$1.5 point in raw numbers — this bucket's runs are the ones whose tape gave no directional tell.

2.2 Why I did not backtest on ten runs

Ten runs over five days cannot support a robustness battery, and `capture.db` — which is what the census reads — is a **5-trading-day rolling window**. Anything I fit on it would be fit on the sample. So the hunt runs on the **full parquet lake**:

```
[lake] MNQ 5s bars: 438,253 rows, 2026-07-16 00:00 → 2026-08-21 20:59 UTC
      27 trading days · typical 15-min range = 44.2pt · run threshold = 1.5× =
66.4pt
      405 runs ≥ threshold · 60 of them OPEN/NEWS
```

The lake's threshold (66.4pt) is stricter than the frozen week's (55pt) because the lake's typical 15-min range is wider (44.2 vs 37pt). I kept the census's own arithmetic rather than pinning the number, so the run definition is the census's definition on every day it is applied to.

60 OPEN/NEWS runs on 26 of 27 days — 32 UP, 28 DN, 32 in hour 13 and 28 in hour 14, sizes 68pt to 234pt. No minute-of-hour concentration at all (the busiest single minute has 3 of 60, and 13:30 — the cash open itself — has 2). **Whatever this bucket is, it is not a news-release footprint.**

2.3 The unit and the costs

- **U = 44.0pt** — median 15-min hi-lo range, computed causally (trailing 24h rolling median), so no config knows the future. This is the census's own "typical 15-min range" statistic.
- **\$2.00 / point** (MNQ; MGC is \$10 and is not in this section).
- **\$1.50 per ROUND TRIP**. Grepped the harness before reuse per the contract: `engine.py: TICK, VPP, FEE = 0.25, 2.00, 1.50` and `net = gross - FEE` charged **once per trade**. Not \$5 (which would have killed every marginal candidate on a fee $3.3\times$ too high), not per side.
- **1 tick (0.25pt) adverse slippage** on every stop-entry and every stop-exit.
- **Pessimistic intrabar**: within a 5s bar the stop is checked **before** the target. A bar that touches both is scored a loss.

2.4 Tick-honesty audit of the fills

The harness runs on 5s OHLC, so I audited the fills against the **tick tape** (memory: "a wrong-instrument tape PASSES the stop-width sweep" — consistency is not correctness).

CHECK

RESULT

entry price inside its own 5s fill-bar [min(price), max(price)] from ticks (40 sampled trades)	40/40
exit price inside its own 5s exit-bar tick range	40/40
exit mix — the wrong-tape signature is every leg at its exact stop	141 stop / 50 time , not 191/0
median hold	265s (max 1,830s)
instrument asserted	MNQ only, from data/tape/{bars,ticks,book}/MNQ/ — never a SELECT DISTINCT symbol

No off-tape fills, no all-stop degeneracy, right instrument.

3. WHAT THE TAPE ACTUALLY TELLS YOU BEFORE A RUN

Before inventing anything I measured what is knowable. Per the memory "census pre-run stats are a picker artefact", every discriminator below is measured on the **unselected** population — every 60s-step window whose forward 15-min move clears the threshold — **not** on the keep-strongest-deduped run list, which shifts these statistics badly.

3.1 Size is visible. Direction is not.

AUC for "this window is the start of a ≥ 66.4 pt move", 13:00–15:00Z, tick era, n=2,520 windows:

FEATURE	AUC
atr15 (trailing-day median 15m range)	0.658
rng_5m (prior 5-min range)	0.652
nt_z (trade-count z, 1-min vs trailing 120)	0.638
vol_z (volume z)	0.631
rng_30m	0.627
er_30m (efficiency ratio)	0.567
`	flow z
sq_5_30 (5m range ÷ 30m range — the "coil" test)	0.532

Direction, among the windows that did produce a run:

FEATURE	AUC FOR P(UP)
pos_60m (where in the 60-min range price sits)	0.554
pos_30m	0.548
er_30m	0.538
d_vwap	0.494
rng_5m	0.475
sq_5_30	0.423

and the crudest possible direction rule — does the run continue the last 15 minutes? — agrees on **569 of 1,049** in-window runs (54.2%), against a **53.4%** base rate on all in-window windows. **0.8 percentage points**.

Two consequences I obeyed for the rest of the page:

1. **sq_5_30 = 0.532 refutes the coil-break idea before it is built.** There is no compression footprint ahead of these runs. I built COILBRK anyway (\$4.2) because a refuted idea with an empty grave is worth less than a refuted idea with a measured one — and it produced one trade at the tight setting, which is itself the answer.
2. **A directional gate cannot be armed on a 0.55-AUC direction call.** The only honest shape is to let the tape choose the side — a two-sided stop-entry — or to accept a 54% momentum coin. Both are tested.

3.2 ★ The escalation: does a footprint appear at SIZE?

The brief says to escalate — full sat-out set, then top-25, then top-15 — because narrowing may reveal a footprint the marginal runs wash out. The clean way to do that with enough n is to sweep the size threshold itself. 13-15Z, tick era, unselected windows:

Detection AUC vs run-size threshold

THRES HOLD	N WINDO WS ≥	VOL_Z	NT_Z	RNG_5M	RNG_30M	ATR15	FZ	SQ_5_30
55pt	988	0.618	0.623	0.648	0.629	0.650	0.558	0.524
66pt	805	0.631	0.638	0.652	0.627	0.658	0.557	0.532
80pt	597	0.655	0.662	0.656	0.620	0.651	0.565	0.543
100pt	393	0.671	0.678	0.661	0.617	0.671	0.569	0.546
120pt	259	0.674	0.681	0.694	0.626	0.703	0.576	0.573
150pt	137	0.711	0.718	0.710	0.628	0.728	0.598	0.571
200pt	43	0.842	0.848	0.791	0.618	0.718	0.668	0.666
250pt	8	0.895	0.901	0.932	0.867	0.799	0.800	0.681

The escalation DOES reveal a footprint, and it is the volume channel — the one the label cannot see. nt_z climbs monotonically 0.623 → 0.848. rng_30m, by contrast, is flat (0.629 → 0.618): this is not simply "wide days have wide runs". | fz| — the label's own instrument — climbs far more slowly (0.558 → 0.668), which is the same message as §1.2 from the other side.

Direction AUC vs size threshold (among windows whose |move| clears the threshold):

THRESHOLD	N	UP%	POS_60M	PREV15 (MOMENTUM)	D_VWAP	NETFLOW_Z
55pt	988	47.5	0.596	0.612	0.516	0.496
100pt	393	41.5	0.621	0.627	0.541	0.476
150pt	137	40.9	0.634	0.620	0.608	0.509
200pt	43	37.2	0.697	0.655	0.799	0.451

Direction improves too — but slowly, and from a much worse start.

3.3 ★★ ...and the honest n behind that footprint is ELEVEN

Those AUCs are computed on 60s-step windows carrying a 15–30 minute forward horizon. **They overlap by ~95%.** An n of 2,520 is not 2,520 observations; it is roughly 84 independent 30-minute blocks over 27 days. So I recomputed the headline claim on **non-overlapping** blocks:

P(30-min move in the momentum direction ≥ X), 13-15Z, one observation per non-overlapping 30-minute block, n=84 blocks:

X	ALL BLOCKS	NT_Z ≥ 1.5 BLOCKS	LIFT
80pt	19.0% (n=84)	54.5% (n=11, ±29.4pp 95% CI)	2.86×
120pt	10.7% (n=84)	54.5% (n=11, ±29.4pp)	5.09×
150pt	8.3% (n=84)	45.5% (n=11, ±29.4pp)	5.45×

The volume footprint is real, it is large, and its independent sample size is eleven. A 95% confidence interval of ±29 percentage points around a 54% hit rate spans 25%–84%. That is not a footprint you can size a position on; it is a hypothesis that needs a quarter of tape.

For completeness, the same measurement on the raw overlapping 5s bars (n=30,169 in-window, 7,441 armed) gives 15.6% → 27.8% at X=120 — a 1.78× lift. **Both numbers are in this report because the gap between them is the finding.** Anyone quoting "n=7,441" off this study is quoting 5-second bars, not observations.

4. THE GRAVEYARD — every attempt, by name

Standing rule: every attempt gets reported, nulls and graves included, each with its stats and a one-line cause of death. Common config for the table below unless stated: stop **0.5×U = 22pt**, no target, 30-minute time cap, 1-tick slippage, \$1.50 round trip, 13:00–15:00Z, 15-minute cooldown.

GATE	N	NET \$	\$/TRADE	WIN%	STRIP-3	STRIP-5	TOP-3 % OF NET	LONG \$/ TR	SHORT \$/TR
ORB30	27	+418.5	+15.50	18.5	−26.98	−46.00	255%	+5.18	+26.62
COILBRK q0.60	9	+492.5	+54.72	22.2	−46.00	−46.00	156%	−46.00	+105.08
COILBRK q0.75	23	+693.5	+30.15	21.7	−25.38	−46.00	173%	−8.50	+65.58
PULSE	191	+2,167.0	+11.35	24.6	+2.63	−1.29	77%	+20.45	+1.13
PULSE-13h	95	+2,194.0	+23.09	26.3	+5.67	−1.89	76%	+30.80	+15.22
FADE1 (PULSE mirror)	211	−2,754.0	−13.05	13.7	−20.25	−24.19	—	−19.56	−7.30
RNGX60	81	−974.0	−12.02	14.8	−23.71	−30.47	—	−5.41	−18.80
VWRC	41	−352.0	−8.59	22.0	−24.46	−32.56	—	−6.69	−10.07
OAB	73	+1,330.5	+18.23	20.5	−2.59	−12.17	114%	+10.26	+26.42
MOMBRK	52	+1,709.0	+32.87	23.1	+4.02	−11.37	89%	+36.59	+29.91
BOOKLEAN	9	+108.0	+12.00	44.4	−29.58	−46.00	264%	+9.43	+21.00
NULL- CLOCK 13:05 — no trigger	26	+1,229.5	+47.29	34.6	+6.37	−15.76	88%	+49.38	+45.50
NULL-CLOCK 13:30 — no trigger	27	−363.5	−13.46	11.1	−46.00	−46.00	—	−16.50	−11.03
NULL-CLOCK 14:05 — no trigger	27	+642.0	+23.78	29.6	−16.29	−29.30	161%	+52.79	+0.57

Read the last three rows first. They are in §9 and they are why nothing on this page ships.

Note on win%. Per the operator's 2026-08-01 judging rule, no candidate here is killed on win rate. PULSE-13h wins 26.3% of the time and I did not hold that against it — the live desk booked +\$625 this week at 39%. Every kill below names a robustness test.

4.1 ORB30 — pre-open range break — −\$26.98/tr after strip-3

Take the 13:00–13:30Z range (the half-hour into the US cash open), rest a buy-stop 1 tick above and a sell-stop 1 tick below, first touch wins, other cancelled. 27 events, 27 triggers — it fires every single day, which is at least a well-specified

gate. Raw +\$15.50/tr looks alive. **Top 3 trades are 255% of the net** — strip them and it is **−\$26.98/tr**; strip 5 and every remaining trade is a full stop. 5 winners out of 27. **Cause of death: strip-the-best-trades.**

4.2 COILBRK — compression break — 1 trade at the setting the theory wanted

Arm when the 15-min range is $\leq q \times U$, then two-sided break. §3.1 predicted this would fail (sq_5_30 AUC 0.532) and it failed in the most literal way available:

Q	15M RANGE MUST BE UNDER	EVENTS	TRADES	\$/TRADE	
0.40	17.6pt	1	1	1	−46.00
0.50	22.0pt	5	5	5	−46.00
0.60	26.4pt	13	9	9	+54.72
0.75	33.0pt	32	23	23	+30.15

The 13:00–15:00Z window essentially **never coils**. To get a usable n you must relax the compression test until it stops meaning compression ($q=0.75$ is a 33pt 15-minute range — barely below the 44pt median). And at $q=0.60$ the whole \$492.50 is three trades (156% of net; strip-3 = −\$46.00, i.e. every survivor is a stop) and the **long side loses on every one of its trades** while the short side makes \$105/tr — $n=9$ split 2/7. **Cause of death: n , and parameter sweep — the edge exists only where the compression test has been relaxed into meaninglessness.**

4.3 PULSE — 1-minute impulse continuation

Trigger: a 1-minute close-to-close move $\geq k \times U$ inside 13–15Z. Direction = the impulse's sign. Entry: stop order 1 tick beyond that minute's extreme. This is the largest- n candidate on the page and got the fullest battery — \$5.

4.4 FADE1 — the exact mirror — −\$13.05/tr

Identical trigger, direction reversed, market entry. −\$13.05/tr over 211. Not a failure so much as a **control that passed**: PULSE's +\$11.35 and FADE1's −\$13.05 are close to symmetric around −\$1.50 of fee, which says the two are genuine mirrors and the harness is not manufacturing money from the entry mechanics. That matters — it means PULSE's direction rule is doing something. §5.4 shows it is not doing enough.

4.5 RNGX60 — 60-minute structure break — $-\$12.02/\text{tr}$

Two-sided break of the trailing 60-minute high/low. 216 events, 81 triggers.

Negative raw, negative at every strip, negative on both sides. **Cause of death: negative expectancy, unconditionally.** By 13:00Z the 60-minute extremes are the overnight range's edges, and MNQ spends the US open chopping across them.

4.6 VWRC — session-VWAP cross — $-\$8.59/\text{tr}$

Enter in the direction of a cross back through the running session VWAP. 42 events, 41 trades, $-\$8.59/\text{tr}$, negative both sides, $-\$24.46$ after strip-3. d_vwap had already scored **AUC 0.494** for direction in §3.1 — a coin. **Cause of death: negative expectancy; predicted by the direction AUC.**

4.7 OAB — volume-ignition two-sided break — no parameter plateau

The first gate built deliberately on the §1.5 orthogonal channel. Arm when raw 1-minute **volume** $z \geq V$ and $|\text{flow } z| < 1$ — a **two-sided auction**: heavy trade, balanced aggression, exactly the state the census's net-flow instrument is blind to. Then a two-sided break of the 5-minute range.

The arm itself is real (§3, and lift $1.66\times$ at $\text{vol}_z > 2.5$ even restricted to flow-quiet minutes). The gate is not. **192-cell sweep** over $V \in \{1.0, 1.5, 2.0, 2.5\} \times \text{wait} \in \{1, 2, 3, 5\} \text{ min} \times \text{stop} \in \{0.375, 0.5, 0.75\}U \times \text{target} \in \{0.75, 1.0, 1.5, \text{None}\}U$:

overall mean $\$/\text{trade}$ across all 192 cells: -5.15 fraction of cells positive: 0.234

mean $\$/\text{trade}$ by arm strength $\times$ trigger window

$vz \setminus \text{wait}$	1min	2min	3min	5min
1.0	-2.26	-2.57	-6.50	-4.90
1.5	-10.32	-7.69	-7.82	-5.05
2.0	-11.89	-6.84	-10.75	-6.00
2.5	-5.23	+2.55	-1.47	+4.29

trades <- the only positive row, $n=24..30$

77% of the surface is negative, and the positive corner is the corner where n collapses to 24-30 trades. $V=2.0$ is negative in all four wait cells; $V=2.5$ flips positive. Adjacent parameter cells flip sign. **Cause of death: parameter sweep — no plateau; edge appears only as n collapses. That is the curve-fit tell named in the brief.**

4.8 MOMBRK — trade-count arm + momentum-direction break

The best-scoring gate I built. Full battery in §6.

4.9 BOOKLEAN — L2 imbalance — full L2 null in §7

9 trades. Reported in the table for completeness; the channel-level result in §7 is the finding.

5. PULSE — the largest-n candidate, killed

Spec (as it would ship):

GATE pulse_open

SESSION13:00 <= UTC hour < 15:00

TRIGGER|close[t] - close[t-60s]| >= 0.30 x U(U = 44.0pt, causal 24h median 15m range)

DIRECTIONd = sign(close[t] - close[t-60s])

ENTRYresting STOP at (high[t] + 1 tick) if d>0 else (low[t] - 1 tick); valid 3 minutes

STOPentry - d x 0.5 x U (22pt)

EXITtime cap 30 minutes; no target, no trail

COOLDOWN15 minutes

COSTS1 tick slippage in and out; \$1.50 per round trip

Base: n=191, net +\$2,167, +\$11.35/trade, win 24.6%, median −\$46.00, best + \$695.50.

5.1 Parameter sweep — 144 cells

$k \in \{0.20, 0.25, 0.30, 0.35, 0.40, 0.50\} \times \text{stop} \in \{0.25, 0.375, 0.5, 0.75\}U \times \text{target} \in \{0.5, 0.75, 1.0, 1.5, 2.0, \text{None}\}U$, 15-min cap:

mean \$/trade by stop x target

stop\targ	0.50U	0.75U	1.00U	1.50U	2.00U	none
0.250	-3.30	-2.12	-2.33	-3.88	-4.66	-3.08
0.375	-1.57	-1.08	-0.78	-3.80	-4.82	-0.80
0.500	-1.51	-0.11	-0.76	-3.40	-3.18	-0.30
0.750	-1.37	+0.50	-0.39	-3.33	-4.01	-0.86

mean \$/trade by trigger size k

0.20	-0.85	0.25	-4.06	0.30	-0.39	0.35	-2.99	0.40	-1.65
0.50	-2.81								

23 of 24 exit cells are negative. And k is not monotone: $0.20 \rightarrow 0.25 \rightarrow 0.30$ gives $-0.85, -4.06, -0.39$. Adjacent trigger thresholds flip sign by \$3.7/trade. The best single cell (k=0.20, stop 0.75U, target 1.0U) prints +\$7.08/tr and is surrounded by cells averaging under zero.

5.2 The time cap is not an exit rule, it is a dial

The +\$11.35 headline comes from lengthening the hold, so I swept it (arm A=1.5 variant, same story for PULSE):

TIME CAP	5M	10M	15M	20M	30M	40M	60M	90M
\$/trade	+0.70	-7.91	+11.16	+22.19	+32.87	+41.26	+50.92	+51.51

It rises essentially monotonically with hold length (with a sign flip at 10m). **That is not "an exit rule works", that is "hold longer, capture more of whatever the tape did"**. A gate whose P&L is a monotone function of how long you stay in is not an entry edge; it is exposure.

5.3 Strip-the-best, day concentration, leave-one-out, OOS

TEST	RESULT	VERDICT
base	+\$11.35/tr, n=191	—
strip 3 best	+\$2.63/tr	top 3 trades = 77.2% of net
strip 5 best	-\$1.29/tr	FAIL
leave-one-day-out (27 folds)	all 27 folds positive, range +\$8.23 ... +\$13.53	pass
day balance	16 profitable days / 11 losing; top day 30.2% of net	pass
long / short symmetry	LONG +\$20.45 (n=101, +\$2,065) · SHORT + \$1.13 (n=90, +\$102)	FAIL — 95% of net is one side
OOS (IS 07-16...08-14 / OOS 08-17...08-21)	IS +\$10.69 (n=155) · OOS +\$14.18 (n=36)	pass

The long/short split is the one that should worry us most: MNQ fell **-318pt** over the sample and **-542pt of that accrued inside 13:00-15:00Z**, so the long side is not being paid by drift — it is being paid against drift, on 101 trades, while the short side in a falling tape returns \$1.13/trade. That asymmetry has no mechanism behind it. It is the signature of a handful of large long winners.

5.4 ★ The placebo — and a correction to how I first ran it

The brief's test: shift the signal series, keep every other rule; if a fake signal books most of the money, the real one is not the edge.

I initially ran this with shifts in both directions and I was wrong to. A negative shift feeds the gate the impulse that happens **5 minutes in the future**

— that is not a placebo, it is a look-ahead oracle. Both directions are reported below, correctly labelled, because the look-ahead leg is a useful harness check (it must beat the real signal; if it did not, the harness would already be leaking).

	−30M	−15M	−5M	REAL	+5M	+10M	+15M	+30M	+60M
	look-ahead	look-ahead	look-ahead		stale	stale	stale	stale	stale
PULSE 13-15Z	+8.05	+4.57	+30.32	+11.35	−12.57	+8.14	−3.46	+6.45	+8.21
PULSE-13h	+10.16	+18.40	+48.45	+23.09	−9.51	+20.76	+18.81	+23.95	+18.47

- Harness check: the −5m look-ahead oracle books +\$30.32 vs the real + \$11.35 (and +\$48.45 vs +\$23.09 on the 13h variant). **The harness is not leaking** — knowing the future is worth $\sim 2.7\times$, as it should be.
- PULSE 13-15Z: stale placebos average **+\$1.35/tr = 12% of real**, 0 of 5 beat it. **Placebo passes.**
- PULSE-13h: stale placebos average **+\$14.50/tr = 63% of real**, and the +30m stale signal books **+\$23.95 — more than the real signal's +\$23.09. Placebo FAILS.**

A signal whose 30-minute-stale copy earns as much as it does is not a trigger. It is a slow-moving state variable being read at an arbitrary moment.

5.5 PULSE-13h — the home-segment variant, and its own battery

\$8's regime work shows PULSE's money is entirely in the **first hour**: 13:00–14:00Z **+\$23.09/tr (n=95, +\$2,194)** vs 14:00–15:00Z **−\$0.28/tr (n=96, −\$27)**. So I ran the battery on the home segment as its own candidate.

TEST	RESULT	VERDICT
base	+\$23.09/tr , n=95, win 26.3%	—
strip 3 best	+\$5.67/tr (top 3 = 76.2% of net)	weak
strip 5 best	−\$1.89/tr	FAIL
leave-one-day-out	27/27 folds positive, +\$14.74 ... +\$26.13	pass
day balance	17 up / 10 down, top day 38.2%	pass
long / short	+\$30.80 (n=48) / +\$15.22 (n=47) — both positive, both sides trade	pass
placebo (stale only)	mean +\$14.50 = 63% of real ; 1 of 5 beats it	FAIL
OOS 08-17...08-21	IS +\$30.49 (n=78) · OOS −\$10.82 (n=17)	FAIL
day-block bootstrap 95% CI	[−\$2.79, +\$52.71] , $P(\leq 0) = 0.044$	inconclusive

PULSE-13h is the most attractive thing on the page — it is the only candidate that passes long/short symmetry with both sides positive. It still dies three ways.

Cause of death (PULSE and PULSE-13h): the placebo test (a 30-minute-stale copy of the signal books more money than the signal), confirmed by strip-5-best going negative and by the OOS leg (−\$10.82/tr).

6. MOMBRK — the best number on the page, killed hardest

Built directly on §3.2's finding: nt_z is the strongest size discriminator and it strengthens with run size. So: arm on trade-count z, take the momentum side, break-entry.

```
GATE mombrk_open
SESSION      13:00 <= UTC hour < 15:00
ARM          nt_z >= 1.5      (1-min trade COUNT vs trailing 120 non-overlapping
1-min buckets)
DIRECTION    d = sign(close[t] - close[t-15min])
ENTRY        resting STOP 1 tick beyond the 5-minute extreme on side d; valid 3
minutes
STOP         0.5 x U (22pt)    EXIT  time cap 30 min      COOLDOWN 15 min
```

Base: n=52, net +\$1,709, +\$32.87/trade, win 23.1%.

6.1 The arm ladder — jumpy, not a plateau

NT_Z ≥	0.0	0.5	0.75	1.0	1.25	1.5	1.75	2.0	2.5	3.0
n	102	88	77	71	63	52	46	39	32	28
\$/trade	−0.73	+0.53	+4.50	+22.23	+20.53	+32.87	+8.55	+16.81	+6.97	−14.21

With the arm off entirely (A=0.0) the gate is −\$0.73/tr, so the arm is unambiguously doing the work. But the ladder is not a plateau: 1.5 → 1.75 → 2.0 reads +32.87, +8.55, +16.81, and 3.0 is −14.21. A quarter-of-a-standard-deviation move in the arm threshold changes expectancy by \$24/trade.

Swapping the arm column barely matters (nt_z +32.87, vol_z +24.02, netflow_z +30.98) — the three are near-collinear busy-tape meters, which is the \$5.4 point again: this is a state, not a trigger.

6.2 The battery

TEST	RESULT	VERDICT
base	+\$32.87/tr, n=52	—
strip 3 best	+\$4.02/tr — top 3 = 88.5% of net	FAIL
strip 5 best	−\$11.37/tr	FAIL

day balance	7 profitable days vs 14 losing. Top day = 50.1% of net. Two days (07-27 +\$801, 07-31 + \$857) = 97% of the +\$1,709	FAIL
leave-one-day-out	21/21 folds positive, +\$18.13 ... +\$37.69	passes — see below
leave-TWO-days-out	210/210 folds still positive, but worst fold + \$1.19/tr (drop 07-27 & 07-31)	fragile
long / short	+\$36.59 (n=23) / +\$29.91 (n=29) — symmetric	pass
OOS 08-17...08-21	IS +\$44.62 (n=42) · OOS -\$16.50 (n=10)	FAIL
placebo (stale +5/+15/+30/+60/+120m)	+23.91, +4.54, +25.20, -5.61, +30.13 → mean +\$15.63 = 48% of real , 0 of 5 beat it	pass
random-arm placebo (200 draws, same count, same window)	mean +\$2.61, sd \$14.87, p95 +\$27.96 → empirical p = 0.025	pass

6.3 ★ Why leave-one-day-out passed and should not have reassured anyone

All 21 LODO folds are positive. Two days carry 97% of the P&L. Both statements are true at once, and that is a property of the test, not of the gate: with 21 days and 2 money days, removing one day always leaves the other. **LODO with k money days cannot fail until you leave k days out.** I therefore ran leave-two-out — 210 folds — and the worst fold (drop 07-27 and 07-31 together) collapses to **+\$1.19/trade**, i.e. break-even.

That is worth carrying forward as a rule: **quote the number of profitable days alongside any LODO result.** 7-up/14-down with a passing LODO is a warning, not a clearance.

Cause of death (MOMBRK): strip-the-best-trades (+\$32.87 → +\$4.02 at strip-3, -\$11.37 at strip-5; top 3 trades are 88.5% of net) and the OOS leg (-\$16.50/trade). Confirmed by the day distribution: 7 winning days out of 21, two of which are 97% of the money.

7. THE L2 CHANNEL — a clean null, and one inverted surprise

The book is the one channel `cluster()` genuinely cannot see (\$1.5): `run_census.py` computes `book_depletion()` and prints it in the book column, but never passes it to `cluster()`. Per the data contract, `capture.db.book` is MNQ-only and event-driven at 41ms — correct for this section — and I read it from the lake (`data/tape/book/MNQ/`, **2026-07-31 → 08-24**, so 16 trading days inside the study window, not 27).

Aggregated to per-minute 3-deep and 10-deep bid/ask size, restricted to 13:00–15:00Z, n=22,979 5s bars:

FEATURE	WHAT IT SHOULD PREDICT	AUC
$imb3 = (bid_3 - ask_3) / (bid_3 + ask_3)$	direction of the next 15 min	0.508
imb10	direction of the next 15 min	0.496
imb3	direction of the next 30 min	0.509
`	imb3	` (magnitude of the lean)
depth_z (3-deep size z)	size,	fwd30
depth3 (raw 3-deep total size)	size,	fwd30

Book LEAN is a coin. 0.508 and 0.496 are as close to nothing as this measurement gets, and the gate built on it (BOOKLEAN, §4.9) produced **9 trades** — the threshold that gives a meaningful lean almost never fires inside the US open. On non-overlapping 30-minute blocks in the book era (n=64), $|imb3| \geq 0.1$ occurs **4 times** and ≥ 0.2 **once**. There is no sample here.

But depth3 at AUC 0.285 is not nothing — it is inverted. Read the other way, a THIN 3-deep book predicts a ≥ 120 pt move at AUC 0.715, which is better than volume-z (0.674) and better than any book lean measure by a mile. That is the same message as §3.2 from a third instrument: the tape tells you **how big**, never **which way**. And like nt_z it is a state variable — thin books cluster in time — so it inherits the same §5.4 problem and I did not build a gate on it.

Cause of death (BOOKLEAN): n=9. Cause of death (the L2 direction channel): AUC 0.508 — no directional information at any depth or horizon.

8. ★ THE SIZE THRESHOLD — this is the deliverable

The brief: report the size-threshold at which a footprint becomes tradeable, if any — that IS the finding. Here it is, three ways.

8.1 The money threshold: ~120pt of follow-through

MOMBRK's 52 trades, bucketed by the **realised** 30-minute move in the traded direction:

REALISED FOLLOW-THROUGH	TRADES	% OF EVENTS	NET \$	\$/TRADE
≤ -60 pt (went hard against)	14	26.9%	−644.0	−46.00

-60 ... -20pt	6	11.5%	-276.0	-46.00
-20 ... +20pt (nothing happened)	9	17.3%	-346.0	-38.44
+20 ... +60pt	2	3.8%	+39.5	+19.75
+60 ... +120pt	5	9.6%	-20.5	-4.10
≥ +120pt	16	30.8%	+2,956.0	+184.75

A move of +60 to +120 points in your favour still loses money. -\$4.10/trade. The 22pt stop gets tagged on the way, or the 30-minute cap arrives before the move does, or the \$1.50 and the two ticks of slippage take the rest. **Only follow-through beyond ~120pt pays**, and when it does it pays +\$184.75.

The arithmetic that makes this a rule rather than an observation: with +\$184.75 on a hit and -\$44 on a miss, **break-even hit rate = $44/(184.75+44) = 19.2\%$** . MOMBRK's arm delivered **30.8%**. That 11.6-point margin is the entire edge — and §6 shows it does not survive strip-3 or OOS.

This lands on the same number as the desk's existing greenfield finding — "the size threshold is 120pt; ≥120pt boarding pays +\$69.38/tr" — from a completely independent cluster and a different harness. Two independent studies converging on ~120pt on MNQ is the most durable thing on this page.

8.2 The detection threshold: a footprint appears above ~120pt, sharpens above 200pt

From §3.2: nt_z AUC runs 0.623 (55pt) → 0.638 (66) → 0.662 (80) → 0.678 (100) → **0.681 (120)** → 0.718 (150) → **0.848 (200)**. The curve is gentle to 120 and steep after 150. **The escalation the brief asked for did reveal a footprint that the marginal runs wash out** — and it is in the volume channel, not the price channel (rng_30m is flat at 0.62 across the whole range).

8.3 The threshold that matters most: ~11 independent observations

§3.3. On non-overlapping blocks, the ≥120pt armed sample is **n=11 at 54.5% ±29.4pp**. So:

The tradeable size threshold on MNQ in this window is ≈120pt of 30-minute follow-through. The footprint that predicts it is real (trade-count z, AUC 0.68 at 120pt rising to 0.85 at 200pt). The sample behind that footprint is eleven independent observations, and eleven is not enough

to arm a gate on. The instrument we need is not a new measurement — it is more tape.

9. ★★★ THE CONTROL THAT KILLS THE WHOLE SECTION

Every candidate that scored well shared three ingredients: the 13:00–15:00Z window, a momentum-derived direction, and a ~30-minute hold. So I stripped out the only remaining ingredient — **the trigger** — and ran what is left.

NULL-CLOCK: no arm, no signal, no condition of any kind.

Enter at a FIXED clock time every trading day.

Direction = $\text{sign}(\text{close} - \text{close}[15 \text{ min ago}])$. Stop $0.5 \times U$. Hold 30 min. Market entry.

ENTRY TIME	TRADES	NET \$	\$/TRADE	WIN%
13:05Z	26	+1,229.5	+47.29	34.6
13:15Z	27	+171.0	+6.33	14.8
13:30Z	27	-363.5	-13.46	11.1
13:45Z	27	-125.0	-4.63	14.8
14:05Z	27	+642.0	+23.78	29.6
14:30Z	27	-389.0	-14.41	14.8

A gate with no information in it books +\$47.29/trade — double the best triggered candidate on this page (PULSE-13h, +\$23.09) and 44% more than MOMBRK (+\$32.87).

And the same trigger-free rule swings from **+\$47.29 to -\$13.46 purely on which minute of the hour you pick it**, across 26–27 trades each. Same rule. Same direction logic. Same stop. Same hold. Twenty-five minutes apart.

For scale, the coin-flip floor: hold the 13:05Z entry times fixed and replace the momentum direction with a fair coin, 100 draws → **mean +\$2.05/tr, sd \$15.21, p95 +\$23.70**. So the momentum direction rule is worth something (+\$47 vs +\$2 is about 3 sd) — but **p95 of pure noise is +\$23.70, which is above PULSE-13h's +\$23.09 and within touching distance of MOMBRK's +\$32.87**.

Every candidate on this page lives inside the noise band of a control that contains no information. That is not a statement about any one gate. It is a

statement about what 27 trading days of a two-hour window can and cannot resolve.

10. HOW MUCH TAPE WOULD IT TAKE

STATISTIC	PULSE (13-15Z)	PULSE-13H
n trades	191	95
mean \$/trade	+11.35	+23.09
sd of \$/trade	120.78	143.26
standard error	8.74	14.70
t-statistic	1.30	1.57
day-block bootstrap (5,000, blocks = trading days) mean	+11.08	+23.33
day-block bootstrap 95% CI	[-\$4.14, +\$26.27]	[-\$2.79, +\$52.71]
P(true expectancy ≤ 0)	0.075	0.044

Neither clears $t=2$. Both bootstrap intervals contain zero. And to settle it:

TARGET 95% CI HALF-WIDTH	TRADES NEEDED	AT 7.1 TRADES/DAY
±\$30	63	9 trading days
±\$20	141	20 trading days
±\$10	561	79 trading days

Roughly sixteen weeks of tape to distinguish a \$10/trade edge from nothing in this window. We have five and a half. That is the honest reason nothing here can be armed, and it is not fixable by a cleverer gate.

11. DID ANYTHING CATCH THE TEN RUNS?

Against the ten sat-out OPEN/NEWS runs of the frozen census (\$2.1), ceiling **\$1,768**. Counted as caught if the gate opened a **correctly-signed** position between 5 minutes before and 15 minutes after the run's start:

GATE	CAUGHT	NET \$ ON THOSE RUNS	CEILING CONVERSION
PULSE	6 / 10	+\$150	8.5%
FADE1 (PULSE's exact mirror)	6 / 10	+\$46	2.6%
VWRC	2 / 10	+\$50	2.8%
MOMBRK	1 / 10	-\$46 —	
ORB30	1 / 10	-\$46 —	
RNGX60	1 / 10	-\$46 —	

★ **PULSE and the gate that does the exact opposite of PULSE catch the same six of ten runs.**

That single line is the cluster's whole character. Something fires into these runs reliably — a 1-minute impulse of $\geq 0.30 \times U$ happens near 6 of the 10 — so **detection is not the problem**. But a gate and its perfect mirror both "catch" 60% of them, and the better of the two converts **8.5% of the ceiling**. The run is there; the side is a coin; the money is in neither.

Across the whole lake, MOMBRK caught 17 of the 60 OPEN/NEWS runs (28%) for +\$884, and 7 of the 23 that were $\geq 120\text{pt}$ for +\$921 — **essentially all of its run-money is in the $\geq 120\text{pt}$ runs**, which is \$8.1 restated on the run set rather than the trade set.

12. REGIME SEGMENTATION AND THE ROUTER

The backtest discipline forbids scoring a static config across the whole tape. Regimes are tagged **causally** per bar: ATR percentile-rank (trailing 2 days) $\times$ efficiency ratio over 30 minutes, in the router's own vocabulary.

A bug I fixed mid-study, reported because it would have silently deleted two regimes. My first ER summed 1-minute absolute changes at 5-second spacing — 360 overlapping legs — inflating the path $\sim 12\times$ and driving every ER to ~ 0.03 . Under that ER, clean-trend and building **did not exist anywhere in 13-15Z**, and the regime table showed only the three chop buckets. Fixed to sum non-overlapping legs (median ER-30m now 0.167). Every regime number below is post-fix.

13-15Z bar counts: building 12,987 · violent-whipsaw 8,679 · dead-chop 8,505 · normal-chop 3,927 · clean-trend 2,575 · trend-thin 2,201.

PULSE by regime (n=191):

REGIME	N	NET \$	\$/TRADE	WIN%
normal-chop	16	+563.0	+35.19	25.0
clean-trend	11	+374.5	+34.05	36.4
dead-chop	36	+866.5	+24.07	27.8
violent-whipsaw	46	+333.5	+7.25	21.7

building	67	+150.0	+2.24	23.9
trend-thin	15	−120.5	−8.03	20.0

PULSE by regime × sub-session (\$/trade, n in brackets):

REGIME	13:00-14:00Z	14:00-15:00Z
normal-chop	+ 157.12 (4)	−5.46 (12)
clean-trend	+ 72.25 (6)	−11.80 (5)
dead-chop	+ 63.53 (17)	−11.24 (19)
violent-whipsaw	+4.22 (18)	+9.20 (28)
building	+0.12 (37)	+4.85 (30)
trend-thin	−2.19 (13)	−46.00 (2)

MOMBRK by regime (n=52): dead-chop +\$73.12 (25) · violent-whipsaw +\$5.24 (21) · normal-chop −\$38.17 (6). By slot: 13h +\$51.14 (42) · 14h −\$43.90 (10).

12.1 What the regime table actually says

The **time-of-day split is far stronger than the regime split**. PULSE is + \$23.09/tr in 13:00–14:00Z and −\$0.28/tr in 14:00–15:00Z; MOMBRK is +\$51.14 and −\$43.90. Within the first hour, every regime except trend-thin is positive; within the second, four of six are negative. Whatever this is, it is a **first-hour-of-the-US-cash-session** effect, not a volatility-regime effect.

And the regime cells that look best are the ones with no sample: normal-chop 13h is **+\$157/trade on 4 trades**; clean-trend 13h is **6 trades**. Those are not home segments, they are anecdotes. The only regime cell with both a decent n and a decent number is dead-chop 13h, +\$63.53 on 17 trades — and §9 has already shown that a no-trigger rule entering at 13:05Z makes +\$47.29 on 26 trades, so even that cell is not clear of the null.

12.2 ★ The router rule — stated for something that did not survive

A gate that is only viable behind a router condition we cannot yet measure is a finding, not a failure. Nothing here is viable, but the honest statement of what a survivor would need is:

```
# WOULD-BE ROUTER RULE — NOT FOR DEPLOYMENT. No candidate in this section earned it.
ARM  pulse_open  WHEN  session == US_OPEN
                                AND  13:00 <= utc_hour < 14:00                # the SECOND hour is
where it dies
                                AND  regime IN (dead-chop, normal-chop, clean-trend)  # n = 17 /
4 / 6 — NOT a sample
                                AND  nt_z(1min, 120-bucket) >= 1.5            # the size footprint, AUC
0.68 at 120pt
```

```
BENCH pulse_open WHEN utc_hour >= 14:00
                   OR regime == trend-thin
                   OR untradeable_meter == TRUE
```

Two of those four arm conditions rest on 4 and 6 trades respectively. **The router condition we cannot yet measure is not an instrument — it is sample size.** We have every input this rule needs (`nt_z` is computable from ticks in one query, `ER` and `ATR-rank` already exist in the router's vocabulary, and the session split is free). What we do not have is 79 trading days of it.

If the desk wants this line of enquiry alive, the cheapest useful thing is **not** another gate: it is to start logging `nt_z`, `depth3` and the realised 30-min follow-through at 13:00–14:00Z every day, so that by December the ≥ 120 pt question has `n` in the hundreds instead of eleven.

13. WHAT I DID NOT DO, AND WHY

Stated so the absences are not read as findings.

- **No cross-asset (MGC) leg.** MGC is \$10/point and gets its own invented-fresh gates in `gf_MGC` ([the gold section below](#)); the contract forbids porting an MNQ gate onto gold and this section is MNQ.
- **No V5 archive extension.** `gaz:v5archive` would add MNQ history before 07-16, but the V5→V7 seam has a **real tick hole 07-17 → 07-24** (`lake.py` warns about it explicitly), and every candidate here needs `nt_z/vol_z`, which need ticks. Extending bars-only would have widened the sample for the price features while leaving the volume features — the only ones that showed a footprint — on the short sample. I judged a clean 27-day tick-complete sample better than a contaminated 40-day one. **This is the single biggest thing that could change the answer**, and if the seam is ever bridged, §8's eleven observations is the number to re-run first.
- **No walk-forward re-optimisation.** With 27 days and a 2-hour window there is not enough tape to split into fit/select/test three ways; the IS/OOS 2-way split in §5-6 is already thin (OOS `n`=10-36).
- **No book-pressure gate beyond BOOKLEAN.** The L2 direction channel measured AUC 0.508 (§7). I did not build further on a coin.
- **The census was not re-run.** `census_summary.json` and `census_stdout.txt` are read frozen, as instructed. Everything in §1-3 re-

implements the census's functions against the lake; it does not re-cut the census's run list.

14. ONE-LINE CAUSES OF DEATH

GATE	\$/TRADE	CAUSE OF DEATH
ORB30	+15.50	strip-the-best — top 3 trades are 255% of net; strip-3 → −\$26.98
COILBRK q0.40	−46.00	n = 1. The US open does not coil; sq_5_30 AUC 0.532 predicted it
COILBRK q0.60	+54.72	parameter sweep + n — edge only where compression is relaxed to 26pt; strip-3 → −\$46.00, long side loses every trade
COILBRK q0.75	+30.15	strip-the-best — strip-3 → −\$25.38, strip-5 → −\$46.00
PULSE	+11.35	placebo (30m-stale copy earns 63–100% of real on the home segment) + strip-5 → −\$1.29 + long/short asymmetry (95% of net on one side)
PULSE-13h	+23.09	placebo (+30m stale books +\$23.95 > real + \$23.09) + strip-5 → −\$1.89 + OOS −\$10.82
FADE1	−13.05	negative expectancy — reported as PULSE's mirror control, which it passed
RNGX60	−12.02	negative expectancy unconditionally, both sides, every strip
VWRC	−8.59	negative expectancy; d_vwap direction AUC 0.494 predicted it
OAB	+18.23	parameter sweep — 77% of a 192-cell surface negative, sign flips between adjacent arm levels, positives only where n falls to 24–30
MOMBRK	+32.87	strip-the-best (top 3 = 88.5% of net; strip-3 → +\$4.02, strip-5 → −\$11.37) + OOS −\$16.50 + 7 winning days of 21, two of them 97% of the money
BOOKLEAN	+12.00	n = 9. L2 lean carries no directional information (AUC 0.508)
the section as a whole	—	the NULL-CLOCK control (§9): a trigger-free rule books +\$47.29/tr, beating every gate built; and the day-block bootstrap (§10) puts both leaders' 95% CI across zero

15. VERDICT

NOTHING SURVIVED. Nothing ships — not to live, not to shadow.

What killed the two leaders, individually and by name:

- **MOMBRK (+\$32.87/trade, n=52)** — killed by **strip-the-best-trades** and the **OOS leg**. The top 3 of 52 trades are 88.5% of the net; removing them takes +\$32.87 to +\$4.02, and removing 5 takes it to −\$11.37. Out of sample (08-17 → 08-21) it made **−\$16.50/trade**. Seven of twenty-one days were profitable and two of them are 97% of the money. Its leave-one-day-out passed all 21 folds, which is a property of running LODO with two money days, not a clearance.
- **PULSE-13h (+\$23.09/trade, n=95)** — killed by the **placebo/shuffle test**, confirmed by **strip-5-best** and the **OOS leg**. A copy of the signal delayed by 30 minutes — same window, same direction rule, same stop, same hold, just a stale trigger — books **+\$23.95/trade, more than the real signal's + \$23.09**. Stale placebos average 63% of the real signal's money. Strip 5 trades and it is −\$1.89. Out of sample it made −\$10.82/trade.

What killed the section: §9. A gate containing **no signal at all** — "enter at 13:05Z, direction = sign of the last 15 minutes, stop $0.5 \times U$, hold 30 minutes" — books **+\$47.29/trade over 26 trades**, double the best gate I built, and the identical rule 25 minutes later books −\$13.46. Everything on this page sits inside the noise band of that control. §10 says the same thing in confidence intervals: $t = 1.30$ and 1.57 , bootstrap 95% CIs of $[-\$4.14, +\$26.27]$ and $[-\$2.79, +\$52.71]$, and **561 trades $\approx$ 79 trading days** needed to resolve a \$10/trade edge in this window. We have 27.

The deliverables, which are not gates:

1. ★ **The cluster label is a clock with an anti-predictive filter bolted on.** The 08-15 fix correctly demoted OPEN/NEWS from checked-first to checked-last, taking it from 100% to **58.7%** of its own window. But its lift **over the bare clock is $0.921 \times$** — the flow and amplitude tests remove 41% of the window and keep the less run-dense half. Same answer in the tick era ($0.946 \times$). Per the brief's own instruction, this finding outranks every signal on the page. **Recommendation:** either drop the two footprint tests from the OPEN/NEWS fallback and call it what it is (US-OPEN-WINDOW, $3.09 \times$ lift on 8.87% of tape), or route the hunt through VOL-EXPANSION instead — **$4.18 \times$**

lift on 1.95% of tape, the sharpest cluster the census produces and the one it uses least.

1. ★ **The tradeable size threshold is ≈ 120 points of 30-minute follow-through.** Below it the gate loses even when right: +60...+120pt of follow-through still books **−\$4.10/trade**. Above it, **+\$184.75/trade**. Break-even hit rate is 19.2%; the best arm delivered 30.8%. This is the same ~ 120 pt number the desk's independent greenfield work reached from a different cluster with a different harness — the most durable result on this page.
1. ★ **Detection has a real footprint and it is VOLUME, not price; direction has none; and the footprint's honest sample is eleven.** Trade-count z separates ≥ 200 pt runs at AUC **0.848**, rising monotonically from 0.623 at 55pt, while 30-minute range is flat at 0.62 throughout — so this is not "wide days have wide runs". Direction tops out at AUC 0.55–0.65 and the crude momentum rule agrees with runs 54.2% of the time against a 53.4% base. On **non-overlapping** 30-minute blocks the armed sample for the ≥ 120 pt claim is **$n=11$ at 54.5% ± 29.4 pp**. And the cluster's own character, in one line: **PULSE and its exact mirror catch the same 6 of the 10 sat-out runs**. Detection is solved. Selection is not.
1. **A test-design finding worth keeping.** Leave-one-day-out cannot fail when k days carry the money and you only remove one. MOMBRK passed 21/21 LODO folds while two of twenty-one days held 97% of its P&L. **Always quote profitable-days-out-of-total next to a LODO result**, and run leave-two-out when the day distribution is that skewed (MOMBRK's worst leave-two fold: +\$1.19/tr).
1. **A harness finding.** A signal shifted backwards is a look-ahead oracle, not a placebo. Only forward (stale) shifts are legitimate placebos; the backward shifts belong in the report as a harness leak-check, which this harness passed (look-ahead earns $2.7\times$ the real signal, as it should). Reported because I ran it the wrong way first and it would have produced a false kill.

Full working, every sweep and every null: this file. Tick-honesty audit \$2.4 — 40/40 entries and 40/40 exits inside their own bar's tick range, MNQ only, \$1.50 per round trip, \$2.00 per point.

FLOW-LED — aggressor-driven continuation

NOT RUN THIS WEEK · dossier 2026-08-25

Runs the tape pushed. Nine candidates built off the flow print.

GREENFIELD HUNT — cluster FLOW-LED

Full working, all attempts, all graves

MNQ · tick-honest over the full Parquet lake, 2026-07-05 → 2026-08-25 (35 tape days, 45,466,604 ticks) Costs: MNQ \$2.00/point, fee **\$1.50 per round trip**, 1 tick (0.25pt) adverse slippage on entry and on every stop.

0. The headline, before any signal

The cluster label is not a footprint. FLOW-LED and VACUUM are a coin flip, and the coin is flipped by the future.

The census assigns FLOW-LED when the prior-60s net aggressor flow is unusual ($|flow-z| \geq 1.0$) and its sign agrees with the FORWARD 15-minute move; VACUUM is the identical test with the signs disagreeing. The second half of that definition is only knowable after the run has happened. At the moment a gate would have to fire, both labels look exactly the same.

Measured on 43,982 minute bars of lake tape that carry a valid flow-z and a resolvable +15min close:

TEST	RESULT
base rate of the flow-event half ($ fz \geq 1.0$) on all bars	20.58 %
given a flow event, flow agrees with the next 15 min	49.62 % (n = 9,022)
control — bars with $ fz < 1.0$, same question	49.29 % (n = 34,786)
⇒ information carried by the flow event about direction	0.33 percentage points
FLOW-LED base rate on all bars	10.21 %
VACUUM base rate on all bars	10.37 %

The flow event does **not** meet the "true on most of the tape" failure mode — 20.6 % is a real filter, and the 2026-08-01 z-score fix that replaced $abs(flow) > 50$ (66.5 % of bars) did its job. The label fails for a different and worse reason: **the filter fires on a fifth of the tape and then splits that fifth into two almost**

exactly equal halves on a coin toss. FLOW-LED vs VACUUM is not two tape behaviours, it is one population cut by its own outcome.

Directly:

E[fwd15 in the flow's direction]			
fz >=1.0	-0.299 pt	n=9,053	t = -0.61
fz >=1.5	+0.408 pt	n=4,594	t = +0.54
fz >=2.0	+0.067 pt	n=2,655	t = +0.06
fz >=2.5	-0.029 pt	n=1,660	t = -0.02

Zero at every threshold, with the sign wandering. The census's own 4 FLOW-LED / 13 flow-event runs (30.8 % agreement) is **not** evidence of a contrarian tape: against the lake's 49.62 % null, $P(\leq 4 \text{ of } 13) = 0.139$. Expected FLOW-LED among 68 census runs = 9.0, observed 4; expected VACUUM = 9.6, observed 9. Both inside noise. **The cluster has n=4 because 68 runs is too few to populate a 10 %-base-rate label, not because the tape did something.**

0b. Flow does predict magnitude — but that is a volume channel wearing a flow costume

CONDITION	E FWD15	VS BASE
base (all bars)	26.70 pt	1.000×
fz ≥ 1.5	33.64 pt	1.260×
fz ≥ 3.0	41.73 pt	1.563×
volume-z ≥ 2.0	41.75 pt	1.564×
flow unusual, volume ORDINARY (fz ≥1.5 & volz <1.0)	25.70 pt	0.963×
volume unusual, flow BALANCED (volz ≥1.5 & fz <0.5)	32.56 pt	1.220×

Strip the volume out of the flow event and the magnitude lift vanishes entirely (0.963× — below base). Strip the flow out of the volume event and the lift survives (1.220×). **The census's flow column is a volume proxy.** Net aggressor imbalance adds nothing over "a lot traded".

That is the strongest statement this section can make, and it outranks everything below.

1. Method

Everything derives from ticks in `gazbot7.lake` (2026-07-05 → 08-25; the documented 07-18→07-23 V5/V7 seam is absent, weekends absent — 35 tape days, 44,847 minute buckets). `capture.db` was never touched: it holds 5 trading days and would have shrunk the sample 7×.

- **No lookahead.** A minute row at `ts` covers ticks `[ts, ts+60)`, so it is only known at `ts+60`. Every entry is stamped `ets = ts+60`. ATR is `.shift(1)`-ed off closed 5-min bars; every z-score uses `closed='left'` on a trailing 120-min window with ≥ 30 buckets, exactly the census recipe.
- **Contiguity guard.** `er30/brk30` use `.shift(30)`; across a weekend or the V5/V7 seam that silently compares two different tapes. 97.5 % of rows pass the guard; the rest are dropped, not used. (The two largest "minute moves" in the tape, 408.5pt and 295.75pt, are both 00:00:00 Monday weekend gaps — they are excluded by this guard, not traded.)
- **Tick-honest exits.** Entry at the first print $\geq ets$, filled 1 tick adverse. Stops fill 1 tick adverse to the stop price; targets fill at the limit; the time stop fills at the print. **Stop is tested before target on every print** (the pessimistic ordering). One position at a time.
- **Regimes** keyed on ATR level + ER + range-break, not the clock: DEAD-CHOP (`atr_rel<0.70 & er30<0.30`), VIOLENT-WHIPSAW (`atr_rel $\geq$ 1.6 & er30<0.35`), CLEAN-TREND (`er30 $\geq$ 0.55`), BUILDING (`0.35 $\leq$ er30<0.55`), NORMAL-CHOP otherwise. Sessions ASIA / LONDON / US / LATE.

Engine integrity check (run before believing any positive number)

Random-direction entries on the **identical** row set, five seeds:

seed 0	n=221	\$ -5.27/tr	seed 3	n=222	\$ -13.59/tr
seed 1	n=219	\$ -11.05/tr	seed 4	n=224	\$ -2.24/tr
seed 2	n=220	\$ -1.30/tr			

All negative, as a healthy engine must be — a coin flip pays the fee and the slippage. There is no drift bug manufacturing the numbers below.

2. Every candidate, blanket screen

Stop $1.0 \times \text{ATR}$, target $1.5 \times \text{ATR}$, 15-min time stop (the census horizon). **The blanket number is context, never the verdict** — per-regime follows.

#	CANDIDATE	SPEC	N	NET	\$/TR	WIN%
FL-1	flow_impulse_cont	$ fz \geq 1.5 \rightarrow$ trade in flow's direction	2,306	−\$4,280.30	−1.86	44.5 %
FL-2	flow_impulse_fade	$ fz \geq 1.5 \rightarrow$ trade AGAINST flow (the VACUUM mirror)	2,323	−\$8,244.89	−3.55	43.7 %
FL-3	flow_persist_cont	3 consecutive same-sign flow minutes, $ 3\text{-min } z \geq 1.5$	1,411	−\$3,080.93	−2.18	44.6 %
FL-4	flow_ignition	$ fz \geq 1.5$ and price already broke $0.5 \times \text{ATR}$ in flow's direction	1,496	+\$1,409.59	+0.94	45.6 %
FL-5	flow_absorb_fade	$ fz \geq 1.5$ and $ 1\text{-min move} \leq 0.15 \times \text{ATR}$ (absorption) $\rightarrow$ fade	832	+\$452.23	+0.54	46.2 %
FL-6	imb_cont	z of the imbalance ratio (flow/volume), $\geq 1.5 \rightarrow$ continuation	2,343	−\$7,451.77	−3.18	44.4 %
FL-7	vol_ignition_dir	volume- $z \geq 1.5$, direction from the bar's own sign (the volume control)	1,774	−\$3,850.50	−2.17	42.6 %
FL-8	flow_x_vol_cont	$ fz \geq 1.5$ and volume- $z \geq 1.5$	1,287	−\$1,637.89	−1.27	43.3 %
FL-9	flow_lot	net aggressor flow restricted to prints $\geq L$ lots ($L = 2/5/10/25/50$)	see §5	—	−1.63 ... −2.85	~44 %

Six of nine are outright negative blanket. The two that clear zero (FL-4 +\$0.94, FL-5 +\$0.54) clear it by less than a tick. **Per the standing judging rule these are not killed on win-rate** — 42-46 % is normal for this desk, which booked +\$625 this week at 39 %. They are killed, below, on the robustness battery.

3. Per-regime — the cells that had a pulse

40 candidate×regime cells were scored. Four came back with a live-looking expectancy:

CELL	N	NET	\$/TR	WIN%
FL-1 flow_impulse_cont @ BUILDING	451	+\$2,495.07	+5.53	47.9 %

FL-4 flow_ignition @ VIOLENT-WHIPSAB	170	+\$2,070.12	+12.18 52.4 %
FL-8 flow_x_vol_cont @ CLEAN-TREND	50	+\$630.91	+12.62 52.0 %
FL-5 flow_absorb_fade @ CLEAN-TREND	28	+\$706.32	+25.23 57.1 %

△ These four were selected out of 40 cells. At a 50/50 null you would expect several positive cells for free. The selection is why the battery below is the deliverable and the table above is not.

4. The battery — every grave, by name and cause of death

💀 FL-1 flow_impulse_cont @ BUILDING — KILLED BY THE OOS LEG

home (full tape) n=451 net=+\$2,495.07 \$/tr= +5.53
 IS (first 20d) n=243 \$/tr=+11.72 OOS (last 15d) n=208 \$/tr= -1.69
 strip 3 best \$/tr= +3.57 strip 5 best \$/tr= +2.48
 LONG n=210 \$/tr=+0.76 SHORT n=241 \$/tr=+9.69 <- asymmetric, all of it is short-side
 leave-one-day-out \$/tr min +3.70 / med +5.59 / max +6.69 (20/33 days profitable)
 PLACEBO (flow shifted ±390..1560 min) \$/tr -6.65..+6.23, 1/6 placebos beat the real signal

Cause of death: OOS sign flip (+\$11.72 → −\$1.69/tr), with a failed long/short symmetry test as the corroborating wound — 94 % of the net is short-only, which is a directional accident of a 35-day tape that fell, not a footprint. A placebo also beat it 1-in-6.

💀 FL-8 flow_x_vol_cont @ CLEAN-TREND — KILLED BY THE PLACEBO AND STRIP-THE-BEST

n=50 net=+\$630.91 \$/tr=+12.62
 top-3 trades = 80.5% of net ; strip 5 best -> \$/tr = -3.14 (NEGATIVE)
 IS \$/tr=+25.27 (n=31) OOS \$/tr= -8.03 (n=19)
 PLACEBO: the shifted-flow control scored +\$20.06/tr, BEATING the real signal

Cause of death: the placebo out-earned the real signal, and stripping 5 trades turns it negative. Triple-killed — OOS is negative too.

💀 FL-5 flow_absorb_fade @ CLEAN-TREND — KILLED BY STRIP-THE-BEST

n=28 net=+\$706.32 \$/tr=+25.23
 top-3 trades = 96.1% of net ; strip 5 best -> \$/tr = -7.63 (NEGATIVE)

Cause of death: strip-the-3-best. 96.1 % of the money is three trades; the remaining 25 trades are worth +\$27.84 in total, i.e. +\$1.11/tr, which the fee eats. n=28 across 17 days is below the threshold at which a placebo test even means anything, so this never reached one.

☠ **FL-2, FL-3, FL-6, FL-7 — KILLED AT THE BLANKET SCREEN, no regime cell rescued them**

- **FL-2 flow_impulse_fade** −\$3.55/tr blanket. The direct test of "the census saw 9 VACUUM vs 4 FLOW-LED, so fade the flow". It is the worst candidate in the suite. Its only positive cell, CLEAN-TREND, is n=55 at +\$5.77/tr, i.e. inside the same 40-cell selection noise. **Cause of death: negative expectancy blanket and in 4 of its 5 regimes.**
- **FL-3 flow_persist_cont** −\$2.18/tr. Requiring three consecutive same-sign flow minutes (a working order, not one print) reduced n from 2,306 to 1,411 and left expectancy unchanged. **Cause of death: negative expectancy; persistence adds no information.**
- **FL-6 imb_cont** −\$3.18/tr. Volume-normalising the flow (imbalance ratio instead of net contracts) makes it worse, not better. **Cause of death: negative expectancy.**
- **FL-7 vol_ignition_dir** −\$2.17/tr. The volume control from \$0b. Confirms that volume's magnitude lift does not convert into a directional edge either. **Cause of death: negative expectancy.**

☠ **FL-4 flow_ignition @ VIOLENT-WHIPSAW — the one that got all the way to the end**

This was the only candidate to survive strip-the-best, long/short symmetry, leave-one-day-out and a 6-shift placebo. It took the full battery to kill it, so it is documented in full.

Spec (as it would have shipped): at each minute close, if $|\text{flow-z}| \geq 1.5$ **and** the 30-minute signed range-break $(\text{close} - \text{close}_{-30})/\text{ATR}$ is $\geq +0.5$ in the flow's direction, enter at market on the next print in the flow's direction. Stop $1.0 \times \text{ATR}$, target $1.5 \times \text{ATR}$, flat at 15 minutes. Armed only when $\text{atr_rel} \geq 1.6$ and $\text{er30} < 0.35$.

Survived:

home regime	n=170	net=+\$2,070.12	\$/tr=+12.18	win=52.4%	median +\$3.75
strip 3 best		\$/tr= +8.08	strip 5 best	\$/tr= +5.49	(top-3 = 34.8% of

```

net)
LONG  n=70  $/tr= +8.51    SHORT n=100 $/tr=+14.75    <- symmetric
leave-one-day-out  $/tr min +8.68 / med +12.23 / max +15.04  (22/33 days profitable)
net without the best day  +$1,415.11  (of +$2,070.12)
6-shift placebo      0/6 beat it
parameter plateaus stop 1.0-2.0xATR all +$10..+16 ; target 0.75-3.0xATR all
positive ;
                                time stop 5-45 min all positive ; break threshold 0.0-1.5 all
positive

```

Died on:

(1) The 40-shift placebo — empirical $p = 0.150$. Shifting only the flow series $\pm 120 \dots 2400$ minutes and keeping every other rule identical:

```

real FL-4  $/tr = +12.18
placebo null (n=40)  mean +2.60  sd 8.37  min -13.81  MAX +16.25
placebos >= real: 6/40    ->  p = 0.150

```

Six fake flow series matched or beat it, and the best fake beat it outright. That is the operator's own stated kill criterion: if a FAKE signal books most of the money, the real one is not the edge.

(2) The paired ablation — the flow term contributes nothing out of sample. Holding the breakout population fixed and splitting it by what flow was doing:

BREAKOUT POPULATION, SPLIT BY FLOW	N	\$/TR	IS	OOS
flow AGREES ($ fz \geq 1.5$) — this is FL-4	72		+13.15	+39.43 -11.71
flow QUIET ($ fz < 1.0$)	338		+4.92	+8.13 +1.44
flow OPPOSES	30		-1.88	-4.28 +1.72

In-sample the ordering is beautiful and monotone (agrees $\gg$ quiet $>$ opposes) — which is exactly what a fitted term looks like. Out of sample it **inverts**: the flow-agreement bucket is the worst of the three, and the only stable positive bucket is the one where flow is doing nothing at all.

(3) Five-fold walk-forward — 3/5, and worse than the flow-free version.

fold		FL-4 (flow+break)	break only, NO flow term
F1 07-05..07-12	n= 25	+\$26.19/tr +\$654.82	n= 65 \$+19.30/tr +\$1,254.50
F2 07-13..07-27	n= 42	\$ -4.02/tr -\$168.71	n=123 \$ +1.20/tr +\$147.59
F3 07-28..08-05	n= 25	+\$62.54/tr +\$1,563.38	n= 91 \$+10.16/tr +\$924.70
F4 08-06..08-14	n= 38	\$ -9.13/tr -\$347.00	n=112 \$ +0.49/tr +\$54.38

F5 08-17..08-25	n= 40	\$ +9.19/tr	+\$367.64	n=100	\$ +5.49/tr	+\$548.52
folds profitable: 3/5				5/5		

Adding the flow term costs two folds and 65 % of the sample.

(4) Block bootstrap CI straddles zero. 10,000 day-block resamples: \$/tr = +12.18, **95 % CI [-2.71, +27.70]**, $P(\$/tr > 0) = 0.945$. It cannot clear a two-sided bar even before the $p=0.150$ placebo is accounted for.

(5) The regime it was armed for is half fiction. Sweeping the home definition:

```
atr_rel >= 1.2 +$7.92 | 1.4 +$13.02 | 1.6 +$12.18 | 1.8 +$10.08 | 2.0 -$0.62
er30 < 0.25 +$16.35 | 0.35 +$12.18 | 0.45 +$13.05 | 0.55 +$12.62 | 1.01 (no ER
filter at all) +$12.66
```

The ATR half is a genuine plateau. **The ER half does nothing** — removing the ER condition entirely changes \$/tr by 4 cents. So "VIOLENT-WHIPSAW" was never the home regime; "high ATR" was, and half the router condition was decoration.

(6) It is a US-session-only signal whose US-session OOS is negative. 192 of 212 trades sit in 13:30–20:00 UTC: US $n=192$, \$/tr +\$13.44, **IS +\$32.07 → OOS −\$8.13.**

CAUSE OF DEATH — FL-4: the PLACEBO test ($p = 0.150$, 6/40 shifted-flow controls beat it, best fake +\$16.25 vs real +\$12.18), confirmed by the paired OOS ablation in which the flow term's contribution flips from +\$39.43/tr in-sample to −\$11.71/tr out-of-sample.

5. The escalation the brief ordered — and the size-threshold answer

5a. Hunt the full sat-out set → top-25 → top-15

Against the frozen census's 68 runs (62 sat out), asking how many FL-4 caught in the correct direction inside a $\pm 5/+10$ min window:

	CENSUS RUNS CAUGHT	ON THOSE TRADES
FL-4 flow_ignition	10 / 68	+\$254.14
the flow-free breakout	18 / 68	+\$1,323.80
FL-4 on the 4 FLOW-LED runs (its own cluster)	1 / 4	—
FL-4 on the top-15 biggest runs	3 / 15	—

Narrowing did **not** reveal a footprint the marginal runs washed out — it went the other way. FL-4 catches its own cluster 1-in-4 and the biggest runs 3-in-15, worse than its 10/68 overall rate.

5b. The size threshold, asked on both sides

Predictor side — raise the flow bar and read expectancy (home = atr_rel≥1.6, break≥0.5):

FLOW-Z BAR	N	\$/TR	IS	OOS
≥1.0	296	+2.32	+11.81	−8.99
≥1.5	212	+12.66	+26.37	−3.29
≥2.0	157	+14.46	+40.61	−12.71
≥2.5	122	+4.26	+27.40	−18.89
≥3.0	96	+3.96	+25.07	−17.14
≥3.5	69	−5.39	+12.69	−22.95
≥4.0	43	+1.90	+26.90	−21.96

Every single threshold is positive in-sample and negative out-of-sample. The OOS column does not merely decay with size, it gets worse. That is the signature of a fitted term, not a footprint that sharpens.

Outcome side — flow agreement as runs get bigger, over all 35 days:

RUN SIZE	N	FLOW-LED	VACUUM
fwd15 ≥ 55pt	1,367	48.4 %	51.6 %
≥ 80pt	677	49.5 %	50.5 %
≥ 100pt	401	48.6 %	51.4 %
≥ 120pt	249	50.2 %	49.8 %
≥ 150pt	147	53.7 %	46.3 %
≥ 200pt	57	50.9 %	49.1 %
≥ 250pt	14	too few	

The only departure from 50/50 is 53.7 % at ≥150pt on n=147 — **0.9 sigma**, and it does not survive to the ≥200pt bucket. There is no size at which the tape starts telling you which way it is going.

💀 5c. FL-9 flow_lot — KILLED BY THE PARAMETER SWEEP (no plateau)

The census counts every contract equally: one 250-lot sweep scores the same as 250 one-lots. If a footprint hides anywhere it should hide in the blocks.

directional accuracy of large-lot flow, z >=1.5, all bars					
lot>=2	n=4,622	agree 48.55%	E[fwd15 in dir]	+0.74pt	t=+0.98
lot>=5	n=4,732	agree 48.77%		+0.74pt	t=+1.02

lot>=10	n=4,571	agree	49.60%	+0.81pt	t=+1.11
lot>=25	n=4,028	agree	50.15%	-0.13pt	t=-0.16
lot>=50	n=1,580	agree	50.70%	+0.89pt	t=+0.62

tick-honest replay, continuation

lot>=2	blanket	-\$2.49/tr	home(atr_rel>=1.6)	-\$0.85/tr	
lot>=5	blanket	-\$2.85/tr	home	-\$2.58/tr	
lot>=10	blanket	-\$2.30/tr	home	-\$2.45/tr	
lot>=25	blanket	-\$1.71/tr	home	+\$6.05/tr	<- the only positive
cell					
lot>=50	blanket	-\$1.63/tr	home	-\$1.75/tr	

Cause of death: the parameter sweep. Every lot threshold is negative blanket, agreement never leaves 48.5–50.7 %, and the single positive cell (lot≥25 on the high-ATR home, +\$6.05/tr) is an isolated spike with −\$2.45 immediately below it and −\$1.75 immediately above. An edge that exists at 25 lots but not at 10 or 50 is a bucket, not a plateau.

6. The residue — and why it is not this section's survivor

The flow-free half of FL-4 is the only thing on this page with a pulse after the battery:

plain 30-min range-break, $\geq 0.5 \times \text{ATR}$, in high ATR ($\text{atr_rel} \geq 1.6$). No flow term whatsoever. n = 491, net +\$2,929.68, \$/tr +\$5.97, win 50.5 %, 5/5 walk-forward folds profitable, 18/68 census runs caught for +\$1,323.80. Exit mix: TIME 285 (+\$20.6/tr) · TGT 102 (+\$150.8/tr) · STOP 158 (−\$112.9/tr).

I am **not** reporting this as a survivor, for three reasons, and the operator should treat it as a lead for the structure-break phase rather than a gate:

1. Its block-bootstrap 95 % CI is **[−2.12, +14.36]** — it straddles zero exactly as FL-4 did.
2. Its own IS/OOS split is +\$10.38 → **+\$1.04/tr**. Positive, but \$1.04 is a third of the fee.
3. Its US-session leg, which is 483 of its 491 trades, goes **IS +\$13.64 → OOS −\$2.15**.

More importantly: **it is not a flow gate, and it is not this cluster**. Reporting a breakout gate as the FLOW-LED cluster's finding would be the exact error the census made — attaching a flow name to something the flow had no part in.

Router rule, if it is ever picked up by another phase (stated in live-router vocabulary so it is falsifiable): ARM when `atr_rel` ≥ 1.4 (ATR-14 on 5-min bars vs its trailing 24h median) **and** `session` = US; BENCH otherwise; no ER condition — the ER term was measured to be worth 4 cents. Note that `atr_rel` is **not currently an instrument the live router reads** — the router's LAST HR ATR is a mean range, not the true-range ATR-14 this is keyed to, so arming on the router's existing number would arm on the wrong measurement. That is the instrument we would need to build first.

7. What would have to be true for this cluster to be worth revisiting

The FLOW-LED hypothesis is not refuted in principle — it is refuted with the instrument we have. The census's flow is a 60-second net-aggressor count from `ticks.aggressor`, and `$0b` shows it is a volume proxy. Three measurements would make it a fresh question rather than a repeat of this one:

- 1. **Resting-liquidity flow, not print flow.** `capture.db.book` is 41 ms event-driven and MNQ-only. The question worth asking is not "who crossed the spread" but "did the far side get pulled" — that is a book measurement, and it is the one the census's book column already gestures at.
 - 2. **Flow at a level, not flow in a clock minute.** A fixed 60-second bucket splits a single institutional order across two buckets and merges two unrelated ones. Volume-clock or order-imbalance-at-price buckets are the standard fix, and the tick data supports both.
 - 3. **A run population large enough to populate a 10 %-base-rate label.** 68 runs cannot; the lake's 5,066 bars at ≥ 55 pt can. Any future cluster work should be labelled on the lake and only illustrated on the week's census.
-

8. Summary table

CANDIDATE	BEST CELL	\$/TR	FATE	CAUSE OF DEATH
FL-1 <code>flow_impulse_cont</code>	BUILDING	+5.53	👻	OOS leg (+11.72 $\rightarrow$ -1.69), long/short asymmetry
FL-2 <code>flow_impulse_fade</code>	—	-3.55	👻	negative expectancy blanket + 4/5 regimes
FL-3 <code>flow_persist_cont</code>	—	-2.18	👻	negative expectancy; persistence adds nothing

FL-4 flow_ignition	VIOLENT-WHIPSAW	+12.18 🐼	placebo p=0.150 ; paired OOS ablation flips +39.43 → −11.71
FL-5 flow_absorb_fade	CLEAN-TREND	+25.23 🐼	strip-the-best (top-3 = 96.1 % of net; strip-5 → −7.63)
FL-6 imb_cont	—	−3.18 🐼	negative expectancy
FL-7 vol_ignition_dir	—	−2.17 🐼	negative expectancy (volume control)
FL-8 flow_x_vol_cont	CLEAN-TREND	+12.62 🐼	placebo beat it (+20.06 vs +12.62); strip-5 → −3.14; OOS −8.03
FL-9 flow_lot (5 lot sizes)	atr_rel≥1.6, lot≥25	+6.05 🐼	parameter sweep — isolated cell, no plateau
(residue, not a survivor) break-only, high ATR	US session	+5.97 —	bootstrap CI [−2.12, +14.36]; OOS +1.04; not a flow gate

VERDICT

NOT SURVIVED. No FLOW-LED entry signal reached a tradeable expectancy. Nine candidates were built and all nine are dead: FL-2/3/6/7 on negative expectancy at the blanket screen, FL-1 on the **OOS leg**, FL-5 and FL-8 on **strip-the-3-best** (with FL-8 also beaten outright by its own **placebo**), FL-9 on the **parameter sweep**, and the last survivor FL-4 flow_ignition on the **PLACEBO/SHUFFLE TEST — empirical p = 0.150, with 6 of 40 time-shifted fake flow series matching or beating it and the best fake (+\$16.25/tr) beating the real signal (+\$12.18/tr) outright** — corroborated by a paired ablation in which the flow term's contribution inverts from +\$39.43/tr in-sample to −\$11.71/tr out-of-sample, and by 3/5 walk-forward folds against 5/5 for the same gate with the flow term deleted.

There is no size threshold at which a FLOW-LED footprint becomes tradeable. On the predictor side every flow-z bar from 1.0 to 4.0 is positive in-sample and negative out-of-sample. On the outcome side, flow agreement with the run is 48.4 %–53.7 % from 55pt to 200pt and never departs from a coin flip by as much as one sigma.

★ **The finding that outranks all of the above: the cluster label itself is not a footprint.** The flow-event filter fires on 20.58 % of all bars and then splits that fifth into FLOW-LED (10.21 % of tape) and VACUUM (10.37 %) on a 49.62 % coin toss — 0.33 percentage points of directional information versus the $|fz| < 1$ control. Whatever lift the flow event carries is a **magnitude** lift, and that lift is entirely a

volume proxy: flow unusual with volume ordinary gives $0.963 \times$ base $E|fwd15|$ (no lift at all), while volume unusual with flow balanced gives $1.220 \times$. The census's flow column measures how much traded, not who was buying.

Recommendation: stop splitting the run census into FLOW-LED/VACUUM. The two labels are one population cut by its own outcome, and the week's 4-vs-9 split is noise (expected 9.0 vs 9.6; $P(\leq 4 \text{ of } 13) = 0.139$ against the lake null).

VACUUM — the snap-back / stop-run NOT RUN THIS WEEK · dossier 2026-08-22

Runs that moved AGAINST the tape.

GREENFIELD HUNT — cluster VACUUM (census 2026-08-16 → 2026-08-21)

The job. Ignore every gate we own. Invent a brand-new entry signal that shows up for the runs the census filed under **VACUUM**, spec it mechanically, backtest it tick-honest over the **full Parquet lake** at the **true \$1.50 round-trip fee**, escalate to the biggest runs if it fails, and try hard to kill it. Working below; every attempt is named, including the graves.

THE THREE ANSWERS, IN ORDER OF HOW MUCH THEY MATTER

1. ★ The VACUUM label is not a footprint — and this week I found where it comes from. Last Friday's section established that the label is a coin flip. This week's larger sample (33 trading days, 43.2M ticks, 42,143 tape-minutes) confirms that and goes one step further. The flow event the label is built on fires on **20.41% of every minute of tape we own**. Given that event, the direction test comes up "VACUUM" **49.90%** of the time — against **49.91%** on minutes where no flow event happened at all. That is a **−0.01pp** difference. But the census still reports VACUUM on 15.7% of its runs, well above the 10.2% base rate, and I traced the whole of that excess to a **selection artefact in run_census.py's own run-picker**. The picker takes every 60s-slid 15-min window over threshold and dedupes "keep-strongest", so the surviving minute is by construction the local extremum of the move. Rebuilding that picker on the lake: the prior-5-minute price direction agrees with the run **48.66% of the time on unselected big minutes and 22.55% after the dedup — a −26.11pp shift created purely by the selection rule**. The census's own 65 sat-out runs read 24.6%, matching the rebuilt picker almost exactly. Push the VACUUM condition through the same three

populations and it goes **10.45% → 14.01% → 21.87%**. **VACUUM** is the name the census gives to the fact that it picked a local bottom.

2. Detection is real, it is size-graded, and it is the answer to the size-threshold question — but direction is a coin flip and I could not break it. A volume-z / ATR-expansion detector separates the biggest runs from ordinary tape at **AUC 0.69 ($p < 0.0001$)** on the 113 lake-wide runs of 120pt+, rising monotonically from 0.557 at 55-70pt. On the census's own top-15 the volume-z AUC is **0.730, $p = 0.0004$** . Turned into a bracket gate (**VAC-IGN-BRK**) it catches **9 of the 65 sat-out runs and 5 of the top-15 on only 48 fires in the census week — 3.6× the random-fire baseline**. And it still cannot be traded, because **the sign-flip control earns exactly as much as the signal** (+\$3.08 vs +\$3.21 per trade). We can see the runs coming. We cannot tell which way they go.

3. Last week's shadow candidate failed its forward test, and last week's one open stone is now closed. **VAC-ABS** reproduced to the dollar on the in-sample leg (+\$24.04/tr vs the +\$24.91 reported) and then lost **-\$529.50 over 27 trades (-\$19.61/tr)** on the five fresh days. Its strip-5 is **-\$427**. It is dead, not shadowed. And the L2 far-side-depletion discriminator that last week's report nominated as "the one stone still unturned" is **refuted**: on 21 days and 246.9M book rows it predicts the next 15 minutes' direction at **49.44% ±2.52pp**.

△ **A method note that changed every headline in this section.** My first pass let overlapping entries run concurrently. Five separate shorts booked the same 2026-07-31 impulse and were worth **\$3.9k of a \$10.3k headline**. A live gate takes one lot and cannot re-enter until flat. Everything below is scored **one position at a time**; where the unconstrained number is shown it is labelled as such and is never the verdict. Under the honest accounting **VAC-IGN-BRK** falls from **+\$10,316 / +\$15.49 per trade to +\$911.55 / +\$3.21 — 91% of the apparent edge was one impulse counted five times**.

1. What the census labelled, and what the label is made of

`scripts/run_census.py::cluster()` calls a run **VACUUM** when, in the 60 seconds before it starts, net aggressor flow was unusual for that moment ($|z| \geq 1.0$ against the trailing two hours of the same statistic) **and pointed the**

opposite way to the move that followed. Plain English: the tape was full of buyers and then price fell.

This week: **11 VACUUM runs of 70**, of which **9 sat out**, carrying **\$1,595** of 1-lot hindsight ceiling out of the week's **\$9,986** total sat-out ceiling.

#	RUN START (UTC)	DIR	MOVE	\$ CEILING	60S FLOW	AMP%	BOOK	US
1	08-17 07:27	UP	+57	\$114	−216	0.05	0.42	sat out
2	08-18 01:38	DN	−101	\$202	+458	0.07	0.49	sat out
3	08-18 13:40	DN	−138	\$276	+485	0.29	0.49	caught (exhaustion_sh ort_A, −\$208)
4	08-18 13:56	DN	−70	\$140	+1850	0.21	0.34	caught (day_rider, + \$32)
5	08-18 14:11	UP	+100	\$200	−1738	0.32	0.52	sat out
6	08-19 12:32	UP	+150	\$300	−131	0.12	0.56	sat out
7	08-19 13:59	DN	−108	\$215	+741	0.23	0.52	sat out
8	08-19 18:03	DN	−89	\$178	+552	0.19	0.54	sat out
9	08-20 05:54	UP	+56	\$112	−128	0.05	0.44	sat out
10	08-20 15:02	DN	−80	\$161	+1542	0.37	0.49	sat out
11	08-21 14:42	UP	+56	\$113	−819	0.14	0.49	sat out

Read the definition again. **The label uses the forward move to decide the label.** You cannot know a minute is VACUUM until you already know where price went for the next fifteen minutes. That is fine for a taxonomy and fatal for a trigger. The only thing knowable in advance is the **flow z-score**, so that is what I tested — and then I tested the label itself.

2. ★ THE LABEL INTERROGATION — the finding that outranks everything else here

reports/friday_v7/vacuum/label_test.py, run over the whole lake: every one of 42,143 tape-minutes scored with exactly the census's own arithmetic (60s prior net aggressor flow, z-scored against the trailing 2h of the same statistic, ≥30 non-empty buckets required).

VALUE	
tape-minutes scored	41,182
the flow event ($ z \geq 1.0$) fires on	20.41% of all minutes
VACUUM (event and direction disagrees) on all minutes	10.18%

P(direction disagrees a flow event happened)	49.90% (95% CI 48.84–50.97)
P(direction disagrees no flow event)	49.91% (95% CI 49.37–50.46)
information the flow event adds to the direction test	–0.01pp

One minute in five is a "real flow event", and knowing that tells you nothing whatsoever about which way the next fifteen minutes go. **VACUUM is a normal flow minute with a coin landed on it.**

2.1 So why does the census see 15.7% and not 10.2%?

Because the run-picker selects local extrema. `stage5_artifact.py` rebuilds the picker on the lake — every minute with $|fwd15| \geq 55pt$, then "keep-strongest" dedup at 15-min spacing — and compares three populations:

FEATURE (SIGN AGREEMENT WITH THE RUN)	ALL-BIG (NO SELECTION)	PICKED (CENSUS PICKER)	SHIFT
prior 5-min price change	48.66% (n=4,930)	22.55% (n=683)	–26.11pp
prior 15-min price change	50.39% (n=4,932)	31.38% (n=682)	–19.01pp
flow z (the VACUUM leg)	48.92% (n=4,928)	41.38% (n=679)	–7.54pp
5-min cumulative-delta z	48.95% (n=4,887)	40.75% (n=670)	–8.20pp

Verification: the census's own 65 sat-out runs read **24.6%** on prior-5-min agreement — the rebuilt picker says 22.55%, the unselected tape says 48.66%. The picker, not the tape, is doing the work.

And the label rate itself, decomposed:

POPULATION	VACUUM RATE	= FLOW-EVENT RATE	× DISAGREEMENT RATE
all scored minutes	10.45%	20.72%	50.42%
all-big minutes ($ fwd15 \geq 55pt$, unselected)	14.01%	27.05%	51.79%
PICKED (the census's own population)	21.87%	28.43%	76.92%
the census's reported week	15.71% (11/70)		

The disagreement rate goes 50.4% → 51.8% → **76.9%** purely by selecting local extrema. **The VACUUM cluster does not describe a state of the tape. It describes the census's dedup rule.** ★ This finding outranks every signal in this document and is the reason none of them was ever going to work as specified.

What this does not say. The census is not broken as a taxonomy — "where was the money and did we show up" is answered correctly, and the sat-out ceiling of

\$9,986 stands. What is unsafe is treating a cluster name as a hunting ground, and any pre-run price-path statistic quoted off the run list.

3. Every candidate, tick-honest, one position at a time

Harness (reports/friday_v7/vacuum/vaclab.py). 1-minute bars built **from the trade ticks**, so price and the aggressor split are the same object. Entry is the **first tick after the trigger bar closes** — no same-bar fill, no lookahead. Stops and targets are walked **tick by tick** over all 43,186,167 MNQ lake ticks, stop checked **before** target on every tick. Costs charged in every number: **\$1.50 per round trip** (the true venue fee) **plus 0.25pt slippage each way**, MNQ at **\$2.00/point**. Verified by hand-replaying the largest trade against the raw tick array — entry, stop price, exit and net all reconcile to the cent.

Serial accounting. One position at a time. See the ⚠ note in the summary.

CANDIDATE	N	NET	\$/TR	WIN%	T	SAT-OUT CAUGHT	TOP-15	CAUSE OF DEATH
VAC-Z1 — trade the label literally	1,768	−\$4,150.79	−2.35	35.7%	—	16/65	3/15	Refuted outright. The label has no edge; unconstrained it loses — \$32,184 over 8,589 trades.
VAC-ABS — 08-15 spec, forward-tested	97	+\$1,319.21	+13.60	38.1%	1.39	0/65	0/15	OOS leg — \$529.50 / — \$19.61 per trade on the 5 fresh days, strip-5 — \$427 , t=1.39. Dead.
VAC-ABSGO — its sign-flip control	96	−\$2,290.43	−23.86	22.9%	—	—	—	Working as intended — confirms the mechanism is directional even though the edge is gone.
VAC-SNAP — absorption → structural break	410	+\$370.64	+0.90	34.9%	0.17	6/65	2/15	Strip-3 — \$1,436 (top-5 trades = 738% of net), OOS — \$18.02/tr, LOO goes negative, 12/33 days green.

VAC-SNAP/GO — its sign-flip control	428	−\$625.71	−1.46	39.0%	—	—	—	Control behaves; the corpse is still a corpse.
VAC-CUMDIV z1.5 / z2.0	506 / 264	−\$1,814.64 / −\$1,668.71	−3.59 / −6.32	32%	—	4/65	2/15	Refuted outright, both parameterisations, both directions of the sweep.
VAC-IGN-BRK — the detection bracket	284	+\$911.55	+3.21	32.4%	0.45	9/65	5/15	SIGN-FLIP: the fade earns + \$3.08/tr against the signal's + \$3.21. Both directions win = volatility, not edge. Also strip-3 −\$915, no plateau.
CTRL-UNCOND — same break, arm every minute	2,404	−\$4,657.54	−1.94	29.7%	—	30/65	7/15	Control. Proves the arm is load-bearing for the detection , and that the break rule alone loses.
VAC-ABS placebo +240m	92	+\$239.00	+2.60	37.0%	—	—	—	A fake trigger still earns — further evidence VAC-ABS's remaining number is noise.
L2 far-side depletion (direction discriminator)	—	—	—	—	—	—	—	Refuted. 49.44% ±2.52pp on unselected big moves. §6.

Nulls that never reached a spec, recorded so nobody re-walks them: **flow-sign continuation** (z5 agreement 48.95% on unselected big moves), **near-30-min-extreme position** (49.05% on all minutes), **coil ratio** (AUC 0.501 on the sat-out set — no detection at all).

3.1 The three mechanical specs, for the record

VAC-SNAP — invented here, refuted here. On completed 1-min bars: $\text{cumflow5} = \Sigma \text{ netflow over the last 5 bars}$; z5 = its z-score against the trailing 2h of the same statistic. **ARM** when $\text{z5} \geq +1.5$ (buyers) and the 5-bar price change is $\leq 0.35 \times \text{ATR}(1\text{m}, 14)$ — heavy one-sided aggression that failed to move price. **TRIGGER** within 10 bars when a bar closes **below the arming window's low** → **SHORT** (mirror for sellers absorbed → LONG on a close above the window high). **ENTRY**

first tick after that bar closes. **STOP** $2.0 \times \text{ATR}(1\text{m},14)$. **EXIT** stop or 30-min time-stop.

VAC-IGN-BRK — invented here, the one that detects. $\text{volz} = \text{z-score of bar volume against the trailing 120 bars}$; $\text{atr_ratio} = \text{ATR}(1\text{m},14)$ averaged over 5 bars $\div$ over 60 bars. **ARM** when $\text{volz} \geq 2.0$ **and** $\text{atr_ratio} \geq 1.15$. **TRIGGER** within 10 bars when a bar closes outside the 5-bar range — **above the high $\rightarrow$ LONG, below the low $\rightarrow$ SHORT**. Non-directional by construction: the tape picks the side, the gate only picks the moment. **ENTRY** first tick after that bar closes. **STOP** $1.5 \times \text{ATR}(1\text{m},14)$. **EXIT** stop or 30-min time-stop.

VAC-ABS — last week's spec, unaltered, forward-tested. $|\text{z}(\text{netflow})| \geq 3.0$ **and** $|\text{close} - \text{prev close}| \leq 2.75\text{pt} \rightarrow$ fade the aggressors. Stop $2.0 \times \text{ATR}$, 30-min time-stop.

4. VAC-ABS — the forward test, in full

This is the honest part of last Friday's deliverable coming due. The spec was frozen on 2026-08-15; the lake now holds five trading days it has never seen.

LEG	N	NET	\$/TR	WIN%
v5 archive, $\leq 07-17$	17	+\$439.14	+25.83	—
in-sample, 07-24 ... 08-15	53	+\$1,409.57	+26.60	—
OUT OF SAMPLE, 08-17 ... 08-21	27	-\$529.50	-19.61	28.6%
all	97	+\$1,319.21	+13.60	38.1%

The in-sample leg reproduces last week's $+\$24.91/\text{tr}$ almost exactly ($+\$26.60$ here; the small gap is the serial constraint dropping 6 overlapping fires). **The forward leg is negative and it is negative in the home regime too** — the $\text{ATR} \geq 15\text{pt}$ cell that last week's router rule was written around books 4 OOS trades for **$-\$154.64$** . Add **strip-5** = **$-\$427$** and **t = 1.39** and there is nothing to defend.

Cause of death: the out-of-sample leg, confirmed by strip-5. Not the win rate — 38.1% would have been perfectly acceptable per the operator's judging rule if the expectancy had held up. It did not.

5. ★ THE SIZE THRESHOLD — the escalation, and the one place it pays off

The mandate asks for the escalation and for the size at which a footprint becomes tradeable. Running the census picker across the **whole lake** gives **686 runs** instead of 70, so the bands have real n. AUC is against a random lake minute; p is a 4,000-draw permutation of the same n.

BAND (MOVE)	N	VOLUME-Z AUC	ATR AUC	FLOW Z AUC	PRIOR-5MIN DIRECTION
55-70 pt	257	0.557 (p=0.0003)	0.616 (p<0.0001)	0.558 (p=0.0005)	19.8%
70-90 pt	188	0.542 (p=0.026)	0.670 (p<0.0001)	0.488 (p=0.73)	26.9%
90-120 pt	128	0.567 (p=0.004)	0.801 (p<0.0001)	0.552 (p=0.021)	20.5%
120 pt +	113	0.690 (p<0.0001)	0.867 (p<0.0001)	0.634 (p<0.0001)	23.9%

And on the census's own week, permutation-tested at n=15/25/65:

FEATURE	SAT-OUT 65	TOP-25	TOP-15
volume-z	0.597 (p=0.0029)	0.637 (p=0.0089)	0.730 (p=0.0004)
ATR(1m,14)	0.553 (p=0.071)	0.583 (p=0.072)	0.676 (p=0.0081)
flow z	0.522 (p=0.27)	0.525 (p=0.34)	0.679 (p=0.0088)
ATR expansion ratio	0.529 (p=0.21)	0.508 (p=0.44)	0.674 (p=0.0093)
ER15		0.442	0.442 0.403 — big runs start from chop, replicating last week

The answer to the size-threshold question: yes, there is one, and it is real — but it is a DETECTION threshold, not a tradeability threshold. Three of the four detectors are indistinguishable from noise on the full sat-out set and all four become significant at the **top-15 band, |move| ≥ 87pt**; lake-wide the effect keeps climbing to **AUC 0.867 on ATR at 120pt+**. Narrowing revealed a footprint exactly as the mandate hoped it might.

And it converts into catches. VAC-IGN-BRK fired **48 times in the census week** and caught **9 of the 65 sat-out runs and 5 of the top-15**, against a random-fire baseline of **2.5** — a **3.6× lift**. That is the first thing in two Fridays of greenfield work that demonstrably shows up for the big runs.

It just cannot be traded, because the direction column above is the picker artefact and nothing else. The prior-5-min "20-27% agreement" looks like a screaming fade signal until you notice it reads **48.66% on unselected big moves**. Strip the selection and there is no direction anywhere:

DISCRIMINATOR, ON UNSELECTED FWD15 ≥ 55PT MINUTES		
prior 5-min price change		48.66%
prior 15-min price change		50.39%
flow z / 5-min cumulative-delta z	48.92% / 48.95%	
position in the 30-min range		49.89%
L2 far-side depletion		49.44%

6. ★ THE STONE LAST WEEK LEFT UNTURNED — L2, and it is now closed

Last Friday's section ended: "revive ATR-IGNITION with an L2 far-side-depletion discriminator once MNQ book coverage ≥ 30 days". Coverage is now **21 days (2026-07-31 ... 2026-08-21), 246,934,712 book rows, 22,119 book-minutes** — short of 30 days but enough for a verdict at this effect size. Ask-share = top-3 ask depth $\div$ (ask + bid), z-scored against its own trailing 2h so "thin" means thin for now. Convention: asks thin $\rightarrow$ should telegraph UP.

POPULATION	N	DIRECTION AGREEMENT
all book minutes	21,249	49.72% (± 0.67 pp)
fwd15 ≥ 30 pt	5,009	50.37% (± 1.38 pp)
fwd15 ≥ 55 pt (unselected big moves)	1,515	49.44% (± 2.52 pp)
fwd15 ≥ 90 pt	418	49.76% (± 4.79 pp)
thin book only (ask-share z ≥ 1)	344	47.38% (± 5.28 pp)
census sat-out 65 / top-15 / VACUUM	65 / 15 / 9	40.0% / 26.7% / 22.2%

The census-population rows look like strong **inverse** information — until you note they are the same -26 pp local-extremum artefact as every other pre-run price statistic. On the honest unselected population the book reads **49.72% on 21,249 minutes with a ± 0.67 pp error bar**. That is as close to a definitive null as this desk can measure.

△ Also worth recording: the census's own book column **cannot be read as a forecast**. It is far/(far+near) where far is chosen using the realised direction. It is a description, not a prediction, and quoting it as one would be the third instance of this same class of error in this document.

Verdict on the stone: REFUTED. MNQ L2 top-3 depth does not supply direction. Do not revive ATR-IGNITION on it at 30 days either — the null is not n-limited.

7. ★ THE ROUTER — what would be armed, and the honest answer that nothing is

The mandate is explicit that a blanket cross-tape number is a bug and that a surviving gate must come with its home regime and an arm rule in the live router's vocabulary. **Nothing survived**, so what follows is the segmentation evidence and the rule that would have been written, stated so it is falsifiable rather than aspirational.

The regime split is consistent across all three candidates and is the one genuine policy finding here — **every one of them is an ATR-high, US-session animal and every one of them bleeds in low ATR:**

	VAC-ABS	VAC-SNAP	VAC-IGN-BRK
ATR low tercile	−\$5.26/tr (n=33)	−\$8.10/tr (n=137)	−\$11.64/tr (n=95)
ATR mid tercile	−\$5.99/tr (n=31)	−\$10.09/tr (n=136)	+\$3.24/tr (n=94)
ATR high tercile	+\$50.86/tr (n=33)	+\$20.82/tr (n=137)	+\$18.03/tr (n=95)
US session 13:00–21:00Z	+\$51.24/tr (n=26)	+\$22.89/tr (n=104)	+\$16.33/tr (n=110)
overnight / pre-open	−\$0.18/tr (n=71)	−\$6.57/tr (n=306)	−\$5.08/tr (n=174)

ER is not a discriminator here, in either direction. Serially, VAC-IGN-BRK books −\$0.49/tr in dead chop (ER15 < 0.15, n=83), **+\$17.85/tr in the 0.15–0.35 middle** (n=93) and −\$6.55/tr in clean trend (n=108) — a middle bucket carrying the whole of a \$911 net across three buckets is noise, not a regime. △ Recorded because it matters: the unconstrained run said +\$22.28 in chop and +\$13.79 in trend, and that was the overlap-inflated ledger — exactly the number this section warns against quoting.

What **is** a router finding, and it comes from the detection work in §5 rather than from any backtest: the biggest runs **start from chop** — ER15 is the only feature measured whose AUC sits below 0.5, and it falls further as the runs get bigger (**0.442** on the sat-out 65, **0.403** on the top-15). Every ER-floor arm condition the live router owns is standing gates down at the moment the largest moves begin. That belongs to the router section, not to a gate.

The rule that would have been written, had the sign-flip control not killed it:

ARM vac_ign_brk when $ATR(1m, 14) \geq 21.0pt$ on MNQ **and** $13:00 \leq t < 21:00Z$. **BENCH** below 19.0pt (2-point hysteresis, matching the existing 2-mark convention) with a **≥ 5 -minute dwell on the arm side** — ATR crossing a bare threshold on a partial bar is exactly the oscillation that produced 20 switch changes for 4 trades on 08-07. **No ER condition**, for the reason above. Expectancy on that home regime only: **+\$18.03/trade over 95 trades**. Blanket, as context and never as the verdict: **+\$3.21 over 284**.

Do not arm it. The home-regime number is the same number the fade control earns; conditioning on ATR-high selects trades with more range in both directions, and a gate that makes money whichever way you point it is a volatility harvester with a stop, not an edge.

What we would need to measure to make this tradeable. A **direction discriminator that is not price, not aggressor flow, and not top-3 book depth** — all three are ruled out above at tight error bars. The concrete candidates, in order of how cheaply we could instrument them: (i) **full-depth book imbalance beyond level 3**, including queue-position and cancel/replace rate — depth.db carries the wide snapshot but at 250ms it cannot see the fleeting quotes capture.db.book does at 41ms, so this needs the 41ms feed widened past level 3, which is an IBKR subscription question, not a code question; (ii) **cross-instrument lead-lag** — we already capture MGC L1+L2 and MES/M2K depth, and none of this week's work looked at whether gold or the breadth complex turns first; (iii) **a trade-size distribution split** (large-lot vs small-lot aggressor), which the tick stream already carries in size and which no census or gate on this desk currently uses. (iii) is free and is the one I would run first next Friday.

MGC: deliberately not attempted. The scope is explicit that gold gates are **invented fresh, never ported or re-tuned from an MNQ gate**, and the VACUUM cluster is a property of the MNQ census. Porting VAC-IGN-BRK to gold would have been the wrong kind of easy number — and at \$10.00/point a mis-priced transfer would have looked five times more exciting than it was. Gold is gf_MGC's work, and it is [directly below this section](#).

8. Robustness, laid out test by test

Every candidate went through the full battery. The table records what each test did, not just whether it passed.

TEST	VAC-ABS	VAC-SNAP	VAC-IGN-BRK
sign-flip	−\$23.86/tr ☐ passes	−\$1.46/tr ☐ passes	+\$3.08/tr vs +\$3.21 ☹ KILL
time-shifted placebo (+30/+90/+240/−90m)	+240m earns +\$2.60 △	—	+90m +\$1.06, +240m +\$0.06 △
random-arm bootstrap (40 draws, same arm count)	—	—	mean −\$2.92, max +\$1.98, base + \$3.21 — beaten 100% ☐
unconditional control (arm every minute)	—	—	−\$1.94/tr ☐ the arm is load-bearing
parameter sweep / plateau	—	edge only as n collapses (+\$0.90 @ n=410 → +\$7.02 @ n=63) ☹	4 of 15 cells negative; −\$1.92 @ n=813 → +\$13.26 @ n=143 — step, not plateau ☹
strip-the-3-best	+\$118 (from +\$1,319) △	−\$1,436 ☹	−\$915 ☹
strip-the-5-best	−\$427 ☹	−\$2,364 ☹	−\$1,685 ☹
leave-one-day-out	min +\$9.53 ☐	min −\$2.55 ☹	min −\$0.26 ☹ (drop 07-31)
out-of-sample leg (08-17... 08-21)	−\$529.50 / −\$19.61 ☹	−\$1,189.64 / −\$18.02 ☹	+\$184.57 / +\$3.85 ☐
v5 archive leg (07-05...07-17)	+\$439.14 ☐	−\$758.14 ☹	−\$1,904.73 / −\$19.44 ☹
long/short symmetry	+\$27.60 vs −\$2.58 ☹ 10:1	+\$8.76 vs −\$6.57 ☹	+\$6.74 vs −\$0.47 △
big-moves-caught	0/65 sat-out, 0/15 top-15 ☹	6/65, 2/15	9/65, 5/15 — 3.6× baseline ☐
t-statistic on \$/trade		1.39	0.17 0.45
days green	15/28	12/33	15/32

Two things in that table deserve to be read rather than skimmed. **VAC-IGN-BRK is the only thing in two Fridays of VACUUM work that passes the placebo family, the unconditional control and the bootstrap, and catches the big runs at 3.6× baseline** — and the sign-flip control still kills it stone dead. **And VAC-ABS is the only candidate whose sign-flip control is convincingly negative** — a real directional mechanism, forward-tested, and it lost money anyway.

9. Files

FILE	WHAT
reports/friday_v7/vacuum/vaclab.py	shared tick-honest harness (walk, <code>walk_serial</code> , costs)
reports/friday_v7/vacuum/label_test.py	★ the label base-rate interrogation
reports/friday_v7/vacuum/stage5_artifact.py	★ the run-picker selection artefact + permutation AUCs
reports/friday_v7/vacuum/stage6_ignbrk.py	detection by size band (686 lake runs), IGN-BRK surface
reports/friday_v7/vacuum/stage7_kill.py	controls, unconstrained (kept to document the overlap trap)
reports/friday_v7/vacuum/stage8_serial.py	everything re-scored one position at a time

reports/friday_v7/vacuum/stage9_l2.py	★ the L2 direction test that closes last week's open stone
reports/friday_v7/vacuum/stage10_final.py	strip / LOO / OOS / symmetry battery
reports/friday_v7/vacuum/ser_*.csv, stage*.json	per-trade ledgers and machine-readable results

VERDICT

The VACUUM cluster did NOT survive, and for the second Friday running it was killed by the LABEL itself rather than by a backtest — but this week the mechanism is identified, not just the symptom. The census's flow event fires on **20.41% of every tape-minute we own**, and conditional on it firing the VACUUM direction test comes up VACUUM **49.90%** of the time against **49.91%** on non-event minutes — **−0.01pp**, inside a ± 1.1 pp error bar. The apparent excess in the census (15.7% of runs vs a 10.2% base rate) is **entirely a selection artefact of run_census.py's keep-strongest run-picker**: prior 5-min price direction agrees with the run **48.66% of the time on unselected big moves and 22.55% after the dedup**, and the VACUUM rate climbs **10.45% → 14.01% → 21.87%** across the same three populations. VACUUM is the name we give to having picked a local bottom. **Traded literally (VAC-Z1) it loses −\$4,150.79 over 1,768 trades serial, −\$32,184 over 8,589 unconstrained.**

Four candidates were invented and all four are graves, each with a named cause of death. **VAC-SNAP** (absorption → structural break, +\$370.64 / n=410) — killed by **strip-the-3-best** (−\$1,436) and the **OOS leg** (−\$18.02/tr). **VAC-CUMDIV — refuted outright**, negative at every threshold. **VAC-ABS** (last week's shadow candidate, forward-tested unaltered, +\$1,319.21 / n=97 / +\$13.60 per trade net of \$1.50 per round trip) — killed by its **OUT-OF-SAMPLE LEG**, −\$529.50 over 27 trades on the five days it had never seen, confirmed by **strip-5 = −\$427**; it is now **RETIRED, not shadowed**. **VAC-IGN-BRK** (the detection bracket, +\$911.55 / n=284) — killed by the **SIGN-FLIP CONTROL**: fading the break earns **+\$3.08/trade against the signal's +\$3.21**, so both directions win and it is volatility harvesting, not edge; the parameter surface is a step rather than a plateau and 4 of 15 cells are negative.

The size-threshold question has a real answer this week, and it is a detection threshold, not a tradeability one. Below $|\text{move}| \approx 87\text{pt}$ (the top-15 band) three of four detectors are indistinguishable from noise; at and above it all four go significant — volume-z **AUC 0.730, p=0.0004** on the census's top-15,

rising lake-wide to **AUC 0.867** on ATR for the **113 runs of 120pt+**. **VAC-IGN-BRK** converts that into **9 of 65 sat-out runs** and **5 of the top-15 caught on 48 fires** — **3.6× the random-fire baseline**, the first thing in this workstream that demonstrably shows up for the big runs. It is still not a gate, because **detection without direction is a coin flip with a stop**, and the sign-flip control proves it.

Last week's one open stone is closed: the L2 far-side-depletion discriminator is REFUTED — **49.44% ±2.52pp** on unselected 55pt+ moves over 21 days and 246.9M book rows, **49.72% ±0.67pp** on all 21,249 book-minutes. Do not revive it at 30 days; the null is not n-limited.

△ **And one methodological finding that should outlive this section: the first pass of this backtest let overlapping entries run concurrently, and five shorts into the single 2026-07-31 impulse were worth \$3.9k of a \$10.3k headline. Scored one position at a time** — the only way a live gate can trade — **VAC-IGN-BRK falls from +\$15.49 to +\$3.21 per trade. 91% of the apparent edge was one impulse counted five times. Every greenfield harness on this desk should be checked for it.**

Dispositions: NOTHING to LIVE. NOTHING to SHADOW. VAC-ABS retired from shadow. Two findings handed onward — (1) the census run-picker's local-extremum artefact, which invalidates every pre-run price-path statistic quoted off the run list and should be fixed or flagged in `run_census.py`; (2) the biggest runs start from CHOP (ER15 AUC 0.403), so every ER-floor arm condition the router owns is standing gates down at the moment the largest moves begin. The single cheapest next experiment is a large-lot vs small-lot aggressor split — the size field is already in the tick stream and no gate or census on this desk uses it.

GOLD — the MGC greenfield (gf_MGC), and the blocker that did not survive thirty-two days

This section was missing from Rev 1 and it should not have been. The gold phase ran on 2026-08-28 and wrote four artifacts at 23:58Z; the report then shipped with no gold greenfield section at all, while three other

places pointed at “gf_MGC” as though it were in the document. Gold is the operator's #1 stated new-instrument priority. Everything below is computed at render time from those artifacts — scripts/
friday_v7_gold_section.py.

THE GOLD ANSWER IN FOUR LINES

1 — The bar-source blocker is DEAD, and it was never the variable.

Last cycle's gold section led with “the sign flips on the bar source alone” — +\$3.06/trade on lab bars against −\$8.96 on production bars — measured on **six** days. On **32** days both arms are the same sign and both are negative: −**\$1.21/trade** on lab bars (506 trades) and −**\$0.41/trade** on production bars (299). On the 19 days both tapes cover it is −\$1.76 against −\$0.41. There is no flip at any level.

2 — What the bar source really controls is HOW OFTEN it fires, not how well. The lab's mid-bar arm takes 2,459 raw fires against production's 1,451, and **178 of the lab arm's 506 trades (35.2%) land in minutes where the trade tape printed nothing at all.** A mid quote flickers when nobody trades; a trade bar cannot. The lab arm was not reading a better price series — it was admitting a population of dead-tape breaks that cannot be filled.

3 — Activity is the real axis, and it does not survive being fitted. Split by how busy the tape is, the busiest quartile carries everything (production bars: **+\$27.11/trade** in Q4 against −\$11.85 in Q1). Fit an activity floor in-sample and it looks superb — +\$40.80/trade at floor 40. Apply that same floor FORWARD and it is **−\$11.92/trade on 12 trades.** The floor is a fit, not a filter.

4 — Nothing gold-side goes anywhere near a shadow arm this week, and the cause of death is named. Every arm here is negative pooled, and each one gets worse when its three best trades are removed. That is the strip-the-best-trades test, and failing it is the one result a thin sample can still deliver honestly.

G1 The three gold arms, pooled and stress-tested

32 trading days, 2026-07-16 to 2026-08-28. Identical gate (60-minute break, 0.10 ATR margin, obstacle-free hole, 45-minute per-side cooldown), identical exit (3.0 ATR stop behind a 2.0/2.0 chandelier, 480-minute backstop), raced on the same 250 ms quote tape with both legs crossing. **MGC is \$10.00/point and the \$1.50 round trip is already inside every figure.**

ARM	TRADES	DAYS	NET \$	\$/TRADE	WIN %	STRIP BEST 3	WORST LODO	GREEN DAYS
DEPTH-MID bars — the LAB's series	506	32	−\$611.00	−\$1.21	52.2%	−\$1,659.50	−\$1,070.50 drop 2026-08-12	18/32
TRADE bars — what PRODUCTION folds	299	19	−\$121.50	−\$0.41	54.2%	−\$999.00	−\$636.50 drop 2026-08-12	12/19
WALL arm — break into a resting wall	270	19	−\$966.00	−\$3.58	53.3%	−\$1,753.50	−\$1,354.00 drop 2026-08-05	6/19

Read the strip-best-3 column, not the win rate. Win rates here are 52–54% — respectable, and irrelevant. Every arm's whole book is a handful of trades: removing three costs +\$787.50 on the arm it hurts least and +\$1,048.50 on the arm it hurts most. **An arm that is negative pooled and more negative without its three best trades has no expectancy to find.**

G2 Walk-forward — the part that decides it

The artifact splits the 32 days into the first 22 (in-sample), the next 7 (F1) and the last 3 (F2, this report's own week). Nothing was re-fitted between them.

ARM	IS N	IS \$/TR	F1 N	F1 \$/TR	F2 N	F2 \$/TR
DEPTH-MID bars — the LAB's series	345	+\$0.80	115	−\$1.96	46	−\$14.39
TRADE bars — what PRODUCTION folds	158	+\$6.16	110	−\$4.43	31	−\$19.63
WALL arm — break into a resting wall	126	−\$6.35	110	−\$0.99	34	−\$1.68

Both break arms decay monotonically out of sample and both are worst in the most recent period. That is the ordering you would expect from a fitted edge and the opposite of the one you would want. F2 is three days and 46 / 31 trades, so it is not on its own a verdict — but it agrees with F1, which agrees with

the pooled book, which agrees with the strip-best test. **Four instruments, one direction.**

G3 The bar-source claim, re-quoted properly

CUT	LAB (DEPTH-MID) \$/TR	PRODUCTION (TRADE) \$/TR	SAME SIGN?
Last cycle's claim — 6 days, gf3_mgc_barsource.json, the “hole PRIMARY” sub-cell (n=56 / 53)	+\$3.06	−\$8.96	no — this is the flip that was reported
32 days pooled, gf4_mgc_barsource.json (n=506 / 299)	−\$1.21	−\$0.41	yes — both negative
The 19 days BOTH tapes cover (n=316 / 299)	−\$1.76	−\$0.41	yes — both negative

The six-day cell was real and it was six days. **The rule the gold play itself proposes — name your bar source in the same sentence as your number — needs a second clause: name your SAMPLE too.** A sub-cell of 56 trades on 6 days flipped sign and became the gating blocker for an entire instrument's programme for a week.

G4 What the bar source actually changes: dead tape

Raw fires: **2,459** on mid bars, **1,451** on trade bars, **1,278** on both. The mid arm fires about 1.7× as often, and the extra fires are not better — they are untradeable: **178 of 506 (35.2%)** of its trades open in a minute where the trade tape printed nothing. Production's arm has 0 such trades.

BAR SOURCE	ACTIVITY QUARTILE	MEDIAN PRINTS/MIN	N	NET \$	\$/TRADE	WIN %
DEPTH-MID (lab)	Q2	0.0	252	−\$997.00	−\$3.96	49.6%
DEPTH-MID (lab)	Q3	15.3	127	−\$322.00	−\$2.54	55.1%
DEPTH-MID (lab)	Q4 busiest	36.9	127	+\$708.00	+\$5.58	54.3%
TRADE BARS (production)	Q1 quietest	8.6	75	−\$888.00	−\$11.85	48.0%
TRADE BARS (production)	Q2	14.7	74	−\$427.00	−\$5.77	54.1%
TRADE BARS (production)	Q3	23.1	75	−\$840.00	−\$11.19	46.7%
TRADE BARS (production)	Q4 busiest	54.3	75	+\$2,034.00	+\$27.11	68.0%

The gradient is not monotone — it is a CLIFF at the top quartile. On production bars Q1, Q2 and Q3 read −\$11.85, −\$5.77 and −\$11.19 a trade with

no ordering between them, and then Q4 reads **+\$27.11 at a 68.0% win rate**. Three-quarters of the fires are noise around zero-minus-costs and the whole book lives in the busiest quarter. That is the single most interesting number gold has produced — and a step with nothing under it is also exactly the shape a fitted threshold makes, which is why the next section is a warning rather than a proposal.

G5 The activity floor: fitted in-sample, dead forward

BAR SOURCE	FLOOR FITTED	IS N	IS \$/TR	IS WIN	FWD N	FWD \$/TR	FWD N, NO FLOOR	FWD \$/TR, NO FLOOR	
DEPTH-MID (lab)		40	42	+\$25.83	76.2%	14	−\$3.43	161	−\$5.51
TRADE BARS (production)		40	40	+\$40.80	80.0%	12	−\$11.92	141	−\$7.77

This is the whole lesson of the gold week in one table. Choosing the floor on the data that chose it gives +\$25.83 and +\$40.80 a trade at a 76–80% win rate. Carry the same floor forward and it makes **−\$3.43** and **−\$11.92** a trade on 14 and 12 trades — worse than taking every forward fire with no floor at all on the production arm (−\$11.92 filtered against −\$7.77 unfiltered). And note what the floor does to the sample: it keeps **42 of 345** fires in-sample (12%) and **14 of 161** forward (9%), so the forward test that would grade it honestly needs months, not days.

G6 The one cell that looked like a candidate, and what killed it

The **double hole** — a break with no obstacle in front of it and no resting support behind it — is positive on BOTH bar sources, which is exactly the agreement the bar-source scare said we could never have. It is also positive in-sample and positive in F1. It is graded here rather than promoted, because it fails the one test that a sample this thin can still run.

BAR SOURCE	N	NET \$	\$/TRADE	WIN %	IS \$/TR	F1 \$/TR	F2 \$/TR	NET, STRIP BEST 3
DEPTH-MID bars — the LAB's series	311	+\$921.50	+\$2.96	54.3%	+\$3.76	+\$4.73	−\$11.75	−\$109.00
TRADE bars — what PRODUCTION folds	154	+\$468.00	+\$3.04	59.1%	+\$6.12	+\$4.42	−\$24.92	−\$328.50

Cause of death: strip-the-best-trades. Both arms go from positive to negative when three trades are removed — from +\$921.50 to −\$109.00 on lab bars, and from +\$468.00 to −\$328.50 on production bars. Three trades out of 311 and 154 are the entire result. It also turns negative in F2 on both. **Not refuted — a thin sample never is — but nowhere near a shadow arm, and the specific thing it needs is more trades, not more tuning.**

GOLD — VERDICT, AND WHAT TO DO ABOUT IT

Gold is an EMPTY SEAT, not a bleed. There is no live gold gate, so the census ceiling gold prints each week is the size of the opportunity, not money forgone. Nothing here arms, nothing here shadows, and no MNQ gate is ported across — the scope's standing rule that gold gates must be invented fresh is unchanged and is reinforced by G4: the two instruments do not even have the same relationship to their own tape.

What actually moved this week. The bar-source blocker that gated the entire gold programme is gone — retired on 32 days after being raised on 6. That is a real deliverable: the next gold cycle does not have to spend itself reconciling two bar builders. **The blocker it is replaced by is smaller and more useful:** gold's edge, if it has one, lives in the busiest quartile of the tape, and every attempt so far to express that as a threshold has been a fit. The next experiment is an activity measure chosen a priori and run forward for a stated number of sessions — the same discipline the Open Rider's pre-registered ATR filter is under, applied to gold.

△ **Honest limits, all of them.** 32 days is one summer of one instrument. F2 is three days. The double-hole cell is 311 and 154 trades and dies on three of them. Every figure is a REPLAY on the 250 ms quote tape with both legs crossing — it is the best counterfactual available and it is not a fill. And gold has never traded live on this desk, so nothing here has ever met a real queue.

★★ ADDENDUM, 2026-09-04 — the three hunts that re-ran tonight, on more tape, and what they overturn

★ THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.

movement3_fresh_0904.html was rebuilt **2026-09-04 23:08 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2–3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

★★ REV2 (2026-09-04, AFTER THESE HUNTS RAN) — THE SAT-OUT POPULATION BELOW IS THE PRE-FIX ONE. ITS DENOMINATORS ARE ABOUT 16% TOO LARGE, AND NONE OF ITS VERDICTS TURNS ON THAT

Every number in the three dossiers below is scored against **73 sat-out runs and a \$10,515 sat-out ceiling**, which is what census_summary.json said when they ran. Rev 2 then found and fixed an attribution bug in run_census.py: it attributed trades over start ± 300 s while a run is 15 minutes long, so any entry more than five minutes into a run's own body was invisible and the run was scored "sat out" while the desk was in it. On the corrected freeze the same week reads **10 caught, 2 fought, 64 sat out, \$8,845 of sat-out ceiling — nine of the 73 runs these hunts treat as absences were runs the desk actually traded**, including the 09-03 13:36 +130pt run that grind_long bought at 13:42:05 and was stopped out of at 13:56. Part 4 §7 has the full reconciliation.

What that does and does not change here. Every "% of the \$10,515" figure below is computed against a denominator that is ~16% too large, so those percentages are understated — the same dollars against \$8,845 are a larger share. **No verdict below moves**, and it is worth being explicit about why: every candidate in these three dossiers was killed by an out-of-sample test — the fit-then-test protocol, a fixed-clock control, a units check, or a forward arm — and none of them by the size of the prize. A candidate that fails out of sample fails at any denominator. The one thing to read differently is any sentence of the form "the detector boards 71 of the 73 sat-out runs": nine of those 73 were not sat out, so boarding them was not an opportunity the desk missed.

★★ THESE THREE HUNTS ARE NEWER THAN THE REST OF THIS REPORT — READ THEM LAST, AND LET THEM WIN

Everything above this box in Movement 3 was rendered on **2026-08-29** from labs that ran against the census frozen on 2026-08-28. The three dossiers below ran **tonight**, finishing at **2026-09-04 21:39 UTC**, against a census frozen at 21:12Z covering **2026-08-30 22:00 → 2026-09-04 20:59 UTC** and against the parquet lake out to 2026-09-04. They are the same three questions asked again on more tape, and they are **27,104 words** of finished, tick-honest working.

Two of the three overturn a verdict printed above them, and one overturns a headline finding of this whole report. Each is named in its own header box below. Where a dossier here disagrees with the editorial board earlier in this movement, this section wins — not because it is louder, but because it is later evidence on strictly more data, which is the only reason a verdict should ever move.

They are carried here rather than folded into the section above on purpose. Rebuilding Movement 3 from tonight's dossiers would have put tonight's appendix underneath last week's summary of it — a map that describes a different territory from the one printed beneath it. That failure has shipped on this desk before. A dated addendum cannot do it.

TONIGHT'S THREE, AND WHAT EACH ONE MOVED

1. **GOLD — MGC, seventeen live shadow days, and the landmine under every prior gold study**

The frozen board above grades gold **SHADOW** on the double-hole arm (n=200, +\$16.52/tr). Tonight's hunt is a **hard null on all four cells** — six candidates, every one beaten by shorting one lot at a fixed time of day — and it finds that **22% of the lake's gold year is the wrong contract**, which puts last week's thirteen-months-of-gold conclusions out of commission.

2. **THE POOLED RUN-CATCHER — re-derived on 60 days, and graded against its own live shadow arm**

The frozen board above PARKS the pooled rider on an out-of-sample leg worth +\$14. Tonight's run reaches the same disposition by a harder road and adds the number that settles it: rider_w5, promoted to shadow last week on a +\$19.29-per-trade backtest, has since booked **108 live shadow trades at −\$25.95 each** — a \$45-a-trade swing between fit and forward.

3. UNCLASS — the biggest bucket, and the death of the size-threshold hypothesis

The frozen board above PARKS UNCLASS on the placebo test. Tonight's run spends the exact revival condition that PARK was granted — a board-time size predictor at $r \geq 0.40$ on ≥ 300 runs — and it comes back **$r = 0.15$ on 4,830 boards**. It also refutes this report's own **size-threshold** finding: measured in ATR units rather than points, every discriminator collapses to $AUC \approx 0.50$.

GOLD — MGC, seventeen live shadow days, and the landmine under every prior gold study

The second instrument, \$10.00 a point, with the gold shadow book now graded on 225 tick-repriced live trades rather than on a model.

What this one moves: The frozen board above grades gold **SHADOW** on the double-hole arm ($n=200$, +\$16.52/tr). Tonight's hunt is a **hard null on all four cells** — six candidates, every one beaten by shorting one lot at a fixed time of day — and it finds that **22% of the lake's gold year is the wrong contract**, which puts last week's thirteen-months-of-gold conclusions out of commission. Source: *gf_MGC.md*, written 2026-09-04 21:39 UTC, 8,472 words, reproduced in full and unedited below.

THE GOLD GATE HUNT — MGC, 2026-09-04

Garrath — the whole week in one paragraph. The gold shadow desk has now been running live for seventeen days and has recorded 225 tick-repriced trades, so for the first time we are not arguing about a model. The book as a whole is — **\$1,671**, and almost all of that is the unfiltered control arm losing steadily — which is itself the answer to the question last Friday said would decide the thesis. The book-filtered LONG arm is **+\$485 across 19 trades** and has improved every week; the SHORT arm is **-\$376 across 20**. Separately, I went hunting the four cells you asked for (momentum and reversion, long and short) across a year of gold and **I did not find one that survives**. Six candidates, built fresh from gold's own distribution, all four cells covered — every single one is beaten by simply **shorting one lot at a fixed time of day** with the same exit, and five of the six

lose money out of sample. That is a hard null and I am not dressing it up. But two things came out of the night that are worth more than another hopeful gate:

- ★★★ **The year of gold in the lake is 22% the wrong contract.**
backfill_1min.parquet keeps the earliest expiry that has a print at each instant, which stops being the front month about five weeks before expiry. On 77 of 266 sessions the lake quotes a contract with a **median minute volume of zero lots**, at prices a **median 30.1 points — \$301 a lot — away** from where gold actually was. **Last Friday's "thirteen months of gold" section ran on that file.** Its conclusions, including the one that killed a router rule, are not safe.
- ★★ **Gold's exit wants the opposite of MNQ's.** On every gold entry I built, the desk's own dual-slot shape (Lot A tight scalp + Lot B chandelier) is **negative**, and a **single lot with a very wide stop held to a clock** is the only exit family that pays. On the lead cell: dual slot **−\$1,649**, single wide lot **+\$9,202**, same entries. If a gold gate is ever promoted, it must not inherit the live slot config.

Everything below is shadow-only. Nothing here is a live promotion and I am not proposing one.

1. THE RULER — and a landmine that has been under every gold study for three weeks

1.1 ▲▲ The year tape is the wrong contract for a fifth of its minutes

This is the most important thing in the section, so it goes first.

gazbot7.lake builds gold's bars view by unioning backfill_1min.parquet. That file is made by backfill_to_lake.py, which pulls seven separate expiries and dedupes them on (symbol, timeframe, bar_ts) **keeping the first**. Its own docstring explains the reasoning: "the FIRST — the front-month print for any instant." The files sort by expiry date, so "first" means "the earliest expiry that has a print at this minute", and that is only the front month until liquidity rolls forward. Gold's volume rolls to the next contract about **five weeks before** expiry, and for those five weeks the file keeps the **dying** contract.

I rebuilt the series from the per-expiry files directly, taking for each session the contract that actually carried that day's volume. The two tapes disagree far more than I expected:

THE LAKE'S BACKFILL_1MIN UNION VS THE TRUE FRONT MONTH		
minutes present in both		352,319
minutes where the lake's price is a different contract	77,926 — 22.1%	
median size of that difference	30.10 points = \$301.00 a lot	
90th percentile / worst	51.90pt / 170.30pt	
sessions affected	77 of 266	
median minute volume on those sessions — lake	0 lots	
median minute volume on those sessions — front month	180 lots	
minutes with zero true range — lake		25.3%
minutes with zero true range — front month		5.2%

Read the volume row twice. On those seventy-seven sessions the lake is quoting a contract in which, in the median minute, **nobody traded at all**. Any ATR, any sixty-minute high, any efficiency ratio computed there is measuring a ghost. It shows up immediately in a monthly median ATR-14:

MONTH	MEDIAN ATR-14, LAKE TAPE	MEDIAN ATR-14, FRONT MONTH
2025-10	0.000	2.314
2025-12	0.000	1.679
2026-04	0.000	2.452
2026-05	2.057	2.150
2026-06	0.214	2.293

Last Friday's \$5 — "THIRTEEN MONTHS OF GOLD — the caveat that could not be answered, answered" — read this file. gf3_mgc_year.py line 52 points straight at backfill_1min.parquet and filters only on timeframe='1min'. So the section that used the year to kill last week's router rule, and to put "a much colder light on the trigger", was reading a tape that is a fifth ghost contract. I am not claiming its conclusions are wrong — I am saying they are **unverified**, and they should be labelled that way until they are re-run on a rolled series.

It also compounds a landmine that section already flagged: its own \$1.2 warned that an unfiltered bars query now unions 1-minute through **daily** bars into one series. That warning was correct and is still live. This is a second, independent defect in the same file, and it is the more dangerous of the two because filtering the timeframe does **not** protect you from it.

The fix is four lines in `backfill_to_lake.py`: rank the per-expiry rows for each `bar_ts` by that day's contract volume rather than by filename order, and keep the winner. It is a build item and it is in the play list at the end.

1.2 The tape this section actually used

Everything below is measured on the rebuilt front-month series:

source	the seven <code>data/backfill/MGC_<expiry>_1min.parquet</code> files, front month by daily volume, plus our own 5-second lake capture resampled to minutes after 2026-08-28
span	2025-07-28 ... 2026-09-04, 277 sessions
span actually studied	2025-10-01 ... 2026-09-04, 226 sessions
why the trim	before October the only contract IBKR will still serve is a back month: 15.4% of 2025-08 minutes have zero true range and its ATR sits at 1.7 basis points against 4-7bp for every later month. That is not gold's tape, it is a quiet contract's tape
session day	22:00Z → 21:00Z , the CME gold day. Every lookback is computed inside one session, so no window spans the daily halt or a roll
bar source	TRADE bars
cost	\$1.50 a round trip plus 0.10pt of slippage a leg = \$3.50 all-in. Swept: at zero slippage the lead cell reads \$27.47 and at 0.30pt it reads \$21.47, so nothing here turns on the assumption
multiplier	\$10.00 a point

On the bar source, and why I did not use the depth mid. Last Friday's headline was that gold results flip sign between depth-mid bars and trade bars, and its single recommendation was to point the shadow's bar builder at the mid. **That build has not shipped.** `gazbot7-shadow-mgc.service` has been up continuously since 2026-08-18 17:49 UTC — two weeks and three days without a restart — and `shadow_mgc.py` still carries its own warning that "this service folds `md`'s TRADE bars". So the live gold book is a trade-bar book, and the honest ruler for anything I might want to promote is trade bars. Section 2 has changed my mind about how urgent that build is.

1.3 The frozen census, for context

`census_summary_MGC.json` (frozen 08-25, 08-17...08-21): **70 gold runs, 70 sat out, a hindsight ceiling of \$11,401**, clustered UNCLASS 35 / OPEN-NEWS 15 / VACUUM 13 / FLOW-LED 6 / VOL-EXPANSION 1. The biggest run of that week was 08-19 12:31 UP 55 points. I have not re-run it and nothing below depends on it — it is the standing reminder of what gold is leaving on the table.

2. ★★★ THE LIVE GOLD SHADOW — seventeen days, 225 trades, and last week's decisive question answered

gazbot7-shadow-mgc.service, up since 2026-08-18, writing to **data/shadow_mgc.db** (not shadow.db — its own store, so gold can never be repriced at MNQ's \$2 a point). Every trade below is tick-repriced by `repricer.reprice()` against real bid and ask, and all 225 are marked filled. The database's own `mtime` reads 15 August; that is a WAL artefact and means nothing — the write-ahead log is 2.8MB and was touched at 21:00 today.

ARM	N	SESSION S	NET	\$/TRADE	MEDIAN	WIN%	MEDIAN HOLD	5-95% BAND
mgc_break_fade_nobook — the control , no book filter	186	16	−\$1,781	−\$9.58	−\$1.50	47%	18m	−\$16.92 ... − \$2.08
mgc_holebreak_fade_long	19	12	+\$486	+\$25.55	+\$4.50	63%	25m	−\$8.97 ... + \$64.56
mgc_holebreak_fade_short	20	12	−\$376	−\$18.80	−\$18.50	40%	22m	−\$45.45 ... + \$9.60
all three	225	16	−\$1,672	−\$7.43	−\$1.50	48%	18m	−\$16.84 ... + \$2.72

(band = a session block bootstrap, 2,000 resamples of whole trading days.)

2.1 The comparison that was supposed to decide the thesis

Last week's control arm was shipped with an explicit instruction attached: "If the two hole arms do not beat this one, the book is decorative and the whole thesis is wrong." At $n=7$ a side it could not be answered and last week's honest answer was "they do not, but I would not conclude anything from 14 trades." It can be answered better now — **and the control is two-sided, so it has to be split by side before anything is compared to it.** Comparing a LONG arm against a bidirectional control mixes in the control's shorts and is not a comparison at all.

SIDE	BOOK-FILTERED ARM	UNFILTERED CONTROL, SAME SIDE	WHAT THE BOOK GATE ADDS
LONG	+\$25.55/trade (n=19)	−\$6.18/trade (n=90)	+\$31.73
SHORT	−\$18.80/trade (n=20)	−\$12.76/trade (n=96)	−\$6.04

The book gate is doing real work on the long side and is actively harmful on the short side. That is a genuinely new result and it is the opposite of what a one-week read said. It is also exactly the kind of asymmetry that a two-sided control hides.

2.2 Week by week, because the trend matters more than the total

ISO WEEK	NOBOOK CONTROL	HOLEBREAK_LONG	HOLEBREAK_SHORT
wk34 (08-18...08-23)	-\$21.11 (n=46)	-\$62.50 (n=2)	-\$49.25 (n=4)
wk35 (08-24...08-30)	-\$7.80 (n=69)	+\$33.20 (n=10)	+\$5.70 (n=5)
wk36 (08-31...09-04)	-\$3.82 (n=71)	+\$39.79 (n=7)	-\$18.86 (n=11)

The long arm has been positive for two straight weeks and the control's bleed is shrinking. I would not read a trend into three weeks, but it is the right direction and it is free to keep watching.

2.3 What I will NOT claim from it

The long arm **fails the strip test**. Nineteen trades: strip the best three and it goes from +\$25.55 a trade to **-\$5.75**; strip its best session and it drops to +\$11.94. Its whole result is a handful of trades, which at n=19 is precisely what you would expect from noise **and** from a real fat-tailed edge, and 19 trades cannot tell those apart. **That is a reason to keep shadowing it, not a reason to promote it and not a reason to kill it.** Same for the short arm in the other direction: at n=20 with strip-best-3 at -\$37.26, it is bad but not disproven.

The **control**, though, is now well powered: 186 trades over 16 sessions, -\$9.58 a trade, with a bootstrap band of -\$16.92 to -\$2.08 that excludes zero. **Fading a sixty-minute level break in gold, unfiltered, loses money.** That is a real finding and it closes a question.

2.4 The sim's marks are still fantasy, and the repricer is still right

the sim's own ceiling_pnl, all 225 trades	-\$1,222
tick-repriced real_pnl	-\$1,672
mean absolute gap per trade	\$28.52 (median \$9.00, worst \$415.00)

A \$415 disagreement on a single gold trade is what happens when a bar-close mark meets a real bid/ask race. Never score a gold arm on ceiling_pnl.

2.5 So how urgent is the depth-mid build now?

Last Friday: "that is the single highest-value change on the gold line and nothing else here should be acted on before it." I am **downgrading it**, and the reason is section 2.1. The claim was that the trade tape was the worse gate, and that the lab's positive number would never be reproduced in production. Seventeen days later the long arm on the trade tape is **+\$25.55 a trade against a side-matched control at -\$6.18**. The trade tape is not obviously the worse gate. It is

still worth building the mid-bar arm — but as a **parallel arm for comparison**, not as a blocker, and not before the contract-roll fix in §1.1, which is cheaper and affects every gold study we own.

3. THE FOUR CELLS — how I hunted them, and what the tape said

3.1 The method: map first, invent second

Rule 1 says every gold gate must be invented fresh, and the 08-14 leads file says a directional momentum clone is a grave already dug. So I did not start by writing gates. I started by asking the tape a single question at every minute of 226 sessions:

given that the last 30 minutes moved k ATRs, what do the next 30 / 60 / 120 / 240 minutes do?

That one measurement answers all four cells at once, because the four cells are two triggers seen from two sides. When price has risen, **going with it is momentum-long and fading it is reversion-short**; when price has fallen, **going with it is momentum-short and fading it is reversion-long**. Everything is reported in dollars a trade at \$10 a point net of the \$3.50 round trip, so a map number is directly comparable with a gate number.

Blanket first, because the discipline says a blanket number is a bug and I want to show why. Across all 226 sessions, 72 combinations of lookback / threshold / horizon: **not one cell's 5-95% block-bootstrap band clears the cost of trading**. The largest gross drift in any of them is about 0.4 points. Blanket gold is a coin flip and the map's job is to find where it is not.

3.2 The conditional map, and why its survivor list is worth nothing on its own

I segmented on volatility (today's ATR against the median of the previous five sessions — causal, dimensionless), on efficiency (ER over 60 minutes), and on session (Asia 00:00–07:00Z, London 07:00–13:30Z, US 13:30–20:00Z, Late 20:00–24:00Z). That is 4 horizons $\times$ 4 sessions $\times$ 3 ATR buckets $\times$ 3 ER buckets $\times$ 2 triggers = **288 cells**, of which about 40 had a bootstrap band excluding zero.

At a 90% band you get roughly 29 of those from pure noise. The survivor list is a lottery ticket stub. Two tests noise cannot pass:

TEST	RESULT	VERDICT
A. sign holds across all four horizons	27 of 58 cells (47%) against a 12.5% coin	✓ ...but see below
B. split-half — does the sign found in the first 113 sessions survive into the second 113	49.3% across 213 cells. A coin.	✗

Test A is not the corroboration it looks like. The four horizons are nested — the 240-minute forward return contains the 120, which contains the 60 — so of course their signs agree. Nested legs are not independent evidence and I should not have counted them as such; the honest reading of Test A is "the measurement is internally consistent", not "the effect is real".

Test B is the real one, and it fails, with one exception:

SESSION	CELLS	SIGN HELD INTO THE SECOND HALF	MEAN \$/TRADE, FIRST HALF	SECOND HALF
ASIA	72	52.8%	+\$1.72	+\$0.16
LONDON	72	36.1%	+\$4.34	−\$4.05
LATE	8	37.5%	+\$5.86	−\$3.12
US	61	62.3%	−\$9.72	−\$6.12

London does worse than a coin — its sign systematically reverses between halves, which is what you see when a first-half pattern is noise that the second half happens to contradict. That kills the strongest-looking thing in the full-sample table, which was "fade London's down-moves": in the full sample it reads +\$41 a trade at a four-hour hold, and it is a first-half artefact.

The US session is the only place that holds, and the sign is negative in both halves — going with a US-session extension loses money in both halves, so the reversion side is where to look.

3.3 The US session, split by side — and the asymmetry that matters

Fading the move, held for h minutes, no stop, \$/trade:

	FULL	FIRST HALF	SECOND HALF	N	HELD?
REVERSION-SHORT (fade an up-move)					
h=60	+\$7.59	+\$10.92	+\$4.06	28,097	✓
h=120	+\$13.20	+\$16.96	+\$8.69	24,922	✓
h=60	−\$12.27	−\$16.70	−\$8.16	28,143	

REVERSION-LONG (fade a down-move)					✓ (negative in both)
h=120	-\$11.95	-\$10.72	-\$13.22	25,146	✓ (negative in both)

So gold's US session is **short-biased on both counts**: its up-extensions revert, and its down-extensions do not. Reversion-long has no home here, and momentum-long is the mirror of the reversion-short row and is therefore deeply negative (−\$14.59 at h=60).

And this is where the night turns. That table looks like a finding. It is not, and the thing that shows it is not is the control.

4. ★★★ THE CONTROL THAT KILLED EVERYTHING — "anything new must clear the best CONSTANT"

Your rule from the 08-14 file was that a gold candidate has to beat the best constant, not a coin. For a short gate the constant is "short one lot, same exit, no trigger at all." I ran it two ways: short at **every** minute of the home window (which is what the map's control was), and short **once a session at a fixed clock time**, which is what a desk would actually do.

US SESSION, 7×ATR STOP, 8-HOUR CAP, \$/TRADE	
the gate — short after a 30-min up-extension	+\$25.47 (n=347)
short at every US minute	+\$12.19
short once a session at 13:35Z	+\$27.65
short once a session at 14:30Z	+\$35.82
short once a session at 14:00Z	+\$41.57

The trigger is worth minus two dollars a trade. Everything the gate appeared to earn, and more, is available by shorting a lot at two o'clock and going away.

And it is not gold's drift, which is the first thing I checked: over the studied sample gold went **3,892.6 → 4,492.2**, so a lot held short across the whole year **loses \$5,996**. This is an intraday pattern — gold rises overnight and gives it back in the US afternoon — not a bet on the metal falling.

4.1 So is the clock itself the finding? I spent the rest of the night trying to kill it, and I did.

Swept every half hour of the 24-hour session, both sides, several stops and caps (clock.csv). Shorting into an 8-hour hold is positive across a broad 07:30–15:00 band — but **almost all of that band is a first-half phenomenon**:

ENTER SHORT AT	N	\$/TRADE	FIRST HALF	SECOND HALF	MONTHS POSITIVE	STRIP BEST 3 SESSIONS
07:30Z	224	+\$47.66	+\$67.61	+\$27.72	9/12	+\$11.47
08:30Z	224	+\$46.50	+\$84.74	+\$8.27	8/12	+\$21.08
09:30Z	224	+\$14.35	+\$73.32	–\$44.62	5/12	–\$8.38
12:30Z	224	+\$40.44	+\$79.23	+\$1.65	8/12	+\$12.05
13:00Z	224	+\$36.71	+\$86.96	–\$13.54	7/12	+\$13.16
14:00Z	225	+\$41.57	+\$53.65	+\$29.59	9/12	+\$22.89
14:30Z	225	+\$35.82	+\$26.74	+\$44.82	8/12	+\$14.55
15:00Z	225	+\$21.65	+\$6.93	+\$36.24	7/12	+\$5.99
16:00Z	225	–\$4.00	–\$64.57	+\$56.03	5/12	–\$21.74

Only **14:00–14:30Z** is positive in both halves, positive in 8–9 months of 12, and still positive after stripping the best three sessions. That is a narrow, specific, mechanistically plausible window — 14:00 UTC is 10:00 New York, and it is also **15:00 in London during British Summer Time, which is the LBMA PM gold fix**. The fix has a well-known reputation for downward pressure. If that is what I had found, it would be a real structural edge.

So I tested it, and the test is one that no amount of curve-fitting can fake. Britain's clocks change by act of parliament, not by anything in the price. The PM fix is 15:00 London, which is **14:00 UTC on BST days and 15:00 UTC on GMT days**. If the effect is the fix, it must move by exactly one hour when the clocks do.

ENTER SHORT AT (UTC)	BST DAYS (N=128) \$/TRADE	GMT DAYS (N=97) \$/TRADE
13:00	–\$9.09	+\$97.78
14:00	+\$21.08	+\$68.60
14:30	+\$23.18	+\$52.50
15:00	+\$16.51	+\$28.43
15:30	+\$29.29	–\$14.87
16:00	+\$29.88	–\$48.71

It moves the wrong way. On BST days the peak is at 15:30–16:00Z; on GMT days it is at 13:00Z. The fix hypothesis predicted BST 14:00 and GMT 15:00 and got neither — it got the opposite displacement. **REFUTED, by a test whose boundary is set outside the market.**

What the table actually shows is that the two sub-samples have completely different best hours, and the "robust" 14:00Z cell is just where those two unstable peaks happen to overlap. It is an average of two different things, not a stable hour.

The clock's own out-of-sample leg says the same. Holding out everything from 2026-07-01:

ENTER SHORT AT	IN-SAMPLE (180 SESSIONS)	OUT OF SAMPLE (46 SESSIONS)	LAST MONTH (21 SESSIONS)
13:35Z	+\$38.91	−\$17.37	−\$19.23
14:00Z	+\$46.08	+\$23.51	+\$2.44
14:05Z	+\$60.22	+\$15.51	−\$6.68
14:30Z	+\$46.27	−\$5.99	−\$0.30
15:00Z	+\$25.62	+\$5.77	+\$15.87

14:00Z holds out of sample. Its neighbours thirty minutes either side do not. **That is a spike, not a plateau**, and your own standing rule is that a plateau is the evidence and a peak is not. Between the DST refutation and the spike shape, **the clock is PARKED, not promoted** — but it is free to shadow, and six weeks of live data settles it in a way another year of backtest never will.

5. RULE 4 — THE FOUR CELLS, EACH SCORED ON ITS HOME REGIME AGAINST ITS OWN CONSTANT

Six mechanical gates, one per promising cell from the map plus two extras. Each was given the **full exit matrix** (below), the best config chosen by **strip-best-3-sessions** — never by the raw peak, because the raw peak is always the least robust cell — then scored on 180 in-sample sessions, on 46 genuinely held-out sessions from 2026-07-01, and against **the best fixed-time constant in its own home window with the identical exit**.

Mechanics, deliberately pessimistic: entry at the **open of the bar after** the signal bar (no left-edge replay); stop checked against the bar's low/high and filled at the level; **if a bar touches both the stop and the target, the stop wins**; \$1.50 fee plus 0.10pt slippage a leg; a 30-minute cooldown and one position at a time.

CELL	GATE	HOME	N	\$/ TRADE IS	1ST HALF	2ND HALF	STRIP BEST 3	OOS (46	BEST CONST ANT	TRIGGE R ADDS
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							SESSIO NS	SESSIO NS)		
REVERSIO N-SHORT	revS_US_fa deup	US 13:30- 20:00Z	276	+\$33.34	+\$11.22	+\$58.55	+\$16.97	-\$5.13	+\$60.22 @14:05	-\$26.88
MOMENTU M-SHORT	momS_US_go dn	US 13:30- 20:00Z	241	+\$30.80	+\$8.80	+\$56.58	+\$12.15	-\$7.73	+\$56.96 @14:05	-\$26.15
MOMENTU M-SHORT	momS_ASIA_ godn	Asia 00:00- 07:00Z	794	+\$6.06	+\$4.01	+\$8.04	-\$0.26	+\$5.66	+\$19.26 @02:05	-\$13.20
MOMENTU M-LONG	momL_LDN_g oup	London 07:00- 13:30Z	1,143	-\$0.31	+\$0.49	-\$1.13	-\$1.88	-\$7.19	+\$2.01 @07:05	-\$2.32
REVERSIO N-LONG	revL_LDN_f adedn	London 07:00- 13:30Z	1,233	-\$2.92	-\$3.32	-\$2.55	-\$4.09	-\$4.07	+\$3.95 @07:05	-\$6.87
REVERSIO N-LONG	revL_US_fa dedn	US 13:30- 20:00Z	737	+\$0.63	+\$4.84	-\$3.41	-\$3.94	-\$5.82	+\$13.35 @16:05	-\$12.72

Every single one is beaten by its own constant, and five of six lose money out of sample. Not one cell has a gate. That is the honest state of the gold hunt tonight.

And it is the same story on the fairer per-session ruler, which asks "which would you rather own for a day" rather than "which is better per trade":

	\$/SESSION IN SAMPLE	\$/SESSION OUT OF SAMPLE	TRADES A SESSION
revS_US_fadeup	+\$51.12	-\$7.92	1.53
momS_US_godn	+\$41.24	-\$9.74	1.34
constant short at 14:00Z	+\$46.08	+\$23.00	1.00

5.1 The one cell that survived out of sample, and why it is still not a gate

momS_ASIA_godn — short the Asian session when the last 30 minutes fell more than 1 ATR — is the only candidate positive out of sample (**+\$5.66/trade over 207 trades on 45 sessions**, in sample +\$6.06 with both halves positive). It also **loses to its own constant by \$13.20**, its strip-best-3 sessions is -\$0.26, and **the live desk is config-blocked from opening between 00:00 and 07:00Z anyway**, so even a validated Asia gate could not trade today without a config change. **PARKED, with a named revival condition** in the disposition table.

6. RULE 3 — THE FULL EXIT MATRIX, and the most actionable structural result of the night

Every entry got three exit families swept and the grid is in reports/friday_v7/gf6/grid_*.csv. Best cell of each family, in sample, ranked by strip-best-3-sessions:

ENTRY	FAMILY	BEST CONFIG	N	\$/TRADE	MEDIAN	WIN%	STRIP BEST 3 SESSIONS
revS_US_fadeup	naked + clock	stop 7×ATR, cap 480m	276	+\$33.34	−\$56.50	45%	+\$16.97
revS_US_fadeup	scalp	stop 5×ATR, target 1.5R, cap 240m	493	+\$11.20	−\$11.50	48%	+\$4.70
revS_US_fadeup	chandelier	stop 5×ATR, arm 3 / trail 1, cap 480m	695	+\$3.72	+\$57.23	65%	+\$0.20
momS_US_godn	naked + clock	stop 10×ATR, cap 480m	241	+\$30.80	−\$23.50	47%	+\$12.15
momS_US_godn	chandelier	stop 5×ATR, arm 2 / trail 1, cap 480m	847	+\$8.83	+\$39.27	75%	+\$6.08
momS_US_godn	scalp	stop 5×ATR, target 2R, cap 240m	475	+\$14.28	−\$26.50	46%	+\$4.07
momS_ASIA_godn	scalp	stop 3×ATR, target 2R, cap 240m	794	+\$6.06	−\$54.70	37%	−\$0.26
momL_LDN_goup	scalp	stop 3×ATR, target 0.5R, cap 240m	1,143	−\$0.31	+\$23.77	69%	−\$1.88
revL_LDN_fadedn	naked + clock	stop 10×ATR, cap 60m	669	−\$0.59	−\$2.50	49%	−\$7.07
revL_US_fadedn	naked + clock	stop 5×ATR, cap 60m	737	+\$0.63	−\$4.50	49%	−\$3.94

6.1 The stop sweep, and proving it is a plateau rather than a censored ruler

The first grid improved monotonically from a 2× to a 5× ATR stop, which is exactly what a **censored** sweep looks like: if every stop tested sits inside the trade's normal adverse excursion, the widest one always wins and you have proved nothing except that your stops were too tight. So I extended it to 7×, 10× and **no stop at all** (revS_US_fadeup, \$/trade):

STOP \ CAP	60M	120M	240M	480M	720M
2×ATR	−\$1.94	−\$1.80	−\$3.45	−\$2.89	−\$2.89
3×ATR	−\$1.54	−\$5.48	−\$2.94	−\$3.63	−\$3.71
5×ATR	+\$3.50	+\$4.78	+\$8.67	+\$13.62	+\$14.22
7×ATR	+\$0.04	+\$7.30	+\$14.56	+\$25.47	+\$26.20

10×ATR	+\$0.61	+\$1.99	+\$13.05	+\$22.82	+\$24.04
no stop	−\$0.05	+\$0.99	+\$10.89	+\$30.78	+\$29.42

After stripping the best three sessions, 7× at 480m still reads **+\$12.41** while no-stop reads +\$10.62 — so there **is** an interior optimum and it is not censored. The fraction of trades that hit the stop falls 78% → 68% → 52% → 41% → 29% → 0% across the column, which is the shape you want to see. **That part of the work is sound; it is the entry that has no edge, not the exit.**

6.2 ★★ Gold does not want MNQ's slot configuration

Rule 3(c) asked specifically for the desk's dual-slot shape — Lot A on a tight scalp, Lot B on a chandelier — because that is what live MNQ actually runs. On the same entries, same sessions:

ENTRY	LOT A (2×ATR STOP, 1R TARGET)	LOT B (5×ATR STOP, 2/2 CHANDELIER)	DUAL SLOT TOTAL	ONE LOT, 7×ATR STOP, 8H CAP
revS_US_fadeup	n=1,079, −\$0.66/tr, −\$713	n=775, −\$1.21/tr, −\$936	−\$1,649	+\$9,202
momS_ASIA_godn	n=1,195, −\$0.59/tr, −\$703	n=910, −\$6.33/tr, −\$5,765	−\$6,467	+\$3,475

The dual slot is not a small loss on gold, it is a catastrophe, and it is the opposite sign to the single wide lot on the identical entries. Lot A's tight stop takes gold out of positions that need hours, and the chandelier is shaken out of a grind — which is precisely what the 08-14 leads file found from the other direction ("a trail is shaken out of a grind; the exit was the cap, not the filter"). This is independent confirmation of that, on a different tape, a year long, and it is the single most portable thing in this section: **if a gold gate is ever promoted, give it ONE lot, a very wide stop and a clock. Do not inherit the live slot config.**

Note what that costs you in comfort: the winning configs have a **negative median trade** (−\$56.50 on the lead cell) and win 45% of the time. The money is in a minority of large winners. That is the same shape the live MNQ desk runs at 39%, and by your standing rule win-rate is not the criterion — but it does mean a gold arm will look wrong for weeks at a time.

7. EVERYTHING THAT DIED THIS WEEK, BY NAME

#	ATTEMPT	WHAT IT WAS	N	RESULT
---	---------	-------------	---	--------

					CAUSE OF DEATH
1	blanket forward-return map	72 lookback × threshold × horizon cells, all tape	58k–121k per cell	no cell's bootstrap band clears the \$3.50 cost	correct by construction — a blanket number is a bug, shown not asserted
2	conditional 288-cell map	session × ATR × ER × trigger × horizon	226 sessions	~40 "significant" cells against ~29 expected from noise	split-half: sign held 49.3%, a coin
3	London reversion (fade London down-moves)	the biggest number in the full-sample table, + \$41/tr at 4h	5,991	full sample strong, second half reversed	split-half: London sign held 36.1%, worse than a coin
4	LATE-session momentum	+\$18.67/tr long at 20:00Z entry	5,474	flips sign against its constant at the 120m hold	beaten by its own constant; sign flips with hold length
5	revS_US_fadeup	REVERSION-SHORT: fade a 1-ATR US up-extension	276	+\$33.34/tr in sample	beaten by the constant by \$26.88; OOS −\$5.13
6	momS_US_godn	MOMENTUM-SHORT: follow a US down-extension	241	+\$30.80/tr in sample	beaten by the constant by \$26.15; OOS −\$7.73
7	momS_ASIA_godn	MOMENTUM-SHORT: follow an Asian down-extension	794	only OOS survivor, + \$5.66	beaten by the constant by \$13.20; strip-best-3-sessions −\$0.26 — PARKED, not dead
8	momL_LDN_goup	MOMENTUM-LONG: follow a London up-extension	1,143	−\$0.31/tr in sample	negative in sample AND out of sample (−\$7.19)
9	revL_LDN_fadedn	REVERSION-LONG: fade a London down-extension	1,233	−\$2.92/tr	negative in both halves and out of sample
10	revL_US_fadedn	REVERSION-LONG: fade a US down-extension	737	+\$0.63/tr	second half −\$3.41, OOS −\$5.82, beaten by constant by \$12.72
11	the LBMA PM fix hypothesis	the 14:00Z short is the London fix	225	BST peak 15:30–16:00Z, GMT peak 13:00Z	REFUTED by the DST natural experiment — the effect moves the WRONG way when London's clocks change
12	the dual-slot exit on gold	Lot A scalp + Lot B chandelier, live MNQ's shape	1,854 legs	−\$1,649 vs +\$9,202 for one wide lot	negative, and opposite in sign to the single lot on identical entries
13	the tight-R scalp family	targets 0.5R–2.0R against matched stops	858–1,649	best cell +\$11.20 vs +\$33.34 naked	dominated by the naked-and-wide family on every entry
14	the chandelier family on gold	arm 1–3 × ATR, trail 1–3 × ATR	983–1,396	best cell +\$8.83	dominated; the trail is shaken out of a grind
15	mgc_break_fade_nobok (LIVE)	unfiltered 60-min level-break fade	186 live	−\$9.58/tr, band −\$16.92...−\$2.08	live, well powered, band excludes zero — the unfiltered fade loses
16	mgc_holebreak_fade_short (LIVE)	book-filtered short arm	20 live	−\$18.80/tr, worse than its side-matched control	thin — PARKED, not refuted

8. WHAT COULD STILL BE WRONG WITH THIS SECTION

1. **My constant is the best of about a dozen fixed times, chosen in sample.** That flatters the control and therefore hurts the candidates. It is the conservative direction, but if you want the candidates given the benefit of the doubt, use the mean constant instead: on the lead cell that changes the sign of the answer. On the lead cell over the whole 226 sessions: the gate makes +\$25.47 a trade, the BEST fixed time makes +\$51.28 (14:05Z) and the MEAN of the seven fixed times tested makes +\$13.42. Against the best constant the trigger is worth -\$25.81; against the mean constant it is worth +\$12.05. **I do not believe the generous reading rescues it** — the held-out leg from 2026-07-01 is still -\$5.13 a trade and -\$7.92 a session — but you should know the number moves with the choice of control, and by how much.
2. **226 sessions is one macro regime for gold:** a year in which the metal ran from 3,892 to 5,037 and back to 4,492. Anything conditioned on direction is conditioned on that story.
3. **The last six sessions (2026-08-28 onward) are our own 5-second capture resampled to minutes, not IBKR's 1-minute bars,** so the tail of the sample is built differently from the 220 sessions before it. The joint is not visible in the ATR series, but it is a seam. There is also a genuine IBKR hole at **2026-01-09 ... 2026-01-22** which no contract fills.
4. **I have not touched L2 at all this week.** Rule 2 called the book the most promising unexplored angle for gold, and the depth data exists. I ran out of night after the contract-roll discovery forced a rebuild of everything. **That is the one stone still unturned** — see §11.
5. **My exit walk arms the chandelier using the entry bar's own high/low,** which is mildly optimistic. The stop-first tie-break pulls in the other direction. Neither matters for the verdicts, since the chandelier family lost anyway.

9. ★ PER-CELL VERDICT TABLE

CELL	BEST CANDIDATE	HOME REGIME IT NEEDS	ROUTER RULE THAT SHOULD ARM IT	EXPECTANC Y ON ITS HOME REGIME	BLANKET, FOR CONTEXT	VERDICT
MOMENTUM-LONG	momL_LDN_goup — buy a 1-ATR 30-min up-extension	London 07:00–13:30Z, any ATR	session==LONDON and ret30 >= 1.0*ATR60*sqrt(30/14)	–\$0.31/trade (n=1,143, 179 sessions); OOS – \$7.19	–\$2.45/tr (all tape, h=60)	NO GATE. Negative in sample and out. The one cell where I found nothing even worth parking on its own terms
MOMENTUM-SHORT	momS_ASIA_godn — sell a 1-ATR 30-min down-extension	Asia 00:00–07:00Z, any ATR	session==ASIA and ret30 <= -1.0*ATR60*sqrt(30/14)	+\$6.06/trade (n=794, 180 sessions); OOS + \$5.66	+\$0.02/tr (all tape, h=60)	PARKED. The only OOS survivor, but loses to its own constant by \$13.20, strip-best-3-sessions –\$0.26, and the desk cannot open in the Asia block
REVERSION-LONG	revL_US_fadedn — buy a 1-ATR 30-min down-extension	US 13:30–20:00Z, calm/chop	session==US and ret30 <= -1.0*ATR60*sqrt(30/14) and ER60 < 0.20	+\$0.63/trade (n=737, 180 sessions); OOS – \$5.82	–\$7.02/tr (all tape, h=60)	NO GATE. Both attempts (US and London) negative out of sample. Gold's down-extensions do not revert — they continue
REVERSION-SHORT	revS_US_fadeup — sell a 1-ATR 30-min up-extension	US 13:30–20:00Z, any ATR	session==US and ret30 >= 1.0*ATR60*sqrt(30/14)	+\$33.34/trade (n=276, 180 sessions); OOS – \$5.13	–\$4.55/tr (all tape, h=60)	PARKED. The strongest in-sample number in the section and the most thoroughly killed: the constant beats it by \$26.88 and it is negative out of sample
(not one of the four, but it beat all four)	the 14:00Z clock short — sell one lot at 14:00Z, no signal	US, once a session	hhmm == 14:00Z	+\$41.57/trade (n=225); OOS + \$23.51; strip-best-3 +\$22.89	n/a	SHADOW. Survives split-half and OOS, but the DST test refuted its mechanism and its neighbours 30 minutes either side fail. A spike, not a plateau — settle it forward

10. SHADOW-READY BLOCKS

10.1 The build item that gates the rest

BUILD — scripts/backfill_to_lake.py, the contract-roll dedup.
Today: rows are deduped on (symbol,timeframe,bar_ts) keeping the FIRST file in name order,
i.e. the earliest expiry with a print. That is the front month only until

liquidity

rolls, about five weeks before expiry.

Effect: 22.1% of MGC's 352,319 year-tape minutes are a different contract, median 30.10pt

(\$301/lot) away, on 77 of 266 sessions whose median minute volume in the lake is 0 lots.

Fix: rank each bar_ts's candidate rows by that DAY's contract volume and keep the winner.

Then: re-run gf3_mgc_year.py and re-label last Friday's section 5 conclusions.

△ MNQ has the same seven-expiry structure and the same dedup. It is very probably affected too

and I have not measured it.

10.2 The gold slate, as I would leave it

```
# — MGC GOLD, 2026-09-04 revision. Runs in the MGC service only (cfg.symbol="MGC",
#   cfg.value_per_point=10.0, fee_rt=1.50, its own store data/shadow_mgc.db).
#   Never in the MNQ slate: reprice_pending() takes ONE value_per_point.

# — UNCHANGED. 17 live days say the LONG arm is +$25.55/trade against a side-
#   matched control at
#   -$6.18, and that it has been positive two weeks running. n=19 and it fails
#   strip-best-3, so
#   it keeps accumulating and is NOT promoted.
ShadowVariant("mgc_holebreak_fade_long", "level_break", symbol="MGC", side="LONG",
              params={"look_min": 60, "margin_atr": 0.10, "fade": True,
                    "book_band_pt": 1.0, "obstacle_max": 0, "cooldown_min": 45},
              **MGC_EXIT),

# — UNCHANGED but now on notice. -$18.80/trade over 20, WORSE than its side-matched
#   control
#   (-$12.76). Thin n is never a kill, so it runs – but if it is still below its
#   control at
#   n=50 the book gate is long-only and this arm comes off.
ShadowVariant("mgc_holebreak_fade_short", "level_break", symbol="MGC", side="SHORT",
              params={"look_min": 60, "margin_atr": 0.10, "fade": True,
                    "book_band_pt": 1.0, "obstacle_max": 0, "cooldown_min": 45},
              **MGC_EXIT),

# — THE CONTROL. KEEP IT, and note it has now done its job: 186 trades, -$9.58/
#   trade, bootstrap
#   band -$16.92...-$2.08. The unfiltered fade LOSES, which is what makes the long
#   arm's +$31.73
#   side-for-side gap meaningful rather than decorative.
ShadowVariant("mgc_break_fade_nobook", "level_break", symbol="MGC", side="",
              params={"look_min": 60, "margin_atr": 0.10, "fade": True,
                    "book_band_pt": 1.0, "require_book": False,
                    "cooldown_min": 45}, **MGC_EXIT),

# — ★NEW, and the only new arm I am proposing. THE 14:00Z CLOCK SHORT.
#   It is not one of the four cells; it is the CONSTANT that beat all four, and it
#   belongs in the
#   book because it is the benchmark every future gold gate has to clear. +$41.57/
#   trade over 225
#   sessions, +$23.51 out of sample, +$22.89 after stripping the best three
#   sessions.
```



```

#   △ Its mechanism is REFUTED (the DST test says it is not the London fix) and its
neighbours at
#   13:35 and 14:30 fail out of sample. It is a spike. Shadow settles that; a
backtest will not.
#   GATE KIND: "clock_rider" is 90% of the way there – it has the window, the
cadence and
#   time_cap_s. It is NOT sufficient: _rider_entry takes its SIDE from sign(close -
close[-15]),
#   and ShadowVariant.side only FILTERS that, so side="SHORT" would fire only on
sessions where
#   the previous 15 minutes happened to fall. That is a different and untested
gate.
#   REQUIRED: one new field, `rider_fixed_side: str = ""`, read only by
_rider_entry – when set,
#   skip the momentum sign and return that side. Four lines, and __post_init__
should refuse it
#   on any gate that is not clock_rider, exactly as it already refuses rider_win_*.
ShadowVariant("mgc_clock_short_1400", "clock_rider", symbol="MGC", side="SHORT",
               rider_fixed_side="SHORT",          # ← needs the new field above
               rider_cadence_min=1,
               rider_win_start_s=14 * 3600, rider_win_end_s=14 * 3600 + 60,
               stop_atr_mult=7.0, target_r=99.0,    # effectively no target: it is
stop-and-clock
               chandelier=False, time_cap_s=480 * 60,
               params={"cooldown_min": 600}),

# — ★NEW CONTROL for the arm above, and it is the whole point of shipping it. If
14:00Z is a real
#   hour, these two lose and it wins. If it is a spike in a noisy surface, all
three wander
#   together. 13:35Z was +$38.91 in sample and -$17.37 out; 15:00Z was +$25.62 and
+$5.77.
ShadowVariant("mgc_clock_short_1335", "clock_rider", symbol="MGC", side="SHORT",
               rider_fixed_side="SHORT", rider_cadence_min=1,
               rider_win_start_s=13 * 3600 + 2100, rider_win_end_s=13 * 3600 + 2160,
               stop_atr_mult=7.0, target_r=99.0, chandelier=False,
               time_cap_s=480 * 60, params={"cooldown_min": 600}),
ShadowVariant("mgc_clock_short_1500", "clock_rider", symbol="MGC", side="SHORT",
               rider_fixed_side="SHORT", rider_cadence_min=1,
               rider_win_start_s=15 * 3600, rider_win_end_s=15 * 3600 + 60,
               stop_atr_mult=7.0, target_r=99.0, chandelier=False,
               time_cap_s=480 * 60, params={"cooldown_min": 600}),

```

Router rule for the clock arms: none, deliberately. They are a clock, and a router condition would make the spike-versus-plateau question unanswerable. The only condition they need is the one they already have — `symbol == MGC`, and never in the MNQ slate.

What ShadowSim._entry can and cannot express here:

ARM	EXPRESSIBLE TODAY?
the three <code>level_break</code> arms	✓ already running
<code>mgc_clock_short_*</code>	✗ needs <code>rider_fixed_side</code> , four lines in <code>shadow.py</code> — see the comment above

10.3 The MGC exit block, revisited

MGC_EXIT currently ships stop_atr_mult=3.0 with a locked chandelier at arm 2.0 / trail 2.0 and an 8-hour cap. **My year-long grid says that is the wrong shape** — a 3×ATR stop is the single worst column in the sweep (negative at every cap) and the chandelier family is dominated everywhere. If you want the gold book measured on the exit the tape actually prefers, the honest block is stop_atr_mult=7.0, chandelier=False, target_r=99.0, time_cap_s=480*60. **I am NOT proposing you change the three running arms to it** — they have seventeen days of history under the current exit and changing it now destroys the comparison. Ship it as a fourth arm if you want the answer.

11. DISPOSITION TABLE

LEAD	VERDICT	WHAT IT NEEDS
mgc_holebreak_fade_long (live)	SHADOW	keep accumulating. Review at n=50 : does it still beat its side-matched control, and does it survive strip-best-3?
mgc_break_fade_nobook (live control)	SHADOW — keep	it is not trying to win. It has just earned its place by making the long arm's +\$31.73 gap interpretable
mgc_holebreak_fade_short (live)	PARKED	revive if it beats its side-matched control at n=50 . If it does not, the book gate is long-only and this arm comes off
the 14:00Z clock short	SHADOW	ship it with the 13:35Z and 15:00Z controls. If 14:00Z separates from both over ~40 sessions it is an hour; if all three wander together it is noise and it comes off
momS_ASIA_godn (momentum-short)	PARKED	revive if (a) it beats a fixed-time Asian constant on another 100 sessions, and (b) the desk's 00:00-07:00Z open block is lifted for gold. Neither is true today
revS_US_fadeup (reversion-short)	PARKED	revive only with a conditioner that is not the clock — something that says which up-extensions revert. Its raw form is dead: beaten by a constant, negative out of sample
momL_LDN_goup (momentum-long)	PARKED	nothing in the 30-minute-stretch family works here. Revive only from a different trigger shape entirely — the level/absorption route in §12
revL_* (reversion-long, both)	PARKED	gold's down-extensions continue rather than revert on this year's tape. Revive on a year where gold trends up , or from an L2 mechanism rather than a price-stretch one
the LBMA PM fix mechanism	REFUTED	the DST natural experiment. The effect moves the wrong way when London's clocks change; there is no reformulation that survives a boundary set by parliament

the blanket / conditional forward-return map	REFUTED as a source of gates	split-half sign agreement 49.3% across 213 cells. It remains useful as a map
the dual-slot exit on gold	REFUTED	−\$1,649 against +\$9,202 for one wide lot on identical entries, both lead cells, a year of tape
the lake's gold year tape	BROKEN — build item	\$10.1. Until it is fixed, no year-length gold conclusion should be quoted, including last Friday's

There is no LIVE candidate and I am not proposing one. There are two SHADOW items — one existing arm that has earned another month, and one new benchmark arm with its own controls.

12. THE ONE STONE STILL UNTURNED

The order book. Rule 2 said gold's book is far thinner than MNQ's, so absorption, depletion and level-reclaim mechanics that wash out in Nasdaq depth may be visible in gold — "that is the most promising unexplored angle". **I did not get to it**, and I want to be plain about why rather than pretend the night was spent optimally: the contract-roll discovery in §1.1 forced me to rebuild the entire tape before any of the price work could be trusted, and that ate the book time.

It is also the angle with the best prior. The one thing on this desk that **is** working in live gold is `mgc_holebreak_fade_long`, and its edge over its own control is **entirely a book condition** — `obstacle_max: 0`, "nothing resting in the way". +\$31.73 a trade, side for side, on the live tape. Every price-shaped thing I built tonight died; the one book-shaped thing we own is the only survivor. **That is where next week goes**, and specifically:

1. Rebuild the front-month year tape first (§10.1), so nothing else is measured on a ghost.
2. Take obstacle/support off `depth.db.depth_snap` for the full L2 history and ask, per side, whether "nothing in the way" predicts anything **without** the level break attached to it — because right now we cannot tell whether the level or the book is carrying that +\$31.73.
3. Then and only then build a book-native gate for the two cells that have no candidate at all, momentum-long and reversion-long.

Files

Everything in this section is reproducible from `reports/friday_v7/gf6/`:

SCRIPT	WHAT IT DOES
<code>scripts/gf6_mgc_base.py</code>	rebuilds the front-month minute tape (the contract-roll fix)
<code>scripts/gf6_mgc_map.py</code>	the blanket and conditional forward-return maps
<code>scripts/gf6_mgc_signcheck.py</code>	the horizon-consistency and split-half audit of the map
<code>scripts/gf6_mgc_us.py</code>	the US session split by side and by half
<code>scripts/gf6_mgc_const.py</code>	the all-minute constant control and placebo
<code>scripts/gf6_mgc_gate.py</code>	the mechanical gate lab — entry, stop, the exit families
<code>scripts/gf6_mgc_exitgrid.py</code>	Rule 3, the full exit matrix
<code>scripts/gf6_mgc_stopsweep.py</code>	the extended stop × clock sweep and the censoring check
<code>scripts/gf6_mgc_verdict.py</code>	the trade-level battery: constant, placebo, split-half, strip
<code>scripts/gf6_mgc_fourcells.py</code>	all four cells, in sample, out of sample, versus their constants
<code>scripts/gf6_mgc_clock.py</code>	the 24-hour clock sweep, both sides
<code>scripts/gf6_mgc_fix.py</code>	the LBMA-fix DST natural experiment
<code>scripts/gf6_mgc_drift.py</code>	pricing the drift against the gate
<code>scripts/gf6_mgc_shadow_ledger.py</code>	the live gold shadow book

The prior week's section is preserved verbatim at `gf_MGC_2026-08-25.md`.

THE POOLED RUN-CATCHER — re-derived on 60 days, and graded against its own live shadow arm

Ignore the cluster labels. Board any run late, ride it wide, filter it causally — the same hypothesis, on 904,075 five-second bars.

What this one moves: The frozen board above PARKS the pooled rider on an out-of-sample leg worth +\$14. Tonight's run reaches the same disposition by a harder road and adds the number that settles it: `rider_w5`, promoted to shadow last week on a +\$19.29-per-trade backtest, has since booked **108 live shadow trades at – \$25.95 each** — a \$45-a-trade swing between fit and forward. Source: *gf_full_RIDER_ALL.md*, written 2026-09-04 21:39 UTC, 7,840 words, reproduced in full and unedited below.

GREENFIELD SYNTHESIS — ONE RIDER FOR EVERY SAT-OUT RUN

MNQ · full parquet lake 2026-06-19 → 2026-09-04 · 904,075 five-second bars · 60 trading days · \$2.00/point · \$1.50 per round trip

THE SHORT VERSION, BEFORE ANY NUMBERS

Garrath — you asked whether one direction-agnostic run-catcher could take some of the money we watch walk past every week. I built one, pooled all the sat-out runs and ignored the cluster labels exactly as you asked. Here is the honest answer.

The rider SEES the money. It cannot KEEP it.

A dumb momentum detector — "price has moved $1.5 \times \text{ATR}$ in the last five minutes, get on" — boards **71 of the 73 sat-out runs**, a median of **3.4 minutes in, with about three-quarters of the move still ahead**. Detection is genuinely solved. But to catch those 71 runs it takes **2,332 trades in five days** and loses **\$8,802** doing it, because the same rule fires 466 times a day on ordinary tape. Filtering the chop out is where it dies, and it dies with a named cause of death that I did not have to go looking for.

The strongest single piece of evidence is not my backtest. It is our own shadow book.

Last week's version of this exact idea — rider_w5, the board gate, promoted to shadow on a **+\$4,841 / +\$19.29-per-trade** backtest — has now been running live in shadow since 08-17. Its record is **108 trades, -\$2,803, -\$25.95 per trade, 21% win rate**. That is a **\$45-a-trade swing** between a fitted backtest and forward reality, on the same hypothesis I was asked to re-derive. My own out-of-sample test found the same thing independently: across the 81 configurations I tried, the **median out-of-sample result is -\$4.26 a trade**.

So: **VERDICT = PARKED**, died at **STEP 3 (FILTER)**, killed by the honest fit-then-test out-of-sample protocol and confirmed by 98 live shadow trades. One narrow leg survives everything I could throw at it and is worth a shadow slot — but it is **not a rider**, it is a short-side momentum scalp, and it takes **2.5% of the ceiling, not 60%**.

△ FIRST, A CORRECTION TO MY OWN BRIEF

My instructions said "68 MNQ runs at or above $1.5 \times \text{ATR}$, we took EIGHT, we sat out SIXTY, worth \$8,025 of ceiling against \$1,198 on the ones we traded." Those

are **last week's numbers**. The frozen census I was told to read says something different, and I have used the frozen file throughout:

BRIEF SAYS		FROZEN CENSUS SAYS
runs $\geq 1.5 \times \text{ATR}$	68	76
sat out	60	73
ceiling on the sat-out runs	\$8,025	\$10,515
caught / fought	8	1 caught (+\$586) · 2 fought (+\$262)
the gate that showed up at all	—	day_rider, \$+847

census_summary.json rounds the ceiling to \$10,512; parsing census_stdout.txt row by row gives \$10,515. I use \$10,515. Nothing below depends on the three dollars. **The money on the table this week is 31% bigger than the brief thought**, which makes the question more urgent, not less.

The cluster split of the 73 sat-out runs: UNCLASS 40, OPEN/NEWS 13, VACUUM 12, FLOW-LED 7, VOL-EXPANSION 1. As instructed I pooled all of them and never used a label as an input.

STEP 1 — BOARD IT. (the good news, and it is real)

The rule is deliberately stupid, because it has to be causal. At every 5-second bar I look back W minutes, and if price has travelled $K \times \text{ATR}$ I get on in whatever direction it went. ATR is a true-range ATR-14 built on **one-minute** bars — the gate's ATR, not the router's "LAST HR ATR" mean-range, which is a different and more generous number. Everything is read from the **last closed minute** and filled at the **next** 5-second bar's open, so there is no look-ahead anywhere.

Could we have known, and what was left?

LOOKBACK	THRESHOLD	FIRE D ON	MEDIAN MINUTES INTO THE RUN	MEDIAN % OF THE MOVE ALREADY SPENT	MEDIAN POINTS STILL TO COME
3 min	$0.8 \times \text{ATR}$	71/73 (97%)	1.7	13%	49.5
3 min	$1.5 \times \text{ATR}$	70/73	3.5	28%	39.6
5 min	$1.5 \times \text{ATR}$	71/73 (97%)	3.4	24%	42.2
5 min	$2.0 \times \text{ATR}$	70/73	4.3	35%	33.8
8 min	$2.0 \times \text{ATR}$	71/73	4.9	32%	33.0
10 min	$2.0 \times \text{ATR}$	71/73	5.0	35%	32.8

You asked for the distribution, not an average. At $W=5\text{min}$ / $K=1.5\times\text{ATR}$, across the 71 runs it boards:

	MIN	P25	MEDIAN	P75	MAX
minutes into the run when it fires	0.1	1.8	3.4	5.5	13.7
% of the 15-min move already done	−2%	12%	24%	38%	88%
points left to the census run-end	−1.5	32.1	42.2	54.3	115.8
best excursion in the next 60 min	26.8	55.4	78.8	109.6	253.2
worst excursion in the next 60 min	−0.0	−5.2	−15.5	−35.8	−232.2

- **90%** of the boarded runs still had ≥ 20 points to give.
- **85%** offered ≥ 50 points of favourable excursion within the hour.
- **69 of 71** offered at least $2\times\text{ATR}$ of excursion.

This is a better boarding curve than the gold one. On MGC the same shape fired a median 19 minutes in; on MNQ it fires at 3.4 minutes with 76% of the move ahead. And it boards every cluster equally well, which is the empirical justification for ignoring the labels:

CLUSTER	N	BOARDED	MEDIAN MINS IN	MEDIAN 60-MIN MFE	MEDIAN POINTS LEFT
UNCLASS	40	38	3.0	77.0	42.4
OPEN/NEWS	13	13	3.8	66.2	43.2
VACUUM	12	12	5.0	90.7	35.3
FLOW-LED	7	7	3.4	80.2	51.8
VOL-EXPANSION	1	1	3.5	73.8	22.0

There is no cluster the rider is blind to. **Pooling is safe and the labels add nothing at the entry.** Step 1 passes cleanly.

⚠ **What Step 1 does NOT say.** Every number above is measured inside a run window we already know was a run. It is a conditional-on-success measurement and it is the single easiest place in this whole study to fool yourself. Step 2 is where the detector meets the other 23 hours.

STEP 2 — EXIT. (and a methodology bug I had to fix mid-flight)

△ The bug: my first sweep was not an exit sweep at all

My first pass let position occupancy decide when the next entry could happen — you cannot enter while you are already in. That means a tight exit frees you up sooner and takes **more** trades than a wide one. The consequence:

n ranged from 1,414 to 10,173 across the 56 cells of a "same entries, different exit" grid.

Every cell was scoring a **different entry set**. I had even written "entries are identical across every cell" into the output, which was simply false. That sweep is void and I have thrown it away.

The rebuilt engine computes the signal list **once** from the entry rule alone — fire, then re-arm only when the displacement falls back under the threshold — and applies every exit to that same frozen list. Trades may overlap. That is the price of a clean exit comparison and I am stating it rather than hiding it.

Two other disciplines, both from things this desk has been bitten by before:

- **Stops are evaluated on bar HIGH/LOW, never on the close.** A close-only stop never sees the wick and understates every loss.
- **The target is in ATR, not in R.** Sweeping a stop while the target is quoted in R silently moves the target too, and the resulting sign flip is an artefact of the coupling, not a finding.

The frozen entry list

28,664 fires over 60 days = 478 a day. One every three minutes. That is the real base rate of "price moved 1.5×ATR in five minutes", and it is the whole problem in one number.

The sweep, done properly — net \$, 60-minute cap, same 28,664 entries in every cell

STOP ↓ / TARGET →	1.5	2.0	3.0	4.0	6.0	8.0	CAP
1.0×ATR	−71,997	−81,207	−86,979	−79,500	−82,240	−84,764	−77,750
1.5×ATR	−87,550	−98,381	−92,959	−91,513	−97,834	−96,239	−92,469
2.0×ATR	−84,791	−101,159	−97,968	−95,921	−104,058	−99,196	−95,115

2.5×ATR	−93,746	−114,445	−115,590	−118,083	−131,960	−127,275	−126,787
3.0×ATR	−95,062	−113,343	−117,688	−123,388	−139,907	−132,062	−128,788
4.0×ATR	−79,655	−95,238	−108,105	−117,085	−138,700	−133,319	−131,047
6.0×ATR	−78,541	−89,697	−96,792	−103,760	−132,893	−129,984	−127,882

Same at 30-minute and 120-minute caps: **every one of the 147 cells is negative**. Not one is close.

The cleanest read in this whole document

Strip the exit design away entirely — no stop, no target, just "where is price N minutes after a displacement?" Breakeven needs **+0.75 points a trade** to pay the \$1.50 round trip.

Gross points per trade — negative everywhere:

THRESHO LD	FIRES/ DAY	2 MIN	5 MIN	10 MIN	15 MIN	30 MIN	60 MIN	120 MIN
K=1.0	679	−0.15	−0.33	−0.15	−0.19	−0.12	+0.09	−0.12
K=1.5	478	−0.40	−0.80	−0.59	−0.93	−1.24	−1.24	−0.83
K=2.0	300	−0.45	−0.66	−0.11	+0.10	−0.35	−0.81	−1.53
K=2.5	170	−0.11	−0.18	+0.43	+0.07	−0.17	−1.87	−3.76
K=3.0	86	−0.16	−0.39	+0.60	+0.26	−0.23	−0.62	−2.32

A raw MNQ displacement does not continue. It mildly mean-reverts. The biggest thresholds reverse hardest at the longest horizons — K=2.5 at two hours gives back 3.76 points a trade. This is consistent with the fact that the desk's own profitable gates (rgv_long, capitulation_long, exhaustion_short) are **faders**. We already make our money fading extension; riding it is the same coin the other way up.

A note on the fade, so nobody over-reads it. Fading the same signal reads +1.24 pt / +\$27,952. That is **not independent evidence** — with no stop, fade is exactly minus ride, so the number is arithmetically forced. And at 478 trades a day it is \$0.97 a trade before a penny of slippage, which the next section shows is fatal.

So: is "the exit is the game" wrong?

No — but it does not rescue this entry, and when it does help it helps in the OPPOSITE direction to the one the brief predicted. I go through the wide-exit rescue properly in Step 3. Flagging it here so the finding is not buried:

the cells that survive are **wide stop + TIGHT target**, not wide stop + far target. The exhaustion_short lesson did not transfer.

INTERLUDE — THE HAYSTACK, MEASURED

detector fires inside the census window (4.7 days)	2,332
of those, landing inside a sat-out run the right way round	414 (17.8%)
distinct sat-out runs touched	71 of 73
ceiling on the runs it touched	\$10,273 of \$10,515
trades per day required	466
commission alone on those trades	\$3,498
net actually booked over the census week	-\$8,802

The detector puts a finger on **97.7% of the money**. It just brings 1,918 losing tickets along for the ride. Commission alone is 0.3× the entire ceiling — survivable on its own — but the trading losses are not.

Are run-fires distinguishable BEFORE entry?

This is the question the whole thing turns on. All features causal, read from the last closed minute:

FEATURE	RUN FIRES (N=414)	OTHER FIRES (N=1,918)
ATR-14 (1 min)	12.73 (p25 9.93)	8.71 (p25 6.74)
ER-30	0.21 (p25 0.09)	0.15 (p25 0.07)
ER-15	0.31 (p25 0.15)	0.22 (p25 0.10)
trailing 60-min range / ATR		7.91 7.53
hour (UTC)	13:00	10:00

There is separation. Runs happen when ATR is higher, the tape is cleaner and the clock is in the US session. It is genuinely there — it is just nowhere near strong enough, as Step 3 shows.

STEP 3 — THEN FILTER. (where it dies)

3a. The obvious filters, each placebo-controlled

Every filter is scored against **discarding the same NUMBER of fires at random**, 200 draws. A filter that merely trades less is not a filter. Baseline:

28,664 fires, stop 3×ATR, hold to a 60-minute cap, **−\$128,788**, containing 10,370 winners and 1,793 big (>+\$200) winners.

FILTER	KEPT	NET \$	\$/TRADE	BIG WINNERS KEPT	PLACEBO MEAN (5-95%)	VERDICT
ATR ≥ 10	18,811	−53,008	−2.82	1,664/1,793	−84,688 (−101,145, −70,298)	beats placebo, still deeply red
ATR ≥ 12	14,149	−47,805	−3.38	1,410/1,793	−62,756 (−80,700, −46,070)	inside band
ATR ≥ 15	8,895	−38,796	−4.36	1,131/1,793	−38,394 (−54,978, −21,351)	inside band
ER-30 ≥ 0.20	11,729	−39,370	−3.36	816/1,793	−53,004 (−72,750, −33,984)	inside band
ER-30 ≥ 0.30	5,890	−11,773	−2.00	440/1,793	−25,782 (−39,751, −10,330)	inside band
ER-15 ≥ 0.35	8,687	−35,861	−4.13	562/1,793	−39,358 (−55,003, −22,200)	inside band
US session 13–21Z	9,816	−59,314	−6.04	915/1,793	−43,812 (−60,630, −26,776)	inside band
ATR≥12 and ER-15≥0.35	4,415	−21,290	−4.82	423/1,793	−20,234 (−31,311, −7,629)	inside band
ATR≥12 and US session	6,980	−44,869	−6.43	841/1,793	−31,555 (−46,030, −14,777)	inside band
ATR≥12 & ER-15≥.35 & US	2,028	−10,684	−5.27	240/1,793	−8,600 (−17,026, +960)	inside band
SHORT side only	14,472	−27,041	−1.87	1,044/1,793	−64,803 (−81,083, −46,896)	beats placebo, still deeply red

Two beat their placebo and neither gets anywhere near zero. And look at the column you asked me never to leave out — **what the cut costs in forgone winners**. ER-30 ≥ 0.30 looks like the best percentage improvement on the list and it throws away **1,353 of the 1,793 big winners** to get there. That is the 15/17-winners rule failing: a filter that wins by dropping the winners.

3b. The lever that actually works: DWELL

The fix comes from Musing (4) in the scope — threshold noise. drift's efficiency oscillates across its own floor, and my detector has exactly the same disease: the displacement wobbles over and under 1.5×ATR and re-arms every time, manufacturing fires out of nothing. **Requiring the condition to HOLD for a while is the single most effective thing in this study.**

Gross points per trade, hold-to-cap, no stop:

K	DWELL	N	/DAY	15 MIN	30 MIN	60 MIN	120 MIN
	1.5 5s	28,664	478	−0.93	−1.24	−1.24	−0.83

1.5 60s	5,454	91	+0.82	+0.03	−1.19	−1.73
1.5 180s	1,788	30	+1.18	+0.92	+0.54	−1.58
1.5 300s	653	11	+3.46	+3.52	+2.85	−2.72
2.0 300s	244	4	+3.01	+3.61	+3.68	−3.41
2.5 60s	1,433	24	+1.40	+0.93	−0.26	−3.18
2.5 120s	620	10	+4.50	+5.53	+5.40	+3.07
2.5 180s	328	5	+4.51	+5.20	+5.98	+4.04
2.5 240s	174	3	+8.50	+7.49	+10.83	+2.72
3.0 120s	258	4	+8.35	+9.07	+8.07	+1.05
3.0 180s	128	2	+13.26	+16.12	+9.72	+8.56

This is a monotone plateau, not a spike. Gross rises smoothly in both K and dwell, in every horizon column. That is exactly the parameter-plateau signature a robust optimum is supposed to have, and it is why I did not stop here. It is also why the next section matters so much.

3c. The robustness battery — three candidates, three graves

Exit held at stop 3×ATR, hold to a 60-minute cap.

TEST	RIDER-A (K2.5 DWELL120S)	RIDER-B (K3.0 DWELL120S)	RIDER-C (K1.5 DWELL240S)
n		620	258
net		+\$3,600	+\$2,696
\$/trade		+5.81	+10.45
win%		37%	37%
strip best 5		−\$840	−\$715
strip best 10		−\$3,597	−\$3,034
strip best 1/3 of days		−\$6,987	−\$4,938
top-3 days as % of net		120%	112%
green days	29/60	27/57	22/58
weeks positive	7/12	7/12	5/12
leave-one-day-out	60/60 pass	57/57 pass	42/58 pass
bootstrap 95% CI on \$/trade	[−5.47, 18.01]	[−7.50, 30.20]	[−7.64, 8.42]

All three dead. Remove five trades from six hundred and the profit is gone. The top three days are more than the entire net. The bootstrap cannot separate any of them from zero.

★ **The LODO paradox, live.** Look at that table: **60/60 leave-one-day-out folds "pass" while the top three days are 120% of the net.** Leave-one-day-out cannot fail when several days carry the money — dropping one still leaves the other two. It is a test this desk already knows is weak in exactly this situation, and here it is producing a clean 60/60 pass on a strategy that strip-

the-best kills with five trades. **Do not accept a LODO pass as robustness evidence on its own.**

3d. The wide-exit rescue — and the hypothesis inverts

Before killing an entry the brief requires sweeping the exit wide. Done, on RIDER-A's 620 entries. A cell only counts as surviving if **net > 0 AND strip-best-5 > 0**:

STOP	TARGET	NET \$	\$/TRADE	STRIP-BEST-5	TOP-3-DAY SHARE	BOOTSTRAP CI LOW
2.0	6.0	4,763	7.68	+1,999	50%	−1.05
3.0	2.0	3,053	4.92	+2,067	62%	−1.28
4.0	2.0	3,835	6.18	+2,846	49%	−0.10
6.0	2.0	5,444	8.78	+4,456	40%	+1.29
99.0	2.0	5,732	9.24	+4,743	39%	−0.49
3.0	4.0	1,012	1.63	−830	204%	−6.16
4.0	4.0	−377	−0.61	−2,219	−478%	−9.32

The wide exit does help — but the surviving corner is wide stop + TIGHT target, the exact opposite of the wide-stop-and-far-target shape the brief predicted from exhaustion_short. The far-target cells (3.0/4.0, 4.0/4.0, 4.0/6.0) are the worst on the board. The exhaustion_short result does not generalise to this entry, and that is a finding in its own right: our house rule "wide stop, far target" is a property of fader entries with a snap-back shape, not a universal law.

Exactly **one cell of 48** has a bootstrap CI excluding zero. With 48 cells at 95%, you expect about 2.4 by chance. I carried it forward anyway — but that is the honest prior on it.

3e. RIDER-A-WIDE, and the confound that nearly fooled me

stop 6×ATR / target 2×ATR / 60-min cap, on RIDER-A's 620 entries:

n / net / \$ per trade	620 · +\$5,444 · +\$8.78
win rate	73%
strip best 1 / 3 / 5 / 10	+5,228 / +4,819 / +4,456 / +3,636
top-1 day / top-3 days	21% / 40% of net
green days	39/60 (65%)
weeks positive	10/12
first half / second half	+\$2,781 (\$10.38/tr) / +\$2,663 (\$7.57/tr)
bootstrap 95% CI on \$/trade	[+1.67, +16.36] — excludes zero
exit mix	TARGET 437 · STOP 95 · CAP 78 · HALT 10
LONG	n=300 · −\$384
SHORT	n=320 · +\$5,828

This one survives strip-the-best, it is not day-concentrated, and its confidence interval clears zero. It is also **entirely the short side** — and over these 60 days **MNQ fell 988 points with only 28 of 60 days closing up**. A short bias on a falling tape is a beta read, not a gate. So I went after it with matched placebos.

PLACEBO 1 — random entry times, same count, same side mix, same exit, 200 draws:

NET	
real signal	+\$5,444
random-time placebo, mean	−\$835
random-time placebo, 5-95%	[−\$4,684, +\$2,933]
best of 200 random draws	+\$4,173
draws beating the real signal	0 / 200 — p = 0.000

PLACEBO 2 — isolating the drift, short-only vs long-only:

SIDE	REAL	RANDOM-ENTRY PLACEBO MEAN (5-95%)	P	READING
SHORT	+\$5,828	+\$256 [−2,078, +2,747]	0.000	signal adds edge
LONG	−\$384	−\$826 [−3,681, +2,136]	0.415	signal adds nothing

The down-drift alone is worth \$256, not \$5,828. The signal is carrying the rest. And a backward shift would be an oracle, so I used forward/stale shifts, which are proper placebos:

SIGNAL USED	NET	\$/TRADE
on time (real)	+\$5,444	+8.78
5 minutes stale	+\$3,824	+6.17
15 minutes stale	−\$336	−0.54
30 minutes stale	−\$4,842	−7.81
60 minutes stale	−\$3,802	−6.13

Clean monotone decay. The edge is time-local, which is what a real signal looks like and what a data artefact does not. **RIDER-S (the short leg) is a genuine signal.** It is just not a rider, and — as the next section shows — it does not hold up out of sample.

3f. The honest out-of-sample test, and the actual cause of death

Everything above used all 60 days to choose as well as to score. That is the sin this whole battery exists to catch. So: **fit on days 1-40 (06-19 → 08-10), test on**

days 41-60 (08-10 → 09-04), and let the in-sample search pick the winner with no knowledge of the test window.

Short-only family, 81 configurations:

CONFIG	STOP	TARGET	IS NET	IS \$/TR	OOS NET	OOS \$/TR	OOS N
K2.0 dwell60s ← the in-sample pick	6.0	CAP	10,804	10.12	−716	−1.61	445
K2.0 dwell60s	3.0	CAP	9,655	9.04	−524	−1.18	445
K2.0 dwell60s	99.0	CAP	8,338	7.81	−335	−0.75	445
K2.0 dwell60s	6.0	3.0	7,021	6.57	−2,820	−6.34	445
K2.5 dwell120s	99.0	CAP	5,843	27.05	+996	+9.58	104
K2.0 dwell60s	99.0	3.0	5,687	5.32	−3,172	−7.13	445
K2.5 dwell120s	6.0	2.0	5,261	24.35	+567	+5.45	104
K2.0 dwell60s	6.0	2.0	5,239	4.91	−1,470	−3.30	445

- The config the search would actually have picked returns −\$1.61 a trade out of sample.
- Only 3 of the top 10 in-sample configs stayed positive out of sample.
- Median out-of-sample result across all 81 configurations tried: −\$4.26 a trade.

Yes, the specific cell I liked (K2.5 dwell120s, stop 6 / target 2) does hold at +\$5.45 a trade out of sample. But **I only know that cell is the good one because I looked at the whole sample**. Under the honest protocol it was not the pick. That is the difference between a result and a hindsight.

CAUSE OF DEATH: STEP 3 (FILTER), killed by the fit-then-test out-of-sample protocol. Secondary: strip-the-best-third-of-days takes RIDER-A from + \$3,600 to −\$6,987.

★ THE DECISIVE EVIDENCE — WE ALREADY SHIPPED THIS AND IT IS LOSING

I would be presenting the above as an interesting-but-inconclusive backtest if the desk had not already run the experiment for real. **It has.**

gazbot7/shadow.py line 929 carries ShadowVariant("rider_w5", "board", ...) — the board gate, k=2.0, w=5, US session 13:00–20:00Z, stop

3.0×ATR, 2R target, 120-minute cap, 15-minute cooldown. That is **this hypothesis**, boarded late, promoted to shadow on a backtest of **+\$4,841 / +\$19.29 per trade**. It has been live in the shadow book since 2026-08-17.

Its actual record:

ALL ROWS		FROM THE 08-18 17:49 CONFIG BOUNDARY THE CODE ITSELF NAMES AS THE HONEST START	
trades		108	98
net (real_pnl)		-\$2,803	-\$2,515
\$ per trade		-\$25.95	-\$25.66
win rate		21%	21%
LONG	n=56 · -\$1,102		
SHORT	n=52 · -\$1,701		
exit mix		ABSORPTION_CUT 39 · STOP 33 · TARGET 19 · TIME_CAP 7	

Both shadow evidence filters are clean — every row is fill_status='filled' and no row carries a data_quality flag — so none of this is a quarantine artefact.

The backtest said **+\$19.29 a trade**. Reality is **-\$25.66**. That is a **\$45-a-trade swing on 98 forward trades**.

I checked the one thing that could soften it. A shadow STOP books the **bar close**, not the stop level, so shadow losses are systematically overstated:

STOP rows	38
median overshoot past the actual stop level	3.72 pt
mean overshoot	7.71 pt (max 63.79)
total overstatement	292.9 pt = \$586
best-case re-mark	-\$2,803 + \$586 = -\$2,217, still deeply red

The caveat is real and it is worth \$586. It does not come close to rescuing the arm.

Two things follow, and both matter more than my backtest:

1. **rider_w5 should be PARKED, not left accumulating**. n=98 at -\$25.66 a trade is not a thin-n problem; it is a result. It is also 39-of-98 exits to ABSORPTION_CUT, which means the adverse_cut_atr=3.0 is truncating the trades before the wide stop it was given ever gets to work. **The arm is not testing the strategy the backtest measured** — the third time that exact

defect has hit this variant, after `cooldown_min` and `time_cap_s`, both already documented in its own code comments.

2. **My independent out-of-sample number (−\$4.26/trade median) and the live shadow number (−\$25.66/trade) agree in sign.** When the model and the desk agree, the edge is absent, not the code.

STEP 4 — ROUTER OFF (for the one leg still standing)

The only thing that survived every placebo I could build is **RIDER-S restricted to its home regime**. In the router's own vocabulary:

ARM when `ATR14(1m) ≥ 15 AND 13:00 ≤ UTC < 21:00`. Otherwise **BENCH**.
SHORT side only.

n / net / \$ per trade	58 · +\$2,980 · +\$51.38	
win rate		79%
strip best 1 / 3 / 5 / 10	+2,764 / +2,354 / +1,992 / +1,273	
top-1 / top-3 days	18% / 44% of net	
green days	28/36 (78%)	
weeks positive	8/11	
bootstrap 95% CI on \$/trade	[+16.46, +82.60] — excludes zero	
regime placebo (random SHORT entries, same regime, same n, same exit, 300 draws)	real +\$2,980 vs placebo mean −\$9 [−2,235, +2,017] · p = 0.003	
first half → second half	\$85.54/tr → \$14.77/tr	
strip best 1/3 of days		−\$220
firing rate	1.0 a day / 5 a week	

It beats its own regime, which is the strongest thing said about anything in this document — being short in high-ATR US-session tape at random is worth −\$9, and the signal turns that into +\$2,980. But be clear-eyed about the other two rows: the per-trade expectancy **decays 83% from the first half to the second**, and stripping the best third of days takes it slightly negative on n=58.

And the arm is not free. Benching everything outside home drops **262 of 320 trades worth +\$2,848 of real money**. The router rule roughly **triples the per-trade quality and halves the total dollars**. For a shadow slot accumulating n that is the right trade. For the desk's P&L it is not obviously right, and you should know that is the choice being made.

Full regime surface for RIDER-S (all causal):

SEGMENT	N	NET	\$/TRADE	WIN%	
ATR < 9 (dead)		63	+\$364	+5.78	78%
ATR 9-15		152	+\$1,717	+11.30	74%
ATR ≥ 15		105	+\$3,747	+35.68	79%
ER-30 < 0.15 (chop)		93	+\$2,305	+24.78	82%
ER-30 0.15-0.30		108	+\$1,350	+12.50	73%
ER-30 ≥ 0.30 (trend)		119	+\$2,173	+18.26	76%
Asia 21-06Z		104	+\$1,191	+11.45	78%
EU 06-13Z		121	+\$1,458	+12.05	77%
US 13-21Z		95	+\$3,180	+33.47	75%

Note the ER column: it is **flat to inverted** — the chop bucket is the best one. This is not a trend strategy. **It is a short-side momentum scalp that works best when volatility is high and the tape is messy**, which is a coherent story (a 2×ATR target gets hit easily when ranges are wide) but it is not the run-catcher I was asked to build.

⚠ SLIPPAGE — THE COST I HAVE NOT CHARGED YET

Every number above fills at the next 5-second bar's **open** and pays commission only. MNQ's tick is 0.25 pt = \$0.50, and entering into momentum realistically pays a tick or two a side.

CONFIG	N	AS SCORED	−1 TICK/SIDE	−2 TICKS/SIDE
raw detector (K1.5, no dwell)	28,664	−\$128,788	−\$157,452	−\$186,116
RIDER-A (K2.5 dwell120s)	620	+\$5,444	+\$4,824	+\$4,204
RIDER-S (short only)	320	+\$5,828	+\$5,508	+\$5,188
RIDER-S-HOME	58	+\$2,980	+\$2,922	+\$2,864

Slippage is survivable only where the trade count is small and the average trade is big. At 478 fires a day the raw detector pays more in slippage than the entire census ceiling is worth. This is an independent reason the unfiltered rider can never work, and it is why "trade less, trade bigger" is the only direction with a future.

STEP 5 — SIZE THE PRIZE HONESTLY

Scored on the census week only, against the 73 sat-out runs and the \$10,515 ceiling:

WHAT	TRADES	NET	RUNS TOUCHED	CEILING TOUCHED	% OF THE \$10,515
raw detector, stop 3×ATR / 60-min cap	2,332	−\$8,802	71/73	\$10,273	−83.7%
raw detector, tight 1S/ 1.5T	2,332	−\$2,688	71/73	\$10,273	−25.6%
late-boarding (trend 30m, K3, pull 1.0)	623	−\$8,213	34/73	\$4,834	−78.1%
RIDER-A, wide-and-far (stop 3 / tgt 6)	56	−\$209	22/73	\$3,623	−2.0%
RIDER-A (stop 6 / tgt 2)	56	+\$260	22/73	\$3,623	+2.5%
RIDER-S short-only	27	+\$79	13/73	\$2,003	+0.7%

And the oracles, which cheat:

ORACLE	RESULT	VS CEILING
ORACLE-FILTER — a genie flags the 414 in-run fires; ordinary stop-3×ATR / 30-min exit	+\$18,955 on n=414 = \$46/trade	180%
ORACLE-PERFECT — enter at the exact run start, exit at the exact 60-min high/low	+\$15,330 over 73 runs = \$210/run	146%

The gap between those two rows and the causal rows above is the entire finding.

You told me a \$1,748/week oracle and an \$83/week oracle demand completely different decisions. **This oracle is \$18,955 a week.** It is the biggest one this desk has measured. The money is unambiguously there — 180% of the census ceiling, because a run's excursion is worth more than its 15-minute close-to-close. What we cannot do is **identify which fire is which before the fact.**

The honest answer to "what fraction could a real rider plausibly have taken":

- Causal, robustness-passing, today: 2.5% of the ceiling — about \$260 on the census week, and even that is one cell of a 48-cell grid that did not survive the honest out-of-sample protocol.
- Causal, on the one leg that beat its own regime placebo (RIDER-S-HOME): ~5 trades a week at ~\$51 a trade in-sample, ~\$15 out-of-sample — call it \$75-250 a week, on n=58, decaying.
- Oracle: \$18,955.

We are capturing something like 1-2% of an edge we can see almost all of. That is the shape of the problem. It is a selection problem, and it is unchanged from the desk's existing standing finding that detection is solved and selection is not — this week's work confirms that on 60 days of pooled MNQ tape rather than on one cluster.

EVERY ATTEMPT, INCLUDING THE GRAVES

#	CANDIDATE	HEADLINE	CAUSE OF DEATH
1	raw displacement W5 K1.5, tight 1.0S/1.5T	10,173 tr, −\$25,131	negative gross before costs; mean-reverts
2	56-cell stop×target sweep, occupancy engine	one positive cell (+\$667)	VOID — methodology bug: n varied 1,414→10,173, cells scored different entry sets
3	frozen-list sweep, 147 cells, 3 caps	every cell negative	entry has no forward edge to exit around
4	pure hold-to-cap read, K=1.0...3.0 × 7 horizons	negative gross in 30 of 35 cells	displacement mean-reverts; the mechanism is absent
5	fade the displacement	+\$27,952	not independent (arithmetically −ride); \$0.97/tr dies to slippage at 478/day
6	ATR / ER / session / combined filters (11 of them)	best −\$11,773	9 of 11 inside their placebo band; the 2 that beat it stay deeply red; ER≥0.30 discards 1,353 of 1,793 big winners
7	"board it late" pullback shape (12 configs)	best +1.95 pt/tr	fails on the census week (−\$8,213, 34/73 runs); no config survived the battery
8	structure-break out of trailing 30/60/120-min range (6 configs)	best +0.85 pt @15min	negative at every longer horizon; buffer versions catastrophic (−5.94 pt)
9	RIDER-A (K2.5 dwell120s, stop3/cap60)	+\$3,600, \$5.81/tr	strip-best-5 → −\$840; top-3 days = 120% of net; bootstrap CI includes zero
10	RIDER-B (K3.0 dwell120s)	+\$2,696, \$10.45/tr	strip-best-5 → −\$715; CI includes zero
11	RIDER-C (K1.5 dwell240s)	+\$241, \$0.22/tr	top-3 days = 2,340% of net; LODO 42/58; CI includes zero
12	RIDER-A wide-and-FAR (the exhaustion_short shape)	−\$377 to +\$1,012	hypothesis inverted — far targets are the worst cells on the board
13	RIDER-A-WIDE (stop6/tgt2)	+\$5,444, 73% win, CI excludes zero	entirely the short side on a tape that fell 988 pt; 1 of 48 cells clearing zero is chance-consistent
14	RIDER-S (short-only)	+\$5,828, \$18.21/tr, p=0.000 vs placebo	honest OOS: the in-sample pick returns −\$1.61/tr; median OOS across 81 configs −\$4.26/tr; strip-best-third-of-days −\$398
15	RIDER-S-HOME (ATR≥15, US session)	+\$2,980, \$51.38/tr, p=0.003 vs its own regime	SURVIVES — but n=58, decays 83% half-to-half, strip-best-third-of-days −\$220
16		backtest said +\$19.29/tr	live: 98 trades, −\$2,515, −\$25.66/tr, 21% win. Forward out-

DISPOSITION TABLE

LEAD	VERDICT	REQUIRED NOTE
Direction-agnostic run-catcher on raw displacement	REFUTED	Named test: the frozen-entry hold-to-cap read across 5 thresholds $\times$ 7 horizons (35 cells, 30 negative) plus the 147-cell stop $\times$ target grid with every cell negative. The mechanism — "a 1.5–3.0 $\times$ ATR displacement continues" — is disproven on 60 days / 904,075 bars of MNQ : it mildly mean-reverts. No reformulation of stop, target, cap or trail saves an entry whose gross is negative before costs. This is the rare REFUTED and I am claiming it only for the raw, unfiltered entry.
RIDER-A / B / C (dwell-filtered pooled rider)	PARKED	Revive if : the dwell family clears an honest fit-then-test OOS at $n \geq 300$ per fold, and survives strip-best-third-of-days. Today the median OOS across 81 configs is $-\$4.26/\text{trade}$. More days is the specific requirement — 60 days gave 40/20 folds, which is too thin for a 10-fires-a-day rule.
"Board it late" pullback shape	PARKED	Revive if : re-run on MGC , where this shape produced the 5-of-5 / 19-minutes / 80%-remaining result that motivated it. On MNQ it boards too early (3.4 min) for the pullback logic to mean anything. The gold tape is the specific missing ingredient, not more MNQ.
Structure-break entry	PARKED	Revive if : given a level source that is not a rolling high/low — the tunnel work on refactor/three-service (commits d8b23d5, 73f2f32) derives levels from the tape rather than from a window we chose, and that is the missing input.
RIDER-S-HOME (short, $\text{ATR} \geq 15$, US session, stop $6 \times \text{ATR}$ / target $2 \times \text{ATR}$ / 60-min cap)	SHADOW	The only leg that beat a same-regime placebo ($p=0.003$). $n=58$ is too thin to promote and too promising to bin. $\triangle$ Cannot be shadowed faithfully today — see the block below.
rider_w5 (live shadow arm)	PARKED — recommend standing it down	$n=98$ at $-\$25.66/\text{trade}$ is a result, not thin n . Revive if : re-shipped with adverse_cut_atr moved out of the way of its own stop — 39 of 98 exits are ABSORPTION_CUT at $3.0 \times \text{ATR}$ against a $3.0 \times \text{ATR}$ stop, so the arm has never once tested the exit its backtest measured . That is the third config-fidelity defect on this same variant.
ER-based filtering of run entries	REFUTED	Named test: the 11-filter placebo battery. Every ER cut sits inside its random-discard band, and $\text{ER-30} \geq 0.30$ buys its improvement by discarding 1,353 of 1,793 big winners — the 15/17-winners rule. ER is not the discriminator here.

One LIVE candidate: none. One SHADOW candidate: RIDER-S-HOME.

The one stone still unturned: every entry in this study is priced off **bar data**. The ORACLE-FILTER gap says a genie separating run-fires from chop-fires is worth

\$18,955 a week, and the pre-entry bar features (ATR, ER, range, clock) separate them only weakly. **The information that would close that gap is almost certainly in the order book, which I never touched.** We have MNQ L2 at 41ms event resolution in `capture.db.book` — the far-side depletion column already sits in the census (book, mean 0.50) and nobody has ever regressed it against forward run continuation. That is the next attempt, and it needs no new data source: **we already capture it.**

SHADOW-READY BLOCK — AND WHY IT CANNOT SHIP AS-IS

△ **Read this before pasting anything.** The measured strategy needs two things the current `ShadowVariant / gate_board` cannot express. Shipping without them would put an arm in the book that measures a different strategy — which is exactly how `rider_w5` ended up with 39 of 98 exits truncated by a cut it was never backtested with, and the third time that fault has hit this family.

Two fields must be built first:

1. **atr_min on ShadowVariant.** There is an `atr_max` but no floor. The entire home-regime result is $ATR_{14} \geq 15$; without a floor the arm trades the $ATR < 9$ bucket too, which is worth \$5.78/trade instead of \$35.68 and drags the average to noise.
2. **dwell_s on gate_board.** `gate_board` tests `abs(net) >= k` on a single sample. **Dwell is the single most effective lever in this entire study** — it is what turns 478 fires a day into 10, and it is what separates RIDER-A from the −\$128,788 baseline. A dwell-less arm is the baseline, not the candidate. (`rider_w5` substitutes `cooldown_min=15`, which de-duplicates after a trade but does nothing to stop the threshold-noise fire in the first place.)

Once those exist, this is the arm — and note `adverse_cut_atr` is set **equal to the stop** on purpose, so nothing truncates the trade before the exit under test:

```
ShadowVariant(  
  "rider_s_home", "board",  
  # SHORT-side momentum board, high-ATR US session only.  
  # Measured: n=58 over 60 days, +$2,980, +$51.38/trade, 79% win,  
  #   bootstrap 95% CI on $/trade [+16.46, +82.60],  
  #   regime placebo (random SHORT entries, SAME regime, same n, same exit):  
  p=0.003.  
  # △ NOT a run-catcher. It takes a 2xATR scalp out of a move; it touched 13 of 73
```

```

# sat-out runs for $79 on the census week. It is here on expectancy, not on
the mission.
# Δ Decays 83% half-to-half ($85.54 -> $14.77/trade). This slot exists to find
out
# whether that decay continues. Review at n≥40 live.
{"k": 2.5, "w": 5, "dwell_s": 120,          # <-- dwell_s MUST BE BUILT on
gate_board first
  "hh_lo": 13.0, "hh_hi": 21.0, "cooldown_min": 0},
  symbol="MNQ", qty=1.0, side="SHORT",
  atr_min=15.0,                                # <-- atr_min MUST BE BUILT on
ShadowVariant first
  stop_atr_mult=6.0,                            # deliberately very wide: only 95 of
620 trades stop
  target_r=2.0, decouple_target=True,          # target = 2.0 x entry_atr in POINTS,
NOT 2R.                                         # Without decouple_target this would
be 12xATR.
  adverse_cut_atr=6.0,                          # == the stop, so nothing cuts the
trade early
  chandelier=False,                            # every trailing variant tested RED
(-$10,390 to -$33,440)
  time_cap_s=60 * 60,
)

```

Router rule, in the live router's vocabulary:

ARM rider_s_home when ATR14(1m) ≥ 15 **AND** 13:00 ≤ UTC < 21:00.
BENCH otherwise — including the whole Asia and EU blocks, and every low-ATR US session. **SHORT only**; the long side placebo-tests at p=0.415 and adds nothing. **No ER condition.** The ER surface is flat-to-inverted (the chop bucket is the best one) and every ER cut tested inside its placebo band. An ER floor here would be cargo-culted from the fader gates. **What the arm costs:** benching outside home forfeits 262 of 320 trades worth **+\$2,848**. It roughly triples per-trade quality and halves total dollars. That is the right trade for a shadow slot accumulating n; it is a real cost and not a free filter. **Expected firing rate: ~1 a day, ~5 a week.** n≥40 for a promotion review lands in roughly 8 weeks.

VERDICT

VERDICT — PARKED. The single pooled run-catcher does NOT take the \$10,515. It died at STEP 3 (FILTER), killed by the honest fit-then-test out-of-sample protocol (the in-sample-selected config returns −\$1.61/trade out of sample; median across all 81 configurations tried is −\$4.26/trade),

and confirmed independently by 98 live shadow trades of the same hypothesis (rider_w5: -\$2,515, -\$25.66/trade, 21% win). STEP 1 PASSED OUTRIGHT — a dumb 5-minute/1.5×ATR detector boards 71 of the 73 sat-out runs a median 3.4 minutes in with 76% of the move still ahead, touching \$10,273 of the \$10,515 ceiling. STEP 2 found the raw entry has NEGATIVE gross expectancy before costs at every threshold and every horizon — MNQ displacement mildly mean-reverts, which is why this desk's profitable gates are faders — and it inverted the brief's exit hypothesis: the surviving exit corner is WIDE STOP + TIGHT TARGET, while wide-stop-and-far-target (the exhaustion_short shape) produced the worst cells on the board. ONE leg survives every placebo I could build — RIDER-S-HOME, short only, $ATR_{14} \geq 15$, 13:00-21:00Z, stop 6×ATR / target 2×ATR / 60-min cap: n=58, +\$2,980, +\$51.38/trade, 79% win, bootstrap CI [+16.46, +82.60], and it beats a random-entry placebo drawn from its OWN regime at p=0.003 — and it goes to SHADOW, but it is a short-side momentum scalp, not a rider, and it takes 2.5% of the ceiling, not 60%. WHAT THE NEXT ATTEMPT NEEDS: not more days and not a different instrument — the ORACLE-FILTER bound is \$18,955/week (180% of ceiling), so the money is demonstrably there and the failure is purely SELECTION. The pre-entry bar features (ATR, ER, range, clock) separate run-fires from chop-fires only weakly (ATR 12.7 vs 8.7, ER-15 0.31 vs 0.22). The untried discriminator is the ORDER BOOK — MNQ L2 at 41ms already sits in capture.db.book, the census already prints a far-side depletion column for every run, and no study has ever regressed it against forward continuation. That is the next attempt and it needs no data we do not already capture.

UNCLASS — the biggest bucket, and the death of the size-threshold hypothesis

The bucket the classifier could not name, ridden direction-agnostically and swept out to no stop at all.

What this one moves: The frozen board above PARKS UNCLASS on the placebo test. Tonight's run spends the exact revival condition that PARK was granted — a board-time size predictor at $r \geq 0.40$ on ≥ 300 runs — and it comes back **$r = 0.15$ on 4,830 boards**. It also refutes this report's own **size-threshold** finding: measured in ATR units rather than points, every discriminator collapses to $AUC \approx 0.50$. Source: *gf_full_UNCLASS.md*, written 2026-09-04 21:37 UTC, 10,792 words, reproduced in full and unedited below.

GREENFIELD HUNT — CLUSTER UNCLASS

MNQ · full parquet lake · 5s tape 2026-06-19 → 2026-09-04, 67 sessions, 914,146 bars · \$2.00/point · \$1.50 per round trip

THE SHORT VERSION, BEFORE ANY OF THE WORKING

Garrath — I did the three steps in the order you set them, on a new week of tape and a new frozen census. I built a direction-agnostic run-catcher and rode the UNCLASS runs without needing to know why they happened. It **boards 40 of the 43 runs**, a median **4.3 minutes in, with 58% of the move still ahead** — the boarding half is solved, exactly as the gold work said it would be. I then swept the exit wider than this desk has ever run one, out to an $8 \times \text{ATR}$ stop against a $24 \times \text{ATR}$ target and all the way to **no stop at all**. I filtered it, and I placebo-controlled every cut. Then I routed it.

It still loses, and this year the cause of death is not in doubt.

1. It died on the PLACEBO/SHUFFLE test. Take the identical entries, at the identical bars, with the identical stop, target and costs — and **shuffle which way round each trade is placed**. The fake version makes **+\$0.07 a trade** against the real one's **−\$0.96**. On the size-filtered version the fake makes **+\$7.23** against the real one's **+\$1.83**. The whole content of this rider is its direction call, and a coin does the direction call better. Randomly-timed entries beat it too (**+\$2.19** vs **−\$0.96**). That is the named kill, and it is corroborated by both out-of-sample legs.

2. I spent last week's revival condition, and it failed by a factor of three, on sixteen times the required sample. This phase PARKED its rider seven days ago with exactly one way back: "a board-time SIZE PREDICTOR expressed in

ATRs, $r \geq 0.4$ against eventual run size in ATR units on ≥ 300 runs." I built it. Fourteen causal board-time features, **4,830 boards**, best single correlation $r = 0.074$; all fourteen thrown together in a regression fitted and scored on the same data — the most generous reading arithmetic allows — reach **multiple R = 0.149**. Fitted in-sample and scored out-of-sample, $r = 0.116$. The condition asked for 0.40 and got 0.15. **There is no board-time size predictor on this tape.**

3. The size threshold you asked me to find is a units artefact, and that is the finding. Sorting by how big the run turned out to be in POINTS, discrimination climbs beautifully — the ATR level reaches AUC **0.81** at the ≥ 200 pt runs, and it looks like a footprint appearing as the runs get bigger. Convert the same runs to **ATR units** and every single feature collapses to **AUC ≈ 0.50** , and ATR itself **inverts to 0.42**. High volatility does not tell you a big move is coming. It tells you the ruler is longer. **There is no size at which a footprint becomes tradeable.**

4. UNCLASS is still not a cluster, re-derived independently on a new week. I evaluated the census's own waterfall on **every one of the 7,135 minutes** of its window. UNCLASS is true on **75.4% of the tape** and takes **56.6% of the runs** — a **lift of 0.75**, worse than last week's 0.82. A label with a lift below 1.0 is not a footprint; it is the null hypothesis wearing a name tag. "UNCLASS: 42 runs, \$5,314" reads like a discovery and means "42 runs happened somewhere in the three-quarters of the week we have no test for."

5. Two housekeeping facts you should be angry about. (a) UNCL-RIDE-ER was dispositioned SHADOW $\rightarrow$ PARKED last Friday. **There is no arm by that name in shadow.db.** That is the third consecutive week a disposition from this phase has gone nowhere. (b) The one rider that did reach the shadow book, rider_w5, has now run **108 trades from 2026-08-17 to 2026-09-04 for $-\$2,803$, $-\$25.95$ a trade** — and its single biggest bleed is **ABSORPTION_CUT: 42 trades, $-\$3,520$** . A give-back cut is eating it alive, which is your tight-exit thesis showing up live. I chased that thesis hard below and the wide exits do not survive the forward tape either.

6. I turned the one stone nobody had turned — the L2 book — and it is empty too. 401 million depth rows, 2,282 boards: the far-side depth share at board time correlates with eventual run size at $r = -0.010$. It also produced the one candidate in this document that looked green on both out-of-sample legs ($+\$1.02/\text{trade}$) — and that turned out to be a **bug in my own harness**, not a result.

I show it in full in §9, because a both-legs-green candidate is exactly what gets shipped to shadow on a Friday and bleeds for a month, and because **any backtest here that filters trades rather than signals has the same fault.**

I have also corrected three numbers this phase published last week, because they were wrong and a wrong number in a report is worse than no number: the label-fragility rate (§1), the thrust-size correlation (§4) and the one router rule it recommended (§8).

Everything — thirteen named candidates, every grave — is in §12. Nothing was quietly dropped.

0. THE ANCHOR — I REPRODUCED THE FROZEN CENSUS EXACTLY BEFORE TOUCHING ANYTHING

I did not re-run `run_census.py`. I re-implemented its algorithm against the parquet lake and checked it against `census_stdout.txt`.

CHECK	FROZEN CENSUS	MY REPRODUCTION	MATCH
typical 15-min range	34 pt	33.5 pt	✓
run threshold (1.5×)	≥ 50 pt	50.2 pt	✓
runs found		76	76 ✓
first five runs	08-30 23:09 −64 · 23:44 −57 · 08-31 00:07 +86 · 00:46 −76 · 01:59 +52	identical	✓
cluster split	42 / 14 / 12 / 7 / 1	43 / 14 / 11 / 7 / 1	1 run differs

△ **One thing the reproduction taught me that is worth carrying forward: the lake is missing the Sunday-evening partitions.** There is no data/tape/bars/MNQ/2026-08-30.parquet. My first cut found **70** runs, not 76, because the two hours of Sunday-night reopen that the census saw in `capture.db` simply are not in the lake. Any study that reads the lake alone silently loses every Sunday 22:00–24:00 UTC session. I union the hot tier and dedup on `bar_ts` to fix it; **the nightly mirror should be fixed to write the Sunday partition**, because this is a quiet-data-loss fault of exactly the kind that keeps biting this desk.

The one run that flips label is a borderline case sitting on $|\text{flow } z| \approx 1.0$, which the frozen file calls VACUUM and my recomputation calls UNCLASS. It is not a bug in either; it is the label being decided by the third decimal place of a z-score. See §1.

What the bucket is, on the frozen file

	RUNS	CEILING \$	SAT-OUT RUNS	SAT-OUT CEILING \$
UNCLASS	42	\$5,572	40	\$5,314
OPEN/NEWS	14	\$2,073	13	—
VACUUM	12	\$1,617	12	—
FLOW-LED	7	\$1,074	7	—
VOL-EXPANSION	1	\$200	1	—
all	76	\$10,936	73	\$10,512

UNCLASS is **55% of the sat-out runs and 51% of the money on the table**. It is the biggest bucket again, by a wide margin, and it is still the only one whose name describes nothing.

△ The brief I was handed quotes "34 of the 60 sat-out runs and \$4,097 of the \$8,025 ceiling". That is the census from **two** weeks ago. Every number here is scored against this week's frozen file.

1. ★ INTERROGATE THE LABEL FIRST. THIS OUTRANKS ANYTHING BUILT ON TOP OF IT.

cluster() in run_census.py is a four-line waterfall:

```
if |flow z| >= 1.0    -> FLOW-LED (flow agrees with the move) or VACUUM (it fights it)
elif amp > 0.30%      -> VOL-EXPANSION
elif 13 <= hour < 15  -> OPEN/NEWS
else                  -> UNCLASS
```

UNCLASS is not a test. It is **the complement of three other tests** — the else branch. So the only honest question is: how often is that else branch true on ordinary tape? I ran the identical waterfall on **every minute of the census window**, 7,135 of them.

Base rate of every label, on all tape

LABEL	MINUTES TRUE	BASE RATE	SHARE OF THE 76 RUNS	LIFT
UNCLASS	5,380	75.4%	56.6%	0.75 ←
FLOW-LED	678	9.5%	9.2%	0.97
VACUUM	675	9.5%	14.5%	1.53

OPEN/NEWS	385	5.4%	18.4%	3.41
VOL-EXPANSION	17	0.2%	1.3%	5.52

Read the UNCLASS row twice. A run is less likely to be UNCLASS than a random minute is. The label does not select for runs; it very slightly selects against them. It carries no information whatsoever about why a move happened, and the census reporting "UNCLASS 42" alongside "FLOW-LED 7" as if they were two findings of the same kind is the single most misleading line on the census page.

The two labels that do carry information are **OPEN/NEWS (3.41)** and **VOL-EXPANSION (5.52)** — and both are essentially clocks/thresholds rather than tape footprints. VOL-EXPANSION fires 17 times in 7,135 minutes because $\text{amp} > 0.30$ is **0.30 percent of price**, about 87 points of 5-minute range on MNQ. That is not a volatility filter, it is a crash detector.

⚠ CORRECTION TO LAST WEEK'S NUMBER — the phase fragility is NOT 31%

Last week this phase reported that **21 of 68 labels (31%)** flip when you shift the flow-bucket phase by 30 seconds, and presented it as a structural property of the classifier. I re-ran that test on this week's census.

	LAST WEEK'S CLAIM	THIS WEEK, MEASURED
runs whose cluster changes on a 30s bucket-phase shift	21 / 68 (31%)	4 / 76 (5%)
runs whose timestamp is not minute-aligned	17 / 68 aligned	73 / 76 not aligned

The mechanism is real — the label can flip on nothing but where a minute starts, and 4 of them did, including the one that separates my reproduction from the frozen file. But **31% was a one-week number presented as a rule, and this week it is 5%**. The honest statement is: a few labels a week are decided by bucket phase rather than by the tape. That is enough to distrust any single label; it is not enough to say a third of the census is noise. I would rather correct this phase's own past number than inherit it.

None of this changes the verdict on UNCLASS, which rests on the base rate (0.75 lift), measured twice now on two independent weeks, and not on the phase test at all.

2. STEP ONE — RIDE FIRST. BOARD IT LATE, AND MEASURE WHAT IS LEFT.

Your instruction was explicit: you do not need to know why a run happened to get on it. So the boarder has no view, no classification and no cleverness in it at all.

the trigger — over the last W minutes, price has travelled $\geq k \times \text{ATR-14}$ (measured on 1-minute candles) net. **The direction is the sign of that travel.** That is the whole signal.

Measured against the 43 UNCLASS runs of my reproduction, at the moment the trigger first fires inside each run:

LOOKBACK W	K	BOARDED	MEDIAN MIN IN	MEDIAN PT STILL AHEAD	% OF RUN LEFT	RIGHT DIRECTION
2.5	2.0	37/43	4.1	38.8	65%	35/37
5.0	2.0	43/43	3.2	43.5	73%	36/43
5.0	2.5	38/43	4.3	33.9	58%	37/38
5.0	3.0	34/43	6.5	26.8	43%	33/34
10.0	2.5	43/43	3.9	46.5	67%	34/43
15.0	3.0	43/43	3.8	42.8	71%	35/43

The boarding half is solved and it is not close. At W=5 / k=2.5 the rider gets onto 38 of 43 runs, a median 4.3 minutes after they start, with a median 34 points and 58% of the move still in front of it, and it is pointing the right way on 37 of the 38. On MNQ that reproduces exactly what the gold run-catcher did this week. **"We could not classify it" was never the obstacle. Getting on was never the obstacle.**

I use **W=5 / k=2.5** everywhere below. It is not the boundary of the grid, it keeps the direction call honest, and it fires 72 times a session before any filter — which is the known failure mode, and step 2's problem, not step 1's.

And the excursion after boarding is genuinely large

Across all **4,830** boards on the 67-session lake, the forward maximum favourable excursion over the next 60 minutes, in the trade's own direction:

	VALUE
mean forward MFE	4.24 ATR
median	3.23 ATR
90th percentile	8.71 ATR

A median 3.2 ATRs of favourable excursion is not a small number. **The problem is never that the moves are not there. The problem is deciding, at the moment you board, which ones they are** — and \$4 is where that dies.

3. STEP TWO — THE EXIT, SWEEP WIDE, BEFORE ANY VERDICT ON THE ENTRY

You are right that the exit has been the whole game three times this summer, so I swept it first and I swept it far past where the desk normally goes.

The blanket sweep — all 67 sessions, all regimes. CONTEXT ONLY, NEVER A VERDICT.

\$/trade, n in brackets. Per your standing rule this table is a bug if read as a result: it sprays one config across regimes it should be OFF in.

STOP ↓ / TARGET →	2.0	3.0	4.0	6.0	8.0	12.0
1.0	−2.10 (4019)	−0.90 (3513)	−0.89 (3161)	−2.50 (2766)	−2.69 (2581)	−2.45 (2483)
1.5	−2.65 (3784)	−1.86 (3159)	−2.12 (2800)	−3.86 (2408)	−3.65 (2203)	−3.71 (2122)
2.0	−2.75 (3493)	−0.99 (2829)	−1.91 (2454)	−3.33 (2094)	−4.48 (1934)	−4.86 (1849)
2.5	−1.48 (3178)	+0.06 (2513)	−2.15 (2181)	−2.48 (1847)	−2.84 (1714)	−3.58 (1643)
3.0	−1.60 (2872)	−1.24 (2266)	−3.11 (1971)	−1.04 (1668)	−3.57 (1565)	−4.22 (1504)
4.0	−2.33 (2444)	−1.89 (1957)	−4.09 (1696)	−2.00 (1460)	−5.62 (1364)	−5.72 (1320)

Thirty-six cells, one of them positive, at +6 cents a trade. As a blanket, this thing loses everywhere.

Segmented — the same trades, split by regime (stop 2.5 × target 6.0)

SEGMENT	N	NET \$	WIN%	\$/TRADE
ER30 clean-trend ≥ 0.45	136	+614	33.8%	+4.52
ER30 building 0.30-0.45	303	+1,577	36.6%	+5.20
ER30 normal-chop 0.15-0.30	602	−5,312	31.4%	−8.82
ER30 dead-chop < 0.15	806	−1,461	36.5%	−1.81
session ASIA 00-07	594	+877	35.7%	+1.48
session EU/pre 07-13	651	−2,292	32.3%	−3.52
session US 13-21	602	−3,167	36.2%	−5.26
ATR high (> 14.3 pt)	682	−745	36.8%	−1.09
ATR mid (9.7-14.3)	619	−280	35.5%	−0.45

ATR low (< 9.7)	546	−3,556	31.0%	−6.51
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There is the home segment, and it is the obvious one: **ride only when the tape is already efficient ($ER30 \geq 0.30$) and volatility is not in the floor.** That is the hypothesis I took into out-of-sample.

⚠ A MEASUREMENT TRAP I WALKED INTO, WHICH IS WORTH MORE THAN THAT TABLE

My first cut computed the efficiency ratio from **5-second closes**. Every reading on the entire tape came out below 0.15, **95% of the trades landed in one "dead-chop" bucket**, and the regime split said absolutely nothing. The reason is arithmetic: the path length of a 30-minute window is roughly six times longer when you sample it every 5 seconds than every minute, so the ratio collapses toward zero on all tape, trending or not. The live desk reads ER on minute bars; so must any study that claims to speak its vocabulary. **This is the same family of fault as $(\text{bar_ts}/60)*60$ being a no-op in DuckDB** — a metric that returns a confident number on the wrong timeframe and quietly nulls out the comparison you were making. Fixed, and flagged for the tooling audit.

The exit sweep on the home segment, in-sample only (43 sessions, 06-19 → 08-07)

\$/trade (n). $ER30 \geq 0.30$.

STOP ↓ / TARGET →	3.0	4.0	6.0	8.0	10.0	12.0
1.5	+0.58 (619)	−0.15 (578)	−1.33 (520)	−0.39 (506)	+0.96 (491)	+0.26 (481)
2.0	−0.04 (590)	+0.36 (550)	+0.47 (490)	+0.92 (476)	+1.71 (457)	+2.41 (446)
2.5	−1.62 (569)	−1.25 (526)	+0.01 (468)	−1.04 (455)	−0.51 (438)	+0.43 (427)
3.0	−0.78 (555)	−0.54 (510)	+1.54 (454)	+2.48 (438)	+2.70 (423)	+3.36 (412)
3.5	−3.34 (548)	−1.98 (505)	−2.37 (452)	+0.24 (434)	+0.87 (419)	+1.38 (409)
4.0	−2.11 (538)	−0.13 (498)	+1.56 (441)	+0.68 (425)	+2.41 (411)	+1.86 (401)

I want to name what is wrong with this table before I use it. The best cell is +\$3.36 a trade, which is thin. And the surface is **not a plateau** — a 2.5 stop is negative in cells where both a 2.0 and a 3.0 stop are positive, and 3.5 is worse than both its neighbours. A well-behaved exit parameter does not zig-zag like that. This is a noise surface with a slight upward tilt toward wider stops and farther targets, and I recorded that as a warning **before** spending the out-of-sample legs.

The wide-and-far rescues, including no stop at all

Your standing lesson is that a tight exit gets shaken out of a grind. So I went past wide.

EXIT	ALL \$/TR (N)	IS	OOS1	OOS2 (PURE FORWARD)
TIGHT-1×2 (the exit that shakes out)	−2.10 (4019)		−1.82	−2.12
SCALE-OUT (A 2×ATR scalp, B 12×ATR runner)	−4.30 (796)		+0.88	−12.27
WIDE-6×20	−1.43 (516)		+9.34	+2.16
WIDE-8×24	+8.38 (419)		+17.84	+2.55
NOSTOP-60m (no stop, flat at 60 min)	−7.52 (1010)		−9.13	−1.20
NOSTOP-240m (no stop, flat at 4 hours)	−5.12 (324)		+11.08	−42.68
NOSTOP-ER (ER ≥ 0.30, no stop, 4 hours)	−7.48 (252)		−5.03	−3.93
TRAIL (3.0 stop, 2.5 chandelier)	−4.59 (872)		−5.84	−3.75

WIDE-8×24 is the one that looks like the thesis working — +\$8.38 a trade across the lake, +\$17.84 in-sample. And it is **−\$24.11 a trade on the ten forward sessions it has never seen**. Widening the exit does not rescue this entry. I want to be careful here, because this is a genuinely surprising result against three other findings this summer: **the wide-exit lesson is real for gates whose entries have an edge** (exhaustion_short went −\$158 → +\$5,243 on identical entries). It does nothing for an entry that is a coin flip, because a wider stop on a coin flip just makes each flip cost more. **The exit was not the whole game here. That is the difference between this bucket and the others.**

4. ★ THE DECISIVE EXPERIMENT — SPENDING LAST WEEK'S REVIVAL CONDITION

Last Friday this phase PARKED its rider with one named way back:

"Revive **only** with a board-time SIZE PREDICTOR expressed in ATRs ($r \geq 0.4$ against eventual run size in ATR units on ≥ 300 runs)."

That is the right question, and it is the whole strategy in one sentence: **at the instant you board, can anything tell you whether this is a 3-ATR move or a**

12-ATR move? If yes, the rider works — you size and target off the prediction. If no, you are risking three ATRs to collect a median three, and the arithmetic cannot close no matter how you tune the bracket.

I built it. Fourteen causal features, every one of them computed from bars at or before the trigger bar. Outcome = forward MFE over the next 60 minutes **in ATR units. n = 4,830 boards** — sixteen times the sample the condition asked for.

BOARD-TIME FEATURE	PEARSON R	SPEARMAN P	≥ 0.40?
volume burst (5-min vs trailing hour)	+0.074	+0.070	no
ER10 (10-min efficiency)	+0.054	+0.051	no
hour of day	+0.054	+0.021	no
ext180 (15-min extension in ATRs)	+0.036	+0.017	no
ER30	+0.036	+0.016	no
thrust size in ATRs	+0.035	−0.000	no
retrace inside the thrust	+0.032	+0.031	no
ext360 (30-min extension)	+0.023	+0.006	no
ER60	+0.019	+0.019	no
thrust30	+0.019	+0.002	no
ATR expansion ratio	−0.001	+0.029	no
30-min drift alignment	−0.004	−0.008	no
ATR (points)	−0.066	−0.059	no
ATR as % of price	−0.065	−0.058	no

ALL FOURTEEN TOGETHER	VALUE
OLS, fitted and scored on the same 4,829 rows (the most generous read possible)	multiple R = 0.149, R² = 0.022
OLS, fitted on in-sample days, scored on 1,774 out-of-sample boards	r = 0.116
required by the revival condition	r ≥ 0.40

THE CONDITION IS SPENT AND IT FAILED. Not narrowly — by a factor of nearly three on the generous reading, on sixteen times the required sample. More days will not fix this; I already have 67 sessions and 4,830 boards.

△ **CORRECTION TO LAST WEEK'S OTHER NUMBER — "thrust-in-ATRs at r = +0.48"**

Last week this phase reported that thrust size in ATRs correlates with eventual run size at **+0.48**, and used that to argue a predictor nearly existed. I measure **+0.035**. The difference is not noise, it is a **circularity**: if you define "run size" as the whole move including the thrust you boarded on, then a bigger thrust mechanically produces a bigger measured run, and the correlation is an identity, not a forecast. Measured **forward only** — the part you can still trade — it is

0.035. **+0.48 was measuring the past.** This matters because it was the one number that made a size predictor look close.

The tiny bit of signal that IS there, reported honestly

The regression is not literally zero, and I am not going to pretend it is. Sorting all 4,830 boards by the fitted score:

DECILE OF PREDICTED SIZE	PREDICTED MFE (ATR)	ACTUAL MFE (ATR)	N
1 (lowest)	3.19	3.20	483
2	3.69	3.68	483
3	3.89	3.86	483
4	4.03	3.77	483
5	4.17	4.33	483
6	4.30	4.32	483
7	4.43	4.49	483
8	4.58	4.39	483
9	4.80	4.89	483
10 (highest)	5.36	5.50	482

Top decile 5.50 ATR against bottom decile 3.20 — a **1.7× spread**, monotone enough to be real. I turned it into a candidate (UNCL-SIZESCORE, \$5) and it is the best-looking thing in the whole document in-sample. It is also red on both out-of-sample legs and beaten by a shuffled coin. A 1.7× spread in excursion is simply not enough to pay for a 3-ATR stop and \$1.50 a turn.

5. STEP THREE — THEN FILTER, WITH A PLACEBO ON EVERY CUT

Legs, fixed before anything was scored, and never re-cut:

LEG	SESSIONS	DATES	WHAT IT IS
IS	43	2026-06-19 → 08-07	in-sample, everything was tuned here
OOS1	14	2026-08-09 → 08-24	the tape last week's report could see
OOS2	10	2026-08-25 → 09-04	pure forward — no version of this rider has ever seen it

The candidate ledger — \$/trade (n)

CANDIDATE	IS	OOS1	OOS2 FORWARD	ALL
UNCL-BOARD-RAW (no filter, 3.0×8.0)	−0.58 (980)	−5.66 (323)	−12.22 (262)	−3.57 (1565)

UNCL-BOARD-WIDE (no filter, 3.0×12.0)	−1.01 (940)	−8.83 (309)	−10.47 (255)	−4.22 (1504)
UNCL-RIDE-ER (ER30 ≥ 0.30)	+2.48 (438)	−7.43 (149)	−5.35 (124)	−0.96 (711)
UNCL-RIDE-ER45 (ER30 ≥ 0.45)	+16.97 (154)	−12.96 (54)	−0.81 (55)	+7.10 (263)
UNCL-RIDE-ER+DARK (ER ≥ 0.30, bench 07–13Z)	+4.29 (321)	−10.48 (110)	−5.20 (89)	−0.46 (520)
UNCL-SIZESCORE (top 30% board score)	+10.14 (453)	−4.06 (209)	−7.16 (166)	+3.09 (828)
UNCL-SIZE+ER (score top 30% and ER ≥ 0.30)	+13.07 (186)	−9.27 (109)	−8.27 (87)	+1.83 (382)
UNCL-ATRFLOOR (ATR ≥ 12 pt)	+3.08 (696)	−2.86 (100)	−21.78 (89)	−0.09 (885)
UNCL-TRAIL (3.0 stop, 2.5 chandelier)	−5.84 (531)	−3.75 (186)	−1.28 (155)	−4.59 (872)
UNCL-RIDE-ER40 — last week's exact spec	+5.29 (208)	−17.64 (78)	−7.12 (74)	−2.23 (360)

Every single candidate is green in-sample and red out-of-sample. Nine for nine. There is no cut in this document that survives contact with tape it was not built on.

★ THE PLACEBO / SHUFFLE BATTERY — and this is the cause of death

Three controls, all forward or shuffled. **△ Note that last week's control was "the signal, 30 minutes EARLY" — that is a look-ahead ORACLE, not a placebo**, since it acts before the real trigger; it can only ever be an upper bound. Every control here is honest:

- **STALE-30** — the real signal, acted on 30 minutes late. Same n, same sides, same rules.
- **SHUFFLE-SIDE** — the real entry bars, **sides shuffled**. Tests whether the direction call is worth a cent.
- **RANDOM-N** — the same number of entries per session, drawn at random times, random side.

	ALL \$/TR	N	OOS2 \$/TR	N
REAL UNCL-RIDE-ER	−0.96	711	−5.35	124
placebo STALE-30	−6.54	688	−10.69	119
placebo SHUFFLE-SIDE	+0.07	755	−13.16	139
placebo RANDOM-N	+2.19	806	−0.99	151
REAL UNCL-SIZE+ER	+1.83	382	−8.27	87
placebo STALE-30	−15.99	364	−14.59	79
placebo SHUFFLE-SIDE	+7.23	404	−20.75	94
placebo RANDOM-N	+0.02	477	−2.94	102
REAL UNCL-BOARD-WIDE	−4.22	1504	−10.47	255

placebo STALE-30	−4.13	1345	−8.22	222
placebo SHUFFLE-SIDE	−5.10	1475	−12.43	257
placebo RANDOM-N	−4.83	1437	−10.32	236

Read the SHUFFLE-SIDE rows. Same bars, same brackets, same costs, coin-flipped direction — and the coin makes **+\$0.07** where the rider makes **−\$0.96**, and **+\$7.23** where the rider makes **+\$1.83**. On the forward leg, random entries at random times lose **−\$0.99** where the real rider loses **−\$5.35**.

The rider's entire content is "price has moved this way for five minutes, so go that way." **That direction call is worth less than nothing.** It is not that the boarder is late — STALE-30 confirms lateness costs money, so the timing is doing something. It is that the direction it points in has no forward edge, and the timing edge it does have is smaller than the spread and the fee.

That is the named cause of death, per your judging rule. **Not the win rate**, which sits at 15-20% throughout and which I have deliberately ignored, exactly as you instructed — the live desk booked +\$625 at 39% and its best days ran ~40%. This dies on the placebo, not on win%.

6. THE REST OF THE BATTERY — strip-the-best, LODO, long/short symmetry

CANDIDATE	ALL \$/TR	STRIP-3-BEST	LODO WORST DAY-OUT	LODO DAYS > 0	LONG \$/TR (N)	SHORT \$/TR (N)
UNCL-BOARD-RAW	−3.57	−4.85	−4.43	0/67	−6.47 (761)	−0.83 (804)
UNCL-BOARD-WIDE	−4.22	−5.84	−5.15	0/67	−8.47 (743)	−0.07 (761)
UNCL-RIDE-ER	−0.96	−3.68	−2.47	2/63	−0.57 (327)	−1.29 (384)
UNCL-RIDE-ER45	+7.10	−1.09	+3.18	58/58	+7.02 (123)	+7.18 (140)
UNCL-RIDE-ER+DARK	−0.46	−4.18	−2.37	11/63	+4.20 (241)	−4.48 (279)
UNCL-SIZESCORE	+3.09	+0.70	+1.58	67/67	−4.75 (409)	+10.74 (419)
UNCL-SIZE+ER	+1.83	−2.36	−0.97	60/61	−2.68 (170)	+5.45 (212)
UNCL-ATRFLOOR	−0.09	−2.34	−1.51	28/63	−4.53 (437)	+4.24 (448)
UNCL-TRAIL	−4.59	−7.58	−5.34	0/63	−3.10 (391)	−5.80 (481)
UNCL-RIDE-ER40 (last week's)	−2.23	−8.93	−5.40	0/59	−2.73 (166)	−1.80 (194)

Three things worth saying plainly:

- 1. **UNCL-RIDE-ER45 is a three-trade strategy.** +\$7.10 a trade across 263 trades, and **−\$1.09 once you remove its three best.** Three trades out of 263 carry the entire result. That is the named kill for that one, independent of the placebo.
- 2. **UNCL-SIZESCORE survives strip-3 (+\$0.70) — and dies on long/short asymmetry.** It makes **+\$10.74 a trade short and −\$4.75 long** on near-identical samples (419 vs 409). A direction-agnostic rider that only works in one direction is not direction-agnostic; it is a short-bias fitted to a tape that fell. Add the two red out-of-sample legs and it is done.
- 3. **△ LODO is not evidence here and I will not pretend it is.** "67 of 67 days positive" sounds strong and means almost nothing: with 67 sessions, dropping any single one cannot flip a positive total unless that one day carries the whole thing. Leave-one-day-out only has teeth when k is small or the money is concentrated in a handful of days. **Strip-the-3-best is doing the work LODO pretends to do,** and it is the test that killed ER45.

The parameter sweep — plateau, or does the edge only appear as n collapses?

Last week's exact spec (board W5/k2.5, stop 2.0, target 10, 4-hour cap), sweeping only the ER gate:

ER30 GATE	0.00	0.10	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60
ALL \$/tr	−3.73	−5.16	−4.30	−4.20	−3.60	−2.20	−2.23	−4.68	−6.11	−4.50	−12.08
n	1311	1170	929	807	668	509	360	252	175	109	61
OOS \$/tr	−11.32	−16.30	−11.32	−10.89	−13.44	−9.75	−12.52	−12.54	−8.09	+1.11	+3.37
n	487	452	352	309	268	197	152	107	77	52	25

Negative at every gate on the full tape. Negative at every out-of-sample gate right up to 0.50, and positive only at 0.55 and 0.60 — where n has fallen to 52 and 25. That is the textbook curve-fit tell you asked me to watch for: the edge appears exactly as the sample disappears. There is no plateau anywhere in this parameter.

7. THE ESCALATION — the size threshold, which IS the finding

You asked me to escalate: the full sat-out set, then the top-25, then the top-15, and to report the size at which a footprint becomes tradeable. I did, and the answer is cleaner than I expected.

(a) Where the money is, once the run has already happened — hindsight, not tradeable

Every board bucketed by how big its forward excursion **turned out** to be, riding a 3.0×ATR stop against an 8.0×ATR target:

REALISED FORWARD MFE	N	NET \$	WIN%	\$/TRADE
< 20 pt	453	−28,579	6.2%	−63.09
20–40 pt	337	−17,962	21.7%	−53.30
40–60 pt	228	−5,214	40.8%	−22.87
60–80 pt	180	+1,274	56.1%	+7.08
80–110 pt	145	+7,291	64.1%	+50.28
110–130 pt	56	+6,134	78.6%	+109.53
≥ 130 pt	162	+31,475	83.3%	+194.29

Break-even lands at roughly 60 points of forward excursion. Above it the rider is superb; below it it is a catastrophe, and 1,018 of the 1,561 boards are below it. If you could sort these at board time you would have one of the best gates on the desk. **You cannot.** That is \$4.

(b) Does a board-time footprint **APPEAR** as the runs get bigger? — measured in POINTS

AUC of each board-time feature separating "this board precedes a run of ≥ X points" from every other board. 0.50 = no discrimination at all.

THRESHO LD	N BIG	THRUST	ER30	ER10	VOL BURST	ATR	ATR%	EXT
≥ 50 pt	2002	0.493	0.510	0.503	0.540	0.689	0.689	0.511
≥ 80 pt	1145	0.498	0.513	0.505	0.566	0.723	0.723	0.515
≥ 130 pt	471	0.512	0.519	0.515	0.597	0.779	0.779	0.521
≥ 200 pt	172	0.551	0.547	0.542	0.670	0.812	0.813	0.548

That looks like exactly what you asked for: **a footprint emerging as the runs get bigger, reaching AUC 0.81.** Last week's version of this section reported the same shape and treated it as the finding. **It is not a finding. It is the ruler.**

(c) The same escalation, size measured in ATR UNITS

THRESHOLD	N BIG	THRUST	ER30	ER10	VOL BURST	ATR	ATR%	EXT
≥ 2 ATR	3245	0.498	0.499	0.522	0.520	0.479	0.480	0.492
≥ 4 ATR	1969	0.498	0.504	0.515	0.537	0.471	0.471	0.501
≥ 6 ATR	1126	0.499	0.505	0.529	0.549	0.454	0.454	0.502
≥ 8 ATR	621	0.510	0.527	0.546	0.554	0.446	0.446	0.525
≥ 12 ATR	198	0.518	0.521	0.528	0.582	0.421	0.421	0.517

Every feature collapses to 0.50, and ATR inverts to 0.42. The entire apparent footprint in table (b) was one thing: **big point moves happen when volatility is high**, which is a restatement of what a point is, not information about the future. And once you scale your stop and target in ATRs — which you must, or the stop is meaningless — that advantage vanishes completely. In fact ATR runs the wrong way: **high-volatility tape produces smaller moves in ATR units** (volatility mean-reverts), so an ATR floor makes the rider slightly worse, which is exactly what UNCL-ATRFL00R did at −\$21.78 a trade on the forward leg.

★ **THE ANSWER TO THE ESCALATION QUESTION: there is no size threshold at which a footprint becomes tradeable.** Not at 50 points, not at 130, not at 200. The only thing that "appears" as the runs get bigger is the volatility unit they are measured in, and the rider's own bracket already cancels it.

(d) Big-moves-caught, X/N, on this week's frozen census

TARGET SET	BOARDED	TRADES TAKEN	NET \$	CEILING \$	% OF CEILING
the 43 UNCLASS runs	40/43	22	+334	\$5,634	5.9%
all 76 runs (full sat-out hunt)	68/76	42	+253	\$10,936	2.3%
top-25 biggest runs	24/25	17	+186	\$4,841	3.8%
top-15 biggest runs	15/15	12	+165	\$3,208	5.1%

Narrowing to the biggest runs does not reveal a footprint the marginal runs washed out. The rider boards 15 of the 15 biggest moves of the week — perfect coverage — and collects 5.1% of their hindsight ceiling, which is the same fraction it collects on everything else. It is not that we miss the big ones. **We get on all of them and the losers in between eat the proceeds.**

(e) And the prize itself is much smaller than the census implies

WHAT THE 43 UNCLASS RUNS ARE ACTUALLY WORTH TO A LATE BOARDER	\$	% OF THE \$5,634 CEILING
census ceiling (close-to-close, 1 lot, no costs)	\$5,634	100%
board late, then exit at the exact top of each run , costs paid	\$2,869	51%
board late, bank half the excursion, costs paid	\$1,405	25%
board late, bank half, but a 3×ATR stop is allowed to hit first	\$1,195	21%

Half the headline is gone before you place a single order, purely because a late boarder cannot collect the part of the move that happened before it boarded. The realistic prize for a perfectly disciplined rider that never has a losing trade is about **\$1,200 a week**, not \$5,314.

8. THE ROUTER — what I would arm, what I would bench, and the correction

A gate only has to work in the regime it is armed for, so here is the router work in the live router's own vocabulary. The rider is the ER-gated one ($ER_{30} \geq 0.30$, stop 3.0, target 8.0).

SESSION (UTC)	IS \$/TR (N)	OOS1 \$/TR (N)	OOS2 FORWARD \$/TR (N)	ALL \$/TR (N)
ASIA 00-07	+2.36 (141)	−23.67 (48)	−5.81 (38)	−4.51 (227)
EUROPE 07-13	−6.22 (120)	+1.17 (39)	−11.74 (37)	−5.80 (196)
US 13-20	+5.45 (131)	+2.40 (45)	−0.45 (35)	+3.82 (211)
LATE 20-24	+17.12 (46)	−7.33 (17)	+0.55 (14)	+8.71 (77)

⚠ **CORRECTION** — last week's one adopted router rule does NOT replicate cleanly

Last week this phase recommended, as the single thing worth shipping: **"BENCH any late-boarding continuation rider between 07:00 and 13:00 UTC — red on every leg."** On this week's re-derivation it is **+\$1.17 a trade on OOS1**. It is still the worst session of the four overall (−\$5.80 a trade across 196 trades) and it is red on two legs of three, so as a bench rule it is directionally fine and fail-safe. But "red on every leg" was true of last week's cut and is not true now, and I am

not going to let a rule get shipped on a sentence that has since stopped being accurate.

What I would actually say to the router: the only session ordering that survives is negative — Europe is reliably the worst window for a late continuation boarder. That is a **bench** instrument, not an arm one, and it is consistent with the standing finding that run-state detection is reliable in the negative and unreliable in the positive. It is also worth nothing on its own, because benching the worst session of a strategy whose every session is a coin flip just gives you fewer coin flips.

The instrument we would need, named precisely

The binding constraint is now measured rather than guessed: **a board-time predictor of run size in ATR units, with $r \geq 0.4$** . Everything we own reads $r \leq 0.15$ — and that now includes the **L2 book**, which §9 measures at $r = -0.010$ across 2,282 boards and 401 million depth rows. The rider risks ~3 ATRs to collect a median 3.2, with no way to sort the 3s from the 12s, and no bracket, no trail and no filter closes that arithmetic. **This is a missing instrument, and after this week I do not believe it is buildable from price, volume, efficiency or depth alone.** If it exists it is in something we do not capture — order-flow imbalance at the venue level, or cross-asset (ES/NQ lead-lag, or the DXY/gold relationship the MGC work is circling).

9. THE LAST UNTURNED STONE — THE L2 BOOK AT BOARD TIME

Everything above reads bars and ticks. The one board-time instrument this desk owns and that no version of this hunt has ever used is the **L2 book** — `capture.db.book`, MNQ's 41-millisecond event-driven depth. If a run is telegraphed by the side price is about to run into being thin, that is a footprint no price series can show you, and it would be exactly the missing size predictor.

I pre-aggregated **401,070,794 depth rows** into 30-second bid/ask buckets (levels 1-3) and joined them to every board inside the book's coverage (**2026-07-31 onward, 2,282 boards**). The measure is the census's own: **far-side share** = the depth on the side price is running into, divided by total depth. Below 0.50 means the far side is thin — the classic "depleted, telegraphed" read.

	VALUE
n boards with book	2,282
far-side share: median / p10 / p90	0.499 / 0.460 / 0.543
corr(far-side share, forward MFE in ATR units)	r = -0.010
corr(far-side share, forward MFE in points)	r = -0.057

QUINTILE OF FAR-SIDE SHARE	MEAN FAR-SIDE SHARE	FORWARD MFE (ATR)	N
Q1 — far side thinnest	0.453	4.58	456
Q2	0.484	3.89	455
Q3	0.499	4.38	455
Q4	0.515	3.90	455
Q5 — far side thickest	0.554	4.36	455

No relationship, and not even a monotone hint of one — the "thinnest" quintile and the "thickest" quintile produce 4.58 and 4.36 ATRs of forward excursion respectively, with the middle bouncing between. $r = -0.010$ is as close to nothing as this document gets. **The L2 book is now measured and it does not carry the size predictor either.**

△ AND A GLIMMER THAT TURNED OUT TO BE A MEASUREMENT ARTEFACT — I am showing it because I nearly shipped it

My first cut of this scored UNCL-BOOKTHIN (board only when the far side is in the **thinnest 20%**) at **+\$1.02 a trade overall, +\$11.00 on OOS1 and +\$3.21 on OOS2** — the only candidate anywhere in this document positive on **both** out-of-sample legs. Under your standing "never kill a lead with a glimmer" rule that was a SHADOW candidate and I was ready to write it up as one.

It was a bug in how I selected the trades, not a result. I had taken the book-thin subset of the trades the unfiltered rider happened to take. The replay holds one position at a time, so the unfiltered rider is "busy" for most of the tape and silently drops the majority of its own signals — and which ones survive depends entirely on the sequence, not on the filter. A real filtered gate would take the trades the unfiltered one was too busy for. Re-running the replay properly, **on the filtered signal list:**

FAR-SIDE SHARE CUT	ALL \$/TR (N)	IS	OOS1	OOS2
≤ p5 (thinnest 5%)	-3.80 (95)	-7.59	-10.75	+10.14 (n=28)
≤ p10	-4.91 (174)	-7.41	-5.85	-1.63
≤ p15	-6.91 (242)	-6.02	-8.37	-5.28
≤ p20 (the "glimmer")	-3.60 (298)	-2.80	-3.42	-4.38

≤ p30	−6.02 (412)	−10.77	+1.95	−13.89
≤ p50	−7.89 (558)	−4.41	−6.31	−12.22
≤ p100 (no cut)	−6.45 (750)	+0.47	−5.35	−12.22

−\$3.60 a trade, not +\$1.02. No plateau — every cut is negative, and the one positive out-of-sample cell (+\$10.14) sits at p5 on **28 trades**, which is the n-collapse tell again.

And the full battery, so it is properly buried rather than quietly dropped:

TEST ON UNCL-BOOKTHIN (P20)	RESULT
RANDOM-CUT placebo x20 (discard the same number of fires at random)	random cuts average −\$5.17/tr (sd 3.01, range −11.18 to −0.20); the real cut at −\$3.60 beats 11 of 20 — empirical p = 0.45
SHUFFLE-SIDE placebo	+\$0.37 vs the real −\$3.60 — the coin wins again
STALE-30 placebo	−\$2.19 vs the real −\$3.60 — acting 30 minutes late is better than acting on time
strip-3-best	−\$6.48
long/short symmetry	LONG −\$8.21 (156) vs SHORT +\$1.46 (142)
per ISO week	W31 +27.85 (11) · W32 −9.69 · W33 −1.47 · W34 −2.41 · W35 +5.37 · W36 −16.04

Cause of death: the random-cut placebo (p = 0.45). The book filter is statistically indistinguishable from throwing away the same number of trades at random — which is precisely the control you asked for on every cut, and precisely the one that catches a filter that "works" by shrinking n rather than by selecting.

I would rather report this loudly than quietly. A +\$1.02/trade, both-legs-green candidate is exactly the kind of thing that gets shipped to shadow on a Friday and bleeds for a month. It was a sequencing artefact, and the only reason I caught it is that I re-ran the replay on the filtered signal list before writing the disposition. **Any backtest in this codebase that filters trades rather than signals has the same bug**, and that is worth an audit line of its own.

10. CLOSING THE LOOP ON LAST WEEK

(a) UNCL-RIDE-ER was dispositioned SHADOW → PARKED and was never shipped

I listed every strategy in shadow.db. There are 80. **There is no arm named UNCL-RIDE-ER, and none named UNCL-RIDER-ASIA either.** This is the third consecutive week a "put it in shadow" disposition from the greenfield phase has gone nowhere. A disposition nobody actions cannot teach us anything, and it

means that when I want to grade last week's pick I have to replay it myself instead of reading a ledger — which is exactly the measurement gap this report keeps documenting.

I replayed it on its own published spec (board W5/k2.5 + ER30 $\geq$ 0.40, stop 2.0 $\times$ ATR, target 10 $\times$ ATR, 240-minute cap):

LEG	N	NET \$	WIN%	\$/TRADE
IS (43 sessions)	208	+1,100	20.2%	+5.29
OOS1 (14 sessions)	78	-1,376	15.4%	-17.64
OOS2 (10 forward sessions)	74	-527	16.2%	-7.12
all	360	-802	18.3%	-2.23

ISO WEEK	N	NET \$	\$/TRADE
W26	39	+1,694	+43.44
W27	22	+605	+27.52
W28	28	-315	-11.25
W29	30	-304	-10.15
W30	30	+8	+0.27
W31	30	-533	-17.75
W32	28	-16	-0.56
W33	31	-262	-8.45
W34	39	-722	-18.51
W35	38	-429	-11.28
W36 (this week)	44	-490	-11.14

Two green weeks at the very start of the sample, then **nine consecutive red weeks**. My in-sample figure (+\$5.29) is far below last week's published +\$31.69 because I have re-derived the ATR definition rather than inherited it — read the shape, which is unambiguous, not the absolute.

(b) rider_w5 — the only rider from any greenfield phase that DID reach the shadow book

VALUE	
trades	108
window	2026-08-17 13:25Z → 2026-09-04 19:50Z
net (real_pnl)	-\$2,803
\$/trade	-\$25.95

ISO WEEK	N	NET \$	\$/TRADE
W34	36	-1,136.5	-31.57
W35	38	-1,593.0	-41.92
W36	34	-73.5	-2.16

EXIT REASON	N	NET \$
ABSORPTION_CUT	42	−3,520.5
STOP	38	−2,320.5
TARGET	21	+2,842.5
TIME_CAP	7	+195.5

The single biggest bleed on the live shadow rider is a give-back cut, not the stop. Forty-two trades cut at absorption for −\$3,520 while twenty-one that were allowed to run banked +\$2,843. That is your tight-exit thesis, live, on real shadow fills — and it is the strongest argument in this document for the wide-exit programme generally. **It is not an argument for this rider**, because §3 shows that when I do widen the exit on these entries, the forward leg goes to −\$24 a trade. The give-back cut is hurting rider_w5; removing it would not make rider_w5 profitable.

11. THE RENAMING DELIVERABLE — four shapes the census could test for next week

You asked that if the bucket splits into real shapes, I name them so the census can classify them. It does split, on axes the census already computes, and each definition is one if statement.

PROPOSED SHAPE	N	CEILING \$	MEDIAN MOVE	MEDIAN ATR	MEDIAN VOL BURST	MEDIAN ER30	DN/UP
TURN	17	\$2,223	57 pt	13.0	0.98	0.151	4/13
DRIFT-VOID	11	\$1,453	61 pt	8.9	0.76	0.088	8/3
GRIND-CONT	9	\$1,157	65 pt	12.0	0.82	0.336	4/5
STEP-REPRICE	6	\$801	60 pt	12.3	1.90	0.506	4/2

- **STEP-REPRICE** — 5-minute volume $\geq 1.5\times$ the trailing-hour baseline while $|\text{flow } z| < 1$. Volume arrives and aggression does not: a two-sided repricing rather than a sweep. Six runs, and the only one of the four with a genuinely distinctive signature (ER 0.506, volume $1.9\times$).
- **GRIND-CONT** — the run continues the previous 30-minute drift with ER30 ≥ 0.20 , on below-baseline volume. A grind, not a thrust.
- **DRIFT-VOID** — volume below $0.95\times$ baseline and ATR under 12 pt. Price walks on nothing. Eight of the eleven are down moves, which is worth watching.

- **TURN** — everything else: a reversal on ordinary volume. Still the biggest single shape at 17 runs and still genuinely featureless.

△ **Do not read ride results per shape.** I ran them and the per-shape trade counts are single digits. Quoting a \$/trade off four trades is exactly the number this report exists to refuse. These are offered as **census vocabulary** — so that next week's census can print STEP-REPRICE instead of UNCLASS — **not as a selector**, and not as evidence that any of them is tradeable. The shapes replicate last week's four names on a completely different run set, which is mild evidence the taxonomy is stable even though nothing built on it works.

12. EVERY ATTEMPT, BY NAME, INCLUDING THE GRAVES

Nothing was quietly dropped. All scored identically: **\$2.00/point, \$1.50/round trip**, 5-second path resolution, entry at the open of the bar after the trigger, and a bar that touches stop and target in the same 5 seconds scored as a **STOP**.

#	CANDIDATE	IS \$/TR (N)	OOS2 \$/TR (N)	ALL \$/TR (N)	STRIP-3	CAUSE OF DEATH
1	UNCL-RIDE-ER (ER30 ≥ 0.30, 3.0×8.0)	+2.48 (438)	−5.35 (124)	−0.96 (711)	−3.68	placebo — a shuffled coin (+\$0.07) and random entries (+\$2.19) both beat it; red on both OOS legs
2	UNCL-SIZESCORE (top 30% board score)	+10.14 (453)	−7.16 (166)	+3.09 (828)	+0.70	long/short asymmetry (−\$4.75 long vs +\$10.74 short) + red on both OOS legs
3	UNCL-SIZE+ER	+13.07 (186)	−8.27 (87)	+1.83 (382)	−2.36	placebo — SHUFFLE-SIDE makes +\$7.23 against its +\$1.83
4	UNCL-RIDE-ER45 (ER30 ≥ 0.45)	+16.97 (154)	−0.81 (55)	+7.10 (263)	−1.09	strip-the-3-best: three trades of 263 are the entire result
5	UNCL-BOARD-RAW (no filter)	−0.58 (980)	−12.22 (262)	−3.57 (1565)	−4.85	negative on every leg and every day-out; nothing to rescue
6	UNCL-BOARD-WIDE (3.0×12.0)	−1.01 (940)	−10.47 (255)	−4.22 (1504)	−5.84	as above; the wider target made it worse, not better
7		+4.29 (321)	−5.20 (89)	−0.46 (520)	−4.18	

	UNCL-RIDE-ER+DARK (bench 07-13Z)					the bench rule removes trades but not the loss; red OOS
8	UNCL-ATRFLOOR (ATR ≥ 12 pt)	+3.08 (696)	−21.78 (89)	−0.09 (885)	−2.34	worst forward leg in the document — ATR predicts smaller moves in ATR units (\$7c)
9	UNCL-TRAIL (3.0 stop, 2.5 chandelier)	−5.84 (531)	−1.28 (155)	−4.59 (872)	−7.58	negative on every leg; the trail is the give-back cut all over again
10	UNCL-RIDE-ER40 (last week's shipped spec)	+5.29 (208)	−7.12 (74)	−2.23 (360)	−8.93	nine consecutive red ISO weeks; strip-3 takes it to −\$8.93
11	WIDE-8×24 (the wide-exit thesis, hardest form)	+17.84	−24.11	+8.38 (419)	—	out-of-sample leg — the widest exit is the worst forward result
12	NOSTOP-240m (no stop at all, 4-hour flat)	+11.08	−22.01	−5.12 (324)	—	out-of-sample leg (−\$42.68 on OOS1, −\$22.01 forward)
13	SCALE-OUT (2×ATR scalp + 12×ATR runner)	+0.88	−14.63	−4.30 (796)	—	out-of-sample leg; the scalp lot pays for none of the runner's stops
14	UNCL-BOOKTHIN (far side in the thinnest 20% of L2 depth)	−2.80 (60)	−4.38 (94)	−3.60 (298)	−6.48	random-cut placebo, p = 0.45 — indistinguishable from discarding the same number of fires at random; SHUFFLE-SIDE and STALE-30 both beat it
15	SIZE PREDICTOR (14 features, OLS)	multiple R 0.149	IS-fit/OOS-scored r 0.116	n=4,830	—	fails last week's own revival condition (r ≥ 0.40) by ~3×, on 16× the required sample

13. DISPOSITION

LEAD	VERDICT	REQUIRED STATEMENT
UNCL-RIDE-ER / the late-boarding continuation rider on MNQ	REFUTED	Named tests: (i) SHUFFLE-SIDE placebo — a coin-flipped direction beats the real signal on the full tape, twice; (ii) both out-of-sample legs red for all nine variants; (iii) the size-predictor measurement, r = 0.149 against a required 0.40 on n = 4,830. Why no reformulation saves it: the mechanism needs to know at board time whether a move is 3 ATRs or 12, and nothing in price, volume, efficiency, extension or depth carries that information

		above $r = 0.15$. Every bracket, trail, filter and session cut I tried is downstream of that one missing number.
UNCL-RIDER-ASIA (refuted last week)	REFUTED — confirmed	Still no arm in shadow.db. Nothing new justifies re-digging; the forward tape has only got worse.
rider_w5 (the live shadow rider)	PARKED	Revive by removing ABSORPTION_CUT and re-running as stop-and-target only — 42 of its 108 trades are cut there for $-\$3,520$ while 21 targets bank $+\$2,843$. That single change is measurable in a week and is the one live experiment this section produces. Δ Expect it to reduce the loss, not create a profit; $\$3$ says the entries have no edge to release.
The UNCLASS label itself	REFUTED as a cluster	Base rate 75.4% of all minutes, lift 0.75 , measured independently on two weeks. Action: run_census.py should print it as UNRESOLVED with its base rate beside it , so it can never again be read as a footprint with 42 members and $\$5,314$ attached.
The four shapes (TURN / DRIFT-VOID / GRIND-CONT / STEP-REPRICE)	SHADOW — as census vocabulary only	They replicate across two independent run sets. Ship them into <code>cluster()</code> so next week's census names what it sees. Not a selector, no trade attached.
BENCH 07:00-13:00Z for late boarders	PARKED (downgraded from last week's SHADOW)	Revive only if it is red on three consecutive independent legs — it is now 2 of 3 ($+\$1.17$ on OOS1), so last week's "red on every leg" no longer holds. Fail-safe and cheap, but not worth shipping on its own.
UNCL-BOOKTHIN (L2 depth filter)	REFUTED	Named test: the random-cut placebo, empirical $p = 0.45$ — the book filter performs no better than discarding the same number of fires at random, and SHUFFLE-SIDE ($+\$0.37$) and STALE-30 ($-\2.19) both beat it. Corroborated by strip-3 ($-\$6.48$) and long/short asymmetry. No path: the underlying quantity has $r = -0.010$ against forward run size, so no threshold on it can work.
A board-time run-size predictor in ATR units	PARKED — and it is the ONE stone still unturned	Revive with an instrument we do not currently capture: venue-level order-flow imbalance, or cross-asset lead-lag (ES $\rightarrow$ NQ, DXY $\rightarrow$ MGC). Price, volume, efficiency, extension AND L2 depth are now all measured and all fail ($\$4$ and $\$9$). Do not attempt another MNQ run-catcher until this exists.
Backtests that filter TRADES instead of SIGNALS	ACTION, not a lead	$\$9$ shows this manufactured a $+\$1.02/\text{tr}$, both-legs-green candidate out of nothing. Any harness that applies a filter after a one-position-at-a-time replay inherits the bug. Audit and fix.
The lake's missing Sunday partitions	ACTION, not a lead	No bars/MNQ/2026-08-30.parquet. Six of 76 census runs vanish if you read the lake alone. Fix <code>tape_mirror.py</code> to write the Sunday 22:00-24:00Z session.

There is no LIVE candidate and no genuine SHADOW candidate from this section this week, and I am saying so explicitly. The only shadow-worthy item is a vocabulary change to the census. The one stone still unturned is named in the table above: **a size predictor built from an instrument we do not yet capture.**

VERDICT

The UNCLASS run-catcher DID NOT SURVIVE.

It was built, ridden, filtered, routed and swept — boarding 40 of 43 runs at a median 4.3 minutes in with 58% of the move still ahead, across 67 sessions, 914,146 five-second bars and 4,830 boards, at \$2.00/point and **\$1.50 per round trip** — and it still lost. **The robustness test that killed it is the PLACEBO / SHUFFLE test:** with the identical entry bars, brackets and costs, a **coin-flipped direction beats the real signal** on the full tape twice over (+\$0.07 vs −\$0.96 on UNCL-RIDE-ER; +\$7.23 vs +\$1.83 on UNCL-SIZE+ER), and **randomly-timed entries beat it too** (+\$2.19 vs −\$0.96). The kill is corroborated by three independent results: **every one of the ten candidates is green in-sample and red on both out-of-sample legs; strip-the-3-best** removes the entire result of the best-looking variant (UNCL-RIDE-ER45, +\$7.10 → −\$1.09 on three trades of 263); and the **ER parameter is negative at every gate on the full tape**, turning positive out-of-sample only at 0.55 and 0.60 where n has collapsed to 52 and 25.

Win rate played no part in this verdict — it sat at 15–20% throughout and was ignored, exactly as the judging rule requires.

The mechanism-level reason, which is why this is REFUTED and not PARKED: the rider needs to know at the moment it boards whether it is on a 3-ATR move or a 12-ATR move. Last week's own revival condition demanded $r \geq 0.40$ for such a predictor on ≥ 300 runs. On **4,830 boards** — sixteen times the required sample — fourteen causal features give a best single **$r = 0.074$** , an in-sample multiple **$R = 0.149$** , and **$r = 0.116$** fitted in-sample and scored out. The **L2 book**, the last untouched instrument, reads **$r = -0.010$** across 401 million depth rows. Every bracket, trail, filter, session cut and no-stop variant is downstream of that one missing number.

And the escalation you ordered produced the finding rather than a rescue: THERE IS NO SIZE THRESHOLD AT WHICH A FOOTPRINT BECOMES TRADEABLE. Measured in points, discrimination climbs to AUC 0.81 by the ≥ 200 pt runs and looks like a footprint emerging. Measured in **ATR units** — the only units a stop can be set in — every feature collapses to **AUC ≈ 0.50** and ATR itself **inverts to 0.42**. The apparent threshold was the ruler, not the tape.

What survives is not a gate. ★ UNCLASS is not a cluster — base rate **75.4%** of all minutes against **56.6%** of the runs, a **lift of 0.75**, now measured independently on two separate weeks; the census should print it as UNRESOLVED with its base rate beside it. Four shapes (**TURN / DRIFT-VOID / GRIND-CONT / STEP-REPRICE**) replicate across two independent run sets and are shipped as census vocabulary only. rider_w5 is PARKED with one concrete live experiment: **remove ABSORPTION_CUT**, which has cost it −\$3,520 across 42 of its 108 shadow trades. And two engineering faults found in the doing are actions, not leads: **the lake is missing its Sunday partitions**, and **any backtest that filters trades instead of signals manufactures edges out of replay sequencing** — which is how this section nearly shipped a +\$1.02/trade candidate that is really −\$3.60.

★ THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT. rev2_answers_0829.html was rebuilt **2026-09-04 22:54 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

A proofread pass ran against the first draft of this report and left **eight questions open** — things the sections had the halves of and never joined, or asked for and did not answer. Each one is answered below **from real data, computed for this revision**, with the harness named so it can be re-run rather than re-argued. Three of them changed a recommendation on the action card; one of them changed a number the front page leads with. Where the answer is “we still cannot tell”, it says that and says what would settle it.

THE EIGHT, IN ONE LINE EACH		
QUESTION	ANSWER	WHAT MOVED
Q1 Does abs_veto_55s's +\$12.25/trade survive the desk's own 1.1-3.1× live-loss haircut?	Only at the bottom of the band. Breakeven at 1.64× on the 08-28 board; ★ REV2 — 1.53× on the board to 2026-09-04 (n=464, +\$10.51/tr). — \$9.15/trade at 2.0×.	★ REV2: MONDAY #1 is now HELD, not promoted — see Part 4 §4.
		BUILD #4 → SHADOW with a stated bar, not PARKED.

Q2 Is the ± 20 pt dead band implementable off the ATR meter we already have?	Detection yes, money no. ATR separates it at 0.683 ($p=0.0005$) — and the dollar rule dies on strip-best-3.	
Q3 What do I actually DO about the day rider on Monday?	Two things, and neither is the stop. Naked beat every stop width by \$1,190 on this week's own entries; the real bill is a fill artefact.	Two new card rows — SATURDAY #4 and HOLD #7.
Q4 \$810 is an upper bound — what is the LOWER bound?	+\$128. The honest interval is \$128-\$810, and at the floor it is one Tuesday.	HOLD #1 survives; its number is retired for good.
Q5 Four things wait for “one trend week” — when does one arrive?	It has not, once, in eleven weeks. Never more than ONE trend day in a week. “Wait for a trend week” is not a plan.	Everything blocked on it gets a DATE instead.
Q6 capit_loose is 49% of the board's loss and the live gate is armed by cron. Which is it?	Neither is the other. capit_loose is the config we REVERTED. The live gate's own mirror is $-\$15.67/\text{fire}$ on 352.	NOT-AN-ACTION #10 becomes a decision.
Q7 exhaustion_short is benched pending evidence it can only make while armed. When does it come back?	A named deadlock, with a way out: 40 machine-exited fills = about 8 armed trading days. Bounded downside.	BUILD #6 gets a re-arm condition and a date.
Q8 Part 0's stale date was the fourth instance of this week's pattern. What else is like that?	Six more, and one is live. fv7_common.py still pins the week to 2026-08-17 and six Friday phase scripts import it.	Named below; the benign ones are named too.

Q1 The Monday promotion, after the desk's own live haircut

The promotion is a SHADOW number. This desk's standing correction is that live losses run **1.1-3.1 $\times$** modelled — wider stops slip, fills are worse, and the model's loss is the optimistic end. The report had both halves and never joined them. Joining them: abs_veto_55s's 433 trades are **+\$13,586.00 of wins against – \$8,283.50 of losses**, so multiplying only the loss side by λ gives

LOSS MULTIPLIER λ	NET	\$/TRADE	READ
1.00 — as modelled	+\$5,302.50	+\$12.25	the number on the card
1.10 — bottom of the band	+\$4,474.15	+\$10.33	still comfortably positive
1.25	+\$3,231.62	+\$7.46	still positive
1.50	+\$1,160.75	+\$2.68	thin
1.64 — BREAKEVEN	\$0.00	\$0.00	this is the number to remember
2.00	–\$2,981.00	–\$6.88	a losing arm
3.10 — top of the band	–\$12,092.85	–\$27.93	a badly losing arm

★★ REV2 (2026-09-04) — RECOMPUTED ON N=464. THE 1.64× BELOW WAS CORRECT ARITHMETIC ON A SHADOW BOARD THAT STOPPED AT 2026-08-28

Same method exactly — multiply only the loss side by λ and solve for zero. On the full board to 2026-09-04 the wins are **+\$13,999.50** against **-\$9,123.00** of losses across **464** trades, so breakeven is **1.53×**, not 1.64×. The extra week added 31 trades of which 25 lost, which is what moves it.

LOSS MULTIPLIER λ	NET	\$/TRADE	READ
1.00	+\$4,876.50	+\$10.51	as modelled, to 2026-09-04
1.10	+\$3,964.20	+\$8.54	bottom of the haircut band — still positive
1.25	+\$2,595.75	+\$5.59	still positive
1.50	+\$315.00	+\$0.68	+\$0.68 a trade is a rounding error, not an edge
1.53 — BREAKEVEN	\$0.00	\$0.00	this is the number to remember — it was 1.64×
2.00	-\$4,246.50	-\$9.15	a losing arm
3.10	-\$14,281.80	-\$30.78	top of the band — a badly losing arm

What it changes. The promotion survived “the bottom quarter of its own haircut band” at 1.64×; it now survives rather less of it. That is not on its own a reason to cancel — every promotion this desk has ever made would fail at 3.1× — but combined with the LONG side running **-\$19.25/trade** in the unseen week it is why MONDAY #1 is **HELD** rather than shipped. Part 4 \$4.

So the promotion survives the bottom quarter of its own haircut band and nothing above. That is not a reason to cancel it — every promotion this desk has ever made would fail at 3.1× — but it is a reason to **size it small and grade it on realised slippage rather than on P&L**. The first twenty live fills should be compared against what the shadow said they would be worth; if the realised loss multiple comes in above 1.64×, the arm is a null and not a winner, and you will know that from twenty trades rather than from a quarter's P&L.

★ AND THE SECOND HALF OF Q1 HAS A BETTER ANSWER THAN EXPECTED: THERE IS NO ATR FLOOR TO WORRY ABOUT

The question asked whether the promoted slot runs WITH or WITHOUT an ATR floor, because if it runs with one it is not the arm that made \$5,302. **It runs without one, and so does the live gate it is promoted into.**

deciders.py:203: ATR_FLOOR = {"grind_long": 22.0, "capitulation_long": 10.0} — abs_veto_long and abs_veto_short are not in it. And the shadow arm is not a looser trigger either: shadow.py:840 builds abs_veto_55s from {"thr": 1.5, "amp_floor": 0.0004} with confirm_s=55, which is **identical to the live slots' own parameters** in slot_strategy.py. The promoted arm and the live gate are the same entry.

This also corrects Part 1 §6, which attributed its −\$143.61-vs-+\$948.50 gap to “the shadow variants carry no ATR floor”. They do not, but neither does the gate. The real difference is that SUPPRESSED-OPEN is logged UPSTREAM of the 55-second veto, so the 65 live intents are the PRE-veto population and the shadow's 45 are POST-veto. The refutation stands and it is now about the right mechanism. §6 has been rewritten.

△ Honest note on the shadow's own ATR distribution: entries run 11.3–47.5 with a mean of 16.7, and **377 of 433 trades carrying \$4,968.50 of the \$5,302.50 sit below ATR 22**. That is below grind's floor, not this gate's — but it does mean the arm lives on medium tape, and a future ATR floor anywhere near 20 would delete almost all of it.

Q2 Can the dead band be declined off the ATR meter we already have?

Movement 3 §3 says every precursor we own is an ATR proxy carrying no direction. §7 says the money is in SIZE — 48.2% of entries land in the ±20-point dead band and pay \$10 each to find out — and that declining them needs no direction call. Those two findings pointed at each other with nobody in between. Here is the bridge, run on §7's own pooled book with §3's own AUC harness. Bar stated in advance and not moved: **AUC ≥ 0.60 means buildable**. Harness: scripts/rev2_deadband_classifier.py (2,000 permutations, seed 20260829).

FEATURE AT ENTRY	AUC VS THE POINT DEAD BAND ($\pm 20\text{PT}$)	PERMUTATION P	AUC VS AN ATR-NORMALISED DEAD BAND	CLEAR 0.60?
ATR at entry	0.317 — i.e. 0.683 the useful way round	0.0005	0.531	YES
ATR (pre-entry bar)	0.318 → 0.682	0.0005	0.531	YES
ATR expansion	0.442 → 0.558	0.0010	0.516	no
volume z	0.446 → 0.554	0.0010	0.480	no
trade-count z	0.430 → 0.570	0.0005	0.455	no
aggressor flow z	0.452 → 0.548	0.0025	0.494	no
ER(15)	0.519	0.1749	0.535	no — and not even significant

The answer to the question as asked is **YES**. ATR at entry separates dead-band entries at **0.683** with a permutation p of **0.0005**. Low ATR predicts “nothing is going to happen in points”, which is close to a tautology and is exactly why it works.

And §3's finding shows up again, and this time it does not bite. Against an ATR-normalised dead band the same feature reads **0.531** — a coin flip, for the third time in this report. But §7's band is defined in POINTS and the friction is paid in DOLLARS, and dollars are points $\times$ \$2.00. **The desk does not pay its bills in ATR.** That is the distinction the two sections needed: §3's collapse is fatal when you want to predict a MOVE, which is scale-free; it is not fatal when you want to predict SIZE IN MONEY.

★ BUT THE MONEY LEG FAILS, AND THE CAUSE OF DEATH IS NAMED

Detection is not a strategy. Declining the lowest-ATR 48.2% of the book takes it from **−\$2,813.99 (−\$1.64/trade)** to **+\$1,517.61 over 886 trades (+\$1.71/trade)** — a \$4,331 swing and a sign flip with no direction call at all. Then the robustness battery:

TEST	RESULT	VERDICT
Strip the best 3 trades from the kept book	+\$1,517.61 → −\$266.64	FAILS — three trades of 886 are the whole result
Walk-forward — fit the cut on the first half (ATR 12.84), apply it to the second	first half kept $−\$2.82/\text{tr}$ (vs $−\$1.97$ unfiltered); second half kept $+\$7.04/\text{tr}$ (vs $−\$1.32$)	FAILS — it loses in the half it was fitted on
Oracle ceiling (decline the TRUE dead band)	+\$5,715.26 ($+\$6.39/\text{tr}$)	the ATR rule captures 27% of it

So: BUILD #4 is not closed and it is not promoted. Its own kill criterion said “close it loudly if $\text{AUC} < 0.60$ ” — the AUC is 0.683, so it is not closed. But the dollar rule dies on strip-best-3, which is this desk's own standing kill test, so it is not built either. **It ships as a SHADOW arm with a threshold pre-registered before the data**, exactly like the Open Rider's ATR filter: pick the cut a priori, run it forward for a stated number of sessions, and grade it on strip-best-3 and a genuine out-of-sample leg rather than on the swing. A rule that inverts on three trades of 886 is a rule about three trades.

Q3 What to do about the day rider on Monday

Answered in full in **Part 1 §3b**, which is new in this revision, and on the card as **SATURDAY #4** and **HOLD #7**. The short version, because it is the bigger book and it deserves a straight answer:

- **Keep the no-stop decision.** Re-priced on this week's own tape at seven stop widths off the ATR the rider itself froze at entry: on the **twelve positions this week actually opened**, naked made **+\$572.50** and every stop width lost money, the best of them at $−\$617.44$. Your 2026-08-20 call wins by **\$1,189.94** on this week's evidence, not just on the 231 sessions it was taken on.
- **The one trade that argues otherwise is not a stop argument.** The 08-21 carry is $−\$2,149.44$ naked against $−\$34.56$ at the tightest stop — and it was carried through a 49-hour CME halt, where no stop can fill. The rule that

would have prevented it is the standing flat-by-20:40Z one, and it failed for an unrelated reason.

- ★ **And the real bill is somewhere else entirely.** Every rider order this week filled as a split whose residual leg was priced **base × (1 ± 0.001), adverse**, at a price the tape never printed. ★ **REV2 — the figure in this bullet was “twelve events, 20 of 116 lots, \$1,170.50, 74% of the loss” and it is superseded.** Part 1.6 §6(b) adjudicates it on a stated window and a stated unit, and the settled figure for 08-24→08-28 is **11 split orders / 15 residual legs / 18 lots / \$1,053.50 = 66.8%** of the rider's −\$1,576.94. The direction of this bullet is unchanged and the conclusion is stronger, not weaker: re-run over the following week (08-31→09-04) it is **7 more split orders, 15 lots, \$876.00, 0 of 7 on the tape**. That is SATURDAY #4, it outranks the stop question by a wide margin, and it is now the largest fixable item on this desk in two consecutive weeks.

The standing instruction that falls out: re-price the no-stop decision every Friday. It takes one script (scripts/rev2_rider_stop_week.py) and about a minute. The difference between a standing decision and an unwatched one is whether anybody checks it.

Q4 The bench saving's LOWER bound — and does the policy hold there?

Part 1 §7 prices the bench at **+\$810.44 over 127 signals** and then says, against its own number, that SUPPRESSED-OPEN is logged upstream of the 55-second veto and both absorption confirms — so for four of the six gates the benched set contains trades the gate's own veto would have refused. That makes \$810 a one-sided figure. Here is the other side.

HALF OF THE SAMPLE	SIGNALS	SAVING	CONTAMINATED BY THE VETO?
exhaustion_short +\$442.65 · abs_veto_short +\$253.60 · abs_veto_long +\$61.42 · rgv_short −\$75.12	108	+\$682.55	YES — all four have a post-suppression confirm stage
grind_long +\$207.44 · capitulation_long −\$79.55	19	+\$127.89	no — measured clean
All six	127	+\$810.44	the published upper bound

The veto's pass rate, measured rather than assumed. The 55-second absorption confirm's effect is directly visible on the shadow board, where

abs_veto_55s is the vetoed arm and thrust_loose is its un-vetoed parent on the same tape. This week: **45 fires against 91** — a **49.5% pass rate**. All-time: **433 against 783 = 55.3%**. So roughly half the contaminated set is signals the gate would never have taken.

BOUND	DESK-WIDE SAVING	SIGNALS	THE ASSUMPTION BEHIND IT
UPPER — as published	+\$810.44	127	every suppressed intent would have become a trade. It would not.
NEUTRAL — the veto refuses at random with respect to P&L	≈ +\$466	≈ 72	scale the contaminated half by the 49.5% pass rate
LOWER — the adversarial floor	+\$127.89	19	assume every refused signal was one of the losers the bench “saved” us from, so the whole contaminated saving is fictitious

The honest interval is \$128 to \$312 (★ REV2 — the upper end was \$810 and is retired; see the note below), and the policy survives at the floor — but not because of the money. At +\$127.89 the entire remaining saving is grind_long's +\$207 minus capitulation_long's −\$80, and \$7's own leave-one-day-out already kills the grind_long half (drop Tuesday and it costs \$35). **At the lower bound the saving is not distinguishable from zero.** That is not new information — it is what the bootstrap said — and it is why HOLD #1 already says keep the policy and stop quoting the number. This revision retires the \$810 as a quotable figure entirely. ★ **REV2 (2026-09-04) — and the central estimate moved with it.** The \$810/127 population was counted off SUPPRESSED-OPEN journal lines and cannot be rebuilt (the systemd journal is a ring buffer), so Part 1 §6 re-derives the saving on the durable signal_journal population: +\$312.35 over 117 signals, bootstrap band −\$623.74 to +\$1,208.31, **P(≤ 0) = 0.253**. The interval this section reports therefore reads **\$128 (this floor) to \$312 (the re-derived central estimate)**, not \$128 to \$810.

★ **And the thing worth saying out loud, which the proofread spotted:** the two gates measured CLEAN are the two we can say least about. grind_long's whole saving is one Tuesday and capitulation_long is n=5 and negative. **The clean half of the sample is the weak half.** The argument for the bench was never the dollar figure: it is that every cut points the same way. ★ **REV2 — the second half of this sentence is DELETED.** It read “and that in aligned trend benching is FREE (50 signals, −\$0.81 each) — which is the actual case for arming aggressively on a confirmed break”. That decomposition does not survive re-derivation on the durable population: chop is **+\$3.99** a signal (n=78) and aligned

trend +\$3.37 (n=35) — the same number — and a 10,000-draw label-shuffle placebo on the gap returns **P = 0.943**. **REFUTED** Nothing about arming may rest on it, HOLD #2 has been re-argued without it, and the \$128 floor stands on its own.

Q5 “Wait for one trend week” — how long is that wait?

Four recommendations are parked on it: the router timing config, the aligned-vs-chop gap, the two-ratchet, and the BIG-TREND state. Nobody has ever counted how often a trend week happens. Here it is, on every session in the lake from **2026-06-19 to 2026-08-28** — 51 days, 11 ISO weeks — using whole-day efficiency ratio (net move divided by summed path) on the desk's own 22:00Z-to-22:00Z day boundary.

★ THE FIRST THING THE COUNT FOUND IS THAT “ER ≥ 0.15” DOES NOT MEAN ANYTHING UNTIL YOU SAY WHICH BARS

PATH MEASURED ON	MEDIAN DAY	BEST DAY IN 51	DAYS WITH ER ≥ 0.15
1-minute bars	0.029	0.106	0 of 51
5-minute bars	0.065	0.222	6 of 51
15-minute bars	0.118	0.356	18 of 51
30-minute bars	0.156	0.526	27 of 51
60-minute bars	0.233	0.674	34 of 51

The same threshold on the same tape gives **0, 6, 18, 27 or 34** trend days depending only on the bar size the path is summed over — and the report quotes “ER 0.017–0.032” for this week without saying which. **A threshold whose meaning moves by 34 days across a rendering choice cannot gate four recommendations.** What IS stable is the ORDERING: 2026-08-04, 2026-07-30, 2026-08-18 and 2026-07-21 are the top four days at every bar size. So the usable definition is a RANK, not a level.

Fixing the definition at **5-minute path, ER ≥ 0.15** (which picks almost exactly the top decile) and counting by week:

ISO WEEK	BEST DAY'S ER	TREND DAYS
W25 (from 06-19)	0.113	0
W26	0.150	1 06-23
W27	0.143	0
W28	0.116	0
W29	0.105	0

ISO WEEK	BEST DAY'S ER	TREND DAYS
W30	0.152	1 07-21
W31	0.182	1 07-30 — the router's go-live
W32	0.222	1 08-04 — the best day on record
W33	0.151	1 08-13
W34	0.157	1 08-18
W35 — this week	0.101	0 the flattest week since W29

Six of eleven weeks contained a trend DAY. Not one contained a trend WEEK. The maximum in any week, across the whole record, is **one**. So four recommendations are waiting for an event that has never occurred, at a rate of **0.55 trend days per week** — and at that rate, ten trend days is **eighteen weeks** away.

WHAT TO DO INSTEAD — AND EVERYTHING THAT IS QUEUED BEHIND IT, IN ONE PLACE

Replace “wait for a trend week” with “accumulate N trend DAYS by a DATE”. A trend day is a day; the shadow arms can collect them one at a time, and the grade happens on the date whether the sample arrived or not — because “we did not get enough” is a result and an open shadow arm with no review date is not.

BLOCKED ON “ONE TREND WEEK”	WHERE	REPLACE WITH
The router timing config (window 45 / step 5 / hold 3)	BUILD #8	shadow it now; grade on 5 trend days or 2026-11-06, whichever comes first
The aligned-vs-chop router value gap	Part 2.6 §3.1, HOLD #2	it is already collecting; state the same date and stop calling it blocked
The two-ratchet runner clip	Part 2 §6	same date; it needs trend days, not a trend week, and it has been running since 07-26
The BIG-TREND exit rung	NOT-AN-ACTION #5	already refuted on 831,379 bars — it is not waiting for anything and should stop being listed as if it were

★ One of the four turns out not to be blocked at all, which is the second-best outcome of counting: it was killed three weeks ago and the queue never noticed.

Q6 capit_loose is 49% of the board's loss and the live gate is armed by cron. Which is it?

Both, and they are not the same mechanism — which is the answer the report owed and did not give.

ARM	CONFIG	N	NET	\$/FIRE	WIN%	WHAT IT IS
capit_loose	climax 2.5 · dom 0.60 · no flip · 1.0R	1,038	−\$16,681.00	−\$16.07	17.2%	the config this desk REVERTED on 2026-08-02
capit_live_mirror	the live gate's own parameters, unbenchd	352	−\$5,515.50	−\$15.67	12.5%	the honest counterfactual for the live gate
capitulation_long LIVE	climax 2.5 · dom 0.6 · flip=True · 2.0R	34	+\$63.00	+\$1.85	38.2%	what the desk actually booked, 07-21 → 08-28

1 — capit_loose is NOT the live mechanism, and Part 2 should stop implying it is. It runs `require_flip=False` at a 1.0R target. That is precisely the configuration this desk shipped on 2026-08-01 and withdrew on 2026-08-02 after finding the statistic behind it (“78% of bounces reach 1R”) was an MFE measure being used as a win rate — the ordering-correct number was 29%, against a 50% breakeven. `slot_strategy.py` records the whole autopsy in place. **So capit_loose's −\$16,681 is not an indictment of the live gate; it is a 1,038-trade gravestone for the config we already rejected, and it is doing its job.** Calling it “the only true relegation case on the board” is right, and it should be relegated — but relegating it says nothing about the live gate.

2 — The live gate's own counterfactual is bad, and that IS the case to answer. `capit_live_mirror` runs the live parameters with no benching: **−\$15.67 a fire over 352 fires at a 12.5% win rate.** The live gate took **34** of those 352 — the 10% the switch file happened to leave armed — and made **+\$63.00**. **The live gate's positive record is what BENCHING selected, not what the gate does.**

3 — And the +\$63.00 is smaller than this week's single win. The gate booked **+\$148.50** on 2026-08-28 at 14:00Z. Its all-time is **+\$63.00**. So on the other 32 fills of its life it is **−\$85.50**. One trade is more than the whole record.

★ **SO NOT-AN-ACTION #10 STOPS BEING A GOVERNANCE OBSERVATION AND BECOMES A DECISION**

reactivate_gates.py arms capitulation_long at 22:00Z nightly and the router does not clean it up, so it accounted for the large majority of the desk's armed hours this week by rota rather than by decision. On the evidence above there are exactly two defensible positions and “leave it as it is” is not one of them:

- **Either the router owns it like the other five** — remove it from the cron's roster and let the tick arm and bench it on the same criteria. Its unbenced counterfactual loses \$15.67 a fire, so leaving the arming decision to a clock is leaving \$15.67 a fire to a clock.
- **Or the asymmetry is written down as deliberate**, with the reason, in reactivate_gates.py itself — because a carve-out nobody can find the rationale for is indistinguishable from a bug, and this one has produced the week's largest single win, which is the most dangerous way for a bug to behave.

△ The honest limit: n=34 live fills over six weeks, and the mirror is a replay rather than a book of fills. Neither number is strong. But the DECISION is not a P&L question — it is about which instrument owns the switch, and that does not need n.

Q7 exhaustion_short — a gate benched pending evidence it can only make while armed

It has been off in gate_switches.env since 2026-08-18 and has not fired since. Two of this report's own instruments point opposite ways on it: Part 2.6 §3.2 prices the UP_OFF rule at **+\$516 of benched winners** over 98 blocked legs, and Part 1 §7 prices its bench at **+\$442.65 saved** over 34 signals but marks it NOT PROVEN because the stop sweep flips at 0.5×. Part 2 §9 asks for 40+ fills on the gate's own exits before the fade-scalp trial can be graded. **It cannot collect them while benched. That is a deadlock and the report should have named it as one.**

ERA	WINDOW	DAYS ARMED	FILLS	MACHINE EXITS	MANUAL_CL AIM	NET	
A — pre-trial, stop 1.0×ATR	07-21 → 08-14		15	146	138	8	−\$158.00

ERA	WINDOW	DAYS ARMED	FILLS	MACHINE EXITS	MANUAL_CL AIM	NET	
B+C — the fade-scalp trial	08-17, 08-18		2	14	7	7	+\$165.50
benched	08-19 → 08-28		0	0	0	0	—

The arithmetic that breaks the deadlock. Era A fired **9.7 fills per armed day**. In the trial, **7 of 14 fills (50%)** ended on the gate's own exits rather than on your claim button. So 40 machine-exited fills needs about **80 fills**, which is about **8.2 armed trading days**.

THE RE-ARM CONDITION, STATED SO IT CAN BE EXECUTED

Arm exhaustion_short for eight trading days: Monday 2026-08-31 through Tuesday 2026-09-15. Grade on **40 machine-exited fills** (MANUAL_CLAIM legs excluded from the grade, counted separately) or that date, whichever comes first. Then decide.

Why the exposure is acceptable, in numbers rather than in nerve. Era A's entire 146-fill history at the OLD config is **−\$158.00**, or **−\$1.08** a fill; 80 fills at that rate is about **−\$86**. Against that, Part 2.6 prices the UP_OFF bench at **+\$516** of benched WINNERS this week alone. **The expected cost of collecting the sample is roughly one bad hour and it is the only way the trial ever gets graded.**

And the alternative is worse than it looks. A gate benched pending evidence it cannot generate is not “parked”; it is retired without a verdict, and next Friday's report will find it in exactly this state again, as this one did. The desk's own standing lesson applies: a lapsed bench is not an arming case — but a bench that makes its own evidence impossible is not a bench either, it is a decision nobody has taken.

△ If you would rather not arm it, that is a legitimate answer — but then **BUILD #6 and the Part 2 §9 grading request should both be withdrawn** rather than carried, because they are asking for something the desk has decided not to produce.

Q8 Part 0's stale date was the fourth instance of this week's pattern. What else is like that?

This report's headline is that instruments fail silently and in the direction of reassurance — a router blind for 3h35m, two nightlies with no timer, two byte-identical shadow arms. Then Part 0 shipped a week stale, inside the report that names the pattern, because AS_OF was a literal. So: a sweep of the report's own generators and the nightly rollups for hard-coded dates and week constants. It took five minutes and it found six more.

FILE · CONSTANT	VALUE	VERDICT
scripts/fv7_common.py · WEEK_LO / WEEK_HI / DAYS / DAYNAME	2026-08-17 → 2026-08-22, and a hard-coded five-day list	★ THE LIVE ONE This is LAST week, and it is a shared helper imported by six Friday phase scripts — fv7_p1_live, fv7_p2_shadow, fv7_p26_router, fv7_p15_rehab, fv7_p25_musings, fv7_m2_idle. They were not used this cycle (the phases wrote their own HTML), but they are on disk, they import cleanly, and anyone who reaches for them gets a confident answer about the wrong week with no error. Same failure as Part 0, one import away from half the report.
scripts/abs_veto_sides.py · WEEK0	2026-07-27	CONFIRMED Already caught by Part 2 this cycle: its column headed “this week” is four and a half weeks and reports +\$4,069 where the truth is +\$948.50. Listed here because it is the same bug, not a separate one.
scripts/friday/progress_page.py · AS_OF	was 2026-08-21	FIXED THIS REVISION Now derived from the report week, with a --as-of override, and the page says so in its own closing paragraph.
scripts/friday_v7_build.py · CYCLE_START	2026-08-28 22:00Z — still, on 2026-09-04	★★ REFUTED IN REV 2 — this row blessed the constant that then broke the build This row said the literal was “the safe shape” because “if it is forgotten the build says so loudly rather than silently — a stale fragment gets a red dated banner”. Six days later the constant was NOT rolled and nothing said anything at all. data/friday_durable.log at 2026-09-04T21:13:24Z declares its cutoff as “artifacts older than 2026-08-28T22:00Z are STALE” — the previous cycle's — and under it every 08-29/08-30 fragment reads as current, so the runner logged HAVE ... skipping ten times (part1_live, day_rider, part2_shadow, part25_musings, part2_6_router, movement2_idle, rehab, run_charts, movement3, gf_report) and rebuilt none of them. No error, no banner, exit clean. That is why this document covers two different weeks. The failure mode a stale literal actually has is not “it screams”, it is “everything looks fresh” — a banner that fires on mtime < CYCLE_START cannot fire when CYCLE_START is the thing that is wrong. The safe shape is to DERIVE

FILE · CONSTANT	VALUE	VERDICT
		the cutoff from the run's own start (the same fix progress_page.py took for AS_OF on 08-29), not to remember to set it. That is SATURDAY #10 — and its shipped-check reads NOT DONE: scripts/friday_v7_build.py:524 still says 2026-08-28.
gf3_*.py, gf4_mgc_barsource.py · IS_END = 2026-08-15 · gf2_mgc_forward.py · SPLIT · gf_mgc_lab.py · OOS_CUT · router_churn.py · CUTOVER	various	CORRECT, AND MUST NOT BE DERIVED These pin an in-sample / out-of-sample boundary or a config change-over. A walk-forward split that moves with the calendar is not a walk-forward split — it leaks. Freezing them is the point.
exit_ladder_lab_v2.py · START · gf_cs_regime.py · START/END · momentum_veto_sweep.py · SINCE · adaptive_r_policy.py · ARCH_END	various 2026-07 dates	AGE HARMLESSLY, BUT AGE These are data-availability floors ("the lake starts here", "depth starts here"). They only ever narrow a sample, never mis-date one, so they cannot produce a wrong answer — but gf_cs_regime.py's END = 2026-08-15 silently drops the two most recent weeks from anything that re-runs it.

The rule the sweep suggests, and it is cheap. A date literal is fine when it names an EVENT (a config change, an out-of-sample boundary, the day a feed started) and dangerous when it names NOW (this week, as-of, the current cycle). The two are trivially distinguishable and nothing on this box distinguishes them. **One convention would close the whole class:** a constant that means “now” is either derived or carries a # EXPIRES: <date> comment that a pre-flight check reads and fails on. This report already has a pre-flight that refuses stale fragments; it has nothing that refuses a stale constant.

★ **And the answer to the second half of Q8 — “did anything check the report's own generators?” — is no.** Nothing did, until a proofread pass noticed that a chart was missing a week. Every instrument failure this report is proud of finding was found by the report; the report's own instruments were found by reading the report. That is one rung further out and it is not a different problem.

★ **THIS SECTION IS A WEEK YOUNGER THAN THE REPORT AROUND IT.** rev2_answers_0904.html was rebuilt **2026-09-04 22:51 UTC** and covers **2026-08-30 22:00 → 2026-09-04 20:59 UTC**. Parts 1 to 3 and Movements 2-3 were written on 2026-08-29/30 and cover the week to Friday 2026-08-28. Both are finished work; **do not line a number here up against a number there** — they are counting different days. The full seam is drawn at the top of this report.

P4 REVISION 2, 2026-09-04 — the ten open questions, answered from data

Garrath — a proofread pass ran against the first draft of this report and returned **20 issues and 10 unanswered questions**. The issues are fixed in the sections that carried them, each marked ★ **REV2**. The ten questions are answered here, in one place, because eight of the ten turn on the same fact: **this document grades the week to 2026-08-28 and is published on 2026-09-04, so a whole week of desk activity happened between the evidence and the reader** — and that week contradicts several of the report's headline conclusions. Lead with what survived: the desk's stop contract, which had never been tested, was tested four times and held; the census attribution bug that was inflating “money left on the table” is fixed and the figure falls by \$1,667; and the week the report could not see was **green on both books**. What did not survive: the Monday promotion, the \$810 bench saving, and the deploy the exit lab was proposing.

★★★ §1 — THERE ARE FOUR LOTS OPEN RIGHT NOW, AND THE REPORT'S FIRST LINE SAID THERE WERE NOT

LONG 4 MNQ from 29,641.25, -114.0 points at the 21:00Z close = – \$912.00 unrealised. Rev 1 opened “the desk is FLAT ... nothing is riding over the weekend” and quoted a desk_reconcile read at 22:26:32Z. That is the **2026-08-28** read, hand-copied forward. The live one says venue ? = tournament +0 + rider +4 → unaccounted ?, with “no venue read — IBKR truth unavailable”.

Why nothing closed it. venue_ok went false at 18:48:06Z and the watchdog has printed SKIP venue read failed every two minutes since. **Both** eod-flatten shots — 20:53Z and 20:57Z, the two 0nCalendar lines on gazbot7-eod-flatten.timer — died inside the same connectAsync TimeoutError, because every flatten path on this desk shares one connection. Port 4002 is refusing connections, which is the gateway being down rather than the 04:28Z hang where the port stays open.

flatten_blocked is true and PLACE_VENUE_STOP is False, so there is no venue stop underneath it either.

§1.1 What the Sunday reopen can actually do — and why the reopen is not the risk

The parquet lake has **no Sunday partitions at all** (that is SATURDAY #11 on the card), so a lake-based reopen study would silently measure Friday-close to Monday-open. capture.db does hold them. Here are all 7 Sunday reopens we own, on 5-second bars:

SUNDAY	FRI CLOSE	SUN OPEN	GAP (PT)	1ST-HOUR LOW	WORST FIRST HOUR FOR A LONG
2026-07-19	28,768.75	28,747.50	-21.25	28,706.75	-62.00
2026-07-26	28,306.75	28,501.50	+194.75	28,501.50	+194.75
2026-08-02	28,284.00	28,567.50	+283.50	28,550.00	+266.00
2026-08-09	29,841.00	29,851.50	+10.50	29,805.00	-36.00
2026-08-16	30,154.75	30,168.50	+13.75	30,166.25	+11.50
2026-08-23	29,370.00	29,392.25	+22.25	29,302.25	-67.75
2026-08-30	29,509.50	29,535.00	+25.50	29,427.00	-82.50

On four lots that is: a median gap worth \$178.00, a worst observed gap DOWN worth −\$170.00, and a worst observed first hour worth −\$660.00. **That is not the danger.** The 2026-08-21 carry — same book, same size, same mechanism — was −121.93 points at its Friday close and the Sunday gapped **+22.25 in its favour**. It was still claimed by hand on the Monday at −267.93 points for **−\$2,149.44**, which is the single largest item in this entire report. **The risk is the Monday session, not the gap.**

The instruction, and it is for Sunday 22:00Z, not Saturday. Nothing can be done before then: CME is shut and the venue is unreachable. At the reopen, get a venue read first and confirm it says 4. Then choose FLAT or MANAGE deliberately. If the read still comes back ?, that is the answer — you are flat-blind and no managing order should go into a book you cannot see. **SATURDAY #0.**

§2 The week's #1 stone is ANSWERED — and the answer kills the proposed fix, not the gate

BUILD #1 named abs_veto_short’s silence the week’s biggest open question: armed for 46m30s of a 328-point one-directional break and zero signals. Its hypothesis was a phase gap — “the router arms when a break is CONFIRMED; a thrust gate needs the moment a break STARTS, and on Friday those were about 25 minutes apart” — and its kill line was “if the measured lag is under ~5 minutes, the hypothesis is wrong”. On 08-28 there were no triggers to measure, so the report was reasoning about an absence. **On 2026-09-01 the gate fired twice and won all four legs.** Four clocks, measured on the 250ms tape (scripts/rev2_break_onset_lag_0904.py):

CLOCK	WHEN	WHAT IT IS
ONSET	07:14Z @ 29,556.50	the 60-minute swing high the move departed from — “the moment the break starts”, the thing a thrust gate is said to need
CONFIRM	08:06Z (+52 min)	the first minute the open-hour watcher’s own break trio is all true — ER-30 0.597 ≥ 0.35, ATR-14 18.2 ≥ 18, new session extreme. This is what the ROUTER waits for
ARM	08:07:29Z (+1 min 29 s)	the router wrote the switch 89 seconds after its own criterion went true. It is not the late one
TRIGGER	08:39:57Z and 09:01:00Z (+34 and +55 min after CONFIRM)	the gate’s own entries — 32 and 55 minutes after it was armed, sitting armed and silent in between

The lag is real and it is enormous: 86 minutes from onset to the first trigger and 107 to the second. But it runs the wrong way for the proposed fix. Arming on break ONSET would have armed the gate 52 minutes earlier and produced exactly the same two signals at exactly the same two times — the gate’s own trigger is the binding constraint, not the arm. **Do not build “arm on break onset”.**

And the trades were good. Both signals, four legs, **+\$163.50, 4 of 4 winners**, entered 86 and 107 minutes into a 328-point break. Three of the four exited on TARGET. Entering late into a confirmed break made money, which is the opposite of what the stone assumed. The router armed the gate three separate times that day (08:07–09:50, 13:16–14:50, 18:46–19:05,

3h36m53s in total) and only the first window produced anything — so the honest reading is that this gate has a very low duty cycle and a decent hit rate when it does fire, not that it is out of phase with the router.

§3 The week this report could not see — 2026-08-31 → 2026-09-04, graded

Everything from Part 1 to Movement 3 grades 08-24→08-28. Here is the week after, on the same ruler. **The desk booked \$804.50 across 47 legs (88 contracts), 76.6% of legs green** — and note that a 76.6% win rate is not why it is green; the four TARGET_100/200 rider exits are.

BOOK	LEGS	NET
The tournament (four gates fired)	12	\$191.00
The day rider	35	\$613.50
Whole desk	47	\$804.50

Per gate — and this is the table that contradicts the body of the report:

GATE	LEGS	NET	WIN %	EXIT MIX	WHAT THE REPORT SAYS ABOUT IT
abs_veto_short	4	\$163.50	100.0%	MANUAL_CLAIM ×1 · TARGET ×3	BUILD #1 called its silence “the week’s #1 stone” — armed for 46m30s of the exact tape it exists to trade, zero signals. It has since fired twice and won every leg.
grind_long	4	\$80.50	50.0%	MANUAL_CLAIM ×2 · STOP ×2	Part 1.5 §0a PARKED it on n and floor-clearance alone (2 lots, + \$63.50). It has since taken 4 more legs and both its losses are STOPS.
exhaustion_short	2	\$11.00	100.0%	TARGET ×2	BUILD #6 asked for it to be armed from Monday 2026-08-31 to produce evidence. The window

GATE	LEGS	NET	WIN %	EXIT MIX	WHAT THE REPORT SAYS ABOUT IT
					opened and it produced 2 fills of the 40 asked for — see §3.1.
abs_veto_long	2	−\$64.00	0.0%	STOP ×2	the aligned-momentum arm (HOLD #2) is the thing that arms this gate. Its only two legs this week both stopped.

Exit mix, whole desk:

EXIT	LEGS	NET
MANUAL_CLAIM	34	\$341.00
TARGET	5	\$146.00
STOP	4	−\$209.00
TARGET_100	3	\$330.00
TARGET_200	1	\$196.50

By day:

DAY	LEGS	DESK NET
2026-08-31	4	\$548.50
2026-09-01	13	−\$218.00
2026-09-02	13	\$1,159.50
2026-09-03	16	−\$344.50
2026-09-04	1	−\$341.00

THE FOUR CLAIMS IN THIS REPORT THAT THIS WEEK TOUCHES

THE CLAIM, AS PRINTED	WHAT 08-31→09-04 SAYS
“Nothing stopped out. Not once.” (Part 1 lede #5, §9, HOLD #3)	Four STOP exits, −\$209.00 — grind_long A/B at 09-03 13:56:18 and abs_veto_long A/B at 09-03 15:40:55. AND THAT IS GOOD NEWS all four filled at the stop: zero STOP_UNFILLED. The stop-contract fix had gone 30 trading days clean but essentially untested; it has now been tested four times and held. HOLD #3's own caveat — “thin and largely untested rather than proven” — is partly answered.
The short side fired nothing all week; abs_veto_short's silence is the #1 stone.	abs_veto_short +\$163.50 on 4 legs, 4 of 4 winners, and exhaustion_short +\$11.00 on 2 legs, both TARGET. The short side was the profitable side of the tournament this week. See §2 for what that does to the stone.
“The desk was switched on for 3.46% of the week.”	6.5% the week after — 46.03 gate-hours of the same 708. Nearly double, and the shape is different: exhaustion_short 20.43h and capitulation_long 17.6h carry it, both armed by reactivate_gates at Paris midnight rather than by the router. 3.46% is a fact about one week, not a policy.

THE CLAIM, AS PRINTED

Churn ran 0.237 gate-value changes an hour.

WHAT 08-31→09-04 SAYS

0.322/hour the week after (38 value changes, 33 from the router and 9 from reactivate_gates). Still nowhere near MONDAY #2's 1.0/hour kill line. The churn problem remains absent.

★ **One methodological note, because it changes how the duty cycle must be measured.** Counting armed time off the router's trial log alone gives **7.89 gate-hours** for this week — and exhaustion_short took a live trade at 09-03 22:59:53Z inside an armed window that log does not contain. gate_switches.env has three writers and the router records only its own. Both writers are replayed in scripts/rev2_armed_time_w36.py; the answer is 46.03 gate-hours, not 7.89.

§3.1 BUILD #6 — exhaustion_short's window opened, and it is not a grade

BUILD #6 asked for the UP_OFF rule to be re-priced on **40 machine-exited fills or 2026-09-15, whichever came first**, with the gate armed from Monday 2026-08-31. The window opened. Here is where it stands:

WHAT WAS ASKED

40 machine-exited fills

WHAT HAPPENED

2 — one signal, two legs, 2026-09-03 22:59:53Z, **both TARGET** at +\$5.50 each

Days armed

20.43h across the week — and all of it from reactivate_gates at 22:00Z, not from a router decision. The gate is armed overnight and benched in the day

Is the window running now?

No. data/gate_switches.env reads exhaustion_short=off at this render, and Friday 22:00Z is VENUE SHUT so reactivate_gates logged "already all off, nothing to do"

Does the -\$86 expected-cost estimate still hold?

Yes, untouched — the gate barely traded, so nothing has happened that could move it

Verdict: RE-ISSUE with the clock restarted, not repeat. Five sessions produced 2 of the 40 fills. That is a duty-cycle problem, not evidence for or against UP_OFF, and re-issuing the identical window would produce the identical answer. The row is re-dated to **2026-09-07 → 2026-09-22** with a new kill line: if the second window also produces fewer than 10 machine-exited fills, withdraw it and

answer the UP_OFF question off the shadow book instead. ★ **The one thing the window did establish:** the gate is not dead. When it fires it exits on its own targets, and both did.

§4 The Monday promotion, re-decided with the eighth week visible — **HELD**

The report's only live action was promoting abs_veto_55s two-sided, on **n=433, +\$5,302.50, +\$12.25 a trade, green in 7 of 7 ISO weeks**, with the stated reason being “another week of FORWARD evidence rather than another week of the same evidence”. That week has now happened. Re-run to 2026-09-04 (scripts/abs_veto_robustness.py, unchanged):

ISO WEEK	N	NET	\$/TRADE	LONG NET	SHORT NET
2026-W29	25	\$355.50	\$14.22	\$30.00	\$325.50
2026-W30	84	\$878.00	\$10.45	\$381.00	\$497.00
2026-W31	132	\$1,266.00	\$9.59	\$607.00	\$659.00
2026-W32	80	\$967.50	\$12.09	\$158.50	\$809.00
2026-W33	21	\$555.00	\$26.43	\$194.50	\$360.50
2026-W34	46	\$332.00	\$7.22	\$80.50	\$251.50
2026-W35	45	\$948.50	\$21.08	\$239.50	\$709.00
2026-W36 ← the week the report does not show	31	−\$426.00	−\$13.74	−\$346.50	−\$79.50

CUT	N	NET	\$/TRADE	VERDICT
As printed in §2.1, the scorecard and the card (to 08-28)	433	\$5,302.50	\$12.25	the promotion case
The unseen week (08-28 22:00Z → 09-04)	31	−\$426.00	−\$13.74	19.4% win. LONG 18 trades at −\$346.50 = −\$19.25/tr, SHORT 13 at −\$79.50
All time, to 2026-09-04	464	\$4,876.50	\$10.51	still green; still beats the un-vetoed thrust_loose by + \$5,600 across the board; both regimes and both walk-forward halves positive

So the decision, in the open. Three things changed and none of them refutes the candidate: (a) “green in 7 of 7 ISO weeks” becomes **7 of 8**; (b) the stated reason for promoting was forward evidence, and the forward evidence arrived arguing the other way; (c) **the card’s own kill criterion — “the LONG side below $-\$5/\text{trade}$ ” — is already tripped**, at $-\$19.25$.

HELD ONE WEEK, NOT KILLED Promoting a candidate the week after its own kill line trips is how a desk learns to ignore its kill lines. Nothing here is a robustness failure — expectancy all-time is $+\$10.51$ a trade and every robustness leg still passes — so the cause is not death, it is timing. Re-read Friday 2026-09-11 and promote only after two consecutive weeks with the LONG side above $-\$5/\text{trade}$.

★ **And the arithmetic downstream moves with it.** Part 3 §Q1 computed the live-loss haircut breakeven at **$1.64\times$** off the stale $+\$12.25$ — multiply only the loss side by λ and solve for zero. On the full 464 trades the wins are $\$13,999.50$ against $-\$9,123.00$ of losses, so breakeven is **$1.53\times$** . The arm now survives live slippage up to $1.53\times$ modelled rather than $1.64\times$, against a desk-wide standing finding that live losses run **$1.1\text{--}3.1\times$** modelled. At $1.50\times$ it is $+\$0.68$ a trade — a rounding error — and at $2.0\times$ it is $-\$9.15$. The headroom was already thin; it is thinner, and it is one more reason the promotion waits.

★ **One caveat that argues the same way.** A shadow mirror steps once per completed 1-minute bar and tests its stop against the bar CLOSE, while the live loss side is a resting venue STP that triggers intrabar. A wick through the stop that closes back inside is invisible to the mirror. The shadow number is therefore an upper bound on this arm, in a direction that gets worse in choppy tape.

§5 Which bench number the desk carries, and what HOLD #2 rests on now

The figure is +\$312.35 over 117 signals. The +\$810.44 over 127 is RETIRED. Not “un-quotable” — retired: its population was counted off SUPPRESSED-OPEN journal lines, the systemd journal is a ring buffer, and it can no longer be rebuilt or checked. Part 1 §6 replaces it with a saved, re-runnable replay over signal_journal, which is durable on disk from 2026-08-04. This revision propagates that result to the four places that were still printing the old one: the “if you read only this” box, the scorecard, HOLD #1, and the standing method-wound paragraph.

TEST	RESULT	READING
Stop-width sweep, 0.5× → 1.5×	+\$624.59 / +\$490.34 / +\$312.35 / +\$73.29 / +\$11.86	SIGN-STABLE which is what earns it a verdict at all — but read the SLOPE: it decays monotonically. Most of what benching “saved” is the desk’s stop being tight enough to guarantee those trades lost
Leave-one-day-out	drop Tue → +\$8.39; drop Thu → +\$16.62	two days of five carry 100% of the gross positive
Strip-the-best-3	+\$312.35 → +\$66.31	three signals of 117 are 79% of the result
Bootstrap, 10,000 draws	−\$623.74 to +\$1,208.31, P(≤0) = 0.253	a direction, not a measurement
Label-shuffle placebo on the regime split	P = 0.943	REFUTED chop is +\$3.99 a signal (n=78) and aligned trend +\$3.37 (n=35) — the same number

What that costs HOLD #2. The whole stated reason for keeping the aligned-momentum arm was that sentence: “in aligned trend, benching is FREE — 50 signals, −\$0.81 each — which is the actual argument for arming aggressively on a confirmed break”. It does not survive its own placebo. So, plainly: **the arm is now held on judgement, with no number behind it.** The honest case is that it is the operator’s standing instruction of 2026-08-05, that the one arm it produced was right, and that no test says it is wrong. That is a reason to leave it alone. It is not a reason to arm harder, and nothing on this card may cite the free-in-trend sentence again. HOLD #2 now says exactly that.

★ The named caveat on any bench figure still stands and is one of the reasons the \$810 is gone: SUPPRESSED-OPEN is logged at tournament.py:306, upstream of the 55-second veto at tournament.py:360, so for four of the six gates the benched population contains trades the gate’s own veto would have thrown away. Anything built on it is an UPPER BOUND. Part 3 §Q4's independent floor is +\$128.

§6 Is the grind MED-TREND branch deployed on Monday? **NO** — one verdict, and the cause of death named

The report gave two opposite instructions about one file on one Monday. Part 2 §10 graded the rung “SHADOW → deploy BANDED” and §11 said to deploy it; the MONDAY window intro said “Part 2 graded twenty of them and all twenty failed, so data/exit_overrides.json is not touched”, and HOLD #4 said change nothing. **They are not contradicting each other about the same test — they are applying different bars, and neither said so.**

TEST	GRIND MED-TREND, A 0.75R / B 1.0R	PASS?
In-sample \$/signal (n=215)	+\$7.70	✓
Strip-the-best-3	+\$5.60	✓
Leave-one-day-out (worst fold)	+\$6.20	✓
Both halves	+\$13.50 / +\$1.90	✓
Parameter plateau	5 adjacent cells positive, monotone gradient; 8 of 48 cells positive	✓ — a plateau, not a spike
Out-of-sample POOL split (the retired V5 desk's real fills)	-\$4.50/signal on n=52	✗
Its own live forward arm (rung_grindA_med_05, running since 08-16)	-\$8.36/trade on n=21, against its control rung_grindA_med_live at — \$9.29	✗

That is the reconciliation. The twenty failed cells are graded on a FOUR-test bar that includes the pool split; §7.2's “PROVEN” is a THREE-test bar. MED-TREND is the one cell that clears three and fails the fourth — so “19 of 20 fail on three tests, 20 of 20 fail on four” and both sentences were true about different things. And the twenty

are four rungs × five families, of which BIG-TREND is only one rung; “the twenty failed cells” was never a statement about BIG-TREND alone.

CAUSE OF DEATH FOR THE DEPLOY: THE OUT-OF-SAMPLE LEG. Two independent out-of-sample legs, one historical and one forward, are both negative, and the forward one is the desk’s own money. The paired comparison is genuinely encouraging — on the 19 entries both A arms share, the proposal makes −\$116.00 where the live ladder makes −\$176.50, a **+\$60.50 edge**, better on 6, worse on 5, identical on 8 — but **the whole rung is losing money in both configurations**. A better exit on a losing entry is an entry problem wearing an exit costume.

ONE verdict, which HOLD #4 and the MONDAY window intro now both repeat: data/exit_overrides.json is not touched this weekend, MED-TREND included. It stays in shadow with `rung_grindA_med_live` as referee. Deploy when EITHER out-of-sample leg turns positive: the V5 pool above \$0/signal, or the forward arm above its control over $n \geq 40$. Not on the in-sample number.

§7 How much of “\$10,512 left on the table” was really sat out? \$8,845 — and nine runs move buckets

The census scored a trade taken inside a run as “sat out”. `run_census.py` attributed trades over `start ± 300s` while a run is `W=180` five-second bars = **15 minutes** long, so any entry more than five minutes into a run’s own body was invisible. The 09-03 13:36 UP +130pt run had an attribution window of 13:31–13:41; `grind_long` bought both lots at **13:42:05** and was stopped at 13:56 — inside the run, scored as an absence.

The fix is one line — attribute over `[start - 300s, start + W×5s]`: five minutes of lead-in for an early entry, then the run’s whole body. Re-frozen on the same tape:

BUCKET	AS PRINTED	ATTRIBUTED OVER THE RUN'S BODY	WHAT MOVED
Caught (aligned)	1	10	nine runs were being scored as absences while the desk was in them
Fought (against)	2	2	unchanged
Sat out	73	64	—
Hindsight ceiling left on the table	\$10,512	\$8,845	the headline overstated the sitting-out by \$1,667, or 16%
What the caught runs actually banked	—	+\$1,912 on a \$1,846 ceiling	104% — see below

The Gate column also stops naming only day_rider on a week when the tournament took 12 legs: abs_veto_short, grind_long and abs_veto_long all now appear against runs they were actually in.

Two template faults fixed alongside it. (1) The lede hard-coded “which is only {conv}% of”, so a week where the caught runs BEAT their ceiling printed as a shortfall — “only 359% of”. It now branches, and says why over 100% is legitimate: the ceiling prices the run’s own 15-minute move at one lot, and a position still open when the run ends goes on earning outside the census window. (2) A positive FOUGHT bucket now explains itself — the bucket is decided by which SIDE the desk was on relative to the run, not by whether the trade won, so a position opened against a run that reverses inside its own 15 minutes books money while still counting as fighting it. Read a green FOUGHT row as “we were on the wrong side and got away with it”, never as a reason to fight runs on purpose.

★ **What this does NOT change.** Movements 2 and 3 are computed on the **2026-08-28** freeze, which is a different week entirely and which this fix has not been applied to. Both sections now say so in a banner at the top rather than claiming their run list matches the Movement 1 printed above them — it does not, and it never did in this document.

§8 The rider’s off-tape residual leg — one figure, and

IT IS STILL HAPPENING

Four different totals for one artefact were in print. The adjudicated one is Part 1.6 §6(b)'s, and it is now the only one anywhere in this report, on the card included:

WINDOW	SPLIT ORDERS	RESIDUAL LEGS	LOTS	COST	SHARE OF THAT BOOK’S P&L
The report week, 08-24 → 08-28	11	15	18	\$1,053.50	66.8% of the rider’s – \$1,576.94
★ The week AFTER, 08-31 → 09-04 — the verification SATURDAY #4 itself asked for	7	7	15	\$876.00	143% of the rider’s +\$613.50 — the rider was green despite it
All time	34	39	50	\$2,914.50	—

The verification asked for “a week with zero off-tape legs, or with the residual priced at the tape”. It got neither. Seven more split orders, seven residual legs, fifteen lots, every one adverse, and **0 of 7 on the tape** within ±30 seconds. Every one is 29.0–29.5 points against, which is the 0.1% factor at this price level:

WHEN	ORDER	SIDE	LOTS	RESIDUAL PRICE	POINTS ADVERSE	COST
09-01 17:28Z	rider-2326	SELL	1	29,105.00	29.0	\$58.00
09-01 18:39Z	rider-2341	BUY	1	29,089.75	29.0	\$58.00
09-02 12:27Z	rider-2401	SELL	1	29,084.50	29.0	\$58.00
09-02 15:04Z	rider-2441	BUY	3	29,171.50	29.0	\$174.00
09-03 10:06Z	rider-2486	BUY	3	29,180.00	29.0	\$174.00
09-03 15:47Z	rider-2516	SELL	3	29,479.50	29.5	\$177.00
09-03 19:48Z	rider-2546	BUY	3	29,551.75	29.5	\$177.00

Every leg carries a distinct, correctly-formatted IBKR execId, so this is not obviously a fabricated row — but 0.001 / 1.001 / 0.999 appears **nowhere** in src/gazbot7/, which is the other half of the question and is still open. Either way it is plumbing, not strategy: **SATURDAY #4 is the largest single fixable item on this desk in two consecutive weeks.**

§9 So what do I actually DO this weekend? The card is re-dated, and every 2026-08-28 row now carries a shipped-check

The card was addressed to a weekend that had gone: five window headers reading “SATURDAY · BEFORE THE SUNDAY 22:00Z REOPEN” and “MONDAY · AT THE DESK”, MONDAY #1-#4 all saying “Arms: Monday”, BUILD #6 asking for an arming window that opened on 2026-08-31. Published 2026-09-04, those point four to six days into the past, and 42 of the 54 rows carried a -0828 suffix with no way to tell which had been actioned.

Both are fixed. Every window is re-dated to Sat 2026-09-05 / Sun 2026-09-06 / Mon 2026-09-07, and every -0828 row carries the verdict of a probe run against this box at render time — a file, a unit, a grep or a database read, never a memory (scripts/rev2_shipped_check_0904.py). Of the 25 shippable rows: 19 NOT DONE, 5 CARRIED, 1 re-measured, and 0 DONE. The other 17 are HOLD or NOT-AN-ACTION rows — standing positions with nothing to ship.

ROW	VERDICT	THE PROBE, AND WHAT IT READ
BUILD #2 Build a DAY-level gate for the Open Rider that reads its feature ONCE at 13:00Z and sets a session flag	NOT DONE	no session-flag file is written at 13:00Z anywhere in data/; the 20-session flag-vs-outcome log the row asks for does not exist
BUILD #4 ★ 48.2% of entries go NOWHERE and pay \$10 each — and REV2 shows ATR can SEE them (AUC 0.683). Ship it as a pre	NOT DONE	no dead-band decline variant appears in the shadow board's strategy list, and no pre-registered threshold file exists
BUILD #5 Two shadow A/Bs are byte-identical and have therefore tested NOTHING — fix the arm wiring	NOT DONE	shadow_trades counts [('cx_clip_brk_live', 388), ('cx_clip_brk_standdown', 388)] — the pair is still running and still matched; no divergence fix is in the shadow wiring
BUILD #7 Trial the untradeable STAY-OUT cutoff at 45 in SHADOW — direction proven, exact value fitted	NOT DONE	no shadow log of the untradeable meter at cutoff 45 beside 65 exists in data/; and Part 2.6 has since graded the meter itself as having no discrimination at any cutoff, which makes this row moot
BUILD #10 ★ The day rider does not MANAGE	NOT DONE	src/gazbot7/day_rider.py still routes the manage path through the RTH-

ROW	VERDICT	THE PROBE, AND WHAT IT READ
positions opened outside 13:30-21:00Z — it carries them		gated drift read; tonight's open position is the demonstration — it was entered 14:15:16Z with entry_atr 6.8 and arm_atr 19.7 and has no trail
BUILD #11 Shadow a dwell / 2-of-3-minute confirmation on the day rider's entry — the floor crossing is a coin flip at th	NOT DONE	no strategy matching 'dwell' has any rows in shadow.db — the rider entry-dwell variant was never armed
BUILD #12 Shadow benching grind_long in NORMAL_CHOP and CLEAN_TREND — together with the 13:00Z hour, never separately	NOT DONE	no strategy matching 'grind_bench' has any rows in shadow.db — the grind regime+hour bench was never armed
BUILD #13 Shadow the SESSION-BOUNDED gold run-boarder — the only MGC candidate that beat both its mirror and its random	NOT DONE	no strategy matching 'run_board' has any rows in shadow.db — the session-bounded gold run-boarder was never armed
MONDAY #1 Promote abs_veto_55s TWO-SIDED — the only arm on a 79-row board that survived everything	NOT DONE	data/gate_switches.env names six gates and abs_veto_55s is not one of them; all six read =off. The shadow arm is still shadow
MONDAY #2 ★ Attack the 20-minute arm lag — the router's dominant error is no longer direction, it is LATENESS	NOT DONE	gazbot7-router-tick.timer reads OnCalendar=*:0/5 and the last 12 trial-log ticks are [4, 5, 49] minutes apart, so the TICK GRID was already 5 minutes and the '15m→5m' half of this row was a no-op — the router's own note at router_tick_durable.py: 343 says so. The half that mattered, the prompt line 'arm on the FIRST confirmed break impulse, not the fifth', is ABSENT from the prompt
MONDAY #3 Make the router quote a WINDOW and a FAMILY whenever it cites the shadow board	NOT DONE	no line in scripts/router_tick_durable.py requires a shadow-board citation to name its date window and arm family; the prompt is unchanged on this point
MONDAY #4 Filter the untradeable stop-rate meter to tournament gates only — it has never once discriminated	NOT DONE	the meter's stop-rate input still reads both books — no filter on gate NOT LIKE 'day_rider%' exists in 64 file(s) that mention the meter. Part 2.6 separately grades the meter itself as dead (rho +0.17 / -0.09 / +0.11), which is why this row is re-issued as WITHDRAW rather than as a tuning
SATURDAY #1 Put router_nightly.py and selector_nightly.py on a timer — both have been silently dead for a fortnight	NOT DONE	zero units matching router-nightly/selector-nightly in /etc/systemd/system; newest router_nightly file 2026-08-28.json, newest selector_nightly 2026-08-28.json — both still stop at the 08-28 report's own rebuild, so nothing has written since
SATURDAY #2 ★ The router went blind for 3h35m and every health instrument on the box said it was fine	NOT DONE	scripts/router_health_check.py DOES read data/router_headless.log and alarms an ABORT streak (ABORT_STREAK_ALARM=3) — that half predates the card (last commit

ROW	VERDICT	THE PROBE, AND WHAT IT READ
		2026-08-13). But the alert text still says "expired login: run `claude` and /login on the box", which is the half the play asked to fix, and the file has not been touched since %ad
SATURDAY #3 Make open_hour_watch.py match its own unit file, or give its prompt header an EXPIRY	NOT DONE	scripts/open_hour_watch.py still contains os.replace(tmp, SWITCHES) — it writes the switch file — while gazbot7-open-hour-watch.service still describes itself as "ALERT-ONLY". Neither side of the contradiction moved
SATURDAY #5 ★★ The day rider's 20:40Z hard flat sits on the edge of a recurring gateway blackout with ZERO margin — it cos	NOT DONE	gazbot7-eod-flatten.timer still reads ['Mon-Fri 16:53 America/New_York', 'Mon-Fri 16:57 America/New_York'] — 20:53Z and 20:57Z, unchanged. No 20:25Z attempt exists and the page is still downstream of the flatten. ★ THIS IS THE ROW THAT BIT TONIGHT: the venue read has failed every two minutes since 18:48:06Z, the 20:40Z flatten died on connectAsync TimeoutError, and four rider lots are open over the weekend
SATURDAY #6 ★★ The Parquet lake silently blends four extra timeframes into its bars view — 44.4% of July's minutes are inf	NOT DONE	src/gazbot7/lake.py still exposes the bars stream without a timeframe filter, so the backfill files still inflate a bar count
SATURDAY #7 Give the router a weekend and holiday check — it armed a gate 38.7 hours before any market opened, on zero bar	NOT DONE	scripts/router_tick_durable.py has no weekday/market-calendar guard, so the router will again evaluate and write across this weekend's ~49h CME halt
SATURDAY #8 Both nightly scorecards only count the modelled side — print realised P&L beside modelled, or stop publishing	NOT DONE	neither nightly has run since 2026-08-28 (see row 1), so neither has been changed to print a realised figure beside its modelled one. This row is blocked by row 1 and should ship with it
BUILD #1 ★★ THE WEEK'S #1 STONE — abs_veto_short was armed for 46m30s of the exact tape it exists to trade and produced	CARRIED	no (break-onset, trigger) table exists on disk. But the gate has since FIRED — twice on 2026-09-01, four legs, +\$163.50, 4 of 4 winners — so the measurement now has live cases instead of an empty tape. Answered in this revision rather than re-issued
BUILD #3 ★★★ THE STRUCTURAL FINDING — every precursor feature on this desk is a proxy for ATR, and that is why selectio	CARRIED	a research instruction, not a shippable change. The bar it set — any new feature must hold its AUC after ATR conditioning — is unchanged and still the right bar
BUILD #6 The exhaustion_short UP_OFF rule is the week's single biggest over-bench	CARRIED	data/gate_switches.env reads exhaustion_short=off RIGHT NOW, and no per-hour reprice of the UP_OFF rule exists on disk. The window did open: the gate took 2 machine-exited leg(s) for \$+11.00 since 2026-08-31 — against the 40 fills the row set as its bar
BUILD #8 Shadow the router timing config	CARRIED	the row's condition is a session with whole-day ER >= 0.15. The

ROW	VERDICT	THE PROBE, AND WHAT IT READ
(window 45m / step 5m / hold 3) — but the grade needs a TREND week		08-31→09-04 week did not supply one either — see the bridge section — so the shadow grade is still waiting on the same trend day it was waiting on
BUILD #9 Gold: 80 runs and \$15,196 of ceiling on the 08-28 freeze, and no gate at all — an empty seat, not a bleed CARRIED		standing research instruction. It has since been sharpened, not shipped: the lake's gold year serves the wrong contract for 22% of its minutes, which sits underneath every gold number the seat was being hunted with
SATURDAY #4 ★★ 74% of the day rider's loss this week is a FILL ARTEFACT — every entry is a split order whose residual leg MEASURED		re-run in this revision — see the off-tape audit below. This is exactly the verification the row asked for ('re-run the audit next Friday')

The one that matters most is SATURDAY #5. “Move the rider’s first flatten attempt earlier and put the page UPSTREAM of the flatten” was issued last weekend, is NOT DONE, and is the row that bit tonight: gazbot7-eod-flatten.timer still fires at 16:53 and 16:57 New York, both shots died in the same timeout, and four lots are open. **If one thing gets done this weekend, it is that one.**

WHAT I COULD NOT CLOSE, SAID PLAINLY

- 1. The 08-31→09-04 week is n=12 tournament legs on 4 gates.**
It contradicts four claims in this report and it is not big enough to REPLACE any of them. It is printed beside them, not instead of them. One good week and one bad week are the same size of evidence.
- 2. Movements 2 and 3 have NOT been recomputed on the corrected census attribution.** They run on the 2026-08-28 freeze, which is a different week; re-freezing it would mean re-running both labs, and neither has run this cycle. The attribution bug is fixed in the script, so it will be right from the next freeze onward, and both sections now name their own freeze rather than claiming to share Movement 1's.
- 3. The bench replay is still in-sample by construction** — it reprices decisions already made, on the tape they were made on

— and every replayed figure is one lot on the Lot-A ruler while the live desk runs two. The Lot-B tail is modelled nowhere.

4. **§2 is n=2 signals.** Two clean cases move a mechanism question in a way a fortnight of silence cannot, and they are still two cases. What they rule out is “arm on onset” as a fix, because the gate’s trigger came after the arm on both of them. They do not establish a median lag.
5. **Nothing here reprices MGC.** The gold half of this report sits on a parquet lake whose year backfill serves the wrong contract for 22% of its minutes; that is SATURDAY #9 and it is unfixed.